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Hasan Khan

Second edition, September 2026

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Complete course notes for IB Mathematics:
Analysis and Approaches, Higher Level

Second edition, September 2026

These notes are designed for DP students aiming for a 7, built on a foundation of clarity and intuitive understanding. My sincere thanks go to the students and colleagues whose questions, feedback, and corrections have shaped every chapter.

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Graphic design and typography are hobbies he takes far too seriously, which is why this book looks the way it does. Away from the desk he is a *pathfinder*, more interested in the route than the arrival.

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Set in TeX Gyre Pagella, TeX Gyre Heros and Latin Modern, using XeLaTeX.

How to use these notes

Every badge the book uses, in one list. Each marks the role of an item, or the kind of thinking it invites.

DEF	A definition to understand and use precisely.
KEY	A key result or formula worth retaining.
KEY · IB	A key formula also printed in the IB formula booklet.
DEF · IB	A definition also printed in the IB formula booklet.
CAUTION	A mistake made here often enough to be worth naming.
TIP	Exam craft: what earns the mark, or what loses it.
GDC	A calculator is required, or genuinely useful.
EXACT	Leave the answer exact: a surd, a fraction, or $k\pi$.
TOK	A Theory of Knowledge question about mathematics.
IM	Mathematics in its historical and cultural setting.
RW	A real-world application or modelling context.
EXT	Beyond the syllabus: enrichment, not examinable.
HL	Additional Higher Level content: examined at HL only.
HL ONLY	A chapter whose whole content is Higher Level.

REFLECTIONS

Every chapter closes with a page of Theory of Knowledge questions, historical and international context, real-world uses, and enrichment. Nothing there is examined, and all of it is worth reading.

PROBLEM COLOURS

3. section skills

17. end-of-chapter questions

38. challenge investigations

BetterMath · For IB Mathematics AA HL

Contents

Symbols and notation

Anything defined only inside one chapter is explained where it appears.

NUMBERS AND SETS

$\mathbb{N}$	natural numbers
$\mathbb{Z}$, $\mathbb{Z}^+$	integers, positive integers
$\mathbb{Q}$	rational numbers
$\mathbb{R}$	real numbers
$\mathbb{C}$	complex numbers
$\in$	is an element of
$\cup$, $\cap$	union, intersection
$\emptyset$	the empty set
$\lvert x\rvert$	modulus of x
$0.\overline{27}$	recurring decimal, $0.2727\ldots$
$\approx$	approximately equal to
$\neq$, $\leq$, $\geq$	not equal to, at most, at least
$\pm$	plus or minus
$\propto$	is proportional to

LOGIC AND PROOF

$\implies$	implies
$\iff$	if and only if
$\equiv$	is identically equal to
$\blacksquare$	end of proof

ALGEBRA AND COUNTING

$n!$	n factorial
${}^nC_r$	combinations: n choose r
${}^nP_r$	permutations
$\sum$	sum
u_n	the nth term
S_n, S_∞	sum of n terms, sum to infinity
d, r	common difference, common ratio
Δ	the discriminant, b^2-4ac

FUNCTIONS

$f(x)$	the value of f at x
f^{-1}	the inverse function
$f\circ g$	composite: f after g
$\lfloor x\rfloor$	the greatest integer $\leq x$
$\to$	tends to
∞	infinity

CALCULUS

$\lim_{x\to a}$	limit as x tends to a
dy/dx	derivative of y with respect to x
f', f''	first and second derivatives
d^ny/dx^n	the nth derivative
$\int$	integral
$\int_a^b$	definite integral from a to b
$+C$	constant of integration
$\dot{x}$, $\ddot{x}$	derivatives with respect to time

GEOMETRY AND ANGLES

$[AB]$	the segment with endpoints A and B
AB	the length of $[AB]$
(AB)	the line through A and B
$A\hat{B}C$	the angle at vertex B
$\triangle ABC$	the triangle ABC
$\perp$, $\parallel$	perpendicular to, parallel to
$^\circ$, rad	degrees, radians

VECTORS

$\mathbf{a}$	a vector
$\lvert\mathbf{a}\rvert$	the magnitude of $\mathbf{a}$
$\overrightarrow{AB}$	the vector from A to B
$\binom{h}{k}$	a column vector; a translation
$\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$	unit vectors along the axes
$\mathbf{a}\cdot\mathbf{b}$	scalar (dot) product
$\mathbf{a}\times\mathbf{b}$	vector (cross) product

COMPLEX NUMBERS

i	the imaginary unit, $i^2=-1$
z^*	the conjugate of z
$\lvert z\rvert$, $\arg z$	modulus and argument of z
Re, Im	real part, imaginary part
$\mathrm{cis}\,\theta$	$\cos\theta+i\sin\theta$

STATISTICS AND PROBABILITY

$\bar{x}$	the sample mean
μ, σ, σ^2	population mean, standard deviation, variance
Q_1, Q_3	lower and upper quartiles
IQR	interquartile range
r	Pearson's correlation coefficient
$P(A)$	the probability of A
A'	the complement of A
$P(A\mid B)$	the probability of A given B
$E(X)$	the expected value of X
$\mathrm{Var}(X)$	the variance of X
$\sim$	is distributed as
$B(n,p)$	binomial distribution
$N(\mu,\sigma^2)$	normal distribution

ABBREVIATIONS

AA HL	Analysis and Approaches, Higher Level
AHL	additional higher level content
GDC	graphical display calculator
s.f., d.p.	significant figures, decimal places
IA	internal assessment
TOK	Theory of Knowledge

NUMBER & ALGEBRA

CHAPTER 1

Exponents & Logarithms

1

SYLLABUS 1.1 1.5 1.7

Japan lies in one of the world's most active seismic zones. Suppose a seismologist asks you to plot two recordings on one axis. The first has an amplitude (the height of the wave) of 10^4. The second is 10^9, a hundred thousand times larger.

An ordinary scale, with evenly spaced marks, cannot show both. If the marks are fine enough for the first, the second lies far beyond the page. If the scale is long enough for the second, the first sits so close to zero that you cannot see it.

So let equal spaces along the axis mean multiplication by 10 rather than addition of 1. The scale then reads $10^0, 10^1, 10^2$, and the two recordings, with exponents 4 and 9, sit five equal steps apart. That exponent is the **logarithm** of the amplitude, and it is the idea this chapter is built on.

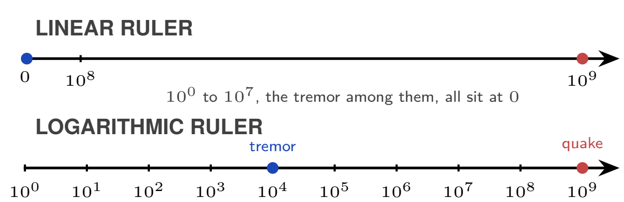


Figure 1.1 A logarithmic scale separates values that collapse near zero on a linear scale.

Earthquake magnitude works this way, and so do pH, decibels, and the brightness of stars. Writing a quantity as a power of ten turns multiplication into addition, so a range too wide to plot fits on an axis you can read. This chapter builds that idea: the laws of exponents, standard form, the move between exponential and logarithmic form, and equations with the unknown in the exponent.

1.1 What an Exponent Counts

An exponent, also called a *power* or an *index*, is shorthand for repeated multiplication. In a^n, a is the **base** and n is the **exponent**. When n is a positive integer,

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}.$$

For example, $a^3 = a \cdot a \cdot a$.

When you multiply powers with the same base, it is tempting to multiply the exponents. Writing out the factors shows what actually happens:

$$a^3 \cdot a^2 = (a \cdot a \cdot a)(a \cdot a) = a^5.$$

There are five factors of a, not six. Since the exponent counts how many times the base appears as a factor, the exponents are **added**:

$$a^m \cdot a^n = a^{m+n}.$$

Division cancels common factors, so the exponents are **subtracted**:

$$\frac{a^5}{a^2} = \frac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a} = a^3 = a^{5-2}.$$

Raising a power to another power repeats the whole power, so the exponents are **multiplied**:

$$(a^2)^3 = a^2 \cdot a^2 \cdot a^2 = a^6 = a^{2\times3}.$$

A power outside a bracket applies to every factor inside:

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

The same laws extend naturally to zero and negative exponents. For $a \neq 0$, the subtraction rule gives $a^3/a^3 = a^{3-3} = a^0$; since any nonzero number divided by itself is 1, this forces $a^0 = 1$. Continuing the same pattern gives $a^{-n} = 1/a^n$. These values are not imposed on us; they are the only ones that keep the laws consistent. The main laws are collected below.

DEF For $a, b \neq 0$ and $m, n \in \mathbb{Z}$:

$a^m \cdot a^n = a^{m+n}$	product of powers: add the exponents
$\dfrac{a^m}{a^n} = a^{m-n}$	quotient of powers: subtract the exponents
$(a^m)^n = a^{mn}$	power of a power: multiply the exponents
$(ab)^n = a^n b^n$	power of a product: raise each factor
$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	power of a quotient: raise top and bottom
$a^0 = 1$	zero exponent: gives 1
$a^{-n} = \dfrac{1}{a^n}$	negative exponent: take the reciprocal

CAUTION Do not confuse $(a^m)^n$ with a^{m^n}. In $(a^m)^n$, the power of a power, you multiply the exponents. In a^{m^n} you start at the top: first work out m^n, then raise a to that result. The two are usually not equal: $(2^3)^2 = 2^6 = 64$, but $2^{3^2} = 2^9 = 512$.

CAUTION A negative exponent means *reciprocal*, not *negative value*. $2^{-3} = \frac{1}{8}$, not -8.

1.1.1 Worked examples

How do these rules combine when a problem has several operations at once?

EXAMPLE 1.1

Simplify $\dfrac{10c^7}{5c^3}$.

Split the coefficient and the variable and handle them separately.

$$\frac{10c^7}{5c^3} = \frac{10}{5} \cdot c^{7-3} = 2c^4.$$

Apply one rule at a time.

EXAMPLE 1.2

$(2x^2y^{-4})^5$ contains several operations: a bracket raised to a power and a negative exponent.

Raise each part in turn: the 2 to the 5th, x^2 to the 5th, and y^{-4} to the 5th.

$$(2x^2y^{-4})^5 = 32x^{10}y^{-20}.$$

The most common error here is forgetting to raise the coefficient.

CAUTION The bracket in $(2x^2)^3$ applies to everything inside it, including the 2. $(2x^2)^3 = 8x^6$, not $2x^6$.

1.1.2 Beyond counting factors

The idea of a^n as repeated multiplication only makes sense when n is a positive integer. For example, what could 2^{-3} mean? You cannot multiply 2 by itself -3 times. And what is $3^{1/2}$, or 4^0? You cannot take half a factor, or no factors at all. Yet each of these still has exactly one sensible value. We choose that value so the exponent laws keep working:

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}, \qquad 3^{1/2} = \sqrt{3}, \qquad 4^0 = 1.$$

The next sections show where these values come from.

1.1.3 Surds and fractional exponents

What is $9^{1/2}$? Since $\frac{1}{2}$ is not an integer, repeated multiplication cannot answer this directly. But the power-of-a-power law gives

$$\left(9^{1/2}\right)^2 = 9^{(1/2)\times 2} = 9,$$

so $9^{1/2}$ must be the non-negative number whose square is 9:

$$9^{1/2} = \sqrt{9} = 3.$$

Notice the value is 3, not ± 3: the symbol $\sqrt{9}$ means the **principal** (non-negative) square root.

DEF For $a > 0$, $n \in \mathbb{Z}^+$ and $m \in \mathbb{Z}$:

$a^{1/n} = \sqrt[n]{a}$	unit fraction: the denominator is the root
$a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$	in general: root then power, in either order

When n is even the symbol $\sqrt[n]{a}$ means the **positive** root, exactly as $\sqrt{9}$ means 3 and not ± 3.

A useful way to remember this:

denominator $n \longrightarrow$ root,	numerator $m \longrightarrow$ power

These are not new laws; they are the meanings chosen so that the familiar exponent laws keep holding.

Using fractional exponents

For $a^{m/n}$, it is usually easier to take the root first, then apply the power:

$$64^{2/3} = \left(\sqrt[3]{64}\right)^2 = 4^2 = 16.$$

Either order gives the same result, $64^{2/3} = \sqrt[3]{64^2} = 16$, but taking the root first keeps the numbers small.

The same laws run backwards. If the unknown carries a fractional exponent, raise both sides to the reciprocal of that exponent: the two multiply to 1, and the power-of-a-power law leaves the unknown on its own. That move keeps every solution only while the unknown is restricted to positive values. Where negative values are allowed and the exponent has an even numerator over an odd denominator, the even power hides a sign and the reciprocal power returns one solution out of two: $x^{2/5} = 4$ says $\left(\sqrt[5]{x}\right)^2 = 4$, so $\sqrt[5]{x} = \pm 2$ and $x = \pm 32$, while raising to $\frac{5}{2}$ offers 32 alone. Fix the restriction on x first, and read the number of solutions from it.

EXAMPLE 1.3

Solve $x^{3/4} = 125$ for $x > 0$.

The exponent $\frac{3}{4}$ is undone by $\frac{4}{3}$, since $\frac{3}{4} \times \frac{4}{3} = 1$. Raise both sides to that power:

$$\left(x^{3/4}\right)^{4/3} = 125^{4/3} \implies x^1 = \left(\sqrt[3]{125}\right)^4 = 5^4 = 625.$$

Check by substituting: $625^{3/4} = \left(\sqrt[4]{625}\right)^3 = 5^3 = 125$. The condition $x > 0$ is what allows the fractional exponent in the first place, since $x^{3/4}$ means $\left(\sqrt[4]{x}\right)^3$ and the fourth root needs a non-negative input.

Working with surds

A **surd** is an irrational root left in exact form, such as $\sqrt{2}$ or $\sqrt[3]{5}$, rather than rounded to a decimal. On a non-calculator paper, surds should be kept exact and simplified using the root laws

$$\sqrt{ab} = \sqrt{a}\,\sqrt{b}, \qquad \sqrt{\tfrac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}},$$

valid wherever the quantities are real. Two techniques cover most questions:

1. **Simplify** a surd by extracting its largest perfect-square factor.
2. **Combine** only like surds, the terms that have the same value under the root.

Two surds that look unrelated often become like surds once each has been simplified, so always simplify before trying to combine.

EXAMPLE 1.4

Simplify $\sqrt{147} - \sqrt{48}$.

Take the largest perfect-square factor out of each root:

$$\sqrt{147} = \sqrt{49 \cdot 3} = 7\sqrt{3}, \qquad \sqrt{48} = \sqrt{16 \cdot 3} = 4\sqrt{3}.$$

Both terms now carry $\sqrt{3}$, so they are like surds and only their coefficients subtract:

$$7\sqrt{3} - 4\sqrt{3} = 3\sqrt{3}.$$

The root laws cover products and quotients but never sums or differences, so $\sqrt{147} - \sqrt{48}$ is not $\sqrt{99}$: one is $3\sqrt{3} \approx 5.20$, the other about 9.95.

1.1.4 Rationalising the denominator

An exact answer is usually written without a surd in the denominator; removing it is called **rationalising the denominator**.

- For a single surd, multiply top and bottom by that surd.
- For a sum or difference, multiply by its **conjugate**, formed by changing the sign between the terms.

EXAMPLE 1.5

Rationalise $\dfrac{1}{\sqrt{3}}$ and $\dfrac{6}{2+\sqrt{3}}$.

For the first, multiply by $\dfrac{\sqrt{3}}{\sqrt{3}}$:

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

For the second, multiply by the conjugate $2 - \sqrt{3}$:

$$\frac{6}{2+\sqrt{3}} = \frac{6(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})} = \frac{6(2-\sqrt{3})}{4-3} = 12 - 6\sqrt{3}.$$

The conjugate works because the product is a difference of two squares, $(2+\sqrt{3})(2-\sqrt{3}) = 2^2 - (\sqrt{3})^2 = 1$, so the surd leaves the denominator.

EXAMPLE 1.6

Rationalise $\dfrac{4+\sqrt{15}}{4-\sqrt{15}}$.

The conjugate of the denominator is $4 + \sqrt{15}$. Multiply top and bottom by it, expanding the numerator as an ordinary bracket:

$$\frac{4+\sqrt{15}}{4-\sqrt{15}} = \frac{(4+\sqrt{15})(4+\sqrt{15})}{(4-\sqrt{15})(4+\sqrt{15})} = \frac{16+8\sqrt{15}+15}{16-15} = 31+8\sqrt{15}.$$

The difference of two squares applies only to the denominator; a numerator that is itself a sum has to be multiplied out in full, and the cross terms it produces do not cancel.

TIP In the real numbers, $(-4)^{1/2}$ is undefined: no real number squares to -4. In general, the radicand, the number inside the root, must be non-negative for an even root to have a real value. Square roots of negative numbers are complex numbers, which are introduced in Chapter 17.

1.1.5 Exponents as continuous growth

There is another way to read a non-integer exponent. Suppose 2^t means growth by a factor of 2 over a time t. Then 2^3 is three complete doublings, while

$$2^{1/2} = \sqrt{2}$$

is the growth factor halfway through one doubling period. Because the growth is exponential, "halfway" refers to the time, not to half the final increase. This reading extends to every real value of t, whether negative, fractional, or irrational:

$$2^{-3}, \qquad 2^{0}, \qquad 2^{1/2}, \qquad 2^{\sqrt{2}}.$$

Read as growth over time t, each has a plain meaning. $2^0 = 1$ is no growth at all: at time zero the population is unchanged, one hundred per cent of where it started. $2^{-3} = \frac{1}{8}$ looks backwards in time, to three periods before now (three seconds ago, if one doubling takes a second), when the population was an eighth of its present size. $2^{1/2} = \sqrt{2}$ is the factor after half a period, and $2^{\sqrt{2}}$ the factor after an irrational stretch of time. For any positive base a, the expression a^x has a meaning for every real exponent x, which is where the exponential function of Chapter 5 begins.

1.1.6 The number *e*

Suppose an account pays 100% interest in a year. Paid once, at the end, an amount of 1 becomes 2. Now suppose the same 100% arrives in two instalments of 50%. After six months the balance is $1 \times 1.5 = 1.5$, and the second 50% is taken on 1.5 rather than on 1, so the year ends at $1.5 \times 1.5 = 2.25$. Splitting the same total rate into more instalments gives more, because an amount added early is itself subject to all the growth that follows.

That leads to a question with a surprising answer. If a finer split always gives more, does the year-end amount grow without limit? In n equal steps the starting amount is multiplied by $\left(1+\frac{1}{n}\right)^n$, so the question is what that expression does as n increases.

instalments in the year, n	amount after one year
1 (once, at the end)	2.00000
2 (every six months)	2.25000
12 (monthly)	2.61304
365 (daily)	2.71457
8760 (hourly)	2.71813
$n \to \infty$ (continuous)	$2.71828\ldots = e$

It does not grow without limit. Each extra split adds less than the split before it, because the money it adds arrives later in the year and so has less of the year left in which to earn anything itself. Going from yearly to six-monthly gains 0.25; going from daily to hourly gains about 0.0036. The totals close in on one value, $2.71828\ldots$, and that number is called e. Like π, it cannot be written exactly as a decimal or as a fraction.

So e answers one specific question: what does 100% growth come to when it acts continuously instead of in steps? Other rates are handled in the exponent. A continuous rate k acting for time t multiplies the starting amount by e^{kt}. This is the model behind population growth, radioactive decay, and cooling. Chapter 13 gives a deeper reason for e: the function e^x grows at a rate equal to its own value.

EXT Formally, $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$. Each n is a number of compounding steps per year, so letting n grow without bound is what "continuously" means. Limits are defined properly in Ch. 13.

YOUR TURN 1.1 What an Exponent Counts

1. Simplify each of the following.

(a) $\dfrac{6c^9}{2c^4}$ (b) $(3x^2y^{-3})^4$

(c) $\dfrac{12a^5b^3}{4a^2b^{-1}}$ (d) $\dfrac{(2x^3)^2 \cdot x^{-4}}{4x^2}$

2. Simplify by dividing each term of the numerator separately.

(a) $\dfrac{4x^3 - 8x^2}{2x}$ (b) $\dfrac{16a^5b^2 - 12ab^4}{4ab^2}$

3. Show that $\dfrac{(a^2b^{-1})^3}{a^{-3}b^2} = \dfrac{a^9}{b^5}$.

4. Evaluate each of the following.

(a) $27^{2/3}$

(b) $81^{-3/4}$

(c) $\left(\dfrac{8}{27}\right)^{-2/3}$

(d) $32^{-2/5}$

5. Write each of the following in the form x^n.

(a) $\sqrt[4]{x^3}$

(b) $\dfrac{1}{\sqrt[3]{x^2}}$

(c) $\dfrac{\sqrt{x^5}}{x^2}$

(d) $\dfrac{x^2 \cdot \sqrt{x}}{x^{3/2}}$

6. Solve each equation for $x > 0$.

(a) $x^{3/2} = 64$

(b) $x^{2/3} = 9$

(c) $x^{-1/2} = \frac{1}{4}$

(d) $x^{5/2} = 32$

7. Simplify each of the following.

(a) $\sqrt{50} - \sqrt{8}$

(b) $\sqrt{12} + \sqrt{27}$

(c) $\sqrt{18} + \sqrt{32} - \sqrt{2}$

(d) $\sqrt{75} - 2\sqrt{12}$

8. Rationalise the denominator of each expression.

(a) $\dfrac{4}{3 - \sqrt{5}}$, in the form $a + b\sqrt{5}$

(b) $\dfrac{6}{\sqrt{3}}$

(c) $\dfrac{1}{\sqrt{7} + \sqrt{5}}$

(d) $\dfrac{2 + \sqrt{3}}{2 - \sqrt{3}}$

1.2 Standard Form

The mass of a proton, written out in full, is

$$0.00000000000000000000000000167 \text{ kg},$$

that is, 26 zeros after the decimal point before the first significant figure. The mass of the Sun runs to the opposite extreme,

$$1{,}990{,}000{,}000{,}000{,}000{,}000{,}000{,}000{,}000{,}000 \text{ kg},$$

a 1 followed by 30 digits. Both numbers are quite difficult to write, let alone compare or calculate with.

There is a shorter way to write both. Instead of all the digits, record *the power of 10* that places the decimal point where it belongs. Mathematicians call the result **standard form** (also known as scientific notation).

DEF A number is in **standard form** when written as $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$.

The proton becomes 1.67×10^{-27} kg. The Sun becomes 1.99×10^{30} kg. The exponent k tells you immediately whether you are dealing with something microscopic or cosmic.

1.2.1 Calculating with standard form

Multiplication and division follow straight from the exponent laws. For a product, multiply the coefficients and add the powers of 10; for a quotient, divide the coefficients and subtract the powers:

$$(a \times 10^m)(b \times 10^n) = (a \times b) \times 10^{m+n}.$$

One check matters afterwards. If the new coefficient falls outside the range $1 \leq a < 10$, shift the decimal point and adjust the power. For example, $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 = 2 \times 10^6$: the coefficient 20 becomes 2, and the exponent rises by one to account for the extra factor of 10.

Addition and subtraction need one more move first. Just as you cannot add metres and kilometres without converting, you cannot add two numbers in standard form until they share the same power of 10. Rewrite the one with the smaller exponent so the powers match, then add the coefficients.

EXAMPLE 1.7

Calculate $(3.5 \times 10^4) \times (2 \times 10^{-7})$.
Group and multiply:

$$(3.5 \times 2) \times 10^{4+(-7)} = 7 \times 10^{-3}.$$

EXAMPLE 1.8

Calculate $(4.8 \times 10^5) + (3.2 \times 10^4)$.
Before adding, you need the same power of 10, just as you cannot add metres and kilometres without converting first.

$$4.8 \times 10^5 + 0.32 \times 10^5 = 5.12 \times 10^5.$$

CAUTION In addition and subtraction, do **not** add the exponents. Only multiplication and division combine powers of 10.

CAUTION Calculator display notation is not acceptable in written answers. If your graphic display calculator (GDC) shows 5.2E30, you must write 5.2×10^{30}.

YOUR TURN 1.2 Standard Form

9. Write 0.000047 in standard form.

10. Write each number in ordinary form.

(a) 3.2×10^5 (b) 8.06×10^{-4}
(c) 1.5×10^7 (d) 9×10^{-1}

11. Calculate $(2.4 \times 10^3) \times (3 \times 10^{-5})$, giving your answer in standard form.

12. Calculate $\dfrac{6.6 \times 10^8}{2.2 \times 10^3}$, giving your answer in standard form.

13. Calculate each of the following, giving your answer in standard form.

(a) $(5.1 \times 10^4) + (3.9 \times 10^3)$ (b) $(7.2 \times 10^{-3}) - (5 \times 10^{-4})$
(c) $(2.5 \times 10^6) + (7.5 \times 10^6)$ (d) $(9.4 \times 10^{-2}) + (6 \times 10^{-2})$

14. GDC The mass of an electron is 9.11×10^{-31} kg. The mass of a proton is 1.67×10^{-27} kg. Find how many times heavier a proton is than an electron. Give your answer in standard form to 3 significant figures.

1.3 The Logarithm: an Exponent Reversed

Many familiar operations come in inverse pairs: addition with subtraction, multiplication with division. So what is the inverse of exponentiation? If $a^x = b$, and you know a and b, how do you find x? That is what a logarithm does. It is not a new idea: it is the exponential equation $a^x = b$ rearranged to make the exponent x the subject.

DEF · IB For $a > 0$, $a \neq 1$, and $b > 0$:

$$a^x = b \iff x = \log_a b$$

The **common logarithm** ($\log x$) uses base 10. The **natural logarithm** ($\ln x$) uses base $e \approx 2.718$, the continuous-growth constant met earlier in this chapter.

The condition $b > 0$ in that definition is not a technicality. A positive base raised to any real power gives a positive result: $10^2 = 100$, $10^0 = 1$, $10^{-3} = 0.001$. No exponent turns 10 into -5, and no exponent turns 10 into 0. So $\log(-5)$ and $\log 0$ do not exist. **Whatever appears inside a logarithm must be positive.** The same holds for every base: **$\log_a x$ is defined only when $x > 0$.**

The base carries conditions of its own. It must be positive, because a negative base has no consistent meaning for fractional exponents, and it cannot be 1, because every power of 1 is 1 and no exponent would give any other value.

The word *logarithm* comes from the Greek *logos* (ratio) and *arithmos* (number).

The graph of $y = \log_a x$ is the mirror image of $y = a^x$ across the line $y = x$, with a vertical asymptote at $x = 0$. These graphs and their transformations are developed in Chapter 5 (Functions).

Because the exponential and logarithm are inverses, each undoes what the other does:

KEY For base 10 and base e:

$\log(10^x) = x$	base 10: true for every real x
$10^{\log x} = x$	the same pair reversed, but only for $x > 0$, since $\log x$ is undefined otherwise
$\ln(e^x) = x$	base e: true for every real x
$e^{\ln x} = x$	reversed, and again only for $x > 0$

The formula booklet prints the general statement $\log_a a^x = x = a^{\log_a x}$; these four lines are its base-10 and base-e cases.

Two special values follow at once from the definition, and both appear constantly: since $a^1 = a$ and $a^0 = 1$,

KEY For any base $a > 0$, $a \neq 1$:

$\log_a a = 1$	since $a^1 = a$
$\log_a 1 = 0$	since $a^0 = 1$

1.3.1 Working with logarithms

Evaluating a logarithm by hand is always the same move in reverse. Since $\log_a b$ is defined as the exponent that turns a into b, every logarithm question is an exponent question asked backwards. Read $\log_a b$ as **"a to what power gives b?"** and in many cases the answer is immediate.

EXAMPLE 1.9

Find $\log_3 81$.

Ask the inverse question: 3 to what power gives 81? You know $3^4 = 81$, so $\log_3 81 = 4$. The logarithm is just the exponent.

EXAMPLE 1.10

Solve $\log_{10}(y+1) = 3$.

The equation says: the exponent you need to get $(y+1)$ from base 10 is 3. So $y + 1 = 10^3 = 1000$, giving $y = 999$.

EXAMPLE 1.11

Without a calculator, find $e^{3\ln 2}$.

Think of $e^{\ln 2}$ first: the exponential and natural log cancel, giving 2. So $e^{3\ln 2} = (e^{\ln 2})^3 = 2^3 = 8$.

CAUTION Solving an equation with a logarithm in it can produce a value that makes an argument negative or zero. Such a value is not a solution, however correct the algebra was. Substitute every answer back into the original equation and keep only those that leave each argument positive.

YOUR TURN 1.3 The Logarithm: an Exponent Reversed

15. Evaluate each of the following without a calculator.

(a) $\log_2 32$

(b) $\log_3 \dfrac{1}{9}$

(c) $e^{2\ln 3}$

(d) $\ln e^5$

16. Rewrite each statement in the other form.

(a) Write $3^4 = 81$ in logarithmic form.

(b) Write $\log_5 x = 3$ in exponential form, and hence find x.

17. State whether each expression is defined. If it is, write down its value.

(a) $\log_5 125$

(b) $\log_5 1$

(c) $\log_5 0$

(d) $\log_5(-25)$

(e) $\log_1 7$

(f) $\log_{1/2} 8$

18. The logarithm and the exponential undo each other, but not on the same set of inputs. Evaluate each of the following, stating any restriction on x needed for your answer to hold.

(a) $\log_7(7^x)$
(b) $7^{\log_7 x}$
(c) $\ln\left(e^{x^2+1}\right)$
(d) $10^{\log(3x)}$

19. Each statement is either true for every value allowed by the definition, or not. State which, and correct those that are not.

(a) $\log_a(a^x) = x$ for every real x
(b) $a^{\log_a x} = x$ for every real x
(c) $\log_a 1 = 0$ for every valid base a
(d) $\log_a 0 = 0$ for every valid base a

20. Solve each equation. Reject any value the definition does not allow.

(a) $\log_4(x+1) = 2$
(b) $\log_3(2x-1) = 4$
(c) $\log_x 49 = 2$
(d) $\log_2(x^2 - 3x) = 2$

1.4 The Laws of Logarithms

A single logarithm is just a number. What makes logarithms worth having is that they trade hard arithmetic for easy arithmetic: multiplication becomes addition, division becomes subtraction, and a power becomes a coefficient. The last of those is the one this chapter is heading towards, because it is what brings an unknown down out of an exponent.

The laws work because a logarithm is an exponent. Multiplying two powers with the same base adds the exponents, $a^p \cdot a^q = a^{p+q}$, and reading that sentence in terms of logarithms is already the first law: multiplication becomes addition.

The first three laws below are the three exponent laws restated for logarithms. A fourth, change of base, is a different kind of rule: it rewrites any logarithm as a ratio of two logarithms in a base of your choice.

KEY · IB For $a, b, x, y > 0$ with $a, b \neq 1$:

$\log_a(xy) = \log_a x + \log_a y$	product: a product becomes a sum
$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$	quotient: a quotient becomes a difference
$\log_a(x^m) = m\log_a x$	power: an exponent becomes a coefficient
$\log_a x = \frac{\log_b x}{\log_b a}$	change of base: rewrite in any base you can compute

All three proofs share one setup. Write $x = a^p$ and $y = a^q$, so that $p = \log_a x$ and $q = \log_a y$ by definition. Each law is then a one-line consequence of the matching exponent law:

- **Product.** $xy = a^p \cdot a^q = a^{p+q}$, so $\log_a(xy) = p + q = \log_a x + \log_a y$.
- **Quotient.** $\frac{x}{y} = a^{p-q}$, so $\log_a\left(\frac{x}{y}\right) = p - q = \log_a x - \log_a y$.
- **Power.** $x^m = (a^p)^m = a^{mp}$, so $\log_a(x^m) = mp = m\log_a x$.

The third law is the most useful in this course. When x appears in an exponent, the power law $\log_a(x^m) = m\log_a x$ moves the exponent into a coefficient. This often converts an exponential equation into a simpler algebraic equation.

1.4.1 Change of base

What do you do when your GDC only has $\log$ (base 10) and $\ln$ (base e), but you need a logarithm in a different base, or you want to rewrite a base-10 logarithm in terms of base e? The change-of-base law above rewrites any logarithm as a ratio of two logarithms you can compute. In practice you take base e, which turns the law into $\log_a x = \frac{\ln x}{\ln a}$.

To prove it, let $\log_a x = y$, so $x = a^y$. Take $\ln$ of both sides: $\ln x = y\ln a$. Solve for y.

EXAMPLE 1.12

Evaluate $\log_3 5$ on a GDC.

Convert to natural logarithms:

$$\log_3 5 = \frac{\ln 5}{\ln 3} \approx \frac{1.609}{1.099} \approx 1.46.$$

One case of the law is worth carrying separately. Take the argument to be b and compute in base b; since $\log_b b = 1$, the numerator disappears:

$$\log_a b = \frac{\log_b b}{\log_b a} = \frac{1}{\log_b a}.$$

KEY For $a, b > 0$ with $a, b \neq 1$:

$$\log_a b = \frac{1}{\log_b a}.$$

The formula booklet does not print this, so quote it with the change-of-base step above, or work from change of base directly.

Swapping a base and an argument is the move to reach for when the unknown sits in the base, because it carries that unknown into an argument, where the ordinary laws apply. Often no swap is needed at all: a logarithm with an unknown base is still an exponent question read backwards.

EXAMPLE 1.13

Solve $\log_x 81 = 4$.

The statement says that x raised to the power 4 gives 81, so

$$x^4 = 81 \implies x = 3 \text{ or } x = -3.$$

A base must satisfy $x > 0$ and $x \neq 1$. That rules out $x = -3$, leaving $x = 3$.

An unknown base is where those two conditions earn their keep: the algebra alone offers two roots, and only the definition of a logarithm removes one.

1.4.2 When one base is a power of another

Questions often mix bases that are powers of one another: $\log_2 x$ alongside $\log_4 x$, or $\log_2 x$ alongside $\log_8 x$. Change of base rewrites them all in one base, and for this particular case the result is short enough to know on sight.

Change $\log_{a^2} x$ to base a:

$$\log_{a^2} x = \frac{\log_a x}{\log_a a^2} = \frac{\log_a x}{2} = \tfrac{1}{2}\log_a x,$$

since $\log_a a^2 = 2\log_a a = 2$ by the power law. Squaring the base halves the logarithm. Replacing a^2 by a^n changes nothing in the argument, so for $a > 0$, $a \neq 1$, $x > 0$ and any $n \neq 0$,

$$\log_{a^n} x = \frac{1}{n}\log_a x.$$

This result is not in the formula booklet. Quote it if you are sure of it, or derive it from change of base; both are accepted.

EXAMPLE 1.14

Solve $\log_2 x + \log_4 x = 6$.
The bases are 2 and $4 = 2^2$, so rewrite the second one in base 2: $\log_4 x = \frac{1}{2}\log_2 x$. Writing $t = \log_2 x$, the equation becomes

$$t + \tfrac{1}{2}t = 6 \implies \tfrac{3}{2}t = 6 \implies t = 4,$$

so $x = 2^4 = 16$. Check: $\log_2 16 = 4$ and $\log_4 16 = 2$, and these sum to 6.

1.4.3 Applying the laws

These laws can be used in either direction: they split one logarithm into several, or gather several back into one. Both directions appear constantly, in simplifying expressions and in solving equations.

EXAMPLE 1.15

Express $\log_5\left(\dfrac{a^3}{b}\right)$ in terms of $p = \log_5 a$ and $q = \log_5 b$.
Apply the quotient law, then the power law:

$$\log_5\left(\frac{a^3}{b}\right) = \log_5 a^3 - \log_5 b = 3\log_5 a - \log_5 b = 3p - q.$$

EXAMPLE 1.16

Write $2\ln a + 6\ln b$ as a single logarithm.
Use the power law in reverse first, then the product law:

$$2\ln a + 6\ln b = \ln a^2 + \ln b^6 = \ln(a^2 b^6).$$

EXAMPLE 1.17

Solve $\log_2 x + \log_2(x-2) = 3$.
The two logs on the left can combine into one product:

$$\log_2[x(x-2)] = 3 \implies x(x-2) = 2^3 = 8.$$

$$x^2 - 2x - 8 = 0 \implies (x-4)(x+2) = 0.$$

So $x = 4$ or $x = -2$. But $\log_2(-2)$ does not exist, so reject $x = -2$. The answer is $x = 4$.
Always check domain when logarithms are involved. It is easy to generate solutions that look valid algebraically but are meaningless.

CAUTION $\log(x+y) \neq \log x + \log y$. The product law is about $\log(xy)$, not a sum inside the log.

YOUR TURN 1.4 The Laws of Logarithms

21. Write each expression as a single logarithm.

(a) $\log_3 5 + \log_3 7$
(b) $\log_2 24 - \log_2 3$, and evaluate it
(c) $3\log_a x + \log_a y$
(d) $2\ln a - 3\ln b + \ln c$

22. Express each logarithm in terms of the quantities given.

(a) $\log_5\left(\frac{a^2}{b}\right)$, in terms of $\log_5 a$ and $\log_5 b$
(b) $\log_3 18$, given $p = \log_3 2$
(c) $\log_{10} 72$, given $\log_{10} 2 = p$ and $\log_{10} 3 = q$
(d) $\log_5 75$, given $\log_5 3 = 0.682$
(e) $\log_4 9$, in terms of natural logarithms
(f) $\log_8 x$, in terms of $\log_2 x$

23. Show that each statement is true.

(a) $\log_a(x^2-1) - \log_a(x+1) = \log_a(x-1)$, for $x > 1$
(b) $\log_a b = \dfrac{1}{\log_b a}$

24. GDC Evaluate $\log_3 50$ to 3 significant figures.

25. Solve each equation, using a law of logarithms first. Reject any value the logarithms do not allow.

(a) $\log_3 x + \log_3(x+6) = 3$
(b) $\log_2(x+3) - \log_2(x-1) = 2$
(c) $2\log_5 x = \log_5 9 + \log_5 4$

26. Change the base in each of the following.

(a) Show that $\log_9 x = \frac{1}{2}\log_3 x$.
(b) Given $\log_2 5 = p$, express $\log_8 5$ in terms of p.
(c) Find the exact value of $\log_4 8$.

1.5 Solving Exponential Equations

An unknown in the exponent is out of reach of ordinary algebra. Nothing you can add, multiply or factorise will get at the x in $5^x = 30$, because x is not being combined with anything: it is counting how many times the base is used. The logarithm is what brings it down, and once it is down the equation is one you already know how to solve. This section is that single move, applied to steadily less obvious situations.

1.5.1 Same base first

Is there ever a way to solve an exponential equation without using a logarithm at all? Before anything else, look for a common base. If both sides can be written as powers of the same number, the exponents must be equal, provided the base is not 0 or 1: $a^p = a^q$ forces $p = q$ for $a > 0,\ a \neq 1$. This is the fastest route whenever it is available, and the intended one on non-calculator papers.

EXAMPLE 1.18

Solve $\left(\frac{1}{3}\right)^x = 9^{x+1}$.

Both sides are powers of 3: $\frac{1}{3} = 3^{-1}$ and $9 = 3^2$. Rewrite and equate exponents:

$$3^{-x} = 3^{2(x+1)} \implies -x = 2x + 2 \implies x = -\tfrac{2}{3}.$$

No logarithm was needed. Check for a shared base before anything else; look for common bases such as $2, 4, 8$ and $3, 9, 27$.

1.5.2 The general strategy

When no common base exists, take a logarithm of both sides, use $\log(x^m) = m \log x$ to bring the exponent down, then solve what is now a linear or quadratic equation. The choice of which logarithm to use ($\log$ or $\ln$) does not matter: pick whichever gives cleaner arithmetic.

EXAMPLE 1.19

Solve $5^x = 30$.

Take $\ln$ of both sides:

$$x \ln 5 = \ln 30 \implies x = \frac{\ln 30}{\ln 5} \approx 2.11.$$

Or equivalently, $x = \log_5 30$.

EXAMPLE 1.20

Solve $7^{x-2} = 5^{x+3}$ exactly.

With different bases on each side, take $\ln$ of both sides and use the power law:

$$(x-2)\ln 7 = (x+3)\ln 5.$$

Collect x on one side:

$$x(\ln 7 - \ln 5) = 3\ln 5 + 2\ln 7 \implies x = \frac{\ln 5^3 + \ln 7^2}{\ln(7/5)} = \frac{\ln 6125}{\ln(7/5)}.$$

TIP Give an exact answer, like $x = \dfrac{\ln 6125}{\ln(7/5)}$, whenever a question says *exact* or on a non-calculator paper (Paper 1). When a decimal is asked for, or on a calculator paper (Paper 2), round to three significant figures unless told otherwise. Three significant figures is the IB default, and rounding too early or giving too few figures costs accuracy marks.

1.5.3 When the unknown appears twice

Taking a logarithm gets nowhere on $2^{2x} - 10 \cdot 2^x + 16 = 0$, because x sits in two exponents and a sum of powers has no log law to break it up. Two patterns handle almost every such equation. When one power is the square of another, a **substitution** turns it into a quadratic. When the exponents differ by a constant, a common factor collapses the terms into one.

EXAMPLE 1.21

Solve $2^{2x} - 10 \cdot 2^x + 16 = 0$.

The first term is the square of the second, since $2^{2x} = (2^x)^2$. Put $u = 2^x$:

$$u^2 - 10u + 16 = 0 \implies (u-2)(u-8) = 0,$$

so $u = 2$ or $u = 8$. Now convert each value back:

$$2^x = 2 \implies x = 1, \qquad 2^x = 8 \implies x = 3.$$

Both roots are positive, so both survive. A negative or zero value of u would have to be rejected instead of converted, since $2^x > 0$ for every real x and no exponent produces it.

EXAMPLE 1.22

Solve $3^{x+2} - 3^x = 72$.

Split the first exponent so that both terms carry 3^x: by the product law, $3^{x+2} = 3^x \cdot 3^2 = 9 \cdot 3^x$.

The equation becomes

$$9 \cdot 3^x - 3^x = 72 \implies 8 \cdot 3^x = 72 \implies 3^x = 9,$$

so $x = 2$. Check: $3^4 - 3^2 = 81 - 9 = 72$. A constant added in the exponent is a constant factor outside it, which is what lets the two terms be gathered into one.

1.5.4 From words to an equation

Many growth and decay problems use one model,

$$A(t) = A_0 \times r^{\,t},$$

where A_0 is the starting amount and r is the multiplier for each period. The wording gives you r. A rise of 6% per year means $r = 1.06$. A fall of 6% means $r = 0.94$. A quantity that doubles each period has $r = 2$.

When the change is continuous rather than in steps, the model uses base e:

$$A(t) = A_0\, e^{kt},$$

where k is the **continuous growth rate** met in §1.1: positive when the quantity grows, negative when it decays.

These are not two models but one, written twice. Since $r = e^{\ln r}$,

$$A_0\, r^{\,t} = A_0 \left(e^{\ln r}\right)^t = A_0\, e^{(\ln r)\, t},$$

so the two forms agree exactly when

$$k = \ln r.$$

A quantity that doubles each period has $k = \ln 2 \approx 0.693$; one falling 6% a year has $k = \ln 0.94 \approx -0.0619$, the negative sign marking decay. Use whichever form the question hands you: a percentage per period gives r, a rate described as *continuous* gives k.

Once the model is written, the logarithm provides the final step.

Decay is often described not by a rate but by a **half-life**: the time in which half of the quantity disappears. One half-life multiplies the amount by $\frac{1}{2}$, and a time t contains t/T half-lives, where T is the half-life itself. So

$$A(t) = A_0 \left(\tfrac{1}{2}\right)^{t/T}.$$

This is the same $A_0\, r^{\,t}$ model with $r = \frac{1}{2}$, and with time counted in half-lives instead of single periods. A **doubling time** works identically, with 2 in place of $\frac{1}{2}$.

EXAMPLE 1.23

Sea ice in a Finnish bay covers 1200 km^2 and shrinks by 6% each year. After how many years will it have shrunk to 800 km^2?

Each year multiplies the area by 0.94, so after t years the area is $A(t) = 1200(0.94)^t$. Set this equal to the target and solve:

$$1200(0.94)^t = 800 \implies (0.94)^t = \tfrac{2}{3} \implies t = \frac{\ln(2/3)}{\ln 0.94} \approx 6.55.$$

The ice passes 800 km^2 partway through the seventh year.

EXAMPLE 1.24

Algae in a Finnish lake grow continuously at 12% per year, starting from 200 grams. When does the mass reach 500 grams?

Continuous growth uses base e, so $A(t) = 200\,e^{0.12t}$. Set it equal to the target:

$$200\,e^{0.12t} = 500 \implies e^{0.12t} = 2.5 \implies t = \frac{\ln 2.5}{0.12} \approx 7.6 \text{ years.}$$

The 12% here is a continuous rate, so it sits in the exponent as 0.12 directly, with no conversion to a yearly factor.

EXAMPLE 1.25

A medical isotope has a half-life of 6 hours: every 6 hours, half of it decays. An 80 mg sample is prepared. How long until only 12 mg remains?

Halving every 6 hours means $A(t) = 80\left(\tfrac{1}{2}\right)^{t/6}$, with t in hours. Set this equal to the target and take a logarithm:

$$80\left(\tfrac{1}{2}\right)^{t/6} = 12 \implies \left(\tfrac{1}{2}\right)^{t/6} = 0.15 \implies t = 6\cdot\frac{\ln 0.15}{\ln 0.5} \approx 16.4 \text{ hours.}$$

The answer need not be a whole number of half-lives: 16.4 hours is a little over 2.7 of them, and the model runs continuously between them.

YOUR TURN 1.5 Solving Exponential Equations

27. Solve each equation, giving your answer as an exact value.

(a) $5^{2x} = 125$

(b) $4^x = 8^{x-1}$

(c) $9^{x+1} = 27^x$

(d) $2^x \cdot 4^{x+1} = 8$

28. GDC Solve each equation, giving your answer to 3 significant figures.

(a) $2^x = 10$

(b) $3^{x+1} = 20$

(c) $5^{2x} = 40$

(d) $7 \cdot 2^x = 200$

29. Solve each equation. *(Hint: in (a) and (b), substitute u for the base raised to x)*

(a) $e^{2x} - 5e^x + 6 = 0$

(b) $2^{2x} - 6 \cdot 2^x + 8 = 0$

(c) $3 \cdot 2^x = 2 \cdot 3^x$

Chapter 1 · Exponents & Logarithms

Reflections

TOK John Napier spent twenty years computing logarithms by hand and published his tables in 1614, not out of mathematical curiosity, but to help sailors and astronomers multiply enormous numbers. He never used the word "logarithm" to mean what we mean today. Does mathematics develop because we need it, or do we find uses for what mathematicians have already built?

TOK The exponent laws force $a^0 = 1$ for every $a \neq 0$, but 0^0 is left undefined, because approaching it along different routes gives different answers. Does leaving 0^0 undefined reveal a limitation of mathematics or the precision of mathematics?

IM Base-10 logarithms feel natural to us because we count in base 10. But the choice is cultural, not mathematical. The Babylonians used base 60 (traces of which survive in our 60-minute hour and 360-degree circle). What does it mean for a mathematical tool to be "natural"?

IM The word *algebra* comes from *al-jabr*, a technique for solving equations described by the Persian mathematician al-Khwarizmi in ninth-century Baghdad; the word *algorithm* is a Latinisation of his own name. His treatises on equations, and the work of mathematicians across the wider Islamic world who followed him, were translated into Latin centuries later and became a foundation of European algebra, including the exponent notation used throughout this chapter. The rules you are applying here developed over many centuries and across many cultures.

RW The decibel (dB) scale for sound is logarithmic: the sound level is $10\log_{10}(I/I_0)$, comparing a sound's intensity I to a fixed reference intensity I_0, so each increase of 10 dB means ten times the intensity. The stellar magnitude scale in astronomy and the octave system in music work the same way. In each case, human perception responds approximately to the logarithm of the physical quantity, not to the quantity itself.

EXT Euler's identity: $e^{i\pi} + 1 = 0$. Five fundamental constants, one equation. It is the exponential function extended into the complex plane, connecting the circular motion of $e^{i\theta} = \cos\theta + i\sin\theta$ to the real exponential growth you have been studying here.

End-of-Chapter Problems — Chapter 1: Exponents & Logarithms

1. Solve $3^{2x+1} = 27^{x-1}$. [3]

2. Show that the equation $2^{x+3} - 2^x = k$ has a real solution only when $k > 0$, and find this solution in terms of k. [5]

3.

(a) Simplify $\left(2x^{-2}y^3\right)^3 \times \left(4x^4y^{-1}\right)^{-1}$, giving your answer with positive indices. [3]
(b) Solve $8^x = 2^{x+6}$. [2]
(c) Show that $\left(a^{1/2} + a^{-1/2}\right)^2 = a + 2 + a^{-1}$, for $a > 0$. [2]

4. GDC The mass of the Earth is 5.97×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg.

(a) Find how many times heavier the Earth is than the Moon, to 3 significant figures. [2]
(b) Find their combined mass, giving your answer in standard form to 3 significant figures. [2]

5. Prove that $\log_a b \cdot \log_b a = 1$ for $a, b > 0$, $a, b \neq 1$. Hence simplify $(\log_2 3)(\log_3 8)$. [4]

6. Given that $\log_a x = 3$ and $\log_a y = 5$, find each of the following.

(a) $\log_a(x^2y)$ [1] (b) $\log_a \sqrt{xy}$ [2]
(c) $\log_a\left(\dfrac{x}{y^2}\right)$ [1]

7. Let $\log_a 2 = p$ and $\log_a 3 = q$.

(a) Write $\log_a 6$ in terms of p and q. [1]
(b) Write $\log_a \sqrt{12}$ in terms of p and q. [2]
(c) Write $\log_a \dfrac{4}{9}$ in terms of p and q. [2]

8.

(a) Show that $\log_a \sqrt[n]{x} = \dfrac{1}{n}\log_a x$. [2]
(b) Hence solve $\log_2 \sqrt{x} + \log_4 x = 6$. [4]

9. Prove that $\log_a b \cdot \log_b c \cdot \log_c a = 1$ for all valid bases and arguments. Hence evaluate $\log_2 3 \cdot \log_3 5 \cdot \log_5 8$ without a calculator. [6]

10. Given that $\log_2 3 = a$, express each of the following in terms of a:

(a) $\log_2 9$ [1]
(b) $\log_4 3$ [2]
(c) $\log_8 27$ [2]
(d) $\log_3 8$ [2]

11. Solve $\log_3(x+5) - \log_3(x-1) = 2$. [4]

12. Solve $\log_6(x+2) + \log_6(x-3) = 1$, justifying any rejection of solutions. [4]

13. Solve the simultaneous equations:

$$\log_2 x + \log_2 y = 5, \qquad \log_2(x-y) = 3.$$

[5]

14. Solve $\log_x 8 = \log_4 x$. [5]

15. GDC Solve $\log_2(x^2 - 2x) = \log_4(x+4)$. [5]

16. GDC

(a) Express $\log_3 x - 2 = \log_9(x+6)$ as a quadratic equation in x, using change of base. [3]
(b) Hence find the value of x. [2]

17. Solve $\log_5(x^2 - 2x + 1) = 2\log_5(x-1) + 1$, or show that no solution exists. [5]

18. The reciprocal of a logarithm is a logarithm with its base and argument swapped.

(a) Show that $\dfrac{1}{\log_a N} = \log_N a$ for all valid a and N. [3]
(b) Hence evaluate $\dfrac{1}{\log_2 36} + \dfrac{1}{\log_3 36}$ without a calculator. [4]

19. Solve $\log_2 x + \log_x 2 = \dfrac{5}{2}$. *(Hint: let $t = \log_2 x$ and use question 18)* [6]

20. Consider the equation $(\log_3 x)^2 = \log_3 x^2 + 3$.

(a) Explain why the right-hand side equals $2\log_3 x + 3$, and state the restriction on x this requires. [2]
(b) Hence solve the equation. [4]
(c) The expression x^2 is defined for every non-zero x, yet negative values of x cannot be solutions. Explain why. [2]

21. Solve $\log_2(\log_3(\log_2 x)) = 0$.

(a) Find x. [4]

(b) Working outwards from the innermost logarithm, state the condition each layer imposes on x, and hence give the values of x for which the left-hand side is defined at all. [4]

22. Evaluate

$$\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdot \ \cdots \ \cdot \log_{31} 32,$$

the product of all thirty factors, without a calculator. Justify your method. [5]

23. Let a, b and c be positive, with a, b and ab all different from 1.

(a) Show that $\log_{ab} c = \dfrac{\log_a c \cdot \log_b c}{\log_a c + \log_b c}$. [5]

(b) Verify your identity with $a = 2$, $b = 5$ and $c = 7$, giving both sides to four decimal places. [2]

24. Solve the simultaneous equations

$$\log_x y + \log_y x = \frac{5}{2}, \qquad xy = 27,$$

where x and y are positive and neither is 1. [7]

25. Solve $7^{x-2} = 5^{x+1}$, giving your answer in exact logarithmic form. [4]

26. Find the exact solution to $e^{2x} = 3e^x + 10$. *(Hint: let $u = e^x$)* [4]

27. Solve the equation $9^x - 4 \cdot 3^{x+1} + 27 = 0$. *(Hint: let $u = 3^x$; note $9^x = u^2$)* [5]

28. GDC The number of bacteria in a culture doubles every 3 hours. Initially there are 500 bacteria.

(a) Write an expression for the number of bacteria after t hours. [2]

(b) Find the time, to the nearest minute, when there are 50 000 bacteria. [3]

29. GDC The population of a city is modelled by $P(t) = 250\,000 \times e^{0.032t}$, where t is the number of years after 2020.

(a) Write down the population in 2020. [1]

(b) Find the population in 2030, to the nearest thousand. [2]

(c) Find the year in which the population first exceeds 400 000. [3]

30. GDC The value V of a car after t years is given by $V = 32\,000 \times (0.85)^t$.

(a) Write down the initial value of the car. [1]
(b) Find the value after 5 years, to the nearest dollar. [2]
(c) Find the number of years until the value falls below $10 000. [3]

31. GDC The pH of a solution is defined by pH $= -\log_{10}[\text{H}^+]$, where $[\text{H}^+]$ is the hydrogen ion concentration in mol/L.

(a) A solution has $[\text{H}^+] = 3.5 \times 10^{-4}$ mol/L. Find its pH to 2 decimal places. [2]
(b) A solution has pH $= 8.3$. Find its hydrogen ion concentration in standard form. [2]
(c) Solution A has pH $= 3$ and solution B has pH $= 7$. Find how many times more acidic solution A is than solution B. [2]

32. Let $f(x) = \log_3(2x - 1)$ and $g(x) = 3^x + 2$. (Inverse functions and graph transformations are developed in Ch. 5; this question looks ahead to them.)

(a) Find the domain of f. [1]
(b) Find $f^{-1}(x)$. [3]
(c) Show that $f^{-1}(x) = \frac{1}{2}g(x) - \frac{1}{2}$, and describe the transformation of g this represents. [3]

33. GDC A radioactive substance decays so that its mass M grams after t years satisfies $M = M_0e^{-kt}$, where M_0 and k are positive constants. After 10 years the mass is halved.

(a) Show that $k = \dfrac{\ln 2}{10}$. [2]
(b) Find the percentage of the original mass remaining after 25 years. [3]
(c) Find the time for the mass to reduce to 10% of its original value. [3]

34. GDC This question is about writing very large and very small quantities in standard form.

(a) A human cell has diameter about 2×10^{-5} m, and a water molecule about 2.8×10^{-10} m. Find how many times wider the cell is, giving your answer in standard form to three significant figures. [3]
(b) The observable universe is about 8.8×10^{26} m across, and the shortest length with any physical meaning, the Planck length, is about 1.6×10^{-35} m. Find the ratio of the two, in standard form. [3]
(c) A sheet of paper is 0.1 mm thick. It is folded in half, then in half again, and so on. Find the thickness after 20 folds, in metres and in standard form. [3]
(d) Find the least number of folds needed for the thickness to exceed the distance to the Moon, 3.84×10^8 m. [4]
(e) Your answer to part (d) is far smaller than most people expect. Explain why, referring to how the thickness grows. [2]

35.

(a) Prove the change of base rule $\log_a b = \dfrac{\log_c b}{\log_c a}$, for valid bases a and c. *(Hint: let $x = \log_a b$, so $a^x = b$, then take $\log_c$ of both sides)* [3]

(b) Hence show that $\log_4 x = \frac{1}{2}\log_2 x$. [2]

(c) Solve $\log_2 x + \log_4 x + \log_{16} x = 7$. [4]

(d) Solve $\log_x 2 + \log_{x^2} 2 = 3$, giving an exact answer. [5]

Challenge Questions

36. GDC The loudness L of a sound in decibels is $L = 10\log_{10}\left(\dfrac{I}{I_0}\right)$, where I is the intensity in W/m^2 and $I_0 = 10^{-12}$ W/m^2 is the threshold of hearing. The scale is logarithmic because human hearing spans a range of intensities far too wide to put on a linear axis.

(a) A concert has loudness 110 dB. Find the intensity I. [2]

(b) A whisper has loudness 30 dB. Show that the concert is 10^8 times as intense as the whisper. [3]

(c) Show that doubling the intensity of *any* sound raises its loudness by $10\log_{10} 2$ dB, whatever the original level was. Give this value to two decimal places. [4]

(d) Hence find how many identical sources, playing together, are needed to raise the level by 10 dB. [3]

(e) Intensity obeys an inverse-square law: at distance r from a source, $I \propto \dfrac{1}{r^2}$. Show that doubling your distance reduces the level by a fixed number of decibels, independent of where you started, and find it. [4]

(f) A workplace safety rule allows 8 hours of exposure at 85 dB, and halves the permitted time for every 3 dB above that. Find the permitted time at 100 dB, and explain what part (c) has to do with the choice of 3 dB. [4]

37. GDC **Counting digits, and the first digit.** This investigation uses logarithms to count digits, and then to explain a pattern that auditors use to detect faked data.

(a) Let N be a positive integer. Show that N has exactly $\lfloor \log_{10} N \rfloor + 1$ digits, where $\lfloor t \rfloor$ is the greatest integer not exceeding t. [3]

(b) Hence find the number of digits in 2^{1000}. [2]

(c) The first digit of N is d exactly when the fractional part of $\log_{10} N$ lies in the interval $[\log_{10} d,\ \log_{10}(d+1))$. Verify this for $N = 2^7$. [3]

(d) Assume the fractional parts of $\log_{10}(2^n)$ are spread evenly over $[0, 1)$ as n increases. Show that the proportion of powers of 2 whose first digit is d is $\log_{10}\left(1 + \dfrac{1}{d}\right)$. [4]

(e) Evaluate this proportion for $d = 1$ and for $d = 9$. Then count the first digits of $2^1, 2^2, \ldots, 2^{30}$ on your GDC and compare. [5]

(f) The pattern in part (d) is known as Benford's law. Comment on how well your counts in part (e) support it, and state one reason why a sample of 30 might mislead. [3]

NUMBER & ALGEBRA

CHAPTER 2

Sequences & Series

2

SYLLABUS 1.2 1.3 1.4 1.8 1.9 1.10

A sheet of paper is about 0.1 mm thick. Fold it once and it is 0.2 mm. Each fold doubles the thickness, so 42 folds would give 0.1×2^{42} mm, which is about $440{,}000$ km. The Moon is about $384{,}000$ km away. Nobody can fold a sheet of paper 42 times, but the arithmetic is correct.

One further fold would reach the Moon and return. The reason lies in how doubling accumulates. Each fold doubles a thickness that has already been doubled, so after 42 folds the paper is not 42 times its original thickness but 2^{42} times it.

Now stack the sheets instead of folding them. Place one sheet at a time, and after 42 sheets the pile is 4.2 mm high. The paper is the same and the number of steps is the same, yet the answer is smaller by a factor of about a hundred billion. Folding multiplies at each step; stacking adds.

Those two operations produce the two kinds of list studied in this chapter. A list built by repeatedly adding a fixed amount is *arithmetic*. A list built by repeatedly multiplying by a fixed factor is *geometric*. The two arise in the same situations, so choosing the wrong one is costly: a salary that rises by a fixed amount each year is arithmetic, a salary that rises by a fixed percentage is geometric, and the two grow far apart over a career.

You will learn to find any term of such a list without writing out the terms before it, and to add the first n terms in a single calculation. You will also see why some sums that never end still have a finite total. The same two patterns govern the value of a savings account, the growth of a population by a fixed percentage, and the expansion of $(a+b)^{12}$ without multiplying out twelve brackets.

2.1 Arithmetic Sequences

A theatre has 20 seats in the front row, 23 in the second, and 26 in the third. Each row has 3 more seats than the one before it. To find how many seats are in row 50, counting up one row at a time would be slow. There is a clear pattern: each row adds 3 seats. A regular pattern like this can be written as a formula, and finding that formula is the point of this section.

A sequence where each term is obtained by adding a fixed constant to the previous one is called **arithmetic** (also called an **arithmetic progression**, or **AP**). That constant is the **common difference** d.

DEF An **arithmetic sequence** has the form $u_1,\ u_1+d,\ u_1+2d,\ \ldots$, where u_1 is the first term and d is the common difference.
For any two consecutive terms: $d = u_{n+1} - u_n$.

2.1.1 The general term

The n-th term is the first term plus $(n-1)$ steps of size d. Starting from u_1, you take one step to reach u_2, two steps to reach u_3, and so on.

KEY · IB

$$u_n = u_1 + (n-1)d$$

EXAMPLE 2.1

An arithmetic sequence has first term 7 and common difference -3. Find u_{20}.
You need 19 steps of size -3 from $u_1 = 7$:

$$u_{20} = 7 + (20-1)(-3) = 7 - 57 = -50.$$

A negative common difference means the sequence is decreasing. The general term can be negative even when the first term is positive.

EXAMPLE 2.2

In an arithmetic sequence, $u_5 = 11$ and $u_9 = 27$. Find u_1 and d.
From u_5 to u_9 is exactly 4 steps:

$$4d = u_9 - u_5 = 27 - 11 = 16 \implies d = 4.$$

Then $u_1 = u_5 - 4d = 11 - 16 = -5$.

EXAMPLE 2.3

Find the number of terms in the arithmetic sequence $5, 9, 13, \dots, 101$.

Here $d = 4$. Set $u_n = 101$:

$$5 + (n-1)(4) = 101 \implies 4(n-1) = 96 \implies n = 25.$$

CAUTION The formula is $u_1 + (n-1)d$, not $u_1 + nd$. For the first term ($n = 1$), you add zero differences, not one.

YOUR TURN 2.1 Arithmetic Sequences

1. Find the term indicated.

(a) u_7 of $3, 7, 11, \dots$

(b) u_{15} of the AP with $u_1 = 12$, $d = -4$

(c) u_{10} of $-5, -1, 3, \dots$

(d) u_{25} of the AP with $u_1 = 100$, $d = -7$

2. Find the quantity indicated.

(a) d, given $u_1 = 5$ and $u_9 = 37$

(b) u_1, given $u_6 = 23$ and $d = 3$

(c) d, given $u_3 = 8$ and $u_{11} = 32$

(d) u_1, given $u_{12} = 50$ and $d = 4$

3. Find the number of terms in each AP.

(a) $4, 7, 10, \dots, 100$

(b) $2, 9, 16, \dots, 100$

(c) $100, 96, 92, \dots, 4$

(d) $-7, -3, 1, \dots, 61$

4. Given $u_4 = 11$ and $u_{10} = 29$, find u_1 and d.

5. Show that the sequence $5, 11, 17, \dots$ has a term equal to 101. State which term it is.

2.2 Arithmetic Series

When you need the total of the first n terms of an arithmetic sequence, adding them one at a time is always possible but rarely necessary. A pattern reduces the whole sum to one short calculation.

Consider adding all integers from 1 to 100. Write the sum forwards and backwards:

$$1 + 2 + 3 + \cdots + 99 + 100$$

$$100 + 99 + 98 + \cdots + 2 + 1$$

Every vertical pair sums to 101, and there are 100 such pairs. So twice the sum equals 100×101, giving $S_{100} = 5050$.

The same pairing works for any arithmetic series: the first and last term always sum to the same value, $u_1 + u_n$. There are n such pairs when you write the series twice, so $2S_n = n(u_1 + u_n)$, and therefore:

KEY · IB

$$S_n = \frac{n}{2}(u_1 + u_n) \qquad = \qquad \frac{n}{2}(2u_1 + (n-1)d)$$

The first form is faster when you know u_n. Substituting $u_n = u_1 + (n-1)d$ gives the second, which works when only u_1 and d are known.

EXAMPLE 2.4

Find the sum of the first 30 terms of the arithmetic sequence $2, 7, 12, \ldots$

Here $u_1 = 2$ and $d = 5$. The last term is not given, so use the second form:

$$S_{30} = \frac{30}{2}(2(2) + 29(5)) = 15(4 + 145) = 15 \times 149 = 2235.$$

EXAMPLE 2.5

Find the sum $4 + 7 + 10 + \cdots + 100$.

First, $d = 3$. Find the number of terms: $4 + (n-1)(3) = 100 \Rightarrow n = 33$.

Then use the first form, since both u_1 and u_{33} are known:

$$S_{33} = \frac{33}{2}(4 + 100) = \frac{33 \times 104}{2} = 1716.$$

EXAMPLE 2.6

The sum of the first n terms of an arithmetic series is given by $S_n = 3n^2 + 5n$. Find u_1, d, and the general term u_n.

$u_1 = S_1 = 3 + 5 = 8$.

$u_2 = S_2 - S_1 = (12 + 10) - 8 = 14$, so $d = u_2 - u_1 = 6$.

For the general term: $u_n = S_n - S_{n-1} = 3n^2 + 5n - 3(n-1)^2 - 5(n-1)$.

Expanding: $u_n = 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5 = 6n + 2$.

Check: $u_1 = 8$ and $u_2 = 14$. Consistent.

CAUTION S_n is the total of the first n terms; u_n is just the n-th term. Confusing the two is the most common error in this topic. A question asking for a "term" wants u_n; a question asking for a "sum" wants S_n.

2.2.1 Application: simple interest

The standard financial application of an arithmetic sequence is **simple interest**: the same fixed amount of interest is added each year, so the balance increases by the same amount each year.

Write the opening balance as u_1 and the yearly interest as d. After n years, n payments of interest have been added, so the balance is

$$u_1 + nd.$$

Note the n here rather than the $n - 1$ of the general term. The opening balance is the amount before any interest has been paid, so n years later there have been n payments, not $n - 1$.

EXAMPLE 2.7

\$2000 is deposited in an account paying 3% simple interest per year. Find the balance after 15 years.

The interest is $3\% \times 2000 = \$60$ every year, so the balances form an arithmetic sequence with $d = 60$. After 15 years, 15 payments of interest have been added:

$$2000 + 15 \times 60 = \$2900.$$

2.2.2 When the data are not quite arithmetic

Real measurements rarely sit exactly on a sequence. A model is still useful if the steps are *close* to constant, and then the common difference has to be estimated rather than read off. The safest estimate uses the whole data set, not two chosen terms: divide the total change by the number of steps.

KEY For data that are approximately arithmetic over n terms,

$$d \approx \frac{u_n - u_1}{n-1}.$$

This is the mean of the successive differences. Those differences telescope, so only the first and last readings survive: the estimate is quick, but it rests entirely on the two endpoints. Fitting a least-squares line, as in Ch. 10, uses every reading instead.

EXAMPLE 2.8

A tree's height is measured at the end of each year: 1.8, 2.5, 3.1, 3.9, 4.6 metres. Show that an arithmetic model is reasonable, estimate the common difference, and predict the height after 8 years.

The successive differences are 0.7, 0.6, 0.8, 0.7. They are not equal, but they are close, so an arithmetic model is reasonable. Over four steps the height rose from 1.8 to 4.6, so

$$d \approx \frac{4.6 - 1.8}{5 - 1} = \frac{2.8}{4} = 0.7 \text{ metres per year.}$$

With $u_1 = 1.8$ and $d = 0.7$, the eighth term is

$$u_8 = 1.8 + 7(0.7) = 6.7 \text{ metres.}$$

Treat that as an estimate, not a fact. Two things limit it. The model was built on five years and year 8 lies outside them, and no tree grows at a constant rate forever, so the model must fail eventually. A prediction is only as good as the range the data covered.

YOUR TURN 2.2 Arithmetic Series

6. Find the sum indicated.

(a) S_{12} for the AP $2, 5, 8, \ldots$

(b) S_{20} for the AP with $u_1 = 100$, $d = -5$

(c) S_{15} for the AP $7, 11, 15, \ldots$

(d) S_{25} for the AP with $u_1 = -12$, $d = 5$

7. Find each sum.

(a) $6 + 10 + 14 + \cdots + 90$

(b) $3 + 8 + 13 + \cdots + 98$

(c) $50 + 46 + 42 + \cdots + 2$

(d) $1 + 1.5 + 2 + \cdots + 20$

8. Find the sum of all integers from 1 to 200 that are multiples of 6.

9. Find n if $S_n = 234$ for the AP $3, 6, 9, \ldots$

10. An AP has $u_1 = 15$, $u_n = -9$ and $n = 13$. Find S_n.

11. Given $S_n = 4n^2 - n$, find u_1, d and the general term u_n.

12. A plant's height is recorded weekly, in centimetres: 4.2, 6.1, 7.8, 9.9, 11.8.

(a) Find the four successive differences, and explain why an arithmetic model is reasonable.
(b) Estimate the common difference from the first and last readings.
(c) Predict the height in week 9, and give one reason to treat the prediction with caution.

2.3 Geometric Sequences

A geometric sequence multiplies by the same number at each step. Start at 1 and double: $1, 2, 4, 8, 16, \ldots$ Many real quantities change in this way. Bacteria double in every generation. A ball bounces to 60% of the height it fell from. Light loses half its intensity for each extra filter. In every case, each new value is the previous value multiplied by a fixed number.

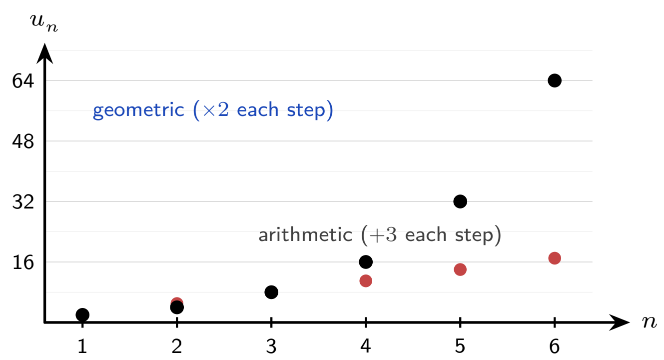


Figure 2.1 Adding a fixed amount grows in a straight line; multiplying by a fixed factor grows explosively.

A sequence where each term is obtained by multiplying the previous term by a fixed constant is called **geometric** (also called a **geometric progression**, or **GP**). That constant is the **common ratio** r.

DEF A **geometric sequence** has the form $u_1,\ u_1r,\ u_1r^2,\ \ldots$, where u_1 is the first term and r is the common ratio.
For any two consecutive terms: $r = \dfrac{u_{n+1}}{u_n}$.

2.3.1 The general term

Starting from u_1, each step multiplies by r. After $(n-1)$ multiplications you reach u_n:

KEY · IB

$$u_n = u_1 \cdot r^{n-1}$$

EXAMPLE 2.9

A geometric sequence has $u_1 = 5$ and $r = 3$. Find u_6.

$$u_6 = 5 \cdot 3^5 = 5 \times 243 = 1215.$$

EXAMPLE 2.10

A geometric sequence has $u_3 = 28$ and $u_6 = 224$. Find u_1 and r.
From u_3 to u_6 is 3 multiplicative steps: $u_6 = u_3 \cdot r^3$, so

$$r^3 = \frac{224}{28} = 8 \implies r = 2.$$

Then $u_1 = u_3/r^2 = 28/4 = 7$.

EXAMPLE 2.11

Find the number of terms in the geometric sequence $2, 6, 18, \ldots, 4374$.
Here $r = 3$. Set $u_n = 4374$:

$$2 \cdot 3^{n-1} = 4374 \implies 3^{n-1} = 2187 = 3^7 \implies n = 8.$$

EXAMPLE 2.12

Given that k, 8 and $k + 12$ are consecutive terms of a geometric sequence, find the possible values of k.
Consecutive terms have a constant ratio, so the ratio across the first gap equals the ratio across

the second:

$$\frac{8}{k} = \frac{k+12}{8} \implies 64 = k(k+12) \implies k^2 + 12k - 64 = 0.$$

Factorising, $(k+16)(k-4) = 0$, so $k = -16$ or $k = 4$.
Both values give a genuine sequence. With $k = 4$ the terms are $4, 8, 16$ and $r = 2$; with $k = -16$ they are $-16, 8, -4$ and $r = -\frac{1}{2}$. Do not discard the negative solution: a negative ratio makes the terms alternate in sign, which is still a geometric sequence.

2.3.2 Recursively defined sequences

Every rule so far has been **explicit**: put n in and the n-th term comes out, with no need for any term before it. A sequence can instead be given by a **recurrence** — a starting value together with a rule that builds each term from the previous one. The sequence $u_1 = 4$, $u_{n+1} = 3u_n - 4$ is defined this way, and it runs $4, 8, 20, 56, \ldots$

A recurrence produces terms one at a time, so reaching u_{50} means computing the forty-nine terms below it. A closed form removes that work. For a rule of the shape $u_{n+1} = au_n + b$, the route to one is a shift: measure the terms from the value the rule leaves unchanged, and what remains is geometric.

EXAMPLE 2.13

A sequence is defined by $u_1 = 4$ and $u_{n+1} = 3u_n - 4$. Let $v_n = u_n - 2$. Show that $\{v_n\}$ is geometric, and find a closed form for u_n.
Subtract 2 from both sides of the recurrence:

$$u_{n+1} - 2 = 3u_n - 6 = 3(u_n - 2), \qquad \text{that is} \qquad v_{n+1} = 3v_n.$$

So $\{v_n\}$ is geometric with $r = 3$, and $v_1 = u_1 - 2 = 2$. Therefore

$$v_n = 2 \cdot 3^{n-1}, \qquad u_n = v_n + 2 = 2 \cdot 3^{n-1} + 2.$$

Check the first three terms: $2+2 = 4$, $6+2 = 8$, $18+2 = 20$, matching 4, $3(4)-4$, $3(8)-4$. The substitution is not a guess. Solving $x = 3x - 4$ gives $x = 2$, the value the rule fixes, and $v_n = u_n - 2$ measures each term from it, which is exactly what cancels the -4.

2.3.3 Where geometric growth shows up

Three contexts account for most examination questions, and each is the same algebra wearing different clothes.

- **Population growth.** A population rising by a fixed *percentage* each year is geometric with $r = 1 + \frac{p}{100}$. A 2% annual rise gives $r = 1.02$.
- **Salary change.** An annual rise of 3% gives $r = 1.03$; a cut of 3% gives $r = 0.97$. Note that a rise then an equal cut does not return you to the start, because $1.03 \times 0.97 =$

0.9991.

- **The spread of a disease.** If each infected person passes it to r others, the number of new cases in successive rounds is geometric. That r is what public health reporting calls the reproduction number, and whether it sits above or below 1 decides everything.

The last one shows why the ratio matters more than the starting value. If $r > 1$ the terms grow without limit, however small u_1 is; if $r < 1$ they shrink to nothing, however large u_1 is. A large outbreak with $r = 0.9$ dies out, and a single case with $r = 1.1$ does not.

EXAMPLE 2.14

A town of $40{,}000$ people grows by 2.5% each year. Find the population after 10 years, and the first year in which it exceeds $60{,}000$.

Here $u_1 = 40{,}000$ and $r = 1.025$. Ten years later is ten multiplications, so the value is $u_1 r^{10}$, not u_{10}:

$$40{,}000 \times 1.025^{10} \approx 51{,}203.$$

For the second part, solve $40{,}000 \times 1.025^n > 60{,}000$, that is $1.025^n > 1.5$. Taking logarithms (Ch. 1),

$$n > \frac{\ln 1.5}{\ln 1.025} \approx 16.4,$$

so the population first exceeds $60{,}000$ during year 17. Round *up* here whatever the decimal: year 16 is not yet enough.

TIP To test whether a sequence is arithmetic, check that differences are constant. To test whether it is geometric, check that ratios are constant. Never apply both tests to the same sequence expecting both to be true.

YOUR TURN 2.3 Geometric Sequences

13. Find the term indicated.

(a) u_6 of the GP $2, 6, 18, \dots$

(b) u_7 of the GP with $u_1 = 96$, $r = \frac{1}{2}$

(c) u_5 of $\frac{1}{4}, \frac{1}{2}, 1, \dots$

(d) u_8 of the GP with $u_1 = 3$, $r = -2$

14. Find the quantity indicated.

(a) r, given $u_1 = 5$ and $u_4 = 40$

(b) u_1, given $u_4 = 24$ and $r = 2$

(c) r, given $u_1 = 81$ and $u_5 = 1$

(d) u_1, given $u_3 = 45$ and $r = 3$

15. Find the number of terms in each GP.

(a) $3, 6, 12, \dots, 768$

(b) $5, 15, 45, \dots, 3645$

(c) $2, -6, 18, \dots, -4374$

(d) $128, 64, 32, \dots, \frac{1}{2}$

16. Given $u_2 = 10$ and $u_5 = 1250$, find u_1 and r.

17. The 3rd term of a GP is 12 and the 6th term is 96. Find the first term.

18. GDC Geometric growth in three contexts.

(a) A population of $25{,}000$ grows by 1.8% each year. Find the population after 12 years.

(b) A salary of \$$48{,}000$ rises by 4% each year. Find the salary in the sixth year.

(c) In an outbreak each infected person infects 1.2 others, and round 1 has 50 new cases. Find the number of new cases in round 6, and state what would happen in the long run if that figure fell to 0.8.

2.4 Geometric Series

Adding the terms of a geometric sequence needs a different method. In an arithmetic series, the first and last terms have the same total as the second and second-to-last terms, so pairing works. In a geometric series these pair totals are all different, so pairing does not help. Instead there is an algebraic method: multiply the sum by r, line the two sums up, and subtract.

Let $S = u_1 + u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1}$. Multiply by r:

$$rS = u_1 r + u_1 r^2 + \cdots + u_1 r^n.$$

Every term appears in both lines except u_1 in the first line and $u_1 r^n$ in the second. Subtract rS from S:

$$S(1-r) = u_1 - u_1 r^n = u_1(1-r^n).$$

Divide both sides by $(1-r)$ to get the formula.

KEY · IB

$$S_n = \frac{u_1(1-r^n)}{1-r} \qquad = \qquad \frac{u_1(r^n-1)}{r-1}, \qquad r \neq 1$$

Both forms are equivalent. When $0 < r < 1$, use the first form: both $1 - r^n$ and $1 - r$ are then positive, so no negative signs appear. When $r > 1$ the second form is tidier, for the same reason. If r is negative, either form works, but r^n changes sign with n, so take care with the arithmetic.

EXAMPLE 2.15

Find the sum of the first 10 terms of the geometric series $5 + 10 + 20 + \cdots$

Here $u_1 = 5$ and $r = 2$:

$$S_{10} = \frac{5(2^{10} - 1)}{2 - 1} = 5 \times 1023 = 5115.$$

EXAMPLE 2.16

A geometric series has $u_1 = 80$ and $r = \frac{1}{2}$. Find the least value of n such that $S_n > 159$.

$$S_n = \frac{80\left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} = 160\left(1 - \frac{1}{2^n}\right).$$

Require $160\left(1 - \frac{1}{2^n}\right) > 159$, so $\frac{1}{2^n} < \frac{1}{160}$, giving $2^n > 160$.
Since $2^7 = 128 < 160$ and $2^8 = 256 > 160$, the least value is $n = 8$.

CAUTION The formula requires $r \neq 1$. If $r = 1$, every term equals u_1 and $S_n = nu_1$. Substituting $r = 1$ into the formula gives $0/0$, which is undefined.

Solving for n in geometric series often requires taking logarithms of both sides. See Ch. 1, §1.5.

YOUR TURN 2.4 Geometric Series

19. Find the sum indicated.

(a) S_8 for the GP $1, 3, 9, \ldots$
(b) S_{10} for the GP with $u_1 = 6$, $r = \frac{1}{3}$
(c) S_6 for the GP $128, 64, 32, \ldots$
(d) S_6 for the GP with $u_1 = 1$, $r = -2$

20. The sum of the first three terms of a GP is 21 and the common ratio is 2. Find the first term.

21. Find n such that $S_n = 63$ for the GP $1, 2, 4, \ldots$

22. Find each sum.

(a) $3 + 6 + 12 + \cdots + 1536$
(b) $2 + 6 + 18 + \cdots + 486$
(c) $1 - 2 + 4 - \cdots + 256$
(d) $96 + 48 + 24 + \cdots + 3$

23. Given $S_3 = 7$ and $S_6 = 63$, find u_1 and r.

2.5 Sigma Notation

By now you have written several sums in the form $u_1 + u_2 + \cdots + u_n$. There is a compact notation for this that appears throughout mathematics, physics, and statistics, and it should be learned properly.

The Greek letter Σ (capital sigma) is used as shorthand for "sum". The expression

$$\sum_{k=1}^{n} u_k = u_1 + u_2 + u_3 + \cdots + u_n$$

is read "the sum of u_k from $k = 1$ to n." The variable k is the **index of summation**: it takes each integer value from the lower limit to the upper limit, the corresponding term is evaluated, and all terms are added.

> DEF $\displaystyle\sum_{k=m}^{n} f(k) = f(m) + f(m+1) + f(m+2) + \cdots + f(n),$ where $m \leq n$.
> The index k is a **dummy variable**: the result depends only on f, m, and n, not on the name chosen for the index.

2.5.1 Properties of sigma notation

These three properties are just familiar rules of arithmetic, written in sigma form. They are used constantly when simplifying or manipulating sums.

> KEY For a constant c and sequences $\{a_k\}$, $\{b_k\}$:
>
> $\displaystyle\sum_{k=1}^{n} c = nc$ — a constant summed n times
>
> $\displaystyle\sum_{k=1}^{n} c\,a_k = c\sum_{k=1}^{n} a_k$ — a constant factor comes outside the sum
>
> $\displaystyle\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$ — a sum splits over addition

EXAMPLE 2.17

Write $3 + 7 + 11 + 15 + 19$ in sigma notation.

The terms form an arithmetic sequence with $u_1 = 3$, $d = 4$: the k-th term is $u_k = 4k - 1$.

$$3 + 7 + 11 + 15 + 19 = \sum_{k=1}^{5} (4k - 1).$$

EXAMPLE 2.18

Evaluate $\sum_{k=1}^{6} 5 \cdot 2^{k-1}$.

This is a geometric series with $u_1 = 5$ and $r = 2$:

$$\sum_{k=1}^{6} 5 \cdot 2^{k-1} = S_6 = \frac{5(2^6 - 1)}{2 - 1} = 5 \times 63 = 315.$$

EXAMPLE 2.19

Evaluate $\sum_{k=3}^{10} (2k+1)$.

The lower limit is 3, not 1. Use the fact that the full sum minus the missing part gives the partial sum:

$$\sum_{k=3}^{10} (2k+1) = \sum_{k=1}^{10} (2k+1) - \sum_{k=1}^{2} (2k+1).$$

The first sum is an arithmetic series: $u_1 = 3$, $u_{10} = 21$, so $S_{10} = \frac{10}{2}(3 + 21) = 120$.

The second sum is just $3 + 5 = 8$.

Result: $120 - 8 = 112$.

CAUTION $\sum_{k=1}^{n} (a_k \cdot b_k) \neq \left(\sum a_k\right)\left(\sum b_k\right)$. You can split a sum over addition but not over multiplication. This is one of the most common misapplications of sigma notation.

YOUR TURN 2.5 Sigma Notation

24. Write each sum in sigma notation.

(a) $5 + 10 + 15 + 20 + 25$

(b) $4 + 8 + 16 + 32 + 64$

(c) $3 + 6 + 9 + \cdots + 30$

(d) $1 + 4 + 9 + 16 + 25$

25. Evaluate each sum.

(a) $\sum_{k=1}^{6} (3k - 2)$

(b) $\sum_{k=1}^{5} 2 \cdot 3^{k-1}$

(c) $\sum_{k=4}^{8} (2k + 1)$

26. Find $\sum_{k=1}^{n} 5$ and $\sum_{k=1}^{n} 5k$ using the sigma properties.

2.6 Infinite Geometric Series

Cut a pizza in half. Eat one half and keep the other. Cut what is left in half again, eat one piece, and keep the rest. If you could continue forever, how much pizza would you eat?

The pieces you eat are $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, ... There are infinitely many of them, and every piece is larger than zero. Adding infinitely many positive numbers sounds like it must give an infinite total. But you began with one pizza, so you can never eat more than one pizza. The total can only be 1.

That is the surprise of this section: *an addition that never ends can still have a finite answer.*

Add the pieces one at a time and record the running total after each step. Each running total is a **partial sum**:

$$S_1 = 0.5, \quad S_2 = 0.75, \quad S_3 = 0.875, \quad S_4 = 0.9375, \quad \ldots$$

Two things happen at once. Every partial sum is larger than the one before it, and every partial sum is still smaller than 1. Look at the distance from 1: it is 0.5, then 0.25, then 0.125. The distance halves at every step, so it never becomes zero, but it becomes as small as you wish. This value that the partial sums approach is called the **limit**, and here the limit is 1.

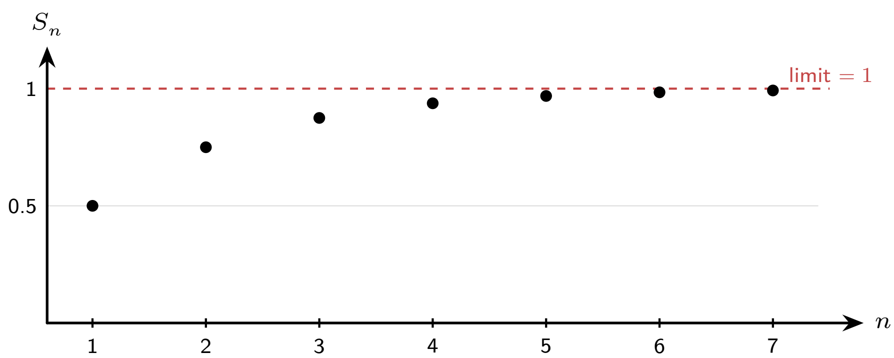


Figure 2.2 The partial sums of $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ climb toward 1 but never pass it.

The pizza works because each piece is smaller than the piece before it. That is exactly what $|r| < 1$ means: every term is a fraction of the previous term, so the terms shrink toward zero.

Now change one number and the behaviour changes completely. Take $1 + 2 + 4 + 8 + \cdots$, where $r = 2$. Each term is larger than the last, the partial sums are $1, 3, 7, 15, \ldots$, and they pass every value you can name. There is no finite total.

A series whose partial sums approach a finite limit is said to **converge**. A series whose partial sums grow without any bound is said to **diverge**. A geometric series converges when $|r| < 1$ and diverges when $|r| \geq 1$.

To turn this into a formula, start from the finite sum and let n grow without bound. When $|r| < 1$, the power r^n shrinks toward 0:

$$S_n = \frac{u_1(1-r^n)}{1-r} \xrightarrow{n\to\infty} \frac{u_1(1-0)}{1-r} = \frac{u_1}{1-r}.$$

KEY · IB For a geometric series with $|r| < 1$:

$$S_\infty = \frac{u_1}{1-r}$$

EXAMPLE 2.20

Find the sum to infinity of $6 + 4 + \frac{8}{3} + \cdots$

The common ratio is $r = 4/6 = \frac{2}{3}$. Since $|r| < 1$ the series converges:

$$S_\infty = \frac{6}{1-\frac{2}{3}} = \frac{6}{\frac{1}{3}} = 18.$$

EXAMPLE 2.21

Express $0.\overline{27} = 0.272727\ldots$ as an exact fraction.

Write as an infinite geometric series with $u_1 = \frac{27}{100}$ and $r = \frac{1}{100}$:

$$0.272727\ldots = \frac{27/100}{1-1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}.$$

EXAMPLE 2.22

A geometric series has first term a and common ratio r. Given $S_\infty = 8$ and $u_2 = 2$, find a and r.

From S_∞: $\dfrac{a}{1-r} = 8$, so $a = 8(1-r)$.

From $u_2 = ar = 2$: substitute a:

$$8(1-r)r = 2 \implies 8r^2 - 8r + 2 = 0 \implies 4r^2 - 4r + 1 = 0 \implies (2r-1)^2 = 0.$$

So $r = \frac{1}{2}$, and $a = 8\left(1 - \frac{1}{2}\right) = 4$.

CAUTION Always verify $|r| < 1$ before applying the infinite sum formula. If $|r| \geq 1$, no finite sum to infinity exists.

YOUR TURN 2.6 Infinite Geometric Series

27. Find S_∞ for each series.

(a) $12 + 4 + \frac{4}{3} + \cdots$
(b) the GP with $u_1 = 10$, $r = \frac{3}{5}$
(c) $9 + 3 + 1 + \cdots$
(d) the GP with $u_1 = 8$, $r = -\frac{1}{4}$

28. Find the quantity indicated.

(a) u_1, given $S_\infty = 20$ and $r = \frac{1}{5}$
(b) r, given $u_1 = 6$ and $S_\infty = 9$
(c) u_1, given $S_\infty = 12$ and $r = \frac{1}{4}$
(d) r, given $u_1 = 15$ and $S_\infty = 12$

29. Express $0.\overline{5} = 0.5555\ldots$ as an exact fraction.

30. State whether $6 + 9 + 13.5 + \cdots$ converges. Give a reason.

31. A GP has $u_1 = 16$ and $u_3 = 4$. Find S_∞.

2.7 Financial Applications

A bank pays you 4% per year. In the first year you earn interest on your deposit. In the second year you earn interest on the deposit *and* on the interest from the first year. The amount added is different every year, so the balances do not form an arithmetic sequence. They form a geometric one: each year the balance is multiplied by 1.04.

This is **compound interest**, and it is why money left alone for a long time grows far more than most people expect. The balance after any number of years is simply a term of a geometric sequence.

Interest is often added more than once a year. If a present value PV earns interest at a nominal annual rate of $r\%$, compounded k times per year, then after n years:

KEY · IB

$$FV = PV \times \left(1 + \frac{r}{100k}\right)^{kn}$$

The quantity $\left(1 + \frac{r}{100k}\right)$ is the common ratio. The total number of compounding periods is kn.

EXAMPLE 2.23

\$5 000 is invested at 4% per year, compounded monthly. Find the value after 3 years.
Here $PV = 5000$, $r = 4$, $k = 12$, $n = 3$:

$$FV = 5000 \times \left(1 + \frac{4}{1200}\right)^{36} \approx \$5636.36.$$

EXAMPLE 2.24

How many complete years does it take for an investment to double at 6% per year, compounded annually?
Set $FV = 2 \cdot PV$:

$$(1.06)^n = 2 \implies n = \frac{\ln 2}{\ln 1.06} \approx 11.9.$$

After 12 complete years the investment has more than doubled.

TIP The Rule of 72: dividing 72 by the annual interest rate gives an approximate doubling time in years. At 6%, that is $72/6 = 12$ years, matching the exact calculation above.

2.7.1 Regular deposits

A single lump sum gives one term of a geometric sequence. A savings plan is different: money goes in every year, and each payment then compounds for however long it has left in the account. The balance is a total of geometric terms, so it is a geometric *series*.

Write the payment as P and the annual rate as a decimal i. With payments made at the **end** of each year for n years, the final payment earns no interest at all, the one before it earns interest for one year, and the first earns interest for $n - 1$ years. Reading the total from the last payment backwards gives a GP with first term P and common ratio $1 + i$.

That is the whole of it. There is no new formula to learn: the sum is the geometric series of §2.4 with $u_1 = P$ and $r = 1 + i$, and every savings-plan question is answered by identifying those two quantities and reaching for the sum you already have.

Payments made at the **start** of each year each spend one extra year in the account. Rather than remembering a correction factor, change the first term: the last payment now earns a year's interest, so $u_1 = P(1 + i)$ and everything else is unchanged.

EXAMPLE 2.25

\$800 is paid into an account at the end of each year for 12 years. The account pays 5% per year, compounded annually. Find the value immediately after the twelfth payment.
The payment made at the end of year 12 has earned nothing, the one at the end of year 11 has earned interest for a single year, and the first payment has earned interest for 11 years. Totalling

from the last payment backwards:

$$800 + 800(1.05) + 800(1.05)^2 + \cdots + 800(1.05)^{11}.$$

This is a geometric series with $u_1 = 800$, $r = 1.05$ and $n = 12$ terms:

$$S_{12} = \frac{800(1.05^{12} - 1)}{1.05 - 1} \approx \$12\,733.70.$$

Had the same payments been made at the *start* of each year, every one of them would earn a further year of interest, so $u_1 = 800(1.05)$ and the same sum gives

$$S_{12} = \frac{800(1.05)(1.05^{12} - 1)}{0.05} \approx \$13\,370.39.$$

Settle which end of the year the payments land on before writing down u_1; the two answers differ by a full year's interest on the whole plan.

2.7.2 Depreciation

Some things lose value instead of gaining it. A car, a laptop, or a machine may lose a fixed *percentage* of its value each year. This is called **depreciation**.

No new formula is needed. Use the same compound-interest formula with a **negative** rate r. A loss of 12% per year means $r = -12$, so the common ratio becomes

$$1 + \frac{r}{100} = 1 - \frac{12}{100} = 0.88,$$

a number less than 1. Multiplying by a ratio less than 1 at every step makes the values shrink, so the sequence of values is still geometric.

EXAMPLE 2.26

A car bought for $28 000 loses 12% of its value each year. Find its value after 5 years.

Here $r = -12$ and $k = 1$, so each year the value is multiplied by 0.88:

$$FV = 28\,000 \times (0.88)^5 \approx \$14\,776.$$

After five years the car has lost nearly half its value. Notice that the value never reaches zero: each year removes 12% of a smaller amount than the year before.

2.7.3 Inflation and real value

There is a second geometric process working against your savings. **Inflation** is the yearly rise in prices. Interest increases the number in your account, while inflation reduces what that number can buy. If your account grows by 2% in a year and prices rise by 3% in the same year, you hold more money but you can buy less with it.

The **real value** of an investment is its future value adjusted for the price rises over the same period. It measures the gain in purchasing power rather than the gain in the number.

Prices climb geometrically too, so the correction is a second geometric factor, this time dividing rather than multiplying.

KEY With interest producing a future value FV over n years, and prices rising at $i\%$ per year:

$$\text{real value} = \frac{FV}{\left(1+\frac{i}{100}\right)^n}.$$

Divide by the inflation factor, because the same goods cost $\left(1+\frac{i}{100}\right)^n$ *times as much after* n *years.*

EXAMPLE 2.27

\$10 000 is invested for 5 years at 4.5% per year, compounded annually. Prices rise at 2.5% per year. Find the future value and the real value of the investment.

The balance grows by a factor of 1.045 each year:

$$FV = 10\,000 \times (1.045)^5 \approx \$12\,461.82.$$

Now adjust for five years of price rises:

$$\text{real value} = \frac{12\,461.82}{(1.025)^5} \approx \$11\,014.43.$$

The account shows a gain of about \$2460, but in purchasing power the investor is only about \$1014 better off. An interest rate means little until you compare it with inflation.

TIP Your GDC's built-in Finance (TVM) solver handles compound interest, depreciation, and unknown rates or times directly: enter the known values of N, $I\%$, PV, FV, and the compounding frequency, and solve for the missing one. In examinations, state the values you entered.

Solving for n in financial problems requires exponential equations. See Ch. 1, §1.5.

YOUR TURN 2.7 Financial Applications

32. GDC Find the future value of each investment.

(a) \$3 000 at 4% p.a., compounded annually, for 5 years

(b) \$8 000 at 6% p.a., compounded monthly, for 3 years

(c) \$500 at 3.6% p.a., compounded monthly, for 10 years

(d) \$2 000 at 5% p.a., compounded quarterly, for 4 years

33. GDC Find the annual interest rate if \$5 000 grows to \$5 828.84 in 3 years, compounded annually.

34. GDC How many complete years does it take \$2 000 to double at 9% p.a., compounded annually?

35. GDC Find the present value needed to accumulate \$15 000 in 8 years at 5% p.a., compounded quarterly.

36. GDC A car valued at \$25 000 depreciates at 15% per year. Find its value after 4 years.

2.8 Counting Principles HL

A four-digit PIN is built from the digits 0 to 9, and any digit may repeat. There are 10 choices for the first digit, 10 for the second, 10 for the third and 10 for the fourth, so the number of PINs is

$$10 \times 10 \times 10 \times 10 = 10\,000.$$

Nobody reaches ten thousand by writing the list out. The total comes from the way the choices are made, and two ordinary words decide what to do with the counts.

And multiplies. When a result needs one choice *and* then another, the counts multiply. A menu offering 3 starters and 2 mains gives $3 \times 2 = 6$ meals.

A tree shows why. Each starter has the same number of branches below it, so the total is one count multiplied by the other rather than added.

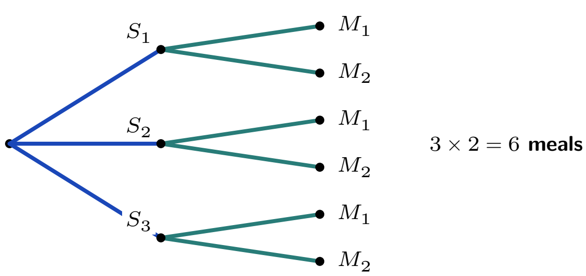


Figure 2.3 Three starters, two mains for each. Every path is a starter *and* a main, and there are $3 \times 2 = 6$ of them: that is what "and multiplies" looks like.

Add a third course and the same reasoning applies again: with 4 desserts the count becomes $3 \times 2 \times 4 = 24$. Every extra choice multiplies.

Or adds. Not every count is a chain of choices. Sometimes the answer is one thing *or* another, and the two cannot both happen: a password of length 3 *or* length 4, a team with three women *or* four. Count each possibility separately and add.

Those two words are the whole decision. Everything else in this section is about counting a single choice correctly once you know which word applies.

DEF HL **And multiplies** (the **multiplication principle**). If a result needs a first choice *and* a second, made in m and n ways, there are $m \times n$ results. Any number of choices chains the same way.
The condition: n must be the same number whichever way the first choice went. It may differ from m — 10 people, then 9 — but it may not depend on which one you picked.

Or adds (the **addition principle**). If the result is one case *or* another, add the counts.
The condition: the cases must not overlap. Nothing may be counted twice.

CAUTION Everyday English uses *or* inclusively: "students who take French or Spanish" includes those taking both. That "or" does not simply add — the students taking both would be counted twice. In this section *or* always means the exclusive one: this case or that case, never both. The overlapping kind is handled in Ch. 11, §11.3, where $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ subtracts the double count.

Arranging n different objects in a row is one long chain of *and*s: n choices for the first position, $n-1$ for the second, and so on down to 1. The product is n **factorial**.

KEY HL

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1, \qquad 0! = 1$$

The value $0! = 1$ is set by definition, not derived. There is one way to arrange no objects at all, and fixing $0! = 1$ keeps every formula below correct at its end points.

2.8.1 Permutations and combinations

There are two standard questions about selecting r objects from n different ones. If the *order* of selection matters, the count is a **permutation**. If only the final collection matters, it is a **combination**.

For a permutation, stop the factorial early. There are n choices for the first object, $n-1$ for the second, and so on for r factors in total. Multiplying and dividing by the remaining $(n-r)$ factors turns this product into a ratio of factorials:

$${}^{n}P_{r} = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

The combination count follows from the permutation count. Choose 3 letters from A, B, C, D, E. The selection $\{A, B, C\}$ can be written in $3! = 6$ orders: ABC, ACB, BAC, BCA, CAB, CBA. A permutation count records six results here; a combination count records one. Every combination is therefore counted $r!$ times among the permutations, so dividing by $r!$ removes the duplicates:

$$^nC_r = \frac{^nP_r}{r!}.$$

KEY · IB HL Selecting r from n distinct objects:

$$^nP_r = \frac{n!}{(n-r)!} \quad \text{(ordered)}, \qquad ^nC_r = \frac{n!}{r!\,(n-r)!} \quad \text{(unordered)}$$

The symbol ^nC_r is read "n choose r". Other books write the same number as $\binom{n}{r}$, so both forms should be recognised; the formula booklet uses ^nC_r, and a GDC labels the two counts nPr and nCr.

The question	The count	Example
Arrange all n objects	$n!$	6 books in a row
Arrange r of the n, order matters	^nP_r	3 medals for 8 runners
Choose r of the n, order does not matter	^nC_r	a committee of 3 from 12

Three steps are enough for almost every counting question in this course.

1. Read the question for *and* or *or*. *And* multiplies, *or* adds. Check the condition each one carries.
2. Within a single choice, ask whether the order matters. If it does, use ^nP_r. If it does not, use ^nC_r.
3. If the restriction is awkward to count directly, count everything and subtract what breaks it.

Two restrictions appear often enough to name. When certain objects must stay together, treat the whole group as one object, arrange the objects, then arrange the group internally and multiply the two counts. When a question asks for *at least one* of something, step 3 is usually shorter: subtract the arrangements that have none from the total.

EXAMPLE 2.28

Eight runners compete. In how many ways can gold, silver, and bronze be awarded?

The three medals are distinguishable, so order matters:

$$^8P_3 = 8 \times 7 \times 6 = 336.$$

EXAMPLE 2.29

A committee of 5 is chosen from 6 boys and 5 girls. How many committees contain exactly 2 girls?

The committee needs 2 girls *and* 3 boys, so the two counts multiply. A committee has no internal order, so each count is a combination:

$$^5C_2 \times {}^6C_3 = 10 \times 20 = 200.$$

EXAMPLE 2.30

Five people sit in a row. In how many ways can they sit if two of them refuse to sit next to each other?

Count the seatings that place the two together, then subtract. Treat the pair as one object: four objects give $4! = 24$ arrangements, and the pair can sit in $2! = 2$ orders inside their block, so

$$24 \times 2 = 48$$

seatings place them together. All five can be seated in $5! = 120$ ways, so

$$120 - 48 = 72$$

seatings keep them apart. Counting the forbidden cases and subtracting is shorter here than building the allowed ones directly.

CAUTION Ask one question before every counting problem: *does order matter?* Medals, codes and rankings: yes, so use ^nP_r. Committees, hands of cards and subsets: no, so use ^nC_r.

TIP A GDC computes both, under the labels nPr and nCr. Two kinds of counting are not required in this course: arrangements in which some of the objects are identical, and arrangements in a circle.

YOUR TURN 2.8 Counting Principles HL

37. Find each count.

(a) arrangements of 6 different books in a row
(b) 4-digit codes using the digits 1 to 9, no digit repeated
(c) committees of 3 chosen from 12 people
(d) groups of 2 men and 2 women chosen from 7 men and 6 women

38. Six different books are arranged in a row. Find the number of arrangements in which two particular books

(a) stand next to each other
(b) do not stand next to each other

39. A 4-digit code uses the digits 1 to 9 with no digit repeated. Find the number of such codes that end in an even digit.

40. A committee of 3 is chosen from 12 people, one of whom is Ana. Find the number of committees that

(a) include Ana
(b) do not include Ana

41. Each of these is an *or*: split the count into parts that cannot both happen, then add.

(a) A group of 4 is chosen from 7 men and 6 women. Find the number of groups containing at least 3 women.
(b) A password is 3 or 4 letters long, chosen from the 26 lowercase letters with no letter repeated. Find the number of possible passwords.

42. Solve ${}^nC_2 = 45$.

2.9 The Binomial Theorem

Expanding $(a+b)^2$ is quick. Expanding $(a+b)^3$ takes longer. Expanding $(a+b)^{12}$ by multiplying out twelve brackets is not realistic. Fortunately the answer follows a pattern, so no bracket multiplication is needed at all.

Look at the small cases:

$$(a+b)^1 = a+b, \qquad (a+b)^2 = a^2+2ab+b^2, \qquad (a+b)^3 = a^3+3a^2b+3ab^2+b^3.$$

Two features are regular. The power of a falls from n to 0 while the power of b rises from 0 to n, and in every term the two powers add to n. The coefficients form a row of **Pascal's**

triangle, in which each entry is the sum of the two entries above it:

$$\begin{array}{ccccccccccc} &&&&& 1 &&&&& \\ &&&& 1 && 1 &&&& \\ &&& 1 && 2 && 1 &&& \\ && 1 && 3 && 3 && 1 && \\ & 1 && 4 && 6 && 4 && 1 & \\ 1 && 5 && 10 && 10 && 5 && 1 \end{array}$$

Counting explains why. Multiplying out $(a+b)^n$ means choosing a or b from each of the n brackets. A term $a^{n-r}b^r$ arises once for every way of choosing which r brackets contribute a b, and that count is exactly ${}^nC_r$. The coefficients of the expansion *are* the combinations of the previous section.

KEY · IB For $n \in \mathbb{N}$:

$$(a+b)^n = a^n + {}^nC_1\, a^{n-1}b + \cdots + {}^nC_r\, a^{n-r}b^r + \cdots + b^n,$$

$$\text{where} \quad {}^nC_r = \frac{n!}{r!\,(n-r)!}.$$

Pascal's triangle is fastest for small n. When n is large, or when a question asks for one coefficient only, use the single term ${}^nC_r\, a^{n-r}b^r$ shown in the middle of the expansion above and choose the r you need.

TIP The booklet shows that single term inside the expansion without naming it. The **general term** is

$$T_{r+1} = {}^nC_r\, a^{n-r}b^r,$$

so $r = 0$ gives the first term, $r = 1$ the second, and r gives term number $r+1$. The subscript is $r+1$ because r counts how many factors of b the term contains, and the first term contains none. In an examination, write down this term, find the r that produces the power you want, then substitute.

EXAMPLE 2.31

Expand $(x+2)^4$.

Row 4 of the triangle gives coefficients $1, 4, 6, 4, 1$, with the powers of 2 increasing as the powers of x decrease:

$$(x+2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16 = x^4 + 8x^3 + 24x^2 + 32x + 16.$$

EXAMPLE 2.32

Find the coefficient of x^3 in $(2x-3)^5$.

The general term is ${}^5C_r\,(2x)^{5-r}(-3)^r$. For x^3, take $5-r=3$, so $r=2$:

$$ {}^5C_2\,(2x)^3(-3)^2 = 10\times 8x^3\times 9 = 720x^3. $$

The coefficient is 720. Keep the sign inside the bracket with $b=-3$: the $(-3)^r$ produces the alternating signs automatically.

EXAMPLE 2.33

Find the term independent of x in $\left(x^2+\dfrac{2}{x}\right)^6$.
Track the power of x in the general term:

$$ T_{r+1} = {}^6C_r\,(x^2)^{6-r}\left(\frac{2}{x}\right)^r = {}^6C_r\,2^r\,x^{12-3r}. $$

Independent of x means $12-3r=0$, so $r=4$:

$$ {}^6C_4\,2^4 = 15\times 16 = 240. $$

TIP ${}^nC_r$ can also be read from a GDC table. To solve ${}^6C_r = 20$, generate ${}^6C_r$ for $r=0,\dots,6$ and read off $r=3$. Symmetry, ${}^nC_r = {}^nC_{n-r}$, halves the search.

YOUR TURN 2.9 The Binomial Theorem

43. Expand or write down the terms requested.

(a) Expand $(x+3)^4$.

(b) Write down the first three terms, in ascending powers of x, of $(1+2x)^7$.

(c) Expand $(2x-1)^4$.

(d) Write down the first three terms, in ascending powers of x, of $(1-3x)^6$.

44. Find the coefficient requested.

(a) x^2 in $(3x-2)^5$

(b) x^3 in $(2+x)^6$

(c) x^4 in $(1-2x)^7$

(d) x^5 in $(x+3)^8$

45. Find the term requested.

(a) the term independent of x in $\left(x+\dfrac{1}{x^2}\right)^9$

(b) the term independent of x in $\left(2x-\dfrac{1}{x}\right)^6$

(c) the coefficient of x^3 in $\left(x-\dfrac{2}{x}\right)^7$

(d) the term independent of x in $\left(3x^2 - \dfrac{1}{x}\right)^6$

46. Find the middle term in the expansion of $(x - 2y)^6$.

2.10 The Binomial Series HL

So far n has been a positive whole number, because $(a+b)^n$ meant n brackets multiplied together. But expressions such as $\sqrt{1+x} = (1+x)^{1/2}$ and $\dfrac{1}{1+x} = (1+x)^{-1}$ also have exponents. Can they be expanded too?

They can, and the same coefficient pattern still works for *any* rational n. Two things change. The expansion never ends, and it is only valid when x is small enough.

KEY · IB HL For $n \in \mathbb{Q}$:

$$(a+b)^n = a^n\left(1 + n\frac{b}{a} + \frac{n(n-1)}{2!}\left(\frac{b}{a}\right)^2 + \cdots\right), \qquad \left|\frac{b}{a}\right| < 1.$$

The series never ends, so the condition is part of the result. Factor out the larger term to make it hold.

In practice you first factor the expression into the form $(1 + \text{something})^n$, which turns the booklet formula into the version you actually compute with:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots, \qquad |x| < 1.$$

Why does the expansion stop for whole numbers but not otherwise? Look at the coefficients $n(n-1)(n-2)\cdots$. If $n = 3$, the third factor is $n - 3 = 0$, so every later coefficient is zero and the series ends: that is the finite theorem of §2.9. If $n = \frac{1}{2}$ or $n = -1$, no factor is ever zero, so the terms continue forever.

The condition $|x| < 1$ should look familiar. Put $n = -1$:

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$$

This is a geometric series with first term 1 and common ratio $-x$, and it converges exactly when $|-x| < 1$. The two infinite series of this chapter are the same idea seen from two directions.

EXAMPLE 2.34

Expand $(1 - 3x)^{1/3}$ up to and including the term in x^2, and state the values of x for which the expansion is valid.

Apply the series with $n = \frac{1}{3}$ and x replaced by $-3x$:

$$(1-3x)^{1/3} = 1 + \tfrac{1}{3}(-3x) + \frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2}(-3x)^2 + \cdots = 1 - x - x^2 - \cdots$$

Validity requires $|-3x| < 1$, that is $|x| < \frac{1}{3}$.

When the bracket does not already start with 1, factor out the larger term first, as the booklet formula does. The expansion is then valid for $\left|\frac{b}{a}\right| < 1$.

EXAMPLE 2.35

Expand $\dfrac{1}{2+x}$ in ascending powers of x up to x^2, stating the validity.

Factor out the 2:

$$(2+x)^{-1} = \frac{1}{2}\left(1+\frac{x}{2}\right)^{-1} = \frac{1}{2}\left(1 - \frac{x}{2} + \frac{x^2}{4} - \cdots\right) = \frac{1}{2} - \frac{x}{4} + \frac{x^2}{8} - \cdots$$

valid for $\left|\frac{x}{2}\right| < 1$, that is $|x| < 2$.

EXAMPLE 2.36

By expanding $(1 - x)^{1/2}$ up to x^2 and choosing a suitable value of x, find an approximation to $\sqrt{2}$.

The expansion is

$$(1-x)^{1/2} = 1 - \frac{x}{2} - \frac{x^2}{8} - \cdots$$

Now pick x so that $1 - x$ connects to 2. Take $x = 0.02$: then $1 - x = 0.98 = \frac{49}{50}$, so

$$\sqrt{0.98} = \frac{7}{\sqrt{50}} = \frac{7\sqrt{2}}{10} \implies \sqrt{2} = \frac{10}{7}\sqrt{0.98}.$$

The series gives $\sqrt{0.98} \approx 1 - 0.01 - 0.00005 = 0.98995$, hence

$$\sqrt{2} \approx \frac{10}{7}(0.98995) \approx 1.41421,$$

agreeing with the true value to five decimal places, from just three terms. The key is small x: the closer x is to 0, the faster the series converges.

CAUTION Always state the interval of validity alongside an HL binomial expansion; the series is meaningless outside it. And the expansion needs the form $(1+\bullet)^n$: factor first, then expand.

At HL these expansions return as Maclaurin series in Ch. 16, where the same coefficients arise from repeated differentiation.

YOUR TURN 2.10 The Binomial Series HL

47. Expand each up to and including the term in x^3, stating the validity.

(a) $(1+x)^{-1}$

(b) $(1-x)^{-2}$

48. Expand each up to and including the term in x^2, stating the validity.

(a) $(1+4x)^{1/2}$

(b) $(1+x)^{1/3}$

49. Expand $\dfrac{1}{3-x}$ in ascending powers of x up to x^2, stating the validity.

50. Expand $(8+x)^{1/3}$ in ascending powers of x up to x^2, stating the validity.

51. Use the expansion of $(1+x)^{1/2}$ up to the term in x^2, with $x = 0.1$, to estimate $\sqrt{1.1}$ to four decimal places.

Chapter 2 · Sequences & Series

Reflections

TOK Fold a sheet of paper 42 times and the stack reaches the Moon. The arithmetic is exact, but no one has folded a sheet more than twelve times, so the object the calculation describes has never existed. Can a statement be mathematically true and physically impossible at the same time? What is the calculation actually telling us about?

TOK The sum $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots=1$ is an equality, yet no finite number of steps ever reaches 1. We accept the answer because we define the sum to be the limit of the partial sums. Is that a discovery about infinity, or a decision we made so that the addition has an answer?

IM The triangle of binomial coefficients carries a different name in almost every place it was found. Indian scholars met these numbers counting the patterns of short and long syllables in poetry and set them out as a staircase; Persian mathematicians used them to expand powers, and Iran still calls it the Khayyam triangle; China calls it Yang Hui's triangle, Italy Tartaglia's, and much of Europe names it after Pascal, who came to it last by more than a thousand years. The same pattern was discovered independently again and again, and each culture named it after whoever reached it first there. What does that tell you about who a mathematical result belongs to?

RW Around 450 BCE Zeno argued that a runner can never finish a race. First he must reach the halfway point, then half of what remains, and so on without end. Each step of the argument looks correct, and it troubled mathematicians for two thousand years. The answer is a single result from this chapter: $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots=1$. Infinitely many steps can have a finite total, provided the steps shrink fast enough.

EXT A geometric series with $|r|<1$ is the simplest example of a **power series**, a sum of increasing powers of x:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \qquad |x|<1.$$

This one identity produces many others. Differentiate each term, or integrate each term, and you obtain the series for $\ln(1+x)$, for $\arctan x$, and for many further functions. In calculus, the topic of this chapter is *the seed from which the analysis of functions grows.*

End-of-Chapter Problems — Chapter 2: Sequences & Series

1. An arithmetic sequence has first term a and common difference d. The 5th term is 17 and the sum of the first 10 terms is 155.

(a) Write down two equations in a and d. [2]
(b) Find a and d. [2]
(c) Find the first negative term of the sequence. [2]

2. An arithmetic sequence has $u_3 = 7$ and $u_9 = 25$.

(a) Find u_1 and d. [3]
(b) Find S_{20}. [2]
(c) Find the least value of n such that $S_n > 600$. [3]

3. An arithmetic series has $u_1 = 2$ and $S_{10} = 155$.

(a) Find the common difference. [2]
(b) Find the sum $\sum_{k=11}^{20} u_k$. [4]

4. An arithmetic sequence has 20 terms. The sum of the odd-positioned terms $u_1 + u_3 + u_5 + \cdots + u_{19}$ is 160, and the sum of the even-positioned terms $u_2 + u_4 + u_6 + \cdots + u_{20}$ is 140. Find the first term u_1 and the common difference d. [6]

5. The sum of the first n terms of a series is given by $S_n = n^2 + 3n$.

(a) Find u_1 and u_2. [2]
(b) Show that the series is arithmetic and state the common difference. [3]
(c) Find the value of n such that $u_n = 44$. [2]

6. An AP has $u_1 = 120$ and $d = -3$.

(a) Find the first term that is negative. [3]
(b) Find the value of n that gives the maximum partial sum S_n, and find that maximum. [4]

7. Consider the two arithmetic sequences $4, 7, 10, \ldots$ and $3, 7, 11, \ldots$ Both sequences contain the value 7.

(a) Find the general term of each sequence. [2]
(b) Find the next three values that appear in both sequences. [4]

(c) Show that the common terms form an arithmetic sequence and state its common difference. [2]

8. A geometric sequence has first term a and common ratio r, where $r > 0$. Given $u_3 + u_4 = 12$ and $u_5 + u_6 = 3$, find a and r. [5]

9. The first three terms of a geometric sequence are k, 6, and $k + 5$.

(a) Show that $k^2 + 5k - 36 = 0$, and hence find the possible values of k. [4]
(b) For the value of k that gives a convergent series, find S_∞. [3]

10. The first, second, and fifth terms of an arithmetic sequence with first term 3 are, in that order, the first, second, and third terms of a geometric sequence.

(a) Given the arithmetic sequence is non-constant, show that its common difference is $d = 6$ and the geometric sequence has common ratio $r = 3$. [4]
(b) Determine whether the geometric series $3 + 9 + 27 + \cdots$ has a sum to infinity, giving a reason. [2]

11. A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n + 1$ for $n \geq 1$.

(a) Write down u_2, u_3, and u_4. [2]
(b) Let $v_n = u_n + 1$. Show that v_n is a geometric sequence. [3]
(c) Find an explicit formula for u_n. [2]

12. An arithmetic sequence has first term a and common difference d, where $a > 0$ and $d \neq 0$. Its 1st, 4th and 13th terms form a geometric sequence.

(a) Show that $3d = 2a$. [4]
(b) Given that $a = 3$, find d and write down the first four terms of the arithmetic sequence. [3]
(c) Find the common ratio of the geometric sequence. [2]
(d) Find the sum of the first 20 terms of the arithmetic sequence. [3]
(e) Find the sum of the first 8 terms of the geometric sequence. [3]

13.

(a) Show that $\displaystyle\sum_{k=1}^{n} r^{k-1} = \frac{1 - r^n}{1 - r}$ for $r \neq 1$ by multiplying S by r and subtracting. [3]
(b) Hence find $\displaystyle\sum_{k=1}^{10} \left(\tfrac{2}{3}\right)^{k-1}$, giving your answer as an exact fraction. [3]

14.

(a) Find $\sum_{k=1}^{n}(4k-3)$ in terms of n. [3]

(b) Find the value of n such that $\sum_{k=1}^{n}(4k-3) = 780$. [3]

15. GDC A geometric series has $u_1 = 3$ and $S_\infty = 12$.

(a) Find the common ratio. [2]

(b) Show that $S_n = 12\left(1 - \left(\frac{3}{4}\right)^n\right)$. [2]

(c) Find the least value of n such that $S_n > 11.9$. [3]

16. Consider the geometric series $1 + 2x + 4x^2 + 8x^3 + \cdots$

(a) Write down the common ratio. [1]

(b) Find the values of x for which the sum to infinity exists. [2]

(c) For these values, find S_∞ as a function of x. [2]

17. A geometric sequence has all positive terms. The sum of the first two terms is 10 and the sum to infinity is 25. Find u_1 and r. [5]

18. The sum to infinity of a geometric series is $\frac{3}{2}$ times the sum of the first 4 terms. Find the common ratio. [5]

19. The repeating decimal $0.\overline{142857} = 0.142857142857\ldots$ can be written as an infinite geometric series.

(a) Write down u_1 and the common ratio r. [2]

(b) Find the exact fractional value of $0.\overline{142857}$. [3]

(c) Verify your answer by converting the fraction from part (b) back to a decimal using long division. [1]

20. A ball is dropped from a height of 10 m. After each bounce it reaches 70% of the height it fell from.

(a) Find the height reached after the 3rd bounce. [2]

(b) Find the total vertical distance travelled by the ball before it comes to rest. [4]

21. GDC A machine is bought for \$60 000 and depreciates at 18% per year.

(a) Find its value after 4 years. [2]

(b) Find the least number of complete years until the value falls below one quarter of the purchase price. [3]

22. GDC \$8000 is invested for 6 years at 5% per year, compounded quarterly. Over the same period, inflation runs at 3% per year.

(a) Find the future value of the investment. [2]
(b) Find the real value of the investment after 6 years. [2]
(c) Comment on what your answers say about the investment's true growth. [1]

23. GDC An investment account pays 5% per year, compounded annually. A sum P is invested.

(a) Write an expression for the value of the account after n years. [1]
(b) Find the value of P if the account is worth \$20 000 after 10 years. [3]
(c) Find the number of complete years for the account to triple in value. [3]

24. GDC The value of a car decreases by 18% each year. A new car costs \$32 000.

(a) Find the value of the car after 3 years. [2]
(b) Find the year in which the value first falls below \$10 000. [3]
(c) Find the total loss in value over the first 5 years. [3]

25. GDC A student can invest \$3000 in one of two accounts. Account A pays 5% per year *simple* interest; account B pays 4% per year, compounded annually.

(a) Write down an expression for the value of each account after n years. [3]
(b) Find the value of each account after 10 years. [2]
(c) Using a table or graph on your GDC, find the least whole number of years after which account B is worth more than account A. [3]

26. GDC A savings plan pays \$500 into an account at the start of each year. The account earns 4% per year, compounded annually.

(a) Show that after 3 years the total value is $500(1.04)^3 + 500(1.04)^2 + 500(1.04)$. [2]
(b) Write this as a geometric series and find the total after 10 years. [4]
(c) Find the number of complete years until the total exceeds \$8 000. [3]

27. GDC An amount P is invested. Bank A pays 4% interest compounded annually. Bank B pays 3.9% compounded monthly, so its value after t years is $P\left(1+\dfrac{0.039}{12}\right)^{12t}$.

(a) Write down an expression for the value in Bank A after t years. [2]
(b) Show that Bank B's value can be written as Pr^t, and find r to five decimal places. [3]
(c) Determine which bank gives more after 10 years, and by how much as a percentage of P. [3]
(d) Use logarithms to find how long an investment in Bank A takes to double. [4]
(e) Bank C pays 4% compounded continuously, giving $Pe^{0.04t}$. Show that Bank C always beats Bank A, and find the difference after 10 years as a percentage of P. [3]

28. A school committee of 6 is to be chosen from 9 teachers and 7 students.

(a) Find the number of committees containing exactly 4 students. [2]
(b) Find the number of committees containing at least 4 students. [3]

29.

(a) Show that ${}^nC_2 = \dfrac{n(n-1)}{2}$, and explain the connection with the sum $1+2+\cdots+(n-1)$. [3]
(b) Hence find n such that ${}^nC_2 = 190$. [2]

30. In the expansion of $(1+kx)^8$, the coefficient of x^2 is 112.

(a) Find the possible values of k. [3]
(b) For $k>0$, find the coefficient of x^3. [2]

31. Consider the expansion of $\left(x+\dfrac{2}{x^2}\right)^{12}$.

(a) Find the term independent of x. [4]
(b) Find the coefficient of x^3. [2]

32. Consider the expansion of $\left(2x-\dfrac{3}{x}\right)^{10}$.

(a) Write down an expression for the general term of the expansion. [2]
(b) Find the term that is independent of x. [4]
(c) Find the coefficient of x^4. [3]
(d) Explain why the expansion contains no term in x^3. [2]
(e) By substituting a suitable value of x, find the sum of all the coefficients. [2]
(f) State how many terms the expansion has, and which powers of x occur. [2]

33.

(a) Expand $(1+2x)^{-2}$ in ascending powers of x up to and including the term in x^3, and state the values of x for which the expansion is valid. [4]
(b) Use your expansion with $x=0.01$ to estimate $\dfrac{1}{1.02^2}$ to 5 decimal places. [2]

34.

(a) Show that $(4+x)^{1/2} = 2\left(1+\dfrac{x}{4}\right)^{1/2}$, and expand it in ascending powers of x up to the term in x^2, stating the validity. [4]
(b) Hence find an approximation to $\sqrt{4.2}$, giving five decimal places. [2]

Challenge Questions

35. GDC **Turning products into sums.** This investigation shows why a logarithm converts a geometric sequence into an arithmetic one.

(a) Let $u_n = \log_2(3^n)$. Show that $\{u_n\}$ is arithmetic and state its common difference. [3]

(b) Find the exact value of $\sum_{k=1}^{20} u_k$. [3]

(c) Find the smallest value of n such that $u_n > 30$. [3]

(d) Let $v_n = \log_b(a^n)$, with $a > 0$ and $b > 1$ fixed. Show that $\{v_n\}$ is arithmetic, and state its common difference. [3]

(e) Let $\{w_n\}$ be a geometric sequence with positive terms and common ratio r. Show that $\{\log_b w_n\}$ is arithmetic, and state its common difference. [4]

(f) Use part (e) to explain, in one or two sentences, why a quantity that grows by multiplication is plotted on a logarithmic scale. Name one such quantity. [3]

36. **Counting multiples.** Each set of multiples below is an arithmetic sequence, so its total is an arithmetic series. Throughout, work with the integers from 1 to 1000.

(a) Find the sum of those divisible by 3, and the sum of those divisible by 5. [3]

(b) Explain why adding your two answers counts some integers twice, and find the sum of those divisible by both. [3]

(c) Hence find the sum of those divisible by 3 or by 5. [2]

(d) Find the sum of those divisible by 3 or by 5, but not by both. [3]

(e) Let p and q be distinct primes and let N be a positive integer. Show that the sum of the integers from 1 to N divisible by p or q but not both is

$$S_p + S_q - 2S_{pq}, \qquad \text{where } S_m = m\,\frac{t(t+1)}{2}, \quad t = \left\lfloor \frac{N}{m} \right\rfloor.$$

[4]

(f) Use the result of part (e) with $p = 7$, $q = 11$ and $N = 1000$. Verify your answer. [3]

37. **A sum that is not a sum.** You may have seen the claim that $1+2+3+4+\cdots = -\frac{1}{12}$. Every term is positive, so the statement cannot mean what addition normally means. This question takes the popular argument apart.

Write $G = 1 - 1 + 1 - 1 + \cdots$ and $T = 1 + 2 + 3 + 4 + \cdots$

(a) Write down the first six partial sums of G. Explain why G has no sum in the usual sense. [3]

(b) Bracketing the terms as $(1-1)+(1-1)+\cdots$ gives 0; bracketing them as $1-(1-1)-(1-1)-\cdots$ gives 1. State what this shows about applying the ordinary rules of algebra to such a series. [3]

(c) Applying $S_\infty = \dfrac{u_1}{1-r}$ to G with $u_1 = 1$ and $r = -1$ gives $\frac{1}{2}$. State the condition under which that formula was derived, and explain why it does not hold here. [3]

(d) Write down the nth partial sum of T and show that it increases without bound. [3]

(e) The popular argument runs: assume $G = \frac{1}{2}$; deduce $1 - 2 + 3 - 4 + \cdots = \frac{1}{4}$; subtract this from T term by term to obtain $4 + 8 + 12 + \cdots = 4T$; conclude $T - \frac{1}{4} = 4T$, so $T = -\frac{1}{12}$. Identify the step that is not valid, and say why. [4]

(f) Take the partial sums of G from part (a) and average the first n of them. Compute these averages for $n = 1$ to 6, and state what value they appear to approach. [4]

(g) Part (f) shows that $\frac{1}{2}$ is not arbitrary: it is what G gives under a *different* definition of "sum". Using this, explain in one or two sentences what a statement such as $1 + 2 + 3 + \cdots = -\frac{1}{12}$ actually asserts, and why writing it with an equals sign is misleading. [4]

NUMBER & ALGEBRA

CHAPTER 3

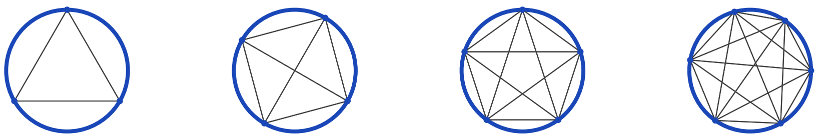
Proofs

3

SYLLABUS 1.6 1.15

Mark some points on a circle. Join every pair of points with a straight line. Then count the regions inside the circle. Place the points carefully, so that no three lines cross at the same spot. The first few cases give 1 region, then 2, then 4, then 8, then 16.

The pattern seems clear: each new point doubles the number of regions, so six points should give 32. In fact six points give **31**.



3 points, **4 regions** 4 points, **8 regions** 5 points, **16 regions** 6 points, **31 regions**

Figure 3.1 Every pair of points is joined, and no three chords meet at one point.

Five cases followed the doubling rule and the sixth did not. The first five gave no sign of it, and checking further cases would not have helped.

This is the difference between seeing a pattern and knowing that it always holds. A pattern is a guess. The word for such a guess is a **conjecture**: a statement believed true but not yet proved. The doubling rule was a conjecture, and the sixth case disproved it.

A **proof** is an argument in which each step follows from the one before, so the conclusion cannot fail, however many points you take.

By the end of this chapter you will be able to prove a statement directly from definitions, disprove one with a single counterexample, argue by contradiction, and prove a result about every positive integer by induction. Checking a rule on a few cases shows only that it has not failed yet. A proof shows that it cannot fail.

3.1 Deductive Proof

A **proof** is a chain of logical steps. It starts from things already accepted, such as definitions, results proved earlier, and the rules of algebra, and it ends at the statement you want to establish. Each step must follow with certainty from the steps before it. If the chain is correct, the conclusion is not just likely. It cannot be false.

This is **deductive** reasoning: you apply general principles to reach a specific, guaranteed result. The oldest example is not mathematical at all:

- Every human being is mortal. (a general principle)
- Socrates is a human being. (a specific case)
- Therefore Socrates is mortal. (the conclusion)

The conclusion is not a guess, and no amount of further evidence could improve it. If the first two statements are true, the third cannot be false. Mathematics uses arguments of exactly this shape:

- Every multiple of 4 is even.
- The number 132 is a multiple of 4.
- Therefore 132 is even.

Notice that you did not divide 132 by 2 to check. The conclusion was forced by the two statements above it.

The circle problem worked in the opposite direction. It used a few particular cases to guess a general rule. Reasoning in that direction, from examples to a general claim, is called induction in the everyday sense of the word, and it can lead to a false conclusion. Deduction cannot.

It may help to picture the two directions. Deduction works *outside in*: it begins with a rule that covers everything and ends at the single case in front of you. Induction works *inside out*: it begins with the few cases you have seen and ends at a rule about all of them. Only the first direction is safe. Mathematical induction, in §3.4, makes the inside-out direction safe, and it is the last method this chapter builds.

3.1.1 Equality versus identity

Before you prove a statement about an equation, you must know which kind of statement it is. The equals sign is used for two very different claims.

DEF An **equation** is a statement that two expressions are equal for *some particular* values of the variable. An **identity** is a statement that they are equal for *every* value for which both sides are defined. An identity is written with the symbol $\equiv$:

$$(x+1)^2 \equiv x^2 + 2x + 1.$$

The equation $x^2 = 4$ is true only when $x = 2$ or $x = -2$. The identity $(x+1)^2 \equiv x^2 + 2x + 1$ is true for every real x. Solving an equation means finding the values that make it true. Proving an identity means showing that no value makes it false. The three-bar symbol $\equiv$ marks which of the two is meant: where it appears, the claim is "for all x", not "find x".

You already know one important identity from trigonometry:

$$\sin^2 x + \cos^2 x \equiv 1.$$

This is true for every angle x, not for some particular angles, which is why it is written with $\equiv$. It is proved and used in Ch. 8.

3.1.2 Proving an identity

The two sides of an identity have names. The expression to the left of the $\equiv$ sign is the **left-hand side**, written **LHS**. The expression to the right is the **right-hand side**, written **RHS**.

To prove an identity you take one side and transform it, step by step, into the other, using only valid algebra. Start from the more complicated side and simplify it: write down the LHS, change it one line at a time, and finish with the RHS. Write the labels LHS and RHS in your working, so the direction of the argument is clear to the marker.

CAUTION Do not start by assuming the identity is true and then manipulate both sides until you reach something obvious like $0 = 0$. That assumes what you are trying to prove. Transform *one* side into the other instead.

EXAMPLE 3.1

The difficulty here is avoiding the common error of doing the same thing to both sides. Prove that $(a+b)^2 - (a-b)^2 \equiv 4ab$.

Start from the LHS and expand each square:

$$\text{LHS} = (a+b)^2 - (a-b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2).$$

Now remove the brackets. Take care with the signs in the second bracket:

$$= a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab = \text{RHS}. \quad \blacksquare$$

Nothing but expansion was used, and the LHS turned into the RHS. The two sides were never assumed to be equal. That they are equal is the result, not the starting point.
A quick **check** takes about five seconds, and most errors show up in it. At $a = 3$, $b = 2$ the left side is $25 - 1 = 24$ and the right side is $4(3)(2) = 24$. A check is not a proof. But if a check fails, something is certainly wrong.

EXAMPLE 3.2

Show that $(x-3)^2 + 5 \equiv x^2 - 6x + 14$.
The LHS is the more complicated side, so start there and expand the square:

$$\text{LHS} = (x-3)^2 + 5 = (x^2 - 6x + 9) + 5.$$

Collect the constant terms:

$$= x^2 - 6x + 14 = \text{RHS}. \quad \blacksquare$$

Check at $x = 1$: the LHS is $(-2)^2 + 5 = 9$, and the RHS is $1 - 6 + 14 = 9$.

TIP Always check your own results by substituting a number, in proofs and everywhere else. Examiners expect you to be able to verify a result you have just derived, and an algebra error will usually show up at one well-chosen value.

EXAMPLE 3.3

Show that $\dfrac{1}{4} + \dfrac{1}{12} = \dfrac{1}{3}$, and prove the algebraic generalisation

$$\frac{1}{m+1} + \frac{1}{m^2+m} \equiv \frac{1}{m}.$$

The numerical case is direct: $\frac{1}{4} + \frac{1}{12} = \frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$. Notice it is the claimed identity at $m = 3$: $m + 1 = 4$ and $m^2 + m = 12$.
For the general statement, start from the left side and factor the second denominator, $m^2+m = m(m+1)$:

$$\frac{1}{m+1} + \frac{1}{m(m+1)} = \frac{m}{m(m+1)} + \frac{1}{m(m+1)} = \frac{m+1}{m(m+1)} = \frac{1}{m}. \quad \blacksquare$$

One calculation confirmed a single case. The algebra covers every $m \neq 0, -1$ at once. This "verify, then generalise" structure is exactly how such questions are phrased in examinations.

3.1.3 Proving a general statement

Most proofs are not about identities. They are about claims such as "the sum of two odd numbers is even". The first step is always the same: write the general object in algebra, so that one expression stands for every case at once.

The table below lists the standard forms. In every row, k and n are integers.

In words	In algebra
an even number	$2k$
an odd number	$2k + 1$, or $2k - 1$
two consecutive integers	n and $n + 1$
three consecutive integers	$n - 1$, n, $n + 1$
two consecutive even numbers	$2k$ and $2k + 2$
two consecutive odd numbers	$2k + 1$ and $2k + 3$
a multiple of 3	$3k$
a number that leaves remainder 1 when divided by 3	$3k + 1$
a square number	k^2
a rational number	p/q, with p, q integers and $q \neq 0$

Choosing the right form matters. To prove a statement about *any* two odd numbers, use $2a + 1$ and $2b + 1$ with two different letters. Writing $2a + 1$ twice would only prove the statement for two *equal* odd numbers.

Once the general case is written in symbols, the rest is ordinary algebra.

EXAMPLE 3.4

A single example such as $3+5 = 8$ confirms nothing about *all* odd numbers. Prove that the sum of any two odd numbers is even.

Let the two odd numbers be $2a + 1$ and $2b + 1$, where a and b are integers. Their sum is

$$(2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

Since $a + b + 1$ is an integer, the sum is $2 \times (\text{integer})$, which is by definition even. ■

EXAMPLE 3.5

Prove that the product of two consecutive integers is always even.

Take the consecutive integers n and $n + 1$. One of any two consecutive integers is even. If n is even, the product $n(n + 1)$ has an even factor. If n is odd, then $n + 1$ is even, and the product has an even factor again. Either way $n(n+1)$ is a multiple of 2, so it is even. ■

EXAMPLE 3.6

Divisibility by a composite number is proved one prime factor at a time. Prove that $n(n+1)(2n+1)$ is divisible by 6 for every integer n.

Since $6 = 2 \times 3$, find a factor 2 and a factor 3 separately.

A factor of 2. One of any two consecutive integers is even, so one of n and $n+1$ is even, and the product is divisible by 2.

A factor of 3. Every integer leaves remainder 0, 1 or 2 on division by 3, so n has one of the forms $3k$, $3k+1$ or $3k+2$, where k is an integer. If $n = 3k$, then the first factor is a multiple of 3. If $n = 3k+1$, then $2n+1 = 6k+3 = 3(2k+1)$. If $n = 3k+2$, then $n+1 = 3k+3 = 3(k+1)$. The three cases cover every integer, and in each one of the three factors is a multiple of 3, so the product is divisible by 3.

The product is divisible by 2 and by 3. Since 2 and 3 are **coprime**, sharing no factor larger than 1, these are separate factors, and the product is divisible by $2 \times 3 = 6$. ■

The coprime condition earns its place: a number divisible by 2 and by 4 need not be divisible by 8, as 12 shows.

CAUTION One example can never prove a statement that begins "for all." It can only *disprove* one, which is the subject of the next section. The proofs above work because the algebra stands for every case at once.

YOUR TURN 3.1 Deductive Proof

1. State whether each of the following is an equation or an identity.

(a) $3x = 12$

(b) $(x+2)^2 = x^2 + 4x + 4$

2. Prove each identity.

(a) $(a+b)^2 = a^2 + 2ab + b^2$

(b) $(a-b)^2 = a^2 - 2ab + b^2$

(c) $\dfrac{1}{x} - \dfrac{1}{x+1} = \dfrac{1}{x(x+1)}$

(d) $(x+3)(x-3) = x^2 - 9$

(e) $\dfrac{1}{x-1} - \dfrac{1}{x+1} = \dfrac{2}{x^2-1}$

3. Prove each statement about odd and even integers.

(a) The sum of any two even integers is even.
(b) The sum of an even integer and an odd integer is odd.
(c) The product of any two odd integers is odd.

4. Prove each statement for every integer n.

(a) $(2n+1)^2$ is odd.
(b) $n^2 - n$ is even.
(c) $(2n+1)^2 - 1$ is divisible by 4.
(d) $n^3 - n$ is divisible by 6.

3.2 Disproof by Counterexample HL

A universal statement claims something about every member of a collection. To show that such a claim is false, you do not need to discuss the whole collection. You need one member for which the claim fails.

DEF A **counterexample** is one specific case for which a general statement fails. A single counterexample is enough to prove that the statement is false.

The logic is exact. "Every n has property P" is false as soon as one value of n does not have property P. This is what happened with the circle problem. The claim "the number of regions doubles each time" was a universal statement, and $n = 6$ was a counterexample to it.

EXAMPLE 3.7

A famous formula, n^2+n+41, produces a prime number for $n = 0, 1, 2, \dots, 39$, forty primes in a row. Decide whether it produces a prime for *every* non-negative integer n.
Test the claim at the first value the pattern does not cover. At $n = 40$:

$$40^2 + 40 + 41 = 1600 + 40 + 41 = 1681 = 41^2.$$

This is 41×41, not prime. So $n = 40$ is a counterexample, and the statement "$n^2 + n + 41$ is always prime" is false. ■
Forty cases agreed with the claim. One case was enough to make it false. Examples can make a claim look true. They can never make it certain.

EXAMPLE 3.8

Disprove the claim: "every multiple of 3 is odd."

The number 6 is a multiple of 3, and 6 is even. That single case is a counterexample, so the claim is false. ■

EXAMPLE 3.9

Show that this statement is not always true: "there are no positive integer solutions to the equation $x^2 + y^2 = 10$."

Take $x = 1$ and $y = 3$. Both are positive integers, and

$$1^2 + 3^2 = 1 + 9 = 10.$$

So a positive integer solution does exist, and the statement is false. ■

Notice what the answer contains: the values, the substitution, and the conclusion. Giving only "$x = 1$, $y = 3$" would not earn full marks.

TIP A counterexample must be *explained*, not just named. State the case, substitute it into the claim, show the calculation, and say in words why this makes the claim false. The IB is explicit that stating the counterexample alone is not enough.

Two more claims fall to counterexamples. The first is "if $a^2 > b^2$ then $a > b$". Take $a = -3$ and $b = 1$: then $9 > 1$ while $-3 < 1$, because squaring removes the sign. The second is "the product of two irrational numbers is irrational". Take $\sqrt{2} \times \sqrt{2} = 2$, where the two irrational parts cancel exactly.

RW In 1640 Fermat studied the numbers $F_n = 2^{2^n} + 1$ and found that $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$ and $F_4 = 65\,537$ are all prime. He conjectured that every F_n is prime. Almost a century later, in 1732, Euler tested the next one:

$$F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417.$$

One counterexample was enough to disprove the conjecture. No larger Fermat prime has ever been found.

CAUTION A counterexample only *disproves* a claim. Finding no counterexample after many attempts is not a proof. The circle problem at the start of this chapter shows why: five cases agreed, and the sixth did not.

YOUR TURN 3.2 Disproof by Counterexample HL

5. Disprove each claim by giving a counterexample.

(a) Every integer is positive.
(b) Every prime number is odd.
(c) If n is divisible by 4 then n is divisible by 8.
(d) The sum of two prime numbers is always even.
(e) Every odd number is prime.
(f) If $a > b$ then $a^2 > b^2$.

6. Disprove each claim by giving a counterexample.

(a) $n^2 \geq 2n$ for all integers n.
(b) $x^2 > x$ for all real x.
(c) If n^2 is even then n is odd.
(d) $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$.
(e) If p is prime then $2^p - 1$ is prime.

3.3 Proof by Contradiction HL

Suppose a friend tells you they have written down the largest whole number that exists. You do not need to see it to know they are wrong. Whatever the number is, add 1 to it. The result is a whole number, and it is larger. The claim disproves itself.

You have just proved something without ever finding the answer. You did not search for the largest whole number. You assumed it existed and showed that the assumption leads to an impossible situation.

This is the idea of **proof by contradiction**. Assume the statement you want to prove is *false*. Reason correctly from that assumption until you reach something impossible. Correct reasoning cannot turn a true assumption into an impossible result, so the assumption must have been false. The original statement is therefore true.

DEF In **proof by contradiction**, you assume the negation of what you want to prove, then derive a logical contradiction. The contradiction forces the assumption to be false, which establishes the original claim.

The method has a Latin name, *reductio ad absurdum*, meaning "reduction to the absurd". The method is useful because the assumption itself is a starting point. A direct proof that no fraction equals $\sqrt{2}$ would mean checking infinitely many fractions. Assuming that such a fraction exists produces an equation instead, and an equation can be worked on. The two proofs below are among the most famous in mathematics, and both work this way.

EXAMPLE 3.10

Prove that $\sqrt{2}$ is irrational, that is, it cannot be written as a fraction of two integers.
Assume the opposite: that $\sqrt{2}$ *is* rational. Then we can write

$$\sqrt{2} = \frac{p}{q},$$

where p and q are integers **sharing no common factor**, so the fraction is fully cancelled. The contradiction at the end of the proof depends on this condition, so state it explicitly. Squaring both sides gives $2 = \frac{p^2}{q^2}$, hence

$$p^2 = 2q^2.$$

So p^2 is even, which forces p itself to be even (the square of an odd number is odd). Write $p = 2m$. Then $p^2 = 4m^2$, and substituting,

$$4m^2 = 2q^2 \implies q^2 = 2m^2,$$

so q^2 is even, and therefore q is even too. But now p and q are both even, sharing the factor 2, contradicting our choice of a fully cancelled fraction. The assumption that $\sqrt{2}$ is rational is impossible, so $\sqrt{2}$ is irrational. ■

EXAMPLE 3.11

Prove that there are infinitely many prime numbers. This argument is due to Euclid, around 300 BCE.
Assume the opposite: that there are only finitely many primes, and list them all as $p_1, p_2, \dots, p_n$. Build the number

$$N = p_1 \times p_2 \times \cdots \times p_n + 1,$$

the product of every prime, plus one. Dividing N by any prime on the list leaves remainder 1, so N is divisible by none of them. Yet every integer greater than 1 has at least one prime factor. That factor cannot be on our list, so our list was not complete, a contradiction. No finite list can hold all the primes, so there are infinitely many. ■

The next example is not itself an argument by contradiction. It is a direct proof, worked by exhausting the possible cases, and it sits in this section because a contradiction proof of the kind above needs a lemma of exactly this shape as soon as the prime changes.

EXAMPLE 3.12

The $\sqrt{2}$ proof moved from "p^2 is even" to "p is even" using the square of an odd number is odd. That sentence is about 2 and says nothing about 5, so the $\sqrt{5}$ version needs its own

argument. Prove that if n^2 is a multiple of 5, then n is a multiple of 5.
Nothing is assumed here and no contradiction is reached; the cases are simply exhausted. Every integer lies within 2 of a multiple of 5, so n is $5k$, $5k \pm 1$ or $5k \pm 2$ for some integer k. Square each form:

$$(5k)^2 = 5(5k^2), \qquad (5k \pm 1)^2 = 5(5k^2 \pm 2k) + 1, \qquad (5k \pm 2)^2 = 5(5k^2 \pm 4k) + 4.$$

Only the first of these is a multiple of 5; the others leave remainder 1 or 4. So n^2 can be a multiple of 5 only when n has the form $5k$, which is to say only when n is a multiple of 5. ∎
A small result proved on the way to a larger one is a **lemma**, and this is the lemma that replaces the parity step when the $\sqrt{2}$ proof is rerun for $\sqrt{5}$: the contradiction is then reached exactly as before, with 5 in place of 2. The same case split serves any prime, with the remainders paired off by the $\pm$ as they were here.

EXAMPLE 3.13
Prove that $(1+\sqrt{2})^2$ is irrational. You may use the theorem that $\sqrt{2}$ is irrational.
Expand first, so that the known irrational stands alone in a single term:

$$(1+\sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}.$$

Assume the opposite: that this number is rational. Then

$$3 + 2\sqrt{2} = \frac{p}{q},$$

where p and q are integers with $q \neq 0$. Isolate $\sqrt{2}$ by subtracting 3 and halving:

$$\sqrt{2} = \frac{1}{2}\left(\frac{p}{q} - 3\right) = \frac{p-3q}{2q}.$$

Now $p - 3q$ and $2q$ are integers and $2q \neq 0$, so the right-hand side is a ratio of two integers, and $\sqrt{2}$ is rational. That contradicts the theorem that $\sqrt{2}$ is irrational, so the assumption fails and $(1+\sqrt{2})^2$ is irrational. ∎
Both middle steps are instances of **closure**: add, subtract, multiply or divide two rational numbers, never dividing by zero, and the result is rational again. Subtracting 3 and dividing by 2 each move from one rational number to another, so neither could have carried a rational number to an irrational one.
The contradiction here is with a result already established, not with anything internal to $(1+\sqrt{2})^2$. Expanding is what made the argument possible, because it put the expression in terms of $\sqrt{2}$: identify the known irrational first, then aim the algebra at it.

CAUTION State your assumption clearly at the start (“assume …is rational”) and name the contradiction when you reach it. A proof by contradiction with no clearly identified contradiction proves nothing.

YOUR TURN 3.3 Proof by Contradiction HL

7. Prove each statement by contradiction.

(a) There is no largest integer.
(b) There is no smallest positive real number.

8. Let n be an integer. Prove each statement by contradiction.

(a) If n^2 is odd then n is odd.
(b) If n^2 is even then n is even.

9. Prove each statement by contradiction.

(a) There are no integers x and y with $2x + 4y = 5$.
(b) The sum of a rational number and an irrational number is irrational.
(c) $\sqrt{3}$ is irrational.
(d) If x is irrational and $x \neq 0$, then $\frac{1}{x}$ is irrational.

3.4 Proof by Induction HL

Picture an endless line of dominoes. You cannot knock over infinitely many by hand. But suppose you can show two things: that the first domino falls, and that *whenever* a domino falls it topples the next. Then every domino in the line must fall, however long the line.

Mathematical induction is exactly this idea, used to prove a statement P_n for every integer n from some starting value onward. The subscript in P_n matters: it marks the n-th claim in an infinite list $P_1, P_2, P_3, \ldots$, not a function being evaluated. You verify the first claim, then prove that each claim forces the one after it.

KEY **Principle of mathematical induction.** To prove that P_n holds for all integers $n \geq 1$:

1. Show that P_1 is true.
2. Assume that P_k is true for **some** $k \geq 1$ (the *inductive hypothesis*).
3. Show that P_k implies P_{k+1}.
4. Conclude: *since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$.*

(If the statement starts later, at n_0, begin at P_{n_0} instead of P_1.)

Every induction proof follows these four steps, and the sentence in step 4 should be written out in full. It is not decoration. It is the step that turns the two facts you proved into a conclusion about every n, and it carries its own mark in examinations.

CAUTION In step 2, assume P_k for **some** k, never for *all* k. Assuming it for all k assumes the very thing you are proving, and examiners withhold the reasoning marks for it. The hypothesis is one domino falling, not the whole line.

3.4.1 Why induction works

The domino image can be made precise. Suppose steps 1 and 3 are done, and pick any n you like, say $n = 1000$. Then P_1 is true; P_1 forces P_2; P_2 forces P_3; and after 999 applications of the inductive step, P_{1000} is true. The same finite chain reaches *any* n, so no value of n is left out, even though only two things were ever proved.

There is a second way to see it, and it connects induction to the previous section. Suppose the two steps are done, but P_n failed for some values of n. Among the failures there would be a *smallest* one, say P_m. Now $m \neq 1$, because step 1 checked P_1. So $m - 1 \geq 1$, and P_{m-1} is true, since m was the smallest failure. But step 3 says that P_{m-1} forces P_m, which contradicts the failure of P_m. So induction is itself a proof by contradiction, and the two techniques of this chapter are closely connected.

Induction is not only for sums. It also proves divisibility statements and inequalities below, De Moivre's theorem in Ch. 17, and formulas for n-th derivatives in Ch. 13. Whenever a claim has the form "for every integer n", induction is the first method to consider.

3.4.2 Induction for a sum

How to approach it. Here P_n says that a sum equals a formula. Add the next term of the sum to both sides of the assumption P_k. Then rearrange the right-hand side until it becomes the original formula with $k + 1$ written wherever n appeared. Write that target expression down before you begin, so you know what you are aiming at.

EXAMPLE 3.14

Prove that $3 + 7 + 11 + \cdots + (4n - 1) = n(2n + 1)$ for all integers $n \geq 1$.

Let P_n be the statement $3 + 7 + \cdots + (4n - 1) = n(2n + 1)$.

Step 1 (P_1): the left side is 3; the right side is $1(2(1) + 1) = 3$. They agree, so P_1 is true.

Step 2: assume P_k is true for some $k \geq 1$, that is,

$$3 + 7 + \cdots + (4k - 1) = k(2k + 1).$$

Step 3: add the next term, $4(k + 1) - 1 = 4k + 3$, to both sides:

$$3 + 7 + \cdots + (4k - 1) + (4k + 3) = k(2k + 1) + 4k + 3.$$

Expand the right side and factorise it:

$$= 2k^2 + 5k + 3 = (k + 1)(2k + 3) = (k + 1)(2(k + 1) + 1).$$

This is exactly P_{k+1}, the original formula with n replaced by $k + 1$. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$. ■

3.4.3 Induction for divisibility

The same four steps prove divisibility statements.

How to approach it. Here P_n says that an expression E_n is divisible by a fixed number d. Turn the assumption into an equation: divisible by d means $E_k = dm$ for some integer m. Then rewrite E_{k+1} so that E_k appears inside it, replace that part by dm, and take d outside as a common factor. The aim is to reach the form $d \times$ (integer).

EXAMPLE 3.15

Prove that $8^n - 1$ is divisible by 7 for all integers $n \geq 1$.

Let P_n be the statement "$8^n - 1$ is divisible by 7."

Step 1 (P_1): $8^1 - 1 = 7$, which is divisible by 7. So P_1 is true.

Step 2: assume P_k is true for some $k \geq 1$, so $8^k - 1 = 7m$ for some integer m.

Step 3: consider the next case:

$$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(8^k - 1) + 8 - 1 = 8(7m) + 7 = 7(8m + 1).$$

Since $8m + 1$ is an integer, $8^{k+1} - 1$ is divisible by 7, which is P_{k+1}. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$. ■

Notice where the hypothesis was used: rewriting $8^k - 1$ as $7m$ is the only reason the final line factors cleanly.

EXAMPLE 3.16

In $8^n - 1$ the previous expression appeared as soon as a factor was taken out at the front. Prove that $n^5 + 4n$ is divisible by 5 for all integers $n \geq 1$, where it has to be extracted from an expansion instead.

Let P_n be the statement "$n^5 + 4n$ is divisible by 5."

Step 1 (P_1): $1^5 + 4(1) = 5$, which is divisible by 5. So P_1 is true.

Step 2: assume P_k is true for some $k \geq 1$, so $k^5 + 4k = 5m$ for some integer m.

Step 3: expand the next case and collect $k^5 + 4k$ out of it:

$$(k+1)^5+4(k+1) = k^5+5k^4+10k^3+10k^2+5k+1+4k+4 = (k^5+4k)+5k^4+10k^3+10k^2+5k+5.$$

The hypothesis replaces the bracket by $5m$, and every remaining term has a factor 5:

$$= 5m + 5(k^4 + 2k^3 + 2k^2 + k + 1) = 5(m + k^4 + 2k^3 + 2k^2 + k + 1).$$

Since $m+k^4+2k^3+2k^2+k+1$ is an integer, $(k+1)^5+4(k+1)$ is divisible by 5, which is P_{k+1}. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction P_n is true for all integers $n \geq 1$. ■

Expand fully first, then look for the previous expression inside the result. If what is left over after removing it does not carry the required factor, the grouping is wrong, not the statement.

3.4.4 Induction for an inequality

How to approach it. Here P_n says that $A_n > B_n$. Inequalities are proved by chaining: if $a > b$ and $b \geq c$, then $a > c$. Start from A_{k+1} and rewrite it so that A_k appears, then use the assumption to replace A_k by something smaller. That gives the first link. Now prove separately that this new quantity is at least B_{k+1}, which gives the second link. The two links join into a single chain from A_{k+1} down to B_{k+1}.

EXAMPLE 3.17

Prove that $2^n > n^2$ for all integers $n \geq 5$. The base case is not $n = 1$ here, and that matters: the inequality is false for $n = 2, 3, 4$.

Let P_n be the statement $2^n > n^2$.

Step 1 (P_5): $2^5 = 32$ and $5^2 = 25$, and $32 > 25$. So P_5 is true.

Step 2: assume P_k is true for some $k \geq 5$, that is, $2^k > k^2$.

Step 3: then

$$2^{k+1} = 2 \cdot 2^k > 2k^2.$$

It is now enough to show $2k^2 \geq (k+1)^2$. Expanding, this asks whether $2k^2 \geq k^2 + 2k + 1$, that is, $k^2 - 2k - 1 \geq 0$. For $k \geq 5$ this certainly holds, since $k^2 - 2k - 1 = (k-1)^2 - 2 \geq 16 - 2 > 0$. Therefore

$$2^{k+1} > 2k^2 \geq (k+1)^2,$$

which is P_{k+1}. So $P_k \Rightarrow P_{k+1}$.

Step 4: since P_5 is true, and P_k true implies P_{k+1} true for $k \geq 5$, by the principle of mathematical induction $2^n > n^2$ for all integers $n \geq 5$. ■

3.4.5 Induction for a recurrence

How to approach it. Here the sequence is given by a rule linking each term to the one before, with no formula in n to prove. Compute the first few terms, conjecture a **closed form** for u_n, and let P_n be that closed form. In the inductive step the hypothesis is fed through the recurrence rather than added to a sum: substitute the assumed u_k into the rule for u_{k+1}, then simplify until the closed form with $k+1$ in place of n appears.

EXAMPLE 3.18

A sequence is defined by $u_1 = 1$ and $u_{n+1} = \dfrac{u_n}{1+2u_n}$ for $n \geq 1$. Find u_2, u_3 and u_4, conjecture a formula for u_n, and prove it by induction.

Each term comes from the one before:

$$u_2 = \frac{1}{1+2} = \frac{1}{3}, \qquad u_3 = \frac{1/3}{1+2/3} = \frac{1/3}{5/3} = \frac{1}{5}, \qquad u_4 = \frac{1/5}{1+2/5} = \frac{1/5}{7/5} = \frac{1}{7}.$$

The terms so far are $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}$, which suggests the conjecture $u_n = \dfrac{1}{2n-1}$. Let P_n be that statement.

Step 1 (P_1): the definition gives $u_1 = 1$, and the formula gives $\dfrac{1}{2(1)-1} = 1$. They agree, so P_1 is true.

Step 2: assume P_k is true for some $k \geq 1$, that is, $u_k = \dfrac{1}{2k-1}$.

Step 3: substitute the hypothesis into the recurrence:

$$u_{k+1} = \frac{u_k}{1+2u_k} = \frac{1/(2k-1)}{1+2/(2k-1)}.$$

Multiply numerator and denominator by $2k-1$:

$$= \frac{1}{(2k-1)+2} = \frac{1}{2k+1} = \frac{1}{2(k+1)-1},$$

which is the formula with n replaced by $k+1$, so $P_k \Rightarrow P_{k+1}$.

Step 4: since P_1 is true, and P_k true implies P_{k+1} true, by the principle of mathematical induction $u_n = \dfrac{1}{2n-1}$ for all integers $n \geq 1$. ■

The four terms produced the conjecture and established nothing; the induction established it. A question of this shape marks the two jobs separately, so give the terms, state the conjecture, and then prove it.

CAUTION An induction proof fails in two ways. One is forgetting the base case, so the dominoes can topple each other yet never start. The other is proving the inductive step *without using* the hypothesis. If you never used P_k, you have not built a chain, only restated the problem.

YOUR TURN 3.4 Proof by Induction HL

10. Prove each result by induction, for all $n \geq 1$.

(a) $1+2+3+\cdots+n=\dfrac{n(n+1)}{2}$

(b) $1+3+5+\cdots+(2n-1)=n^2$

(c) $1^2+2^2+\cdots+n^2=\dfrac{n(n+1)(2n+1)}{6}$

(d) $1^3+2^3+\cdots+n^3=\dfrac{n^2(n+1)^2}{4}$

11. Prove each result by induction, for all $n \geq 1$.

(a) $2+4+6+\cdots+2n=n(n+1)$

(b) $1+2+4+\cdots+2^{n-1}=2^n-1$

(c) $1\cdot 2+2\cdot 3+\cdots+n(n+1)=\dfrac{n(n+1)(n+2)}{3}$

(d) $\dfrac{1}{1\cdot 2}+\dfrac{1}{2\cdot 3}+\cdots+\dfrac{1}{n(n+1)}=\dfrac{n}{n+1}$

12. Prove each divisibility result by induction, for all $n \geq 1$.

(a) 3^n-1 is divisible by 2.

(b) 4^n-1 is divisible by 3.

(c) 5^n-1 is divisible by 4.

13.

(a) Prove by induction that $2^n \geq n+1$ for all $n \geq 0$.

(b) Explain why the base case here is $n=0$ rather than $n=1$.

Chapter 3 · Proofs

Reflections

TOK A proof reduces a claim to things already accepted. But those starting points, the axioms, are assumed rather than proved. In 1931 Kurt Gödel showed that any system rich enough to describe arithmetic must contain true statements it cannot prove. Mathematical certainty is real, then, but not absolute. It is certainty *relative* to a chosen foundation. Does that make mathematics more like a landscape discovered, or a structure built?

IM How did the Pythagoreans discover that $\sqrt{2}$ is irrational? They began from the belief that any two lengths share a common unit, so that every ratio of lengths is a ratio of whole numbers. Their own theorem defeated them: the diagonal of a unit square has length $\sqrt{2}$, and no such ratio gives it. The argument is the one used in §3.3, and it is a proof by contradiction: assume $\sqrt{2} = p/q$ in lowest terms, and an impossibility follows. The discovery is credited to Hippasus in the fifth century BCE, and later writers report that his school received it badly. The result stood regardless, and Greek mathematics had to widen its idea of number.

RW Proof is not confined to pure mathematics. In software engineering, *formal verification* proves that a program meets its specification exactly. No amount of testing can give that guarantee, so the method is used where failure would be catastrophic: aircraft control, medical devices, spacecraft. Cryptography rests on proved properties of prime numbers, so the security of online banking is, in the end, a theorem.

EXT The four-colour theorem says any map can be coloured with four colours so that no two neighbouring regions match. In 1976 it became the first major theorem whose proof needed a computer. The machine checked nearly two thousand cases, far more than a person could. That raised a new question. If a proof is too long for anyone to read, but every step has been checked by machine, is it still a proof? What is a proof *for*: certainty, or understanding?

A gallery of conjectures

Every theorem began as a conjecture. The seven below are a small selection, chosen because each can be stated in a single sentence. Those marked open have been neither proved nor disproved; the remaining two show what settling a conjecture looks like, in opposite directions. None is examinable, and far more exist than are listed here, so you are invited to search out others of your own.

Goldbach conjecture (1742): open. Every even number greater than 2 is the sum of two primes: $28 = 5 + 23$, $100 = 3 + 97$. Checked for every even number up to 4×10^{18}, and still unproved.

Twin prime conjecture: open. There are infinitely many primes p for which $p + 2$ is also prime, such as $11, 13$ and $101, 103$. In 2013 Yitang Zhang proved there are infinitely many prime pairs at most 70 million apart. Later work brought that down to 246. Closer, but 246 is not 2.

Collatz conjecture: open. Take any positive integer. Halve it if even; triple it and add 1 if odd. Repeat. Does every starting number reach 1? Try 27 on your GDC: it takes 111 steps and climbs to 9232 on the way. Checked for a vast number of starting values, proved for none.

Odd perfect numbers: open for ∼2300 years. A number is perfect if it equals the sum of its proper divisors, such as $6 = 1+2+3$ and $28 = 1+2+4+7+14$. Every perfect number ever found is even. Whether an odd one exists may be the oldest unsolved problem in mathematics. Any example must be larger than 10^{1500}.

Legendre's conjecture: open. Between any two consecutive squares n^2 and $(n+1)^2$ there is always at least one prime. It is true in every case ever checked, and still unproved.

Fermat's Last Theorem (1637): proved, 1995. No positive integers satisfy $x^n + y^n = z^n$ for $n \geq 3$. Fermat wrote in a book's margin that he had a proof, but that the margin was too small to hold it. The first accepted proof came 358 years later, from Andrew Wiles, and it uses mathematics far beyond anything Fermat could have known. Conjectures can be settled, and this one was.

Pólya's conjecture (1919): disproved. At least half of the numbers up to any N have an odd number of prime factors. Checked for millions of cases and believed for forty years, until it was shown to fail near $N = 906{,}150{,}257$. This is the circle problem from the start of the chapter on a larger scale: millions of cases agreed, one did not, and the claim is false.

TOK These conjectures press on what mathematical certainty means. Goldbach has been verified for every even number up to 4×10^{18} and is still not certain. Pólya's conjecture survived millions of checks and was false. Does certainty in mathematics admit of degrees, or does a claim stay wholly uncertain until the moment it is proved?

End-of-Chapter Problems — Chapter 3: Proofs

1. Prove that the product of any three consecutive integers is divisible by 6. [5]

2. Consider the cube of a binomial.

(a) Prove the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. [3]

(b) Hence prove that $(n+1)^3 - n^3 = 3n^2 + 3n + 1$ for every integer n. [3]

3. Disprove the claim "$n^2 + 1$ is never divisible by 5" by giving two counterexamples. [3]

4. Prove by contradiction that $\sqrt{3}$ is irrational. [6]

5. Let a be a non-zero rational number and b an irrational number.

(a) Prove by contradiction that ab is irrational. [4]

(b) Hence state, with a reason, whether $\dfrac{\sqrt{2}}{5}$ is rational or irrational. [2]

6.

(a) Prove by contradiction that $x^2 - 6x + 10 = 0$ has no real solution, without using the discriminant. [3]

(b) Prove by contradiction that if n is an integer and n^2 is a multiple of 3, then n is a multiple of 3. [3]

7. Prove that $\sqrt{2} + \sqrt{3}$ is irrational. You may assume that $\sqrt{6}$ is irrational. [6]

8. Prove by contradiction that $\sqrt[3]{5}$ is irrational. [6]

9. Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$. [5]

10. Prove by induction that $6^n + 4$ is divisible by 5 for all $n \geq 1$. [5]

11. Prove by induction that $n! > 2^n$ for all $n \geq 4$, stating why the base case is $n = 4$. [5]

12. Prove by induction that $5^{2n} - 1$ is divisible by 24 for all $n \geq 1$. [5]

13. Prove by induction that $2^n > n$ for all $n \geq 1$. [5]

14. Prove by induction that $9^n - 1$ is divisible by 8 for all $n \geq 1$. [5]

15. Prove by induction that $n^3 + 2n$ is divisible by 3 for all $n \geq 1$. [5]

16. Prove by induction that $\displaystyle\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$. [6]

17. Prove by induction that $\displaystyle\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$ for all $n \geq 1$. [6]

18. Prove by induction that $\displaystyle\sum_{r=1}^{n} r(r+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \geq 1$. [6]

19. Prove by induction that $\displaystyle\sum_{r=1}^{n} (3r-2) = \frac{n(3n-1)}{2}$ for all $n \geq 1$. [6]

20. Prove by induction that $\displaystyle\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}$ for all $n \geq 1$. [6]

21. A student claims that $n^2 - n + 11$ is prime for every positive integer n.

(a) Verify the claim for $n = 1$ and $n = 2$. [2]

(b) Disprove the claim with a counterexample. [3]

22. Consider parity of squares.

(a) Prove that if n is odd then n^2 is odd. [2]

(b) Hence prove by contradiction that if n^2 is even then n is even. [3]

23. Prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ in two ways:

(a) directly, using the sum of an arithmetic sequence; [3]

(b) by induction. [3]

24. Prove that $\log_2 3$ is irrational. [6]

25. Prove by induction that $\displaystyle\sum_{r=1}^{n} r \cdot r! = (n+1)! - 1$ for all $n \geq 1$. [6]

26.

(a) Prove by induction that $\displaystyle\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ for all $n \geq 1$. [5]

(b) You may assume that $\displaystyle\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$. Find $\displaystyle\sum_{r=1}^{n} r(r-1)$, giving your answer in fully factorised form. [4]

(c) Hence evaluate $\sum_{r=1}^{20} r(r-1)$. [3]

(d) Show that $\sum_{r=1}^{n} r(r-1) = 2\,{}^{n+1}C_3$, and explain why this means the sum is always even. [3]

27.

(a) Prove that if n^2 is a multiple of 3, then n is a multiple of 3. [3]

(b) Hence prove by contradiction that $\sqrt{3}$ is irrational. [5]

(c) Explain precisely where the argument in part (b) breaks down if you attempt to use it to prove that $\sqrt{9}$ is irrational. [3]

(d) A student writes: "*$\sqrt{3}$ is irrational, so $\sqrt{3}\times\sqrt{3}$ must be irrational too.*" Explain the error, and state what it shows about products of irrational numbers. [3]

28. A sequence is defined by $u_1 = 1$ and $u_{n+1} = \dfrac{u_n}{1+u_n}$ for $n \geq 1$.

(a) Find u_2, u_3 and u_4. [3]

(b) Write down a conjecture for u_n in terms of n. [1]

(c) Prove your conjecture by induction. [5]

(d) A student's whole proof reads: "*Assume the statement is true for $n = k$. Then it is true for $n = k+1$. Hence it is true for all n.*" State what is missing, and explain why the argument establishes nothing without it. [3]

(e) Give a statement $P(n)$ for which $P(k) \Rightarrow P(k+1)$ is true for every k, but $P(n)$ is false for every n. Justify both claims. [3]

The remaining questions use the proof techniques of this chapter on material introduced later in the book. They appear in the order those topics are covered, so return to them as you go.

29. Let n and r be integers with $1 \leq r \leq n$. Starting from ${}^{n}C_r = \dfrac{n!}{r!\,(n-r)!}$, prove Pascal's identity

$$ {}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r. $$

[5]

30. Using Pascal's identity, prove by induction that for every integer $n \geq 1$,

$$ (a+b)^n = \sum_{r=0}^{n} {}^{n}C_r a^{n-r} b^r. $$

[7]

31. You may assume the compound-angle formulae for $\cos(A+B)$ and $\sin(A+B)$. Prove by induction that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

for every integer $n \geq 1$. [6]

32. You may assume that $\dfrac{d}{dx}(x) = 1$ and the product rule. Prove by induction that

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

for every integer $n \geq 1$. [6]

33. You may assume $\lim_{h\to 0} \dfrac{\sin h}{h} = 1$ and $\lim_{h\to 0} \dfrac{\cos h - 1}{h} = 0$. Working from the definition

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

prove that the derivative of $\sin x$ is $\cos x$. [6]

34. Let $I_n = \displaystyle\int_0^{\pi/2} \sin^n x\, dx$ for integers $n \geq 0$.

(a) Use integration by parts to prove that $I_n = \dfrac{n-1}{n} I_{n-2}$ for every integer $n \geq 2$. [5]

(b) Write down I_0 and I_1, and hence find I_4 and I_5. [4]

Challenge Questions

35. Counting the regions of a circle. This investigation completes the problem from the start of the chapter. Throughout, n points are placed on a circle so that no three chords meet at a single interior point.

(a) By drawing, verify that $n = 1, 2, 3, 4, 5$ give $1, 2, 4, 8, 16$ regions. [2]

(b) Explain why each interior intersection point corresponds to exactly one choice of 4 of the n points, and deduce that there are ${}^nC_4$ interior points. [3]

(c) The chords and arcs form a network. Show that it has $V = n + {}^nC_4$ vertices and $E = n + {}^nC_2 + 2\,{}^nC_4$ edges. *(Hint: each interior point adds one extra segment to each of the two chords through it)* [4]

(d) Euler's formula for a connected planar network states $V - E + F = 2$, where F counts all faces including the region outside the circle. Use it to prove that the number of regions inside the circle is ${}^nC_4 + {}^nC_2 + 1$. [4]

(e) Hence find the number of regions for $n = 6$ and $n = 10$. [2]

(f) If the n points are equally spaced, the count for $n = 6$ is 30 rather than 31. Explain what changes. [3]

36. Fermat's conjecture. In 1640 Fermat studied $F_n = 2^{2^n} + 1$.

(a) Show that F_0, F_1, F_2, F_3 and F_4 are prime. [3]

(b) Suppose m has an odd factor $d > 1$, so $m = de$. Using the factorisation

$$x^d + 1 = (x+1)\left(x^{d-1} - x^{d-2} + \cdots - x + 1\right) \quad (d \text{ odd}),$$

with $x = 2^e$, prove that $2^m + 1$ is not prime. [5]

(c) Deduce that if $2^m + 1$ is prime, then m must be a power of 2. [2]

(d) Verify that 641 divides $F_5 = 4\,294\,967\,297$, and state what this shows about Fermat's conjecture. [3]

37. A proof that names no example. Consider the number $\sqrt{2}^{\sqrt{2}}$.

(a) Suppose $\sqrt{2}^{\sqrt{2}}$ is rational. Explain why this immediately produces an irrational number raised to an irrational power that is rational. [2]

(b) Suppose instead $\sqrt{2}^{\sqrt{2}}$ is irrational. Show that $\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = 2$, and explain why this also produces such a pair. [4]

(c) Conclude that there exist irrational a and b with a^b rational. [2]

(d) This argument proves that such a pair exists without telling you which case is the true one. Compare it with the proof that $\sqrt{2}$ is irrational, and comment on what each proof gives you. [4]

38. Fibonacci identities. The Fibonacci numbers are defined by $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$.

(a) Prove by induction that $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$ for all $n \geq 1$. [5]

(b) Prove by induction that $F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$ for all $n \geq 1$. [5]

(c) Part (b) has a picture proof: squares of sides $F_1, F_2, \ldots, F_n$ tile a rectangle. State the dimensions of that rectangle and explain how it proves the identity. [3]

Chapter 4

Lines & Quadratics

4

SYLLABUS 1.16 2.1–2.4 2.6 2.7

On 17 January 1995 an earthquake struck near Kobe, in western Japan. The Akashi Kaikyō Bridge was under construction. Its two towers already stood in the strait, but the roadway between them did not yet exist. The earthquake moved the towers apart.

The engineers did not begin the work again. They measured the new gap and changed the design to match it. The bridge opened in 1998 with a main span of 1991 metres, one metre longer than the 1990 originally planned.

This was possible because the design existed as equations rather than as fixed drawings. The main cables of a suspension bridge carry a level roadway of even weight, and they hang in the curve called a *parabola*, the same curve as a thrown ball. If one value in the equations changes, every other measurement follows from it.

That is the value of a **function**. A function takes an input and returns exactly one output. A whole structure therefore becomes a single rule, and a rule can be graphed, studied and recalculated. This chapter builds the two functions on which the rest of the course depends: lines, which change at a steady rate, and quadratics, which give the cable its curve.

By the end of this chapter you will give the domain, range and inverse of a function, and find the equation of a line from the information you are given. You will solve systems of linear equations, write a quadratic in its factorised, vertex and expanded forms, and use the discriminant to count the roots of an equation.

4.1 Functions

A function is a rule you can rely on. Give it an input and it returns one definite output, and it returns the same output every time. That is what makes it possible to reason about.

DEF A **function** f assigns exactly one output $f(x)$ to each input x. The set of inputs it accepts is the **domain**; the set of outputs that actually occur is the **range**.

The phrase "exactly one" carries the definition. A rule that could return two different outputs for the same input is not a function. On a graph this becomes the **vertical line test**: a curve represents a function only if no vertical line crosses it more than once.

The letter f carries no meaning of its own. A function is named for what it measures, and the letter inside the bracket for what it depends on. A height that changes with time is written $h(t)$, a velocity $v(t)$, a cost that depends on how many items you make $C(n)$. Read $C(n)$ as "the cost of making n items", not as C multiplied by n. Examination questions name their functions this way, so read the notation as a sentence rather than as algebra.

That naming points at what functions are usually for. A **mathematical model** is a function chosen to describe something real: the letters stand for measured quantities, and the rule claims that one determines the other. Modelling adds a question to every problem that pure algebra never asks, which is whether the answer means anything.

Two habits follow from that. The domain is set by the situation, not by the algebra alone: $h(t) = 20t - 5t^2$ is defined for every real t, but a stone in flight only exists from the moment it is thrown until it lands, so the model holds on $0 \leq t \leq 4$ and nowhere else. And a value that solves the equation can still be rejected, because a negative length or a negative time is arithmetic the situation does not allow. Every model is an approximation, so ask what it leaves out before trusting it far beyond the data it came from.

4.1.1 Reading the domain

A function need not accept every number. Three operations put limits on what you may substitute. Learn the three and you can read the domain of almost any function on the syllabus.

Where x appears	Condition	Reason
In a denominator	$\neq 0$	you cannot divide by zero
Inside an even root	≥ 0	a negative number has no real square root
Inside a logarithm	> 0	zero and negative numbers have no logarithm

Note the difference between the last two. A square root allows zero, because $\sqrt{0} = 0$. A logarithm does not, because no power of the base gives zero. That single distinction, $\geq$ against $>$, is the most common slip in this topic.

Function	Condition	Domain
$\sqrt{x+3}$	$x + 3 \geq 0$	$x \geq -3$
$\sqrt{5-x}$	$5 - x \geq 0$	$x \leq 5$
$\dfrac{1}{x-2}$	$x - 2 \neq 0$	$x \neq 2$
$\ln(x-4)$	$x - 4 > 0$	$x > 4$
$\dfrac{1}{\sqrt{x-1}}$	$x - 1 > 0$, since it is both under a root and in a denominator	$x > 1$

When a function contains more than one of these, every condition must hold at once. Write them all down, then take the values that satisfy all of them.

EXAMPLE 4.1

State the domain of $f(x) = \dfrac{\sqrt{x+4}}{x-3}$.

There are two restrictions here. The square root needs $x + 4 \geq 0$, so $x \geq -4$. The denominator needs $x - 3 \neq 0$, so $x \neq 3$.

Both must hold, so the domain is

$$x \geq -4, \quad x \neq 3.$$

Notice that $x = 3$ satisfies the first condition but fails the second, so it must still be removed.

EXAMPLE 4.2

State the domain of $g(x) = \ln(9 - x^2)$.

A logarithm needs a positive argument, so $9 - x^2 > 0$, that is $x^2 < 9$. This is a quadratic inequality, solved in §4.7: the parabola $x^2 - 9$ lies below the axis between its roots. So

$$-3 < x < 3.$$

Both ends are excluded, because at $x = \pm 3$ the argument is zero and $\ln 0$ does not exist.

CAUTION Solve the condition, do not guess it. For $\sqrt{5-x}$ the domain is $x \leq 5$, not $x \geq 5$. Write the inequality down and rearrange it every time, however obvious it looks.

4.1.2 Reading the range

The range is the set of outputs that actually occur. There is no single rule, but four facts cover most functions you will meet, and each says that some expression can never go below zero.

Almost every range argument rests on one fact: some part of the function can never be negative. A square is never negative, so $x^2 \geq 0$; a modulus is never negative, so $|x| \geq 0$; the square root symbol means the non-negative root, so $\sqrt{x} \geq 0$; and a positive base raised to any power stays positive, so $a^x > 0$.

To use them, build the function up from the part you know. Ask what that part can be, then follow the operations one at a time.

Function	**Reasoning**	**Range**
$y = x^2 + 1$	$x^2 \geq 0$, then add 1	$y \geq 1$
$y = \lvert x\rvert - 3$	$\lvert x\rvert \geq 0$, then subtract 3	$y \geq -3$
$y = \sqrt{x} + 2$	$\sqrt{x} \geq 0$, then add 2	$y \geq 2$
$y = 3 - \lvert x+1\rvert$	$\lvert x+1\rvert \geq 0$, so subtracting it takes 3 downward	$y \leq 3$
$y = -(x-2)^2 + 5$	$(x-2)^2 \geq 0$, and the minus sign turns the inequality round	$y \leq 5$

CAUTION A minus sign in front reverses the direction. Since $(x-2)^2 \geq 0$, it follows that $-(x-2)^2 \leq 0$, so $-(x-2)^2 + 5 \leq 5$. Multiplying an inequality by a negative number always flips the sign.

For a quadratic, complete the square first. The vertex form shows the smallest or largest output immediately.

EXAMPLE 4.3

State the range of $f(x) = x^2 - 12x + 40$.

Complete the square:

$$x^2 - 12x + 40 = (x-6)^2 - 36 + 40 = (x-6)^2 + 4.$$

Now use the first fact. Since $(x-6)^2 \geq 0$, adding 4 gives $f(x) \geq 4$. The range is $y \geq 4$. The smallest value, 4, occurs at $x = 6$, which is the vertex.

EXAMPLE 4.4

State the domain and range of $g(x) = \sqrt{x-2}$.

Domain. The square root needs $x - 2 \geq 0$, so $x \geq 2$.

Range. As x runs from 2 upward, $x - 2$ runs from 0 upward, and so does its square root. So $g(x) \geq 0$.

The domain comes from the inputs the rule permits. The range comes from the outputs those inputs give. Always ask both questions.

EXAMPLE 4.5

State the domain and range of $f(x) = \sqrt{2-x}$.

The square root needs $2 - x \geq 0$, that is $x \leq 2$. So the domain is $x \leq 2$.

As x decreases from 2, the quantity $2 - x$ increases from 0 upward, and so does its square root. The range is $f(x) \geq 0$.

Compare this with $\sqrt{x-2}$ above. The two look similar but their domains point in opposite directions, so read the inequality from the expression each time rather than from memory.

TIP On Paper 2, check a range on your GDC. Graph the function and look at the lowest and highest points the curve reaches. The algebra is still expected in the working, but the graph catches a sign error in seconds.

4.1.3 Reading a graph

Most of what you need from a function can be read straight from its graph. Five features matter, and each has its own name.

Feature	What it is	How to find it
x-intercepts (also called zeros, or roots)	the points where the curve meets the x-axis	set $y = 0$ and solve
y-intercept	the point where the curve meets the y-axis	set $x = 0$
Turning points	the local maximum and minimum points, where the curve changes direction	GDC, or the vertex form for a quadratic
Asymptote	a line that the curve gets closer and closer to, without ever reaching it	look for values the function cannot take
Intersection	a point that lies on both of two graphs	solve the two equations together

Your GDC finds all five directly. The graph menu has zero, min/max and intersect tools. On Paper 2 these are the intended method. Graph the function, read the feature, and write down what you asked the calculator to find. Many equations have no algebraic solution at all. For those the graph is not a faster method. It is the only method.

EXAMPLE 4.6

Use a GDC to find the y-intercept, the zeros and the turning points of $f(x) = x^3 - 2x^2 - 5x + 4$. Give coordinates to three significant figures where they are not exact.

The y-intercept needs no calculator. Setting $x = 0$ leaves the constant term, so the curve meets the y-axis at $(0, 4)$.

Graph the function over a window wide enough to show every feature; $-3 \leq x \leq 4$ works here. The curve crosses the x-axis three times, so run the zero tool three times, once on each crossing:

$$(-1.86,\ 0), \qquad (0.678,\ 0), \qquad (3.18,\ 0).$$

The maximum and minimum tools return both coordinates at once:

$$\text{local maximum } (-0.786,\ 6.21), \qquad \text{local minimum } (2.12,\ -6.06).$$

The turning points check the zeros. The local maximum lies above the axis and the local minimum below it, so the curve is forced to cross three times, and three is what the zero tool found.

TIP Know the difference between the command terms: **sketch** means show the shape with key features labelled, not to scale; **draw** means an accurate, to-scale plot. A sketch of a parabola needs the vertex, the intercepts, and the correct opening direction, and nothing else.

4.1.4 Inverse functions

An **inverse function** reverses the rule. If f sends 3 to 7, then f^{-1} sends 7 back to 3. An inverse exists only when no two inputs give the same output. If two did, the way back would have two answers, and a function may return only one.

DEF The **inverse** f^{-1} of a function f undoes it: $f^{-1}(f(x)) = x$. It exists when f is **one-to-one**, meaning no two inputs share an output (the *horizontal* line test).

The inverse swaps input and output. Two things follow. Its graph is the mirror image of f in the line $y = x$, and its domain and range are those of f exchanged. To find a formula, write $y = f(x)$, swap x and y, then solve for y.

EXAMPLE 4.7

Find the inverse of $f(x) = 2x + 3$.

Write $y = 2x + 3$, then swap x and y:

$$x = 2y + 3.$$

Solving for y,

$$y = \frac{x-3}{2}, \qquad \text{so} \qquad f^{-1}(x) = \frac{x-3}{2}.$$

Check: f doubles and adds three; f^{-1} subtracts three and halves. Each step of f is undone, in reverse order.

When x appears twice, once on top and once underneath, you cannot undo the rule step by step. Swap the letters anyway, clear the fraction, and collect the y terms.

EXAMPLE 4.8

Find f^{-1} for $f(x) = \dfrac{4x+3}{x-2}$, and state its domain.

Write $y = \dfrac{4x+3}{x-2}$ and swap x and y:

$$x = \frac{4y+3}{y-2}.$$

Multiply both sides by $y - 2$ to clear the fraction, then gather every term containing y on one side:

$$x(y-2) = 4y + 3 \implies xy - 2x = 4y + 3 \implies xy - 4y = 2x + 3.$$

Both terms on the left carry a factor of y, so factorise and divide:

$$y(x-4) = 2x + 3 \implies f^{-1}(x) = \frac{2x+3}{x-4}, \qquad x \neq 4.$$

The domain of f^{-1} is the range of f, and this confirms it: $f(x) = 4$ would need $4x + 3 = 4x - 8$, which no x satisfies, so f never outputs 4.

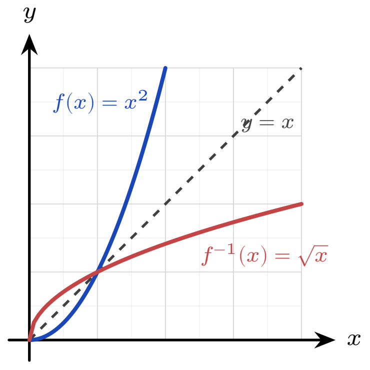


Figure 4.1 A function and its inverse are mirror images in the line $y = x$.

The inverse also gives another way to read an equation. Asking "for which x does $f(x) = 10$?" is the same as asking "what does f^{-1} send 10 to?"

KEY Solving $f(x) = 10$ is equivalent to evaluating $f^{-1}(10)$.

EXAMPLE 4.9

For $f(x) = 2x + 3$, solve $f(x) = 10$ in two ways.

Directly: $2x + 3 = 10$ gives $2x = 7$, so $x = \frac{7}{2}$.

Using the inverse found above, $f^{-1}(x) = \dfrac{x-3}{2}$, so

$$f^{-1}(10) = \frac{10-3}{2} = \frac{7}{2}.$$

The two routes must agree, because they ask the same question. When a question gives you f^{-1} already, the second route is the shorter one.

CAUTION The notation f^{-1} means the inverse, *not* the reciprocal. $f^{-1}(x)$ is not $\dfrac{1}{f(x)}$.

YOUR TURN 4.1 Functions

1. State the domain of each function.

(a) $\sqrt{x+3}$

(b) $\sqrt{7-x}$

(c) $\dfrac{1}{x-2}$

(d) $\dfrac{1}{x+5}$

(e) $\ln(x-4)$

(f) $\ln(2x+1)$

(g) $\dfrac{1}{\sqrt{x-1}}$

(h) $\dfrac{1}{x^2-9}$

2. State the domain of each function. Each has more than one restriction.

(a) $\dfrac{\sqrt{x+2}}{x-1}$ (b) $\dfrac{\sqrt{x}}{x-9}$

3. State the range of each function on the domain given.

(a) $x^2+1, \quad x \in \mathbb{R}$ (b) $|x|-3, \quad x \in \mathbb{R}$
(c) $\sqrt{x}+2, \quad x \geq 0$ (d) $3-|x+1|, \quad x \in \mathbb{R}$

4. Find the range of each quadratic by completing the square. Each is defined for all real x. If you have not met this method yet, it is set out in §4.5.

(a) $x^2-6x+11$ (b) x^2-4x
(c) $-(x-2)^2+5$ (d) $-x^2+6x-4$

5. For $f(x)=x^2-3x$, find each value.

(a) $f(2)$ (b) $f(-1)$
(c) $f(0)$ (d) the values of x with $f(x)=0$

6. Find the inverse of each function.

(a) $f(x)=3x-6$ (b) $f(x)=\dfrac{x+1}{2}$
(c) $f(x)=2x+7$ (d) $f(x)=\dfrac{1}{x+2}$
(e) $f(x)=\dfrac{2x+1}{x-3}$ (f) $f(x)=\dfrac{x-4}{x+2}$

7. Consider $f(x)=\dfrac{3x+2}{x-1}$.

(a) State the domain of f. (b) Find $f^{-1}(x)$.
(c) State the domain of f^{-1}. (d) Verify that $f^{-1}(f(2))=2$.

8. Decide whether each rule defines y as a function of x, and justify your answer.

(a) $y^2=x$ (b) $y=x^2$

9. The function $f(x)=x^2$ has no inverse over all real numbers. State a restricted domain on which an inverse does exist, and give that inverse.

4.2 Linear Functions

The simplest function that actually changes is the line: its output rises at a constant rate. That rate is the **gradient**, the rise divided by the run.

KEY · IB The gradient of the line through (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

A gradient and one point on the line are enough to fix the line completely. From that fact come three ways to write its equation. All three describe the same line.

KEY · IB A straight line can be written

$$y = mx + c, \qquad y - y_1 = m(x - x_1), \qquad ax + by + d = 0.$$

Here m is the gradient and c the y-intercept.

Use the first form when you know the gradient and the y-intercept. Use the second when you know the gradient and any point. The third is the tidy general form, with whole-number coefficients.

Two lines are **parallel** when they rise at the same rate. They are **perpendicular** when you get one gradient from the other by turning the fraction upside down and changing the sign.

KEY For two lines with gradients m_1 and m_2:

$$m_1 = m_2 \text{ (parallel)}, \qquad m_1 m_2 = -1 \text{ (perpendicular)}.$$

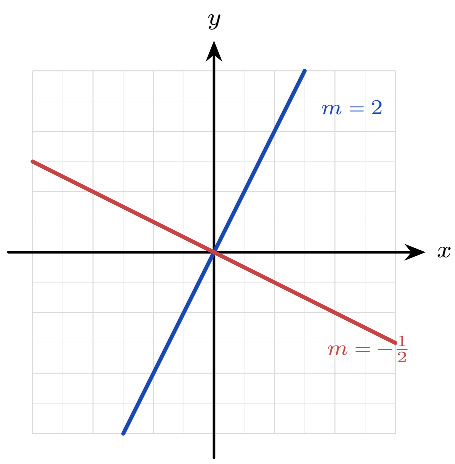


Figure 4.2 Perpendicular gradients: turn the fraction upside down and change the sign.

EXAMPLE 4.10

Find the equation of the line through $(1, 2)$ and $(4, 8)$.

First the gradient:

$$m = \frac{8-2}{4-1} = \frac{6}{3} = 2.$$

Then use the point $(1, 2)$ in point–gradient form:

$$y - 2 = 2(x - 1) \implies y = 2x.$$

EXAMPLE 4.11

Find the equation of the line through $(0, 4)$ perpendicular to $y = 3x - 1$.

The given line has gradient 3, so the perpendicular gradient is $-\frac{1}{3}$. Through $(0, 4)$,

$$y = -\tfrac{1}{3}x + 4.$$

For a perpendicular gradient, turn the fraction upside down and change the sign.

Two points fix more than a gradient. They also have a distance between them and a point halfway along, and both come from the coordinates directly. The distance is Pythagoras applied to the horizontal and vertical gaps; the midpoint is the average of the coordinates.

KEY · IB For the points (x_1, y_1) and (x_2, y_2), the distance between them and the midpoint of the segment joining them are

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}, \qquad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

The order of the points does not matter in either formula: the differences are squared, and the sums are symmetric. Both formulas extend to three dimensions by adding a z-term to each; see Ch. 7.

Together with a perpendicular gradient these give the **perpendicular bisector** of a segment, the line at right angles to it through its midpoint.

EXAMPLE 4.12

The points $A(2, -1)$ and $B(6, 7)$ are given. Find the length of AB, the midpoint of AB, and the equation of the perpendicular bisector of AB.

The horizontal gap is 4 and the vertical gap is 8, so

$$AB = \sqrt{(2-6)^2 + (-1-7)^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}.$$

The midpoint is
$$M = \left(\frac{2+6}{2}, \frac{-1+7}{2}\right) = (4, 3).$$
The gradient of AB is $\dfrac{7-(-1)}{6-2} = 2$, so the bisector has gradient $-\frac{1}{2}$. Through $M(4, 3)$,
$$y - 3 = -\tfrac{1}{2}(x-4) \implies y = -\tfrac{1}{2}x + 5.$$
Every point of this line is the same distance from A as from B. Take $(0, 5)$ on it: the distance to A is $\sqrt{4+36} = \sqrt{40}$, and to B it is $\sqrt{36+4} = \sqrt{40}$.

YOUR TURN 4.2 Linear Functions

10. Find the gradient of the line through each pair of points.

(a) $(1, 3)$ and $(5, 11)$
(b) $(2, 1)$ and $(4, 7)$
(c) $(-1, 4)$ and $(3, -4)$
(d) $(0, 5)$ and $(2, 5)$

11. Find the equation of each line, giving your answer as $y = mx + c$.

(a) gradient 2, through $(0, -3)$
(b) through $(2, 1)$ and $(4, 7)$
(c) gradient $-\frac{1}{2}$, through $(4, 0)$
(d) through $(0, 5)$, perpendicular to $y = -2x$

12. Write each equation as $y = mx + c$ and state the gradient.

(a) $2x + 3y = 6$
(b) $4x - 2y = 8$

13. Answer each question about parallel and perpendicular lines.

(a) Write down the gradient of a line perpendicular to $y = 4x + 1$.
(b) Write down the gradient of a line parallel to $3x + y = 7$.
(c) Are $y = 2x + 1$ and $y = 2x - 3$ parallel? Give a reason.
(d) Are $y = 3x - 1$ and $y = -\frac{1}{3}x + 2$ perpendicular? Give a reason.

4.3 Systems of Linear Equations HL

Two lines in a plane usually cross at one point. Finding that point means finding the values of x and y that satisfy *both* equations at once, a **system** of equations. The standard tool is **elimination**: combine the equations to cancel one variable, solve for the other, then substitute back.

EXAMPLE 4.13

Solve $2x + y = 7$ and $x - y = 2$.
Adding the two equations cancels y:

$$(2x + y) + (x - y) = 7 + 2 \implies 3x = 9 \implies x = 3.$$

Substituting into $x - y = 2$ gives $3 - y = 2$, so $y = 1$. The lines cross at $(3, 1)$.

Not every system has a single answer, and the picture shows why. Two lines can do three things. They cross once, which gives a **unique solution**. They stay parallel and never meet, so there is **no solution**. Or they are the same line drawn twice, which gives **infinitely many solutions**.

CAUTION If two equations have the same gradient but different intercepts, such as $2x + y = 3$ and $4x + 2y = 10$, the lines are parallel and the system has *no* solution. Elimination then produces a false statement such as $0 = 4$.

The same idea extends to three equations in three unknowns, where each equation is now a plane in space. You eliminate one variable at a time to reduce a 3×3 system to a 2×2 one, or read the solution directly from a GDC.

EXAMPLE 4.14

Solve the system

$$x + y + z = 6, \qquad 2x - y + z = 3, \qquad x + 2y - z = 2.$$

Eliminate z in pairs. Subtracting the first equation from the second:

$$(2x - y + z) - (x + y + z) = 3 - 6 \implies x - 2y = -3.$$

Adding the first and third equations:

$$(x + 2y - z) + (x + y + z) = 2 + 6 \implies 2x + 3y = 8.$$

Now solve this 2×2 system. From $x - 2y = -3$ we get $x = 2y - 3$; substituting,

$$2(2y-3) + 3y = 8 \implies 7y = 14 \implies y = 2,$$

so $x = 1$. Finally the first equation gives $1 + 2 + z = 6$, hence $z = 3$. The solution is $(x, y, z) = (1, 2, 3)$.

Eliminating one variable at a time, working down the system, is called **row reduction**. It is what your GDC's simultaneous-equation solver does inside. Use the solver to check your work. You still need the algebra in an examination, and you need it when the solver returns only an error message.

4.3.1 The augmented matrix

Look again at the elimination above and notice what you were really changing: never the letters x, y, z, only the numbers in front of them. The letters only marked the columns, so you can stop writing them. Reduce the system to its coefficients and constants, written in a grid called the **augmented matrix**:

$$\begin{cases} x + y + z = 6 \\ 2x - y + z = 7 \\ x + 2y - z = 3 \end{cases} \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 7 \\ 1 & 2 & -1 & 3 \end{array}\right)$$

The vertical bar shows where the equals signs were. Nothing new is happening: the matrix records the same elimination in a shorter form, with fewer symbols to copy incorrectly.

Elimination becomes three **row operations**. None of them changes the solution set. Each one is reversible, and each does to a whole equation only what you were always allowed to do:

KEY The three elementary row operations:

$$R_i \leftrightarrow R_j \ \text{(swap two rows)}, \qquad R_i \to kR_i \ \text{(multiply a row by } k \neq 0\text{)},$$

$$R_i \to R_i + kR_j \ \text{(add a multiple of one row to another)}.$$

Swapping reorders the equations, scaling multiplies one through by a constant, and the third adds a multiple of one equation to another: the same three moves as elimination, written compactly.

The goal is a step shape, with zeros below the first entry of each row. The last row then contains one unknown, the row above it two, and so on. This shape is called **row echelon form**. Once you reach it, work upwards from the last row, finding one unknown at a time. That is **back-substitution**.

EXAMPLE 4.15

Solve, by row reduction,

$$x + y + z = 6, \qquad 2x - y + z = 7, \qquad x + 2y - z = 3.$$

Work on the augmented matrix, clearing the first column below the pivot. $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 7 \\ 1 & 2 & -1 & 3 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -5 \\ 0 & 1 & -2 & -3 \end{array}\right)$$

A leading entry of 1 gives simpler arithmetic than -3, so swap $R_2 \leftrightarrow R_3$. Then clear below it with $R_3 \to R_3 + 3R_2$:

$$\to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -3 \\ 0 & -3 & -1 & -5 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & -7 & -14 \end{array}\right)$$

The step shape is complete. Now read it from the bottom up. The last row says $-7z = -14$, so $z = 2$. The middle row says $y - 2z = -3$, so $y = 4 - 3 = 1$. The top row gives $x = 6 - y - z = 3$. The solution is $(x, y, z) = (3, 1, 2)$.

Every step here is the elimination you already know. The matrix keeps the working short and the columns lined up, and misaligned columns are the usual cause of mistakes.

Once a system is in row echelon form, the last row determines the outcome. There are only three possible outcomes, and every 3×3 system gives exactly one of them.

Last row reduces to	What it means	The three planes
$(0\ 0\ a \mid b)$, $a \neq 0$	one unique solution	meet at a single point
$(0\ 0\ 0 \mid c)$, $c \neq 0$	no solution: the row says $0 = c$, which is false	have no common point; two are parallel, or the three meet in a triangular prism
$(0\ 0\ 0 \mid 0)$	infinitely many solutions: the row says $0 = 0$, which is true but tells us nothing	share a whole line, or a whole plane if two rows vanish

Count the rows that vanish. No vanishing row gives one solution. One vanishing row gives a line of solutions, described by one free variable. Two vanishing rows mean all three equations describe the same plane, and the solutions form that whole plane, described by two free variables.

CAUTION Read the constant on the right before deciding. The rows $(0\ 0\ 0 \mid 0)$ and $(0\ 0\ 0 \mid 5)$ look almost identical, and they give opposite answers: infinitely many solutions, and none at all.

4.3.2 When a system has no unique solution

Each equation in three unknowns is a plane in space. Three planes usually meet at exactly one point. That point is the unique solution found in the example above.

Two other arrangements are possible. If two planes are parallel, or if the three meet in three parallel lines like the faces of a triangular prism, no point lies on all three. The system is **inconsistent**, and there is no solution. If instead the three planes share a whole line, like pages sharing a spine, every point of that line is a solution, so there are infinitely many.

You can separate the two cases by algebra alone, without a picture. If elimination produces a false statement such as $0 = 4$, the planes have no common point, and there is no solution. If it produces $0 = 0$, one equation repeats information already contained in the other two. There are infinitely many solutions, and one variable may take any value.

EXAMPLE 4.16

Consider the system

$$x + y + 2z = 2, \qquad 2x - y + z = 1, \qquad x + 4y + 5z = k.$$

Find the value of k for which the system has infinitely many solutions, and give the general solution. Show that for every other value of k there is no solution.

The question asks for infinitely many solutions, so expect a row that reduces to zeros. Eliminate x using the first equation. Second minus twice the first:

$$(2x - y + z) - 2(x + y + 2z) = 1 - 4 \implies -3y - 3z = -3 \implies y + z = 1.$$

Third minus the first:

$$(x + 4y + 5z) - (x + y + 2z) = k - 2 \implies 3y + 3z = k - 2 \implies y + z = \frac{k-2}{3}.$$

The same left side, $y+z$, has appeared twice. The two remaining equations are consistent only if

$$1 = \frac{k-2}{3} \implies k = 5.$$

If $k \neq 5$, subtracting them gives $0 = \frac{k-5}{3} \neq 0$, a false statement: the system is inconsistent, and no solution exists for any such k.

If $k = 5$, the third equation gives no new information, and the system reduces to two equations. Let the free variable be $z = t$. Then $y = 1 - t$, and the first equation gives

$$x = 2 - y - 2z = 2 - (1 - t) - 2t = 1 - t.$$

The general solution is

$$(x, y, z) = (1-t,\ 1-t,\ t), \qquad t \in \mathbb{R},$$

a whole line of solutions: the line where the three planes meet. Check with $t = 0$: the point $(1, 1, 0)$ satisfies all three original equations. So does $(0, 0, 1)$, from $t = 1$. One free parameter gives one line, so the geometry and the algebra give the same result.

CAUTION "No unique solution" covers two different cases, so always separate them. A row that reduces to $0 = 0$ gives infinitely many solutions. A row that reduces to $0 = c$, with $c \neq 0$, gives none. Examiners ask for both, and for the general solution in terms of a parameter.

YOUR TURN 4.3 Systems of Linear Equations HL

14. Solve each system by elimination.

(a) $x + y = 5, \quad x - y = 1$
(b) $3x - y = 5, \quad x + y = 3$
(c) $x + y = 4, \quad 2x + y = 7$
(d) $2x + 3y = 12, \quad x - y = 1$

15. Solve each system by substitution.

(a) $y = 2x, \quad x + y = 6$
(b) $y = x - 1, \quad 2x + y = 8$

16. State how many solutions each system has, and give a reason.

(a) $2x + y = 4, \quad 4x + 2y = 9$
(b) $y = x + 1, \quad y = x + 3$
(c) $x + y = 3, \quad 2x + 2y = 6$
(d) $x + y = 3, \quad x - y = 1$

17. Solve each system of three equations.

(a) $x + y + z = 6, \quad x - y + z = 2, \quad x + y - z = 0$
(b) $x + y + z = 4, \quad 2x - y + z = 8, \quad x + 2y - z = -3$
(c) $x + y + z = 2, \quad 2x - y + 3z = -1, \quad x - 2y + z = -1$
(d) $2x + y - z = -4, \quad x + 3y + 2z = 3, \quad 3x - y + z = 9$

18. For parts (a) to (c), write the augmented matrix, reduce it to row echelon form, and state the number of solutions.

(a) $x + y + z = 6, \quad x + 2y - z = 2, \quad 2x + 3y + z = 11$
(b) $x + y + z = 3, \quad 2x + 2y + 2z = 7, \quad x - y + z = 1$
(c) $x + y + z = 6, \quad x - y + z = 2, \quad 2x + 2z = 8$
(d) For $x + y + z = 3, \quad 2x + y + 3z = 7, \quad 3x + 2y + 4z = k$, find the value of k for which the system has infinitely many solutions, and state what happens for every

other value of k.

4.4 Quadratic Functions

A **quadratic** is a function whose highest power is x^2. Its graph is the **parabola**, the curve of Galileo's cannonball: a single smooth arch, symmetric about a vertical line through its turning point.

The same quadratic can be written in three forms, and each form shows a different feature immediately.

KEY A quadratic can be written in three forms:

$$y = ax^2 + bx + c, \qquad y = a(x-p)(x-q), \qquad y = a(x-h)^2 + k.$$

Standard form shows the y-intercept c; **factored** form shows the roots p and q; **vertex** form shows the turning point (h, k).

The sign of a sets the direction. If $a > 0$ the parabola opens upward and the turning point is a minimum. If $a < 0$ it opens downward and the turning point is a maximum. The curve is symmetric, and its mirror line passes through the vertex.

KEY · IB The graph of $f(x) = ax^2 + bx + c$ has axis of symmetry

$$x = -\frac{b}{2a}.$$

To move from standard form to vertex form, **complete the square**. This means rewriting $x^2 + \frac{b}{a}x$ as a perfect square, then subtracting whatever that added. No single step in this topic is used more often.

EXAMPLE 4.17

Write $y = x^2 - 10x + 21$ in vertex form, and state its vertex and axis of symmetry.

Take the $x^2 - 10x$ part and complete the square. Half of -10 is -5, and $(-5)^2 = 25$:

$$x^2 - 10x = (x-5)^2 - 25.$$

So

$$y = (x-5)^2 - 25 + 21 = (x-5)^2 - 4.$$

The vertex is $(5, -4)$ and the axis of symmetry is $x = 5$. Because $a = 1 > 0$, this vertex is the minimum point.

EXAMPLE 4.18

Sketch $y = x^2 - 2x - 3$, showing the vertex, axis of symmetry, and all intercepts.

Factor for the roots: $x^2 - 2x - 3 = (x-3)(x+1)$, so the x-intercepts are $x = -1$ and $x = 3$. The y-intercept is $c = -3$. Completing the square, $y = (x-1)^2 - 4$, so the vertex is $(1, -4)$ and the axis of symmetry is $x = 1$, which is exactly halfway between the two roots.

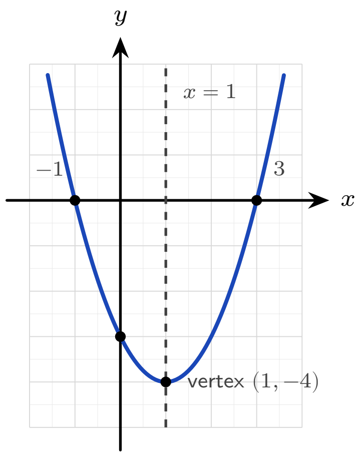


Figure 4.3 $y = x^2 - 2x - 3$, with its vertex, axis of symmetry and intercepts.

The three forms also work in reverse. Given points on a parabola, choose the form that already contains what you know, then use the remaining point to fix a. Two roots and one further point call for the factored form.

EXAMPLE 4.19

A parabola passes through $(0, -8)$, $(-1, 0)$ and $(4, 0)$. Find its equation.

Two of the points lie on the x-axis, so they are the roots. The factored form gives

$$y = a(x+1)(x-4),$$

with a still unknown: infinitely many parabolas share a pair of roots, and they differ only in how steeply they open. Substitute the third point, $(0, -8)$, to pin it down:

$$-8 = a(0+1)(0-4) = -4a \implies a = 2.$$

Expanding,

$$y = 2(x+1)(x-4) = 2x^2 - 6x - 8.$$

Check the three points: at $x = 0$, $y = -8$; at $x = -1$ and $x = 4$, $y = 0$. Had the curve passed through $(0, -4)$ instead, the same substitution would give $a = 1$, halving every y-coordinate while leaving the roots where they are.

YOUR TURN 4.4 Quadratic Functions

19. Write each quadratic in vertex form and state its vertex.

(a) $y = x^2 + 4x + 1$
(b) $y = x^2 - 4x$
(c) $y = x^2 - 6x + 11$
(d) $y = x^2 + 2x + 5$

20. State the vertex and the axis of symmetry of each quadratic.

(a) $y = (x-2)^2 + 5$
(b) $y = x^2 - 8x + 3$
(c) $y = -(x+1)^2 + 2$
(d) $y = 2(x-3)^2 - 1$

21. Consider $y = (x-1)(x+5)$.

(a) Write down its roots.
(b) Find its y-intercept.
(c) Find its axis of symmetry.
(d) Find its vertex.

22. State whether each quadratic has a maximum or a minimum, give a reason, and find that maximum or minimum value.

(a) $y = -x^2 + 2x$
(b) $y = 3x^2 - x + 1$
(c) $y = x^2 - 6x + 5$
(d) $y = 5 + 4x - x^2$

4.5 Quadratic Equations

Setting a quadratic equal to zero is the same as asking where its parabola crosses the x-axis. There are three ways to find those crossings, and all three give the same answers. Choosing the right method saves time in an examination.

Method	Use it when	Be careful with
Factorisation	the quadratic factorises with whole numbers	many quadratics do not factorise, and searching for factors takes time
Completing the square	you also want the vertex, or an exact surd answer	the arithmetic is harder when $a \neq 1$
The formula	always. It works for every quadratic	slower than factorising when factors are obvious

4.5.1 Solving by factorisation

If the quadratic factorises, you can read the roots immediately. The reason is the zero product rule: a product is zero only when one of its factors is zero.

EXAMPLE 4.20

Solve $x^2 - 8x + 15 = 0$.
Find two numbers that multiply to 15 and add to -8. They are -3 and -5:

$$(x-3)(x-5) = 0.$$

A product is zero only when a factor is zero, so $x - 3 = 0$ or $x - 5 = 0$, giving $x = 3$ or $x = 5$.

CAUTION The zero product rule needs a zero on the right. From $(x-2)(x-3) = 2$ you may *not* conclude $x - 2 = 2$. Expand, collect everything on one side, and factorise again.

4.5.2 Solving by completing the square

Completing the square works whether or not the quadratic factorises, and it gives exact answers in surd form. Rewrite the quadratic as a perfect square plus a constant, then take square roots of both sides.

EXAMPLE 4.21

Solve $x^2 + 4x - 12 = 0$ by completing the square.
Half of 4 is 2, and $2^2 = 4$, so $x^2 + 4x = (x+2)^2 - 4$. The equation becomes

$$(x+2)^2 - 4 - 12 = 0 \implies (x+2)^2 = 16.$$

Take the square root of both sides, remembering both signs:

$$x + 2 = \pm 4 \implies x = 2 \text{ or } x = -6.$$

EXAMPLE 4.22

Solve $x^2 - 4x - 1 = 0$, giving exact answers.
Half of -4 is -2, and $(-2)^2 = 4$:

$$(x-2)^2 - 4 - 1 = 0 \implies (x-2)^2 = 5 \implies x - 2 = \pm\sqrt{5}.$$

So $x = 2 \pm \sqrt{5}$. This quadratic does not factorise with whole numbers, which is why the answer contains a surd.

TIP When $a \neq 1$, divide the whole equation by a first. For $2x^2 - 8x + 5 = 0$, divide by 2 to get

$x^2 - 4x + 2.5 = 0$, then complete the square as usual.

4.5.3 Solving by formula

The formula works for every quadratic, and it is in the formula booklet.

KEY · IB The solutions of $ax^2 + bx + c = 0$, with $a \neq 0$, are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Substitute the three coefficients and the formula returns both roots at once.

The formula is not a rule to memorise without understanding. It is what **completing the square** gives when the coefficients are letters instead of numbers, so you can derive it yourself from $ax^2 + bx + c = 0$ at any time.

Divide through by a, then complete the square on $x^2 + \frac{b}{a}x$:

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \implies \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0.$$

Move the two constants to the right and write them over the common denominator $4a^2$:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

Take the square root of both sides, which is where the $\pm$ enters, and then subtract $\frac{b}{2a}$:

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a} \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The $\pm$ is the algebra recording what the picture already shows: the parabola is symmetric, with one root on each side of its axis. The expression under the root, $b^2 - 4ac$, decides how many roots there are, and that is the subject of the next section.

EXAMPLE 4.23

Solve $2x^2 - 5x - 12 = 0$.

Here $a = 2$, $b = -5$, $c = -12$. The formula gives

$$x = \frac{5 \pm \sqrt{25 + 96}}{4} = \frac{5 \pm 11}{4},$$

so $x = 4$ or $x = -\frac{3}{2}$.

YOUR TURN 4.5 Quadratic Equations

23. Solve each equation by factorisation.

(a) $x^2 - 5x + 6 = 0$
(b) $x^2 - x - 12 = 0$
(c) $x^2 + x - 20 = 0$
(d) $2x^2 - 5x - 3 = 0$

24. Solve each equation by completing the square, giving exact answers.

(a) $x^2 + 8x + 7 = 0$
(b) $x^2 - 2x - 5 = 0$
(c) $x^2 + 6x + 2 = 0$
(d) $2x^2 - 8x + 5 = 0$ *(Hint: divide by 2 first)*

25. Solve each equation with the quadratic formula, giving exact answers.

(a) $x^2 + 2x - 1 = 0$
(b) $x^2 + 6x + 4 = 0$
(c) $2x^2 - 3x - 2 = 0$
(d) $3x^2 + 2x - 1 = 0$

26. For each equation, state which of the three methods you would choose, and give a reason. You do not need to solve the equations.

(a) $x^2 - 7x + 12 = 0$
(b) $x^2 + 4x - 2 = 0$
(c) $5x^2 - 3x - 1 = 0$
(d) $x^2 - 9 = 0$

4.6 The Discriminant

The quadratic formula answers more than it is asked. Before you calculate a single root, the quantity under the square root sign already tells you how many real roots there are. That quantity has its own name.

KEY · IB The **discriminant** of $ax^2 + bx + c$ is

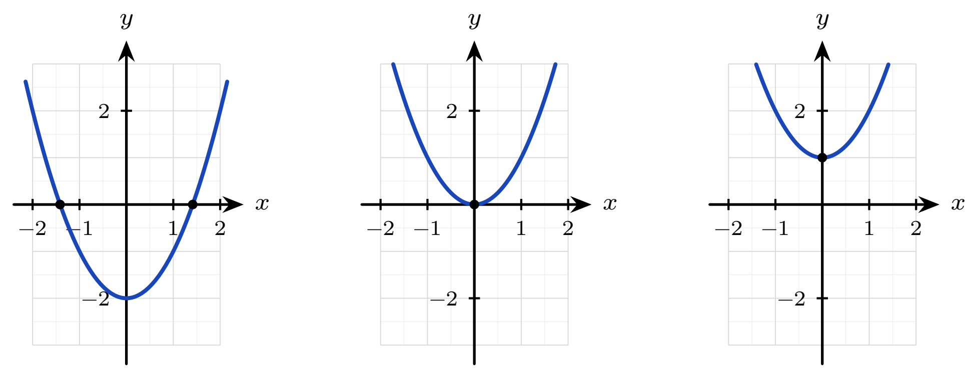
$$\Delta = b^2 - 4ac.$$

Its sign tells you how many real roots the equation has, without solving it.

The sign of Δ determines how many times the parabola meets the x-axis. The reason is that a negative number has no real square root, so $\pm\sqrt{\Delta}$ has no real value.

Discriminant	Real roots	Parabola and x-axis
$\Delta > 0$	two distinct	crosses twice
$\Delta = 0$	one repeated	touches once
$\Delta < 0$	none	does not meet it

The three cases are three positions of the same parabola relative to the x-axis.



(a) $\Delta > 0$: crosses twice (b) $\Delta = 0$: touches once (c) $\Delta < 0$: never meets

Figure 4.4 The same upward parabola in three positions: $y = x^2 - 2$, $y = x^2$ and $y = x^2 + 1$. Lowering it past the axis gives two roots, resting it on the axis gives one, and lifting it clear gives none. The discriminant measures which of the three has happened.

A downward parabola behaves the same way, with the picture turned over: the sign of a decides which way it opens, and Δ still counts the crossings.

EXAMPLE 4.24

Find the values of k for which $x^2 + kx + 9 = 0$ has exactly one (repeated) root.
One repeated root means $\Delta = 0$:

$$k^2 - 4(1)(9) = 0 \implies k^2 = 36 \implies k = \pm 6.$$

At these two values the parabola touches the x-axis at one point. Notice that the discriminant answers a question about the roots without finding them.

EXAMPLE 4.25

For the equation $3kx^2 + 2x + k = 0$, find the values of k that give two distinct real roots, two equal real roots, and no real roots.
Here the unknown constant appears in two places: $a = 3k$ and $c = k$, with $b = 2$. So

$$\Delta = b^2 - 4ac = 4 - 4(3k)(k) = 4 - 12k^2.$$

Now consider the three cases separately.

$$\Delta > 0: \; 4 - 12k^2 > 0 \implies k^2 < \tfrac{1}{3} \implies -\tfrac{1}{\sqrt{3}} < k < \tfrac{1}{\sqrt{3}}.$$

$$\Delta = 0: \; k^2 = \tfrac{1}{3} \implies k = \pm\tfrac{1}{\sqrt{3}}.$$

$$\Delta < 0: \; k^2 > \tfrac{1}{3} \implies k < -\tfrac{1}{\sqrt{3}} \text{ or } k > \tfrac{1}{\sqrt{3}}.$$

One value of k must be checked separately. If $k = 0$ the equation becomes $2x = 0$. That is linear, not quadratic, and it has the single root $x = 0$. So the answer for two distinct real roots is $-\frac{1}{\sqrt{3}} < k < \frac{1}{\sqrt{3}}$, *with* $k \neq 0$.

CAUTION Before using the discriminant, check that the equation really is quadratic. If the coefficient of x^2 contains the unknown constant, the value that makes it zero must be excluded and treated separately.

TIP A question that says "two distinct roots", "equal roots", "a repeated root", "touches the x-axis", or "no real solutions" is asking about the discriminant. Recognising the wording tells you which method to use.

4.6.1 A line meeting a parabola

The discriminant does more than count where a parabola meets the x-axis. A point common to two graphs satisfies both equations, so setting the two expressions for y equal gives a quadratic whose real roots are the x-coordinates of the intersections. The discriminant counts them. When there is exactly one, the line touches the curve without crossing it, and is a **tangent**.

EXAMPLE 4.26

Substitute $y = mx + c$ into $y = x^2 - 2x + 5$ and read the three cases geometrically. Hence find the values of m for which $y = mx + 4$ is a tangent to that parabola.

At any intersection both equations hold, so

$$x^2 - 2x + 5 = mx + c \implies x^2 - (m+2)x + (5-c) = 0.$$

Its coefficients are 1, $-(m+2)$ and $5-c$, so

$$\Delta = (m+2)^2 - 4(1)(5-c) = (m+2)^2 + 4c - 20.$$

The constant term is $5 - c$, not c; the line's intercept is subtracted when it moves across. Each real root is one intersection point, so $\Delta > 0$ gives two intersections, $\Delta = 0$ gives one and the line is a tangent, and $\Delta < 0$ means the line misses the parabola entirely.

For $y = mx + 4$ take $c = 4$ and impose tangency:

$$\Delta = (m+2)^2 - 4 = 0 \implies m + 2 = \pm 2 \implies m = 0 \text{ or } m = -4.$$

Check $m = 0$: the quadratic is $x^2 - 2x + 1 = (x-1)^2$, a repeated root at $x = 1$, so the horizontal line $y = 4$ touches at the vertex $(1, 4)$. For $m = -4$ the quadratic is $x^2 + 2x + 1 = (x+1)^2$, touching at $(-1, 8)$.

A tangent shows up as a repeated root, so factorising the quadratic at each value of m both confirms the answer and hands you the point of contact.

4.6.2 The sum and product of the roots

The roots of a quadratic are linked to its coefficients before you solve anything. If $ax^2 + bx + c$ has roots α and β, with $a \neq 0$ so that the expression really is quadratic, then it factorises as

$$a(x-\alpha)(x-\beta) = ax^2 - a(\alpha+\beta)x + a\,\alpha\beta.$$

Comparing the coefficient of x gives $b = -a(\alpha+\beta)$, and comparing the constant term gives $c = a\,\alpha\beta$. Dividing each by a, which the condition $a \neq 0$ permits, leaves the sum of the roots as $-\dfrac{b}{a}$ and their product as $\dfrac{c}{a}$. Nothing in that expansion depended on there being two brackets, so the same comparison settles a polynomial of any degree, and the result is stated properly in Ch. 6.

YOUR TURN 4.6 The Discriminant

27. Evaluate the discriminant and state the nature of the roots.

(a) $x^2 - 4x + 4$
(b) $2x^2 + 3x + 5$
(c) $3x^2 - 2x - 1$
(d) $x^2 + 5x + 7$

28. Find the value or values of k in each case.

(a) $x^2 + 6x + k = 0$ has equal roots
(b) $x^2 + kx + 16 = 0$ has equal roots
(c) $x^2 - 3x + k = 0$ has no real roots
(d) $kx^2 + 4x + k = 0$ has equal roots. State any value of k that must be excluded.

29. The equation $x^2 + (k+1)x + 9 = 0$ has two distinct real roots. Find the set of values of k.

4.7 Quadratic Inequalities

A quadratic **inequality** is a different question. It does not ask where the parabola meets the axis. It asks where the parabola lies above the axis, or below it. The method is always the same three steps.

KEY To solve a quadratic inequality:

1. Collect every term on one side, so the other side is 0.
2. Find the roots of the matching equation. These are the only places where the sign can change.
3. Sketch the parabola and read the intervals where it lies on the side you want.

The sketch is the method, so here it is in full. The roots cut the axis into three stretches, and the curve keeps one sign right across each of them.

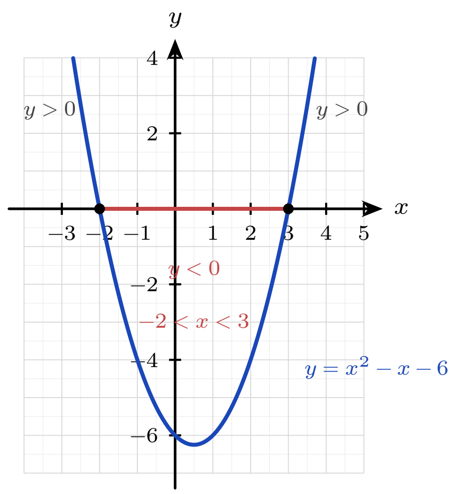


Figure 4.5 $y = x^2 - x - 6$, with roots at $x = -2$ and $x = 3$. The curve lies below the axis on the scarlet stretch between the roots and above it on both sides. So $x^2 - x - 6 < 0$ gives $-2 < x < 3$, and $x^2 - x - 6 > 0$ gives $x < -2$ or $x > 3$.

Read the picture, not a rule. An upward parabola is negative between its roots and positive outside them; a downward one is the other way round. Sketching takes a few seconds and never has the sign backwards.

EXAMPLE 4.27

Solve $x^2 - x - 6 > 0$.

Factor: $x^2 - x - 6 = (x - 3)(x + 2)$, so the roots are $x = -2$ and $x = 3$. The parabola opens upward. It is therefore *above* the axis outside the roots, and below the axis between them. So

$$x^2 - x - 6 > 0 \quad \text{when} \quad x < -2 \ \text{ or } \ x > 3.$$

A rough sketch of the parabola is faster and safer than memorising sign rules.

EXAMPLE 4.28

Solve $3x^2 \leq 5x + 2$.
First collect everything on one side:

$$3x^2 - 5x - 2 \leq 0.$$

Factorise: $(3x+1)(x-2) = 0$ gives roots $x = -\frac{1}{3}$ and $x = 2$. The parabola opens upward, so it lies *below* the axis between the roots. The sign $\leq$ includes the roots themselves:

$$-\tfrac{1}{3} \leq x \leq 2.$$

CAUTION Never divide an inequality by a term containing x. Its sign is unknown, and dividing by a negative quantity reverses the inequality. Collect on one side and factorise instead.

EXAMPLE 4.29

Solve $x^2 + 3x + 4 > 0$.
The discriminant is $9 - 16 = -7 < 0$, so the parabola never meets the x-axis. Since $a = 1 > 0$ it opens upward, so it lies entirely above the axis. The inequality holds for *every* real x.
When the discriminant is negative, a quadratic inequality has either every real number as its answer or none at all. The sign of a decides which.

YOUR TURN 4.7 Quadratic Inequalities

30. Solve each inequality.

(a) $x^2 - 9 < 0$
(b) $x^2 + 2x - 15 < 0$
(c) $x^2 - 4x + 3 \geq 0$
(d) $-x^2 + 4x > 0$

31. Solve each inequality. Two of them are special cases, so justify your answers.

(a) $2x^2 \leq x + 6$
(b) $x^2 \geq 16$
(c) $x^2 + x + 1 > 0$
(d) $x^2 + 4 < 0$

32. A rectangle has width x and length $x + 2$, both in centimetres.

(a) Show that its area is $x^2 + 2x$.
(b) Find the values of x for which the area is less than 15.

Chapter 4 · Lines & Quadratics

Reflections

TOK The same parabola appears in pure algebra, as the graph of x^2, and in the physical world, as the path of every thrown object, the cross-section of a satellite dish, and the cable of certain bridges. Why should a curve defined by an abstract equation also describe falling stones? This is what the physicist Eugene Wigner called the "unreasonable effectiveness of mathematics." Is mathematics invented to fit the world, or discovered as something already present in it?

TOK Descartes showed that a geometric problem can be solved with algebra, and an algebraic one with a picture. A line is an equation; an equation is a line. Neither one is the real object and the other its shadow. Does changing the representation give you new knowledge, or only a new view of knowledge you already had?

RW A parabola reflects every incoming parallel ray to a single point, called the focus. Satellite dishes and radio telescopes use this to collect weak signals at one point. Car headlights and torches use it in reverse: a bulb at the focus produces a parallel beam. Solar concentrators use it to focus sunlight and boil water. Quadratics also model optimisation everywhere: the maximum area for a fixed fence, the price that maximises profit, the angle at which a ball travels farthest.

RW A gradient is a rate, so any line drawn between two quantities has one. A road sign showing 12% is a gradient, and so is the limit on a wheelchair ramp, about 1 in 12 in most building codes. In physics the gradient of a distance-time graph is speed, and the gradient of a speed-time graph is acceleration. In economics it is elasticity: how much demand changes when the price changes.

EXT The discriminant divides quadratics into three cases. When $\Delta < 0$ the parabola does not meet the x-axis and there are no *real* roots, but the quadratic formula still returns answers, now containing $\sqrt{-1}$. Keeping these "impossible" numbers instead of rejecting them leads to the **complex numbers** of Chapter 17, where every quadratic has exactly two roots and the discriminant only tells you whether those roots are real.

End-of-Chapter Problems — Chapter 4: Lines & Quadratics

1. Consider $f(x) = 2x - 1$.

(a) Find $f^{-1}(x)$. [2]

(b) Verify that $f(f^{-1}(x)) = x$. [2]

2. Consider $f(x) = 3 - 2x$.

(a) State the x- and y-intercepts. [2]

(b) Find $f^{-1}(x)$. [2]

3. Consider $f(x) = \dfrac{2x+1}{x-3}$, $x \neq 3$.

(a) State the domain. [1]

(b) Find $f^{-1}(x)$. [4]

4. Find the equation of the line through $(3, -1)$ parallel to $2x - y = 4$. [3]

5. Consider the points $A(-1, 2)$ and $B(3, 10)$.

(a) Find the midpoint of AB. [1]

(b) Find the gradient of AB. [1]

(c) Find the length of AB. [2]

6. The line $2x + 3y = 12$ is given.

(a) Find its x- and y-intercepts. [2]

(b) State its gradient. [2]

7. A line passes through $A(1, 2)$ and $B(5, 4)$.

(a) Find the gradient of AB. [1]

(b) Find the equation of AB. [2]

(c) Find the equation of the perpendicular bisector of AB. [4]

8. Solve the system $3x + 2y = 7,\quad x - 2y = -3$. [3]

9. Solve the system

$$x + 2y + z = 4, \qquad 2x + y - z = 2, \qquad x - y + 2z = 2.$$

[6]

10. Consider the system

$$x + y + z = 3, \qquad x + 2y + 3z = 4, \qquad 2x + 3y + 4z = k.$$

(a) Find the value of k for which the system has solutions. [3]
(b) For this value of k, find the general solution. [3]
(c) Describe the configuration of the three planes when $k = 10$. [1]

11. A quadratic has roots 2 and -5.

(a) Write it in factored form (with leading coefficient 1). [1]
(b) Expand it into standard form. [2]

12. A parabola passes through $(0, 3)$, $(1, 0)$ and $(3, 0)$. Find its equation. [5]

13. Consider $f(x) = x^2 + 2x - 3$.

(a) Factorise $f(x)$ and state its roots. [2]
(b) Find the vertex. [2]
(c) Sketch the graph, showing all intercepts. [2]

14. Consider $f(x) = x^2 - 6x + 5$.

(a) Write $f(x)$ in vertex form. [2]
(b) State the vertex and the axis of symmetry. [2]
(c) Find the x- and y-intercepts. [3]

15. Solve the inequality $x^2 - 2x - 8 \leq 0$. [3]

16. The equation $x^2 - 4x + c = 0$ has no real roots. Find the set of values of c. [3]

17. Solve the inequality $x^2 + x - 6 > 0$. [3]

18. The equation $x^2 + kx + 4 = 0$ has two distinct real roots. Find the set of values of k. [4]

19. Find the values of k for which $x^2 + kx + (k + 3) = 0$ has equal roots. [4]

20. Two numbers have a sum of 10 and a product of 21. Find the numbers. [4]

21. The roots of $x^2 + bx + c = 0$ sum to 6 and differ by 4. Find b and c. [5]

22. Show that $x^2 + (k+1)x + k = 0$ has real roots for every real k, and find the value of k for which the roots are equal. [5]

23. The roots of $x^2 - px + q = 0$ are α and 2α. Show that $2p^2 = 9q$. [5]

24. Find the points of intersection of $y = x^2$ and $y = 2x + 3$. [4]

25. Consider the line $y = 2x - 5$ and the parabola $y = x^2 - 2x + 2$.

(a) Show that they do not intersect. [3]
(b) Explain what this means geometrically. [1]

26. A ball is thrown so that its height in metres after t seconds is $h(t) = 20t - 5t^2$.

(a) Find the time at which the ball lands. [2]
(b) Find the maximum height reached. [3]

27. Find the value of k for which the line $y = x + k$ is a tangent to the parabola $y = x^2$. [5]

28. A rectangle has length 3 more than its width w.

(a) Write its area A as a function of w. [2]
(b) Given the area is 40, find w. [3]

29. A rectangle has perimeter 20. Show that its area is $A = x(10-x)$, where x is one side, and find the maximum possible area. [5]

30. Prove that the line $y = mx + c$ is a tangent to the parabola $y = x^2$ if and only if $c = -\dfrac{m^2}{4}$. [6]

31. GDC Consider $f(x) = x^3 - 4x + 1$.

(a) Using your GDC, sketch the graph of f for $-3 \leq x \leq 3$, labelling all key features. [2]
(b) Find the zeros of f, giving your answers to 3 significant figures. [2]
(c) Find the coordinates of the local maximum and local minimum points. [2]
(d) Solve $f(x) = 2x - 1$, giving your answers to 3 significant figures. [2]

32. GDC A bridge has a parabolic arch. The arch is 40 m wide at road level and 10 m high at its centre. Take the road as the x-axis, with the centre of the arch at $x = 0$.

(a) Show that the arch is modelled by $y = 10 - \frac{x^2}{40}$. [4]
(b) Find the height of the arch 12 m from the centre. [2]
(c) A lorry 8 m wide and 9.5 m high drives along the centre of the road. Determine whether it passes under the arch. [4]
(d) Find the greatest width of a lorry 9 m high that can pass under the arch, giving your answer to three significant figures. [4]
(e) The model gives a negative value of y when $|x| > 20$. Explain what this means for the model. [2]

33. Consider the line $y = mx + 3$ and the parabola $y = x^2 + 5x + 7$.

(a) Show that at any point of intersection, $x^2 + (5 - m)x + 4 = 0$. [3]
(b) Find the values of m for which the line is a tangent to the parabola. [4]
(c) Find the values of m for which the line does not meet the parabola. [3]
(d) For the smaller value of m found in part (b), find the coordinates of the point of tangency. [3]
(e) State the values of m for which the line meets the parabola twice. [2]

34. HL Consider the system of equations

$$x + y + z = 2, \qquad x + 2y + 3z = 5, \qquad 2x + 3y + az = b,$$

where a and b are real constants.

(a) Write the system as an augmented matrix and row reduce it. [4]
(b) Find the value of a for which the system does not have a unique solution. [3]
(c) For that value of a, find the value of b for which the system is consistent. [2]
(d) For those values of a and b, find the general solution of the system. [4]
(e) Describe how the three planes are arranged in each of the three cases: a not equal to the value in part (b); a equal to it with b consistent; and a equal to it with b not consistent. [3]

Challenge Questions

35. **Tangents to a parabola.** A line is a tangent to a curve when it meets the curve exactly once. This investigation turns that idea into a condition on the discriminant, then uses it to count the tangents from a point.

(a) Show that the line $y = mx + c$ meets $y = x^2$ where $x^2 - mx - c = 0$, and hence that the line is a tangent exactly when $c = -\frac{m^2}{4}$. [4]

(b) Find the two tangents to $y = x^2$ that pass through the point $(0, -4)$, and state where each one touches the parabola. [4]

(c) Repeat part (a) for the parabola $y = ax^2$, with $a \neq 0$, and show that the tangency condition becomes $c = -\dfrac{m^2}{4a}$. [3]

(d) Using part (c), find how many tangents to $y = x^2$ pass through the point $(0, k)$, for each value of k. State the three cases and explain each one from the picture. [5]

(e) A student claims that every point in the plane has either two tangents to $y = x^2$ or none. Decide whether the claim is true, and justify your answer. [4]

36. Sum and product of roots. A quadratic can be read from its roots without ever solving it. This investigation builds the relationships and then uses them to answer questions that would be slow by the formula.

(a) Let the roots of $x^2 - Sx + P = 0$ be α and β. By expanding $(x - \alpha)(x - \beta)$, show that $\alpha + \beta = S$ and $\alpha\beta = P$. [3]

(b) Show that $(\alpha - \beta)^2 = S^2 - 4P$, and hence write down the condition on S and P for the roots to be real. [3]

(c) The roots of a quadratic sum to 6 and differ by 4. Use part (b) to find the quadratic, without finding the roots first. [3]

(d) Show that $\alpha^2 + \beta^2 = S^2 - 2P$ and that $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{S}{P}$. Verify both for $x^2 - 6x + 5 = 0$. [4]

(e) Find the condition on S and P for the two roots to be reciprocals of each other, and give one quadratic with that property and irrational roots. [3]

FUNCTIONS

Chapter 5

Functions & Graphs

5

SYLLABUS 2.5 2.8–2.11 2.13–2.16

Hold a card in front of a lamp and look at its shadow on the wall. Slide the card and the shadow slides. Move it closer to the lamp and the shadow grows. Turn it over and the shadow flips. The card itself never changes. Only what you do to it changes.

Graphs work the same way, and that is what makes this part of the course manageable. There are only a few shapes to learn by heart, a line, a parabola, a square root, a reciprocal, together with a short list of moves that slide them, stretch them and reflect them. Every other curve you will meet is one of those few shapes with some of those moves applied to it.

So this chapter has two halves. The first is **composition** and the list of moves: how the output of one function becomes the input of the next, and what each move does to an equation. The second is the small set of shapes those moves are applied to, and in particular the ones whose graphs have a line they approach but never reach. Reading that behaviour from a formula is the second skill this chapter builds.

By the end of this chapter you will be able to compose two functions and find the domain of the result, apply translations, stretches and reflections, and test a function for symmetry. You will sketch the related graphs $|f(x)|$, $f(|x|)$ and $\dfrac{1}{f(x)}$. You will find the asymptotes of a rational function, solve a rational inequality with a sign table, and treat the exponential and the logarithm as inverses of each other. What ties these together is that every number in a formula corresponds to something you can see. Change the number and the graph moves in a way you can predict. A formula and a picture are two views of one function, and this chapter is about reading each from the other.

5.1 Composite Functions

To work out $\sqrt{3x+1}$ you do three things, in order. Multiply by 3. Add 1. Take the square root. Nearly every formula in this course is built like that, from a few simple steps joined end to end. Reading a formula as a chain of steps is what lets you undo it, and later what lets you differentiate it. **Composition** is the notation for that chain.

Two functions can be joined end to end. Send a number into the first function. Take the number that comes out. Send that number into the second function. The single rule that does both steps at once is called a **composite** function.

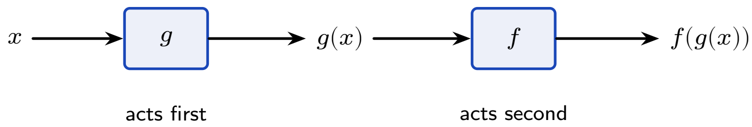


Figure 5.1 Composition sends x through g first, then sends the result into f. The inner function acts first, which is why $f \circ g$ is read from the inside out.

> DEF The **composite** $f \circ g$ is the function $(f \circ g)(x) = f(g(x))$: apply g first, then f. The **identity function** $I(x) = x$ returns every input unchanged.

The symbol $\circ$ is read "of", so $f \circ g$ is "f of g". It is also read "f after g", which gives the order in which the two functions act. It is not multiplication. Read the notation from right to left: the function written closest to x acts on x first.

5.1.1 Substituting a Whole Function

Composition looks unfamiliar because a function is being written where a number normally goes. The rule for f does not change at all. Only what you write in the slot changes.

The x in $f(x)$ is a **placeholder** (a marker showing where the input goes). It is not a fixed value, and it does not have to be a single letter.

Read f as a rule with an empty slot. If $f(x) = x^2 + 1$, the rule says: square the input, then add one. Whatever you write in the slot is squared, and then one is added.

$$f(\square) = \square^2 + 1 \qquad \Longrightarrow \qquad f(3) = 3^2 + 1, \quad f(2x) = (2x)^2 + 1, \quad f(g(x)) = \big(g(x)\big)^2 + 1.$$

The last one is composition. You are substituting a whole function exactly as you would substitute a number.

To build $(f \circ g)(x)$:

1. Write the formula for $f(x)$.
2. Substitute the expression $g(x)$ for **every** x in that formula.
3. Simplify.

5.1.2 Order of Composition

Putting on socks and then shoes is not the same as putting on shoes and then socks. Composition behaves in the same way, so $f \circ g$ and $g \circ f$ are usually two different functions. Deciding which function acts first is therefore the first thing to do in any composition question.

EXAMPLE 5.1

For $f(x) = x^2 + 1$ and $g(x) = 2x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

For $f \circ g$, apply g first. Its output is $2x$, so substitute $2x$ for every x in $f(x) = x^2 + 1$, then simplify:

$$(f \circ g)(x) = f(2x) = (2x)^2 + 1 = 4x^2 + 1.$$

For $g \circ f$, apply f first. Its output is $x^2 + 1$, so substitute $x^2 + 1$ for every x in $g(x) = 2x$, then simplify:

$$(g \circ f)(x) = g(x^2 + 1) = 2(x^2 + 1) = 2x^2 + 2.$$

The two answers are different. In general $f \circ g \neq g \circ f$, so always check which function acts first.

One input makes the difference concrete. Send $x = 3$ through both composites and follow it step by step.

Composite	Input	After the first function	After the second function
$f \circ g$	3	$g(3) = 6$	$f(6) = 37$
$g \circ f$	3	$f(3) = 10$	$g(10) = 20$

5.1.3 The Domain of a Composite

Each part of a composite can be defined at a number while the composite is not. The reason is that f is never applied to x. It is applied to $g(x)$, and $g(x)$ may be a value that f does not take. So two conditions must hold, and the second one is easy to miss.

KEY The **domain** of $f \circ g$ is the set of values x for which

1. x is in the domain of g, **and**
2. $g(x)$ is in the domain of f.

Find the domain from the two original functions, before simplifying. Simplification is an algebraic step, and a simplified formula may be defined at values where the composite is not.

EXAMPLE 5.2

For $f(x) = x^2$ and $g(x) = \sqrt{x - 1}$, find $(f \circ g)(x)$ and state its domain.

Apply g first, then square its output:

$$(f \circ g)(x) = f(\sqrt{x-1}) = \left(\sqrt{x-1}\right)^2 = x - 1.$$

Now take the two conditions in order.
Condition 1: the domain of g requires $x - 1 \geq 0$, so $x \geq 1$.
Condition 2: the domain of f is all real numbers, so this adds nothing here.
The domain of $f \circ g$ is therefore $x \geq 1$. The composite is the line $y = x - 1$ with everything to the left of $x = 1$ removed.

CAUTION A simplified formula is not the composite. $\left(\sqrt{x-1}\right)^2$ simplifies to $x-1$, which is defined for every real x, but the composite exists only for $x \geq 1$. Always return to the original functions to find the domain.

EXAMPLE 5.3
For $f(x) = \frac{1}{x}$ and $g(x) = x - 3$, find $(f \circ g)(x)$ and state its domain.

$$(f \circ g)(x) = f(x-3) = \frac{1}{x-3}.$$

Condition 1: the domain of g is all real numbers, so it removes nothing.
Condition 2: 0 is not in the domain of f, so the output $g(x) = x - 3$ must not equal 0. This removes $x = 3$.
The domain of $f \circ g$ is every real number except $x = 3$.

The two examples restrict the domain for different reasons, which is why both conditions have to be checked every time.

The functions	**1. x in the domain of g**	**2. $g(x)$ in the domain of f**	**Domain of $f \circ g$**
$f(x) = x^2$ $g(x) = \sqrt{x-1}$	$x \geq 1$	no extra condition	$x \geq 1$
$f(x) = \frac{1}{x}$ $g(x) = x - 3$	every real x	$x - 3 \neq 0$, so $x \neq 3$	$x \in \mathbb{R}$, $x \neq 3$

5.1.4 Composition and Inverses

Some pairs of functions undo each other. Composition is how that relationship is written down, and composing the two is how you check that an inverse is correct.

The **inverse** of f, written f^{-1}, is the function that reverses f. If f sends 3 to 10, then f^{-1} sends 10 back to 3. Apply f and then apply f^{-1}, and you arrive back at the number you started with. The composite is therefore the identity function.

KEY For a function f with inverse f^{-1},

$$(f^{-1} \circ f)(x) = x \qquad \text{and} \qquad (f \circ f^{-1})(x) = x.$$

An inverse exists exactly when f is **one-to-one**: no two inputs share an output.

To find an inverse, undo the function one operation at a time.

1. Write $y = f(x)$.
2. Solve for x, so that x stands alone on one side.
3. Exchange the letters, so the inverse is written as a function of x.
4. State the domain of f^{-1}, which is the range of f.

Composition is also how you check the answer. Take $f(x) = 2x + 1$, whose inverse is $f^{-1}(x) = \dfrac{x-1}{2}$. Substituting one into the other gives

$$f^{-1}(f(x)) = \frac{(2x+1)-1}{2} = \frac{2x}{2} = x,$$

which confirms the pair. The clearest pair of inverse functions in this chapter is the exponential and the logarithm; see §5.5.

A quotient of two linear expressions needs one extra move at step 2. The variable appears twice, once above the line and once below it, so it has to be collected into a single term before it can be isolated. The range of such a function, which is the domain of its inverse, is worked out in §5.4.1.

EXAMPLE 5.4

Find the inverse of $f(x) = \dfrac{3x-1}{x-1}$, and state its domain.
Write $y = f(x)$ and clear the fraction by multiplying both sides by the denominator:

$$y = \frac{3x-1}{x-1} \implies y(x-1) = 3x-1 \implies xy - y = 3x - 1.$$

Now x sits in two terms. Gather those on one side and everything else on the other, then factorise and divide:

$$xy - 3x = y - 1 \implies x(y-3) = y-1 \implies x = \frac{y-1}{y-3}.$$

Exchange the letters, so the inverse is written as a function of x:

$$f^{-1}(x) = \frac{x-1}{x-3}, \qquad x \neq 3.$$

The excluded value is not guesswork. The domain of f^{-1} is the range of f, and f never produces 3: written as $f(x) = 3 + \dfrac{2}{x-1}$, its second term is never zero, so the output is never

exactly 3.
Check one point: $f(0) = \dfrac{-1}{-1} = 1$, and $f^{-1}(1) = \dfrac{0}{-2} = 0$, which returns the input.

YOUR TURN 5.1 Composite Functions

1. For $f(x) = x + 3$ and $g(x) = 2x$, find each composite.

(a) $(f \circ g)(x)$
(b) $(g \circ f)(x)$
(c) $(f \circ f)(x)$
(d) $(g \circ g)(x)$

2. For $f(x) = x^2$ and $g(x) = x - 1$, find each of the following.

(a) $(f \circ g)(x)$
(b) $(g \circ f)(x)$
(c) $(f \circ g)(3)$
(d) $(g \circ f)(3)$

3. For $f(x) = x + 5$ and $g(x) = x^2$, evaluate each of the following.

(a) $(g \circ f)(2)$
(b) $(f \circ g)(2)$
(c) $(g \circ f)(-3)$
(d) $(f \circ g)(-3)$

4. For $f(x) = \dfrac{1}{x}$ and $g(x) = x + 1$, find each composite and state its domain.

(a) $(f \circ g)(x)$
(b) the domain of $f \circ g$
(c) $(g \circ f)(x)$
(d) the domain of $g \circ f$

5. Find the inverse of each function.

(a) $f(x) = 3x - 2$
(b) $f(x) = \dfrac{x}{4} + 1$
(c) $f(x) = \dfrac{x-5}{2}$
(d) $f(x) = \dfrac{1}{x-1}$

6. Let $f(x) = 2x + 1$.

(a) Find $f^{-1}(x)$.
(b) Show that $(f \circ f^{-1})(x) = x$.
(c) Show that $(f^{-1} \circ f)(x) = x$.
(d) State what these two results tell you about f^{-1}.

7. Let $f(x) = x^2$ and $g(x) = x + 2$.

(a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, and show that they are not the same.
(b) Find the value of x for which the two composites are equal.

5.2 Transformations of Graphs

Plotting points is slow, and for most curves it is unnecessary. A small change to an equation moves the graph in one predictable way. Once you know which change produces which move, you can sketch a new curve from a familiar one in a few seconds. You can also work the other way, and write down the equation of a curve you are shown.

There are three moves: a **translation** (a slide), a **reflection** (a flip) and a **stretch** (a scaling). Each one matches one change in the equation, and each one sends the point (p, q) to a new point you can write down.

Move	Equation	Acts on	(p, q) becomes	Effect
Translation	$y = f(x) + k$	output	$(p,\ q+k)$	k up
	$y = f(x-h)$	input	$(p+h,\ q)$	h right
Reflection	$y = -f(x)$	output	$(p,\ -q)$	in the x-axis
	$y = f(-x)$	input	$(-p,\ q)$	in the y-axis
Stretch	$y = a\,f(x)$	output	$(p,\ aq)$	vertical, factor a
	$y = f(bx)$	input	$(\frac{p}{b},\ q)$	horizontal, factor $\frac{1}{b}$

The table is three pairs, one pair for each move. Read the **Acts on** column and a single pattern accounts for all six rows. Everything depends on whether the change sits outside the bracket or inside it.

> KEY **Outside** the bracket, the change acts on the output: it is **vertical**, and it does what it says. **Inside** the bracket, the change acts on the input: it is **horizontal**, and it does the **opposite**.

The first line behaves as you would expect. Adding k raises every output by k, so the graph moves up by k. Multiplying by a multiplies every output by a, so the graph stretches vertically by a.

The second line needs an explanation, because the reversal is not obvious. Take $y = f(x-h)$

and evaluate it at $x = p + h$:

$$f((p + h) - h) = f(p),$$

which is the value the original graph had at $x = p$. So the point that was at $x = p$ now appears at $x = p + h$. Every point moves h units to the *right*, not left. The same calculation works for $y = f(bx)$: at $x = \frac{p}{b}$ it returns $f(b \cdot \frac{p}{b}) = f(p)$, so the point at p moves to $\frac{p}{b}$. That is a horizontal stretch of factor $\frac{1}{b}$, which brings every point closer to the y-axis whenever $b > 1$.

5.2.1 Translations

A translation slides the whole graph. Its shape and its size do not change, only its position.

Vertically, $y = f(x) + k$ adds k to every output, so every point moves up by k. A negative k moves the graph down.

Horizontally, $y = f(x - h)$ moves the graph right by h. This sign is easy to reverse. Use the reason above rather than memorising the result. The change is inside the bracket, so it acts on the input, and its effect is the opposite of its sign.

In examination questions a translation is usually written as a vector. The graph of $y = f(x - h) + k$ is the graph of $y = f(x)$ translated by $\binom{h}{k}$.

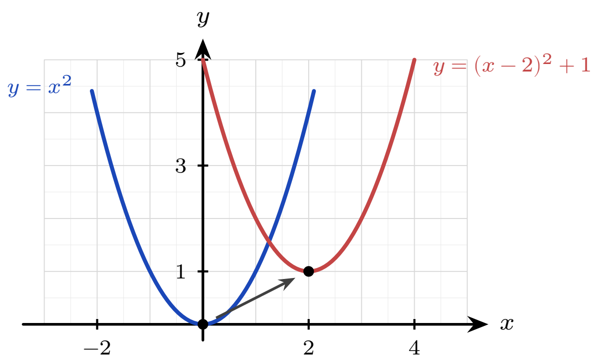


Figure 5.2 The graph of $y = (x - 2)^2 + 1$ is the graph of $y = x^2$ translated 2 units right and 1 unit up. The vertex moves from $(0, 0)$ to $(2, 1)$, and the shape is unchanged.

EXAMPLE 5.5

Describe how the graph of $y = (x - 2)^2 + 1$ is obtained from $y = x^2$.

Start with the bracket. The -2 sits inside it, so it acts on the input and moves the parabola 2 units *right*.

Now take the rest. The $+1$ sits outside the bracket, so it acts on the output and moves the parabola 1 unit *up*.

Together these send the vertex from $(0, 0)$ to $(2, 1)$, which is exactly what the completed-square form predicts. Transformations and the completed-square form are two descriptions of the same thing.

5.2.2 Reflections

A reflection flips the graph across an axis. Every point keeps its distance from that axis and moves to the other side of it.

For $y = -f(x)$ the sign of every output changes, so the point (p, q) moves to $(p, -q)$. That is a reflection in the x-axis. A maximum becomes a minimum, and any point already on the x-axis stays where it is.

For $y = f(-x)$ the sign of every input changes, so the point (p, q) moves to $(-p, q)$. That is a reflection in the y-axis. The y-intercept does not move, because changing the sign of 0 leaves it as 0.

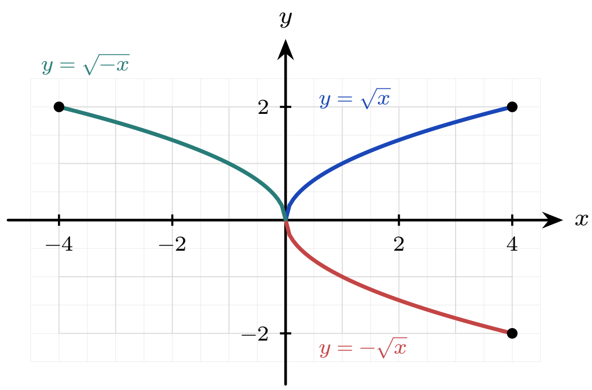


Figure 5.3 Reflections of $y = \sqrt{x}$. Changing the sign of the output gives $y = -\sqrt{x}$, and the point $(4, 2)$ moves to $(4, -2)$: a reflection in the x-axis. Changing the sign of the input gives $y = \sqrt{-x}$, and $(4, 2)$ moves to $(-4, 2)$: a reflection in the y-axis.

EXAMPLE 5.6

Describe how the graph of $y = -(x-1)^2$ is obtained from $y = x^2$.

There are two changes. The -1 is inside the bracket, so it is horizontal: translate 1 unit right.

The minus sign is outside the bracket, so it is vertical: reflect in the x-axis.

One change is horizontal and the other is vertical, so they do not affect each other and may be done in either order.

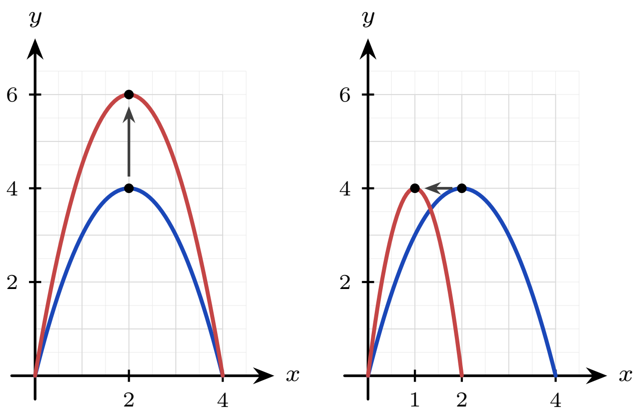
The vertex moves from $(0, 0)$ to $(1, 0)$, and the parabola now opens downward.

5.2.3 Stretches

A stretch changes the size of the graph in one direction only. One line stays fixed while every other point moves away from it or towards it.

For $y = a\,f(x)$ every output is multiplied by a, so (p, q) moves to (p, aq). This is a vertical stretch of factor a. Points on the x-axis do not move, because $a \times 0 = 0$.

For $y = f(bx)$ every input is divided by b, as shown above, so (p, q) moves to $(\frac{p}{b}, q)$. This is a horizontal stretch of factor $\frac{1}{b}$. Points on the y-axis do not move.



(a) $y = 1.5\,f(x)$ (b) $y = f(2x)$

Figure 5.4 The original curve $y = f(x)$ is cobalt and the stretched curve is scarlet. In (a) every y-coordinate is multiplied by 1.5, so the maximum moves from $(2, 4)$ to $(2, 6)$ while both roots stay on the x-axis. In (b) every x-coordinate is halved, so the maximum moves from $(2, 4)$ to $(1, 4)$ and the root at $x = 4$ moves to $x = 2$.

CAUTION A factor smaller than 1 is still called a stretch. The graph of $y = \frac{1}{2}f(x)$ is a vertical stretch of factor $\frac{1}{2}$, even though it makes the graph shorter. Write "stretch, factor $\frac{1}{2}$", not "squash".

EXAMPLE 5.7

The graph of $y = f(x)$ has a maximum point at $(6, 4)$. State the coordinates of that maximum point on the graph of (a) $y = 3f(x)$, (b) $y = f(3x)$.

(a) The 3 is outside the bracket, so it is a vertical stretch of factor 3. It multiplies the y-coordinate and leaves the x-coordinate alone: $(6, 12)$.

(b) The 3 is inside the bracket, so it is a horizontal stretch of factor $\frac{1}{3}$. It divides the x-coordinate and leaves the y-coordinate alone: $(2, 4)$.

Each stretch changes one coordinate and leaves the other unchanged. Checking which coordinate moved is a quick way to confirm you have used the right one.

5.2.4 Composite Transformations

Most graphs are several moves away from a familiar shape, applied one after another. This is a **composite transformation**. Describing one is a matter of taking the changes in a reliable order.

Start by writing the target equation in one standard form:

$$y = a\,f\big(b(x - h)\big) + k.$$

Then a is the vertical stretch factor, b gives the horizontal stretch factor $\frac{1}{b}$, h is the horizontal translation and k is the vertical translation. Every value can be read straight from the equation.

For a quadratic this standard form is the **completed-square form**. Writing $y = a(x-h)^2+k$ names the vertex (h,k) and the two moves that put it there, so completing the square and describing a transformation are two ways of doing the same work.

KEY To describe a composite transformation:

1. Split the changes into two groups: those **inside** the bracket, which are horizontal, and those **outside** it, which are vertical.
2. Do the two groups in either order. A horizontal move and a vertical move do not affect each other.
3. Inside each group the order matters. Apply the **stretch before the translation**.

Step 3 has a reason, and understanding it is safer than memorising it. A stretch multiplies every value it is applied to, including a translation that has already been carried out. Do the stretch first and there is no translation for it to multiply. The second example below shows what goes wrong in the other order.

Whatever sequence you write down, check it by tracking a single point. Choose a point you know on the original graph, such as a vertex, an endpoint or an intercept. Apply your moves to its coordinates. Then check that the result satisfies the target equation.

CAUTION Factorise inside the bracket first. In $y = f(2x-6)$ the translation is *not* 6. Written as $y = f(2(x-3))$, it is a horizontal stretch of factor $\frac{1}{2}$ followed by a translation of 3 to the right. Forgetting to factorise is a common error here.

EXAMPLE 5.8

Describe a sequence of transformations that maps $y = \sqrt{x}$ onto $y = 3\sqrt{x-2}+1$.

Take the change inside the bracket first. The -2 acts on the input, so it translates the graph 2 units *right* and gives $\sqrt{x-2}$.

Now take the vertical changes, stretch before translation. The factor 3 stretches the graph vertically by scale factor 3 and gives $3\sqrt{x-2}$. The $+1$ then translates it 1 unit *up*.

So the sequence is: translate 2 right, stretch vertically by factor 3, translate 1 up.

Check it by tracking the endpoint. The graph of $y = \sqrt{x}$ begins at $(0,0)$. The translation carries it to $(2,0)$. The stretch leaves it there, because $3 \times 0 = 0$. The final translation carries it to $(2,1)$. That is exactly where $y = 3\sqrt{x-2}+1$ begins.

Two moves **commute** (give the same result in either order) when swapping them leaves the final graph unchanged. A horizontal move and a vertical move always commute, because they change different coordinates. A stretch and a translation in the *same* direction do not commute, unless the stretch factor is 1 or the translation is 0, so that one of the two moves does nothing. The next example shows the two different answers they produce.

EXAMPLE 5.9

Starting from $y = x^2$, apply a vertical stretch by factor 2 and a translation 1 unit up, in both orders. Are the results the same?

Stretch first, then translate. The stretch doubles every value, giving $2x^2$; the translation then adds 1:

$$x^2 \longrightarrow 2x^2 \longrightarrow 2x^2 + 1.$$

Translate first, then stretch. The translation adds 1, giving $x^2 + 1$; the stretch then doubles everything, *including* the 1:

$$x^2 \longrightarrow x^2 + 1 \longrightarrow 2(x^2 + 1) = 2x^2 + 2.$$

The two curves are different. A vertical stretch multiplies whatever is already there, so a translation applied first is multiplied as well. A vertical stretch and a vertical translation commute only when one of them does nothing, that is when the stretch factor is 1 or the translation is 0. In every other case, state them in the order that actually builds the target curve.

TIP In examinations, "describe the sequence of transformations" expects named moves with their values: translation by $\binom{-3}{0}$, vertical stretch scale factor 2, translation by $\binom{0}{-1}$. Vague words such as "moved" or "squashed" do not earn the marks.

YOUR TURN 5.2 Transformations of Graphs

8. Describe the single transformation that takes $y = x^2$ to each graph.

(a) $y = x^2 + 3$
(b) $y = (x + 4)^2$
(c) $y = -x^2$
(d) $y = (x - 1)^2$

9. Describe the single transformation that takes $y = f(x)$ to each graph.

(a) $y = f(-x)$
(b) $y = -f(x)$
(c) $y = 3f(x)$
(d) $y = f(2x)$

10. The point $(2, 5)$ lies on $y = f(x)$. Find the coordinates of its image on each graph.

(a) $y = f(x - 3) + 1$
(b) $y = 3f(x)$
(c) $y = f(2x)$
(d) $y = -f(x) + 2$

11. The point $(-2, 3)$ lies on $y = f(x)$. Find the coordinates of its image on each graph.

(a) $y = f(2x)$
(b) $y = f(x) - 4$
(c) $y = f(-x)$
(d) $y = \frac{1}{2}f(x)$

12. Describe the sequence of transformations that maps $y = \sqrt{x}$ onto each graph.

(a) $y = \sqrt{x+3}$
(b) $y = 2\sqrt{x}$
(c) $y = 2\sqrt{x+3} - 1$
(d) $y = \sqrt{2x-6}$ *(Hint: factorise inside the root first)*

13. Start from $y = x^2$.

(a) Apply a vertical stretch of factor 3, then a translation 2 units up. Write the resulting equation.
(b) Apply the same two moves in the opposite order, and write the resulting equation.

14. Consider $y = f(4x - 8)$.

(a) Write it in the form $y = f(b(x - h))$.
(b) Describe the sequence of transformations from $y = f(x)$.

5.3 Symmetry and Modulus HL

Half of a sketch is half the work, and for some graphs half is all you need. A graph has a **symmetry** when some motion leaves it looking exactly as it did. Reflect it, rotate it or slide it, and the picture is unchanged. One half of the curve then fixes the other half, and the symmetry gives you a way to check a sketch against the formula.

Three motions are common enough to be named, and each one has a short algebraic test.

Type	Test	Motion that leaves the graph unchanged	Examples
Even	$f(-x) = f(x)$	reflection in the y-axis	$x^2,\quad x^4,\quad \lvert x\rvert,\quad \cos x$
Odd	$f(-x) = -f(x)$	rotation of 180° about the origin	$x,\quad x^3,\quad \dfrac{1}{x},\quad \sin x$
Periodic	$f(x+p) = f(x)$	translation of p along the x-axis	$\sin x,\quad \cos x,\quad \tan x$
Neither	no test holds	none of the three	$x^2 + x,\quad x+1,\quad e^x$

Most functions have none of these symmetries, so "neither" is a complete and correct answer when the tests fail.

The third row is the one you meet again in trigonometry. A function is **periodic** if its graph repeats at regular intervals, and the smallest positive p with $f(x+p) = f(x)$ is called the **period**. The graphs of $\sin x$ and $\cos x$ both have period 2π.

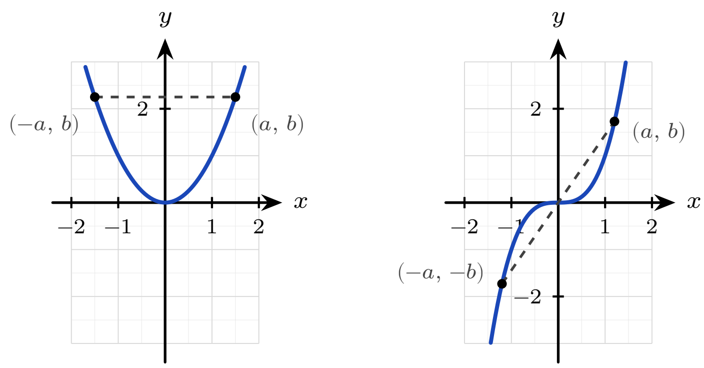
5.3.1 Testing for Even and Odd

The names come from the powers. The functions $x^2, x^4, \ldots$ have even exponents and satisfy $f(-x) = f(x)$. The functions $x, x^3, \ldots$ have odd exponents and satisfy $f(-x) = -f(x)$.

> DEF A function is **even** if $f(-x) = f(x)$ for every x in its domain, and **odd** if $f(-x) = -f(x)$ for every x in its domain. An even graph is symmetric in the y-axis; an odd graph has rotational symmetry of order two about the origin.

To test a function, substitute $-x$ for x, simplify, and compare the result with the original. There are three possible outcomes:

- the result is $f(x)$, so the function is **even**;
- the result is $-f(x)$, so the function is **odd**;
- the result is neither, so the function has **no symmetry** of this kind.



(a) $y = x^2$ is even (b) $y = x^3$ is odd

Figure 5.5 In (a) the point (a, b) has a partner at $(-a, b)$, directly across the y-axis, so reflecting in that axis leaves the curve unchanged. In (b) the partner is at $(-a, -b)$, directly opposite through the origin, so a half turn about the origin leaves the curve unchanged.

EXAMPLE 5.10

Determine whether $f(x) = x^3 - x$ is odd, even, or neither.
Substitute $-x$ for x, simplify, then look for a factor of -1:

$$f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x).$$

The result is $-f(x)$, so by the definition the function is **odd**. Its graph therefore has rotational symmetry of order two about the origin.

5.3.2 Self-Inverse Functions

A fourth symmetry belongs to functions that undo themselves, and it is usually met as an algebraic expression rather than as a graph. Recall that f^{-1} reverses f. If applying f twice already returns the input, then f reverses itself.

DEF A function is **self-inverse** if $f(f(x)) = x$ for every x in its domain. Then $f = f^{-1}$, and the graph of f is symmetric in the line $y = x$.

That symmetry has a simple reason. Reflecting any graph in the line $y = x$ produces the graph of its inverse, because the reflection swaps the two coordinates of every point. If f is its own inverse, the reflection has to return the same curve.

To test for self-inverse, substitute the whole expression for $f(x)$ in place of x, simplify, and check whether the result is x.

EXAMPLE 5.11

Show that $f(x) = \dfrac{1}{x}$ is self-inverse.

Apply the function twice. Substitute the whole expression $\frac{1}{x}$ in place of x:

$$f(f(x)) = \frac{1}{\frac{1}{x}} = x.$$

Applying f twice returns the input, so f reverses itself and $f = f^{-1}$. Its graph must therefore be symmetric in the line $y = x$, which the figure below confirms.

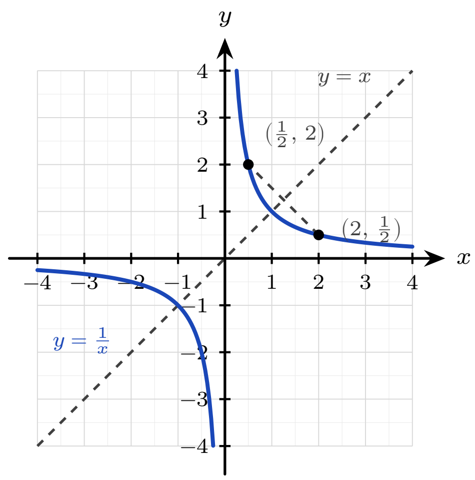


Figure 5.6 The reciprocal is symmetric in the line $y = x$. Each point on the curve has a partner with its two coordinates swapped, so reflecting in $y = x$ returns the same curve.

Self-inverse functions are not rare, and they are not all this simple. The reflection $f(x) = 6 - x$ is self-inverse, and so is $f(x) = \dfrac{2x+1}{x-2}$, which you can check by the same substitution.

5.3.3 Restricting the Domain to Get an Inverse

An inverse exists only when no two inputs share an output. Many familiar functions fail that test on their full domain, and a smaller domain repairs it.

CAUTION A function such as $f(x) = x^2$ is not one-to-one on all of $\mathbb{R}$, since $f(2) = f(-2)$, so it has no inverse there. Restricting the domain to $x \geq 0$ makes it one-to-one, and the inverse is $f^{-1}(x) = \sqrt{x}$.

Restricting a domain is a standard examination technique, and the restriction you choose decides which branch of the inverse you get. Two facts hold for any inverse. The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f.

EXAMPLE 5.12

Let $f(x) = (x-3)^2$ for $x \geq 3$. Find $f^{-1}(x)$, stating its domain.

First check that an inverse exists. On $x \geq 3$ the parabola only rises, so no two inputs share an output and f is one-to-one.

Now find the inverse. Write $y = f(x)$ and solve for x:

$$x - 3 = \pm\sqrt{y} \implies x = 3 \pm \sqrt{y}.$$

Two branches appear, so the restriction must decide between them. Since $x \geq 3$, the expression $x - 3$ cannot be negative, so only the $+$ branch is possible. Hence

$$f^{-1}(x) = 3 + \sqrt{x}, \qquad x \geq 0,$$

where the domain of f^{-1} is the range of f. Had the restriction been $x \leq 3$ instead, the same algebra would give $f^{-1}(x) = 3 - \sqrt{x}$. The algebra is the same until that final choice of sign, but the inverse is different.

5.3.4 The Modulus Graph $|f(x)|$

The remaining graphs in this section are also produced by reflections, so none of them needs a table of values. Each is built from a function f by one operation, and each has a single visual rule.

The modulus of a number is its size without its sign, so $|f(x)|$ is never negative. That gives the rule at once.

KEY To sketch $y = |f(x)|$: keep every part of $y = f(x)$ that lies on or above the x-axis, and reflect every part below the axis upward in the x-axis. The x-intercepts do not move, because $|0| = 0$. A root where f *crosses* the axis becomes a corner; a root where f merely touches the axis, as in $y = x^2$, stays smooth.

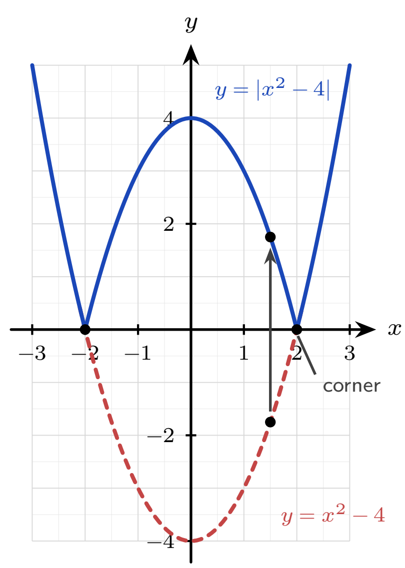


Figure 5.7 $y = |x^2 - 4|$ against $y = x^2 - 4$. The part of the parabola below the x-axis is reflected upward, shown by the arrow, which carries the point $(1.5, -1.75)$ to $(1.5, 1.75)$; everything already above the axis is untouched. The two roots stay at $x = \pm 2$ and become corners.

EXAMPLE 5.13

Sketch $y = |f(x)|$ for $f(x) = x^2 - 6x + 8$, stating the coordinates of the intercepts, the corners and the turning point.

Find the key points of f first. Factorising gives $f(x) = (x-2)(x-4)$, so the roots are $x = 2$ and $x = 4$, and the vertex lies halfway between them at $x = 3$, where $f(3) = 9 - 18 + 8 = -1$. So f has roots $(2, 0)$ and $(4, 0)$ and a minimum at $(3, -1)$.

Only the arc between the roots lies below the x-axis, so only that arc is reflected. Everything outside $2 \leq x \leq 4$ is already above the axis and is left alone.

Now take the key points one at a time. The roots $(2, 0)$ and $(4, 0)$ do not move, because $|0| = 0$, and f *crosses* the axis at each of them, so each becomes a **corner**. The minimum $(3, -1)$ is reflected to $(3, 1)$, and it is now a local **maximum**. The y-intercept is $f(0) = 8$, which is already positive and so is left where it is, at $(0, 8)$.

So the curve falls from the upper left to the corner $(2, 0)$, rises to the peak $(3, 1)$, falls to the corner $(4, 0)$, then rises again to the upper right, and no part of it lies below the x-axis.

The peak sits at height $|-1| = 1$: reflecting in the x-axis changes the sign of the y-coordinate and leaves the x-coordinate where it was.

This shape is also what makes modulus equations solvable, and §5.3.8 uses it.

5.3.5 The Graph of $f(|x|)$

The previous rule applied the modulus to the *output*. Applying it to the *input* instead gives a different graph and a different rule, and telling the two apart is what most questions on this pair are testing.

Since $|x|$ and $|-x|$ are the same number, f receives the same input at x and at $-x$, so the two outputs are equal. The graph is therefore its own mirror image in the y-axis.

KEY To sketch $y = f(|x|)$: keep the part of $y = f(x)$ with $x \geq 0$, and reflect it in the y-axis to make the part with $x < 0$. The original graph to the left of the y-axis is discarded. The result is always an even function.

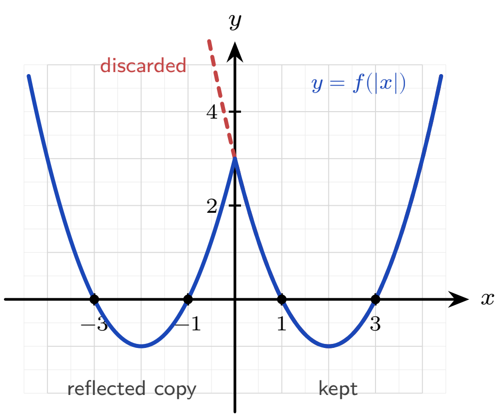


Figure 5.8 $y = f(|x|)$ for $f(x) = x^2 - 4x + 3$. The half with $x \geq 0$ is copied across the y-axis, so the roots at $x = 1$ and $x = 3$ gain partners at $x = -1$ and $x = -3$. The short dashed piece is the part of f that

is discarded.

EXAMPLE 5.14

Sketch $y = f(|x|)$ for $f(x) = x^2 - 6x + 8$, stating the coordinates of the intercepts, the corner and the turning points.

Keep the half of $y = f(x)$ with $x \geq 0$ and discard the rest. That half carries the y-intercept $(0,8)$, the roots $(2,0)$ and $(4,0)$, and the minimum $(3,-1)$.

Reflecting it in the y-axis copies each of those points across. The point $(0,8)$ is on the axis and stays where it is; $(2,0)$ gains a partner at $(-2,0)$; $(4,0)$ gains a partner at $(-4,0)$; and $(3,-1)$ gains a partner at $(-3,-1)$.

So the graph meets the x-axis at $(-4,0)$, $(-2,0)$, $(2,0)$ and $(4,0)$, and it has two minima, at $(-3,-1)$ and $(3,-1)$, both of value -1. The y-intercept is $(0,8)$.

The y-axis is where the two halves join, and they do not join smoothly. Just to the right of it the kept half is falling, with gradient -6 at $x = 0$; just to the left the mirror image is rising, with gradient $+6$. The two pieces therefore meet at a **corner** at $(0,8)$, which is a local maximum of value 8. The result is a W shape, symmetric in the y-axis.

The formula agrees: $f(|x|) = |x|^2 - 6|x| + 8 = x^2 - 6|x| + 8$, which is unchanged when x is replaced by $-x$, so it is even, and the discarded half of f leaves no trace in it.

5.3.6 The Graph of $[f(x)]^2$

Squaring the output also removes the sign, so $[f(x)]^2$ and $|f(x)|$ share a rule: neither is ever negative. They differ in size, because squaring changes the size and a modulus does not.

Every output is squared, so this rule has two parts.

KEY To sketch $y = [f(x)]^2$: no output is negative, so the whole curve lies on or above the x-axis. Each root of f becomes a point where the new curve **touches** the axis instead of crossing it. Where $|f(x)| > 1$ the squared value is larger, so the curve rises. Where $|f(x)| < 1$ the squared value is smaller, so the curve moves closer to the axis.

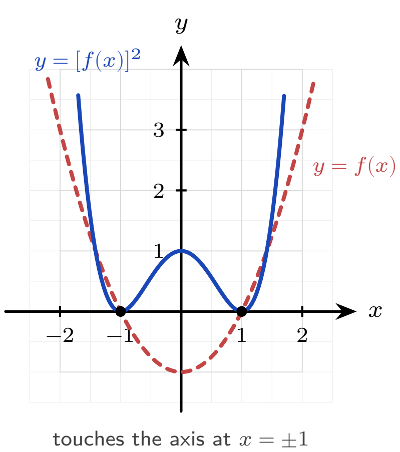


Figure 5.9 $y = [f(x)]^2$ for $f(x) = x^2 - 1$. The roots at $x = \pm 1$ become points where the squared curve

touches the x-axis. Between them $|f(x)| < 1$, so the squared curve lies closer to the axis than $|f(x)|$ does; outside them $|f(x)| > 1$, so it rises faster.

EXAMPLE 5.15

Sketch $y = [f(x)]^2$ for $f(x) = x^2 - 6x + 8$, stating where the curve meets the x-axis and the coordinates of its turning points.

The key points of f are again the roots $(2, 0)$ and $(4, 0)$ and the minimum $(3, -1)$. Squaring acts on the output only, so each of them keeps its x-coordinate.

Each root gives $0^2 = 0$, so the new curve still meets the x-axis at $(2, 0)$ and $(4, 0)$. It cannot cross there, because no output is negative, so at each root it **touches** the axis and turns back: both are minima, of value 0.

The minimum $(3, -1)$ gives $(-1)^2 = 1$, so the curve passes through $(3, 1)$. Between the two roots the squared value rises from 0 to 1 and falls back to 0, which makes $(3, 1)$ a local **maximum**.

Outside $2 \le x \le 4$ the size $|f(x)|$ passes 1 and keeps growing, so the squared curve climbs far faster than f does: at $x = 5$, $f(5) = 3$ while $[f(5)]^2 = 9$.

So the sketch drops steeply to touch at $(2, 0)$, arches up to $(3, 1)$, drops back to touch at $(4, 0)$, then climbs steeply away. Inside that arch $|f(x)| \le 1$, so squaring pulls the curve closer to the axis than $|f(x)|$ lies, and on $2 \le x \le 4$ the two curves coincide only where $|f(x)|$ is 0 or 1, at $x = 2$, $x = 3$ and $x = 4$. Beyond the arch they meet once more on each side, wherever $|f(x)| = 1$ again: solving $x^2 - 6x + 8 = 1$ gives $x = 3 \pm \sqrt{2}$, one point below $x = 2$ and one above $x = 4$. Squaring and taking a modulus agree exactly at height 0 and height 1, and nowhere else.

5.3.7 The Graph of $f(ax + b)$

The last of these graphs needs no new rule at all, only the old one applied carefully. Everything in $f(ax + b)$ happens inside the bracket, so every change is horizontal, and horizontal changes run opposite to what the algebra looks like.

The one step that matters is factorising before you read anything off. Written as $f(ax + b)$ the two numbers hide each other; written as $f(a\left(x + \frac{b}{a}\right))$ they separate, and each can be read straight off.

KEY To sketch $y = f(ax + b)$, first factorise the bracket:

$$f(ax + b) = f\left(a\left(x + \frac{b}{a}\right)\right).$$

Then apply a horizontal stretch of scale factor $\dfrac{1}{a}$, followed by a horizontal translation of $-\dfrac{b}{a}$. The stretch comes first, because it acts on x before the shift does.

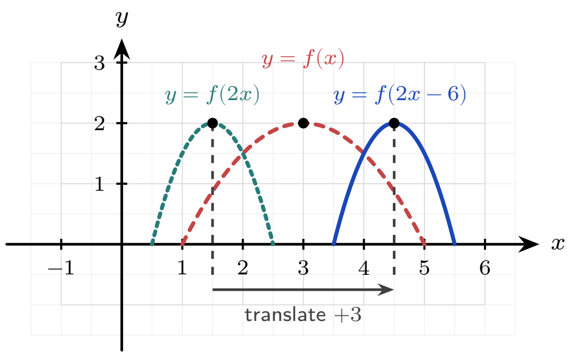


Figure 5.10 $y = f(2x-6)$, read as $f(2(x-3))$. The stretch acts first: $f(2x)$ halves the width and pulls the peak from $x = 3$ to $x = 1.5$. The translation of 3 to the right then carries it to $x = 4.5$. Reading the -6 as a shift of 6 is the standard error: the shift is $-b/a = 3$, and it is applied after the stretch, not before.

EXAMPLE 5.16

Sketch $y = f(2x-6)$ for $f(x) = x^2 - 6x + 8$, stating the images of the roots and of the vertex. Factorise inside the bracket before reading anything off:

$$f(2x-6) = f(2(x-3)).$$

Here $a = 2$ and $b = -6$, so the moves are a horizontal stretch of scale factor $\frac{1}{2}$ followed by a translation of $-\frac{b}{a} = 3$ to the right. Both are horizontal, so no y-coordinate changes.
Now track the key points of f: the roots $(2,0)$ and $(4,0)$ and the vertex $(3,-1)$. The stretch halves each x-coordinate and the translation then adds 3:

$$2 \to 1 \to 4, \qquad 4 \to 2 \to 5, \qquad 3 \to \tfrac{3}{2} \to \tfrac{9}{2}.$$

So the roots move to $(4,0)$ and $(5,0)$, and the vertex moves to $(\frac{9}{2},-1)$: still a minimum, still of depth -1, and still halfway between the roots. The curve is a parabola through those three points, half as wide as f, with y-intercept $f(-6) = 36+36+8 = 80$.
Expanding confirms every number: $f(2x-6) = ((2x-6)-2)((2x-6)-4) = (2x-8)(2x-10) = 4(x-4)(x-5)$, whose roots are 4 and 5 and whose value at $x = \frac{9}{2}$ is $4(\frac{1}{2})(-\frac{1}{2}) = -1$.

5.3.8 Modulus Equations and Inequalities

A modulus throws a sign away. Solving one means putting the sign back, and what sits on the other side decides how.

There are four forms, and they must be separated before any algebra starts. Two of them are safe, one needs a check at the end, and one has no algebraic route at all.

A modulus equal to a number. If $|A| = k$ and k is positive, then A has size k, so it is k or $-k$. Both cases must be written down.

KEY For $k > 0$, $|A| = k$ means $A = k$ or $A = -k$.
For $k = 0$ it means $A = 0$. For $k < 0$ there is no solution, because a modulus is never negative.

EXAMPLE 5.17

Solve $|2x - 1| = 5$.
The right side is positive, so there are two cases:

$$2x - 1 = 5 \implies x = 3, \qquad 2x - 1 = -5 \implies x = -2.$$

So $x = 3$ or $x = -2$.
The graph agrees. The V-shaped curve $y = |2x - 1|$ meets the line $y = 5$ once on each arm.

A modulus equal to a modulus. Two sizes are equal when the numbers are equal or opposite.

KEY $|A| = |B|$ means $A = B$ or $A = -B$.
Squaring both sides gives the same two answers, because $|A|^2 = A^2$.

EXAMPLE 5.18

Solve $|2x - 1| = |x + 4|$.

$$2x - 1 = x + 4 \implies x = 5, \qquad 2x - 1 = -(x + 4) \implies 3x = -3 \implies x = -1.$$

Both work: $|9| = |9|$ and $|-3| = |3|$. So $x = 5$ or $x = -1$.

A modulus equal to an expression in x. This form looks like the first, and it is the one that costs marks. A modulus is never negative, so the other side cannot be negative either. Solving both cases can therefore produce a value that solves neither.

CAUTION For $|A| = B$, solve $A = B$ and $A = -B$ as usual, then **substitute each answer back** and discard any that makes B negative. These are called **extraneous solutions**: they satisfy the squared equation but not the original.

EXAMPLE 5.19

Solve $|x - 1| = 2x - 4$.
Two cases, as before:

$$x - 1 = 2x - 4 \implies x = 3, \qquad x - 1 = -(2x - 4) \implies 3x = 5 \implies x = \tfrac{5}{3}.$$

Now test each against the right side. At $x = 3$: $2(3) - 4 = 2$, which is positive, and $|3 - 1| = 2$. That works.
At $x = \frac{5}{3}$: $2\left(\frac{5}{3}\right) - 4 = -\frac{2}{3}$, which is negative. A modulus cannot equal a negative number, so this value is rejected.
The only solution is $x = 3$.

Inequalities. Read these from the V rather than memorising them. The graph of $y = |A|$ is a V sitting on the x-axis, and a horizontal line $y = k$ cuts it twice. Below the line is the piece between the crossings; above it are the two pieces outside.

KEY For $k > 0$:

$$|A| < k \text{ means } -k < A < k, \qquad |A| > k \text{ means } A < -k \text{ or } A > k.$$

One interval for $<$, two for $>$. Use $\leq$ and $\geq$ to include the endpoints.

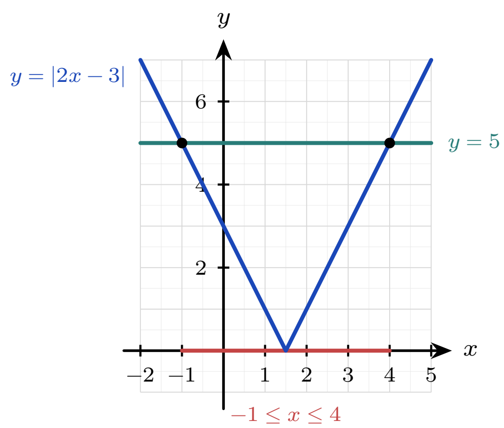


Figure 5.11 The V meets the line $y = 5$ at $x = -1$ and $x = 4$. Between those crossings the V is below the line, so $|2x - 3| \leq 5$ holds on the scarlet stretch. Outside them the V is above the line, so $|2x - 3| > 5$ holds there instead.

EXAMPLE 5.20

Solve (a) $|2x - 3| \leq 5$ and (b) $|3x + 1| > 7$.
(a) One interval, so write the double inequality and work on all three parts at once:

$$-5 \leq 2x - 3 \leq 5 \implies -2 \leq 2x \leq 8 \implies -1 \leq x \leq 4.$$

(b) Two intervals, so write two separate inequalities:

$$3x + 1 > 7 \implies x > 2, \qquad 3x + 1 < -7 \implies x < -\tfrac{8}{3}.$$

So $x < -\frac{8}{3}$ or $x > 2$.

The same two rules work when the expression inside is not linear.

EXAMPLE 5.21

Solve $|x^2 - 4| < 3$.

The right side is positive, so this is the one-interval case:

$$-3 < x^2 - 4 < 3 \implies 1 < x^2 < 7.$$

Now $x^2 > 1$ means $|x| > 1$, and $x^2 < 7$ means $|x| < \sqrt{7}$. Together, $1 < |x| < \sqrt{7}$, which is two intervals on the axis:

$$-\sqrt{7} < x < -1 \qquad \text{or} \qquad 1 < x < \sqrt{7}.$$

The graph of $y = |x^2 - 4|$ from §5.3.4 confirms it. That curve dips below $y = 3$ on exactly those two stretches.

When both sides contain x. An inequality such as $|f(x)| > g(x)$ has no rule of its own, because the number of intervals depends on the two curves. The method is the one from §5.4.5 in a different costume: find where the two sides are equal, and those points cut the axis into intervals on which the answer cannot change.

KEY To solve $|f(x)| > g(x)$ or $|f(x)| < g(x)$:

1. Sketch $y = |f(x)|$ and $y = g(x)$ on the same axes.
2. Find every point where they meet, by solving $|f(x)| = g(x)$ or from a calculator.
3. Read off the intervals where the curve you want is the higher one.

TIP Many modulus inequalities have no algebraic solution at all. The syllabus example $|3x \arccos x| > 1$ is one. Graph both sides on your GDC, use intersect to find the crossing points, and read the intervals from the picture. State the intersection coordinates you used, to three significant figures.

The four forms belong together.

Form	Method	Watch for
$\lvert A\rvert = k$	$A = k$ or $A = -k$	no solution if $k < 0$
$\lvert A\rvert = \lvert B\rvert$	$A = B$ or $A = -B$	nothing; both roots are valid
$\lvert A\rvert = B$	two cases, then substitute back	reject any root making B negative
$\lvert A\rvert \lessgtr k$	$-k < A < k$, or two intervals	one interval for $<$, two for $>$

YOUR TURN 5.3 Symmetry and Modulus HL

15. Determine whether each function is odd, even, or neither.

(a) $f(x) = x^4 - 3x^2$
(b) $f(x) = x^3 + x$
(c) $f(x) = x^2 + x$
(d) $f(x) = x^5$
(e) $f(x) = 5$
(f) $f(x) = \dfrac{1}{x}$

16. Self-inverse functions.

(a) Show that $f(x) = \dfrac{1}{x}$ is self-inverse.
(b) Show that $f(x) = 6 - x$ is self-inverse.
(c) Show that $f(x) = \dfrac{2x+1}{x-2}$ is self-inverse.
(d) Explain why $f(x) = 2x$ is not self-inverse.

17. Restricting a domain so that an inverse exists.

(a) State a domain on which $f(x) = x^2$ has an inverse, and give that inverse.
(b) Do the same for $f(x) = (x-1)^2$.
(c) State the domain of the inverse you found in part (b).
(d) Explain why $f(x) = x^3$ needs none.

18. Solve each equation.

(a) $|x - 3| = 2$
(b) $|2x + 1| = 7$
(c) $|3 - x| = 1$
(d) $|x| = 5$

19. The graph of $y = f(x)$ crosses the x-axis at $x = 2$ and has a minimum point at $(0, -4)$.

(a) State where $y = |f(x)|$ meets the x-axis.
(b) State where $y = [f(x)]^2$ meets the x-axis.
(c) State the point that $(0, -4)$ becomes on $y = |f(x)|$.
(d) Explain why $y = f(|x|)$ is always an even function.

20. Solve each equation. One part has no solution.

(a) $|x - 2| = |2x + 1|$
(b) $|3x| = |x + 4|$
(c) $|x - 1| = 3x - 5$
(d) $|2x + 1| = x - 1$

21. Solve each inequality.

(a) $|x-4|<3$
(b) $|2x+5|\geq 9$
(c) $|3-x|\leq 1$
(d) $|x^2-9|<7$

22. GDC Both sides now contain x.

(a) Sketch $y=|x-2|$ and $y=x$ on the same axes, and hence solve $|x-2|<x$.
(b) Solve $|2x|>x+3$ algebraically.
(c) Use a GDC to solve $|x^3-3x|>1$ for $0\leq x\leq 3$, giving answers to three significant figures.

5.4 Reciprocal and Rational Functions

A line, a parabola and a cubic have a value at every input, and each one rises or falls without limit. Divide by a variable and both of those stop being true. There are now inputs where the function has no value, and there are lines the curve approaches without ever reaching. Finding those lines from the formula is what makes a rational curve quick to sketch. The same work also tells you where the function is positive and where it is negative.

The simplest function of this kind is the **reciprocal** $y=\dfrac{1}{x}$, and it shows all of the new behaviour at once.

It is undefined at $x=0$. As x approaches 0, the output grows without bound, towards $+\infty$ on one side and $-\infty$ on the other. For large $|x|$ the opposite happens: the output becomes very small, and the curve approaches zero. So the reciprocal has two asymptotes, the vertical line $x=0$ and the horizontal line $y=0$.

It is defined for every input except $x=0$, so its **domain** is all real numbers with $x\neq 0$. It produces every output except 0, so its **range** is all real numbers with $y\neq 0$. It is also self-inverse, so its graph is symmetric in the line $y=x$. See §5.3.

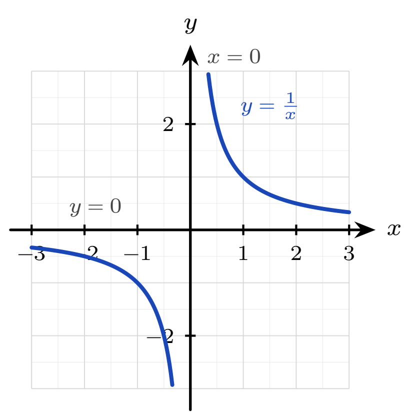


Figure 5.12 The reciprocal $y=\frac{1}{x}$. Its asymptotes are the two axes: the curve approaches $x=0$ and

$y = 0$ without meeting either. The domain is every real number except 0, and so is the range.

DEF An **asymptote** is a line that the graph of a function approaches arbitrarily closely (as close as you like). A **vertical asymptote** occurs at a value of x where the function is undefined and grows without bound. A **horizontal asymptote** is a value the function approaches as $x \to \pm\infty$.

5.4.1 Linear over Linear

Every function of the form $\dfrac{ax+b}{cx+d}$ is a reciprocal curve that has been shifted and stretched, so all of them have the same two-branch shape. Dividing the numerator by the denominator shows that directly: it rewrites the function as a constant plus a multiple of $\dfrac{1}{x-p}$, and the two asymptotes then cross at the centre of that reciprocal curve. Two conditions are needed for that to be true. If $c = 0$ the function is linear and has no asymptote at all. If $ad-bc = 0$ the numerator is a multiple of the denominator and the function is constant.

KEY For $f(x) = \dfrac{ax+b}{cx+d}$, with $c \neq 0$ and $ad - bc \neq 0$:

$$\text{vertical asymptote at } x = -\frac{d}{c}, \qquad \text{horizontal asymptote at } y = \frac{a}{c}.$$

The vertical one is where the denominator is zero. The horizontal one is the ratio of the leading coefficients.

EXAMPLE 5.22

Sketch $f(x) = \dfrac{2x+1}{x-1}$, showing all asymptotes and intercepts.

Find the vertical asymptote first. The denominator is zero at $x = 1$, so the line $x = 1$ is the vertical asymptote.

Next the horizontal asymptote. The ratio of the leading coefficients gives $y = \dfrac{2}{1} = 2$.

Rewriting the function confirms the shape. Divide the numerator by the denominator:

$$\frac{2x+1}{x-1} = 2 + \frac{3}{x-1},$$

a reciprocal curve centred on the crossing of the two asymptotes, $(1, 2)$.

Now find the intercepts. Substitute $x = 0$ for the y-intercept: $f(0) = \dfrac{1}{-1} = -1$. Set the numerator to zero for the x-intercept: $2x + 1 = 0$, so $x = -\frac{1}{2}$. Both are marked on the figure below.

Two asymptotes and two intercepts fix the whole sketch.

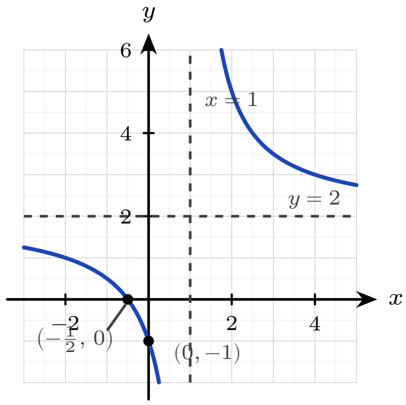


Figure 5.13 The curve $y = \frac{2x+1}{x-1}$, with its two asymptotes dashed and its two intercepts marked. These four labelled features are what a sketch must show.

TIP A sketch question is marked on the asymptotes and the intercepts. Write the equation of each asymptote beside its dashed line, and write the coordinates beside each intercept. An unlabelled curve of the right shape earns few marks.

The divided form answers a second question, and it answers it for any quotient of two linear expressions. A sketch shows which heights the curve reaches, but a question asking for the range wants a reason rather than a picture, and writing the function as a constant plus a fraction supplies one. The function below is the one whose inverse was found in §5.1.4.

EXAMPLE 5.23

State the range of $f(x) = \dfrac{3x-1}{x-1}$, $x \neq 1$, justifying your answer.

Divide the numerator by the denominator, the same first move as in the sketch above:

$$f(x) = 3 + \frac{2}{x-1}.$$

The second term is a fraction with numerator 2, so it is never 0, whatever x is. Adding something that is never 0 to 3 never gives 3, so $f(x) \neq 3$ for every x in the domain.

Every other value is produced. Set $f(x) = k$ and solve back for x:

$$\frac{2}{x-1} = k-3 \implies x-1 = \frac{2}{k-3} \implies x = 1 + \frac{2}{k-3},$$

which is a real input, and never equal to 1, whenever $k \neq 3$. So the range is $y \in \mathbb{R}$, $y \neq 3$.

The one excluded value is the horizontal asymptote, and that is what a horizontal asymptote tells you about the range: the curve closes in on $y = 3$ from both sides without ever landing on it.

5.4.2 Reading the Asymptotes from the Degrees

The two rules above are one case of a general pattern. Which asymptote a rational function has as $x \to \pm\infty$ depends only on how the degree of the numerator compares with the degree of the denominator.

Degrees	Asymptote as $x \to \pm\infty$	Example
numerator $<$ denominator	horizontal, $y = 0$	$\frac{x+1}{x^2-4}$
numerator $=$ denominator	horizontal, at the ratio of leading coefficients	$\frac{2x+1}{x-1}$
numerator $=$ denominator $+1$	oblique, a slanted line	$\frac{x^2-2x+3}{x-1}$

In every case the vertical asymptotes sit at the zeros of the denominator, once any common factor has been cancelled.

The syllabus names one more shape: a linear numerator over a quadratic denominator, $\frac{ax+b}{cx^2+dx+e}$. The denominator has the higher degree, so the horizontal asymptote is $y = 0$, and each real root of the denominator gives its own vertical asymptote. This form can therefore have two.

With two asymptotes and an intercept, the branches can still go several ways. The sign of f on each interval shows which way its branch goes. A rational function changes sign only where it is zero or where it is undefined, so those values cut the x-axis into intervals, and the sign is the same right across each one. Testing one point per interval is therefore enough.

The sign beside a vertical asymptote gives the direction of that branch. Where f is positive the curve climbs to $+\infty$ as it nears the asymptote; where f is negative it falls to $-\infty$. §5.4.5 sets the same work out as a table.

EXAMPLE 5.24

Sketch the key features of $f(x) = \dfrac{x+1}{x^2-4}$.

Factorise the denominator to find the vertical asymptotes: $x^2 - 4 = (x-2)(x+2)$, so there are *two*, at $x = 2$ and $x = -2$.

The numerator has lower degree than the denominator, so the horizontal asymptote is $y = 0$.

For the intercepts, set the numerator to zero to get $x = -1$, and substitute $x = 0$ to get $f(0) = -\frac{1}{4}$.

Now the signs. The three values -2, -1 and 2 cut the axis into four intervals. Take the sign of each factor of $\dfrac{x+1}{(x-2)(x+2)}$, one interval at a time:

$$x < -2 : \ \tfrac{-}{(-)(-)} < 0, \qquad -2 < x < -1 : \ \tfrac{-}{(-)(+)} > 0,$$

$$-1 < x < 2 : \ \tfrac{+}{(-)(+)} < 0, \qquad x > 2 : \ \tfrac{+}{(+)(+)} > 0.$$

Read the branches from those signs. Where f is positive next to a vertical asymptote the curve rises to $+\infty$; where it is negative it falls to $-\infty$. The middle branch is negative to the right of

$x = -1$ and positive to the left of it, and it passes through both intercepts.

A zero of the denominator does not always produce an asymptote. That is what the phrase "once any common factor has been cancelled" is guarding against. If the same factor divides the numerator too, it cancels, and all that is left at that value of x is one missing point: a **hole** (a removable discontinuity). The function still has no value there, so the domain still excludes it, but the curve on either side lines up.

EXAMPLE 5.25

Describe the behaviour of $f(x) = \dfrac{x^2-9}{x-3}$ at $x = 3$, and state its domain. Compare it with $g(x) = \dfrac{x^2-9}{x+4}$ at $x = -4$.

Factorise the numerator first, because that is what decides the answer:

$$x^2 - 9 = (x+3)(x-3).$$

For f the factor $x-3$ appears above and below the line, so it cancels:

$$f(x) = \frac{(x+3)(x-3)}{x-3} = x+3, \qquad x \neq 3.$$

The graph of f is therefore the straight line $y = x+3$ with a single point removed. That point is at $x = 3$, where the line has height $3+3 = 6$, so the hole is at $(3, 6)$. There is no vertical asymptote at all, and the domain is $x \in \mathbb{R}$, $x \neq 3$.

For g the denominator is zero at $x = -4$. Substitute that into the numerator: $(-4+3)(-4-3) = 7$, which is not 0, so nothing cancels. As x approaches -4 the numerator approaches 7 while the denominator approaches 0, so $|g(x)|$ grows without bound and $x = -4$ is a vertical asymptote.

That substitution decides which of the two you have: put the denominator's zero into the numerator, and if the result is not 0 then no common factor exists and the asymptote stands.

5.4.3 Oblique Asymptotes HL

When the numerator has a higher degree than the denominator, there is no horizontal asymptote. The curve approaches a slanted line instead, and finding that line is a division problem.

The useful case is when the numerator's degree is exactly one more than the denominator's. **Polynomial division** then splits the function into a linear part plus a remainder, and that remainder approaches zero for large $|x|$. The linear part is the **oblique** (slant) asymptote. Polynomial division is treated in full in the Polynomials chapter; all that is needed here is the quotient-plus-remainder form.

EXAMPLE 5.26

Find the asymptotes of $f(x) = x + \dfrac{1}{x}$.

The function is undefined at $x = 0$, so the line $x = 0$ is a vertical asymptote.

For large $|x|$ the term $\dfrac{1}{x}$ approaches zero, so $f(x)$ approaches x. The line $y = x$ is therefore the oblique asymptote.

The graph has two separate branches. Each one lies between the y-axis and the line $y = x$.

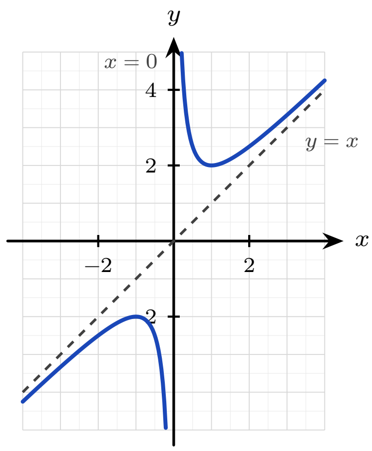


Figure 5.14 The two branches of $y = x + \frac{1}{x}$, each approaching the vertical asymptote $x = 0$ on one side and the oblique asymptote $y = x$ on the other.

In that example the split into a line plus a remainder was already written out. In general you produce it yourself by **polynomial division**.

EXAMPLE 5.27

Find all asymptotes of $f(x) = \dfrac{x^2 - 2x + 3}{x - 1}$.

The denominator is zero at $x = 1$, so there is a vertical asymptote there. The numerator's degree exceeds the denominator's by one, so there is an oblique asymptote, found by dividing:

$$\frac{x^2 - 2x + 3}{x - 1} = x - 1 + \frac{2}{x - 1},$$

since $(x - 1)(x - 1) + 2 = x^2 - 2x + 3$. As $|x|$ grows the remainder term $\frac{2}{x-1}$ approaches zero, so the curve approaches the line

$$y = x - 1,$$

which is the oblique asymptote.

TIP Show the division, not only the answer. The quotient *is* the asymptote, and the division itself carries most of the marks.

5.4.4 The Reciprocal of Any Function HL

Every reciprocal so far has been the reciprocal of something simple: of x, of a linear expression, of one polynomial divided by another. The same reading works for the reciprocal of any function at all. Given a function f, its **reciprocal** is

$$y = \frac{1}{f(x)},$$

defined at every x where f is defined and $f(x) \neq 0$.

You never need to calculate points to sketch it. Every feature of the reciprocal can be read from the graph of f, one feature at a time.

Where f does this	The reciprocal does this	Reason
$f(x) = 0$	vertical asymptote	you cannot divide by zero
$\lvert f(x) \rvert$ **is large**	the curve approaches the x-axis	a large number has a small reciprocal
$f(x) = 1$ **or** -1	the two graphs cross	1 and -1 are their own reciprocals
$f(x) > 0$	$\frac{1}{f(x)} > 0$	the sign is unchanged
a local maximum	a local minimum	and a minimum becomes a maximum

The last row needs a second look. Where f is large the reciprocal is small, so a high point of f becomes a low point of $\frac{1}{f}$. This holds wherever f is not zero, whatever the sign of f.

CAUTION $\dfrac{1}{f(x)}$ is *not* $f^{-1}(x)$. The reciprocal divides 1 by the output; the inverse undoes the function. For $f(x) = x + 3$ the reciprocal is $\dfrac{1}{x+3}$ and the inverse is $x - 3$, which are not the same. The notation f^{-1} never means "one over f".

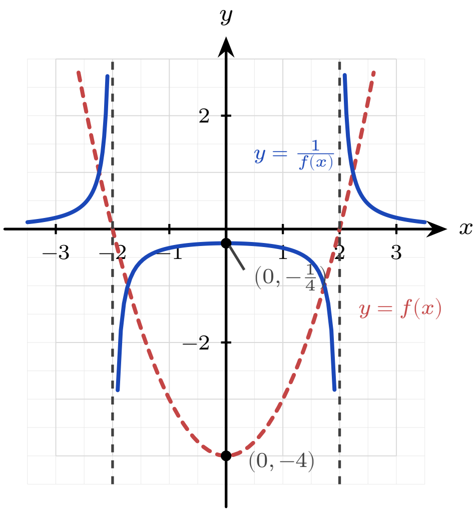


Figure 5.15 $y = \dfrac{1}{x^2 - 4}$ against $y = x^2 - 4$. Zeros of the original become vertical asymptotes at $x = \pm 2$, the minimum $(0, -4)$ becomes the maximum $(0, -\frac{1}{4})$ of the middle branch, and large values of f become values close to zero.

EXAMPLE 5.28

The graph of $y = f(x)$ touches the x-axis at $x = 5$ and has no other root, it has a local minimum at $(2, 4)$, and $f(x) \to +\infty$ as $x \to \pm\infty$. Describe the corresponding features of $y = \dfrac{1}{f(x)}$.

Take the features one at a time.

The root at $x = 5$ makes the denominator zero, so $y = \dfrac{1}{f(x)}$ has a **vertical asymptote at** $x = 5$. Since the curve only touches the axis there, f is positive on both sides of $x = 5$, so the reciprocal runs up to $+\infty$ on both sides of that asymptote.

The minimum at $(2, 4)$ has $f(2) = 4$, so the reciprocal passes through $(2, \frac{1}{4})$. A minimum of f becomes a **maximum** of the reciprocal, so this is a local maximum at $(2, \frac{1}{4})$.

Since $f(x) \to \infty$ at both ends, $\dfrac{1}{f(x)} \to 0$, so $y = 0$ is a **horizontal asymptote**.

No algebra was needed. Every answer came from one feature of f.

TIP Sketching a reciprocal in an examination: mark the roots of f as vertical asymptotes first, then the points where $f = 1$ and $f = -1$, then the turning points with their y-coordinates inverted. Draw the curve through those, one branch at a time.

5.4.5 Rational Inequalities HL

An inequality asks the same question as a sketch, in a shorter form: on which intervals is the expression positive, and on which is it negative?

One move gets it into a shape you can answer. Collect every term on one side, so that the inequality compares a single fraction with zero.

The rest is a **sign table**. A rational expression changes sign only where it is zero or where it is undefined, so those **critical values** cut the x-axis into intervals on which the sign never changes. Give the table one row for each factor and one column for each interval. Fill in the sign of every factor, then multiply down each column to get the sign of the whole expression. The last row is the answer, read off interval by interval.

KEY HL To solve a rational inequality:

1. Collect every term on one side, combine into a single fraction and factorise.
2. Mark the **critical values**, where the numerator or the denominator is zero.
3. Build a sign table on the intervals between them, and read off the intervals with the sign you need.
4. Include a zero of the numerator when the inequality is $\geq$ or $\leq$; never include a zero of the denominator.

CAUTION Do not multiply both sides by an expression whose sign you do not know. Multiplying an inequality by a negative number reverses it, so $x-1$ would split the question into two cases. Move everything to one side instead.

The same table works when there is no fraction at all. A quadratic inequality is the simplest case, and it is the one to try first.

EXAMPLE 5.29

Solve $x^2 \leq 3x + 10$.

Collect every term on one side, then factorise:

$$x^2 - 3x - 10 \leq 0 \implies (x+2)(x-5) \leq 0.$$

The critical values are $x = -2$ and $x = 5$, and they cut the axis into three intervals.

Interval	$x < -2$	$-2 < x < 5$	$x > 5$
$x+2$	$-$	$+$	$+$
$x-5$	$-$	$-$	$+$
$(x+2)(x-5)$	$+$	$-$	$+$

The middle interval gives the sign required. Both endpoints are zeros of a polynomial, not of a denominator, so the inequality allows both. The solution is $-2 \leq x \leq 5$.

EXAMPLE 5.30

Solve $\dfrac{x+3}{x-2} \geq 2$.

Collect everything on one side and combine into a single fraction:

$$\frac{x+3}{x-2} - 2 \geq 0 \implies \frac{(x+3)-2(x-2)}{x-2} \geq 0 \implies \frac{7-x}{x-2} \geq 0.$$

Now the critical values. The numerator is zero when $7-x=0$, that is $x=7$. The denominator is zero when $x=2$. These two values cut the axis into three intervals.

Interval	$x<2$	$2<x<7$	$x>7$
$7-x$	$+$	$+$	$-$
$x-2$	$-$	$+$	$+$
$\frac{7-x}{x-2}$	$-$	$+$	$-$

Only the middle interval gives the sign required. Now decide the two endpoints. Exclude $x = 2$, because the fraction is undefined there. Include $x = 7$, because the inequality allows equality and the fraction is zero there. So

$$2 < x \leq 7.$$

A graph agrees: $y = \dfrac{x+3}{x-2}$ lies above the line $y = 2$ only between the asymptote and the crossing point. The boundary of the solution is a vertical asymptote at one end and a crossing at the other, and a sign table finds both.

TIP A GDC solves any equation or inequality of this kind once you move everything to one side. Rewrite $f(x)=g(x)$ as $f(x)-g(x)=0$, graph the single function $y=f(x)-g(x)$, and read its zeros; for an inequality, read the intervals where that curve is above or below the axis. Reading zeros is quicker and safer than locating two intersection points, and it is the method to use when an equation has no exact solution, such as $e^x = 3-x$.

YOUR TURN 5.4 Reciprocal and Rational Functions

23. State the vertical and horizontal asymptotes of each function.

(a) $f(x) = \dfrac{1}{x-3}$ (b) $f(x) = \dfrac{3}{x+2}$

(c) $f(x) = \dfrac{2x}{x+1}$ (d) $f(x) = \dfrac{x+1}{x-2}$

(e) $f(x) = \dfrac{1}{x-5}$ (f) $f(x) = \dfrac{3x-6}{x+1}$

24. For each function, state the domain, the range, both asymptotes and both intercepts. Where an intercept does not exist, say why.

(a) $f(x) = \dfrac{1}{x-5}$

(b) $f(x) = \dfrac{1}{x} + 4$

(c) $f(x) = \dfrac{x-1}{x+2}$

(d) $f(x) = \dfrac{2x+3}{x-1}$

25. Consider $f(x) = \dfrac{3x-6}{x+1}$.

(a) Write $f(x)$ in the form $A + \dfrac{B}{x+1}$.

(b) Hence state the point on which the two asymptotes cross.

(c) Sketch the curve, showing both asymptotes and both intercepts.

26. For each function, state every vertical asymptote, and state the kind and the equation of the asymptote as $x \to \pm\infty$.

(a) $f(x) = \dfrac{1}{(x-1)(x+3)}$

(b) $f(x) = \dfrac{x}{x^2-9}$

(c) $f(x) = \dfrac{x^2+1}{x^2-4}$

(d) $f(x) = \dfrac{x^2+1}{x-2}$

(e) $f(x) = \dfrac{x^2-9}{x+2}$

27. Write each function as a polynomial plus a remainder, and state the oblique asymptote.

(a) $f(x) = \dfrac{x^2+3}{x}$

(b) $f(x) = \dfrac{x^2}{x-1}$

(c) $f(x) = \dfrac{x^2+1}{x-2}$

(d) $f(x) = \dfrac{x^2+4x+1}{x+1}$

28. Consider $f(x) = x + \dfrac{2}{x}$.

(a) State the vertical asymptote.

(b) State the oblique asymptote.

(c) Explain why the curve never crosses the oblique asymptote.

(d) Sketch the two branches.

29. For each function f, sketch $y = \dfrac{1}{f(x)}$, showing every asymptote and the coordinates of any turning point.

(a) $f(x) = x - 4$

(b) $f(x) = x^2 - 1$

(c) $f(x) = x^2 + 1$

(d) $f(x) = (x-1)(x-3)$

30. The graph of $y = f(x)$ has a single root at $x = 3$ and a local minimum at $(1, 2)$, and $f(x) \to \infty$ as $x \to \pm\infty$.

(a) State the equation of the vertical asymptote of $y = \dfrac{1}{f(x)}$.

(b) Find the coordinates of the turning point of $y = \dfrac{1}{f(x)}$, and state its type.

(c) Describe the behaviour of $y = \dfrac{1}{f(x)}$ as $x \to \pm\infty$.

(d) State the condition on $f(x)$ at a point where $y = f(x)$ and $y = \dfrac{1}{f(x)}$ cross.

31. Let $f(x) = x - 2$.

(a) On one set of axes, sketch $y = f(x)$ and $y = \dfrac{1}{f(x)}$.

(b) Find the coordinates of the two points where the graphs cross. *(Hint: they cross where $f(x) = \pm 1$)*

(c) State the interval on which $\dfrac{1}{f(x)} > f(x)$ for $x > 2$.

32. Solve each inequality. HL

(a) $x^2 < 4$

(b) $x^2 \geq 9$

(c) $x^2 \leq x + 6$

(d) $x^2 > 2x + 3$

(e) $\dfrac{x+2}{x-1} \geq 3$

(f) $\dfrac{2x-1}{x+2} \leq 1$

5.5 Exponential and Logarithmic Functions

Pour a cup of coffee at $90\,^\circ$C and set it down in a room at $20\,^\circ$C. It cools quickly at first, then more slowly, then so slowly that you can barely measure the change. Look at the numbers rather than the coffee. In each fixed interval of time the difference between the cup and the room falls by the same *fraction*, not by the same amount. Isaac Newton measured that and found the rule. No polynomial behaves this way.

An **exponential function** $y = a^x$ does, because it places the variable in the exponent rather than in the base. That one change has a large consequence: an exponential grows at a rate proportional to its own size, so a quantity twice as large grows twice as fast. The same function therefore describes cooling coffee, growing populations and decaying atoms. Ch. 1 sets up the algebra of exponents and logarithms. This section is about their graphs.

For $a > 1$ the graph rises, and rises ever more steeply: this is **growth**. For $0 < a < 1$ it falls towards zero: this is **decay**. Either way the graph passes through $(0, 1)$, stays positive everywhere, and approaches the horizontal asymptote $y = 0$ on one side, so the domain is all real numbers and the range is $y > 0$.

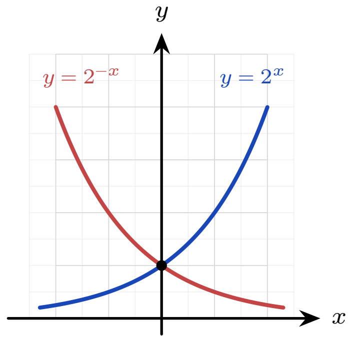


Figure 5.16 Growth and decay are reflections of each other in the y-axis. Both pass through $(0, 1)$ and approach $y = 0$, but on opposite sides.

One base is used more than any other: $e \approx 2.718$, the base whose curve has steepness equal to its height at every point. That property makes the calculus of e^x the simplest of any exponential. It is also why every other exponential is usually rewritten in terms of e. See Ch. 13.

KEY · IB Any exponential can be written with base e, and exponential and logarithm undo each other:

$$a^x = e^{x \ln a}, \qquad \log_a(a^x) = x, \qquad a^{\log_a x} = x \quad (a, x > 0,\ a \neq 1).$$

EXAMPLE 5.31

For $f(x) = 5(0.4)^x$, state whether the function grows or decays, and give its domain, range and asymptote.

Look at the base first. Here $a = 0.4$, and $0 < 0.4 < 1$, so the function **decays**.

The factor 5 stretches the graph vertically and changes nothing else, and the outputs of $(0.4)^x$ are always positive. So the domain is all real numbers, the range is $y > 0$, and the horizontal asymptote is $y = 0$.

5.5.1 Transforming an Exponential Graph

An exponential is rarely met on its own. In a model it usually appears as

$$y = k\,a^x + c,$$

and each constant has a separate effect. The factor k stretches the graph vertically. The constant c moves it up or down.

The vertical shift is the one that changes the asymptote. The graph of a^x approaches $y = 0$, so adding c makes it approach $y = c$ instead. **The horizontal asymptote moves with the graph.**

EXAMPLE 5.32

Sketch $y = 3(2)^x - 5$, stating the asymptote and both intercepts.

Start with the asymptote. The -5 shifts the whole graph down by 5, so the horizontal asymptote is $y = -5$, not $y = 0$.

For the y-intercept, substitute $x = 0$:

$$y = 3(2)^0 - 5 = 3 - 5 = -2.$$

For the x-intercept, set $y = 0$ and solve:

$$3(2)^x = 5 \implies 2^x = \tfrac{5}{3} \implies x = \log_2 \tfrac{5}{3} \approx 0.737.$$

The curve therefore rises from just above $y = -5$ on the left. It crosses the axes at $(0, -2)$ and at $(0.737, 0)$, then rises steeply.

CAUTION A shifted exponential does *not* have the asymptote $y = 0$. In $y = k\,a^x + c$ the asymptote is $y = c$. Reading it as $y = 0$ without checking is a common error.

5.5.2 Exponential Models

Most real quantities are modelled with base e and time as the variable:

$$A(t) = A_0\,e^{kt},$$

where A_0 is the amount at $t = 0$. The sign of k decides the behaviour, $k > 0$ for growth and $k < 0$ for decay, and its size decides the speed. Ch. 1 solves these models algebraically; what is added here is reading them off a graph.

Two data points fix the whole model. The first gives A_0 directly. For the second, substitute its t and A into $A = A_0 e^{kt}$, divide by A_0 and take logarithms:

$$k = \frac{1}{t}\ln\left(\frac{A}{A_0}\right).$$

CAUTION A model fitted to two points assumes the fractional rate of change never varies. Where the real rate is itself drifting, the prediction drifts with it, so report such a figure as what would happen if nothing changed rather than as a forecast.

EXAMPLE 5.33

Japan's population was about 128.1 million in 2010 and about 126.1 million in 2020. Model the population as $P(t) = P_0 e^{kt}$, with t in years after 2010, and use it to estimate the population in 2030.

Take $P_0 = 128.1$, the value at $t = 0$, then use the 2020 figure to find k. Substitute $t = 10$ and $P = 126.1$:

$$126.1 = 128.1\, e^{10k} \implies k = \frac{1}{10}\ln\left(\frac{126.1}{128.1}\right) \approx -0.00157.$$

The sign is negative, which confirms decay: the population falls by about 0.16% each year. For 2030, substitute $t = 20$:

$$P(20) = 128.1\, e^{-0.00157\times 20} \approx 124.1 \text{ million.}$$

Japan's rate of decline is itself increasing, so official projections for 2030 are lower than this figure.

5.5.3 The Logarithm, Its Inverse

Since $y = a^x$ is one-to-one, it has an inverse, and that inverse is the **logarithm** $y = \log_a x$. It answers the reversed question: not "what is a to the x?" but "to what power must I raise a to get x?" Everything about its graph follows from being the mirror image of the exponential in the line $y = x$.

Reflecting swaps the axes' roles. The exponential's horizontal asymptote $y = 0$ becomes the logarithm's *vertical* asymptote $x = 0$; its domain of all reals becomes the logarithm's range, and its range of positive numbers becomes the logarithm's domain. So $\log_a x$ exists only for $x > 0$, passes through $(1, 0)$, and rises without bound, ever more slowly. Its domain is therefore the positive real numbers, $x > 0$, and its range is all real numbers.

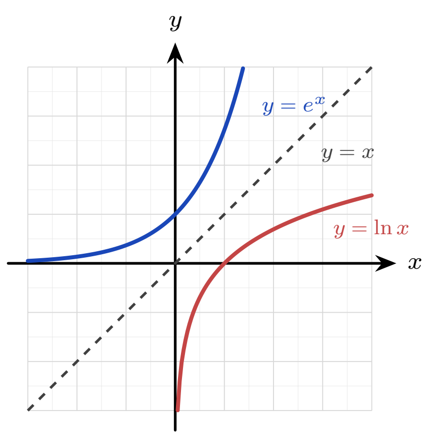


Figure 5.17 The logarithm is the exponential reflected in the line $y = x$. Reflecting exchanges the two axes, which is why one asymptote is horizontal and the other vertical.

CAUTION The logarithm is only defined for positive inputs. Writing $\ln(x - 3)$ silently assumes $x > 3$; forgetting this domain restriction is a common error.

FINDING the domain

Because a logarithm accepts only positive inputs, every logarithmic function carries a hidden condition. To find the domain, set whatever sits inside the logarithm greater than zero, and solve.

EXAMPLE 5.34

State the domain of $f(x) = \ln(x-3)$ and of $g(x) = \log(5-2x)$.
For f, the input to the logarithm is $x-3$. Require it to be positive:

$$x - 3 > 0 \implies x > 3.$$

For g, the input is $5-2x$. Require that to be positive:

$$5 - 2x > 0 \implies 5 > 2x \implies x < \tfrac{5}{2}.$$

Note the direction. Dividing by -2 would have reversed the inequality, so it is safer to move the term across as above.

TRANSFORMING a logarithmic graph

The same shifts apply as for any function, but they act on the *vertical* asymptote rather than a horizontal one.

In $y = \ln(x-h)$ the change is inside the bracket, so it moves the graph h units right. The asymptote travels with it, from $x = 0$ to $x = h$.

The x-intercept moves with it too, and one fact locates it. A logarithm is zero exactly when its input is 1, so $\ln(x-h) = 0$ when $x - h = 1$, that is at $x = h + 1$.

EXAMPLE 5.35

Sketch $y = \ln(x-5)$, stating its asymptote, its domain and its x-intercept.
The input must be positive, so $x - 5 > 0$ and the domain is $x > 5$.
That same condition gives the asymptote. As x approaches 5 from above, $x - 5$ approaches 0 and $\ln(x-5)$ decreases without bound, so $x = 5$ is a vertical asymptote.
For the x-intercept, set $y = 0$. A logarithm is zero when its input is 1:

$$x - 5 = 1 \implies x = 6.$$

So the curve rises from the asymptote at $x = 5$, crosses the axis at $(6, 0)$, and continues upward ever more slowly.

Using the inverse relationship

Because the exponential and the logarithm undo each other, you can use a logarithm to solve for a variable in an exponent, and an exponential to solve for a variable inside a logarithm. That is the step that makes the four-part method of §5.1 work here: at stage 2, where you solve for x, apply a logarithm if x sits in an exponent and an exponential if x sits inside a logarithm.

EXAMPLE 5.36

Find the inverse of $f(x) = 2e^{3x} - 5$, and state its domain.

Write $y = f(x)$, then undo each operation one at a time until x stands alone:

$$y = 2e^{3x} - 5 \implies \frac{y+5}{2} = e^{3x}.$$

The variable is now in an exponent, so apply the natural logarithm to both sides:

$$3x = \ln\left(\frac{y+5}{2}\right) \implies x = \tfrac{1}{3}\ln\left(\frac{y+5}{2}\right).$$

Exchange the letters to write the inverse as a function of x:

$$f^{-1}(x) = \tfrac{1}{3}\ln\left(\frac{x+5}{2}\right).$$

Now the domain. The domain of f^{-1} is the range of f. Since $e^{3x} > 0$ for every x, we have $2e^{3x} > 0$ and so $f(x) > -5$. The domain of f^{-1} is therefore $x > -5$, which agrees with requiring $\frac{x+5}{2} > 0$ inside the logarithm.

TIP State the domain whenever a question asks for an inverse involving a logarithm. It is a separate mark, and the quickest check is that **the domain of f^{-1} is the range of f**.

YOUR TURN 5.5 Exponential and Logarithmic Functions

33. Evaluate each expression.

(a) 2^{-2} (b) e^0
(c) $\log_2 8$ (d) $\log_5 125$
(e) $\ln 1$ (f) $\ln e$

34. For each graph, state its asymptote and the coordinates of the point where it crosses an axis.

(a) $y = 5 \cdot 2^x$ (b) $y = 2^x + 5$
(c) $y = 4(3)^x - 1$ (d) $y = \ln x$
(e) $y = \ln(x-2)$ (f) $y = \log(x+5)$

35. State whether each function grows or decays, and give a reason.

(a) $y = (0.5)^x$
(b) $y = 3^x$
(c) $y = 5e^{-0.2x}$
(d) $y = 2^{-x}$

36. Solve each equation.

(a) $2^x = 8$
(b) $\log_3 x = 2$
(c) $5 \cdot 3^x = 45$
(d) $\ln x = 3$
(e) $3(2)^x = 4$
(f) $\log_2(x-1) = 4$

37. Write each in the required form.

(a) 2^x in terms of base e
(b) 5^x in terms of base e

38. GDC Two exponential models.

(a) A population is modelled by $P(t) = 800e^{0.03t}$, with t in years. State P_0, and say whether the population grows or decays.
(b) Find the population after 10 years.
(c) Find how long it takes the population to reach 1200.
(d) A radioactive sample decays according to $A(t) = 50e^{-0.2t}$, with t in days. Find $A(6)$, correct to three significant figures.
(e) Find the half-life of the sample.

39. State the domain of each function.

(a) $y = \ln(x-2)$
(b) $y = \log(x+5)$
(c) $y = \ln(2x-6)$
(d) $y = \log(4-x)$
(e) $y = \ln(9-x^2)$
(f) $y = \ln(x^2-1)$

40. Find the inverse of each function, and state its domain.

(a) $f(x) = e^x + 2$
(b) $f(x) = \ln(x-1)$
(c) $f(x) = 3e^{2x} + 1$

41. GDC Consider the equation $e^x = 3 - x$.

(a) Explain why it has exactly one real solution.
(b) Find that solution, correct to three significant figures.
(c) State why a graph is the appropriate method for this equation.

42. GDC After a sauna is switched off, its temperature in °C after t minutes is $T(t) = 20 + 70e^{-0.12t}$.

(a) State the temperature at the moment it is switched off.
(b) Find the temperature after 5 minutes.
(c) Find how long it takes to cool to $40\,^\circ$C.
(d) State the temperature the sauna approaches, and explain what it represents.

Chapter 5 · Functions & Graphs

Reflections

TOK Slide the graph of $y = x^2$ two units right and you get $y = (x-2)^2$. Is this the same function or a different one? It has the same shape, the same name ("a parabola"), yet a different rule and different values everywhere. When we call two things "the same" in mathematics, we are always choosing what to ignore. The transformation makes that choice visible.

TOK An asymptote describes a curve approaching a line it never meets, across infinitely many ever-smaller steps. This is the shape of Zeno's ancient paradox: to cross a room you must first cross half, then half of what remains, forever. Motion, Zeno argued, is therefore impossible. Calculus answers him: infinitely many shrinking steps can sum to a finite whole. Does mathematics dissolve the paradox, or merely rename it?

IM In 1872 Felix Klein, aged just twenty-three, proposed in his *Erlangen Programme* a radical reorganisation of geometry: a geometry *is* a set of transformations together with the properties they leave unchanged. Euclidean geometry studies what survives rotation and reflection; other geometries allow other moves. The transformations of this chapter are the same idea in miniature, and the question "what stays the same when I move this graph?" is one of the deepest in mathematics.

RW Exponential decay is the clock of the physical world. Every radioactive isotope falls by half over a fixed *half-life*, from carbon-14's $5{,}730$ years, used to date ancient wood and bone, to isotopes that decay in seconds. The same exponential describes unrestricted population growth and compound interest. When growth is limited by a maximum, it becomes the logistic curve of a spreading rumour or an epidemic. Rational functions model saturation: reaction rates, drug concentrations, and average costs that flatten toward a limit.

EXT The exponential e^x has a *secret identity*. Give it an imaginary exponent and it becomes a rotation: Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ ties the transcendental e to the trigonometry of Chapter 8 and the complex numbers of Chapter 17. Setting $\theta = \pi$ gives $e^{i\pi} + 1 = 0$, a single line linking e, π, i, 1, and 0, often called the most beautiful equation in mathematics.

End-of-Chapter Problems — Chapter 5: Functions & Graphs

1. Let $f(x) = 2x - 3$ and $g(x) = x^2$.

(a) Find $(f \circ g)(x)$. [2]
(b) Find $(g \circ f)(x)$. [2]
(c) Solve $(f \circ g)(x) = 5$. [2]

2. The graph of $y = f(x)$ has a maximum point at $(1, 4)$. State the coordinates of the maximum of:

(a) $y = f(x) + 2$; [1]
(b) $y = f(x - 3)$; [1]
(c) $y = 2f(x)$; [1]
(d) $y = -f(x)$ (state the resulting minimum). [1]

3. For each quadratic, write $f(x)$ in completed-square form and describe the transformations that map $y = x^2$ onto $y = f(x)$.

(a) $f(x) = x^2 - 4x + 5$; state also the range of f. [5]
(b) $f(x) = x^2 - 6x$; state also the vertex. [5]

4. Find the inverse of each function, and state its domain.

(a) $f(x) = \dfrac{x+1}{x-2}, \quad x \neq 2.$ [4]
(b) $f(x) = x^2 + 2, \quad x \geq 0.$ [4]
(c) $f(x) = e^x.$ [3]

5. Solve each inequality.

(a) $x^2 + 2 \leq 3x$ [4]
(b) $|2x - 1| < 5$ [3]
(c) $\dfrac{2x-1}{x+2} \leq 1$ [5]

6. Determine whether each function is odd, even, or neither.

(a) $f(x) = x^3 - 2x$ [1]
(b) $f(x) = |x|$ [1]
(c) $f(x) = x^2 + x^3$ [1]
(d) $f(x) = \dfrac{x}{x^2+1}$ [1]
(e) $f(x) = \dfrac{1}{x^2+1}$ [1]
(f) $f(x) = x + 1$ [1]

7. Let $f(x) = x^2 - 2x$.

(a) Find the roots and the vertex of f. [3]
(b) Sketch $y = |f(x)|$, showing the intercepts and the turning point. [3]
(c) Sketch $y = |x^2 - 1|$, showing the x- and y-intercepts and the shape. [4]

8. Composites of two functions.

(a) Let $f(x) = x + 2$ and $g(x) = x^2$. Solve $(f \circ g)(x) = (g \circ f)(x)$. [4]
(b) For $f(x) = \dfrac{x}{1+x}$, find $(f \circ f)(x)$ and simplify fully. [5]

9. Describing and applying transformations.

(a) The graph of $y = f(x)$ passes through $(0, 1)$, $(2, 0)$ and $(3, 4)$. Find the images of these three points on $y = f(x + 1) - 2$. [3]
(b) The graph of $y = x^2$ is transformed to $y = -(x - 1)^2 + 2$. Describe the sequence of transformations and state the coordinates of the vertex. [4]

10. Inverses of three functions.

(a) Let $f(x) = 3x + 2$. Find $f^{-1}(x)$, then solve $f(x) = f^{-1}(x)$. [5]
(b) Let $g(x) = \sqrt{x - 1}$. State the domain and range of g, then find $g^{-1}(x)$ and state its domain. [5]
(c) Let $h(x) = \dfrac{2}{x-3}$. State both asymptotes, then find $h^{-1}(x)$. [5]

11. Solve each equation.

(a) $|x - 2| = |x + 4|$ [4]
(b) $|x| = 2 - x$, and state the coordinates of the point where $y = |x|$ meets $y = 2 - x$. [3]

12. Reasoning from the definitions of odd and even.

(a) The function f is even with $f(3) = 7$, and g is odd with $g(2) = -5$. Write down the values of $f(-3)$ and $g(-2)$. [2]
(b) Prove that if f and g are both odd functions, then $f \circ g$ is also odd. [5]

13. Finding unknown constants from given conditions.

(a) The function $f(x) = ax + b$ satisfies $f(1) = 5$ and $f(3) = 11$. Find a and b, then find $f^{-1}(x)$. [5]
(b) The quadratic $g(x) = x^2 + bx + c$ has vertex $(2, -1)$. Find b and c, then describe the transformation of $y = x^2$ that gives $y = g(x)$. [5]

14. Let $f(x) = (x + 1)^2 - 4$ for $x \geq -1$.

(a) Explain why f has an inverse on this domain. [1]
(b) Find $f^{-1}(x)$, stating its domain. [4]

15.

(a) Solve $|x-2| = |3x+1|$. [4]
(b) Hence, or otherwise, solve $|x-2| \geq |3x+1|$. [3]

16. The function $f(x) = ax + b$ satisfies $f(f(x)) = x$ for all real x. Find all possible values of a and b. [6]

17. Prove that every function $f : \mathbb{R} \to \mathbb{R}$ can be written as the sum of an even function and an odd function. [6]

18. Show that $f(x) = \dfrac{2x+3}{x-2}$ is self-inverse, that is, $(f \circ f)(x) = x$. [5]

19. Let $f(x) = x^2 - 4x + 3$.

(a) Write $f(x)$ in completed-square form, and state the coordinates of the vertex. [3]
(b) Find the roots of f and the y-intercept. [3]
(c) Sketch $y = f(x)$, showing the vertex and both intercepts. [2]
(d) On separate axes, sketch $y = |f(x)|$ and $y = f(|x|)$, showing the key points of each. [4]
(e) The graph of $y = f(x)$ is translated 2 units to the right and 3 units up to give $y = g(x)$. Find $g(x)$ in the form $ax^2 + bx + c$. [3]

20. Let $f(x) = \dfrac{3x-1}{x+2}$ for $x \neq -2$, and let $g(x) = x + 2$.

(a) State the range of f, justifying your answer. [2]
(b) Find $f^{-1}(x)$, and state its domain. [4]
(c) Find $(f \circ g)(x)$, giving your answer as a single fraction. [3]
(d) State the domain of $f \circ g$. [2]
(e) Solve $(f \circ g)(x) = 2$. [4]

21. HL Let $f(x) = x^2 - 4$.

(a) Sketch $y = f(x)$, showing the intercepts. [2]
(b) Write down the equations of the vertical asymptotes of $y = \dfrac{1}{f(x)}$, and explain how you found them. [3]
(c) Find the coordinates of the maximum point of $y = \dfrac{1}{f(x)}$ on the interval $-2 < x < 2$, and explain why it is a maximum even though the value is negative. [3]
(d) Describe the behaviour of $y = \dfrac{1}{f(x)}$ as $x \to \pm\infty$, and state the horizontal asymptote. [2]

(e) Sketch $y = \dfrac{1}{f(x)}$, showing all asymptotes and the point found in part (c). [4]

22. For each function, state both asymptotes and the coordinates of the y-intercept.

(a) $f(x) = \dfrac{1}{x+2}$ [3]

(b) $f(x) = \dfrac{2x+3}{x-1}$ [3]

(c) $f(x) = \dfrac{4x-1}{2x+1}$ [3]

(d) Sketch the graph in part (b), showing both asymptotes and both intercepts. [3]

23. For each function, state the asymptotes and find the inverse.

(a) $f(x) = \dfrac{x-2}{x+3}$; state also the domain of f. [6]

(b) $f(x) = \dfrac{3x}{x-2}$; show that $f^{-1}(x) = \dfrac{2x}{x-3}$. [6]

24. For each function, state every asymptote, and say what happens at the value of x where the denominator is zero.

(a) $f(x) = \dfrac{x^2-1}{x}$; state also the x-intercepts. [5]

(b) $f(x) = \dfrac{x^2+x-2}{x-1}$; simplify the expression first. [5]

25. A bacterial population is modelled by $P(t) = 500 \cdot 2^t$, where t is in hours.

(a) State the initial population. [1]

(b) Find the population after 3 hours. [2]

(c) Find when the population reaches 8000. [3]

26. GDC Two decay models.

(a) A radioactive sample has mass $M(t) = 80 \cdot (0.5)^{t/5}$ grams, with t in days. State the initial mass and the half-life. [3]

(b) Find the mass after 10 days. [2]

(c) The value of a car after t years is $V(t) = 20000(0.85)^t$. State the initial value, and find the value after 4 years. [3]

(d) State the long-term behaviour of $V(t)$, and name the feature of the graph it corresponds to. [2]

27. Solve each equation.

(a) $e^{2x} - 3e^x + 2 = 0$ [5]

(b) $4^x - 2^{x+1} = 8$ [5]

28. Sketch, on the same axes, $y = 2^x$ and $y = 2^{-x}$, stating the horizontal asymptote and the common point. [4]

29. Show that $f(x) = \dfrac{ax+b}{cx+d}$ is self-inverse (that is, $f(f(x)) = x$) if and only if $a + d = 0$. [6]

30. The curve $y = \dfrac{x^2 + ax + b}{x - 1}$ has oblique asymptote $y = x + 3$.

(a) Find a. [3]

(b) Find the value of b for which the curve has a hole at $x = 1$ instead of a vertical asymptote. [3]

31. HL Consider $f(x) = \dfrac{x^2 - 4x + 5}{x - 2}$.

(a) Show that $f(x) = (x - 2) + \dfrac{1}{x-2}$. [3]

(b) Write down the equations of the vertical and oblique asymptotes. [3]

(c) Find the y-intercept, and show that the curve has no x-intercept. [3]

(d) Show that the curve never meets its oblique asymptote. [2]

(e) Sketch the curve, showing both asymptotes and the y-intercept. [4]

32. GDC The number of people who have heard a rumour t days after it starts is modelled by

$$N(t) = \frac{5000}{1 + 49e^{-0.8t}}.$$

(a) Find the number of people who had heard the rumour at $t = 0$. [2]

(b) Find $N(5)$, giving your answer to the nearest whole number. [2]

(c) Find the time at which 2500 people have heard the rumour, giving an exact answer and a value to three significant figures. [4]

(d) State the value N approaches as t becomes large, and interpret it in context. [3]

(e) Show that $\dfrac{5000 - N}{N} = 49e^{-0.8t}$, and hence explain why plotting $\ln\left(\dfrac{5000 - N}{N}\right)$ against t gives a straight line. State its gradient and its vertical intercept. [5]

33. Let $f(x) = \ln(2x - 4)$.

(a) State the domain of f. [2]

(b) Find the exact x-intercept of the graph of f. [3]

(c) State the equation of the vertical asymptote, and describe the behaviour of $f(x)$ near it. [3]

(d) Find $f^{-1}(x)$, and state its domain and range. [4]

(e) Describe a sequence of transformations that maps $y = \ln x$ onto $y = \ln(2x - 4)$. [3]

34. Let $f(x) = x^2 - 1$.

(a) Sketch $y = |f(x)|$ for $-3 \le x \le 3$, showing the coordinates of both corners. [3]
(b) Hence solve $|x^2 - 1| = 3$. [3]
(c) Solve $|x^2 - 1| < 3$. [3]
(d) Solve $|x - 1| = 3x + 5$, showing clearly why one candidate is rejected. [4]

Challenge Questions

35. Self-inverse rational functions. A function of the form

$$f(x) = \frac{ax + b}{cx + d}, \qquad c \neq 0,$$

is called *self-inverse* when $(f \circ f)(x) = x$ for every x in its domain. This investigation finds exactly which of them are.

(a) Show that $f(x) = \dfrac{x+1}{x-1}$ is self-inverse. [3]
(b) Show that for the general function above,

$$(f \circ f)(x) = \frac{(a^2 + bc)\,x + b(a + d)}{c(a + d)\,x + (bc + d^2)}.$$

[4]
(c) Hence prove that, when $c \neq 0$, the function f is self-inverse if and only if $a + d = 0$. *(Hint: two expressions in x are equal for all x only when their coefficients match)* [5]
(d) Show that if $ad - bc = 0$ then f is constant, and explain why the condition $ad - bc \neq 0$ must therefore be assumed throughout. [3]
(e) Write down two further self-inverse examples with $c \neq 0$, one of which has $b = 0$. Verify one of them. [3]

36. When does the order of two transformations matter? Section 5.2 showed that a vertical stretch and a vertical translation give different curves in different orders. This investigation settles the general question.

Throughout, start from $y = f(x)$.

(a) Apply a vertical stretch of factor a and then a translation of k units up, and write down the result. Do the same in the opposite order. Deduce the exact condition on a and k under which the two agree for every function f. [4]
(b) The horizontal moves are substitutions: move A replaces x by $x - h$, and move C replaces x by bx. Show that A followed by C gives $y = f(bx - h)$, while C followed by A gives $y = f(bx - bh)$. Deduce the condition under which these agree for every f. [5]
(c) Show that any vertical move and any horizontal move commute, whatever their parameters. *(Hint: consider which variable each move acts on)* [3]

(d) Using parts (a) to (c), state a single rule that predicts when two transformations may be applied in either order. Explain why your rule is consistent with all three cases above. [4]

37. Counting the solutions of $|f(x)| = k$. Let $f(x) = x^2 - 4$ and let $k \geq 0$ be a constant.

(a) Sketch $y = |f(x)|$, showing the x-intercepts and the coordinates of the local maximum. [3]

(b) Find the number of real solutions of $|f(x)| = k$ for each of $k = 0$, $k = 3$, $k = 4$ and $k = 6$. [4]

(c) Explain, using your sketch, why the number of solutions changes at $k = 4$ and not at any other value. [3]

(d) State the number of real solutions of $|f(x)| = k$ for every $k \geq 0$, giving your answer as a list of cases. [3]

(e) Now let g be any quadratic with a single minimum point of value m, where $m < 0$. State the corresponding list of cases for $|g(x)| = k$, in terms of m. [3]

38. GDC **Newton's law of cooling.** A body at temperature T in a room at temperature R cools so that the difference $T - R$ decays exponentially:

$$T(t) = R + (T_0 - R)\, e^{-kt}, \qquad k > 0,$$

where T_0 is the temperature at $t = 0$ and t is measured in minutes.

(a) Show that the temperature difference $T(t) - R$ falls by the same factor over any interval of fixed length, and that this factor does not depend on T_0. [4]

(b) Deduce that the difference halves every $\dfrac{\ln 2}{k}$ minutes. [3]

(c) A cup of coffee at $90\,^\circ$C is left in a room at $20\,^\circ$C. After 10 minutes it has cooled to $60\,^\circ$C. Find k in exact form, and hence find the half-life of the temperature difference. [4]

(d) Find, to the nearest minute, how long the coffee takes to reach $40\,^\circ$C. [3]

(e) Explain why the model predicts that the coffee never reaches $20\,^\circ$C, and state what feature of the graph carries that fact. [3]

FUNCTIONS

CHAPTER 6

Polynomials

6

SYLLABUS 1.11 2.12 HL ONLY

Nature rarely changes in sudden jumps. Temperature, water level and the position of a moving object all change smoothly, and polynomials are the simplest functions that change in the same way. The degree matches the situation: distance travelled at a constant speed is linear, a thrown ball follows a parabola, and the volume of a sphere grows with the cube of its radius. Over a short enough interval almost any smooth curve can be matched closely by a polynomial, so measurements are often fitted with one when no exact formula is known. See Ch. 16.

That smoothness comes from the way a polynomial is built. It uses one variable and a few constants, combined by addition and multiplication only. There is no division by the variable, no square root and no trigonometry. A polynomial is therefore defined for every real number, so its graph has no gaps, no asymptotes and no corners.

Describing a polynomial is easy. Solving one is not. An exact formula for the roots exists for degree 2, degree 3 and degree 4, and Niels Abel proved in 1824 that for degree 5 and above no such formula exists. By the end of this chapter you will be able to read a polynomial's graph from its factorised form, divide one polynomial by another, test whether a value is a root and factorise once you have found one, and give the sum and the product of the roots from the coefficients alone.

One idea underlies all of it: a polynomial's coefficients and its roots describe the same object, and nearly every technique here moves between the two. The chapter closes with **partial fractions**, which splits a fraction by factorising its denominator and giving each factor a numerator of its own. That is why it belongs here rather than with the rational functions of §5.4, and it is what makes a rational function possible to integrate later. See Ch. 15.

6.1 Polynomial Functions and their Graphs

Much of the graph of a polynomial can be described before a single point is plotted, because the degree and the coefficients already carry that information. Reading those features from the equation is faster than building a table of values, and they are what a sketch has to show.

DEF A **polynomial function** of **degree** n has the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

where the coefficients are constants and n is a non-negative integer.

6.1.1 Degree, Coefficients and End Behaviour

The graph features in this subsection are read from two things: the degree, and the sign of the lead coefficient.

Polynomials are named by their degree. Degree 1 is **linear**, degree 2 is **quadratic**, degree 3 is **cubic**, degree 4 is **quartic**, and degree 5 is **quintic**. The degree also limits the shape: it limits how many times the curve can meet the x-axis, and how often it can change direction.

The constants $a_n, a_{n-1}, \ldots, a_1, a_0$ are the **coefficients** of the powers of x. The coefficient of the highest power is a_n, called the **lead coefficient** (also written leading coefficient), and the term $a_n x^n$ that contains it is the **leading order term**.

For a factorised polynomial neither has to be multiplied out. The degree is the total number of linear factors, and the lead coefficient is the product of the leading coefficients of those factors.

A coefficient may take any value, with one restriction: the lead coefficient cannot be zero. If a_n were zero, the term $a_n x^n$ would be zero, and what remained would be a polynomial of degree $n - 1$ instead.

The sign of the lead coefficient determines which way the ends of the curve point. A polynomial with $a_n > 0$ is sometimes called a **positive polynomial**, and one with $a_n < 0$ a **negative polynomial**. These names describe the lead coefficient only. They do not mean that the function is positive or negative everywhere: $y = x^2 - 4$ has a positive lead coefficient, yet y is negative for $-2 < x < 2$.

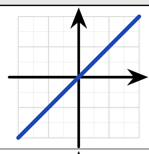
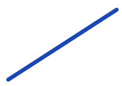
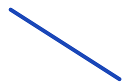
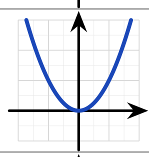
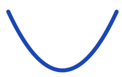
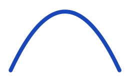
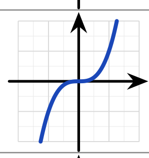
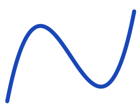
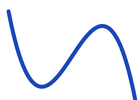
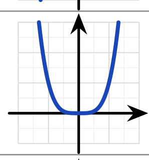
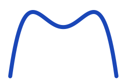
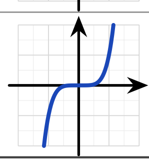
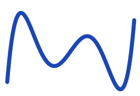
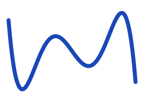
Three features are visible before you plot a single point.

The **end behaviour**, meaning the direction of the curve as $x \to \pm\infty$, is determined entirely by the leading term. If n is odd, the two ends point in opposite directions. If n is even, they point the same way, and the sign of a_n determines whether that direction is upward or downward.

The **roots** are the places where the graph meets the x-axis, and their number can never exceed the degree. A cubic meets the axis at most three times, a quartic at most four.

A **turning point** is a place where the curve stops rising and begins to fall, or stops falling and begins to rise. A polynomial of degree n has at most $n-1$ turning points. The possible numbers then decrease in steps of two: $n-1$, $n-3$, and so on. The number changes by two rather than by one because a maximum and a minimum are always lost together, never one alone.

The table sets out all three features for the degrees you meet most often.

n	$y = x^n$	**Shape when** $a_n > 0$	**Shape when** $a_n < 0$	**Times it meets the** x**-axis**	**Turning points**
1				1	0
2				0, 1 or 2	1
3				1, 2 or 3	0 or 2
4				0, 1, 2, 3 or 4	1 or 3
5				1, 2, 3, 4 or 5	0, 2 or 4

EXAMPLE 6.1

State the degree, the lead coefficient and the two end behaviours of (a) $p(x) = 5 - 4x + 6x^3 - x^2$ and (b) $q(x) = (1-2x)(x+3)^2(x-4)$.

(a) The terms are not written in order of power, so find the highest power rather than reading the first term. It is $6x^3$, so the degree is 3 and the lead coefficient is 6. An odd degree points the two ends in opposite directions, and a positive lead coefficient sends the right-hand end upward:

$$p(x) \to +\infty \text{ as } x \to +\infty, \qquad p(x) \to -\infty \text{ as } x \to -\infty.$$

(b) Nothing needs to be multiplied out. Count the linear factors: $(1-2x)$, two copies of $(x+3)$, and $(x - 4)$, so the degree is 4. The lead coefficient is the product of the four leading coefficients,

$$(-2) \times 1 \times 1 \times 1 = -2.$$

An even degree points both ends the same way, and the negative lead coefficient sends them down: $q(x) \to -\infty$ as $x \to \pm\infty$.

Everything in (b) turns on the -2, which comes from the factor written backwards as $(1 - 2x)$. Reading the coefficient of x in that bracket as 2 and losing the minus sign gives a lead coefficient of $+2$, which reverses both ends.

6.1.2 Roots and Multiplicity

The table gives the shape in outline. What fixes it exactly is what the curve does at each root, and that is determined by whether a factor is repeated.

The **multiplicity** of a root is the number of times its factor appears in the factorised polynomial. In $(x-1)(x-2)^2$ the root $x = 1$ has multiplicity one, and the root $x = 2$ has multiplicity two. A root of multiplicity one is called a **simple root**, and one of multiplicity two a **double root**.

KEY At a root of **multiplicity one** the curve crosses the axis straight. At a root of **even** multiplicity it touches the axis and turns back. At a root of **odd** multiplicity greater than one it crosses, but flattens against the axis as it does so.

Whether the multiplicity is odd or even determines crossing or touching. Its size determines how flat the curve is where it meets the axis.

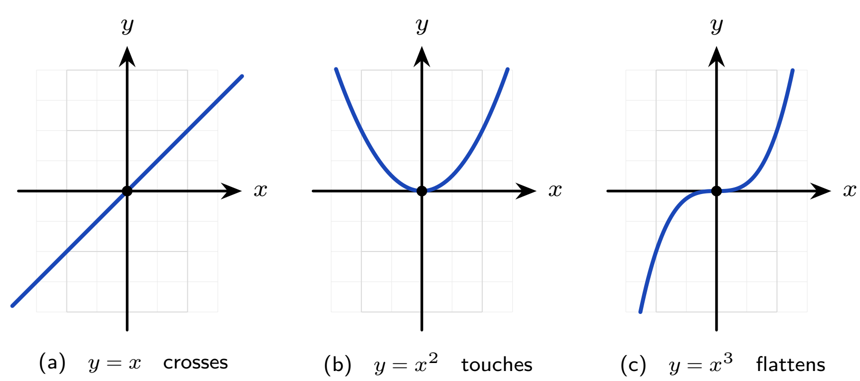


Figure 6.1 The three behaviours at a root, shown at $x = 0$. In (a) the multiplicity is one and the curve crosses straight. In (b) it is two, an even number, so the curve touches and turns back. In (c) it is three, odd, so the curve crosses again, but flattens against the axis as it passes through.

Three readings then give the whole sketch. Take the behaviour at each x-intercept from the roots and their multiplicities, the y-intercept from the value at $x = 0$, and the two ends from the leading term.

Practise both directions: reading the multiplicities from a sketch, and predicting the sketch from the factors.

EXAMPLE 6.2

Sketch $y = (x+1)(x-2)^2$, showing the intercepts and the behaviour at each root.

Read the roots and their multiplicities from the factors. The factor $(x+1)$ is single, so the graph crosses the axis at $x = -1$. The factor $(x-2)^2$ is squared, so the graph touches the axis at $x = 2$ and turns back.

Find the y-intercept. Substitute $x = 0$: $y = (1)(-2)^2 = 4$, the point $(0, 4)$.

Fix the ends from the leading term. Multiplying out leads with x^3: an odd degree with a positive coefficient, so the curve runs from $-\infty$ at the left to $+\infty$ at the right.

These three facts fix the shape.

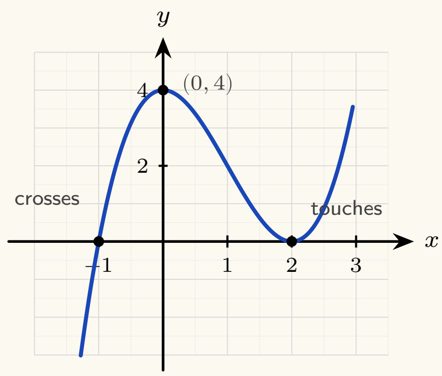


Figure 6.2 A sketch of $y = (x+1)(x-2)^2$. The curve crosses the axis at the simple root $x = -1$ and touches it at the double root $x = 2$. The y-intercept is $(0, 4)$, and the ends follow x^3.

Every factorised polynomial is sketched the same way: the roots and their multiplicities give the behaviour at the x-axis, the value at $x = 0$ gives the y-intercept, and the leading term gives the two ends.

When the polynomial is written out in full rather than factorised, the y-intercept is even quicker to read. Every term except the constant contains x, so substituting $x = 0$ makes all of them zero. The constant term a_0 is what remains.

KEY The graph of $y = a_n x^n + \cdots + a_1 x + a_0$ crosses the y-axis at $(0, a_0)$. *The constant term is the y-intercept.*

For $y = 2x^3 - 3x + 5$, the constant term is 5, so the graph crosses the y-axis at $(0, 5)$.

6.1.3 From the Graph Back to the Equation

The reverse problem is also examined. Given a sketch, you are asked for the polynomial that produced it. The intercepts give you almost everything you need.

Each x-intercept contributes a factor, and the behaviour there gives its multiplicity: a crossing gives a single factor, a touch gives a squared factor. That fixes the polynomial up to one unknown number, the leading coefficient, and any known point that is not an x-intercept determines it. Check that the factors account for the whole degree of the curve; if they do not, an intercept has been missed or a multiplicity is higher than it looks.

KEY **Finding a polynomial from its graph.** Write one factor for each x-intercept, with the multiplicity the graph shows at that point. Then substitute the coordinates of any known point that is not an x-intercept to find the leading coefficient a_n.

EXAMPLE 6.3

The graph below shows a cubic. It touches the x-axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. Find its equation.

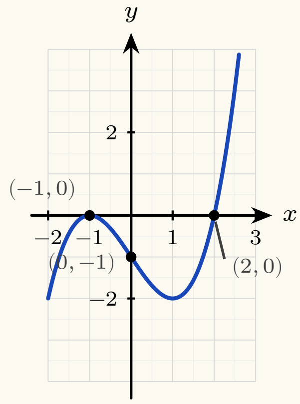


Figure 6.3 The curve touches the x-axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. These three facts determine the equation.

Write the factors from the x-intercepts. The curve touches at $x = -1$, so that root is repeated and contributes $(x + 1)^2$. It crosses at $x = 2$, so that root is single and contributes $(x - 2)$. Those factors already account for degree three, so

$$y = a(x + 1)^2(x - 2)$$

for some constant a.

Find a from the remaining point. Substitute $x = 0$ and $y = -1$:

$$-1 = a(1)^2(-2) = -2a,$$

so $a = \frac{1}{2}$ and

$$y = \tfrac{1}{2}(x + 1)^2(x - 2).$$

Check the answer against the graph. The leading term is $\frac{1}{2}x^3$, which is positive, so the curve rises to the right, and the sketch agrees.

The intercepts fix the factors; one further point fixes the leading coefficient.

YOUR TURN 6.1 Polynomial Functions and their Graphs

1. State the degree and the lead coefficient of each polynomial.

(a) $f(x) = 3x^4 - 2x + 7$
(b) $f(x) = 5 - x^3$
(c) $f(x) = (x-1)(x+2)(x-3)$
(d) $f(x) = -2x^5 + x^2$

2. Describe the end behaviour of each graph as $x \to +\infty$ and as $x \to -\infty$.

(a) $y = x^3$
(b) $y = -x^4$
(c) $y = 2x^5 - x$
(d) $y = 3 - x^2$

3. Answer each question about degree and shape.

(a) State the maximum number of real roots of a degree-5 polynomial.
(b) State the maximum number of turning points of a quartic.
(c) List every possible number of turning points a cubic can have.
(d) Explain why a cubic must have at least one real root.

4. State the roots of each polynomial, and say at each one whether the graph crosses or touches the x-axis.

(a) $y = (x-2)(x+3)(x-5)$
(b) $y = (x-1)^2(x+4)$
(c) $y = x(x-3)^2$
(d) $y = (x+1)^2(x-2)^2$

5. Consider $y = (x+1)(x-3)^2$.

(a) State the roots and their multiplicities.
(b) Find the y-intercept.
(c) State the end behaviour.
(d) Sketch the curve.

6. A cubic touches the x-axis at $x = 2$, crosses it at $x = -1$, and passes through $(0, 4)$.

(a) Write down the factors, with the correct multiplicities.
(b) Find the lead coefficient.
(c) Write the equation of the cubic.

6.2 Polynomial Division

You can already add, subtract and multiply polynomials. Addition and subtraction collect like terms, and multiplication expands every product before collecting. Division is the operation that still needs a method, and it is the one §6.3 and §6.4 are built on.

Factorising a polynomial means writing it as a product. Once you know one factor, the other is found by dividing. So a method for dividing one polynomial by another is needed before a higher-degree polynomial can be factorised from a known factor.

6.2.1 The Division Algorithm

Before dividing anything it helps to know what the answer should look like, and whole-number division already shows that.

Polynomial division follows the same pattern. When you divide 47 by 5 you write

$$47 = 5 \times 9 + 2,$$

which records a divisor, a quotient, and a remainder smaller than the divisor. Polynomial division works the same way, with "smaller" measured by degree instead of size.

> KEY **The division algorithm.** For polynomials $f(x)$ and $d(x)$, with $d(x)$ not the zero polynomial, there are unique polynomials $q(x)$ and $r(x)$ such that
>
> $$f(x) = d(x)\,q(x) + r(x),$$
>
> where the degree of r is less than the degree of d. *Dividend equals divisor times quotient, plus remainder.*

Fix the four names now, because the methods that follow use them. The polynomial being divided, f, is the **dividend**. The polynomial you divide by, d, is the **divisor**. The result q is the **quotient**, and the part that remains, r, is the **remainder**. When $r = 0$ the division is exact, and d is a factor of f.

6.2.2 Long Division

Long division is the method that works for a divisor of any degree. It is three steps repeated until the remainder has lower degree than the divisor.

KEY **Divide, multiply, subtract.**

1. **Divide** the leading term of the dividend by the leading term of the divisor. The result is the next term of the quotient.
2. **Multiply** the whole divisor by that term.
3. **Subtract** the product from the dividend, and repeat on what is left.

Set the work out in columns, one for each power of x: the divisor to the left of the bracket, the dividend inside it, the quotient written on the top line, and the remainder on the bottom line. Write the result as $f(x) = d(x)\,q(x) + r(x)$ when you finish, because multiplying that right-hand side out checks the whole division.

EXAMPLE 6.4

Divide $2x^3 - 3x^2 + x - 5$ by $(x - 2)$.
Set the work out in columns, one column for each power of x.

$$
\begin{array}{r|rrrrl}
 & & 2x^2 & +x & +3 & \text{quotient} \\ \cline{2-5}
x-2\,) & 2x^3 & -3x^2 & +x & -5 & \text{dividend} \\
 & 2x^3 & -4x^2 & & & \\ \cline{2-3}
 & & x^2 & +x & & \\
 & & x^2 & -2x & & \\ \cline{3-4}
 & & & 3x & -5 & \\
 & & & 3x & -6 & \\ \cline{4-5}
 & & & & 1 & \text{remainder}
\end{array}
$$

The divisor $x - 2$ is written to the left of the bracket, and the dividend $2x^3 - 3x^2 + x - 5$ inside it. The quotient is written on the top line, one term at each stage, and the remainder appears on the bottom line.
Each stage repeats the same three steps: divide, multiply, subtract.
Stage 1. Divide the leading term of the dividend by the leading term of the divisor: $2x^3 \div x = 2x^2$. Write $2x^2$ in the quotient. Multiply the divisor by it: $2x^2(x-2) = 2x^3-4x^2$. Subtract that from the dividend, which leaves $x^2 + x$.
Stage 2. Repeat on $x^2 + x$. Divide: $x^2 \div x = x$. Multiply: $x(x - 2) = x^2 - 2x$. Subtract, which leaves $3x - 5$.
Stage 3. Repeat on $3x - 5$. Divide: $3x \div x = 3$. Multiply: $3(x - 2) = 3x - 6$. Subtract, which leaves 1.
The degree of 1 is lower than the degree of $x - 2$, so the division stops.
The quotient is $2x^2 + x + 3$ and the remainder is 1:

$$2x^3 - 3x^2 + x - 5 = (x - 2)(2x^2 + x + 3) + 1.$$

Always write the answer in this form. Multiplying out the right side is a complete check of the division.

CAUTION Write the dividend in descending powers and insert a zero term for every missing power before dividing. Divide $x^3-13x+12$ as $x^3+0x^2-13x+12$. Without that zero term the columns shift, and the subtractions that follow go wrong.

EXAMPLE 6.5

Divide $x^3 - 13x + 12$ by $(x-1)$.
There is no x^2 term, so write it as $x^3 + 0x^2 - 13x + 12$ and keep a column for it.

$$
\begin{array}{r|rrrr}
 & & x^2 & +x & -12 \\
\cline{2-5}
x-1 & x^3 & +0x^2 & -13x & +12 \\
 & x^3 & -x^2 & & \\
\cline{2-3}
 & & x^2 & -13x & \\
 & & x^2 & -x & \\
\cline{3-4}
 & & & -12x & +12 \\
 & & & -12x & +12 \\
\cline{4-5}
 & & & & 0
\end{array}
$$

The remainder is 0, so the division is exact and $(x-1)$ is a factor. The quotient x^2+x-12 factorises further:

$$x^3 - 13x + 12 = (x-1)(x^2+x-12) = (x-1)(x+4)(x-3).$$

A remainder of zero shows that the divisor is a factor. The next two sections are built on that fact.

A divisor of higher degree changes none of the three steps. What it changes is where the division stops: the remainder must have degree lower than the divisor, so dividing by a quadratic can leave a remainder with an x in it. Set the columns out in the same way, and keep a column for every power of x from the highest down to the constant, whether or not the dividend has a term there.

EXAMPLE 6.6

Divide $x^4 + 2x^2 - 3x + 1$ by $(x^2 + x - 1)$.
There is no x^3 term, so write the dividend as $x^4+0x^3+2x^2-3x+1$ and give that power a col-

umn of its own.

$$
\begin{array}{rrrrrr}
 & & & x^2 & -x & +4 \\ \cline{2-6}
x^2+x-1 \,) & x^4 & +0x^3 & +2x^2 & -3x & +1 \\
 & x^4 & +x^3 & -x^2 & & \\ \cline{2-4}
 & & -x^3 & +3x^2 & -3x & \\
 & & -x^3 & -x^2 & +x & \\ \cline{3-5}
 & & & 4x^2 & -4x & +1 \\
 & & & 4x^2 & +4x & -4 \\ \cline{4-6}
 & & & & -8x & +5
\end{array}
$$

Stage 1. Divide the leading terms: $x^4 \div x^2 = x^2$. Multiply the divisor by it: $x^2(x^2+x-1) = x^4+x^3-x^2$. Subtract, which leaves $-x^3+3x^2-3x$.
Stage 2. Repeat. Divide: $-x^3 \div x^2 = -x$. Multiply: $-x(x^2+x-1) = -x^3-x^2+x$. Subtract, which leaves $4x^2-4x+1$.
Stage 3. Repeat. Divide: $4x^2 \div x^2 = 4$. Multiply: $4(x^2+x-1) = 4x^2+4x-4$. Subtract, which leaves $-8x+5$.
The remainder $-8x+5$ has degree one, the divisor has degree two, so the division stops:

$$x^4+2x^2-3x+1 = (x^2+x-1)(x^2-x+4)-8x+5.$$

Multiplying the right side out returns the dividend, which checks every stage at once. A remainder with an x in it is the normal outcome against a quadratic divisor, not a sign that a stage has gone wrong.

YOUR TURN 6.2 Polynomial Division

7. Simplify each expression.

(a) $(x^3+2x-5)+(3x^3-x^2+1)$ (b) $(x^4+3x^2-1)-(2x^3-x^2+2)$
(c) $(x^2+2x-1)(x^2-x+3)$ (d) $(x-2)(x^2+2x+4)$

8. Divide, and state the quotient and the remainder.

(a) x^2+5x+6 by $(x+2)$ (b) x^3-3x^2+2x by $(x-1)$
(c) x^3+2x^2-x+1 by $(x-1)$ (d) x^3-8 by $(x-2)$

9. Divide, and state the quotient and the remainder. Insert a zero term for each missing power first.

(a) $2x^3-x^2-3$ by $(x+1)$ (b) x^3+3x^2-4 by $(x+2)$
(c) $2x^4-3x^3+x-6$ by $(x-2)$

10. Divide by the quadratic divisor.

(a) $x^4 - 1$ by $(x^2 - 1)$

(b) $x^4 - 1$ by $(x^2 + 1)$

6.3 The Remainder and Factor Theorems

6.3.1 The Remainder Theorem

Here are two operations that appear unrelated. *Evaluating* f at 2 gives the height of the graph above one point. *Dividing* f by $(x - 2)$ is the long calculation of the previous section. The remainder theorem states that the two produce the same number, and it is quickly proved.

Take the division algorithm with a linear divisor $(x - a)$:

$$f(x) = (x - a)\, q(x) + r,$$

where the remainder r must be a *constant*, because its degree is less than the degree of $x - a$, and less than one means zero.

Now substitute $x = a$ into that identity. The factor $(x - a)$ becomes zero, so the whole first term is zero, leaving $f(a) = r$. The height of the graph at a *is* the remainder, because everything else in the polynomial is divisible by $(x - a)$ and so contributes nothing at that point.

KEY **Remainder theorem.** When a polynomial $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$. *One substitution replaces the division.*

The theorem holds for any linear divisor, not only for one written as $(x - a)$. Divide by $(bx - c)$ and the remainder is f evaluated at the **zero of the divisor**, meaning the value of x that makes the divisor equal to zero. For $(2x - 1)$ that value is $\frac{1}{2}$, never 1 or 2.

Read the theorem in either direction. Given the coefficients it returns the remainder; given two remainders it returns two linear equations in the unknown coefficients.

EXAMPLE 6.7

Find the remainder when $f(x) = x^3 - 2x + 5$ is divided by $(x - 2)$.

By the remainder theorem, the remainder is

$$f(2) = 2^3 - 2(2) + 5 = 8 - 4 + 5 = 9.$$

No division needed; the answer is a single evaluation.

EXAMPLE 6.8

Find the remainder when $f(x) = 8x^3 + 4x^2 - 6x + 5$ is divided by $(2x - 1)$.
The divisor need not have the form $(x - a)$. The theorem works for any linear divisor, provided you evaluate at the **zero of the divisor**, meaning the value of x that makes the divisor equal to zero. Here $2x - 1 = 0$ when $x = \frac{1}{2}$, so the remainder is

$$f(\tfrac{1}{2}) = 8 \cdot \tfrac{1}{8} + 4 \cdot \tfrac{1}{4} - 3 + 5 = 1 + 1 - 3 + 5 = 4.$$

Evaluate at the zero of the divisor, not at one of its coefficients. Dividing by $(2x-1)$ means substituting $\frac{1}{2}$, never 1 or 2.

EXAMPLE 6.9

The polynomial $f(x) = x^3 + ax + b$ leaves remainder 7 when divided by $(x - 2)$ and remainder -5 when divided by $(x + 1)$. Find a and b.
Each remainder condition is one evaluation, so two conditions give two equations:

$$f(2) = 8 + 2a + b = 7 \implies 2a + b = -1,$$

$$f(-1) = -1 - a + b = -5 \implies -a + b = -4.$$

Subtracting, $3a = 3$, so $a = 1$ and $b = -3$. The remainder theorem has turned two divisions into a pair of linear equations; questions of this kind, working from the remainders back to the coefficients, are common.

The theorem is stated for a linear divisor, but the idea behind it reaches further. Divide by a quadratic and the remainder has degree at most one, so it is $rx + s$: two unknowns instead of one. The division algorithm still records the whole calculation as an identity, and each zero of the divisor makes the divisor term vanish, leaving one equation in r and s. Two distinct zeros give two equations, which is exactly what two unknowns need.

EXAMPLE 6.10

Find the remainder when $f(x) = x^3 + 2x^2 - 5x + 3$ is divided by $(x - 1)(x - 2)$.
The divisor has degree two, so the remainder has degree at most one. Write it as $rx + s$ and record the division as an identity:

$$f(x) = (x - 1)(x - 2)\, q(x) + rx + s.$$

This holds for every value of x, so choose the two values that make the product term disappear.
Substituting $x = 1$,

$$f(1) = 1 + 2 - 5 + 3 = 1 \implies r + s = 1,$$

and substituting $x = 2$,

$$f(2) = 8 + 8 - 10 + 3 = 9 \implies 2r + s = 9.$$

Subtracting the first equation from the second gives $r = 8$, and then $s = -7$. The remainder is

$$8x - 7.$$

Long division confirms it: the quotient is $x + 5$, and $(x - 1)(x - 2)(x + 5) + 8x - 7$ returns $f(x)$. The method depends on the divisor having two *distinct* zeros; against $(x - 1)^2$ there is only one substitution available, and the second equation has to come from comparing coefficients instead.

6.3.2 The Factor Theorem

More follows when the remainder is zero. If dividing by $(x - a)$ leaves no remainder, then $(x-a)$ divides $f(x)$ exactly: it is a factor. That is what turns a test for a root into a method for factorising.

KEY **Factor theorem.** $(x - a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.

To factorise an unfamiliar cubic, search for a root by testing small values. The factors of the constant term are the natural candidates, and the reason is short. If a is an integer root of $a_n x^n + \cdots + a_1 x + a_0$ with integer coefficients, then substituting and moving a_0 across gives

$$a\left(a_n a^{n-1} + a_{n-1} a^{n-2} + \cdots + a_1\right) = -a_0,$$

so a divides a_0. When the leading coefficient is not 1, the candidates are the fractions $\frac{p}{q}$ with p a factor of a_0 and q a factor of a_n. Once one root a is found, $(x - a)$ is a factor, and dividing it out leaves a quadratic you can finish by hand.

That division need not be long division. To **compare coefficients**, write the quotient with unknown coefficients, one degree lower than the dividend, multiply out, and equate the coefficients of matching powers. Each unknown comes from one comparison, and any coefficient left over is a check.

EXAMPLE 6.11

Show that $(x - 1)$ is a factor of $f(x) = x^3 - 7x^2 + 14x - 8$, and factorise f completely.

Test $x = 1$:

$$f(1) = 1 - 7 + 14 - 8 = 0,$$

so by the factor theorem $(x - 1)$ is a factor.

Now divide it out. Instead of long division, *compare coefficients*. The quotient must be a quadratic, so write

$$x^3 - 7x^2 + 14x - 8 = (x - 1)(x^2 + bx + c),$$

then compare the coefficients of matching powers on the two sides. Compare the constant terms: $-c = -8$, so $c = 8$. Compare the coefficients of x^2: $b - 1 = -7$, so $b = -6$. The coefficient of x has not been used, so use it as a check: on the right it is $c - b = 8 + 6 = 14$, which matches the left. ✓ Hence

$$f(x) = (x-1)(x^2 - 6x + 8) = (x-1)(x-2)(x-4).$$

The constant term -8 gives the candidates $\pm1, \pm2, \pm4, \pm8$. Testing $x = 1$ succeeded on the first attempt, and comparing coefficients carried out the division with a check included.

TIP Comparing coefficients is usually faster than long division in an examination. Each unknown comes from one comparison, and any coefficient you did not use provides a check. Long division is equally acceptable if you prefer it.

YOUR TURN 6.3 The Remainder and Factor Theorems

11. Find the remainder in each case.

(a) $x^3 + 2x - 1$ divided by $(x-1)$

(b) $x^3 - 4x^2 + x + 6$ divided by $(x-2)$

(c) $2x^3 - 3x + 1$ divided by $(x+1)$

(d) $x^3 + x^2 + x + 1$ divided by $(x+2)$

12. Find the remainder in each case. The divisor is not of the form $(x-a)$.

(a) $4x^3 - 4x^2 + 5x - 1$ divided by $(2x-1)$

(b) $2x^3 + x^2 - 2$ divided by $(2x+1)$

13. Interpret each statement.

(a) The remainder when $f(x)$ is divided by $(x - 3)$ is 0. What does this tell you?

(b) The remainder when $f(x)$ is divided by $(x-2)$ is 7. State $f(2)$.

(c) $f(-1) = 0$. Write down a factor of $f(x)$.

(d) $x^4 - 1$ is divided by $(x-1)$. Find the remainder without dividing.

14. The polynomial $f(x) = x^3 + ax + b$ leaves remainder 2 when divided by $(x-1)$ and remainder -13 when divided by $(x+2)$.

(a) Write down two equations in a and b.

(b) Solve them to find a and b.

15. Decide whether the given linear expression is a factor. Show your working.

(a) Is $(x-2)$ a factor of $x^3 - 3x^2 + 4$?

(b) Is $(x+1)$ a factor of $x^3 + 2x^2 - x - 2$?

(c) Is $(x-1)$ a factor of x^3+x^2+x+1?

(d) Is $(x+2)$ a factor of x^3+8?

16. Factorise each polynomial completely.

(a) $x^3 - x$

(b) $x^3 - 4x$

(c) $x^3 + 2x^2 - x - 2$

(d) $2x^3 + x^2 - 5x + 2$

17. Answer each part.

(a) One factor of $x^2 - 5x + 6$ is $(x-2)$. State the other.

(b) Show that $x^3 - 3x^2 + 4$ has a repeated factor.

6.4 Solving Polynomial Equations

You already know how to solve linear and quadratic equations. A linear equation is rearranged directly. A quadratic is solved by factorising, by completing the square, or by the quadratic formula.

For a cubic or higher there is no formula of that kind simple enough to use by hand, and from degree five upward none exists at all. The factor theorem takes its place.

KEY To solve a polynomial equation of degree 3 or more:

1. Test the divisors of the constant term until one value a gives $f(a) = 0$.
2. Divide $f(x)$ by $(x-a)$, either by long division or by comparing coefficients.
3. Repeat on the quotient until a quadratic remains, then solve that in the usual way.

Every real root is an x-intercept of the graph, so the algebra and the graph always agree.

EXAMPLE 6.12

Solve $x^3 - 3x^2 - x + 3 = 0$.

Test the divisors of 3. At $x = 1$: $1 - 3 - 1 + 3 = 0$, so $(x-1)$ is a factor. Dividing,

$$x^3 - 3x^2 - x + 3 = (x-1)(x^2 - 2x - 3) = (x-1)(x-3)(x+1).$$

The roots are $x = 1,\ 3,\ -1$.

CAUTION A cubic with real coefficients always has at least one real root, because one end goes to $+\infty$ and the other to $-\infty$, so the curve must cross the axis. If no rational root can be found by testing, one root is irrational, and in this course the equation is then solved with a calculator.

YOUR TURN 6.4 Solving Polynomial Equations

18. Solve each equation. Each is already factorised or factorises at sight.

(a) $(x-1)(x+2)(x-4)=0$
(b) $(x-3)^2(x+1)=0$
(c) $x^3=4x$
(d) $x^3+x^2-2x=0$

19. Solve each equation, using the factor theorem to find the first root.

(a) $x^3-6x^2+11x-6=0$
(b) $x^3-2x^2-x+2=0$
(c) $2x^3+x^2-5x+2=0$

20. Solve each equation.

(a) $x^4-5x^2+4=0$ *(Hint: substitute $u=x^2$)*
(b) $x^3-1=0$, giving only the real root, and explain why there is exactly one.

6.5 Sum and Product of Roots

The coefficients of a polynomial determine the sum and the product of its roots. You can therefore find those two values *without solving the equation at all*. This matters whenever the roots are irrational, or when the equation is too hard to solve, because the sum and the product remain easy to obtain.

6.5.1 The Quadratic Case

The pattern is easiest to see where there are only two roots to follow.

If $ax^2+bx+c=0$ has roots α and β, then the polynomial *is* $a(x-\alpha)(x-\beta)$, and the expansion gives the coefficients:

$$a(x-\alpha)(x-\beta)=a(x^2-(\alpha+\beta)x+\alpha\beta)=ax^2-a(\alpha+\beta)\,x+a\,\alpha\beta.$$

Compare with ax^2+bx+c. The middle coefficient is $b=-a(\alpha+\beta)$ and the constant is $c=a\,\alpha\beta$. Solving each for the sum and the product gives

$$\alpha+\beta=-\frac{b}{a},\qquad \alpha\beta=\frac{c}{a}.$$

The minus sign on the sum is not a convention to memorise. It comes from the brackets, one $-\alpha$ and one $-\beta$.

6.5.2 Every Degree at Once

Nothing in that expansion depended on there being two brackets, so the same argument settles every degree, using the counting argument of Ch. 2. Expanding

$$a_n(x-r_1)(x-r_2)\cdots(x-r_n)$$

means choosing, from each bracket, either the x or the root, exactly as in the binomial theorem. To build the x^{n-1} term you take x from all brackets but one, so each root appears exactly once, each carrying a single minus sign: the coefficient equals $-(r_1+r_2+\cdots+r_n)$, the *sum*. To build the constant term you take the root from *every* bracket: all n roots multiplied together, carrying n minus signs, which is where the $(-1)^n$ comes from. So the outermost coefficients of a polynomial give the sum and the product of its roots.

KEY · IB For the equation $a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$,

$$\text{sum of roots} = -\frac{a_{n-1}}{a_n}, \qquad \text{product of roots} = (-1)^n\,\frac{a_0}{a_n}.$$

The second coefficient is the sum of the roots with its sign changed; the constant term is their product.

These are the outermost of a whole family of relations (Vieta's formulas), but the sum and product are the two you will use most, for checking answers, for building equations with required roots, and for finding an unknown coefficient.

Reading the relations backwards builds an equation from its roots. A quadratic whose roots have sum S and product P is

$$x^2 - Sx + P = 0,$$

so an equation whose roots are a transformation of α and β is found by transforming the sum and the product. The roots themselves are never needed.

One more relation fills the gap between them. Building the x^{n-2} term means taking the root from exactly two brackets and x from the rest, so that coefficient is the sum of the roots multiplied together in pairs. For a cubic $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$ with roots α, β, γ, the three relations together are

$$\alpha+\beta+\gamma = -\frac{a_2}{a_3}, \qquad \alpha\beta+\beta\gamma+\gamma\alpha = \frac{a_1}{a_3}, \qquad \alpha\beta\gamma = -\frac{a_0}{a_3}.$$

The syllabus requires only the first and the last. The middle relation is a short extension, and it is the one to use whenever a question involves all three roots at once.

EXAMPLE 6.13

The equation $2x^2-5x+3=0$ has roots α and β. Find $\alpha+\beta$ and $\alpha\beta$ without solving, then verify.

Here $a = 2$, $b = -5$, $c = 3$, so

$$\alpha + \beta = -\frac{-5}{2} = \frac{5}{2}, \qquad \alpha\beta = \frac{3}{2}.$$

Solving confirms it: the roots are 1 and $\frac{3}{2}$, whose sum is $\frac{5}{2}$ and product is $\frac{3}{2}$. The sum and product were available from the coefficients before the roots were found.

EXAMPLE 6.14

The cubic $x^3 - 2x^2 - 9x + 18 = 0$ has roots α, β, γ. Write down $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$.
With $a_3 = 1$, $a_2 = -2$, $a_0 = 18$ and degree $n = 3$,

$$\alpha + \beta + \gamma = -\frac{-2}{1} = 2, \qquad \alpha\beta\gamma = (-1)^3\,\frac{18}{1} = -18.$$

(The roots happen to be $-3, 2, 3$: sum 2, product -18, as promised.)

6.5.3 Symmetric Combinations of the Roots

The sum and the product are more useful than they appear, because many other expressions in the roots can be rebuilt from them. That is what makes the method valuable when the roots are irrational: the examples below answer precise questions about the roots without ever finding one.

A **symmetric combination** of the roots is an expression whose value does not change when the roots are exchanged. Both $\alpha^2 + \beta^2$ and $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$ are symmetric; $\alpha - \beta$ is not. Every symmetric combination of two roots can be written in terms of $\alpha + \beta$ and $\alpha\beta$. Two rearrangements cover most cases, and a sum of reciprocals reduces by putting it over a common denominator, $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{\alpha + \beta}{\alpha\beta}$.

KEY

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \qquad (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta.$$

Neither is a new formula. Each is the expansion of $(\alpha \pm \beta)^2$ rearranged.

EXAMPLE 6.15

The equation $x^2 - 3x + 1 = 0$ has roots α and β. Without solving, find $\alpha^2 + \beta^2$ and $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$.
The roots are irrational, but their sum and product are not: $\alpha + \beta = 3$ and $\alpha\beta = 1$. For the sum of squares, use the first rearrangement above:

$$(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2 \implies \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 - 2 = 7.$$

For the sum of reciprocals, combine over a common denominator:

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{3}{1}=3.$$

Neither answer required a root. Both were rebuilt from the sum and the product alone.

EXAMPLE 6.16

With α, β as above, find a quadratic equation with integer coefficients whose roots are 2α and 2β.

A quadratic is determined by the sum and product of its roots: $x^2-(\text{sum})x+(\text{product})=0$. The new sum and product follow from the old ones:

$$2\alpha+2\beta=2(\alpha+\beta)=6, \qquad (2\alpha)(2\beta)=4\alpha\beta=4.$$

So the required equation is

$$x^2-6x+4=0.$$

No root was ever computed: transforming an equation's roots means transforming their sum and product, nothing more.

EXAMPLE 6.17

The roots of $x^2+kx+12=0$ differ by 1. Find the possible values of k.

Let the roots be α and β with $\alpha-\beta=1$. The coefficients give $\alpha+\beta=-k$ and $\alpha\beta=12$. Connect the difference to these totals by squaring it:

$$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta \implies 1=k^2-48,$$

so $k^2=49$ and $k=\pm 7$. Check with $k=7$: the equation $x^2+7x+12=0$ has roots -3 and -4, which indeed differ by 1. A condition on the roots has become an equation in the coefficient.

YOUR TURN 6.5 Sum and Product of Roots

21. State the sum and the product of the roots of each equation.

(a) $x^2-7x+10=0$
(b) $2x^2+3x-5=0$
(c) $x^2+4x+4=0$
(d) $3x^2-2x-8=0$

22. For each cubic or quartic, state the sum and the product of the roots.

(a) $x^3-6x^2+11x-6=0$
(b) $2x^3-4x^2+x-3=0$

(c) $x^4 - 3x^3 + 2x - 5 = 0$

23. Build the required equation.

(a) Write a quadratic with lead coefficient 1 whose roots sum to 5 and multiply to 6.
(b) The roots of $x^2 + bx + 12 = 0$ are 3 and one other. Find the other root and b.
(c) Find the value of k for which $x^2 - kx + 9 = 0$ has equal roots.
(d) Write a cubic with lead coefficient 1 whose roots are -1, 2 and 4.

24. The roots of $2x^2-7x+6 = 0$ are α and β. Without solving the equation, find each value.

(a) $\alpha + \beta$ and $\alpha\beta$
(b) $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$
(c) $\alpha^2 + \beta^2$
(d) $(\alpha - \beta)^2$

25. The roots of $x^2 + 5x + 3 = 0$ are α and β.

(a) Write down $\alpha + \beta$ and $\alpha\beta$.
(b) Find the sum and product of $\alpha + 2$ and $\beta + 2$.
(c) Hence find a quadratic equation with integer coefficients whose roots are $\alpha + 2$ and $\beta + 2$.

26. The roots of $x^3 - 4x^2 + x + 6 = 0$ are α, β and γ.

(a) Write down $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$.
(b) Write down $\alpha\beta + \beta\gamma + \gamma\alpha$.
(c) Hence find $\alpha^2 + \beta^2 + \gamma^2$.

6.6 Partial Fractions

Adding two simple fractions produces one complicated fraction. **Partial fractions** reverse that process: they split a single fraction back into a sum of simpler ones, one for each factor of the denominator. The split form is easier to use, and it is what makes a rational function possible to integrate. See Ch. 15.

The syllabus limits the problem to a denominator of **at most two distinct linear factors**, with the numerator of lower degree than the denominator. That is the case treated below. The larger theory is sketched at the end of the section for reference.

The denominator is a product of two different linear factors, and each factor contributes one term with a constant numerator.

KEY For a proper fraction over two distinct linear factors,

$$\frac{px+q}{(x-\alpha)(x-\beta)} = \frac{A}{x-\alpha} + \frac{B}{x-\beta}.$$

To find A and B, multiply through by the denominator, then substitute the values of x that make one factor zero. Each substitution isolates one unknown.

EXAMPLE 6.18

Express $\dfrac{3x+5}{(x-1)(x+2)}$ in partial fractions.

Write

$$\frac{3x+5}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2},$$

and multiply through by $(x-1)(x+2)$:

$$3x+5 = A(x+2) + B(x-1).$$

Setting $x = 1$ isolates A: $8 = 3A$, so $A = \frac{8}{3}$. Setting $x = -2$ isolates B: $-1 = -3B$, so $B = \frac{1}{3}$. Therefore

$$\frac{3x+5}{(x-1)(x+2)} = \frac{8}{3(x-1)} + \frac{1}{3(x+2)}.$$

Choosing the two values that each make one factor zero is what makes the method quick. It turns a pair of simultaneous equations into two one-line calculations.

EXAMPLE 6.19

Express $\dfrac{x+5}{x^2-x-2}$ in partial fractions.

This denominator is not factorised, so factorise it first:

$$x^2 - x - 2 = (x-2)(x+1).$$

Now proceed as before:

$$\frac{x+5}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1} \implies x+5 = A(x+1) + B(x-2).$$

Setting $x = 2$: $7 = 3A$, so $A = \frac{7}{3}$. Setting $x = -1$: $4 = -3B$, so $B = -\frac{4}{3}$. Hence

$$\frac{x+5}{x^2-x-2} = \frac{7}{3(x-2)} - \frac{4}{3(x+1)}.$$

TIP Examination questions usually give the denominator in expanded form. The factorising step is part of the answer, so write it out.

EXT **Beyond the syllabus.** Three things extend the method. None of them is on the AA HL syllabus, although the first is short enough to try in Your Turn Q30. A fraction whose numerator has degree equal to or higher than its denominator is **improper**, and must be divided first: $\dfrac{x^2}{x^2-1} = 1 + \dfrac{1}{x^2-1}$, and only the proper part is split. A **repeated** factor $(x-\alpha)^2$ needs one term for each power, $\dfrac{A}{x-\alpha} + \dfrac{B}{(x-\alpha)^2}$. An **irreducible quadratic**, one with negative discriminant, takes a linear numerator $\dfrac{Bx+C}{x^2+bx+c}$. One rule covers all of them: give each factor a numerator of degree one less than itself, and an extra term for every power of a repeated factor.

Factor in the denominator	**Term(s) it contributes**
linear $(x-\alpha)$	$\dfrac{A}{x-\alpha}$
repeated linear $(x-\alpha)^2$	$\dfrac{A}{x-\alpha} + \dfrac{B}{(x-\alpha)^2}$
irreducible quadratic (x^2+bx+c)	$\dfrac{Bx+C}{x^2+bx+c}$

YOUR TURN 6.6 Partial Fractions

27. Express each fraction in partial fractions.

(a) $\dfrac{1}{(x-1)(x-2)}$ (b) $\dfrac{5}{(x-2)(x+3)}$

(c) $\dfrac{3}{(x+1)(x+4)}$ (d) $\dfrac{x}{(x-1)(x+1)}$

28. Express each fraction in partial fractions.

(a) $\dfrac{2x+1}{x(x+1)}$ (b) $\dfrac{x+4}{x(x-2)}$

(c) $\dfrac{2x}{(x-3)(x+1)}$ (d) $\dfrac{5x-1}{(x-1)(x+2)}$

29. Factorise the denominator first, then express each fraction in partial fractions.

(a) $\dfrac{4x}{x^2-4}$ (b) $\dfrac{x+7}{x^2+x-2}$

30. Consider $\dfrac{x^2}{x^2-1}$.

(a) Explain why this fraction is improper.

(b) Divide to write it as $1 + \dfrac{1}{x^2-1}$.

(c) Hence express the whole fraction in partial fractions.

Chapter 6 · Polynomials

Reflections

TOK Abel and Galois proved something surprising. The roots of a general quintic *exist*, but they cannot be written using only addition, subtraction, multiplication, division and roots. The equation $x^5 - x - 1 = 0$ has a real solution. That number is completely determined, and a computer can calculate it to a thousand decimal places. Even so, no formula of that kind can express it exactly. What does it mean, then, to "know" a number? Is such a root any less real than one you can write down?

IM The history behind this chapter is unusual. In Italy in the 1530s, mathematicians competed in public problem-solving contests, and a loser could lose his university post. Methods were therefore kept secret. Niccolò Tartaglia found a way to solve the cubic and told no one. Gerolamo Cardano persuaded him to share it under an oath of silence, then published it in 1545, together with his student Ferrari's solution of the quartic. Almost three hundred years passed before Niels Abel, who died of tuberculosis at twenty-six, proved that no such formula exists for the quintic. Évariste Galois, killed in a duel at twenty, explained exactly which equations can be solved by a formula and which cannot. To do this he created *group theory*, the mathematical study of symmetry, which is now used in particle physics and cryptography.

RW Polynomials are used constantly in computing, because they need only multiplication and addition. Those are the two operations a processor performs fastest. The curved shapes in fonts and animation are drawn with *Bézier curves*, which are polynomial curves whose shape is controlled by a few chosen points. *Error-correcting codes* are built from polynomials as well. These are methods of adding extra information to a message so that it can still be read after part of it is damaged, which is how a scratched disc still plays and how a faint signal from a spacecraft is recovered. Polynomials are also used to fit smooth models to scattered measurements in science and finance.

EXT The factor theorem raises a question it cannot answer. How many roots does a polynomial really have? Over the real numbers, $x^2 + 1$ has none. The **Fundamental Theorem of Algebra** gives the answer. Over the *complex* numbers of Chapter 17, every polynomial of degree n has exactly n roots, counted with multiplicity. The sum and product formulas then hold for every polynomial, because the roots they describe always exist.

End-of-Chapter Problems

1. Consider $f(x) = x^3 - 2x^2 - 5x + 6$.

(a) Show that $x = 1$ is a root. [2]
(b) Factorise $f(x)$ completely. [3]
(c) Solve $f(x) = 0$. [1]

2. Consider $f(x) = 2x^3 + 3x^2 - 11x - 6$.

(a) Show that $(x - 2)$ is a factor. [2]
(b) Factorise $f(x)$ completely. [3]

3. Find the unknown coefficients in each polynomial.

(a) When $x^3 + ax + b$ is divided by $(x - 1)$ the remainder is 4, and $(x + 1)$ is a factor. Find a and b. [5]
(b) The polynomial $x^3 + ax^2 + bx - 2$ has $(x - 1)$ and $(x - 2)$ as factors. Find a and b. [5]
(c) The remainder when $x^3 + cx + d$ is divided by $(x - 1)$ is 6, and $(x - 2)$ is a factor. Find c and d. [5]

4. The polynomial $x^3 - 3x^2 + kx - 8$ has $(x - 2)$ as a factor. Find k. [3]

5. Sketch each graph, showing all intercepts and describing the behaviour at each root.

(a) $y = (x + 1)(x - 1)(x - 3)$ [4]
(b) $y = x^2(x - 3)$ [4]

6. Build the equation described in each part.

(a) A quadratic $x^2 + px + q$ has roots 2 and -5. Find p and q. [3]
(b) The cubic $x^3 + px^2 + qx + r$ has roots -1, 2 and 4. Find p, q and r. [5]
(c) The quadratic with roots α and β is $x^2 - 4x + 1$. Find a quadratic with leading coefficient 1 whose roots are $\alpha + 1$ and $\beta + 1$. [5]

7. Consider $f(x) = x^3 + 2x^2 - x - 2$.

(a) Factorise $f(x)$ completely. [3]
(b) Solve $f(x) = 0$. [1]

8. Solve each equation.

(a) $x^3 - 7x + 6 = 0$ [4]
(b) $2x^3 + 3x^2 - 3x - 2 = 0$ [5]

9. Each equation has roots α and β. Answer without solving the equation.

(a) $x^2 - 5x + 3 = 0$: find $\alpha^2 + \beta^2$. [4]
(b) $3x^2 - x - 2 = 0$: find $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$. [3]
(c) $x^2 - 6x + 7 = 0$: find $\alpha^2 + \beta^2$ and $(\alpha - \beta)^2$. [5]

10. A cubic has roots 1, 2 and 3.

(a) Write it in factored and expanded form (leading coefficient 1). [3]
(b) State the sum and product of the roots and verify them from the expanded form. [2]

11. Consider $f(x) = x^4 - 5x^2 + 4$.

(a) Solve $f(x) = 0$. [3]
(b) State how many x-intercepts the graph has. [1]

12. Show that $x = 1$ is a root of $x^3 + 4x^2 + x - 6 = 0$, and find the other roots. [4]

13. The equation $x^2 - kx + 9 = 0$ has equal roots. Use the sum and product of roots to find k. [4]

14. Consider $f(x) = 2x^3 - x^2 - 13x - 6$.

(a) Show that $(x + 2)$ is a factor. [2]
(b) Factorise $f(x)$ completely and solve $f(x) = 0$. [4]

15. The polynomial $x^3 - kx + 2$ has $(x - 2)$ as a factor.

(a) Find k. [2]
(b) Find the other two roots, in exact form. [4]

16. The polynomial $x^3 - 2x^2 + x - 2$ factorises as $(x - 2)(x^2 + 1)$.

(a) Verify this by expansion. [2]
(b) State the only real root, and explain why there are no others. [2]

17. When a polynomial $p(x)$ is divided by $(x - 1)$ the remainder is 3, and by $(x - 2)$ the remainder is 5. Find the remainder when $p(x)$ is divided by $(x - 1)(x - 2)$. [6]

18. The roots of $x^3 - 6x^2 + 11x - 6 = 0$ are α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [5]

19. Prove that if a polynomial with integer coefficients has a rational root $\frac{p}{q}$ in lowest terms, then p divides the constant term and q divides the leading coefficient. [6]

20. The cubic $x^3 + px + q = 0$ has a repeated root. Show that $4p^3 + 27q^2 = 0$. [6]

21. Find all real values of k for which $x^3 - 3x + k = 0$ has three distinct real roots. *(Hint: use Q20 to find the values of k that give a repeated root)* [6]

22. Consider $f(x) = 2x^3 - 3x^2 - 11x + 6$.

(a) Show that $(x - 3)$ is a factor of $f(x)$. [2]
(b) Factorise $f(x)$ completely. [4]
(c) Solve $f(x) = 0$. [2]
(d) State the y-intercept, and describe the behaviour of $f(x)$ as $x \to \pm\infty$. [3]
(e) Sketch $y = f(x)$, showing all intercepts. [2]
(f) Hence solve the inequality $f(x) \leq 0$. [3]

23. The polynomial $p(x) = x^3 + ax^2 + bx + 6$ leaves a remainder of 12 when divided by $(x - 2)$, and $(x + 1)$ is a factor.

(a) Show that $4a + 2b = -2$ and $a - b = -5$. [4]
(b) Hence find a and b. [2]
(c) Factorise $p(x)$ as far as possible over the real numbers. [4]
(d) State how many real roots $p(x) = 0$ has, giving a reason. [2]
(e) Find the remainder when $p(x)$ is divided by $(x - 1)(x + 1)$. *(Hint: the remainder has the form $rx + s$)* [4]

24. The cubic equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ.

(a) Write down $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ in terms of p, q and r. [3]
(b) The equation $x^3 - 6x^2 + 3x + 10 = 0$ has roots α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [4]
(c) Find $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. [3]
(d) Given that one root is -1, find the other two. [3]
(e) Write down a cubic equation with integer coefficients whose roots are 2α, 2β and 2γ. [3]

25. Express each fraction in partial fractions.

(a) $\dfrac{7x - 1}{(x - 1)(x + 2)}$ [4]
(b) $\dfrac{1}{x^2 - 4}$ [4]

(c) $\dfrac{x+5}{x^2+x}$ [4]

Challenge Questions

26. Power sums of the roots. The equation $x^2 - px + q = 0$ has roots α and β. For each integer $k \geq 0$ define

$$S_k = \alpha^k + \beta^k.$$

(a) Write down the values of S_0 and S_1. [2]

(b) Show that $S_2 = p^2 - 2q$. [2]

(c) Find S_3 in terms of p and q. [3]

(d) Explain why $\alpha^k = p\,\alpha^{k-1} - q\,\alpha^{k-2}$ for every $k \geq 2$, and deduce that $S_k = p\,S_{k-1} - q\,S_{k-2}$. [4]

(e) The equation $x^2 - 3x + 1 = 0$ has roots α and β. Use the recurrence to find S_2, S_3 and S_4. [4]

(f) The roots of $x^2 - 3x + 1 = 0$ are irrational. Explain why S_k is nevertheless an integer for every k. [3]

27. Palindromic polynomials. A polynomial is *palindromic* if its list of coefficients reads the same forwards as backwards. For example $x^4 + x^3 - 4x^2 + x + 1$ has coefficients $1, 1, -4, 1, 1$.

(a) Explain why $x = 0$ is never a root of a palindromic polynomial. [2]

(b) Let $P(x)$ be palindromic of degree n. Show that $x^n P\left(\dfrac{1}{x}\right) = P(x)$, and deduce that if α is a root of P then so is $\dfrac{1}{\alpha}$. [4]

(c) Divide $x^4 + x^3 - 4x^2 + x + 1 = 0$ through by x^2 and substitute $u = x + \dfrac{1}{x}$. Show that the equation becomes $u^2 + u - 6 = 0$. [4]

(d) Hence find all four roots of $x^4 + x^3 - 4x^2 + x + 1 = 0$ in exact form. [5]

(e) Show that every palindromic polynomial of odd degree has $x = -1$ as a root. [3]

28. Splitting a sum with partial fractions. Partial fractions are usually a step towards integration. This investigation uses them for something else: adding infinitely many terms exactly.

(a) Show that $\dfrac{1}{n(n+1)} = \dfrac{1}{n} - \dfrac{1}{n+1}$ for every positive integer n. [2]

(b) Hence write out the sum $\displaystyle\sum_{n=1}^{N} \frac{1}{n(n+1)}$ term by term, and show that almost every term cancels. Deduce that the sum equals $\dfrac{N}{N+1}$. *(Hint: write the first three and the last two* *terms)* [4]

(c) State the value the sum approaches as $N \to \infty$, and explain what that value means for the graph of the partial sums. [2]

(d) Show that $\dfrac{1}{n(n+2)} = \dfrac{1}{2}\left(\dfrac{1}{n} - \dfrac{1}{n+2}\right)$, and hence find $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$. [4]

(e) Investigate $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+k)}$ for a positive integer k. Find its value for $k = 3$ and $k = 4$, then state a general formula and justify it. [6]

GEOMETRY & TRIGONOMETRY

CHAPTER 7

Shapes & Measures

7

SYLLABUS 3.1 3.2 3.3 3.4

You cannot pace out the Earth. So how did anyone find its size? Not by walking. The answer came from something much smaller: an angle.

Check the fact yourself. On flat ground the horizon is level with your eye. Climb a hill and it drops slightly below your eye line, and the higher you climb, the further it drops. That drop is the **dip**, and it is small: about half a degree from a good hill, the width of the Moon in the sky.

Your line of sight does not cut into the Earth. It touches the surface and passes on, meeting the radius at that point at a right angle. So the hilltop, the horizon and the centre of the Earth form one right-angled triangle, with hill height h and dip θ:

$$\cos\theta = \frac{R}{R+h}, \qquad \text{so} \qquad R = \frac{h\cos\theta}{1-\cos\theta}.$$

Measure the hill, measure the dip, and the radius comes out close to the modern 6371 km, from one hill, in one afternoon. Al-Bīrūnī did exactly this a thousand years ago, from the fort of Nandana in what is now Pakistan.

Size decides more than shape. Double an object and its surface grows fourfold while its volume grows eightfold, so bigger things carry less surface for what they hold. That is why crushed ice melts fast, and why no insect could be scaled up to the size of a horse.

By the end of this chapter you will find the angle between a line and a plane in a solid, work out the volume and surface area of standard solids, solve any triangle with the sine and cosine rules, ambiguous case included, handle elevation, depression and bearings, and measure in **radians**, which turn arc length and sector area into one multiplication each.

> A calculator is assumed throughout this chapter. The EXACT badge marks the questions whose answer must be left exact.

7.1 Three-Dimensional Geometry

The measurements you need from a solid are often not the ones marked on its faces. The diagonal of a room, or the angle a roof strut makes with the floor: neither of these lies along a face. Both run through the inside of the object, and coordinates are how you reach them.

Before any measuring, a word on names. Examination papers use a fixed notation for points, segments, lines and angles, and they use it without explanation.

Notation	Meaning
$A(x, y)$	the point A with coordinates x and y
$[AB]$	the line segment with endpoints A and B
AB	the length of $[AB]$
(AB)	the line through A and B, continuing both ways
$A\hat{B}C$	the angle at B, between $[BA]$ and $[BC]$
$\triangle ABC$	the triangle with vertices A, B and C

The middle letter is the one to hold on to: it names the vertex. So $A\hat{B}C$ and $C\hat{B}A$ are the same angle, while $B\hat{A}C$ is a different one. Note too that the length of a segment carries no bars. Bars are reserved for the magnitude of a vector, which Chapter 9 introduces.

Coordinates in the plane extend to space by adding a third axis, z. The distance formula extends with them. In two dimensions it is Pythagoras' theorem. In three dimensions it is Pythagoras used *twice*: once across the base, then once upward.

> KEY · IB The distance between (x_1, y_1, z_1) and (x_2, y_2, z_2) is
>
> $$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$
>
> The midpoint is the average of the coordinates, $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}, \dfrac{z_1 + z_2}{2}\right)$.

Pythagoras is used twice. Across the base it gives

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Treat that as one side of a new right triangle. Its other side is the vertical rise, $(z_2 - z_1)$, and Pythagoras on this second triangle gives the formula.

The equation of a sphere comes from the same formula. A sphere is the set of points at a fixed distance r from a centre (a, b, c), so its equation is $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$.

7.1.1 The angle between a line and a plane

A line that is not parallel to a plane meets it at an angle. Many angles can be measured at that meeting point, so one of them has to be chosen, and mathematics chooses the smallest. Picture a lamp directly above the line, shining straight down. The line casts a shadow on the plane, and the angle between the line and its own shadow is the smallest angle available. That shadow is called the **projection** of the line onto the plane.

DEF The **angle between a line and a plane** is the angle between the line and its projection on the plane. To find it, drop a perpendicular from a point on the line down to the plane. The line, its projection and that perpendicular form a right triangle, and right-angled trigonometry solves it.

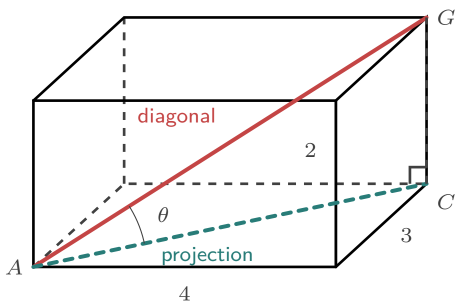


Figure 7.1 The space diagonal AG and its shadow AC on the base. The vertical edge CG meets the base at a right angle, so ACG is a right triangle and θ is the angle between the diagonal and the base.

EXAMPLE 7.1

A cuboid has a 4×3 base and height 2. Find the angle between the space diagonal AG and the base.

The diagonal of the base is $AC = \sqrt{4^2 + 3^2} = 5$. This is the projection of AG onto the base, and $CG = 2$ is vertical, meeting the base at a right angle. In the right triangle ACG,

$$\tan\theta = \frac{CG}{AC} = \frac{2}{5} \implies \theta = \arctan(0.4) \approx 21.8^\circ.$$

The order is the same each time. Find the projection on the plane, then the perpendicular height, then use $\tan$.

7.1.2 The angle between two intersecting lines

Two lines that meet also make an angle, and inside a solid those two lines are usually edges from the same corner. Take the two edges, then join their far ends. The three segments form a triangle, and the angle you want sits at the corner. Measure the three sides with the distance formula, then solve the triangle.

EXAMPLE 7.2

A square-based pyramid has base corners $A(0,0,0)$, $B(6,0,0)$, $C(6,6,0)$, $D(0,6,0)$ and apex $V(3,3,4)$. Find the angle between the slant edges VA and VB.

The two edges meet at V and span the triangle VAB, so the angle we want is $A\hat{V}B$. First measure the triangle. By the distance formula,

$$VA = \sqrt{(3-0)^2 + (3-0)^2 + (4-0)^2} = \sqrt{34},$$

and $VB = \sqrt{34}$ by the same calculation, while $AB = 6$ along the base.

The triangle is isosceles, so cut it in half. The midpoint of AB is $M(3,0,0)$, and VM meets AB at a right angle. In the right triangle VMA, the side $AM = 3$ is opposite the half-angle at V, and the hypotenuse is $VA = \sqrt{34}$:

$$\sin\frac{\theta}{2} = \frac{3}{\sqrt{34}} \implies \frac{\theta}{2} \approx 31.0^\circ \implies \theta \approx 61.9^\circ.$$

The two formulas of this section carry the whole solution: the distance formula for the three sides, the midpoint to cut the triangle in half, then right-angled trigonometry for the angle.

YOUR TURN 7.1 Three-Dimensional Geometry

1. Find the distance between each pair of points.

(a) $(1,2,3)$ and $(4,6,3)$

(b) $(0,0,0)$ and $(2,3,6)$

(c) $(-1,0,2)$ and $(2,-4,2)$

(d) $(2,-1,4)$ and $(5,3,4)$

2. Answer each part.

(a) Find the midpoint of $(1,2,3)$ and $(5,4,1)$.

(b) Find the midpoint of $(2,-1,4)$ and $(6,3,0)$.

3. Write the equation of each sphere.

(a) centre $(0,0,0)$, radius 5

(b) centre $(1,2,3)$, radius 4

(c) centre $(2,-1,3)$, radius 6

(d) centre $(0,0,2)$, radius $\sqrt{7}$

4. A cuboid has base 3×4 and height 12.

(a) Find the length of the diagonal of the base.
(b) Find the length of the space diagonal.
(c) Find the angle between the space diagonal and the base.

7.2 Volumes and Surface Areas of Solids

You already know the cuboid, the prism and the cylinder. Three curved solids appear beside them again and again. What matters is where each formula comes from, because a formula you can rebuild is a formula you will not misapply.

KEY · IB

Sphere, radius r	$V = \frac{4}{3}\pi r^3, \quad A = 4\pi r^2$
Cone, radius r, height h	$V = \frac{1}{3}\pi r^2 h$, curved $A = \pi r l$
Pyramid, base area B, height h	$V = \frac{1}{3}Bh$
Cylinder, radius r, height h	$V = \pi r^2 h$, curved $A = 2\pi r h$

The cylinder is listed in the *Prior learning* table at the front of the booklet rather than beside the other three.

Each formula has a short reason behind it.

- **Cylinder.** Cut the curved side along one vertical line and unroll it into a rectangle of height h and width $2\pi r$, the base circumference. Its area is $2\pi r h$.
- **Cone.** The same cut unrolls the curved side into a sector. Its radius is the **slant height** l, measured from the rim to the tip, and its arc is the base circumference $2\pi r$. A sector's area is half its arc times its radius, so the curved area is $\frac{1}{2}(2\pi r)(l) = \pi r l$.
- **The $\frac{1}{3}$.** A cube cuts into three identical square pyramids. Each one stands on a face of the cube, and all three tips meet at the opposite corner. Each is therefore a third of the cube, and the same fraction holds for every pyramid and every cone: $V = \frac{1}{3}Bh$.
- **Sphere.** Put a sphere inside the smallest cylinder that holds it, of radius r and height $2r$. Archimedes proved the sphere is two thirds of that cylinder. The cylinder holds $\pi r^2(2r) = 2\pi r^3$, so the sphere holds $\frac{4}{3}\pi r^3$. Its surface, $4\pi r^2$, is the area of four **great circles** (a circle round the sphere at its widest).

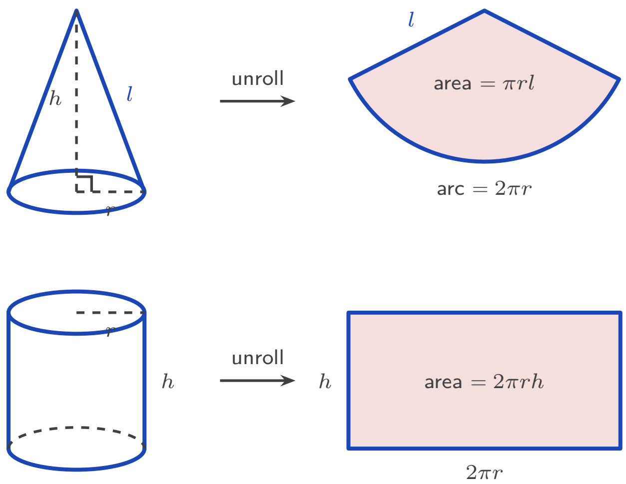


Figure 7.2 Cut the curved surface of a cone along one slant line and it opens into a sector of radius l whose arc is the base circumference. Cut a cylinder the same way and it opens into a rectangle. Each curved-area formula is the area of the flat shape it opens into.

Two mistakes are common. The cone's curved surface uses the slant height l, not the vertical height h; Pythagoras links them by $l = \sqrt{r^2 + h^2}$. And "total surface area" means every face: a solid hemisphere has a curved part $2\pi r^2$ *and* a flat base πr^2, giving $3\pi r^2$, while the same hemisphere as an open bowl has only $2\pi r^2$.

EXAMPLE 7.3

A solid is a cone of base radius 8 and height 6 sitting on top of a hemisphere of radius 8. Find its total surface area.

The exposed surface is the cone's curved part plus the hemisphere's curved part. The two flat circles where they join are hidden inside, so neither counts. The cone has slant height

$$l = \sqrt{8^2 + 6^2} = 10,$$

so its curved area is $\pi r l = 80\pi$. The hemisphere's curved area is $2\pi r^2 = 128\pi$, and the total is

$$80\pi + 128\pi = 208\pi \approx 653.$$

Deciding which faces are exposed is the harder step; the formulas themselves are routine.

7.2.1 Similar solids

Every formula above is a constant times a length squared, for an area, or a length cubed, for a volume. That fixes what happens when a solid is enlarged.

Two solids are **similar** when they have the same shape and all corresponding lengths are in the same ratio. That ratio is the **linear scale factor** k. Multiplying every length by k multiplies $4\pi r^2$ by k^2 and $\frac{4}{3}\pi r^3$ by k^3. Every formula here behaves the same way, because each depends on length raised to one fixed power.

KEY **Scale factors.** If two similar figures have lengths in the ratio k, then

$$\frac{\text{length}_2}{\text{length}_1} = k, \qquad \frac{\text{area}_2}{\text{area}_1} = k^2, \qquad \frac{\text{volume}_2}{\text{volume}_1} = k^3.$$

One ratio of lengths settles every area and every volume.

A cone cut parallel to its base is the case to recognise. The small cone above the cut has the same apex and the same slope, so it is similar to the whole, and one ratio of heights gives its radius, its surface and its volume at once.

EXAMPLE 7.4

A cone has base radius 9 cm and height 12 cm. A cut parallel to the base is made 4 cm below the apex. Find the radius of the circular cut and the volume of the small cone above it.

The small cone is similar to the whole cone, and the two heights are known, so

$$k = \frac{4}{12} = \frac{1}{3}.$$

The radius is a length, so it scales by k:

$$r = \tfrac{1}{3}(9) = 3 \text{ cm}.$$

Similar triangles say the same thing directly: the vertical axis and the radius of the cut form a triangle similar to the one made by the axis and the base radius, so $\dfrac{r}{4} = \dfrac{9}{12}$.

The whole cone holds

$$V = \tfrac{1}{3}\pi(9)^2(12) = 324\pi \text{ cm}^3,$$

and volume scales by k^3, so the small cone holds

$$\left(\tfrac{1}{3}\right)^3(324\pi) = \tfrac{1}{27}(324\pi) = 12\pi \text{ cm}^3.$$

Computing it directly gives $\tfrac{1}{3}\pi(3)^2(4) \quad = \quad 12\pi$, which agrees. The piece left below the cut is not similar to either cone, so no scale factor applies to it; its volume is the difference, $324\pi - 12\pi = 312\pi$ cm^3.

YOUR TURN 7.2 Volumes and Surface Areas of Solids

5. EXACT Find the volume of each solid. Leave your answers in terms of π where appropriate.

(a) sphere of radius 3

(b) cone of radius 3, height 4

(c) pyramid with base area 20, height 9

(d) cylinder of radius 4, height 10

6. EXACT Find the surface area asked for in each case.

(a) the surface area of a sphere of radius 5

(b) the curved surface area of a cone of radius 5, slant height 13

(c) the total surface area of a solid hemisphere of radius 4

(d) the curved surface area of a cylinder of radius 4, height 10

7. EXACT A cone has radius 6 and height 8.

(a) Find its slant height.
(b) Hence find its curved surface area.

8. EXACT A solid is a cone of radius 3 and height 4 standing on a hemisphere of radius 3.

(a) Explain which surfaces are exposed.
(b) Find the total surface area, in terms of π.
(c) Find the total volume, in terms of π.

7.3 The Sine and Cosine Rules

Right-angled triangles are solved with $\sin$, $\cos$ and $\tan$. Name the sides opposite, adjacent and hypotenuse relative to an angle, and $\sin\theta = \frac{o}{h}$, $\cos\theta = \frac{a}{h}$, $\tan\theta = \frac{o}{a}$. Most triangles have no right angle. Two rules extend trigonometry to all of them, and between them they solve any triangle from three known parts.

Both rules use one labelling. A side takes the lower-case letter of the angle *opposite* it, so side a faces angle A.

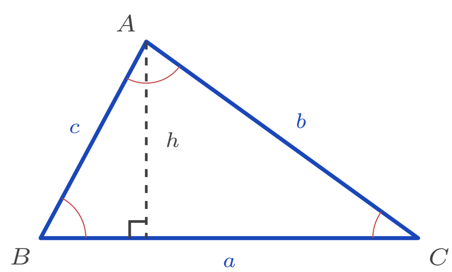


Figure 7.3 Each side is named after the angle opposite it: side a faces angle A, and so on. The dashed altitude h splits the triangle into two right triangles, which is where the sine rule comes from.

7.3.1 The sine rule

Drop a perpendicular of height h from A onto side a, as in the figure above. It cuts the triangle into two right triangles, and in each one h is the side opposite a known angle. On the left, $h = c\sin B$. On the right, $h = b\sin C$. Both expressions measure the same segment, so $c\sin B = b\sin C$, which rearranges to $\frac{b}{\sin B} = \frac{c}{\sin C}$. Dropping a perpendicular from a different corner gives the third ratio.

KEY · IB **Sine rule.** In any triangle,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

The sine rule pairs a side with the angle facing it. Use it whenever you know one complete pair, plus one more side or angle. The standard shorthand writes S for a known side and A for a known angle, in the order they occur round the triangle. These are the cases AAS, ASA and SSA.

7.3.2 The ambiguous case

The sine rule needs one extra check when you use it to find an *angle*.

Suppose you know two sides and an angle that is *not* between them. This is the SSA case. The sine rule gives an equation of the form $\sin B = k$, and between $0°$ and $180°$ that equation has two solutions: the angle B and its supplement $180° - B$. An angle and its supplement have the same sine, so the equation cannot separate them.

Your calculator returns only the first. Test the second yourself. Add it to the angle you were given: if the total is still below $180°$, a positive third angle remains, so a second triangle exists and both answers are correct.

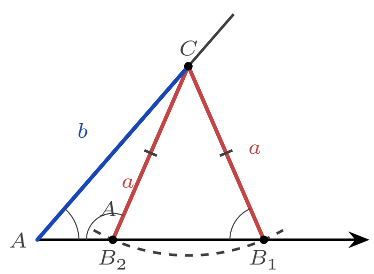


Figure 7.4 Two sides and a non-included angle. Swinging the fixed length a from C meets the base at B_2 and at B_1, so two different triangles share the same a, b and A. The angles at B_2 and B_1 are supplementary — here $113.5°$ and $66.5°$ — which is why the sine rule returns both.

EXAMPLE 7.5

In triangle ABC, $a = 9$, $b = 11$, and $A = 40°$. Find angle B.

By the sine rule, $\dfrac{\sin B}{11} = \dfrac{\sin 40^\circ}{9}$, so

$$\sin B = \frac{11 \sin 40^\circ}{9} \approx 0.786.$$

The calculator gives $B \approx 51.8^\circ$. The supplement is $180^\circ - 51.8^\circ = 128.2^\circ$, so test it: $A + B = 40^\circ + 128.2^\circ = 168.2^\circ$, which is below 180°. A third angle of 11.8° remains, so that triangle exists too.

Both values of B are correct. There are two triangles, and a complete answer reports both.

CAUTION This check is needed only when the sine rule gives an *angle*. If you are finding a *side*, or using the cosine rule, no supplement arises. The cosine rule returns an angle between 0° and 180° directly, because cos is negative for obtuse angles and positive for acute ones, so it distinguishes the two.

7.3.3 The cosine rule

When the known angle sits *between* the two known sides, no side is paired with the angle opposite it, so the sine rule cannot start. The cosine rule covers this case. Read it as Pythagoras' theorem with a correction term for the angle not being 90°.

Coordinates prove it directly. Put the corner C at the origin with side a along the x-axis, so B sits at $(a, 0)$. The point A lies a distance b from the origin, in the direction C. By the definitions of cosine and sine, its coordinates are $(b \cos C,\ b \sin C)$. The distance formula now measures the third side:

$$c^2 = (a - b\cos C)^2 + (b \sin C)^2 = a^2 - 2ab\cos C + b^2\cos^2 C + b^2 \sin^2 C,$$

and $\cos^2 C + \sin^2 C = 1$, so the last two terms add to b^2.

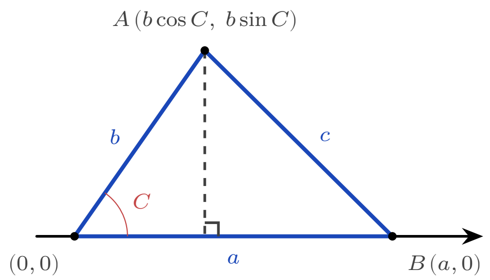


Figure 7.5 Placing C at the origin with a along the axis fixes all three corners. Only A needs trigonometry, and the distance from A to B is then the cosine rule.

KEY · IB **Cosine rule.** In any triangle,

$$c^2 = a^2 + b^2 - 2ab\cos C, \qquad \text{equivalently} \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

The letters rotate. Written about a different corner the same rule reads $a^2 = b^2 + c^2 - 2bc\cos A$, and the pattern is the same in each version: the side opposite the named angle stands alone on the left.

When $C = 90^\circ$, $\cos C = 0$ and the rule becomes $c^2 = a^2 + b^2$, which is Pythagoras' theorem. The first form finds a third side from two sides and the angle between them (SAS). The rearranged form finds an angle from all three sides (SSS).

7.3.4 The area of a triangle

The familiar area $\frac{1}{2}$ base $\times$ height becomes a trigonometric formula as soon as the height is written in terms of a side. Take b as the base. The perpendicular height from B down to it is the opposite side of the right triangle at C, so $h = a\sin C$ and

$$\text{Area} = \tfrac{1}{2}\, b\,(a\sin C).$$

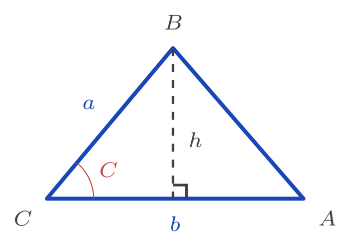


Figure 7.6 The height of the triangle on base b is the opposite side of the right triangle at C, so it is $a\sin C$. Substituting it into $\frac{1}{2} \times$ base $\times$ height gives the formula.

KEY · IB **Area of a triangle.** Area $= \dfrac{1}{2}ab\sin C$, using two sides and the angle between them.

This formula needs exactly what the cosine rule needs, so one sketch usually answers both questions at once.

EXAMPLE 7.6

A triangular field has $AB = 90$ m, $AC = 70$ m, and the angle at A is 110°. Find the third side and the area of the field.

The known angle lies between the two known sides, so the cosine rule finds BC. Rotate the letters to put A in the named position:

$$BC^2 = 90^2 + 70^2 - 2(90)(70)\cos 110^\circ \approx 17\,309 \implies BC \approx 132 \text{ m}.$$

The area needs the same three pieces of information, so no further measurement is required:

$$\text{Area} = \tfrac{1}{2}(90)(70)\sin 110^\circ \approx 2960 \text{ m}^2.$$

Two sides and the angle between them fix a triangle completely. Once you hold that combination, every other part of the triangle can be found.

What you are given	What to use
Two angles and any side (AAS, ASA)	sine rule
Two sides and an angle opposite one (SSA)	sine rule, plus the ambiguous case
Two sides and the angle between them (SAS)	cosine rule, for the third side
All three sides (SSS)	cosine rule rearranged, for an angle
The same (SAS), when the area is wanted	area $= \frac{1}{2}ab\sin C$

Read the table by counting what you know. Three parts of a triangle determine the rest, and the arrangement of those three parts is what chooses the rule.

YOUR TURN 7.3 The Sine and Cosine Rules

9. Use the sine rule to find the required side or angle.

(a) $a = 7$, $A = 40^\circ$, $B = 60^\circ$: find b

(b) $a = 9$, $A = 35^\circ$, $B = 55^\circ$: find b

(c) $A = 45^\circ$, $B = 60^\circ$, $a = 10$: find C and b

(d) $a = 10$, $b = 12$, $A = 35^\circ$: find the acute angle B

10. Use the cosine rule.

(a) $b = 5$, $c = 8$, $A = 60^\circ$: find a

(b) $a = 6$, $b = 9$, $C = 110^\circ$: find c

(c) $a = 6$, $b = 7$, $c = 8$: find angle A

(d) sides 5, 6, 7: find the largest angle

11. Find the area of each triangle.

(a) sides 5 and 8, included angle 30°

(b) $a = 9$, $b = 6$, $C = 90^\circ$

(c) sides 7 and 9, included angle $40°$

12. In triangle ABC, $a = 7$, $b = 9$ and $A = 45°$.

(a) Find the two possible values of angle B.
(b) Show that both give a valid triangle.
(c) Explain why this situation cannot arise when the given angle is included between the two given sides.

7.4 Elevation, Depression and Bearings

Real questions do not arrive as labelled triangles. They arrive as heights, distances and directions: how far away is the boat, what heading takes the ship home. Two conventions turn that language into a triangle you can solve.

An **angle of elevation** is measured upward from the horizontal to a line of sight. An **angle of depression** is measured downward from it. The observer's horizontal and the ground are parallel lines. So the angle of depression from a cliff top down to a boat equals the angle of elevation from the boat up to the cliff top. They are alternate angles, and either one can be used in the triangle.

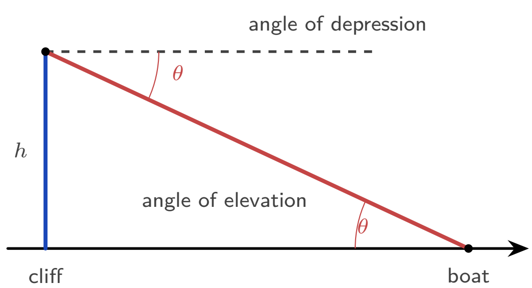


Figure 7.7 The dashed line at the top is the observer's horizontal, and the ground is parallel to it. The line of sight crosses both, so the two marked angles are alternate angles and are equal.

One sighting is never enough to find a height, because the distance to the object is unknown as well: two unknowns, one equation. A second sighting from a known distance nearer supplies the missing equation. The height is the same from both positions, so writing it twice and equating the two expressions eliminates the distance.

EXAMPLE 7.7

From a point on level ground the angle of elevation of the top of a tower is $25°$. From a point 30 m nearer the tower, on the same straight line, the angle of elevation is $40°$. Find the height of the tower.

Let h be the height and let d be the distance from the *nearer* point to the foot of the tower. The far point is then $d + 30$ from the foot. Each sighting gives a right triangle with h opposite the angle and the ground distance adjacent to it, so

$$h = d \tan 40^\circ \qquad \text{and} \qquad h = (d+30)\tan 25^\circ.$$

Both measure the same tower, so set them equal and collect the terms in d:

$$d \tan 40^\circ = (d+30)\tan 25^\circ \implies d\,(\tan 40^\circ - \tan 25^\circ) = 30\tan 25^\circ,$$

$$d = \frac{30\tan 25^\circ}{\tan 40^\circ - \tan 25^\circ} \approx \frac{13.99}{0.3728} \approx 37.5 \text{ m}.$$

The height follows from either equation:

$$h = 37.5\tan 40^\circ \approx 31.5 \text{ m}.$$

Check it against the other sighting: the far point is 67.5 m away, and $\arctan\left(\frac{31.5}{67.5}\right) \approx 25^\circ$, as given. Measuring d from the nearer point is what keeps the second distance as $d + 30$; taking d from the far point instead makes it $d - 30$, and mixing the two is the usual source of an impossible answer.

A **bearing** gives a direction as an angle measured clockwise from north. It runs from 000° to 360° and is always written with three figures, so due east is 090° and not 90°. Three figures keep the reading unambiguous on a chart. Due south is 180°, due west is 270°, and north-west is 315°.

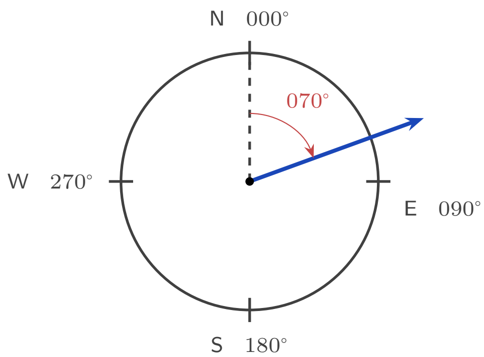


Figure 7.8 A bearing is measured clockwise from north, always with three figures. The arrow here is on a bearing of 070°.

The **back bearing** is the direction of the return journey, and it differs from the outward bearing by 180°. Add 180° when the bearing is below 180°, and subtract 180° when it is 180° or more. Either way the answer stays between 000° and 360°.

EXAMPLE 7.8

A ship sails 60 km from port P on a bearing of 050° to a point Q, then 80 km on a bearing of 140° to R. Find the distance PR.

The two bearings differ by $140°-50° = 90°$, so the ship turns through a right angle at Q: the angle PQR is $90°$. Then PR is a hypotenuse, and Pythagoras gives

$$PR = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100 \text{ km}.$$

Sketching the bearings from a north line at P and at Q is what shows the right angle. The arithmetic then follows from the picture.

Most journeys do not turn so conveniently. When the corner angle is not $90°$, read the two distances and that angle off the same sketch. Find the missing distance with the cosine rule, then find the direction back with the sine rule. Most bearings questions have this shape, so the order of the steps is always the same.

KEY **Solving a bearings problem.**

1. Sketch the journey, and draw a north line at **every** point where a bearing is given.
2. Mark each bearing as an angle clockwise from its own north line.
3. Find the interior angle of the triangle at the corner. Take the back bearing of the first leg, then subtract the second bearing.
4. Use the cosine rule for the missing distance, then the sine rule for the missing angle.
5. Turn that angle back into a bearing by adding it to the bearing of the first leg.

Students often leave out step 1. Without a north line drawn at the corner, step 3 cannot be done at all.

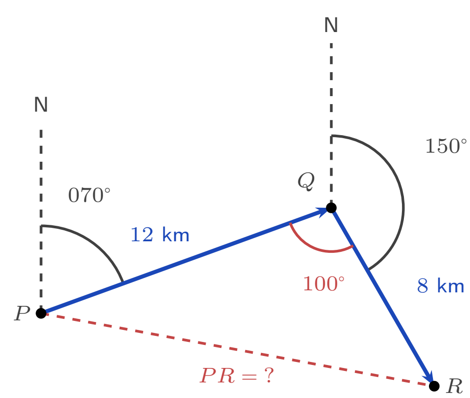


Figure 7.9 The two bearings are measured clockwise from a north line at P and at Q. Together they fix the interior angle $P\hat{Q}R = 100°$, which is what the cosine rule needs.

EXAMPLE 7.9

A hiker walks 12 km from P on a bearing of $070°$ to Q, then 8 km on a bearing of $150°$ to R. Find the distance PR and the bearing of R from P.

First construct the angle at the corner. The bearing back from Q to P is $070° + 180° = 250°$, and the onward bearing is $150°$, so the interior angle is $P\hat{Q}R = 250° - 150° = 100°$. Subtracting

the second bearing from the back bearing is the standard method, and the sketch agrees: the angle at the corner is obtuse.
No right angle, so the cosine rule finds the closing side:

$$PR^2 = 12^2 + 8^2 - 2(12)(8)\cos 100^\circ \approx 241.3 \implies PR \approx 15.5 \text{ km}.$$

For the bearing of R from P, find the angle $Q\hat{P}R$ by the sine rule:

$$\frac{\sin(Q\hat{P}R)}{8} = \frac{\sin 100^\circ}{15.54} \implies \sin(Q\hat{P}R) \approx 0.507 \implies Q\hat{P}R \approx 30.5^\circ.$$

The bearing of R from P is therefore $070^\circ + 30.5^\circ \approx 100.5^\circ$, reported as 100° to the nearest degree. Every step came from the labelled diagram: the bearings give the corner angle, the cosine rule the closing side, and the sine rule the final direction.

YOUR TURN 7.4 Elevation, Depression and Bearings

13. Find the angle of elevation in each case.

(a) a 50 m tower, from 120 m away on level ground

(b) a 45 m tower, from 90 m away on level ground

14. In each case, find the horizontal distance from the base of the tall object to the object below.

(a) From the top of an 80 m cliff, the angle of depression to a boat is 25°.
(b) From the top of a 30 m building, the angle of depression to a car is 40°.
(c) From the top of a 60 m cliff, the angle of depression to a buoy is 20°.

15. Answer each question about bearings.

(a) Write the bearing of due west as a three-figure bearing.
(b) A ship travels on a bearing of 120°. Find its back bearing.
(c) B is on a bearing of 065° from A. Find the bearing of A from B.
(d) Find the back bearing of 200°, and explain why you subtract here rather than add.

16. A ladder 6 m long rests against a vertical wall at 70° to the ground.

(a) Find the height it reaches on the wall.
(b) Find the distance from the foot of the ladder to the wall.

17. A ship sails 40 km from P on a bearing of 030° to Q, then 50 km on a bearing of 120° to R.

(a) Show that angle $PQR = 90^\circ$.
(b) Find the distance PR.
(c) Find the bearing of R from P.

7.5 Radians, Arc Length, and Sector Area

The degree is a human invention. The Babylonians cut the circle into 360 parts, and we inherited the choice. Nothing about a circle requires 360. The circle does suggest a unit of its own: take the radius, bend it round the circumference, and measure the angle it makes at the centre. That angle is one **radian**. Because the circumference is $2\pi r$, exactly 2π radii fit around it, so a full turn is 2π radians and one radian is about 57.3°.

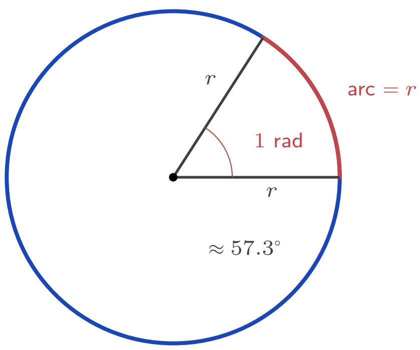


Figure 7.10 One radian is the angle at the centre for which the arc is exactly as long as the radius. A full turn takes 2π of them, because the circumference is 2π radii long.

KEY 2π radians $= 360^\circ$, so π radians $= 180^\circ$. Convert by multiplying: degrees to radians $\times \dfrac{\pi}{180}$; radians to degrees $\times \dfrac{180}{\pi}$.

The angles that appear most often should be known without conversion.

Degrees	30°	45°	60°	90°	180°	360°
Radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

The reason for the new unit appears at once: arc length and sector area become simple fractions of the whole circle. An arc is the same fraction of the circumference as its angle θ is of a full turn, so $\dfrac{l}{2\pi r} = \dfrac{\theta}{2\pi}$, and the two 2π's cancel.

KEY · IB For a sector of radius r and angle θ **in radians**,

$$\text{arc length } l = r\theta, \qquad \text{sector area } A = \tfrac{1}{2}r^2\theta.$$

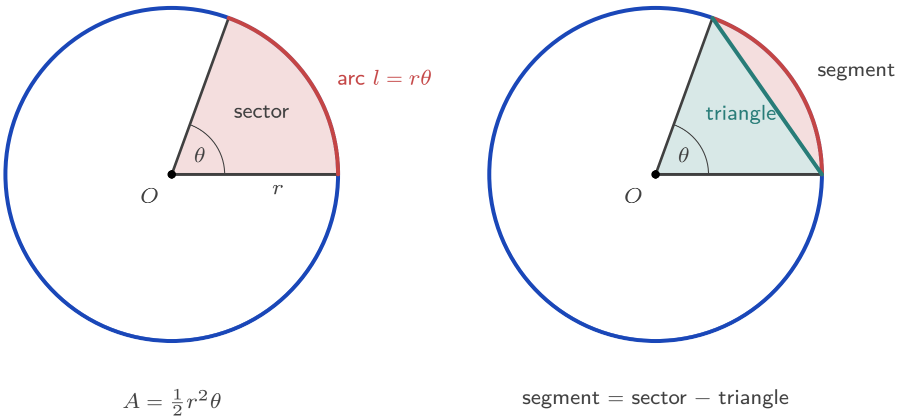


Figure 7.11 **Left:** the *sector* is the pie slice between two radii and the arc. **Right:** the chord cuts that same sector into a triangle and a *segment*, so subtracting the triangle's area from the sector's leaves the segment.

The area works the same way. The sector is the fraction $\dfrac{\theta}{2\pi}$ of the whole disc πr^2, so $A = \dfrac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}r^2\theta$. A **segment** is the region between a chord and its arc. It is a sector with the triangle removed, and that triangle has two sides r with the angle θ between them, so §7.3 gives its area as $\frac{1}{2}r^2 \sin\theta$:

$$\text{segment} = \tfrac{1}{2}r^2\theta - \tfrac{1}{2}r^2 \sin\theta = \tfrac{1}{2}r^2(\theta - \sin\theta).$$

CAUTION Every formula in this section needs θ in **radians**. Substituting an angle in degrees gives an answer that is wrong by a factor of $\frac{180}{\pi}$, and the working looks correct, so the error is easy to miss. Check the mode on your calculator before you start.

EXAMPLE 7.10

A sector has radius 10 cm and angle $\dfrac{2\pi}{5}$ radians. Find its arc length and area.

Directly from the formulas,

$$l = r\theta = 10 \cdot \frac{2\pi}{5} = 4\pi \approx 12.6 \text{ cm}, \qquad A = \tfrac{1}{2}r^2\theta = \tfrac{1}{2}(100)\frac{2\pi}{5} = 20\pi \approx 62.8 \text{ cm}^2.$$

For the segment cut off by the chord, subtract the triangle from the sector:

$$\tfrac{1}{2}r^2(\theta - \sin\theta) = 50\left(\frac{2\pi}{5} - \sin\frac{2\pi}{5}\right) \approx 50(1.2566 - 0.9511) \approx 15.3 \text{ cm}^2.$$

Each formula took the angle in radians, exactly as given. The segment is far smaller than the sector, because most of the sector is triangle.

A sector's boundary is not the arc alone. Walk round the edge and you travel out along one radius, round the arc, and back along the other radius. The perimeter of a sector is therefore the arc plus two radii, and it has to be assembled from $l = r\theta$ rather than looked up, because the formula booklet does not print it.

KEY **Perimeter of a sector** of radius r and angle θ in radians:

$$P = 2r + r\theta = r(2 + \theta).$$

Two straight radii and one curved arc. This one is not in the formula booklet.

EXAMPLE 7.11

A sector has radius 7 cm and angle 1.5 radians. Find its perimeter.
The arc is

$$l = r\theta = 7(1.5) = 10.5 \text{ cm},$$

and the two radii add 14 cm, so

$$P = 2(7) + 10.5 = 24.5 \text{ cm}.$$

A quick sanity check: the two radii alone measure 14 cm, and an angle of 1.5 radians is under a quarter turn, so an arc a little longer than one radius is what the picture should show.

Given the perimeter and the area together, neither formula can be used on its own, because each contains both r and θ. Treat them as simultaneous equations, and eliminate θ.

EXAMPLE 7.12

A sector has perimeter 36 cm and area 45 cm^2. Find its radius and its angle.
Write down both conditions:

$$2r + r\theta = 36, \qquad \tfrac{1}{2}r^2\theta = 45.$$

The second gives $r^2\theta = 90$, so $\theta = \dfrac{90}{r^2}$. Substituting that into the first,

$$2r + r \cdot \frac{90}{r^2} = 36 \implies 2r + \frac{90}{r} = 36.$$

Multiply through by r and collect:

$$2r^2 - 36r + 90 = 0 \implies r^2 - 18r + 45 = 0 \implies (r-3)(r-15) = 0,$$

so $r = 3$ or $r = 15$. Test each against $\theta = \dfrac{90}{r^2}$. With $r = 3$ the angle is $\theta = 10$ radians, which is more than a full turn of $2\pi \approx 6.28$, so no such sector exists. With $r = 15$ the angle is $\theta = 0.4$ radians.

So $r = 15$ cm and $\theta = 0.4$ radians. Check: the arc is $15(0.4) = 6$ cm, giving a perimeter of $30 + 6 = 36$ cm, and the area is $\frac{1}{2}(225)(0.4) = 45$ cm^2. ✓ The quadratic will usually offer two radii, and rejecting the one whose angle exceeds 2π is part of the answer, not an afterthought.

EXAMPLE 7.13

The opener measured the Earth from a hilltop. There is an older method, and it uses this section's formula. Eratosthenes measured the sun's rays at 7.2° to the vertical in Alexandria, while in Syene they were vertical. He took the arc between the towns to be 5000 stadia. Find the Earth's radius and circumference in stadia.

Convert the angle first:

$$7.2^\circ \times \frac{\pi}{180} = \frac{\pi}{25} \text{ radians.}$$

The two towns and the Earth's centre make a sector whose arc is the distance between them, so $l = r\theta$ can be used in reverse:

$$5000 = r \cdot \frac{\pi}{25} \implies r = \frac{125\,000}{\pi} \approx 39\,800 \text{ stadia,}$$

and the circumference is $2\pi r = 250\,000$ stadia, which is the one-fiftieth argument written as a formula. Taking a stadion as roughly 158 m gives a circumference near $39\,500$ km, within a couple of percent of the true $40\,075$ km. One angle, one arc, one formula, and the planet is measured.

YOUR TURN 7.5 Radians, Arc Length, and Sector Area

18. EXACT Convert to radians, in terms of π.

(a) 60°

(b) 135°

(c) 210°

(d) 90°

19. Convert to degrees.

(a) $\frac{3\pi}{4}$

(b) $\frac{2\pi}{3}$

(c) 1 radian, to three significant figures

20. Find the arc length and the area of each sector.

(a) radius 8, angle 1.5 radians

(b) radius 6, angle $\frac{\pi}{3}$ radians

(c) radius 9, angle 2.4 radians

(d) radius 4, angle 90°

21. In each case the angle is unknown. Find it from the information given.

(a) A sector has radius 10 and arc length 25. Find its angle in radians.

(b) A sector has radius 6 and area 54. Find its angle in radians, then find its arc length.

22. A chord subtends an angle of 2 radians at the centre of a circle of radius 8.

(a) Find the area of the sector.

(b) Find the area of the triangle formed by the chord and the two radii.

(c) Find the area of the minor segment.

(d) Find the perimeter of the sector.

Chapter 7 · Shapes & Measures

Reflections

TOK The formulas $l = r\theta$ and $A = \frac{1}{2}r^2\theta$ are this simple only when θ is in radians, and later you will meet another case: the derivative of $\sin x$ is exactly $\cos x$, again only in radians. The degree is a human choice, made by the Babylonians. The radian is not a choice. It is the angle whose arc equals its radius, and the circle fixes it. Is the radian discovered, then, where the degree is invented?

IM Trigonometry was built by many people over many centuries. The Greek astronomer Hipparchus made the first table of chords around 150 BCE. In India, Aryabhata replaced the chord with the half-chord and called it the *jyā*, which is our sine. The word passed into Arabic, was misread there as *jaib*, meaning "bay", and was translated into Latin as *sinus*. Islamic astronomers, al-Bīrūnī among them, extended the sine and cosine rules, partly to find the direction of Mecca from any city.

RW Fixing an unknown position from measured angles is called **triangulation**, and it is still how we map the world. Surveyors once carried chains of triangles across whole countries. Astronomers use *parallax*: as the Earth moves round the Sun a near star seems to shift against the far ones, and that small angle gives its distance. Each one is a triangle solved for a side nobody can reach.

RW Your phone does the same job with distances instead of angles. A GPS receiver times the signal from each satellite and turns each time into a distance, then finds the one point lying at all of those distances at once. Three satellites narrow it to a point on the Earth's surface, and a fourth corrects the receiver's clock. This is **trilateration**, and it is the sphere equation of §7.1 solved several times over.

EXT The ambiguous case is really a statement about congruence. Give a triangle by two angles and a side (ASA or AAS), by three sides (SSS), or by two sides and the angle between them (SAS), and only one triangle fits. Those are the congruence rules. Two sides and an angle that is *not* between them (SSA) is not one of them, so it can leave the triangle undecided. The sine rule's two answers are that gap showing through.

End-of-Chapter Problems — Chapter 7: Shapes & Measures

1. A cuboid has dimensions $6 \times 8 \times 10$.

(a) Find the length of the space diagonal. [2]

(b) Find the angle between the space diagonal and the base. [3]

2. Consider the points $A(2, 0, 0)$, $B(0, 4, 0)$ and $C(0, 0, 4)$.

(a) Find the lengths AB, AC and BC. [3]

(b) Find angle BAC. [3]

3. EXACT A cone has radius 5 and height 12.

(a) Find the slant height. [1]

(b) Find the volume. [2]

(c) Find the total surface area. [2]

4. EXACT A solid consists of a cylinder of radius 3 and height 10 with a hemisphere of radius 3 on top. Find its total surface area. [5]

5. In triangle ABC, $a = 8$, $b = 11$, $c = 15$.

(a) Find the largest angle. [3]

(b) Find the area of the triangle. [2]

6. In triangle PQR, $p = 10$, $q = 7$ and $R = 95^\circ$.

(a) Find r. [2]

(b) Find the area of the triangle. [2]

7. A triangle has two sides of length 9 and 12 with an included angle of 40°.

(a) Find the length of the third side. [2]

(b) Find the area of the triangle. [2]

8. From port P, a ship sails 40 km on a bearing of 030° to Q, then 50 km on a bearing of 120° to R.

(a) Show that angle $PQR = 90^\circ$. [2]

(b) Find the distance PR. [2]

(c) Find the bearing of R from P. [3]

9. A sector has radius 12 cm and angle 1.2 radians.

(a) Find the arc length. [1]
(b) Find the area of the sector. [2]
(c) Find the perimeter of the sector. [2]

10. A sector has area 30 cm^2 and radius 6 cm.

(a) Find the angle of the sector in radians. [2]
(b) Find the arc length. [2]

11. A chord subtends an angle of 1.5 radians at the centre of a circle of radius 10 cm.

(a) Find the area of the sector. [2]
(b) Find the area of the triangle formed by the chord and the two radii. [2]
(c) Find the area of the minor segment. [2]

12. In triangle ABC, $A = 50^\circ$, $a = 6$ and $b = 7$.

(a) Find the two possible values of angle B. [3]
(b) Find the two possible areas of the triangle. [3]

13. EXACT A regular hexagon is inscribed in a circle of radius 8. Find its area. [4]

14. EXACT A cone has volume 100π and radius 5.

(a) Find the height. [2]
(b) Find the slant height. [2]
(c) Find the curved surface area. [1]

15. From a point on level ground, the angle of elevation of the top of a tower is 30°. Moving 40 m closer, the angle of elevation becomes 45°. Find the height of the tower. [5]

16. A square-based pyramid has base side 10 and height 12.

(a) Find the volume. [2]
(b) Find the slant height of a triangular face. [2]
(c) Find the total surface area. [2]

17. The area of a triangle is 24, with $a = 8$ and $b = 10$. Find the two possible values of the included angle C. [3]

18. Two ships leave a port at the same time. One sails on a bearing of 040° at 12 km/h, the other on a bearing of 100° at 16 km/h. Find the distance between them after 2 hours. [5]

19. Points $A(0, 0, 0)$ and $B(4, 4, 7)$ are given.

(a) Find AB. [2]
(b) Find the midpoint of $[AB]$. [1]
(c) Show that B lies on the sphere $x^2 + y^2 + z^2 = 81$. [1]

20. A pendulum of length 0.8 m swings through an angle of 0.5 radians. Find the length of the arc traced by the bob. [2]

21. EXACT Convert to radians, in terms of π.

(a) 30° [1] (b) 225° [1]
(c) 300° [1]

22. Two observers stand 200 m apart on level ground, with a balloon in the vertical plane between them. From one the angle of elevation to the balloon is 40°; from the other it is 55°. Find the height of the balloon. [6]

23. In triangle ABC, $AB = 12$, $BC = 9$ and angle $BAC = 35^\circ$. Find the two possible values of angle BCA. [4]

24. Radians and arc length.

(a) Convert $\frac{5\pi}{6}$ to degrees. [1]
(b) Convert $\frac{7\pi}{4}$ to degrees. [1]
(c) Find the arc length of a sector of radius 9 and angle $\frac{5\pi}{6}$. [2]

25. A sector of a circle of radius 12 and angle 1.5 radians is rolled into a cone, so that the radius becomes the slant height and the arc becomes the base circumference. Find the radius and height of the cone. [6]

26. A sector of a circle has perimeter 20 and area 16. Find the radius r and angle θ. [6]

27. A chord of a circle of radius 10 has length 12. Find the area of the minor segment it cuts off. [6]

28. The area of a triangle is 84, and two of its sides have lengths 13 and 14. Find both possible lengths of the third side. [6]

29. A yacht leaves harbour H and sails 25 km on a bearing of 048° to a buoy A. It then sails 18 km on a bearing of 130° to a second buoy B.

(a) Show that angle $HAB = 98^\circ$. [3]
(b) Find the distance HB. [4]
(c) Find the bearing of B from H. [4]
(d) Find the area of triangle HAB. [3]
(e) The yacht returns directly from B to H. Find the total distance it has sailed. [2]

30. A rectangular box $ABCDEFGH$ has a horizontal base $ABCD$ with $AB = 8$ cm and $BC = 6$ cm, and vertical edges of length 5 cm, so that G is the vertex above C.

(a) Find the length of the base diagonal AC. [2]
(b) Find the exact length of the space diagonal AG. [3]
(c) Find the angle that AG makes with the base $ABCD$. [3]
(d) Find the angle $G\hat{A}B$ between the diagonal AG and the edge AB. [4]
(e) Find the total surface area of the box. [3]

31. EXACT A container is a cone of base radius 6 cm and height 15 cm, held with its point downwards.

(a) Find the volume of the cone when full, in terms of π. [2]
(b) Water is poured in to a depth of 10 cm. Using similar triangles, find the radius of the circular water surface. [3]
(c) Find the volume of water, in terms of π. [3]
(d) Find the fraction of the cone's volume that is filled, and explain why it is not $\frac{2}{3}$ even though the water reaches $\frac{2}{3}$ of the height. [4]
(e) Find the depth of water when the cone is exactly half full, giving your answer to three significant figures. [3]

Challenge Questions

32. How many triangles fit the data? In triangle ABC you are given the side b, the angle A, and the side a opposite A. This is the SSA case. This investigation finds exactly how many triangles fit, for every possible value of a.

Throughout, take A to be acute, and let h be the distance from C to the line AB.

(a) Show that $h = b \sin A$. [2]
(b) Take $b = 10$ and $A = 40^\circ$. Find the number of triangles when $a = 5$, when $a = 8$, and when $a = 12$. [4]

(c) Explain why no triangle exists when $a < b\sin A$. *(Hint: what is the shortest distance from C to the line AB?)* [3]

(d) Explain why exactly one triangle exists when $a = b\sin A$, and state what is special about it. [3]

(e) Show that two triangles exist exactly when $b\sin A < a < b$, and explain what goes wrong at $a = b$. [4]

(f) Now let A be obtuse. State how many triangles exist for $a \le b$ and for $a > b$, and justify each answer. [4]

33. The best cone from a sector. A sector of radius R and angle θ radians is cut from a sheet of paper. The two straight edges are joined to make a cone. This investigation finds the angle that gives the largest cone.

Write $t = \dfrac{\theta}{2\pi}$, so that $0 < t < 1$.

(a) The arc of the sector becomes the circumference of the base. Show that the base radius is $r = Rt$. [3]

(b) Show that the height of the cone is $h = R\sqrt{1-t^2}$. [3]

(c) Hence show that the volume is $V = \frac{1}{3}\pi R^3 t^2\sqrt{1-t^2}$. [3]

(d) Taking $R = 1$, use technology to find the value of t that makes V largest, and hence find θ to the nearest degree. [4]

(e) Show that at that maximum $\dfrac{r}{R} = \sqrt{\dfrac{2}{3}}$, and find the exact value of $\dfrac{h}{r}$. [5]

34. How reliable was al-Bīrūnī's measurement? The opener described his method. From a hill of height h he measured the angle of dip θ to the horizon, and the radius of the Earth follows from one right-angled triangle. This investigation asks how much trust the answer deserves.

(a) The line of sight touches the Earth, so it meets the radius there at a right angle. Show that $\cos\theta = \dfrac{R}{R+h}$, and hence that $R = \dfrac{h\cos\theta}{1-\cos\theta}$. [3]

(b) Al-Bīrūnī recorded $h = 652$ cubits and $\theta = 34'$. Find R in cubits. *(Hint: $34' = \frac{34}{60}$ of a degree)* [3]

(c) Repeat the calculation for $\theta = 33'$ and for $\theta = 35'$. Find the percentage change in R in each case. [4]

(d) For small θ the approximation $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ holds, with θ in radians. Use it to show that $R \approx \dfrac{2h}{\theta^2}$. [3]

(e) Hence explain why an error of about 3% in θ produces an error of about 6% in R. Comment on what this means for how confidently his result can be quoted. [4]

GEOMETRY & TRIGONOMETRY

CHAPTER 8

Trigonometry

8

SYLLABUS 3.5–3.11

Your voice is a wave. So is the light from your screen, the signal carrying this page to your device, the current in the wall socket, the tide in Tokyo Bay, and the beating of your own heart.

They do not look alike. A heartbeat is a sharp spike, a spoken vowel is jagged, a tide is a slow rise and fall over twelve hours. In 1822 Joseph Fourier made a startling claim: a repeating signal, however jagged, can be built up from pure sine waves. Choose enough of them, with the right heights and the right timings, and the shape appears. Your phone uses this idea many times a second to compress your voice.

So the same curve describes all of them. Where does it come from?

Watch one seat on a turning Ferris wheel. As the wheel turns steadily, the seat rises, slows near the top, falls, slows near the bottom, and rises again. Plot its height against time and the sine curve appears. The seat is a point moving round a circle, and its height is the vertical coordinate of that point. The circle and the wave are the same motion seen two ways: one going round, one going along.

That is why this chapter starts with a circle. In Ch. 7 sine and cosine were ratios inside a right-angled triangle, so the angle had to be acute. On the circle they become functions of *any* angle, and they repeat forever.

By the end of this chapter you will be able to read exact values of sine, cosine and tangent from the unit circle. You will use identities to rewrite one expression as another, sketch and transform the graphs of these functions, and work with the reciprocal and inverse functions. You will also solve trigonometric equations, where a single value of the sine or cosine is shared by two different angles, so answers usually have to be looked for in pairs.

8.1 The Unit Circle

How do you find $\sin 120°$? Or $\cos 360°$, or $\tan(-30°)$?

You already have a definition, and it is SOHCAHTOA. In a right triangle,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Now try to draw the triangle. For $120°$ there is none: the three angles add to $180°$, one of them is already the right angle, so the other two must each be under $90°$. For $360°$ there is none either, and $-30°$ is worse, because a triangle has no negative angles. Under SOHCAHTOA these three expressions have no meaning at all.

Yet your calculator answers all three without complaint, and a turn of $120°$ is an ordinary thing to make. So the definition has to widen. Whatever replaces it must agree with SOHCAHTOA wherever SOHCAHTOA works, and keep working where it does not.

The unit circle does exactly that. Take the circle of radius 1 centred at the origin, and a point P on it at angle θ, measured anticlockwise from the positive x-axis.

For an acute θ, drop a perpendicular from P to the x-axis. This makes a right triangle whose hypotenuse is the radius, so the hypotenuse is 1. The old ratios then read

$$\sin\theta = \frac{\text{opposite}}{1} = \text{opposite}, \qquad \cos\theta = \frac{\text{adjacent}}{1} = \text{adjacent},$$

and those two sides are exactly the coordinates of P. So for acute angles the new definition agrees with the old one.

The new form never mentions a triangle. A point can sit anywhere on the circle, at any angle, and it always has coordinates. That is what extends the definition to every real θ.

DEF On the unit circle, for a point P at angle θ:

$$\cos\theta = x\text{-coordinate of } P, \qquad \sin\theta = y\text{-coordinate of } P.$$

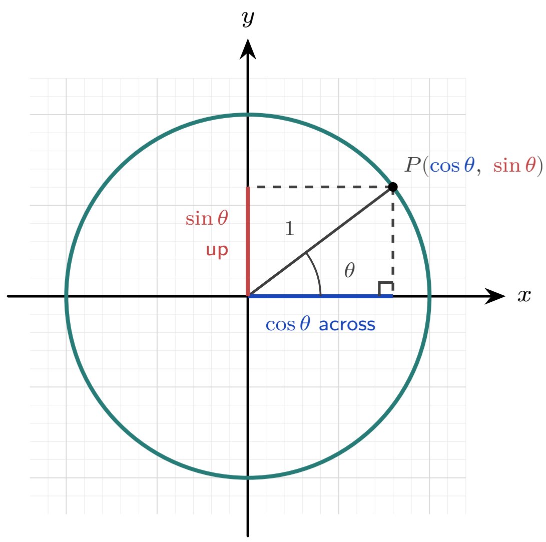


Figure 8.1 On a circle of radius 1 the hypotenuse is 1, so the two shorter sides *are* the coordinates of P. The cobalt side, measured across, is $\cos\theta$; the scarlet side, measured up, is $\sin\theta$.

The tangent is the gradient of the line OP, so $\tan\theta = \dfrac{\text{rise}}{\text{run}} = \dfrac{\sin\theta}{\cos\theta}$. It is undefined wherever $\cos\theta = 0$.

KEY · IB $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

That word *gradient* is not accidental. The line OP passes through the origin, and its gradient is $\tan\theta$, so its equation is $y = x\tan\theta$. This ties the tangent to the straight lines of Ch. 4: a gradient and an angle are two names for the same thing, and the tangent converts between them. A line of gradient 1 makes an angle of $\frac{\pi}{4}$ with the positive x-axis, because $\tan\frac{\pi}{4} = 1$.

KEY A line through the origin making an angle θ with the positive x-axis has equation

$$y = x\tan\theta.$$

Read forwards it gives the gradient from the angle; read backwards, $\theta = \arctan(\text{gradient})$ gives the angle from the gradient.

8.1.1 Exact values

A few angles give exact coordinates, and they come from the 30–60–90 and 45–45–90 triangles.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Only three numbers appear in the shaded row, and they are all that has to be memorised:

$$\frac{1}{2}, \qquad \frac{\sqrt{2}}{2}, \qquad \frac{\sqrt{3}}{2}.$$

The cosine row is those same three in reverse order. That is not a coincidence. Since $\cos\theta = \sin(\frac{\pi}{2} - \theta)$, reading the sine row backwards reads the cosine row forwards. The tangent row needs nothing new at all: $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, so divide one row by the other. At $\theta = \frac{\pi}{2}$ the cosine is zero, which is why the tangent has no value there.

The first-quadrant values repeat into the other three quadrants with signs changed. Collecting them all onto one circle gives the diagram to learn.

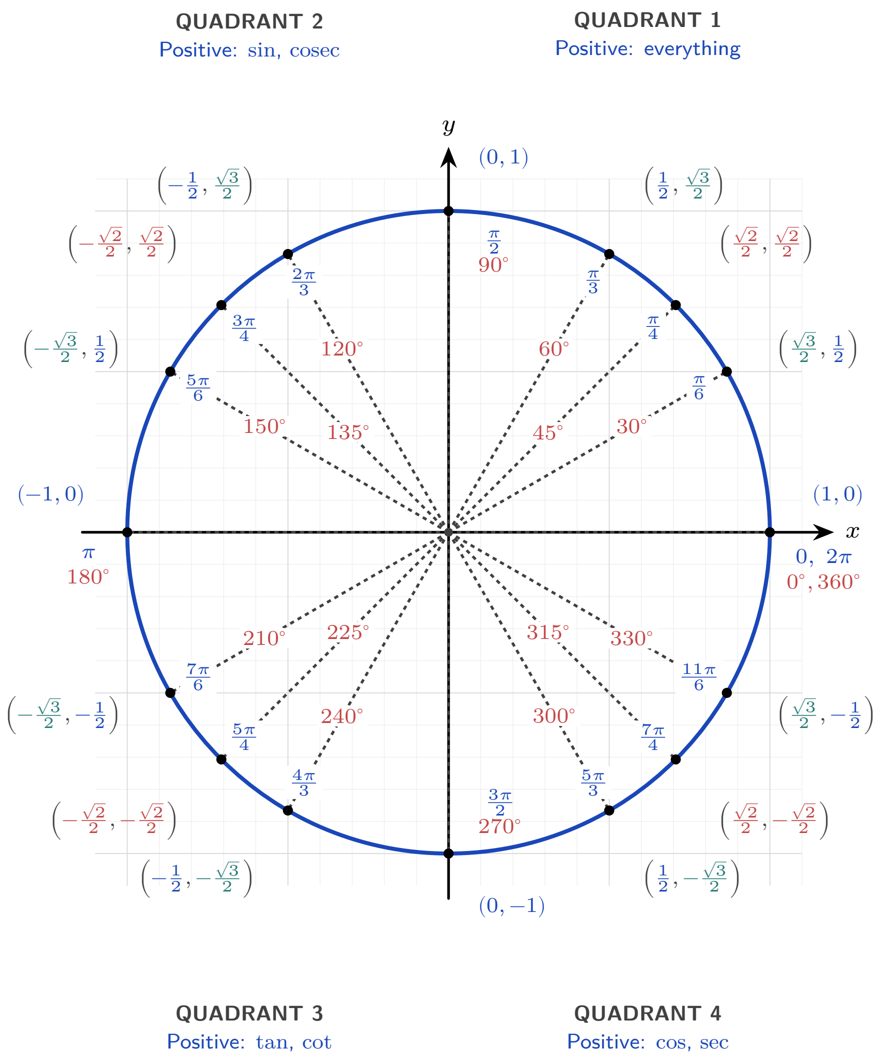


Figure 8.2 The unit circle with every special angle. Each spoke carries the angle twice: in cobalt on the outer ring in radians, and in scarlet on the inner ring in degrees. Outside the circle is the point $(\cos\theta,\ \sin\theta)$, with each coordinate coloured by its size: cobalt for $\frac{1}{2}$, scarlet for $\frac{\sqrt{2}}{2}$, teal for $\frac{\sqrt{3}}{2}$. To read a value, find the angle and take the matching coordinate. The corner labels say which functions are positive in each quadrant; sec, cosec and cot are the reciprocal functions of §8.4.

The colours make the point. Only three numbers appear anywhere on the circle, and every coordinate on it is one of those three with a sign attached. Add the four points on the axes and the diagram is complete.

8.1.2 Symmetry and the sign of each function

Since $\cos\theta$ and $\sin\theta$ are the coordinates of P, their signs are just the signs of x and y in each quadrant. Nothing needs to be memorised separately: in the second quadrant x is negative and y is positive, so $\cos\theta < 0$ and $\sin\theta > 0$, and $\tan\theta = \frac{\sin}{\cos}$ is negative.

Figure 8.2 records the answer at each corner. Reading the four quadrants anticlockwise from the first, the functions that stay positive are all three, then sine alone, then tangent alone, then cosine alone.

The mnemonic **All Students Take Calculus** is the usual way to remember that order. Use it as a check, not as the reason. The reason is always the sign of the coordinates, and if you can picture the point you never need the letters at all.

Each symmetry of the circle now gives an identity, and each one is a statement about where a reflected point lands.

Reflect in the y-axis. The point at angle $\pi - \theta$ is the mirror image, across the y-axis, of the point at angle θ. Reflecting in the y-axis negates the x-coordinate and leaves y alone, so the two points are $(-\cos\theta,\ \sin\theta)$ and $(\cos\theta,\ \sin\theta)$. Comparing coordinates:

$$\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta.$$

Reflect in the x-axis. Measuring $-\theta$ means turning clockwise instead of anticlockwise, which lands on the mirror image of P across the x-axis. That reflection negates y and leaves x alone, so

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta.$$

Turn a full circle. Adding 2π to the angle sends P once round and back to exactly the same point, so both coordinates are unchanged:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This is why both functions are **periodic** with period 2π. Every identity of this kind says the same thing: move the point, see which coordinate changes sign.

You have met the first of these relations before. Two angles, θ and $\pi - \theta$, share the same sine. That shared value is what produced the ambiguous case of the sine rule in Ch. 7. A triangle built from a sine value can sometimes close in two ways, one acute and one obtuse.

An exact value at any special angle now comes out of the circle in two moves: find the acute angle the arm makes with the x-axis, which fixes the size, then look at the quadrant, which fixes the sign.

EXAMPLE 8.1

Find the exact values of $\sin\dfrac{5\pi}{3}$ and $\cos\dfrac{5\pi}{3}$.

Write the angle as $2\pi - \frac{\pi}{3}$. A full turn returns to the positive x-axis, so turning back through $\frac{\pi}{3}$ leaves the point in the fourth quadrant, where the x-coordinate is positive and the y-coordinate negative. The arm makes an acute angle of $\frac{\pi}{3}$ with the axis, so both values have the size they have at $\frac{\pi}{3}$:

$$\sin\frac{5\pi}{3} = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}, \qquad \cos\frac{5\pi}{3} = \cos\frac{\pi}{3} = \frac{1}{2}.$$

Dividing them gives $\tan\frac{5\pi}{3} = -\sqrt{3}$, negative, because only the sine carries a minus sign. That is the fourth quadrant's entry in the mnemonic, recovered from the coordinates rather than remembered.

YOUR TURN 8.1 The Unit Circle

1. Write down the exact value of each.

(a) $\sin\frac{\pi}{6}$ (b) $\cos\frac{\pi}{4}$
(c) $\tan\frac{\pi}{3}$ (d) $\tan\frac{\pi}{4}$

2. Use the symmetry of the circle to find each exact value.

(a) $\sin\frac{5\pi}{6}$ (b) $\cos\frac{2\pi}{3}$
(c) $\sin\frac{4\pi}{3}$ (d) $\cos\frac{7\pi}{4}$

3. State the quadrant or quadrants in which each condition holds.

(a) $\sin\theta > 0$ (b) $\cos\theta < 0$
(c) $\tan\theta > 0$ (d) $\sin\theta < 0$ and $\cos\theta > 0$

4. The point P on the unit circle has angle θ.

(a) State the coordinates of P in terms of θ.
(b) Explain why $\cos^2\theta + \sin^2\theta = 1$ for every θ.
(c) Explain why $\tan\theta$ is undefined at $\theta = \frac{\pi}{2}$.

5. Lines through the origin, using $y = x\tan\theta$. Give each angle in radians.

(a) Write the equation of the line through the origin making an angle of $\frac{\pi}{3}$ with the positive x-axis.
(b) Find the angle a line of gradient 1 makes with the positive x-axis.
(c) Find the anticlockwise angle from the positive x-axis to the line $y = -x$.

8.2 Trigonometric Identities

An **identity** is an equation true for every angle. The most fundamental one is Pythagoras' theorem read off the unit circle: the point $(\cos\theta, \sin\theta)$ is distance 1 from the origin, so $\cos^2\theta + \sin^2\theta = 1$.

KEY · IB $\cos^2\theta + \sin^2\theta = 1.$

8.2.1 Double and compound angles

The functions of a sum of two angles can be expanded, and from the sum formulas everything else follows. These **compound-angle** identities are usually just quoted, but the proof is short and uses only the unit circle and the cosine rule from Ch. 7.

Put two points on the unit circle, P at angle A and Q at angle B. Their coordinates are $P(\cos A,\ \sin A)$ and $Q(\cos B,\ \sin B)$, and the angle $\angle POQ$ at the centre is $A - B$.

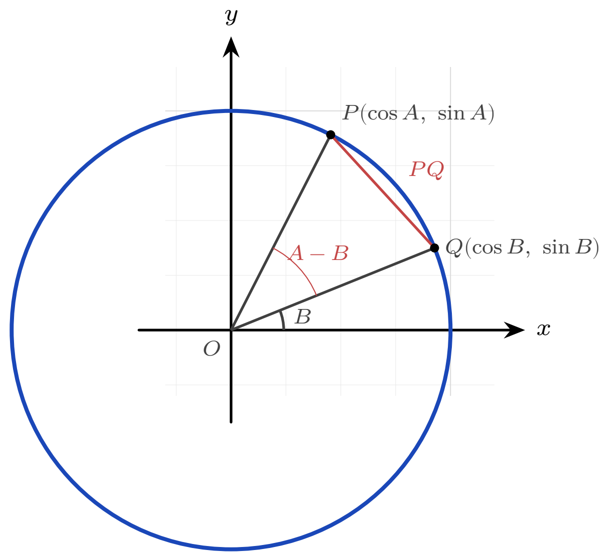


Figure 8.3 Two points on the unit circle, at angles A and B. The angle between the radii is $A - B$, and the chord PQ can be measured either by the distance formula or by the cosine rule.

Now find the distance PQ in two different ways.

By the distance formula:

$$PQ^2 = (\cos A - \cos B)^2 + (\sin A - \sin B)^2.$$

Expanding gives $\cos^2 A - 2\cos A\cos B + \cos^2 B + \sin^2 A - 2\sin A\sin B + \sin^2 B$. Both Pythagorean pairs collapse to 1, so

$$PQ^2 = 2 - 2(\cos A\cos B + \sin A\sin B).$$

By the cosine rule in triangle OPQ, where both sides from the centre have length 1 and the angle between them is $A - B$:

$$PQ^2 = 1^2 + 1^2 - 2(1)(1)\cos(A - B) = 2 - 2\cos(A - B).$$

The two expressions are the same length, so

$$\cos(A - B) = \cos A\cos B + \sin A\sin B.$$

Every other compound-angle formula follows from this one. Replacing B by $-B$, and using

$\cos(-B) = \cos B$ with $\sin(-B) = -\sin B$, gives the formula for $\cos(A+B)$.

A sine is a cosine a quarter turn away: $\sin\theta = \cos(\frac{\pi}{2} - \theta)$. That single fact is all the sine version needs. Write the sum in that form, then group the result as a *difference* of two angles:

$$\sin(A+B) = \cos(\tfrac{\pi}{2} - A - B) = \cos((\tfrac{\pi}{2} - A) - B)\,.$$

That is exactly the shape just proved. Expanding it, and using $\cos(\frac{\pi}{2} - A) = \sin A$ together with $\sin(\frac{\pi}{2} - A) = \cos A$, gives

$$\sin(A+B) = \sin A\cos B + \cos A\sin B.$$

The tangent versions come from dividing the sine formula by the cosine formula.

KEY · IB HL **Compound angles:**

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B, \qquad \cos(A \pm B) = \cos A\cos B \mp \sin A\sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}.$$

Setting $B = A$ reduces each of these to a **double-angle** identity, the most useful family in the topic. Working through the derivation once shows there is nothing new to learn here.

Put $B = A$ in the sine formula:

$$\sin 2A = \sin(A+A) = \sin A\cos A + \cos A\sin A = 2\sin A\cos A.$$

Do the same in the cosine formula:

$$\cos 2A = \cos(A+A) = \cos A\cos A - \sin A\sin A = \cos^2 A - \sin^2 A.$$

That is the first form. The other two come from $\sin^2 A + \cos^2 A = 1$, used to remove one function or the other. Replacing $\sin^2 A$ by $1 - \cos^2 A$ gives

$$\cos 2A = \cos^2 A - (1 - \cos^2 A) = 2\cos^2 A - 1,$$

and replacing $\cos^2 A$ by $1 - \sin^2 A$ instead gives

$$\cos 2A = (1 - \sin^2 A) - \sin^2 A = 1 - 2\sin^2 A.$$

The tangent version follows in the same way, by putting $B = A$ in the tangent formula.

The sine and cosine versions sit in the SL half of the booklet; the tangent version is HL.

KEY · IB **Double angles:**

$$\sin 2\theta = 2\sin\theta\cos\theta, \qquad \cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta.$$

KEY · IB HL $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$.

The three forms of $\cos 2\theta$ are one identity rewritten by $\sin^2 + \cos^2 = 1$. Choose whichever leaves you with only sines or only cosines. That choice is what makes the identity so useful when solving equations.

EXAMPLE 8.2

Find the exact value of $\sin 165^\circ$.

Write $165^\circ = 120^\circ + 45^\circ$ and use the compound-angle formula:

$$\sin 165^\circ = \sin 120^\circ \cos 45^\circ + \cos 120^\circ \sin 45^\circ = \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right)\cdot\frac{\sqrt{2}}{2} = \frac{\sqrt{6}-\sqrt{2}}{4}.$$

Splitting an awkward angle into a sum or a difference of two known ones is the standard method for exact values beyond the basic table. The pieces need not be acute: $\cos 120^\circ = -\frac{1}{2}$ carries a minus sign from the second quadrant, and that minus sign is what turns the sum into a difference.

Identities also let you move between ratios *without ever finding the angle*. Examiners set this often: the angle stays unknown from start to finish, and the Pythagorean identity supplies everything needed.

EXAMPLE 8.3

Given that $\cos x = \dfrac{3}{4}$ and x is acute, find the exact value of $\sin 2x$ without finding x.

From $\cos^2 x + \sin^2 x = 1$, $\sin^2 x = 1 - \frac{9}{16} = \frac{7}{16}$, and since x is acute, $\sin x = \frac{\sqrt{7}}{4}$ (positive). Then

$$\sin 2x = 2\sin x\cos x = 2\cdot\frac{\sqrt{7}}{4}\cdot\frac{3}{4} = \frac{3\sqrt{7}}{8}.$$

The word "acute" fixed the sign of $\sin x$. If x had instead been in the fourth quadrant, the same working would give $\sin 2x = -\frac{3\sqrt{7}}{8}$: always let the quadrant choose the sign before doubling.

Proving an identity is the same game played in reverse: start from one side, substitute known identities, and simplify until the other side appears.

EXAMPLE 8.4

Prove that $\cos(A-B)-\cos(A+B)=2\sin A\sin B$.

Work on the left side, expanding each cosine by the compound-angle formula:

$$\cos(A-B)=\cos A\cos B+\sin A\sin B,\qquad \cos(A+B)=\cos A\cos B-\sin A\sin B.$$

The two expansions differ only in the sign of the second term, so subtracting removes the $\cos A\cos B$ pair and doubles what is left:

$$\cos(A-B)-\cos(A+B)=2\sin A\sin B. \qquad \blacksquare$$

Expanding the more complicated side first is the usual choice. Once both expansions are written down, the terms that have to cancel are visible before any algebra is done.

8.2.2 One wave from two

A sum such as $8\sin x+15\cos x$ mixes a sine and a cosine of the same angle. Both have period 2π, so the sum repeats every 2π as well, and it turns out to be a single sine wave, stretched and shifted. Reading the compound-angle formula backwards shows why.

Expanding $R\sin(x+\alpha)$ gives

$$R\sin(x+\alpha)=R\cos\alpha\,\sin x+R\sin\alpha\,\cos x,$$

which already has the shape $a\sin x+b\cos x$, with $a=R\cos\alpha$ and $b=R\sin\alpha$. Squaring those two and adding removes α, since $\cos^2\alpha+\sin^2\alpha=1$, and leaves $a^2+b^2=R^2$. Dividing them removes R instead and leaves $\tan\alpha=\frac{b}{a}$. So any such sum can be written as one wave, and the two constants can be recovered from a and b alone.

KEY **The wave form.** For constants a and b not both zero,

$$a\sin x+b\cos x=R\sin(x+\alpha),\qquad R=\sqrt{a^2+b^2},\qquad \tan\alpha=\frac{b}{a},$$

with $R>0$ and α taken in the quadrant where $\cos\alpha$ has the sign of a and $\sin\alpha$ the sign of b. The same R and α give the cosine form

$$a\cos x+b\sin x=R\cos(x-\alpha).$$

Since $\sin$ and $\cos$ run between -1 and 1, the expression has maximum R and minimum $-R$.

This form is not printed in the formula booklet, so rebuild it from the expansion of $R\sin(x+\alpha)$ whenever it is needed. Its value is that the variable now appears once instead of twice, which turns a maximum, a minimum or an equation into the ordinary work of §8.6.

EXAMPLE 8.5

Write $8\sin x+15\cos x$ in the form $R\sin(x+\alpha)$ with $R>0$ and $0<\alpha<\frac{\pi}{2}$. State the maximum and minimum values, and solve $8\sin x+15\cos x=6$ for $0\le x\le 2\pi$.

Here $a=8$ and $b=15$. Both are positive, so α is in the first quadrant:

$$R=\sqrt{8^2+15^2}=\sqrt{289}=17,\qquad \tan\alpha=\frac{15}{8}\implies\alpha=\arctan\frac{15}{8}=1.0808,$$

giving $8\sin x+15\cos x=17\sin(x+1.0808)$.

The sine factor runs between -1 and 1, so the maximum is 17 and the minimum is -17.

For the equation, $17\sin(x+\alpha)=6$, so $\sin(x+\alpha)=\frac{6}{17}$. Put $u=x+\alpha$. As x runs over $[0,\ 2\pi]$, u runs over $[1.0808,\ 7.3640]$. The reference angle is $\arcsin\frac{6}{17} = 0.3607$, and $\frac{6}{17}$ is positive, so sine takes it in the first and second quadrants:

$$u=0.3607\qquad\text{and}\qquad u=\pi-0.3607=2.7809.$$

The first sits below 1.0808, so add a turn: $0.3607+2\pi=6.6439$, which is inside. The second is already inside, and a further turn from it would pass 7.3640. So $u=2.7809$ or $u=6.6439$, and subtracting α,

$$x=1.70\qquad\text{or}\qquad x=5.56\quad(3\text{ s.f.}).$$

Transforming the interval along with the substitution is what saved the second answer. Had u been left to run over $[0,\ 2\pi]$, the pair 0.3607 and 2.7809 would have given one negative x and lost 5.56 altogether.

YOUR TURN 8.2 Trigonometric Identities

6. Simplify each expression.

(a) $\sin^2\theta+\cos^2\theta$
(b) $2\sin\theta\cos\theta$
(c) $\cos^2\theta-\sin^2\theta$
(d) $1-2\sin^2\theta$

7. Given $\sin\theta=\dfrac{3}{5}$ with θ acute, find each exact value.

(a) $\cos\theta$
(b) $\tan\theta$
(c) $\sin 2\theta$
(d) $\cos 2\theta$

8. Given $\cos\theta=-\dfrac{3}{5}$ with θ in the second quadrant, find each exact value.

(a) $\sin\theta$
(b) $\tan\theta$
(c) $\sin 2\theta$
(d) $\cos 2\theta$

9. Use a compound-angle formula to find each exact value.

(a) $\cos 15^\circ$, from $45^\circ - 30^\circ$

(b) $\sin 75^\circ$, from $45^\circ + 30^\circ$

10. Prove each identity.

(a) $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$

(b) $\dfrac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$

(c) $(\sin\theta + \cos\theta)^2 = 1 + \sin 2\theta$

(d) $\dfrac{\cos 2\theta}{\cos\theta + \sin\theta} = \cos\theta - \sin\theta$

(e) $\tan\theta \sin 2\theta = 2\sin^2\theta$

(f) $\dfrac{\sin 3\theta}{\sin\theta} - \dfrac{\cos 3\theta}{\cos\theta} = 2$

8.3 Graphs of Trigonometric Functions

The graphs are the circle unrolled. Send the point P round the circle and plot its y-coordinate against the angle: that plot *is* $y = \sin x$. Plot its x-coordinate instead and you get $y = \cos x$.

Every feature of the sine graph can be read off the circle before you plot a single point.

- The y-coordinate of P never leaves -1 to 1, so the curve stays between those two heights. Its **amplitude**, the distance from the middle to the top, is 1.
- P returns to its starting place after one full turn, so the curve repeats every 2π. Its **period**, the width of one complete cycle, is 2π.
- $y = 0$ when P sits on the x-axis, at $\theta = 0$, π and 2π, so the curve crosses the axis at those three angles.
- y is largest at the top of the circle, at $\theta = \frac{\pi}{2}$, so the maximum is 1 and it occurs at $x = \frac{\pi}{2}$.
- $y > 0$ everywhere in the upper half of the circle, so the curve is above the axis for $0 < x < \pi$.

Read the x-coordinate instead and the cosine graph appears the same way. It starts at 1, because P starts on the positive x-axis, and falls to 0 at $\frac{\pi}{2}$, where P is at the top and its x-coordinate is zero.

So both curves have amplitude 1 and period 2π. They differ only in where a cycle begins. The x-coordinate at angle θ is the y-coordinate a quarter turn later, so $\cos\theta = \sin(\theta + \frac{\pi}{2})$, and the cosine curve is the sine curve shifted left by $\frac{\pi}{2}$.

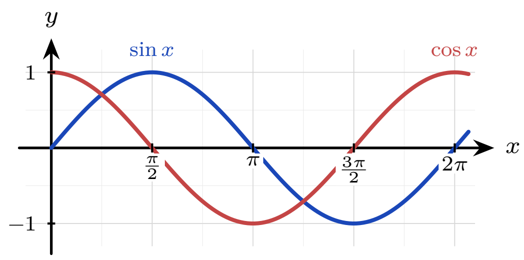


Figure 8.4 One turn of the circle unrolled. Both waves have amplitude 1 and period 2π; the cosine is the sine shifted left by $\frac{\pi}{2}$.

The tangent behaves differently, and the circle explains why. Since $\tan\theta = \frac{y}{x}$, it is undefined wherever $x = 0$, which is at the top and the bottom of the circle. That gives vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.

Its period is π, not 2π. Half a turn sends P to the point diametrically opposite, whose coordinates are $(-x, -y)$. The ratio $\frac{-y}{-x}$ equals $\frac{y}{x}$, so the tangent repeats after only half a revolution.

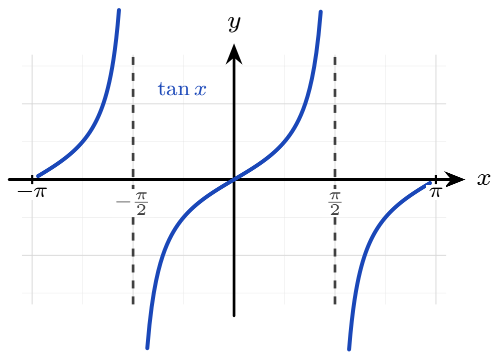


Figure 8.5 The tangent repeats every π, not every 2π, with a vertical asymptote wherever $\cos x = 0$.

That is the basic behaviour, collected in one place.

	$\sin x$	$\cos x$	$\tan x$
Domain	all x	all x	$x \neq \frac{\pi}{2} + n\pi$
Range	$-1 \leq y \leq 1$	$-1 \leq y \leq 1$	all y
Period	2π	2π	π
Amplitude	1	1	none
Zeros	$n\pi$	$\frac{\pi}{2} + n\pi$	$n\pi$

Here n stands for any integer, and the last row gives the values of x at which the curve crosses the axis. The tangent has no amplitude because it has no largest or smallest value. Amplitude measures half the rise and fall, and the tangent rises without limit.

8.3.1 Transforming the wave

The general sinusoid stretches and slides the basic sine, and each parameter has a plain meaning.

KEY For $y = a\sin\big(b(x+c)\big) + d$:

$$\text{amplitude} = |a|, \qquad \text{period} = \frac{2\pi}{|b|},$$

$$\text{horizontal shift} = -c, \qquad \text{vertical shift (midline)} = d.$$

Questions often run the other way, giving you the graph's largest and smallest values and asking for a and d. Since the maximum is $d + |a|$ and the minimum is $d - |a|$, adding and subtracting recovers both constants.

- **Amplitude.** From the equation it is $|a|$. From a graph it is $\dfrac{\text{max} - \text{min}}{2}$, half the total rise and fall.
- **Midline.** From the equation it is the line $y = d$. From a graph it is $y = \dfrac{\text{max} + \text{min}}{2}$, the level halfway between the top and the bottom.
- **Maximum and minimum.** They are $d + |a|$ and $d - |a|$: the midline, plus and minus the amplitude.
- **Period.** From the equation it is $\dfrac{2\pi}{|b|}$. From a graph it is the horizontal distance from one repeat to the next.

Read the first two as sentences and they stop being formulas to memorise. Consider the wave $y = 5\sin(2x) - 3$, whose maximum is 2 and whose minimum is -8. The amplitude is $\frac{2-(-8)}{2} = 5$ and the midline is $\frac{2+(-8)}{2} = -3$, which is what the equation says.

EXAMPLE 8.6

Describe $y = 2\sin(3x) + 1$ and state its range.

The amplitude is 2, so the wave rises and falls 2 either side of its midline. The period is $\frac{2\pi}{3}$, so it completes a full cycle three times as fast as $\sin x$. The midline is $y = 1$. The graph therefore runs from $1 - 2 = -1$ up to $1 + 2 = 3$, giving range $-1 \le y \le 3$.

8.3.2 Symmetry of the graphs HL

Each reflection symmetry of the circle from Section 8.1 becomes a visible symmetry of the graphs. The cosine curve is mirror-symmetric in the y-axis, so $\cos(-x) = \cos x$: cosine is an **even** function. The sine and tangent curves map onto themselves under a half-turn about the origin, so $\sin(-x) = -\sin x$ and $\tan(-x) = -\tan x$: both are **odd**, in the sense of Ch. 5. The same circle symmetries give the supplementary-angle relations.

KEY HL $\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta, \qquad \tan(\pi - \theta) = -\tan\theta.$

You can see all three on the graphs directly: the sine curve is symmetric about the vertical line $x = \frac{\pi}{2}$, while cosine and tangent flip sign across it. In an exam these relations remove the need for a calculator: $\cos\frac{3\pi}{4} = -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2}$.

YOUR TURN 8.3 Graphs of Trigonometric Functions

11. State the amplitude, the period and the midline of each graph.

(a) $y = 3\sin x$

(b) $y = \sin 2x$

(c) $y = 4\sin(3x) - 2$

(d) $y = 5\sin(4x) + 3$

12. State the range of each function.

(a) $y = 2\sin x + 1$

(b) $y = 3\sin(2x) - 1$

(c) $y = 2\cos\left(\frac{x}{2}\right) - 1$

13. Answer each question about $y = \tan x$.

(a) State its period.

(b) State the equations of two of its asymptotes.

(c) Explain why the asymptotes occur where they do.

14. A wheel of diameter 40 m has its lowest point 2 m above the ground and turns once every 20 seconds. A seat starts at the bottom.

(a) State the amplitude and the midline of the seat's height.

(b) Find the value of b in $h(t) = d - a\cos(bt)$.

(c) Write down $h(t)$.

(d) Find the height after 5 seconds.

8.4 Reciprocal Trigonometric Functions HL

Divide 1 by a sine, a cosine or a tangent and three more functions appear. Each has a name of its own, a graph with vertical asymptotes, and an identity that makes certain equations solvable.

KEY · IB $\sec\theta = \dfrac{1}{\cos\theta}, \qquad \operatorname{cosec}\theta = \dfrac{1}{\sin\theta}.$

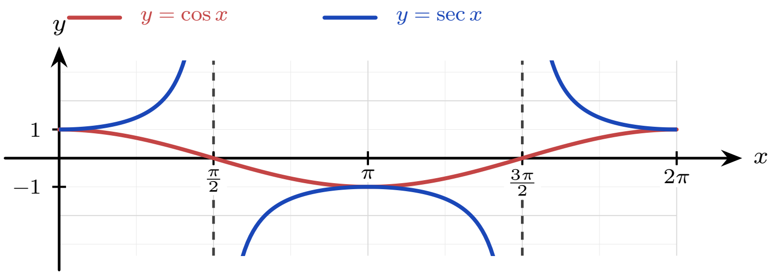
The third is the **cotangent**, $\cot\theta = \dfrac{1}{\tan\theta} = \dfrac{\cos\theta}{\sin\theta}$. Its definition is not printed in the formula booklet, so learn it.

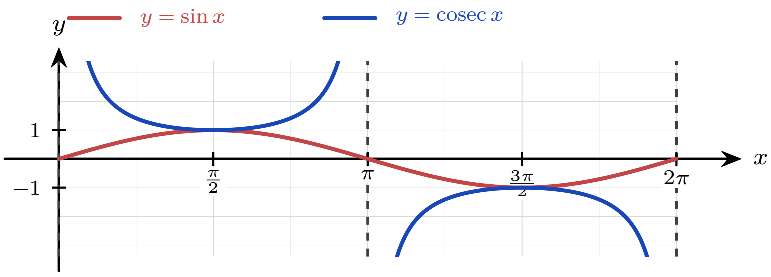
The pairing looks wrong at first: *sec*ant goes with *cos*ine, and *cosec*ant with *sin*e. There is no reason behind it, so use the third letter. In **sec** the third letter is c, for cosine. In **cosec** it is s, for sine. In **cot** it is t, for tangent.

Two facts follow from the reciprocal alone, and between them they fix every graph.

First, a reciprocal has an asymptote wherever the original function is zero, because you cannot divide by zero. So $\sec$ has asymptotes where $\cos\theta = 0$, at $\theta = \frac{\pi}{2} + n\pi$, while cosec and $\cot$ have them where $\sin\theta = 0$, at $\theta = n\pi$.

Second, $\sin$ and $\cos$ never exceed 1 in size, so their reciprocals are never smaller than 1 in size. Neither $\sec\theta$ nor $\operatorname{cosec}\theta$ ever takes a value strictly between -1 and 1. That gap in the middle is the feature examiners test most often.





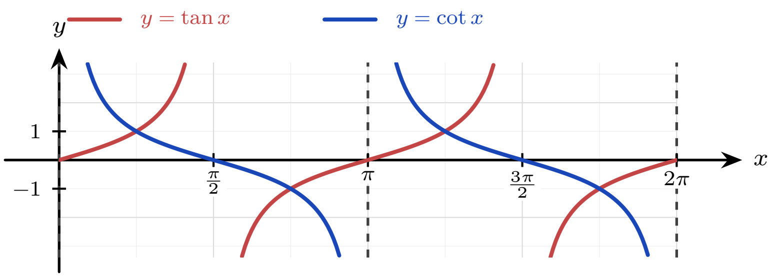


Figure 8.6 In each panel the parent function is scarlet and its reciprocal cobalt. The reciprocal has an asymptote, marked by a dashed line, wherever its parent crosses the axis, and the two curves meet wherever the parent reaches ± 1. Secant and cosecant never enter the band between -1 and 1; cotangent has no such gap.

Taking n to be any integer, the three read as follows.

Function	Domain	Range	Period
$\sec\theta = \dfrac{1}{\cos\theta}$	all θ except $\frac{\pi}{2} + n\pi$	$y \le -1$ or $y \ge 1$	2π
$\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$	all θ except $n\pi$	$y \le -1$ or $y \ge 1$	2π
$\cot\theta = \dfrac{\cos\theta}{\sin\theta}$	all θ except $n\pi$	all real y	π

Every excluded value in the domain column is a zero of the function underneath, and every one of them is an asymptote on the graph.

8.4.1 The two Pythagorean variants

The identity $\cos^2\theta + \sin^2\theta = 1$ has two companions, one for each reciprocal pair, and both come from dividing it through.

Dividing $\cos^2\theta + \sin^2\theta = 1$ by $\cos^2\theta$ gives

$$1 + \tan^2\theta = \sec^2\theta,$$

since $\frac{\sin\theta}{\cos\theta} = \tan\theta$ and $\frac{1}{\cos\theta} = \sec\theta$. Dividing the same identity by $\sin^2\theta$ instead gives the companion result. Deriving them when you need them is safer than trusting your memory of which is which.

KEY · IB $1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \operatorname{cosec}^2\theta.$

These identities exist to turn mixed equations into single-function ones, exactly as the double angles did.

EXAMPLE 8.7

Solve $\tan^2 x + \sec x = 1$ for $0 \le x \le 2\pi$.

The equation mixes $\tan$ and $\sec$; the identity $\tan^2 x = \sec^2 x - 1$ rewrites it in $\sec x$ alone:

$$\sec^2 x - 1 + \sec x = 1 \implies \sec^2 x + \sec x - 2 = 0 \implies (\sec x + 2)(\sec x - 1) = 0.$$

So $\sec x = -2$ or $\sec x = 1$, which is $\cos x = -\frac{1}{2}$ or $\cos x = 1$. From the circle,

$$x = \frac{2\pi}{3},\ \frac{4\pi}{3} \quad \text{or} \quad x = 0,\ 2\pi.$$

Converting back to cos at the end is the safest final step, because the unit circle is read in sine and cosine, not in secant.

YOUR TURN 8.4 Reciprocal Trigonometric Functions HL

15. Evaluate each exactly.

(a) $\sec\dfrac{\pi}{3}$

(b) $\operatorname{cosec}\dfrac{\pi}{2}$

(c) $\cot\dfrac{\pi}{4}$

(d) $\sec\dfrac{2\pi}{3}$

(e) $\cot\dfrac{2\pi}{3}$

(f) $\operatorname{cosec}\dfrac{7\pi}{6}$

16. Simplify each expression.

(a) $\sec^2\theta - \tan^2\theta$

(b) $\operatorname{cosec}^2\theta - \cot^2\theta$

(c) $\dfrac{1}{\cot\theta}$

(d) $\tan\theta\cot\theta$

(e) $\dfrac{\sec\theta}{\operatorname{cosec}\theta}$

(f) $\sin\theta\operatorname{cosec}\theta + \cos\theta\sec\theta$

17. Prove each identity.

(a) $\cot\theta\sec\theta = \operatorname{cosec}\theta$

(b) $\dfrac{\operatorname{cosec}\theta}{\cot\theta + \tan\theta} = \cos\theta$

(c) $\sec^2\theta + \operatorname{cosec}^2\theta = \sec^2\theta\operatorname{cosec}^2\theta$

(d) $\dfrac{1+\cot^2\theta}{1+\tan^2\theta} = \cot^2\theta$

8.5 Inverse Trigonometric Functions HL

Everything so far has gone one way: give an angle, get a ratio. The inverse functions run the other way, recovering the angle from the ratio.

You have already used them without naming them. When you solve $\sin x = \frac{1}{2}$ and press the $\sin^{-1}$ key, that key is the inverse function.

Two notations, one function. The inverse of sine is written either $\arcsin x$ or $\sin^{-1} x$.

They mean exactly the same thing. These notes use $\arcsin$, because the other form invites a serious confusion:

$$\sin^{-1} x \text{ means the } \textit{inverse}, \qquad \text{but} \qquad (\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x.$$

The -1 is not an exponent here. Writing $\arcsin$, $\arccos$ and $\arctan$ removes the ambiguity, and your calculator's $\sin^{-1}$ key is the same function under the other name.

8.5.1 Why a restriction is needed

There is an immediate difficulty. A function has an inverse only if it is one-to-one (Ch. 5), and $\sin$, $\cos$ and $\tan$ are as far from one-to-one as a function can be. Each takes every value in its range infinitely often, so the horizontal line $y = \frac{1}{2}$ meets $y = \sin x$ infinitely many times.

That is a real problem, not a technicality. If you ask "which angle has sine $\frac{1}{2}$?", there is no single answer: $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{13\pi}{6}$ and infinitely many others all qualify. A function must return one output, so it has to choose.

The fix is the one from Ch. 5: cut the domain down until the function is one-to-one, then invert what is left. Which piece to keep is a choice, and mathematicians have agreed on one for each function. Those agreed pieces are shaded in the diagrams below, and they are what fix the ranges in the table.

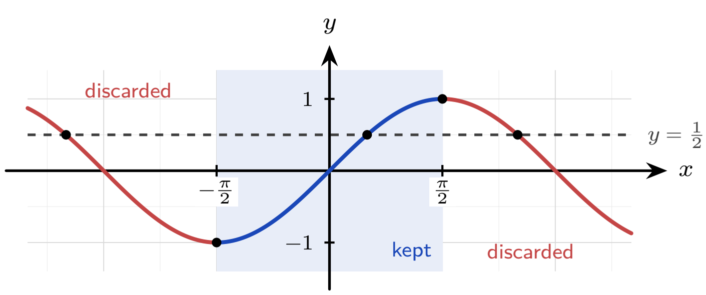


Figure 8.7 The dashed line $y = \frac{1}{2}$ meets $y = \sin x$ again and again, which is why "the angle whose sine is $\frac{1}{2}$" has no single answer. Keeping only the shaded piece from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ leaves one intersection. On that piece the curve rises steadily, so it is one-to-one and every output comes from exactly one input.

The same is done for the other two: cosine keeps $0 \le x \le \pi$, where it falls steadily, and tangent keeps $-\frac{\pi}{2} < x < \frac{\pi}{2}$, one branch between two asymptotes. Those kept pieces become the *ranges* of the inverses, because domain and range swap when a function is inverted.

Function	Domain	Range
$\arcsin x$	$-1 \le x \le 1$	$-\frac{\pi}{2} \le y \le \frac{\pi}{2}$
$\arccos x$	$-1 \le x \le 1$	$0 \le y \le \pi$
$\arctan x$	all real x	$-\frac{\pi}{2} < y < \frac{\pi}{2}$

Each inverse is a reflection of its restricted parent in the line $y = x$ (Ch. 5), so the graphs are the familiar curves turned on their sides. Arcsine climbs steeply from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$. Arccosine falls from $(-1, \pi)$ to $(1, 0)$. Arctangent flattens towards the horizontal asymptotes $y = \pm\frac{\pi}{2}$, which are the vertical asymptotes of tangent after the reflection.

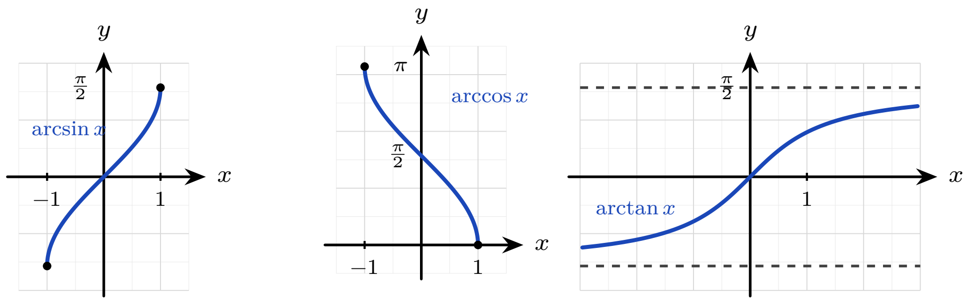


Figure 8.8 Each inverse is its restricted parent reflected in $y = x$. The marked endpoints show the range each one is confined to.

EXAMPLE 8.8

Find the exact values of (a) $\arcsin\left(-\frac{\sqrt{3}}{2}\right)$, (b) $\arccos\left(-\frac{\sqrt{3}}{2}\right)$, (c) $\arcsin\left(\sin\frac{5\pi}{4}\right)$.

(a) The angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is $-\frac{\sqrt{3}}{2}$ is $-\frac{\pi}{3}$. Not $\frac{4\pi}{3}$ and not $\frac{5\pi}{3}$: those solve the equation but lie outside arcsine's range.

(b) The angle in $[0, \pi]$ whose cosine is $-\frac{\sqrt{3}}{2}$ is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Arccosine handles negatives by reaching into the second quadrant.

(c) $\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$, and $\arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}$. The composition returns $-\frac{\pi}{4}$, not $\frac{5\pi}{4}$: arcsine can only answer with angles from its own range.

CAUTION $\arcsin(\sin x)$ does not always give back x. Every answer from arcsin must lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, and 2 does not. So $\arcsin(\sin 2) = \pi - 2$, an angle with the same sine as 2 but inside the range. The restriction is working, not failing.

8.5.2 Compositions the other way round

A composition can also run with the inverse on the inside, as in $\cos(\arcsin k)$. The angle is never found here, and it never needs to be. Name it: let $\theta = \arcsin k$, so $\sin\theta = k$. A right triangle with opposite k and hypotenuse 1 has exactly that sine, and Pythagoras gives its adjacent side as $\sqrt{1-k^2}$, so

$$\cos(\arcsin k) = \sqrt{1-k^2}.$$

The positive root is the right one because arcsin answers in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and cosine is never negative there. The same triangle serves arccos and arctan: place the known ratio on the two sides it names, find the third by Pythagoras, then read off whatever the question asks for.

EXAMPLE 8.9

Find the exact value of (a) $\cos\left(\arcsin\dfrac{20}{29}\right)$, (b) $\sin\left(\arccos\dfrac{9}{41}+\arctan\dfrac{20}{21}\right)$.

(a) Let $\theta = \arcsin\frac{20}{29}$, so $\sin\theta = \frac{20}{29}$: opposite 20, hypotenuse 29. The adjacent side is $\sqrt{29^2-20^2}=\sqrt{441}=21$, so

$$\cos\left(\arcsin\frac{20}{29}\right)=\frac{21}{29},$$

positive because θ lies in $[-\frac{\pi}{2},\ \frac{\pi}{2}]$.

(b) Name both angles: $A=\arccos\frac{9}{41}$ and $B=\arctan\frac{20}{21}$. A 9–40–41 triangle gives $\cos A=\frac{9}{41}$ and $\sin A=\frac{40}{41}$, taken positive because $\arccos$ answers in $[0,\ \pi]$, where sine is never negative. A 20–21–29 triangle, the same one as in part (a), gives $\sin B=\frac{20}{29}$ and $\cos B=\frac{21}{29}$, both positive because $\tan B>0$ puts B in the first quadrant. The sum is now a compound angle:

$$\sin(A+B)=\sin A\cos B+\cos A\sin B=\frac{40}{41}\cdot\frac{21}{29}+\frac{9}{41}\cdot\frac{20}{29}=\frac{840+180}{1189}=\frac{1020}{1189}.$$

Neither A nor B was ever evaluated. The range of each inverse function is what fixes the sign of every side, and that is the only place in the working where a decision is made.

YOUR TURN 8.5 Inverse Trigonometric Functions HL

18. Evaluate each exactly.

(a) $\arcsin\dfrac{1}{2}$

(b) $\arccos 0$

(c) $\arctan(-1)$

(d) $\arccos\left(-\dfrac{1}{2}\right)$

19. An inverse undoes its function only on the piece that was kept.

(a) State the range of $\arcsin x$, of $\arccos x$ and of $\arctan x$.

(b) Evaluate $\arcsin\left(\sin\dfrac{3\pi}{4}\right)$, and explain why the answer is not $\dfrac{3\pi}{4}$.

(c) Evaluate $\arccos\left(\cos\dfrac{4\pi}{3}\right)$.

(d) Evaluate $\arctan\left(\tan\dfrac{5\pi}{6}\right)$.

(e) State every value of x for which $\arcsin(\sin x)=x$.

20. GDC Find the exact value, or write **undefined**. Check each on your GDC.

(a) $\arcsin\left(\sin\dfrac{2\pi}{3}\right)$

(b) $\arccos\left(\cos\dfrac{3}{2}\right)$

(c) $\tan(\arctan 12)$

(d) $\cos\left(\arccos\dfrac{2\pi}{3}\right)$

(e) $\arctan\left(\tan\left(-\dfrac{3\pi}{4}\right)\right)$

(f) $\sin(\arcsin\pi)$

(g) $\sin\left(\arctan\dfrac{3}{4}\right)$
(h) $\cos\left(\arcsin\dfrac{7}{25}\right)$
(i) $\arcsin\left(\tan\dfrac{\pi}{3}\right)$
(j) $\arctan\left(2\sin\dfrac{\pi}{3}\right)$
(k) $\cos\left(\arctan\dfrac{1}{2}\right)$
(l) $\cos(\arcsin 0.6)$
(m) $\sin\left(\arccos\dfrac{3}{5}+\arctan\dfrac{5}{12}\right)$
(n) $\cos\left(\arctan 3+\arcsin\dfrac{1}{3}\right)$

8.6 Solving Trigonometric Equations

An equation like $\sin x = \frac{1}{2}$ has not one solution but infinitely many, because the sine repeats every 2π. Your calculator returns only one of them, the **principal value**. Everything else you must find yourself, and the circle is what tells you how.

Start by counting the solutions in a single turn. Solving $\sin x = k$ means finding the points on the circle whose y-coordinate is k, so draw the horizontal line at height k and see where it meets the circle. Three things can happen.

- If $-1 < k < 1$ the line cuts the circle in **two** places, so two angles in each turn have sine k.
- If $k = 1$ or $k = -1$ the line touches the circle at **one** point, the top or the bottom. The two angles have merged into one.
- If $|k| > 1$ the line misses the circle, and there is **no** solution at all.

The same three cases hold for $\cos x = k$, read with a vertical line at $x = k$ instead. Tangent behaves differently: it takes every real value, so $\tan x = k$ has solutions whatever k is.

Two words are needed next. You will use both constantly.

DEF The **reference angle** $\hat{\theta}$ is the acute angle between the arm OP and the x-axis. It is positive, it is less than $\frac{\pi}{2}$, and it ignores signs entirely: you get it from the size of k alone, as $\arcsin|k|$, $\arccos|k|$ or $\arctan|k|$. It is defined whenever the arm lies off the axes.
The **principal value** is the single angle your calculator returns. It is the one solution lying inside the range of the matching inverse function:

$$\arcsin k \in \left[-\tfrac{\pi}{2},\, \tfrac{\pi}{2}\right], \qquad \arccos k \in [0,\, \pi], \qquad \arctan k \in \left(-\tfrac{\pi}{2},\, \tfrac{\pi}{2}\right).$$

The two are easy to confuse. A principal value can be negative, and for a cosine it can be obtuse; a reference angle is never either. Take $\cos x = -\frac{1}{2}$. The calculator returns the principal value $\frac{2\pi}{3}$, while the reference angle is $\frac{\pi}{3}$, the acute angle the arm makes with the negative x-axis. Work from the reference angle and the calculator's choice of quadrant stops mattering.

The method. Every solution is built from the reference angle and the quadrant it sits in. The sign of k tells you which two quadrants to use, by the rule of §8.1.2, and each quadrant has one construction.

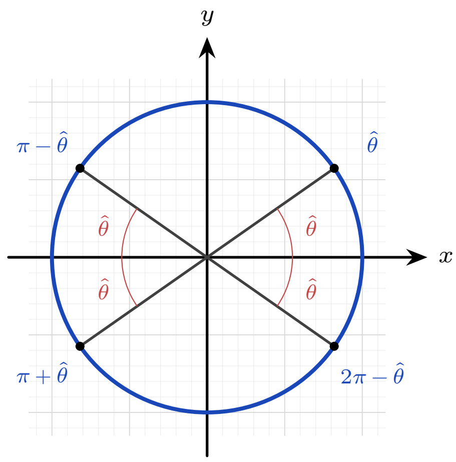


Figure 8.9 One reference angle $\hat{\theta}$, measured from the x-axis, gives an angle in each quadrant. The scarlet arcs are all the same size; only the direction they are measured from changes.

KEY **Solving a trigonometric equation.**

1. Rearrange until a single trigonometric function is alone on one side.
2. If the argument is not plain x, substitute u for it and **transform the interval as well**.
3. Find the reference angle from the *size* of k: $\hat{\theta} = \arcsin|k|$, $\arccos|k|$ or $\arctan|k|$.
4. Use the sign of k to choose the two quadrants, then build the angle in each: $\hat{\theta}$, $\pi - \hat{\theta}$, $\pi + \hat{\theta}$ or $2\pi - \hat{\theta}$.
5. Add or subtract whole turns of 2π until you have every solution the interval asks for.
6. If you substituted, convert back to x at the very end.

There is a shorter route once you already have one solution, which you always do: the calculator has just given you one. Call it θ. Its partner in the same turn is fixed by the function alone, whatever quadrant θ landed in.

KEY **The partner solution.** If θ is one solution, then within $0 \le x \le 2\pi$ the other is

$$\sin x = k: \quad \pi - \theta, \qquad \cos x = k: \quad 2\pi - \theta, \qquad \tan x = k: \quad \pi + \theta.$$

Then add or subtract whole turns of 2π — for the tangent, half turns of π — to reach every solution the interval asks for.

The interval decides how the partner is written, not what it is. On $-\pi \le x \le \pi$ the cosine pair is θ and $-\theta$, because $2\pi-\theta$ is one whole turn away from $-\theta$. That is why a cosine equation on a symmetric interval always answers $\pm$ the same number, and it is worth recognising on sight: $\cos x = \frac{1}{2}$ on $-\pi \leq x \leq \pi$ gives $x = \pm\frac{\pi}{3}$ and nothing else.

Both routes give the same angles. The reference angle is the one to fall back on when the signs get confusing, or when the equation has been rearranged so far that you no longer trust which quadrant you are in.

Two checks belong in every solution. Your two angles should lie in the quadrants where that function has the sign of k; if only one does, you have built the wrong pair. And when $k = \pm 1$ the two constructions land on the same point, so report that angle once.

Step 2 is where marks are most often lost, and step 5 is where solutions go missing.

EXAMPLE 8.10

Solve $\cos x = -\dfrac{1}{2}$ for $0 \leq x \leq 2\pi$.
Take the reference angle from the size of k alone:

$$\hat{\theta} = \arccos\left|-\tfrac{1}{2}\right| = \arccos\tfrac{1}{2} = \frac{\pi}{3}.$$

Cosine is negative in the second and third quadrants, so build one angle in each:

$$\pi - \frac{\pi}{3} = \frac{2\pi}{3}, \qquad \pi + \frac{\pi}{3} = \frac{4\pi}{3}.$$

Both lie in the interval, so $x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.
Notice that the calculator alone would have given only $\frac{2\pi}{3}$. The reference angle gives both, and the sign of k says where they live.

EXAMPLE 8.11

Solve $\sin x = -0.8$ for $0 \leq x \leq 2\pi$, giving the answers to three significant figures.
The constant is not one of the exact values, but the method does not change. The reference angle comes from the size of k: $\hat{\theta} = \arcsin 0.8 = 0.9273$. Sine is negative in the third and fourth quadrants, giving $\pi + 0.9273 = 4.0689$ and $2\pi - 0.9273 = 5.3559$. Both lie in the interval, so

$$x = 4.07 \quad \text{or} \quad x = 5.36 \quad \text{(3 s.f.)}.$$

Always ask which *other* quadrant shares the sign; that second angle is the one candidates most often miss.

TIP On a calculator paper, always check your solution count graphically: plot both sides and count the intersections in the interval. If your algebra found three solutions and the screen shows four, the screen is right.

When the argument is not plain x but $2x$ or $3(x-4)$, substitute for the whole argument and, crucially, transform the interval along with it. A doubled argument sweeps the circle twice as fast, so it collects twice as many solutions.

EXAMPLE 8.12

Solve $\cos 2x = \dfrac{\sqrt{3}}{2}$ for $0 \le x \le 2\pi$.

Let $u = 2x$. As x runs over $[0, 2\pi]$, u runs over $[0, 4\pi]$: two full turns.

The reference angle is $\hat{\theta} = \arccos \frac{\sqrt{3}}{2} = \frac{\pi}{6}$. Cosine is positive, so the first turn gives

$$u = \frac{\pi}{6} \quad \text{and} \quad u = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}.$$

The interval covers a second turn, so add 2π to each:

$$\frac{\pi}{6} + 2\pi = \frac{13\pi}{6}, \qquad \frac{11\pi}{6} + 2\pi = \frac{23\pi}{6}.$$

A third turn would pass 4π, so the list is complete:

$$u = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6} \implies x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}.$$

Four solutions, not two. Note where the doubling acts: the step in u is still 2π, because $\cos u$ has period 2π whatever u came from. What doubling changes is the *interval*, and that is why two extra solutions appear. In x the period has become π.

8.6.1 When the interval is not 0 to 2π

The interval is part of the question, and it is not always $0 \le x \le 2\pi$. The commonest alternative is $-\pi \le x \le \pi$. That is still one whole turn, but measured differently: half a turn clockwise from the positive x-axis, and half a turn anticlockwise.

The method does not change. Build the two angles from the reference angle exactly as before, then shift either one by a whole turn if it falls outside the interval: subtract 2π from an angle above π, add 2π to an angle below $-\pi$. A whole turn returns the point to exactly where it was, so the value of the function is untouched.

EXAMPLE 8.13

Solve $\cos x = -\dfrac{1}{2}$ for $-\pi \le x \le \pi$.

The reference angle is $\hat{\theta} = \arccos \frac{1}{2} = \frac{\pi}{3}$. Cosine is negative, so the two quadrants are the second and the third:

$$\pi - \frac{\pi}{3} = \frac{2\pi}{3}, \qquad \pi + \frac{\pi}{3} = \frac{4\pi}{3}.$$

The first already lies in $[-\pi, \ \pi]$. The second does not, because $\frac{4\pi}{3} > \pi$, so subtract one turn:

$$\frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}.$$

So $x = \frac{2\pi}{3}$ or $x = -\frac{2\pi}{3}$.
The two answers form a $\pm$ pair, and that is not a coincidence. Cosine is even, so on any interval symmetric about 0 its answers are $\pm\alpha$ for a single angle α. Sine is odd, and its pair on the same interval is

$$\hat{\theta} \text{ and } \pi - \hat{\theta} \quad \text{when } k > 0, \qquad -\hat{\theta} \text{ and } -\pi + \hat{\theta} \quad \text{when } k < 0.$$

8.6.2 Using identities

An equation mixing $\sin 2x$ with $\sin x$, or $\cos 2x$ with $\cos x$, cannot be solved as it stands. An identity rewrites everything in terms of a single function, turning the equation into an algebraic one, usually a quadratic.

EXAMPLE 8.14

Solve $\cos 2x + 3\cos x + 2 = 0$ for $0 \leq x \leq 2\pi$.
Choose the form of $\cos 2x$ that gives only cosines, $\cos 2x = 2\cos^2 x - 1$:

$$2\cos^2 x - 1 + 3\cos x + 2 = 0 \implies 2\cos^2 x + 3\cos x + 1 = 0.$$

This factors as $(2\cos x + 1)(\cos x + 1) = 0$, so $\cos x = -\frac{1}{2}$ or $\cos x = -1$. From the circle,

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{or} \quad x = \pi.$$

Picking the form of the double-angle identity that matches the other terms is the key step in these problems.

YOUR TURN 8.6 Solving Trigonometric Equations

21. Solve for $0 \leq x \leq 2\pi$.

(a) $\sin x = \frac{1}{2}$

(b) $\cos x = 0$

(c) $\tan x = 1$

(d) $\cos x = 1$

22. Solve for $0 \leq x \leq 2\pi$.

(a) $\sin x = -\frac{1}{2}$

(b) $2\cos x = \sqrt{3}$

(c) $\sqrt{2}\sin x = -1$

(d) $\tan x = -\sqrt{3}$

23. Solve for $0 \leq x \leq 2\pi$. Transform the interval before solving.

(a) $\sin 2x = 0$

(b) $\cos 2x = \frac{1}{2}$

(c) $\tan 2x = 1$

24. Solve for $0 \leq x \leq 2\pi$, using an identity first.

(a) $2\sin^2 x - \sin x = 0$
(b) $2\cos^2 x + \cos x - 1 = 0$
(c) $\cos 2x + \sin x = 0$
(d) $2\cos^2 x - 3\cos x + 1 = 0$
(e) $\sin 2x = \cos x$
(f) $\tan^2 x - \tan x = 0$

25. Solve for $-\pi \leq x \leq \pi$. Give every answer inside that interval.

(a) $\cos x = \dfrac{\sqrt{3}}{2}$
(b) $\sin x = -\dfrac{1}{2}$
(c) $\tan x = -1$
(d) $2\sin x = \sqrt{2}$

Chapter 8 · Trigonometry

Reflections

TOK A sine wave is periodic: it repeats forever, identical in every cycle. Yet it is built from the endlessly irrational π and takes irrational values at almost every point. How can something so regular be made of something so unruly? The physicist Joseph Fourier then proved that *any* repeating signal, however jagged, is a sum of pure sine waves. So the smooth circle is hidden inside every square wave and every spoken word. Is the sinusoid a human tool, or the alphabet in which nature writes all its rhythms?

IM Trigonometry began as astronomy. Babylonian and Greek observers needed to predict the wandering of the planets, and the sine was born as a tool for the geometry of the celestial sphere. Its greatest early development came in India and in the Islamic world. Astronomers there tabulated sines to many decimal places, to fix the times of prayer and the direction of Mecca. For two thousand years, the functions of this chapter were the mathematics of the heavens.

RW Because a sine wave is what steady rotation looks like from the side, it models every oscillation in science and engineering. Alternating current rises and falls as a sine while the generator spins. Sound is pressure oscillating in the same way, and its pitch is the frequency of the wave. Light, radio, and every signal your phone sends and receives are sinusoids too. Fourier analysis, decomposing a signal into its sine components, is how music is compressed, images are stored, and earthquakes and heartbeats are read.

EXT In radians, $\sin x$ and x almost agree for small x: $\sin 0.1 = 0.09983\ldots$ On the unit circle x is the arc and $\sin x$ is the height, and over a tiny arc there is almost no difference between them. Physicists use this as the **small-angle approximation**: for $|x|$ small and in radians,

$$\sin x \approx x, \qquad \tan x \approx x, \qquad \cos x \approx 1 - \tfrac{1}{2}x^2.$$

At $x = 0.1$ these give 0.1, 0.1003 and 0.995, against true values 0.09983, 0.10033 and 0.99500. The pendulum equation is only solvable because $\sin\theta$ is replaced by θ, which is why a pendulum keeps time for small swings and drifts for large ones. All three fail in degrees, and that is what makes radians the natural unit for calculus. Chapter 13 turns the first into $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$, the limit that gives $\dfrac{d}{dx}\sin x = \cos x$.

End-of-Chapter Problems — Chapter 8: Trigonometry

1. Use the unit circle to write down the exact value of each.

(a) $\sin\dfrac{2\pi}{3}$ [1]

(b) $\cos\dfrac{2\pi}{3}$ [1]

(c) $\tan\dfrac{2\pi}{3}$ [2]

(d) $\cos\dfrac{5\pi}{4}$ [2]

(e) $\sin\dfrac{11\pi}{6}$ [2]

(f) $\tan\dfrac{3\pi}{4}$ [2]

2. The point P lies on the unit circle at angle θ.

(a) State the coordinates of P when $\theta = \dfrac{5\pi}{6}$. [2]

(b) P has coordinates $\left(-\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2}\right)$. Find θ for $0 \le \theta \le 2\pi$. [2]

(c) P lies in the third quadrant. State the sign of $\sin\theta$, of $\cos\theta$ and of $\tan\theta$. [3]

(d) Given $\sin\theta = \dfrac{\sqrt{3}}{2}$, write down the two possible values of θ in $0 \le \theta \le 2\pi$, and explain why there are exactly two. [3]

3. Prove each identity. They are arranged in increasing order of difficulty.

(a) $\cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$ [2]

(b) $\dfrac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ [3]

(c) $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$ [3]

(d) $\cos^4\theta - \sin^4\theta = \cos 2\theta$ [3]

(e) $\dfrac{1-\cos 2\theta}{1+\cos 2\theta} = \tan^2\theta$ [4]

(f) $\dfrac{1}{\sec\theta - \tan\theta} = \sec\theta + \tan\theta$ [4]

(g) $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\operatorname{cosec}\theta$ [5]

(h) $\dfrac{1+\sin 2\theta}{\cos 2\theta} = \dfrac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}$ [5]

4. Given $\cos\theta = -\dfrac{3}{5}$ with θ in the second quadrant, find the exact value of each.

(a) $\sin\theta$ [2]

(b) $\tan\theta$ [1]

(c) $\sin 2\theta$ [2]

(d) $\cos 2\theta$ [2]

5. Given $\tan\theta = 2$ with θ acute, find the exact value of each.

(a) $\sin\theta$ and $\cos\theta$ [2]
(b) $\sin 2\theta$ [2]
(c) $\cos 2\theta$ [2]

6. Use a compound-angle formula to find each exact value.

(a) $\tan 75°$ [4]
(b) $\cos 15°$ [3]
(c) $\sin 105°$ [3]
(d) $\tan 105°$ [4]

7. Prove each result.

(a) $\sin(A+B)+\sin(A-B)=2\sin A\cos B$ [4]
(b) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ [5]
(c) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ [5]

8. Solve for $0 \le x \le 2\pi$.

(a) $\sin x = \dfrac{1}{2}$ [2]
(b) $\sqrt{3}\tan x = 1$ [3]
(c) $2\cos^2 x = 1$ [4]
(d) $\tan^2 x = 3$ [4]

9. Solve for $0 \le x \le 2\pi$. Each becomes a quadratic.

(a) $2\sin^2 x + \sin x - 1 = 0$ [5]
(b) $4\sin^2 x - 3 = 0$ [4]
(c) $\cos 2x = \cos x$ [5]
(d) $\sin 2x = \sin x$ [5]

10. Solve for $0 \le x \le 2\pi$.

(a) $2\sin x\cos x = \dfrac{1}{2}$ [4]
(b) $\cos 2x + 3\sin x = 2$ [6]
(c) $\sin x + \cos x = 1$ [6]

11. Solve each equation for $-\pi \le x \le \pi$. Build every answer from the reference angle, then shift by a whole turn where the interval requires it.

(a) $\sin x = -\dfrac{1}{2}$ [3]
(b) $\cos x = \dfrac{\sqrt{3}}{2}$ [3]
(c) $\tan x = -1$ [3]
(d) $2\cos^2 x - 1 = 0$ [4]
(e) $\sin 2x = \dfrac{\sqrt{3}}{2}$ [5]
(f) $\tan\left(x + \dfrac{\pi}{4}\right) = \sqrt{3}$ [5]

12. Given the identity $1 + \tan^2\theta = \sec^2\theta$, solve $\sec^2 x = 4$ for $0 \le x \le \dfrac{\pi}{2}$. [4]

13. Evaluate, giving exact values.

(a) $\arctan 1$ [1]
(b) $\arcsin\left(-\dfrac{1}{2}\right)$ [1]

(c) $\arccos(-1)$ [1]   (d) $\arcsin\left(\sin\dfrac{5\pi}{6}\right)$ [3]

14. Consider $f(x) = 3\sin(2x) + 1$.

(a) State the amplitude and the period. [2]
(b) State the maximum and minimum values. [2]

15. The graph of $y = a\sin(bx) + d$ has maximum 7, minimum 1, and period π. Find a, b and d. [4]

16. Sketch $y = \cos x - 1$ for $0 \leq x \leq 2\pi$, stating the maximum, the minimum, and the intercepts. [4]

17. The depth of water in a harbour is $d(t) = 5 + 3\sin\left(\dfrac{\pi t}{6}\right)$ metres, t hours after midnight.

(a) State the maximum and minimum depths. [2]
(b) Find the period, and say what it represents. [2]

18. A seat on a Ferris wheel has height $h(t) = 10 - 8\cos\left(\dfrac{\pi t}{15}\right)$ metres, t seconds after the start.

(a) State the minimum and maximum heights. [2]
(b) Find the time for one full revolution. [2]

19. GDC Express $3\sin x + 4\cos x$ in the form $R\sin(x + \alpha)$ with $R > 0$.

(a) Find R and α. [3]
(b) Hence state the maximum value. [1]

20. GDC Find the maximum value M of $5\sin x + 12\cos x$, and the value of x in $[0, 2\pi]$ at which it occurs. [5]

21. Use the symmetry of the unit circle to answer each part.

(a) Given $\sin 0.6 = 0.565$, write down $\sin(\pi - 0.6)$. [2]
(b) Given $\cos 1.2 = 0.362$, write down $\cos(-1.2)$ and $\cos(2\pi - 1.2)$. [3]
(c) Explain why $\sin(\theta + 2\pi) = \sin\theta$ for every θ. [2]
(d) Explain why $\tan(\theta + \pi) = \tan\theta$, but $\sin(\theta + \pi) \neq \sin\theta$. [4]

22. Solve each equation on the interval given. Take care: the same equation has different answers on different intervals.

(a) $\cos x = -\dfrac{\sqrt{2}}{2}$ for $0 \le x \le 2\pi$, and then for $-\pi \le x \le \pi$. [4]
(b) $\sin x = -\dfrac{\sqrt{3}}{2}$ for $-\pi \le x \le \pi$. [3]
(c) $\tan 2x = \sqrt{3}$ for $-\pi \le x \le \pi$. [5]
(d) Explain why an equation of the form $\cos x = k$, with $|k| < 1$, always has two solutions on $-\pi \le x \le \pi$, and why they are negatives of each other. [3]

23. GDC **Daylight in Sapporo.** The number of hours of daylight is modelled by

$$L(t) = 12.2 + 3.4 \sin\left(\frac{2\pi(t-81)}{365}\right),$$

where t is the day of the year, with $t = 1$ on 1 January.

(a) State the maximum and minimum daylight, and find the value of t at which each occurs. [4]
(b) State the period of L, and say what it represents. [2]
(c) Find the daylight on 1 January, to three significant figures. [2]
(d) Find the two days of the year on which the daylight is exactly 12.2 hours, and explain what is special about them. [4]
(e) Find the values of t for which $L(t) \ge 15$, and state how many days of the year have at least 15 hours of daylight. [6]

24. **One function, two forms.** Let $f(x) = 2\sin x + \cos 2x$ for $0 \le x \le 2\pi$.

(a) Show that $f(x) = -2\sin^2 x + 2\sin x + 1$. [3]
(b) Let $u = \sin x$. Write f as a quadratic in u, and state the interval of values u can take. [3]
(c) Hence show that the maximum value of f is $\dfrac{3}{2}$. [4]
(d) Find the values of x in $[0, 2\pi]$ at which that maximum occurs. [3]
(e) Solve $f(x) = 1$ for $0 \le x \le 2\pi$. [5]
(f) State the range of f. [3]

25. Find the exact value where there is one, and write **undefined** where there is not.

(a) $\arcsin\left(\sin \dfrac{7\pi}{6}\right)$ [2]
(b) $\arccos\left(\cos \dfrac{7\pi}{4}\right)$ [2]
(c) $\arctan\left(\tan \dfrac{2\pi}{3}\right)$ [2]
(d) $\sin\left(\arccos \dfrac{5}{13}\right)$ [2]
(e) $\tan\left(\arcsin \dfrac{8}{17}\right)$ [3]
(f) $\cos(\arctan 2)$ [2]
(g) $\arccos(\sin \pi)$ [2]
(h) $\sin(\arcsin 2)$ [2]
(i) $\cos\left(\arcsin \dfrac{3}{5} + \arctan \dfrac{4}{3}\right)$ [4]
(j) $\tan\left(2\arctan \dfrac{1}{3}\right)$ [3]

26. **Ten identities.** Prove each one. They draw on the double-angle, compound-angle and reciprocal results of this chapter, and they are arranged in increasing order of difficulty.

(a) $\sin(A+B) - \sin(A-B) = 2\cos A \sin B$ [3]

(b) $\dfrac{\sin 2\theta}{1-\cos 2\theta} = \cot\theta$ [3]

(c) $\operatorname{cosec} 2\theta = \frac{1}{2}\sec\theta \operatorname{cosec}\theta$ [3]

(d) $\dfrac{\sec\theta + 1}{\tan\theta} = \dfrac{\tan\theta}{\sec\theta - 1}$ [4]

(e) $\tan\left(\theta + \dfrac{\pi}{4}\right) = \dfrac{1+\tan\theta}{1-\tan\theta}$ [4]

(f) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B$ [5]

(g) $\dfrac{1+\tan^2\theta}{1-\tan^2\theta} = \sec 2\theta$ [5]

(h) $\cot\theta - \tan\theta = 2\cot 2\theta$ [5]

(i) $\dfrac{\cos 2\theta}{1+\sin 2\theta} = \dfrac{1-\tan\theta}{1+\tan\theta}$ [6]

(j) $\operatorname{cosec}\theta - \cot\theta = \tan\dfrac{\theta}{2}$ [6]

Challenge Questions

27. GDC **The wave form, and when an equation can be solved.** A sum of a sine and a cosine of the same angle is itself a single shifted sine wave: $a\sin\theta + b\cos\theta = R\sin(\theta+\alpha)$. This investigation proves it and then uses it to decide, without solving, whether an equation has any solutions at all.

Throughout take $a > 0$ and $b > 0$.

(a) Expand $R\sin(\theta+\alpha)$ and compare it with $a\sin\theta + b\cos\theta$ to show that $R\cos\alpha = a$ and $R\sin\alpha = b$. [3]

(b) Hence show that $R = \sqrt{a^2+b^2}$ and $\tan\alpha = \dfrac{b}{a}$. *(Hint: square and add)* [3]

(c) Write $7\sin\theta + 24\cos\theta$ in the form $R\sin(\theta+\alpha)$, and state its maximum and minimum values. [3]

(d) Explain why the equation $a\sin\theta + b\cos\theta = k$ has a solution if and only if $|k| \le R$. [4]

(e) Solve $3\sin x + 4\cos x = 5$ for $0 \le x \le 2\pi$, giving your answer to three significant figures. Explain why this equation has exactly one solution in the interval. [5]

28. $\cos n\theta$ **as a polynomial in** $\cos\theta$**.** You have already met $\cos 2\theta = 2\cos^2\theta - 1$ and, in the end-of-chapter problems, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Each is a polynomial in $\cos\theta$ alone. This investigation shows that this always happens, and finds the pattern.

Write $c = \cos\theta$ throughout.

(a) Use the compound-angle formulas to show that

$$\cos((n+1)\theta) = 2\cos\theta\,\cos(n\theta) - \cos((n-1)\theta).$$

(Hint: expand $\cos(n\theta + \theta)$ and $\cos(n\theta - \theta)$, then add) [5]

(b) Starting from $\cos 0 = 1$ and $\cos\theta = c$, use the recurrence to obtain $\cos 2\theta$ and $\cos 3\theta$ in terms of c. [3]

(c) Use it again to find $\cos 4\theta$ and $\cos 5\theta$ in terms of c. [4]

(d) State the degree of the polynomial for $\cos n\theta$, and the coefficient of its highest power. Justify both from the recurrence. [4]

(e) Explain why no such polynomial in $\cos\theta$ alone exists for $\sin n\theta$ when n is even. *(Hint: compare $\sin(-\theta)$ with $\sin\theta$)* [3]

29. GDC **The window of safe passage.** At a harbour mouth the depth of water is modelled by

$$d(t) = 6 + 2\sin\left(\frac{\pi t}{6}\right) \text{ metres,}$$

where t is the number of hours after midnight. A ferry can enter only while the depth is at least the figure its hull requires. This investigation finds the window of time available, and then finds it in general.

(a) State the greatest and least depths, the period of the tide, and the time of the first high tide. [4]

(b) The ferry needs at least 7 m of water. Solve $d(t) \geq 7$ on $0 \leq t \leq 12$, and state the window as a pair of clock times. [5]

(c) Show that the window is exactly 4 hours long, and explain why the same window opens once in every tidal cycle. [3]

(d) Now take a general tide $d(t) = m + a\sin\left(\frac{2\pi t}{T}\right)$ and a required depth k with $m < k < m + a$. Show that the time available in each cycle is

$$\frac{T}{\pi}\arccos\left(\frac{k-m}{a}\right).$$

(Hint: solve for the two ends in terms of $\arcsin$, subtract, then use $\frac{\pi}{2} - \arcsin c = \arccos c$) [6]

(e) A larger ferry needs 7.5 m. Use the formula to find its window, to the nearest minute. Then explain, from the shape of the arccosine graph, why the last half-metre of clearance costs far more time than the first. [5]

GEOMETRY & TRIGONOMETRY

Chapter 9

Vectors

9

SYLLABUS 3.12–3.18 HL only

A camera tripod has three legs, not four. The choice is deliberate, and the reason is geometry rather than engineering.

Any three points lie in one flat surface, whatever their heights. All three feet of a tripod therefore reach the ground together, on any floor, however uneven. A fourth point need not lie in that surface. When it does not, the stand rocks between two pairs of legs. This is why a stool with four legs wobbles where a stool with three legs does not.

Ordinary arithmetic cannot express that fact. The fact is not about how large anything is. It is about position, direction and flatness. To state it, and to prove it, we need a language in which a direction is something we can calculate with.

A **vector** is a quantity that has both a size and a direction. Speed is a number, but velocity is a vector. Distance is a number, but displacement is a vector. A pilot who flies due north through a wind from the west does not travel due north, and does not travel at the aircraft's airspeed. The motion and the wind point in different directions, and both must be counted.

Once a quantity has a direction, addition changes. Three plus three need not give six. If the two vectors point the same way, the answer is six. If they point in opposite directions, the answer is zero. Any value between those two is possible, and the angle decides which one.

By the end of this chapter you will add and scale vectors, use the scalar product to find angles and to test for perpendicularity, and use the vector product to find areas and normals. You will write the equation of a line and of a plane, decide how two lines or three planes are arranged, and find shortest distances. A vector holds a size and a direction in a single object, which makes it the natural tool for describing position and movement in three dimensions. Physics, computer graphics and navigation all rely on it, from the pilot's crosswind to the satellites overhead.

9.1 Vectors: Magnitude and Direction

A vector has exactly two properties: a **magnitude**, which is its length, and a **direction**. Where it sits in space does not matter. Two arrows of the same length pointing the same way are the same vector.

An arrow drawn to represent a vector is called a **directed line segment**. "Directed" because it has an arrowhead, and "line segment" because it is a piece of a line with two ends. Examination questions use this phrase; it means an arrow and nothing more.

Vectors must be kept apart from ordinary numbers, which in this context are called **scalars**. Three notations are in common use, and they all mean the same thing. Printed books, including this one, use a bold letter $\mathbf{a}$. By hand you cannot write bold type, so you write either a letter with an arrow above it, $\vec{a}$, or an underlined letter, $\underline{a}$. An arrow is also used for a vector named by its two endpoints: $\overrightarrow{AB}$ means the vector from the point A to the point B.

The most useful representation is the **column vector**. It lists how far to travel along each axis to get from the tail to the head.

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}},$$

Here $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are the **base vectors**: arrows of length 1 along the x-, y- and z-axes. The hat is there to record that each one has length 1, a point returned to in the next subsection. Many books print them without hats, as $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$; the two notations are equivalent, and the formula booklet uses only the column form. The column form and the base-vector form describe the same vector, so use whichever is convenient.

9.1.1 The algebra of arrows

Vectors are added **tip to tail**. Draw the second vector starting where the first one ends. The sum is the arrow from the start of the first to the end of the second. In components it is simple: everything happens coordinate by coordinate.

KEY

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}, \qquad k\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix}.$$

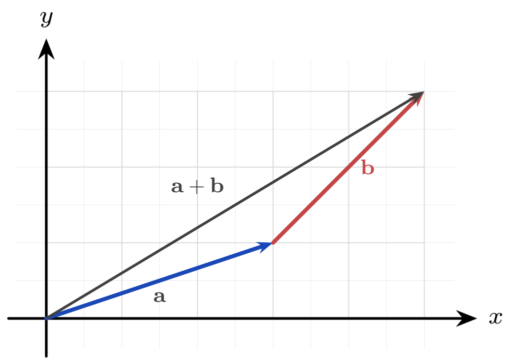


Figure 9.1 Adding tip to tail. Draw $\mathbf{b}$ starting where $\mathbf{a}$ ends; the sum runs from the start of $\mathbf{a}$ to the end of $\mathbf{b}$.

Scalar multiplication by k scales the length by $|k|$ and reverses the direction when k is negative. This gives a clean test: two nonzero vectors are **parallel** exactly when one is a scalar multiple of the other, $\mathbf{a} = k\mathbf{b}$.

Two particular vectors have names of their own.

The **negative** of $\mathbf{v}$, written $-\mathbf{v}$, is the vector of the same length pointing the opposite way. It is the case $k = -1$ of scalar multiplication. Subtraction is then just addition of the negative:

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).$$

The **zero vector** $\mathbf{0}$ has magnitude 0 and no direction at all. It is what you get from $\mathbf{v} + (-\mathbf{v})$, or from $0\mathbf{v}$. It is the only vector whose direction is undefined, and it is written in bold to keep it apart from the number 0.

EXAMPLE 9.1

Let $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$. Find $2\mathbf{a} - \mathbf{b}$, and determine whether it is parallel to $\mathbf{c} = \begin{pmatrix} -8 \\ -12 \\ 10 \end{pmatrix}$ and whether it is parallel to $\mathbf{d} = \begin{pmatrix} 12 \\ 18 \\ -14 \end{pmatrix}$.

Scale first, then subtract, working one coordinate at a time:

$$2\mathbf{a} - \mathbf{b} = \begin{pmatrix} 6 \\ 2 \\ -4 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ -5 \end{pmatrix}.$$

Parallel means one vector is a scalar multiple of the other, so look for a single k that works in all three coordinates. For $\mathbf{c}$, the first coordinate forces $k = -2$, and that value holds throughout:

$$-2\begin{pmatrix}4\\6\\-5\end{pmatrix} = \begin{pmatrix}-8\\-12\\10\end{pmatrix} = \mathbf{c},$$

so $\mathbf{c}$ is parallel to $2\mathbf{a} - \mathbf{b}$, pointing the opposite way because k is negative. For $\mathbf{d}$, the first two coordinates both give $k = 3$, but $3 \times (-5) = -15$, and the third coordinate of $\mathbf{d}$ is -14. No single k works, so $\mathbf{d}$ is not parallel to $2\mathbf{a} - \mathbf{b}$.

Test all three coordinates before answering. Two coordinates that agree are not a near miss but a different direction, and a question is usually built so that the third coordinate is where the test fails.

9.1.2 Magnitude and unit vectors

Three words are used for the same quantity. The **magnitude** of a vector, its **length**, and its **modulus** all mean the size of the arrow, and all are written $|\mathbf{v}|$. These notes use whichever reads more naturally in the sentence. The value comes straight from Pythagoras, extended into three dimensions.

KEY · IB The magnitude of $\mathbf{v} = \begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}$ is $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$.

Dividing a vector by its own length leaves the direction untouched but sets the length to 1. The result is a **unit vector**, written $\hat{\mathbf{v}}$, the pure direction stripped of any magnitude, ready to be scaled to any length you like.

KEY $\hat{\mathbf{v}} = \dfrac{\mathbf{v}}{|\mathbf{v}|}$ is the unit vector in the direction of $\mathbf{v}$.

The base vectors are unit vectors, which is what their hats record: $|\hat{\mathbf{i}}| = |\hat{\mathbf{j}}| = |\hat{\mathbf{k}}| = 1$, one unit along each axis. So writing $\mathbf{a} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}}$ says that $\mathbf{a}$ is built by taking a_1 steps of length 1 along x, then a_2 along y, then a_3 along z.

EXAMPLE 9.2

Find a vector of magnitude 14 in the direction of $\mathbf{v} = \begin{pmatrix}2\\3\\6\end{pmatrix}$.

First the length: $|\mathbf{v}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7$. The unit vector is $\hat{\mathbf{v}} = \frac{1}{7}\mathbf{v}$, and scaling it to length 14 means multiplying by 14:

$$14\hat{\mathbf{v}} = \frac{14}{7}\begin{pmatrix}2\\3\\6\end{pmatrix} = \begin{pmatrix}4\\6\\12\end{pmatrix}.$$

Reduce to a unit vector first, then rescale. Those two steps answer every question of the form "find a vector of magnitude m in the direction of".

YOUR TURN 9.1 Vectors: Magnitude and Direction

1. Find the magnitude of each vector.

(a) $\begin{pmatrix}3\\4\\0\end{pmatrix}$ (b) $\begin{pmatrix}2\\-1\\2\end{pmatrix}$

(c) $\begin{pmatrix}0\\3\\4\end{pmatrix}$ (d) $\begin{pmatrix}1\\1\\1\end{pmatrix}$

2. Given $\mathbf{a} = \begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}0\\-1\\4\end{pmatrix}$, evaluate each of the following.

(a) $\mathbf{a}+\mathbf{b}$ (b) $\mathbf{a}-\mathbf{b}$

(c) $\mathbf{a}+2\mathbf{b}$ (d) $3\mathbf{a}-\mathbf{b}$

3. Unit vectors and scaling.

(a) Find the unit vector in the direction of $\begin{pmatrix}0\\3\\4\end{pmatrix}$.

(b) Find the unit vector in the direction of $\begin{pmatrix}2\\-1\\2\end{pmatrix}$.

(c) Find a vector of magnitude 6 in the direction of $\begin{pmatrix}1\\2\\2\end{pmatrix}$.

(d) Explain why every unit vector has magnitude exactly 1, whatever its direction.

4. Decide whether each pair of vectors is parallel, and justify your answer.

(a) $\begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\begin{pmatrix}2\\4\\6\end{pmatrix}$ (b) $\begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\begin{pmatrix}2\\4\\5\end{pmatrix}$

9.2 Position Vectors and Geometry

Fix the origin O. The **position vector** of a point A is the arrow $\overrightarrow{OA}$ from the origin to A; its components are simply the coordinates of A. Position vectors are the bridge between the free-floating algebra of arrows and the fixed points of a geometry problem.

The arrow from one point to another is found by subtraction. Almost everything that follows rests on this one fact.

KEY $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, the position vector of the head minus that of the tail.

The distance between A and B is then the magnitude of that displacement, $|\overrightarrow{AB}| = |\mathbf{b} - \mathbf{a}|$, and the midpoint of AB has position vector $\frac{1}{2}(\mathbf{a} + \mathbf{b})$.

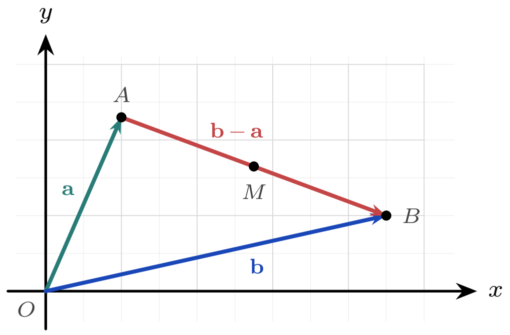


Figure 9.2 Head minus tail. The two position vectors run from the origin; the arrow between the points closes the triangle, so $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. The midpoint M sits at the average of the two position vectors, $\frac{1}{2}(\mathbf{a} + \mathbf{b})$.

EXAMPLE 9.3

Points $A(2, -3, 1)$ and $B(5, 1, 6)$ are given. Find $\overrightarrow{AB}$ and the distance AB.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 5 \\ 1 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}, \qquad AB = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50} = 5\sqrt{2}.$$

Head minus tail gives the arrow; the magnitude of that arrow is the distance. Every three-dimensional distance problem reduces to those two steps.

Three points A, B, C are **collinear** (on one straight line) when $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel, that is, one is a scalar multiple of the other. This is the vector version of "same direction from a shared point", and the equation of a line in the next section is built from it.

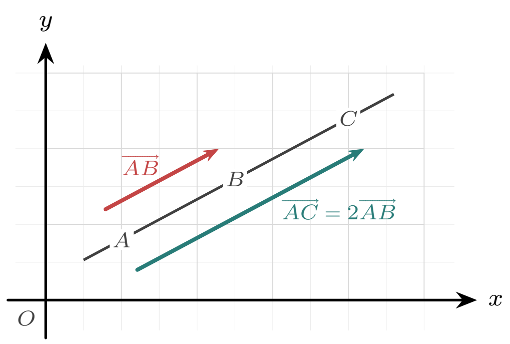


Figure 9.3 Collinear points. Both displacements start at A, and one is a scalar multiple of the other, so

the three points lie on a single line. The two arrows are drawn slightly to either side of that line so they can be told apart.

9.2.1 Proving geometry with vectors

Vectors give a way of proving geometric facts by algebra, with no diagram-chasing and no coordinates. The method is always the same three steps.

KEY **Proving a geometric property with vectors.**

1. Choose one point as the origin, and name the position vectors of the others with single letters.
2. Write every point in the figure in terms of those letters.
3. Show the required relation by algebra.

Choosing the origin well is what keeps the algebra short. Put it at a vertex that many edges meet.

EXAMPLE 9.4

Prove that the diagonals of a parallelogram bisect each other.

Let the parallelogram be $OABC$, with the origin at O. Write $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.

Because $OABC$ is a parallelogram, $\overrightarrow{CB} = \overrightarrow{OA} = \mathbf{a}$, so the fourth vertex is

$$\overrightarrow{OB} = \mathbf{a} + \mathbf{c}.$$

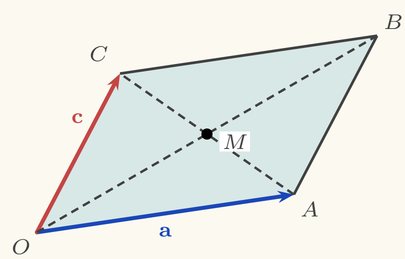


Figure 9.4 The parallelogram $OABC$. The two dashed diagonals meet at the point M, and the proof below shows that M is the midpoint of both.

The two diagonals are OB and AC. Find the midpoint of each.

Midpoint of OB: $\frac{1}{2}(\mathbf{0} + \mathbf{a} + \mathbf{c}) = \frac{1}{2}(\mathbf{a} + \mathbf{c})$.

Midpoint of AC: $\frac{1}{2}(\mathbf{a} + \mathbf{c})$.

The two midpoints are the same point. A single point lying halfway along both diagonals is exactly what "bisect each other" means. ■

Notice that no coordinates were used, and the result holds for every parallelogram, in a plane or in space.

EXAMPLE 9.5

M and N are the midpoints of the sides OA and OB of a triangle. Prove that MN is parallel to AB and half its length.

Take the origin at O, with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The midpoints are then

$$\overrightarrow{OM} = \tfrac{1}{2}\mathbf{a}, \qquad \overrightarrow{ON} = \tfrac{1}{2}\mathbf{b}.$$

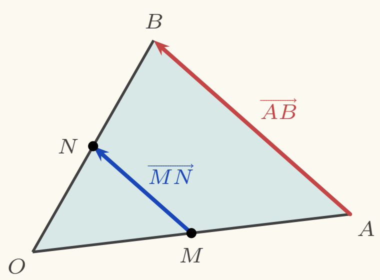


Figure 9.5 The midpoint theorem. Joining the midpoints of two sides gives a segment parallel to the third side and half as long.

Now find the two displacement vectors, each as head minus tail:

$$\overrightarrow{MN} = \tfrac{1}{2}\mathbf{b} - \tfrac{1}{2}\mathbf{a} = \tfrac{1}{2}(\mathbf{b} - \mathbf{a}), \qquad \overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

So $\overrightarrow{MN} = \frac{1}{2}\overrightarrow{AB}$.

One vector is a scalar multiple of the other, so the two segments are parallel, and the scalar is $\frac{1}{2}$, so MN is half the length of AB. ■

That is the entire proof. This result is known as the midpoint theorem.

YOUR TURN 9.2 Position Vectors and Geometry

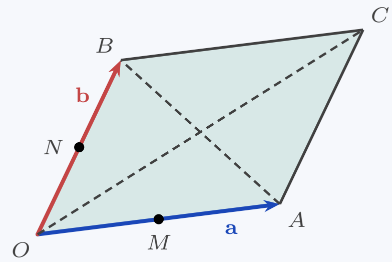
5. Points $A(1, 2, -1)$ and $B(4, 0, 3)$ are given.

(a) Find $\overrightarrow{AB}$.
(b) Find the distance AB.
(c) Find the midpoint of AB.
(d) Find $\overrightarrow{BA}$.

6. Further position-vector work.

(a) A is $(1, 1, 1)$ and $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$. Find the coordinates of B.
(b) B is the midpoint of AC, with $A(2, 1, 3)$ and $B(4, 5, 3)$. Find C.
(c) Show that $A(1, 1, 1)$, $B(2, 3, 5)$ and $C(4, 7, 13)$ are collinear.

7. Proving geometry with vectors. In each part, take the origin at O and write $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$. Parts (a) to (c) refer to the parallelogram below; M and N are the midpoints of OA and OB.



(a) $OACB$ is a parallelogram. Write $\overrightarrow{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
(b) Find the midpoint of the diagonal OC, and the midpoint of the diagonal AB.
(c) Hence prove that the diagonals of a parallelogram bisect each other.
(d) M and N are the midpoints of OA and OB. Prove that $\overrightarrow{MN} = \frac{1}{2}\overrightarrow{AB}$, and state what this shows about the two segments.

9.3 The Scalar Product

There are two ways to multiply vectors, and they answer two different questions. The first, the **scalar product** (or dot product), takes two vectors and returns a number, and that number carries the angle between them. It has two forms, and they always agree.

KEY · IB **Scalar product.** For $\mathbf{a}, \mathbf{b}$ with angle θ between them,

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 \qquad \text{and} \qquad \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|\cos\theta.$$

In the second form, the $\cos\theta$ is not decoration. Ordinary multiplication assumes both quantities are measured along the same line: 3×4 makes sense because both numbers sit on one number line. Two vectors usually point different ways, so multiplying their lengths would measure nothing in particular.

The repair is to bring them onto the same line first. Keep $\mathbf{a}$ as it is, and use only the part of $\mathbf{b}$ that lies along $\mathbf{a}$, the shadow $\mathbf{b}$ casts on $\mathbf{a}$, which has length $|\mathbf{b}|\cos\theta$. Multiplying those two gives

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}|\cos\theta,$$

the length of $\mathbf{a}$ times how much of $\mathbf{b}$ goes in $\mathbf{a}$'s direction. Doing it the other way round, casting $\mathbf{a}$ onto $\mathbf{b}$, gives $|\mathbf{b}| \times |\mathbf{a}|\cos\theta$, the same number. That is why the order does not matter.

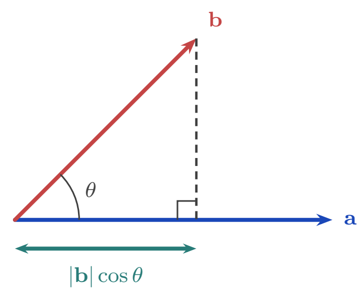


Figure 9.6 The shadow of $\mathbf{b}$ on $\mathbf{a}$. Dropping the tip of $\mathbf{b}$ perpendicularly onto $\mathbf{a}$ leaves a length of $|\mathbf{b}|\cos\theta$, and the scalar product multiplies that by $|\mathbf{a}|$.

The scalar product then obeys the rules you would expect of a multiplication.

- **Commutative:** $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$. The order does not matter.
- **Distributive:** $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$. Brackets expand as usual.
- **Scalars move outside:** $(k\mathbf{a})\cdot\mathbf{b}=k(\mathbf{a}\cdot\mathbf{b})$. A constant factor can be taken out.
- **Dotted with itself:** $\mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$, since $\theta=0$ and $\cos 0=1$. The result is the squared length, and this is the line used most often in proofs.

The base vectors show both extremes at once. Each has length 1 and lies along its own axis, so dotting one with itself gives $1\times1\times\cos 0=1$. Different axes are at right angles, so dotting two different base vectors gives 0.

$$\hat{\mathbf{i}}\cdot\hat{\mathbf{i}}=\hat{\mathbf{j}}\cdot\hat{\mathbf{j}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}=1,\qquad \hat{\mathbf{i}}\cdot\hat{\mathbf{j}}=\hat{\mathbf{j}}\cdot\hat{\mathbf{k}}=\hat{\mathbf{k}}\cdot\hat{\mathbf{i}}=0.$$

This is exactly why the component form works. Expanding $(a_1\hat{\mathbf{i}}+a_2\hat{\mathbf{j}}+a_3\hat{\mathbf{k}})\cdot(b_1\hat{\mathbf{i}}+b_2\hat{\mathbf{j}}+b_3\hat{\mathbf{k}})$ produces nine terms, and every term pairing two *different* base vectors is wiped out by that zero. Only the three matching terms survive, leaving $a_1b_1+a_2b_2+a_3b_3$.

The component form is easy to compute; the cosine form carries the geometry. Setting the two equal is what lets you find the angle between any two directions, and it is used throughout the chapter.

KEY · IB $$\cos\theta=\frac{a_1b_1+a_2b_2+a_3b_3}{|\mathbf{a}|\,|\mathbf{b}|}.$$

EXAMPLE 9.6

Find the angle between $\mathbf{a}=\begin{pmatrix}3\\2\\6\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}2\\6\\3\end{pmatrix}$.

Compute the three ingredients: $\mathbf{a}\cdot\mathbf{b}=6+12+18=36$, $|\mathbf{a}|=\sqrt{9+4+36}=7$, $|\mathbf{b}|=7$. Then

$$\cos\theta=\frac{36}{49}\implies\theta=\arccos\frac{36}{49}\approx 42.7^\circ.$$

Dot product over product of lengths: the recipe never changes, whether the vectors come from a triangle, a pair of lines, or a pair of planes.

The most important special case is a right angle. Since $\cos 90^\circ = 0$, the scalar product of perpendicular vectors is zero, and the converse holds too. This gives a test for perpendicularity that needs no angles at all, just a multiply-and-add.

KEY $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0 \quad$ (for nonzero $\mathbf{a}, \mathbf{b}$).

Parallel vectors sit at the opposite extreme. There θ is 0 or π, so $\cos\theta$ is $+1$ or -1 and the scalar product reaches the largest size it can. Taking the modulus covers both directions at once.

KEY For nonzero $\mathbf{a}, \mathbf{b}$: $\quad \mathbf{a} \parallel \mathbf{b} \iff |\mathbf{a} \cdot \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|$.

So the scalar product recognises both extremes. It is zero when the vectors are perpendicular, and its size is as large as possible when they are parallel. Every angle in between falls somewhere on that scale.

EXAMPLE 9.7

Find the angle between $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$.

The dot product is $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(0) + (2)(-1) = 0$. Because it vanishes, the vectors are perpendicular: $\theta = 90^\circ$. Neither magnitude was needed: a zero dot product settles the angle at once.

YOUR TURN 9.3 The Scalar Product

8. Evaluate each scalar product.

(a) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$

(b) $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

(d) $\begin{pmatrix} 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

9. GDC Find the angle between each pair of vectors.

(a) $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

(b) $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$

(c) $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

10. Perpendicular and parallel vectors.

(a) Given $|\mathbf{a}| = 3$, $|\mathbf{b}| = 4$ and an angle of 60° between them, find $\mathbf{a} \cdot \mathbf{b}$.

(b) Find k so that $\begin{pmatrix} 2 \\ k \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 0$.

(c) Find k so that $\begin{pmatrix} 1 \\ k \\ 2 \end{pmatrix}$ is perpendicular to $\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$.

(d) Explain why $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$ for every vector $\mathbf{a}$.

(e) Given $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$ and $\mathbf{a} \cdot \mathbf{b} = -15$, state what this tells you about the directions of $\mathbf{a}$ and $\mathbf{b}$, and justify your answer.

11. A rhombus has sides $\mathbf{a}$ and $\mathbf{b}$ drawn from one vertex, so $|\mathbf{a}| = |\mathbf{b}|$.

(a) Write the two diagonals in terms of $\mathbf{a}$ and $\mathbf{b}$.

(b) Expand $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$.

(c) Hence prove that the diagonals of a rhombus are perpendicular.

(d) Verify your result for $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

9.4 The Vector Equation of a Line

A line is fixed by two pieces of information: one point it passes through, and a direction it runs along. Start at a known point $\mathbf{a}$ and move any distance along a direction $\mathbf{b}$. As the **parameter** λ takes every real value, this reaches every point on the line.

KEY · IB **Vector equation of a line:** $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, where $\mathbf{a}$ is the position vector of a point on the line and $\mathbf{b}$ is a direction vector.

This differs from the equation of a line met earlier, and the difference matters. An equation such as $y = 2x + 1$ is a *test*: give it a point and it tells you whether that point is on the line. The vector equation is a *recipe*: give it a value of λ and it hands you a point. That is why the vector form survives the move into three dimensions, where no single equation of the form $y = mx + c$ can describe a line at all.

Writing $\mathbf{r} = (x, y, z)$ and reading off each coordinate gives the **parametric form**; eliminating λ between them gives the **Cartesian form**.

KEY · IB **Parametric:** $x = x_0 + \lambda l, \quad y = y_0 + \lambda m, \quad z = z_0 + \lambda n.$

Cartesian: $\dfrac{x - x_0}{l} = \dfrac{y - y_0}{m} = \dfrac{z - z_0}{n}, \quad$ where $\mathbf{b} = \begin{pmatrix} l \\ m \\ n \end{pmatrix}$.

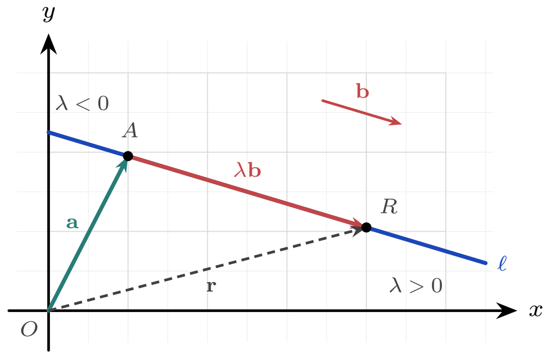


Figure 9.7 A line built from a point and a direction. The vector $\mathbf{a}$ reaches the known point A; travelling $\lambda\mathbf{b}$ from there lands on a general point R, whose position vector $\mathbf{r}$ closes the triangle. Negative λ runs back the other way, so the whole line is covered. The direction $\mathbf{b}$ has no fixed home, so the free copy drawn above is the same vector.

The direction vector carries the geometry of how two lines relate. The **angle between two lines** is the angle between their direction vectors, found by the scalar product (take the acute value). Two lines are **parallel** when their directions are scalar multiples. And in three dimensions there is a possibility with no two-dimensional analogue: lines that are neither parallel nor crossing.

Two lines in space fall into exactly one of four cases. Two questions settle which: are the directions parallel, and do the lines share a point?

- **Coincident.** Directions parallel, every point shared. The same line written two ways.
- **Parallel.** Directions parallel, no point shared. Different lines running the same way.
- **Intersecting.** Directions not parallel, exactly one point shared. They cross once.
- **Skew.** Directions not parallel, no point shared. They pass without meeting, like an overpass above a road.

Only the last case is new. Two lines in a plane must either be parallel or cross, but in three dimensions they can miss each other entirely.

KEY **Classifying two lines.**

1. Compare the directions. If they are scalar multiples, check whether a point of one lies on the other: yes means **coincident**, no means **parallel**.
2. If the directions differ, set the two equations equal and solve for the parameters. Two components fix them; the third decides.
3. A consistent solution means the lines **intersect**, and substituting back gives the point of intersection. An inconsistent system means they are **skew**.

EXAMPLE 9.8

Do the lines $\mathbf{r}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{r}_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ intersect?

The directions are not multiples of each other, so the lines are not parallel. Setting components equal:

$$2 + \lambda = 1 + 2\mu, \qquad 1 + \lambda = 3 - \mu, \qquad \lambda = 2 + \mu.$$

From the first two, $\lambda - 2\mu = -1$ and $\lambda + \mu = 2$, so $\mu = 1$, $\lambda = 1$. Check in the third: $\lambda = 1$ and $2 + 1 = 3$. Since $1 \neq 3$, the system is inconsistent, so the lines are **skew**. Two equations fix the parameters; the third decides whether the lines really meet.

9.4.1 Lines in motion: kinematics

So far the parameter has had no meaning attached to it. Give it one and the line equation describes a moving object. Read λ as *time* t, and $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ says: start at $\mathbf{a}$, and move with constant **velocity** $\mathbf{b}$. The direction of $\mathbf{b}$ is the heading; its magnitude $|\mathbf{b}|$ is the **speed**. This is precisely the pilot from the opening page, with velocity the tip-to-tail sum of airspeed and wind.

EXAMPLE 9.9

An aircraft passes the point $(20, 10)$ (coordinates in km) at $t = 0$ and flies with constant velocity $\begin{pmatrix} 240 \\ 100 \end{pmatrix}$ km h^{-1}. Find its speed, its position at $t = 2$ hours, and whether its path passes through the airport at $(500, 210)$.

The speed is the magnitude of the velocity: $\sqrt{240^2 + 100^2} = 260$ km h^{-1}. The position at time t is

$$\mathbf{r} = \begin{pmatrix} 20 \\ 10 \end{pmatrix} + t \begin{pmatrix} 240 \\ 100 \end{pmatrix}, \qquad \text{so at } t = 2: \quad \mathbf{r} = \begin{pmatrix} 500 \\ 210 \end{pmatrix}.$$

That *is* the airport, reached at exactly $t = 2$. To test any point, solve the first coordinate for t, then check the others. One value of t must satisfy every equation at once. This is the same test used for intersecting lines.

Two objects moving at constant velocity are always some distance apart, and that distance falls as they approach and rises again once they have passed. The moment of **closest approach** is found from the vector between them. Subtracting the position vectors gives $\mathbf{r}_B - \mathbf{r}_A$, whose components are linear in t, so the squared distance

$$d^2 = |\mathbf{r}_B - \mathbf{r}_A|^2$$

is a quadratic in t. Completing the square gives its least value and the time at which that value occurs.

Work with d^2 rather than with d. Squaring clears the square root, and the two are least at the same instant, since squaring preserves order among non-negative numbers.

EXAMPLE 9.10

Two birds fly in straight lines at constant velocity. Taking t in seconds and distances in metres, their position vectors are

$$\mathbf{r}_A = \begin{pmatrix} 8 \\ 5 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \qquad \mathbf{r}_B = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + t \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}.$$

Find the time at which they are closest, and the least distance between them.
Subtract, component by component:

$$\mathbf{r}_B - \mathbf{r}_A = \begin{pmatrix} 2-8 \\ 3-5 \\ 3-1 \end{pmatrix} + t \begin{pmatrix} 3-1 \\ 1-(-1) \\ 2-3 \end{pmatrix} = \begin{pmatrix} 2t-6 \\ 2t-2 \\ 2-t \end{pmatrix}.$$

Its squared length is

$$d^2 = (2t-6)^2 + (2t-2)^2 + (2-t)^2 = (4t^2-24t+36) + (4t^2-8t+4) + (t^2-4t+4) = 9t^2-36t+44.$$

Complete the square:

$$d^2 = 9\left(t^2-4t\right) + 44 = 9(t-2)^2 - 36 + 44 = 9(t-2)^2 + 8.$$

The squared bracket is never negative and is zero only at $t = 2$, so the least value of d^2 is 8, reached at $t = 2$ seconds. The least distance is

$$d = \sqrt{8} = 2\sqrt{2} \approx 2.83 \text{ m}.$$

Check either side: at $t = 1$ and at $t = 3$ the formula gives $d^2 = 17$ both times, larger than 8 and equal to each other, which is what a parabola symmetric about $t = 2$ requires.

YOUR TURN 9.4 The Vector Equation of a Line

12. Writing the equation of a line.

(a) Write a vector equation of the line through $A(1,2,3)$ with direction $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$.
(b) Find a direction vector of the line through $A(1,0,2)$ and $B(4,3,5)$.
(c) Write a vector equation of that line.

13. Forms of the equation.

(a) Write $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ in parametric form.
(b) Write $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ in Cartesian form.

(c) Does $(3, 3, 2)$ lie on $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$? Justify your answer.

14. An aircraft passes $(10, 5)$ at $t = 0$, with velocity $\begin{pmatrix} 150 \\ 200 \end{pmatrix}$ km h^{-1}.

(a) Find its speed.

(b) Write its position vector at time t.

(c) Find its position after 2 hours.

15. A boat leaves the point $(2, -1)$, in km, at $t = 0$ and moves with constant velocity $\begin{pmatrix} 6 \\ 8 \end{pmatrix}$ km h^{-1}.

(a) Find the speed of the boat.

(b) Write down its position vector at time t.

(c) Find its position after 30 minutes.

(d) A lighthouse stands at $(20, 15)$. Show that the boat does not pass through it.

9.5 The Vector Product

The second way to multiply vectors returns not a number but a *vector*. The **vector product** (or cross product) $\mathbf{a} \times \mathbf{b}$ points perpendicular to *both* of the vectors it is built from, and its length measures the area they span. Where the dot product was about alignment, the cross product is about the plane the two vectors sweep out.

KEY · IB **Vector product.** For $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$,

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}, \qquad \mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \sin\theta\, \hat{\mathbf{n}},$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and $\hat{\mathbf{n}}$ is the unit vector perpendicular to both, with direction given by the right-hand screw rule. Taking magnitudes of the second form gives $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$, since $|\hat{\mathbf{n}}| = 1$.

The direction is fixed by the **right-hand screw rule**: turn a screw from $\mathbf{a}$ towards $\mathbf{b}$ and it advances along $\mathbf{a} \times \mathbf{b}$. Four properties are used constantly.

- **Perpendicular to both.** The result is at right angles to the plane of $\mathbf{a}$ and $\mathbf{b}$. This is what makes it the tool for building normal vectors in the next section.
- **Not commutative.** It is **anti-commutative**: $\mathbf{b} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{b})$. The order matters, and

reversing it reverses the answer.

- **Distributive, and scalars come out.** $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$ and $(k\mathbf{v}) \times \mathbf{w} = k(\mathbf{v} \times \mathbf{w})$.
- **Zero for parallel vectors.** If $\mathbf{a}$ and $\mathbf{b}$ are parallel then $\sin\theta = 0$, so $\mathbf{a} \times \mathbf{b} = \mathbf{0}$. In particular $\mathbf{v} \times \mathbf{v} = \mathbf{0}$, the mirror image of $\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2$.

The base vectors again show the pattern most clearly. Each is perpendicular to the other two, so crossing two different ones gives a unit vector along the third axis, and the screw rule fixes which way it points. Crossing one with itself gives $\mathbf{0}$, since the angle is 0 and $\sin 0 = 0$.

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \qquad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \qquad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}, \qquad \hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = \mathbf{0}.$$

Read the first three in the cyclic order $\hat{\mathbf{i}} \to \hat{\mathbf{j}} \to \hat{\mathbf{k}} \to \hat{\mathbf{i}}$: going forwards gives a plus, and going backwards gives a minus, so $\hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}$.

9.5.1 One array, two products

Both products come out of the same nine numbers. Multiply every component of $\mathbf{a}$ by every component of $\mathbf{b}$ and lay the results in a square.

	b_1	b_2	b_3
a_1	a_1b_1	a_1b_2	a_1b_3
a_2	a_2b_1	a_2b_2	a_2b_3
a_3	a_3b_1	a_3b_2	a_3b_3

The **diagonal** is the scalar product. Those are the three terms where a base vector meets itself, and $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$ keeps them; everything off the diagonal is killed by $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$. Adding the shaded entries gives $a_1b_1 + a_2b_2 + a_3b_3$.

The **off-diagonal** entries are the vector product, taken in opposite pairs. Here the diagonal is what dies, since $\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \mathbf{0}$, and each pair of mirror-image entries subtracts to give one component:

$$\underbrace{a_2b_3 - a_3b_2}_{\text{the pair missing } 1}, \qquad \underbrace{a_3b_1 - a_1b_3}_{\text{the pair missing } 2}, \qquad \underbrace{a_1b_2 - a_2b_1}_{\text{the pair missing } 3}.$$

So the component of $\mathbf{a} \times \mathbf{b}$ in a given direction is built from the two rows and columns that leave that direction out, and that is exactly why the answer is perpendicular to both. The two products are the same array read two ways: down the diagonal, or across it.

In practice the array is never written out. For each component, cover its own row in both vectors and multiply the remaining four numbers crosswise. The subtraction then runs one way for the outer two components and the other way for the middle one:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}.$$

Only the middle component reverses. Covering row 1 leaves rows 2 and 3, and covering row 3 leaves rows 1 and 2; both pairs are in the cyclic order $1 \to 2 \to 3 \to 1$. Covering row 2 leaves rows 1 and 3, which run against that order, so the two products swap places before subtracting. Taking all three the same way is the standard error here, and the vector it produces is not perpendicular to $\mathbf{a}$ or to $\mathbf{b}$, so a dot-product check catches it at once. A cross product built correctly answers "find a vector perpendicular to both" immediately; a *unit* vector perpendicular to both needs one further step.

EXAMPLE 9.11

Let $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$. Find $\mathbf{a} \times \mathbf{b}$, and hence a unit vector perpendicular to both.

Each component leaves out its own row, and the middle one subtracts the other way round:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} (1)(1) - (0)(2) \\ (0)(2) - (2)(1) \\ (2)(2) - (1)(2) \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}.$$

The middle entry is $a_3b_1 - a_1b_3$, not $a_1b_3 - a_3b_1$; reading it in the same direction as the other two would give $+2$ there instead of -2.

Check it against both vectors: $(1)(2) + (-2)(1) + (2)(0) = 0$ and $(1)(2) + (-2)(2) + (2)(1) = 0$.

The rejected version fails this test, since $(1)(2) + (2)(1) + (2)(0) = 4$.

Perpendicular is not the same as unit. The magnitude is $\sqrt{1+4+4} = 3$, so divide by it:

$$\hat{\mathbf{n}} = \frac{1}{3}\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}.$$

The vector $-\hat{\mathbf{n}} = \frac{1}{3}\begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}$ is equally correct. It is $\mathbf{b} \times \mathbf{a}$ divided by its magnitude, and it runs the opposite way along the same perpendicular line, so unless the question fixes a direction either answer is complete.

9.5.2 Area of a parallelogram and a triangle

The magnitude has a clean geometric meaning: it is the area of the parallelogram with $\mathbf{a}$ and $\mathbf{b}$ as sides. To see why, read the two factors of $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|\sin\theta$ separately. Take $|\mathbf{a}|$ as the base. Then $|\mathbf{b}|\sin\theta$ is the perpendicular height, because $\mathbf{b}$ is the slanted side and $\sin\theta$ keeps only the part of it at right angles to the base. So the magnitude is base $\times$ height, which is the area.

> KEY · IB Area of a parallelogram $= |\mathbf{a}\times\mathbf{b}|$, where $\mathbf{a}$ and $\mathbf{b}$ are two adjacent sides.

A triangle is half of such a parallelogram, so its area is $\frac{1}{2}|\mathbf{a}\times\mathbf{b}|$; remember the half, because the booklet prints only the parallelogram.

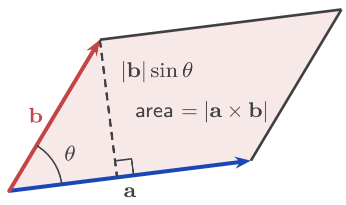


Figure 9.8 The parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$. Its area is $|\mathbf{a}\times\mathbf{b}|$, and a triangle on the same two sides has half that area.

EXAMPLE 9.12

Find the area of the triangle with vertices $A(0,0,0)$, $B(3,1,0)$, $C(1,0,2)$.

Take $\overrightarrow{AB} = \begin{pmatrix}3\\1\\0\end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix}1\\0\\2\end{pmatrix}$. Their cross product is

$$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}(1)(2)-(0)(0)\\(0)(1)-(3)(2)\\(3)(0)-(1)(1)\end{pmatrix} = \begin{pmatrix}2\\-6\\-1\end{pmatrix}.$$

Its magnitude is $\sqrt{4+36+1} = \sqrt{41}$, so the triangle's area is $\frac{1}{2}\sqrt{41}$. The cross product avoids trigonometry here: one calculation and one square root give the area.

YOUR TURN 9.5 The Vector Product

16. Evaluate each vector product.

(a) $\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\0\end{pmatrix}$ (b) $\begin{pmatrix}2\\0\\0\end{pmatrix}\times\begin{pmatrix}0\\3\\0\end{pmatrix}$

(c) $\begin{pmatrix}1\\2\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\2\end{pmatrix}$ (d) $\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}4\\5\\6\end{pmatrix}$

17. Areas and magnitudes.

(a) Given $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$ and an angle of 30°, find $|\mathbf{a}\times\mathbf{b}|$.

(b) Find the area of the parallelogram with sides $\begin{pmatrix}1\\0\\0\end{pmatrix}$ and $\begin{pmatrix}0\\2\\0\end{pmatrix}$.

(c) Find the area of the triangle with vertices $A(0,0,0)$, $B(2,1,0)$, $C(1,0,2)$.

(d) Explain why $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ for every vector $\mathbf{v}$.

18. Let $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$.

(a) Find $\mathbf{a} \times \mathbf{b}$.
(b) Verify that $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.
(c) Find a unit vector perpendicular to both.
(d) Write down $\mathbf{b} \times \mathbf{a}$, without further calculation.

9.6 Planes

A plane is a flat, endless sheet, and like a line it can be built from a point and some directions. A line needed only one direction, because a line is one-dimensional: one number, the parameter, is enough to say where you are on it. A plane is two-dimensional, so it needs two directions and two numbers. Anchor at $\mathbf{a}$, then travel λ lots of $\mathbf{b}$ and μ lots of $\mathbf{c}$. The pair (λ, μ) acts as a coordinate system inside the sheet, and every point of the plane has exactly one such pair. The two directions must not be parallel: if they were, every combination of them would still point along a single line, and the sheet would collapse to that line.

KEY · IB **Vector equation of a plane:** $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$.

Two parameters make this form awkward to use. It is hard to tell whether a point lies on the plane, or whether two such equations describe the same plane. A better description uses not the directions the plane contains but the single direction it does *not*: the **normal vector** $\mathbf{n}$, perpendicular to the whole sheet. Every displacement within the plane is perpendicular to $\mathbf{n}$, and the scalar product states exactly that.

KEY · IB **Normal (scalar-product) form:** $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$,

Cartesian form: $n_1x + n_2y + n_3z = d$, where $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$ and $d = \mathbf{a} \cdot \mathbf{n}$.

The picture behind this is simple. Take any point R on the plane. The displacement $\mathbf{r} - \mathbf{a}$ from the known point A to R lies flat in the sheet, so it is perpendicular to $\mathbf{n}$ and its scalar product with $\mathbf{n}$ is zero. Expanding $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$ gives the form above.

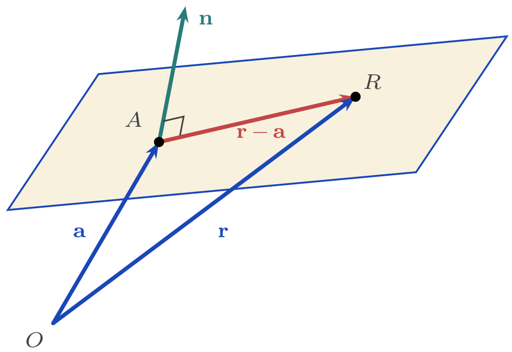


Figure 9.9 A plane held by one point and one perpendicular direction. Every displacement inside the sheet, such as $\mathbf{r}-\mathbf{a}$, is at right angles to the normal $\mathbf{n}$, so $(\mathbf{r}-\mathbf{a})\cdot\mathbf{n}=0$ for every point R of the plane, and for no other point.

A Cartesian plane equation can now be read directly: the coefficients of x, y, z *are* the normal vector. To find a plane through three points, use the cross product of two edge vectors to build $\mathbf{n}$, then fix d from any one point.

EXAMPLE 9.13

Find the Cartesian equation of the plane with normal $\mathbf{n}=\begin{pmatrix}3\\-2\\2\end{pmatrix}$ through $A(4,8,-1)$.

With $\mathbf{n}=(3,-2,2)$ the equation is $3x-2y+2z=d$. Substitute A to find d:

$$d=3(4)-2(8)+2(-1)=12-16-2=-6, \qquad \text{so} \qquad 3x-2y+2z=-6.$$

The normal supplies the coefficients and a single point supplies the constant. That is the method.

When no normal is given, the cross product produces one.

EXAMPLE 9.14

Find the Cartesian equation of the plane through $A(3,0,0)$, $B(0,4,0)$, $C(0,0,2)$.

Two edge vectors span the plane: $\overrightarrow{AB}=\begin{pmatrix}-3\\4\\0\end{pmatrix}$ and $\overrightarrow{AC}=\begin{pmatrix}-3\\0\\2\end{pmatrix}$. Their cross product is normal to both, hence to the plane:

$$\mathbf{n}=\overrightarrow{AB}\times\overrightarrow{AC}=\begin{pmatrix}(4)(2)-(0)(0)\\(0)(-3)-(-3)(2)\\(-3)(0)-(4)(-3)\end{pmatrix}=\begin{pmatrix}8\\6\\12\end{pmatrix}=2\begin{pmatrix}4\\3\\6\end{pmatrix}.$$

Any scalar multiple of a normal is a normal, so use the cleaner $(4,3,6)$. Through A: $d=4(3)=12$, giving $4x+3y+6z=12$. Check with B: $3(4)=12$; and C: $6(2)=12$. Both lie on it. Scaling the normal down before computing d keeps every later number small.

That example settles the stool from the opening page. Any three points not in a straight line give two edge vectors; their cross product is a normal, and a normal with a point fixes a plane. So three points always have exactly one plane through them, and three feet always reach the floor. A fourth point comes with no such guarantee, and the fourth leg is the one left hanging.

9.6.1 Angles with a plane

Two planes meet at an angle, and that angle is the angle between their normals, computed by the ordinary scalar product. To see why, look along the line where the planes meet, so that each plane is seen edge-on and appears as a line. Turning one arm of an angle through 90°, and then the other arm through 90°, leaves the angle unchanged. The normals are exactly the two arms turned in this way, so they carry the same angle.

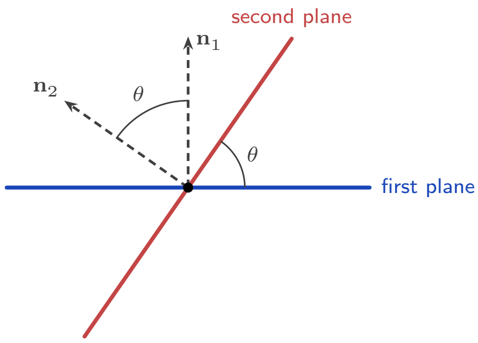


Figure 9.10 Two planes seen end-on, looking straight along the line where they meet. Each plane shows as a line. The angle between the planes and the angle between their normals are the same angle.

Taking the absolute value of the dot product guarantees the acute answer.

KEY Angle between two planes (normals $\mathbf{n}_1, \mathbf{n}_2$): $\cos\theta = \dfrac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1|\,|\mathbf{n}_2|}.$

EXAMPLE 9.15

Find the acute angle between the planes $2x + 2y - z = 4$ and $x - 2y + 2z = 1$.

Read the normals straight off the coefficients: $\mathbf{n}_1 = (2, 2, -1)$ and $\mathbf{n}_2 = (1, -2, 2)$. Then $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 4 - 2 = -4$, and both magnitudes are 3, so

$$\cos\theta = \frac{|-4|}{3 \times 3} = \frac{4}{9} \implies \theta \approx 63.6^\circ.$$

The dot product came out negative, which means the angle between the normals is obtuse. Taking the absolute value gives the acute angle instead, which is what the question asks for.

A line and a plane need one extra step. The angle we want is between the line and its projection *in* the plane: the shadow the line would cast if light fell straight down onto the sheet. But the scalar product gives the angle φ between the line and the *normal*. These two angles add to $90°$, so subtract from $90°$ to get the one we want.

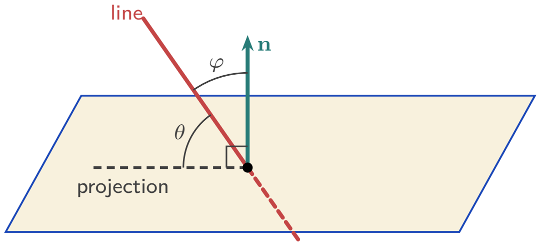


Figure 9.11 The angle between a line and a plane is measured to the line's projection in the plane, marked θ, not to the normal. Since the normal stands at $90°$ to the projection, the two angles at the point of entry satisfy $\theta + \varphi = 90°$.

KEY Angle between a line (direction $\mathbf{d}$) and a plane (normal $\mathbf{n}$): $\sin\theta = \dfrac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}|\,|\mathbf{n}|}$.

Using $\sin$ instead of $\cos$ is what builds the "$90°$ minus" into the formula. The line's angle to the plane is the complement of its angle to the normal.

EXAMPLE 9.16

Find the angle between the line $\mathbf{r} = \begin{pmatrix}1\\0\\-2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\2\end{pmatrix}$ and the plane $2x + 2y + z = 4$.

The direction is $\mathbf{d} = \begin{pmatrix}2\\1\\2\end{pmatrix}$, and the normal is read straight off the coefficients, $\mathbf{n} = \begin{pmatrix}2\\2\\1\end{pmatrix}$. Then $\mathbf{d}\cdot\mathbf{n} = 4 + 2 + 2 = 8$, with $|\mathbf{d}| = 3$ and $|\mathbf{n}| = 3$, so

$$\sin\theta = \frac{|8|}{3\times 3} = \frac{8}{9} \implies \theta = \arcsin\frac{8}{9} \approx 62.7°.$$

The same fraction fed into a cosine would give $27.3°$, and that is the angle to the *normal*, not to the plane. The two add to $90°$, so a solution that reaches for $\arccos$ out of habit lands on the complement of the answer.

9.6.2 Intersections: line and plane, plane and plane

To find where a line meets a plane, put the line's parametric coordinates into the plane's Cartesian equation. That gives one equation in λ. Solve it, then substitute back to get the point.

EXAMPLE 9.17

Find where the line $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ meets the plane $3x + y - z = 9$.

Parametrically, $x = 1 + 2\lambda$, $y = 2 - \lambda$, $z = 3\lambda$. Substituting into the plane,

$$3(1 + 2\lambda) + (2 - \lambda) - 3\lambda = 9 \implies 2\lambda + 5 = 9 \implies \lambda = 2,$$

so the point is $(5, 0, 6)$. Check: $3(5) + 0 - 6 = 9$. If instead the λ terms had cancelled, the line would be parallel to the plane. There is then no solution if the constants disagree, and the whole line lies in the plane if they agree.

Two non-parallel planes do not meet at a point; they meet along a whole **line**, like the spine of an open book. That line runs within both planes, so its direction is perpendicular to both normals, and the cross product builds it directly.

EXAMPLE 9.18

Find a vector equation of the line of intersection of the planes $x+y+2z = 9$ and $3x-y+z = 7$.

The direction is perpendicular to both normals:

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix}.$$

For a point on the line, pick a coordinate and set it to zero. With $z = 0$: $x+y = 9$ and $3x-y = 7$, so $x = 4$, $y = 5$. Hence

$$\mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} + t \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix}.$$

Check the direction: $(3, 5, -4) \cdot (1, 1, 2) = 0$ and $(3, 5, -4) \cdot (3, -1, 1) = 0$. Perpendicular to both normals, as the geometry demands. Setting $z = 0$ fails only if the line never crosses the plane $z = 0$. In that case set $x = 0$ or $y = 0$ instead.

9.6.3 Three planes

Three planes form a system of three linear equations, the same systems solved by row reduction in Ch. 4. Nothing new has to be calculated here. What is new is that every algebraic outcome now has a shape, and the shape is often easier to remember than the algebra.

There are five arrangements. The first two have solutions; the last three do not, and they are told apart by how many of the normals are parallel.

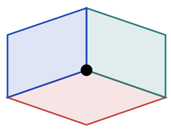
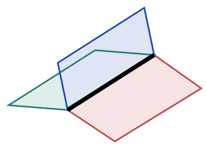
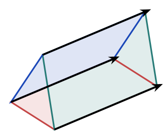
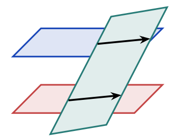
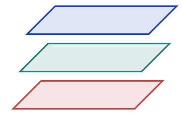
Unique solution	Infinitely many solutions	No solutions (inconsistent system)		
		No normals parallel	Two normals parallel	Three normals parallel
				
Three planes meet at a point	Three planes meet along a line	Three planes form a prism	One plane cutting two parallel planes	Three parallel planes

Figure 9.12 The five arrangements of three planes. Only the first has a single solution; the second has a whole line of them. The last three have none, and the number of parallel normals says which one you are looking at.

One further case is possible but rarely asked: if the three equations are multiples of one another, they all describe the same plane, and every point of that plane is a solution.

Row reduction tells you which case you are in without any picture. A row of zeros with a zero on the right means a dependent system, so the planes share a line. A row of zeros with a non-zero on the right means an inconsistent one, so there is no common point. The pictures only tell you what that answer looks like.

EXAMPLE 9.19

The planes $x + y + 2z = 9$ and $3x - y + z = 7$ from the previous example are joined by a third, $4x + 3z = c$. Describe the configuration of the three planes for $c = 16$ and for $c \neq 16$.

Adding the first two equations gives $4x+3z = 16$, so every point on their line of intersection satisfies $4x+3z = 16$. Indeed, check the line $\mathbf{r} = (4, 5, 0)+t(3, 5, -4)$ in the third plane: $4(4+3t)+3(-4t) = 16 + 12t - 12t = 16$, for every t.

So when $c = 16$ the third plane contains the entire line, and all three planes share it, like an open book with three pages on one spine. When $c \neq 16$ the third plane is parallel to that line but misses it. The three planes then bound a triangular prism: each pair meets in a line, the three lines are parallel, and no point lies on all three. In the language of Ch. 4, the system is dependent when $c = 16$ and inconsistent otherwise; the geometry is what those words look like.

YOUR TURN 9.6 Planes

19. Equations of planes.

(a) Write the Cartesian equation of the plane with normal $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ through the origin.

(b) State a normal vector to the plane $2x - y + 3z = 5$.

(c) Find the Cartesian equation of the plane with normal $\begin{pmatrix}1\\1\\1\end{pmatrix}$ through $(2, 0, 1)$.

(d) Does $(1, 1, 1)$ lie on the plane $x + y + z = 3$?

20. GDC Planes from points, and angles.

(a) Find the Cartesian equation of the plane through $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$.

(b) Find the acute angle between the planes $x + 2y - z = 3$ and $2x - y + z = 1$.

21. GDC Intersections.

(a) Find where the line $\mathbf{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ meets the plane $3x - 2y + 2z = 11$.

(b) Find a direction vector of the line of intersection of $x+y+z = 6$ and $2x-y+z = 3$.

(c) Find the angle between the line in part (a) and the plane $3x - 2y + 2z = 11$.

22. For each system of three planes, decide whether they meet at a single point, in a line, or not at all. Give the point or the line where there is one.

(a) $x + y + z = 6, \quad x - y + z = 2, \quad 2x + y - z = 1$

(b) $x + y + z = 6, \quad 2x + 2y + 2z = 12, \quad x - y + z = 2$

(c) $x + y + z = 6, \quad 2x + 2y + 2z = 10, \quad x - y + z = 2$

9.7 Distances

Two objects that never meet are still some distance apart, and asking for that distance always means the same thing: the *shortest* one. There is a single idea behind every case in this section, and it needs no new formula.

Shortest distance is not itself listed in the syllabus, which asks for intersections and angles. Every step below is built from tools that are listed, and exam questions reach the same answers by asking for the foot of a perpendicular. Treat this section as the natural extension of what comes before, not as separate material.

KEY **The shortest link is perpendicular.** Of all the segments joining two objects, the shortest is the one at right angles to what it joins. Perpendicular means a scalar product of zero, so the geometry turns straight into an equation.

That gives a method rather than a formula to memorise. Nothing here is printed in the formula booklet, so working it out is the point.

KEY **Finding a shortest distance.**

1. Write the unknown point on each object in terms of its parameter.
2. Set the scalar product of the joining vector with each direction it must be perpendicular to equal to zero.
3. Solve for the parameter or parameters, which locates the feet of the perpendicular.
4. Find the length of the joining vector.

9.7.1 From a point to a line

A general point on the line is $F = \mathbf{a}+\lambda\mathbf{b}$. As λ varies, F slides along the line and the length PF changes. It is shortest exactly when $\overrightarrow{PF}$ is perpendicular to the line, so $\overrightarrow{PF}\cdot\mathbf{b} = 0$. That is one equation in one unknown.

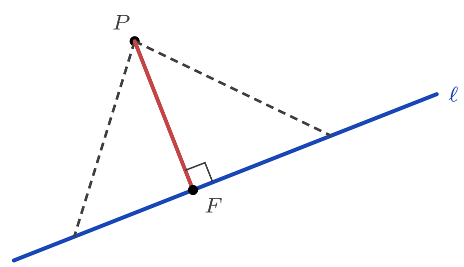


Figure 9.13 Every dashed segment joins P to the line. The shortest of them meets the line at right angles, at the foot of the perpendicular F.

EXAMPLE 9.20

Find the distance from $P(0,1,3)$ to the line $\mathbf{r} = \begin{pmatrix}1\\0\\-1\end{pmatrix} + \lambda\begin{pmatrix}1\\2\\2\end{pmatrix}$.

A general point on the line is $F = (1+\lambda,\ 2\lambda,\ -1+2\lambda)$, so

$$\overrightarrow{PF} = \begin{pmatrix}1+\lambda\\2\lambda-1\\2\lambda-4\end{pmatrix}.$$

Now make it perpendicular to the direction:

$$\overrightarrow{PF}\cdot\begin{pmatrix}1\\2\\2\end{pmatrix} = (1+\lambda)+2(2\lambda-1)+2(2\lambda-4) = 9\lambda-9 = 0,$$

so $\lambda = 1$ and the foot is $F(2,2,1)$. Then $\overrightarrow{PF} = (2,1,-2)$ and the distance is $\sqrt{4+1+4} = 3$. Notice what the method gives you for free: not just the distance but the point on the line that is closest to P, which many questions ask for as well.

9.7.2 From a point to a plane

For a plane the perpendicular direction is already known: it is the normal $\mathbf{n}$. So no scalar product needs to be set to zero. Instead, walk from P along $\mathbf{n}$ until you land on the plane. The general point on that walk is $\mathbf{p} + t\mathbf{n}$, and putting it into $\mathbf{r} \cdot \mathbf{n} = d$ gives one equation in t. The distance travelled is then $|t|\,|\mathbf{n}|$.

EXAMPLE 9.21

Find the distance from $P(1, 1, 1)$ to the plane $2x - y + 2z = 12$.

The normal is $\mathbf{n} = (2, -1, 2)$, so a point on the walk from P is

$$\mathbf{p} + t\mathbf{n} = (1 + 2t,\ 1 - t,\ 1 + 2t).$$

This lies on the plane when

$$2(1 + 2t) - (1 - t) + 2(1 + 2t) = 12 \implies 9t + 3 = 12 \implies t = 1.$$

The foot is $(3, 0, 3)$, and the distance is $|t|\,|\mathbf{n}| = 1 \times 3 = 3$. Check the foot on the plane: $6 - 0 + 6 = 12$. ✓

9.7.3 Between two skew lines

Skew lines never meet, yet there is still a shortest gap between them, and the same idea finds it. Take a general point on each, P_1 on the first and P_2 on the second. The joining vector $\overrightarrow{P_1P_2}$ is shortest when it is perpendicular to *both* directions at once. That is two scalar products set to zero, so two equations in the two unknowns λ and μ.

EXAMPLE 9.22

Find the shortest distance between $l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

General points are $P_1 = (\lambda,\ \lambda,\ 1 + \lambda)$ and $P_2 = (1 + \mu,\ 1,\ \mu)$, so

$$\overrightarrow{P_1P_2} = \begin{pmatrix} 1 + \mu - \lambda \\ 1 - \lambda \\ \mu - 1 - \lambda \end{pmatrix}.$$

Perpendicular to both directions gives two equations:

$$\overrightarrow{P_1P_2} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 - 3\lambda + 2\mu = 0, \qquad \overrightarrow{P_1P_2} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2\mu - 2\lambda = 0.$$

The second gives $\mu = \lambda$; the first then gives $1 - \lambda = 0$, so $\lambda = \mu = 1$. The feet are $P_1(1, 1, 2)$

and $P_2(2,1,1)$, and

$$\overrightarrow{P_1P_2} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \text{distance} = \sqrt{2}.$$

Two ordinary simultaneous equations, and the answer comes with both endpoints of the shortest segment attached.

9.7.4 The remaining cases

Nothing else needs a new method. Every other pairing either meets, and so has distance zero, or is parallel, and then any convenient point reduces it to one of the three cases above.

The two objects	What to do
Two parallel lines	Take any point on one line, then use point to line
Two intersecting lines	They meet, so the distance is 0
Two skew lines	Two scalar products set to zero, as above
A line parallel to a plane	Take any point on the line, then use point to plane
A line meeting a plane	They meet, so the distance is 0
Two parallel planes	Take any point on one plane, then use point to plane
Two non-parallel planes	They meet along a line, so the distance is 0

The first step in any of these questions is therefore not calculation but classification. Decide whether the objects meet, and if they do not, decide which of the three methods the pair reduces to.

EXAMPLE 9.23

Find the distance between the parallel planes $2x - y + 2z = 12$ and $2x - y + 2z = 3$.
Both have normal $(2, -1, 2)$, so they are parallel and never meet. Take any point on the second plane; setting $y = z = 0$ gives $x = \frac{3}{2}$, so $A(\frac{3}{2}, 0, 0)$ lies on it. Now walk from A along the normal to the first plane:

$$2\left(\tfrac{3}{2} + 2t\right) - (-t) + 2(2t) = 12 \implies 9t + 3 = 12 \implies t = 1,$$

so the distance is $|t|\,|\mathbf{n}| = 3$. Any other starting point on the second plane gives the same answer, which is what parallel means.

YOUR TURN 9.7 Distances

23. Distance from a point to a line. Throughout, $l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

(a) Find the foot of the perpendicular from $P(4, 0, 2)$ to l, and hence the distance from P to l.

(b) Find the distance from $Q(1, 4, 4)$ to l.

(c) The line m has equation $\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. Explain why m is parallel to l, and write down the distance between them.

(d) Check your answer to part (c) by repeating the method with the point on m where $t = 1$.

24. Distance from a point to a plane. Throughout, $\Pi : x + 2y + 2z = 14$.

(a) Find the foot of the perpendicular from $P(1, 1, 1)$ to Π, and hence the distance from P to Π.

(b) Find the distance from the origin to Π.

(c) Find the distance between Π and the parallel plane $x + 2y + 2z = 5$.

(d) Show that the line $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$ is parallel to Π, and write down the distance between the line and the plane.

25. Distance between two lines. Throughout, $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

(a) Show that l_1 and l_2 are skew.

(b) Write down general points P_1 on l_1 and P_2 on l_2, and find the values of λ and μ for which $\overrightarrow{P_1P_2}$ is perpendicular to both direction vectors.

(c) Hence state the two feet of the common perpendicular and the shortest distance between l_1 and l_2.

(d) The lines $n_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $n_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ are not skew. Apply the same method to them and explain what your answer means.

Chapter 9 · Vectors

Reflections

TOK Both operations are called products, yet neither behaves like multiplication. The dot product takes two vectors and gives back a number. The cross product changes sign when you swap the order, and returns zero for two vectors that are not themselves zero.

So is either one really multiplication, or have we borrowed a comfortable word for something new? And why these definitions in particular? Nothing forced $|\mathbf{a}||\mathbf{b}|\cos\theta$ on us. It was chosen because it answered a question people wanted answered. How much of mathematics is discovered, and how much is chosen and then found to be useful?

TOK The same plane can be a drawing, a vector equation, or the single line $2x - y + 3z = 7$. The symbols are easier to calculate with. The picture is easier to believe.

If the two disagree in your head, which one do you trust, and what happens in four dimensions, where the picture is no longer available to you?

RW Every three-dimensional game and animated film runs on this chapter. The dot product decides how brightly a surface is lit, by comparing the direction of the light with the direction the surface faces: a value near 1 points at the light and is bright, a value near 0 faces edge-on and stays dark. The cross product builds the normals that say which way each surface points in the first place. At the same time the physics engine adds force and velocity vectors tip to tail, thousands of times a second, which is why a thrown object curves the way your eye expects.

EXT The cross product only exists because we live in three dimensions.

In two dimensions there is no room for a vector perpendicular to two others. In four dimensions or more there is too much room: no single perpendicular direction is picked out, because a whole family of them works. Three dimensions is the only case where two vectors determine exactly one perpendicular line. That is why $\mathbf{a} \times \mathbf{b}$ can be a vector at all.

End-of-Chapter Problems

1. Given $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$:

(a) find $\mathbf{a} + 2\mathbf{b}$; [2]
(b) find $|\mathbf{a}|$; [1]
(c) find the unit vector in the direction of $\mathbf{a}$. [2]

2. For $A(1, -1, 2)$ and $B(3, 2, 4)$:

(a) find $\overrightarrow{AB}$; [1]
(b) find $|\overrightarrow{AB}|$; [2]
(c) find the midpoint of AB. [2]

3. The vectors $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ k \end{pmatrix}$ are perpendicular. Find k. [3]

4. Find the angle between $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. [4]

5. Consider the line $l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$.

(a) Find the point on l when $\lambda = 2$. [2]
(b) Determine whether $P(5, 2, 3)$ lies on l. [2]

6. GDC Find, in degrees, the acute angle between two lines with direction vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$. [4]

7. Given $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$, find $\mathbf{a} \times \mathbf{b}$ and $|\mathbf{a} \times \mathbf{b}|$. [4]

8. Find the area of the triangle with vertices $A(1, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 1)$. [4]

9. A plane Π has normal $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and passes through $A(1, 2, -1)$.

(a) Find the Cartesian equation of Π. [3]
(b) Determine whether $B(2, 1, 0)$ lies on Π. [1]

10. Find the Cartesian equation of the plane through $A(1, 0, 0)$, $B(0, 1, 0)$ and $C(0, 0, 1)$. [4]

11. Given $\mathbf{a} = \begin{pmatrix} 4 \\ -3 \\ 0 \end{pmatrix}$:

(a) find $|\mathbf{a}|$; [1]
(b) find the unit vector $\hat{\mathbf{a}}$; [2]
(c) find the vector of magnitude 10 in the direction of $\mathbf{a}$. [2]

12. Points $A(2,1,3)$ and $B(4,5,3)$ are given. Find the point C such that B is the mid-point of AC. [3]

13. The line $l : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ meets the plane $z = 4$. Find the point of intersection. [4]

14. GDC Find the angle between the vectors $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. [4]

15. Show that the points $A(1,2,3)$, $B(2,4,5)$ and $C(4,8,9)$ are collinear. [4]

16. GDC Given $|\mathbf{a}| = 5$, $|\mathbf{b}| = 3$ and $\mathbf{a} \cdot \mathbf{b} = 9$, find the angle between $\mathbf{a}$ and $\mathbf{b}$. [4]

17. The lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ intersect. Find the point of intersection. [5]

18. The plane Π has equation $2x + y - 2z = 6$.

(a) State a normal vector to Π. [1]
(b) Find where the line $\mathbf{r} = \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ meets Π. [4]

19. Given $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$:

(a) find $\mathbf{a} \cdot \mathbf{b}$; [2]
(b) hence state the angle between $\mathbf{a}$ and $\mathbf{b}$. [2]

20. Find the area of the triangle with vertices $A(0,0,0)$, $B(3,0,0)$, $C(0,4,0)$ using the vector product. [4]

21. Find a vector equation of the line through $A(2,-1,3)$ and $B(4,1,7)$. [4]

22. GDC Find the acute angle between the planes $x + y + z = 1$ and $x - y + z = 3$. [5]

23. Given $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}$, find $\mathbf{a} \times \mathbf{b}$ and $|\mathbf{a} \times \mathbf{b}|$. [4]

24. Find the value of t for which $\begin{pmatrix} 1 \\ t \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \\ t \end{pmatrix}$ are perpendicular. [4]

25. At time $t = 0$ seconds a drone is at the point $(2,-1,10)$ (coordinates in metres) and moves with constant velocity $\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ $\mathrm{m\,s^{-1}}$.

(a) Write down a vector equation for the drone's position at time t. [2]

(b) Find the drone's speed. [2]

(c) Determine whether the drone passes through the point $(17, 19, 10)$, and if so, at what time. [3]

26. Find a vector equation of the line of intersection of the planes $x + y - z = 2$ and $2x - y + z = 1$. [6]

27. Consider the planes $\Pi_1 : x + y + z = 3$, $\Pi_2 : x - y + z = 1$ and $\Pi_3 : x + z = c$, where $c \in \mathbb{R}$.

(a) Find a vector equation of the line of intersection of Π_1 and Π_2. [4]

(b) Show that when $c = 2$ the three planes intersect in this line. [2]

(c) Describe geometrically the configuration of the three planes when $c \neq 2$. [2]

28. Show that the lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ intersect, and find the point of intersection. [6]

29. Find the shortest distance from the point $P(1, 2, 3)$ to the plane $2x - y + 2z = 0$, by finding the foot of the perpendicular from P. [6]

30. Find a unit vector perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$. [5]

31. The points $A(1, 0, 0)$, $B(0, 2, 0)$ and $C(0, 0, 3)$ are given.

(a) Find the Cartesian equation of the plane ABC. [3]

(b) Find the area of triangle ABC. [3]

32. GDC Find the angle that the line $l : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ makes with the plane $x + 2y + 2z = 5$. [6]

33. The points $A(2, 0, 1)$, $B(1, 3, 2)$ and $C(4, 1, 0)$ are given, and O is the origin.

(a) Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$. [2]

(b) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. [3]

(c) Find the exact area of triangle ABC. [2]

(d) Find the Cartesian equation of the plane through A, B and C. [4]

(e) Find the exact shortest distance from O to that plane. [3]

(f) Hence find the volume of the tetrahedron $OABC$. [2]

34. Consider the lines

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

(a) Show that l_1 and l_2 are not parallel. [2]

(b) Determine whether the lines intersect, and if so find the point of intersection. [5]

(c) Find the acute angle between the lines. [3]

(d) Find the Cartesian equation of the plane containing both lines. [4]

(e) Explain why such a plane exists only because of your answer to part (b). [2]

35. GDC Two aircraft fly at the same altitude. At time t hours their position vectors, in km, are

$$\mathbf{r}_A = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 300 \\ 400 \\ 0 \end{pmatrix}, \qquad \mathbf{r}_B = \begin{pmatrix} 600 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 0 \\ 500 \\ 0 \end{pmatrix}.$$

(a) Find the speed of each aircraft. [3]

(b) Find $\overrightarrow{AB}$ at time t. [3]

(c) Show that the distance d between them satisfies $d^2 = 100\,000t^2 - 360\,000t + 360\,000$. [4]

(d) Find the time at which the aircraft are closest, and the least distance between them. [4]

36. Consider the planes

$$\Pi_1 : x + y + z = 1, \qquad \Pi_2 : x + 2y + 3z = 4, \qquad \Pi_3 : x + y + kz = m,$$

where k and m are real constants.

(a) Find a vector equation of the line of intersection of Π_1 and Π_2. [4]

(b) Show that when $k = 2$ the three planes meet at a single point, and find that point in terms of m. [4]

(c) Find the values of k and m for which the three planes meet in a line. [3]

(d) Find the values of k and m for which the three planes have no common point, and describe that arrangement geometrically. [3]

37. The points $A(1, 2, 1)$, $B(3, 1, 2)$ and $C(2, 4, 3)$ are given.

(a) Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$. [2]

(b) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. [3]

(c) Hence find the area of triangle ABC. [3]

(d) Find the Cartesian equation of the plane through A, B and C. [3]

(e) Explain why the vector product gives a normal to that plane. [2]

38. GDC Two drones move with constant velocity. At time t seconds their positions, in metres, are

$$\mathbf{r}_A = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 4 \\ -3 \\ 12 \end{pmatrix}, \qquad \mathbf{r}_B = \begin{pmatrix} 3 \\ -13 \\ 6 \end{pmatrix} + t \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}.$$

(a) Find the speed of each drone. [3]

(b) Show that the two flight paths cross, and find the point where they cross. [5]

(c) Explain why the drones nevertheless do not collide. [3]

(d) Find the distance between the drones at $t = 2$. [3]

(e) Find the acute angle between the two flight paths. [3]

Challenge Questions

39. The shortest distance from a point to a line. Section 9.7 works this out one case at a time. Here you build a formula that does every case, then a second formula, and show the two agree.

Let l be the line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, and P a point not on it.

(a) Explain why the shortest distance from P to l is measured along the perpendicular. [2]

(b) Let $F = \mathbf{a} + \lambda\mathbf{d}$ be the foot of that perpendicular. Explain why $\overrightarrow{PF} \cdot \mathbf{d} = 0$, and use this to show that

$$\lambda = \frac{(\mathbf{p} - \mathbf{a}) \cdot \mathbf{d}}{|\mathbf{d}|^2}.$$

[5]

(c) For $l : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $P(4, 3, 1)$, find λ, the foot F, and the distance. [5]

(d) A second formula is distance $= \dfrac{|\overrightarrow{AP} \times \mathbf{d}|}{|\mathbf{d}|}$, where A is any point on l. Check that it agrees with part (c). [4]

(e) Explain why the second formula works, using $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta$. *(Hint: what is $|\overrightarrow{AP}|\sin\theta$ in the diagram?)* [4]

40. How far apart are two skew lines? Two skew lines never meet, yet there is still a shortest distance between them. This investigation finds it.

Let $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda\mathbf{d}_1$ and $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu\mathbf{d}_2$ be skew.

(a) Show that $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ are skew. [4]

(b) The shortest join between the two lines is perpendicular to both. Explain why its direction must be parallel to $\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2$. [3]

(c) Hence explain why the shortest distance is the length of the projection of $\mathbf{a}_2 - \mathbf{a}_1$ onto $\mathbf{n}$, that is

$$\text{distance} = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot \mathbf{n}|}{|\mathbf{n}|}.$$

[4]

(d) Use the formula on the two lines in part (a), giving an exact answer. [5]

(e) What does the formula give when the two lines intersect? Explain why your answer is what it should be. [3]

41. The triple product and volume. Combining both products gives $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$, a single number called the *scalar triple product*. It measures volume.

Throughout, take $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$.

(a) Find $\mathbf{v} \times \mathbf{w}$, then $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$. [4]

(b) The parallelepiped with edges $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ has base area $|\mathbf{v} \times \mathbf{w}|$. Explain why its volume is $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$. *(Hint: volume = base area × perpendicular height)* [4]

(c) Evaluate $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{u})$ and $\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v})$. State what each result shows about the triple product. [4]

(d) Three vectors are *coplanar* when they all lie in one plane. Explain why this happens exactly when the triple product is zero, and test $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix}$. [4]

(e) A tetrahedron on the same three edges has one sixth of the parallelepiped's volume. Find the volume of the tetrahedron with vertices at the origin and at the three points $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ above. [3]

STATISTICS AND PROBABILITY

Chapter 10

10

Statistics

SYLLABUS 4.1 4.2 4.3 4.4 4.10

Statistics begins with a practical difficulty: a list of numbers is usually too long to read and too varied to summarise by eye.

In the 1880s Francis Galton met exactly this problem. He recorded the heights of 928 adult children and of the 205 sets of parents who raised them, in order to answer a single question: how well does a parent's height predict a child's? Tall parents did tend to have tall children. But the children of very tall parents were, on average, slightly shorter than their parents, and the children of very short parents slightly taller. Each generation drifted toward the middle. Galton called this "regression toward mediocrity". The judgement in that phrase has since been dropped; the name *regression* has not.

Before any of this could be seen, the ordinary work had to be done first. One number says very little, and nine hundred say too much at once. The data had to be summarised, their variation measured, and one list set against another.

Those are the tasks of this chapter. You will collect data by a defensible method, describe a single list by its centre and its spread, and display a distribution as a histogram, a cumulative frequency curve or a box plot. You will then compare two lists, measure how closely they move together, and fit the line that best predicts one from the other.

Each of these tools has a range in which it can be trusted and a range in which it cannot, and knowing where the boundary lies is part of the subject. Correlation does not establish cause. A fitted line says nothing reliable beyond the data it was fitted to. Galton's drift is the standing example: it is not a fact about families, and no biology is needed to explain it. The reason appears at the end of the chapter.

10.1 Sampling and Data

You almost never have all the data. Galton could not measure every family in England, a factory cannot test every light bulb it makes, and a biologist cannot weigh every fish in a lake. So you measure a part and reason about the whole. The honesty of that reasoning depends entirely on how the part was chosen.

The complete group you want to know about is the **population**. The smaller group you actually measure is the **sample**. A number describing the population is a **parameter**; the matching number computed from a sample is a **statistic**. You use the statistic to estimate the parameter you cannot see.

> DEF A **population** is the entire set of individuals under study. A **sample** is a subset of the population from which data are actually collected.
> A **parameter** describes a population; a **statistic** describes a sample.

10.1.1 Types of data

Data come in two broad kinds. **Qualitative** data describe a quality or category, such as eye colour or country of birth. **Quantitative** data are numerical. Quantitative data split again. **Discrete** data take separate values you can count, such as the number of goals in a match. **Continuous** data can take any value in a range, such as height or time, limited only by the precision of your instrument.

The distinction matters because it decides how you may display and summarise the data. You can find the mean number of goals, but "mean eye colour" is meaningless.

10.1.2 Sampling techniques

A good sample resembles the population on a smaller scale. The techniques below are different ways of achieving that. Each one balances fairness against the effort required.

> DEF **Simple random:** every member of the population has an equal chance of selection.
> **Systematic:** order the population and select every k-th member from a random start.
> **Stratified:** divide the population into groups (strata), then sample from each group in proportion to its size.
> **Quota:** like stratified, but members within each group are chosen non-randomly until each quota is filled.
> **Convenience:** select whoever is easiest to reach.

Stratified sampling is the most useful technique for populations with clear subgroups. If a school is 60% junior and 40% senior students, a stratified sample of 50 takes 30 juniors and

20 seniors, preserving the structure.

EXAMPLE 10.1

A school has 800 juniors and 400 seniors. A stratified sample of 60 students is taken. How many should be seniors?

Seniors are $\frac{400}{1200} = \frac{1}{3}$ of the school. The sample must keep that proportion:

$$\frac{1}{3} \times 60 = 20 \text{ seniors} \quad (\text{and } 40 \text{ juniors}).$$

Stratified sampling guarantees both groups are represented in their true ratio, which a simple random sample only achieves on average.

10.1.3 Bias and reliability

A sample is **biased** when its method systematically favours some outcomes over others. The statistic then misses the parameter, no matter how large the sample grows. A survey about reading habits conducted in a library will overstate how much people read. Bias is a fault in the method, not in the sample size; collecting more biased data only makes you more confident in the wrong answer.

Reliability also depends on the records themselves. Real data sets contain missing entries and recording errors. Before any analysis, ask how the data were collected, what might be missing, and whether any value could have been recorded wrongly.

CAUTION A large sample is not automatically a good one. The 1936 *Literary Digest* poll surveyed 2.4 million people and still predicted the wrong US election winner. Its list of names came from telephone and car owners, who were wealthier than the average voter. Size cannot repair a biased method.

10.1.4 Outliers

An **outlier** is a value far from the rest of the data. It may be a genuine extreme value, or it may be a recording error. Either way it can distort a summary badly. We will give it a precise definition once we have quartiles, in §10.4. For now, treat any suspiciously distant value as something to investigate, not something to delete immediately.

YOUR TURN 10.1 Sampling and Data

1. State whether each variable is discrete or continuous.

(a) the number of goals scored in a match

(b) the mass of a parcel

(c) the number of students in a class

(d) the time taken to run 100 m

2. A college has 1500 students, 600 of them male. A stratified sample of 50 is taken.

(a) Find the number of males in the sample.
(b) Find the number of females in the sample.
(c) Explain one advantage of stratified sampling over simple random sampling here.

3. Sampling methods.

(a) A sample of 40 is taken from 2000 people by choosing every kth person. Find k.
(b) Name the sampling method in part (a).
(c) A researcher surveys people at a gym about how much they exercise. Explain why this is biased.
(d) A town of $15\,000$ has 4000 residents over 65. In a stratified sample of 150, find the number over 65.
(e) A researcher is told to interview 20 men and 20 women, choosing whoever is willing to stop. Name this sampling method and give one weakness of it.

10.2 Measures of Central Tendency

Once you have a list of numbers, the first question is always the same: what is a typical value? A single number that stands in for the whole list is a measure of **central tendency**. There are three, and they answer slightly different questions.

The **mean** is the balance point: add everything and share it equally. The **median** is the middle value when the data are ordered. The **mode** is the most frequent value.

10.2.1 The mean

For n raw values, the mean is the total divided by the count. When values repeat, you weight each value by how often it occurs, which is just a shortcut for the same sum.

KEY · IB

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{n}, \qquad \text{where } n = \sum_{i=1}^{k} f_i$$

Here x_i are the distinct values, f_i their frequencies. A table listing each value alongside its frequency is called a **frequency distribution**, and the next example shows one. It is the standard compact way to present discrete data. With no repeats, every $f_i = 1$ and this collapses to $\bar{x} = \frac{\sum x_i}{n}$, the familiar form.

EXAMPLE 10.2

In 20 football matches a team scored the following numbers of goals. Find the mean.

Goals x	**0**	**1**	**2**	**3**	**4**
Frequency f	4	7	5	3	1

Multiply each value by its frequency, then divide by the total count:

$$\bar{x} = \frac{(0)(4) + (1)(7) + (2)(5) + (3)(3) + (4)(1)}{4 + 7 + 5 + 3 + 1} = \frac{30}{20} = 1.5.$$

The mean need not be a value the data can actually take. No team scores 1.5 goals, but 1.5 is the fair share.

10.2.2 Combining two groups

Two sets of data are often summarised separately and then have to be described together. Averaging the two means will not do it, unless the groups happen to be the same size, because the larger group holds more values and must count for more.

Go back to totals. A group of n values with mean $\bar{x}$ has total $n\bar{x}$, so each group's total can be recovered from its summary. Add the totals, add the counts, and divide.

KEY A group of n_1 values with mean $\bar{x}_1$ and a group of n_2 values with mean $\bar{x}_2$ combine to give

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}.$$

The weights need not be counts. A final grade made of 40% coursework and 60% examination is the same calculation with 0.4 and 0.6 in place of n_1 and n_2, over a denominator of $0.4 + 0.6 = 1$.

EXAMPLE 10.3

Twelve students in one group scored a mean of 54 on a test, and the other eight scored a mean of 69. Find the mean for all twenty. Five more students then sit the same test, and the mean of all twenty-five becomes 58. Find the mean of those five.

Recover each total, then combine:

$$\bar{x} = \frac{12(54) + 8(69)}{12 + 8} = \frac{648 + 552}{20} = \frac{1200}{20} = 60.$$

Run the same identity backwards for the missing group. All twenty-five values total $25 \times 58 = 1450$, and the first twenty account for 1200, so the five total $1450 - 1200 = 250$ and have mean $\frac{250}{5} = 50$.
A combined mean always lies between the two group means and nearer the mean of the larger group, so 60 falling between 54 and 69 and leaning towards the twelve is the check. Averaging the two means directly gives 61.5, which leans the wrong way.

10.2.3 Median and mode

The median splits ordered data into two equal halves. For n values, it sits at position $\frac{n+1}{2}$. If n is odd this is a single middle value; if n is even it is the average of the two middle values.

EXAMPLE 10.4

Find the median and mode of $4, 7, 2, 9, 4, 11, 6$.
Order first: $2, 4, 4, 6, 7, 9, 11$. With $n = 7$, the median is at position $\frac{7+1}{2} = 4$, the fourth value: 6.
The mode is the most frequent value: 4 appears twice, everything else once, so the mode is 4.

10.2.4 Grouped data

Continuous data are often recorded in classes, such as "time between 10 and 20 minutes." You no longer know the individual values, only the class each fell into. To estimate the mean, assume every value sits at its class **midpoint**, then apply the frequency formula. The result is an estimate, because the assumption is rarely exact.

EXAMPLE 10.5

Fifty phone calls to a help line had the durations below. Estimate the mean duration.

Duration (min)	$0 \le t < 4$	$4 \le t < 8$	$8 \le t < 12$	$12 \le t < 16$
Frequency	8	18	16	8

Use the midpoint of each class: $2, 6, 10, 14$.

$$\bar{x} = \frac{(2)(8) + (6)(18) + (10)(16) + (14)(8)}{50} = \frac{16 + 108 + 160 + 112}{50} = \frac{396}{50} = 7.92 \text{ min.}$$

The **modal class** is $4 \le t < 8$, the class with the highest frequency. With grouped data you report a modal class, not a single modal value.
The middle is located by a running cumulative count rather than by a formula. With $n = 50$ the median lies between the 25th and 26th values. The first class holds values 1 to 8; the second carries the running total to $8 + 18 = 26$, so it holds values 9 to 26. Both the 25th and the 26th

fall there, and the **median class** is $4 \leq t < 8$.

Again you name a class, not a value: the individual durations are gone, so the most that can be said is which class the middle value fell in. The median class and the modal class coincide here, which is common but not automatic, since one is located by a running total and the other by the largest single frequency.

10.2.5 Which measure to use

The three measures agree when the data are symmetric. They disagree when the data are skewed, and the disagreement is itself information.

The mean uses every value, so one very large value pulls it towards that value. The median depends only on position, so the same value moves it hardly at all. This is why incomes are reported by median, not mean: a few billionaires would make the mean meaningless as a "typical" wage.

The same difference makes the two measures a quick test of shape. A long tail on the right holds a few unusually large values, and those values pull the mean above the median. A long tail on the left does the reverse. Skew is named after the side the tail is on, not the side the peak is on, which is the part most often misremembered.

Comparison	Shape	Usually report
$\bar{x} >$ median	positively skewed, tail to the right	median
$\bar{x} \approx$ median	roughly symmetric	mean
$\bar{x} <$ median	negatively skewed, tail to the left	median

For the single-peaked distributions in this course the comparison is a dependable guide, and it is fast: it needs two numbers, not a graph. It is a guide rather than a proof, since data sets can be built that break it, so check it against a diagram whenever one is available.

CAUTION The mean is sensitive to outliers; the median is resistant to them. When a distribution is strongly skewed or contains a suspect extreme value, the median usually describes the centre better.

YOUR TURN 10.2 Measures of Central Tendency

4. For the data $5,\ 8,\ 8,\ 10,\ 14,\ 15$, find each of the following.

(a) the mean
(b) the median
(c) the mode
(d) the range

5. Find the required measure in each case.

(a) the median of $3, 7, 9, 12, 15, 20$

(b) the mode of $2, 4, 4, 5, 7, 7, 7, 9$

(c) the mean of $3, 7, 7, 8, 10, 11, 12, 14$

(d) the median of $12, 15, 15, 18, 20, 22$

6. Work backwards from a given mean.

(a) The numbers $4, 7, x, 12, 15$ have a mean of 10. Find x.

(b) The mean of six numbers is 14. A seventh number is added and the mean becomes 15. Find the seventh number.

7. The table shows a frequency distribution.

Value x	1	2	3	4	5
Frequency f	3	5	8	4	2

(a) Find n.

(b) Find $\sum fx$.

(c) Find the mean.

(d) State the mode.

8. The grouped data below record a measurement x.

Class	$0 \le x < 10$	$10 \le x < 20$	$20 \le x < 30$	$30 \le x < 40$
Frequency	6	10	9	5

(a) Write down the four class midpoints.

(b) Estimate the mean.

(c) State the modal class.

(d) Explain why the mean is only an estimate.

10.3 Measures of Dispersion

The centre alone does not describe a data set. Consider two classes that both average 60% on a test. In the first, every student scored between 58 and 62. In the second, half scored 30 and half scored 90. The mean is the same; the two classes are not. To capture the difference you need a measure of **spread**.

10.3.1 Range and interquartile range

The simplest measure is the **range**: the largest value minus the smallest. It is easy to find but unreliable. It depends only on the two most extreme values, which are exactly the ones most likely to be outliers.

A better approach is to ignore the extremes and measure the spread of the middle. Just as the median cuts the data in half, the **quartiles** cut it into quarters. Q_1 is the value one quarter of the way through the ordered data, Q_2 is the median, and Q_3 is three quarters of the way. The spread of the central half is the **interquartile range**.

KEY · IB

$$\text{IQR} = Q_3 - Q_1$$

Because it discards the top and bottom quarters, the IQR is unaffected by outliers, which is why it is normally reported alongside the median.

EXAMPLE 10.6

Find the range and interquartile range of

$$3,\ 5,\ 6,\ 8,\ 8,\ 9,\ 11,\ 13,\ 15.$$

The data are already ordered, with $n = 9$. The range is $15 - 3 = 12$.
The median (Q_2) is the 5th value, 8. The lower quartile is the median of the four values below it, $3, 5, 6, 8$:

$$Q_1 = \frac{5+6}{2} = 5.5.$$

The upper quartile is the median of the four values above, $9, 11, 13, 15$:

$$Q_3 = \frac{11+13}{2} = 12.$$

So $\text{IQR} = 12 - 5.5 = 6.5$. Methods for placing quartiles vary slightly between textbooks and calculators; in this course, report the value your GDC gives.

10.3.2 Variance and standard deviation

The IQR describes the middle half and ignores the rest, so two data sets with very different extremes can share an IQR. A measure built from every value avoids that. The natural quantity to start from is the **deviation** of each value from the mean, $x_i - \bar{x}$, which says how far that value sits from the centre and on which side.

Averaging the deviations themselves does not work, and it fails for every data set, not just awkward ones. Adding them gives

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0,$$

because $\sum x_i = n\bar{x}$ is what the mean means. Values above the centre and values below it cancel exactly. The total is 0 for tightly packed data and 0 for widely scattered data, so it separates nothing.

The fix is to remove the signs before averaging, and squaring is the standard way to do it. Each $(x_i - \bar{x})^2$ is positive or zero, squares are far easier to handle in algebra than absolute values, and squaring gives a large deviation more weight than a small one, which is usually what you want from a measure of spread. The mean of the squared deviations is the **variance**, written σ^2.

Squaring changes the units: for times in minutes the variance is in minutes squared, which is not a spread anyone can picture. Taking the square root undoes that and returns to the original units. The result is the **standard deviation** σ, and it is the one normally reported.

KEY · IB HL

$$\sigma^2 = \frac{\sum_{i=1}^{k} f_i(x_i - \bar{x})^2}{n} = \frac{\sum_{i=1}^{k} f_i x_i^2}{n} - \bar{x}^2, \qquad \sigma = \sqrt{\sigma^2}$$

The second form is the same quantity rearranged, and it is quicker by hand because it needs only the totals $\sum f_i x_i^2$ and $\sum f_i x_i$, not a separate subtraction for every value. The booklet writes μ where this chapter writes $\bar{x}$; for a set of data they are the same number.

The standard deviation σ is, roughly, the typical distance of a value from the mean. A small σ means the values lie close to the centre. A large σ means they are widely spread.

CAUTION In this course you find σ with your **GDC** (graphic display calculator), not by hand. Both forms appear in the HL formula booklet, so you are never asked to recall them; they are there so that you understand what the calculator computes. Know what σ *means*.

EXAMPLE 10.7

Return to the two classes, both with mean 60. Class A scored $58, 59, 60, 61, 62$; Class B scored $30, 45, 60, 75, 90$. Compare their standard deviations.

Both means are 60. For Class A the deviations are $-2, -1, 0, 1, 2$:

$$\sigma_A = \sqrt{\frac{4+1+0+1+4}{5}} = \sqrt{2} \approx 1.41.$$

For Class B the deviations are $-30, -15, 0, 15, 30$:

$$\sigma_B = \sqrt{\frac{900 + 225 + 0 + 225 + 900}{5}} = \sqrt{450} \approx 21.2.$$

The means are identical, but the standard deviations differ by a factor of 15. The single number σ captures exactly the difference that the mean alone does not show.

10.3.3 The effect of changing the data

Suppose every value in a data set is changed in the same way, either by adding a constant or by multiplying by a constant. Temperatures convert from Celsius to Fahrenheit; prices rise by a fixed tax. You can predict the new mean and standard deviation without recomputing anything.

Adding a constant c to every value shifts the whole distribution sideways. The centre moves by c, but the spread does not change. Multiplying every value by k stretches the distribution away from zero, scaling both the centre and the spread.

KEY If $y_i = kx_i + c$ for every value, then

$$\bar{y} = k\bar{x} + c, \qquad \sigma_y = |k|\,\sigma_x, \qquad \sigma_y^2 = k^2\sigma_x^2.$$

The constant c does not affect the spread. Moving every value by the same amount changes no distance between them. Only the multiplier k rescales σ.

EXAMPLE 10.8

A set of test marks has mean 54 and standard deviation 8. Every mark is scaled by multiplying by 1.5 and then adding 5. Find the new mean and standard deviation.
Here $k = 1.5$ and $c = 5$:

$$\bar{y} = 1.5(54) + 5 = 86, \qquad \sigma_y = 1.5 \times 8 = 12.$$

The $+5$ moves the mean but does not change the spread. Only the $\times 1.5$ affects σ.

The same transformation carries the median and the quartiles across. For $k > 0$ the map $x \mapsto kx + c$ is increasing, so it preserves order, and whichever value sat in the middle before still sits in the middle afterwards. The new median is therefore $k \times (\text{old median}) + c$, and Q_1 and Q_3 obey the identical rule. When $k < 0$ the order reverses, and the old Q_3 becomes the new Q_1.

Questions often run this backwards: the target mean and standard deviation are given, and the transformation has to be found. Take the standard deviation equation first, because c

does not appear in it.

EXAMPLE 10.9

A set of examination marks has mean 52 and standard deviation 16. The marks are rescaled by $y = ax + b$ with $a > 0$, so that the scaled marks have mean 65 and standard deviation 12. Find a and b. The original median was 50; write down the scaled median.

Only a appears in the standard deviation, so start there. With $a > 0$, $\sigma_y = a\sigma_x$:

$$12 = 16a \implies a = 0.75.$$

Now the mean equation has one unknown left:

$$65 = 0.75(52) + b = 39 + b \implies b = 26,$$

so $y = 0.75x + 26$. Since $a > 0$ the rescaling preserves order, and the median maps by that same rule:

$$\text{median}_y = 0.75(50) + 26 = 63.5.$$

Beginning with the mean equation instead leaves two unknowns in one equation and stalls, so the order of the two steps is forced.

YOUR TURN 10.3 Measures of Dispersion

9. For the data $2,\ 4,\ 5,\ 7,\ 8,\ 10,\ 12,\ 14,\ 16,\ 20$, find each of the following.

(a) the range
(b) Q_1
(c) Q_3
(d) the interquartile range

10. GDC Use your GDC for each part.

(a) Find the mean and standard deviation of $5,\ 8,\ 9,\ 11,\ 13,\ 14$.
(b) Find the mean and standard deviation of the values $10, 20, 30, 40$ with frequencies $2, 5, 8, 3$.
(c) Find the standard deviation of $12,\ 15,\ 15,\ 18,\ 20,\ 22$.

11. Each data set is transformed. Find the new mean and the new standard deviation.

(a) Mean 50, standard deviation 6; every value is multiplied by 3.
(b) Mean 30, standard deviation 5; the value 10 is added to every data point.
(c) Mean 60, standard deviation 12; each value is transformed by $y = 0.5x + 20$.
(d) Mean 40, standard deviation 9; each value is transformed by $y = 2x - 5$.

12. Explain each of the following.

(a) Why adding a constant to every value leaves the standard deviation unchanged.
(b) Why the interquartile range is preferred to the range when the data may contain outliers.

10.4 Visualising Distributions

Numbers summarise a data set; a picture shows its shape. A mean and standard deviation tell you where the data sit and how far they spread. They do not tell you whether the distribution is symmetric, or leans to one side, or is concentrated at one end. Three kinds of display are used in this course.

10.4.1 Histograms

A **histogram** groups continuous data into classes and draws a bar for each, with height equal to the frequency. Unlike a bar chart for categories, the bars touch, because the horizontal axis is a continuous number line. In this course every class has the same width.

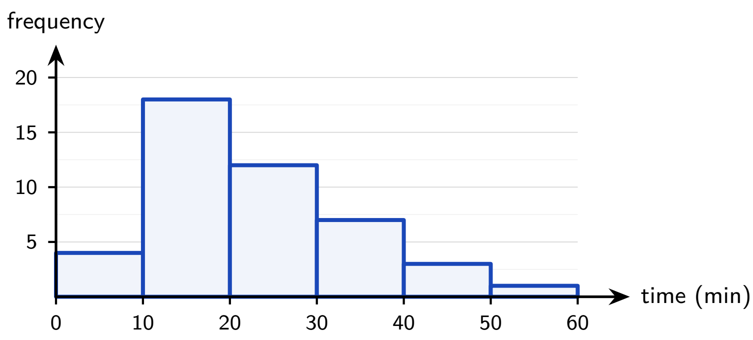


Figure 10.1 A histogram of 45 waiting times in classes of equal width. The bars touch because the horizontal axis is a continuous number line. The long tail to the right is positive skew.

The shape is what a histogram is for. It may be a symmetric peak, a long tail to the right (**positive skew**), as above, or a long tail to the left (**negative skew**).

In a positively skewed distribution the few large values raise the mean above the median. In a negatively skewed one the order reverses. Comparing mean and median is a quick numerical test for skew.

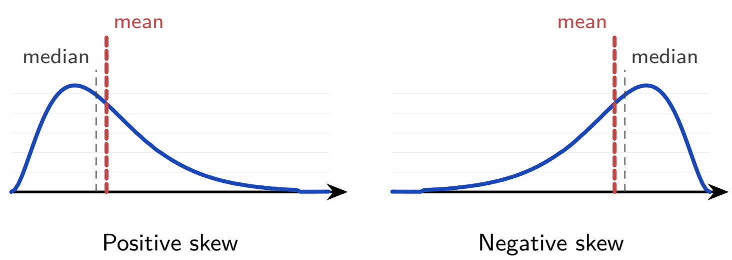


Figure 10.2 The long tail pulls the mean towards it, and the median hardly moves. With a tail to the right the mean sits above the median; with a tail to the left it sits below. Comparing the two numbers therefore tells you which way a distribution leans.

10.4.2 Cumulative frequency

A **cumulative frequency** curve answers "how many values fall below this point?" You plot the running total of frequencies against the *upper* boundary of each class, then join the points with a smooth curve. The result is an S-shape, and the median and quartiles can be read from it directly. Enter from the vertical axis at one quarter, one half and three quarters of the total. Move across to the curve, then drop to the horizontal axis to read Q_1, Q_2 and Q_3.

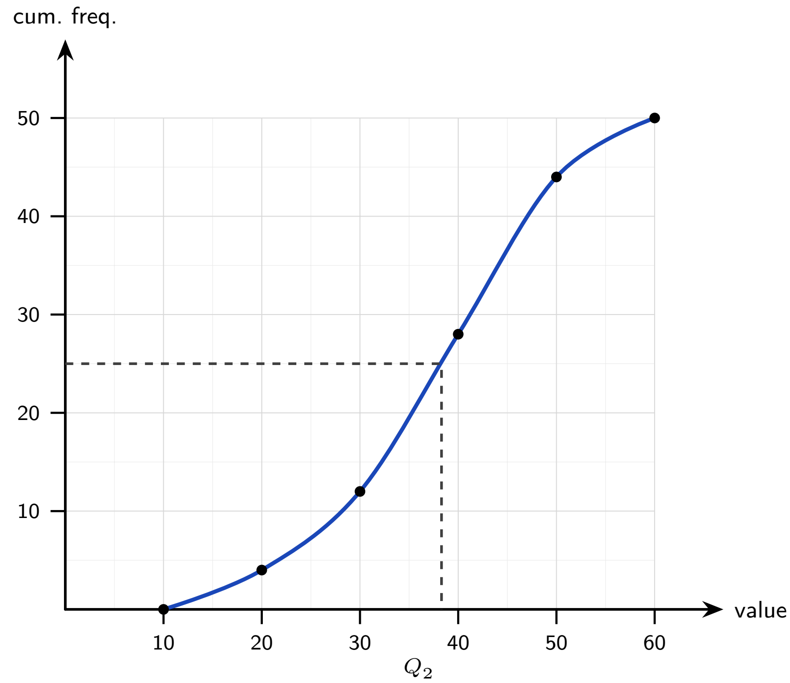


Figure 10.3 A cumulative frequency curve for 50 values. Each point is plotted at the *upper* boundary of its class. The dashed line reads the median: enter at half the total frequency, cross to the curve, drop to the value axis.

The dashed line shows the median. Enter at half the total frequency, here 25 of 50, cross to the curve, then drop to the value axis. Note that each point is plotted at the *upper* boundary of its class, and the curve is drawn through the points.

The same read-off gives any **percentile**: the p-th percentile is the value below which $p\%$ of the data lie. Enter the vertical axis at $\frac{p}{100} \times n$ and read across and down exactly as for the median. The median is the 50th percentile, and the quartiles Q_1 and Q_3 are the 25th and 75th. For the curve above, the 90th percentile is read off at a cumulative frequency of $0.9 \times 50 = 45$, giving a value of roughly 51.

EXAMPLE 10.10

The times taken by 80 competitors to finish a course are recorded below.

Time (min)	$0 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$
Frequency	8	24	28	20

Tabulate the cumulative frequencies, then estimate Q_1, the median, Q_3 and the 90th percentile from the curve.

Add the frequencies as you go, and record each running total against the *upper* boundary of its class. Nobody has finished before $t = 0$, so the table opens at $(0, 0)$.

Time less than	0	10	20	30	40
Cumulative frequency	0	8	32	60	80

Plot those five points and join them with a smooth increasing curve. Every reading then enters from the vertical axis, crosses to the curve, and drops to the time axis.

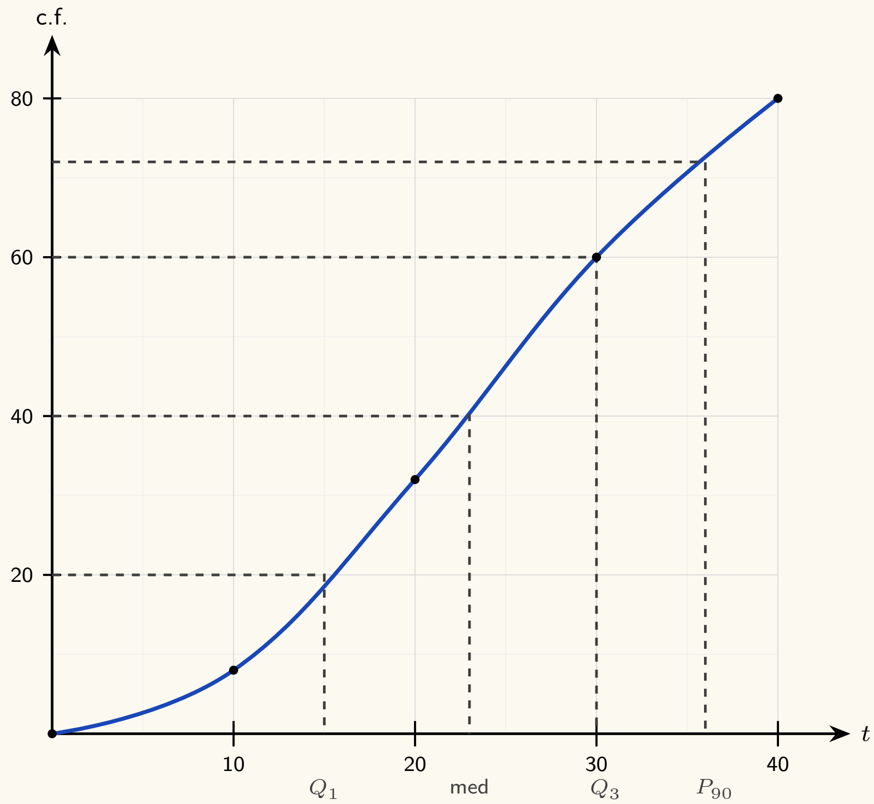


Figure 10.4 The cumulative frequency curve for the 80 finishing times, plotted at the upper class boundaries. Each read-off enters at a height on the c.f. axis, crosses to the curve, and drops to the time axis.

With $n = 80$, enter at $\frac{1}{4}(80) = 20$ for Q_1. That level lies between the plotted points $(10, 8)$ and $(20, 32)$, halfway between them, and the curve is met at about 15 minutes. The median needs $\frac{1}{2}(80) = 40$, a little above $(20, 32)$ and well below $(30, 60)$, giving about 23 minutes. For Q_3, enter at $\frac{3}{4}(80) = 60$, which is the plotted point $(30, 60)$ itself, so $Q_3 = 30$ minutes exactly. Hence

$$\text{IQR} \approx 30 - 15 = 15 \text{ minutes.}$$

The 90th percentile is read the same way, entering at $0.9 \times 80 = 72$. That falls between $(30, 60)$ and $(40, 80)$, giving about 36 minutes, so nine competitors in ten were home within 36 minutes. The upper boundary is the part to get right. A cumulative frequency counts everything that has happened *by* a value, so the 32 belongs at $t = 20$ and not at the midpoint 15; plotting at midpoints drags every quartile you later read off.

10.4.3 Box-and-whisker plots

A **box-and-whisker plot** draws the **five-number summary**: minimum, Q_1, median, Q_3, maximum. The box spans the interquartile range, and a line inside it marks the median. The whiskers reach to the smallest and largest values that are *not* outliers. Any outlier is marked separately with a cross. In one small picture you see the centre, the spread, and the skew.

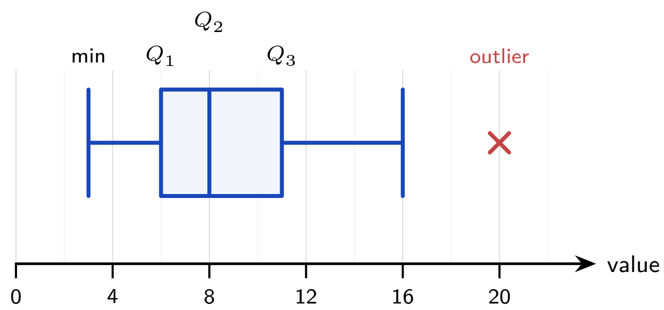


Figure 10.5 A box-and-whisker plot of the five-number summary. The box spans the interquartile range and the inner line marks the median. The right whisker stops at 16, the largest value that is not an outlier; the value 20 is marked with a cross.

A box positioned to the left, with a long right whisker, as above, indicates positive skew. Here the value 20 is an outlier, by the rule below. The whisker therefore stops at 16, the largest value that is not an outlier, and 20 is marked with a cross. Comparing two box plots side by side is the fastest way to contrast two distributions.

A box plot also gives a first hint about the normal distribution of Ch. 12. If the box and whiskers are roughly symmetric about the median, the data *may* be normally distributed. Clear skew shows that they are not.

EXAMPLE 10.11

The masses, in kilograms, of eleven parcels are

$$8,\ 11,\ 13,\ 14,\ 16,\ 16,\ 18,\ 21,\ 22,\ 25,\ 41.$$

Find the five-number summary, determine whether 41 is an outlier, and state where each whisker of the box plot stops.

The values are already ordered, with $n = 11$. The median is the 6th value, 16. The lower quartile is the median of the five values below it, $8, 11, 13, 14, 16$, so $Q_1 = 13$; the upper quartile is the median of the five above, $18, 21, 22, 25, 41$, so $Q_3 = 22$. The five-number summary is

$$8, \quad 13, \quad 16, \quad 22, \quad 41.$$

Now apply the $1.5 \times$ IQR rule stated below. Here IQR $= 22 - 13 = 9$, so the boundaries are

$$Q_1 - 1.5(9) = 13 - 13.5 = -0.5, \qquad Q_3 + 1.5(9) = 22 + 13.5 = 35.5.$$

Since $41 > 35.5$, the heaviest parcel is an outlier, and it is marked with a cross. No value falls below -0.5, so there is no outlier at the low end.
The left whisker therefore reaches the minimum, 8. The right whisker stops at 25, the largest mass that is not an outlier.
The maximum in the summary is still 41: the whisker stops at 25, the data do not. Reading a five-number summary back off a plot therefore means taking the maximum from the cross, not from the end of the whisker.

10.4.4 Comparing two distributions

One box plot describes a data set. Two drawn on the same scale compare them, and that comparison is what examinations ask for. Four things should be read off, and a good answer mentions the centre, the spread, and at least one shape or outlier feature, always in context.

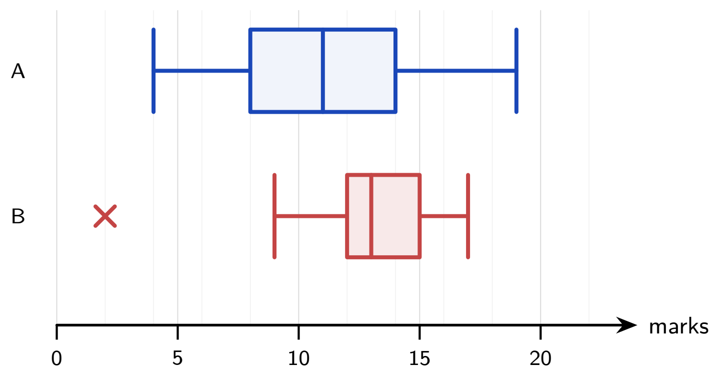


Figure 10.6 Two classes' marks on the same scale. Class B has the higher median and the smaller interquartile range, so it did better and more consistently. Class A is roughly symmetric; class B is negatively skewed and has one low outlier, marked with a cross.

Written out, that comparison reads: class B has the higher median (13 against 11), so B performed better on average. B also has the smaller interquartile range (3 against 6), so B's marks are more consistent. A is roughly symmetric while B is negatively skewed, and B contains one low outlier that A does not. Notice that every claim names a statistic and then says what it means for the classes. A comparison that only says "B is better" earns nothing.

10.4.5 The outlier rule

Now "outlier" can be defined precisely. A value is treated as an outlier if it lies more than 1.5 interquartile ranges beyond the nearer quartile.

KEY A value x is an **outlier** if

$$x < Q_1 - 1.5\,(\text{IQR}) \qquad \text{or} \qquad x > Q_3 + 1.5\,(\text{IQR}).$$

EXAMPLE 10.12

A data set has $Q_1 = 24$, $Q_3 = 40$. Is a value of 70 an outlier?

First IQR $= 40 - 24 = 16$. The upper boundary is

$$Q_3 + 1.5(\text{IQR}) = 40 + 1.5(16) = 40 + 24 = 64.$$

Since $70 > 64$, the value is an outlier. On a box plot it would be marked with a cross, with the whisker stopping at the largest value that is not an outlier.

YOUR TURN 10.4 Visualising Distributions

13. Consider the data $4,\ 6,\ 7,\ 9,\ 10,\ 12,\ 13,\ 15,\ 18$.

(a) Find the median.
(b) Find Q_1 and Q_3.
(c) Write down the five-number summary.
(d) State the interquartile range.

14. Apply the $1.5 \times$ IQR outlier rule.

(a) A data set has $Q_1 = 30$ and $Q_3 = 50$. Determine whether 95 is an outlier.
(b) A data set has $Q_1 = 12$ and $Q_3 = 20$. Determine whether 2 is an outlier.
(c) A data set has $Q_1 = 9$ and $Q_3 = 19$. Find both outlier boundaries.

15. Reading displays.

(a) A box plot has $Q_1 = 12$ and $Q_3 = 24$. Find the interquartile range.
(b) A cumulative frequency curve is drawn for 80 values. State the cumulative frequencies at which you read off Q_1, the median and Q_3.
(c) Explain why the bars of a histogram touch, while those of a bar chart do not.

10.5 Correlation

Everything so far has described one variable. Galton's question was different: he had *two* measurements on each family, parent height and child height, and wanted to know whether they were linked. This is the study of **bivariate** data, meaning data in which two variables are recorded for each individual. The first step is always to look.

10.5.1 Scatter diagrams

A **scatter diagram** plots each pair (x, y) as a point. The cloud of points has a shape, and the shape is the relationship. If y tends to rise as x rises, the correlation is **positive**. If y falls as x rises, it is **negative**. If there is no tilt, there is no linear correlation. The tighter the points cluster around a straight line, the **stronger** the correlation.

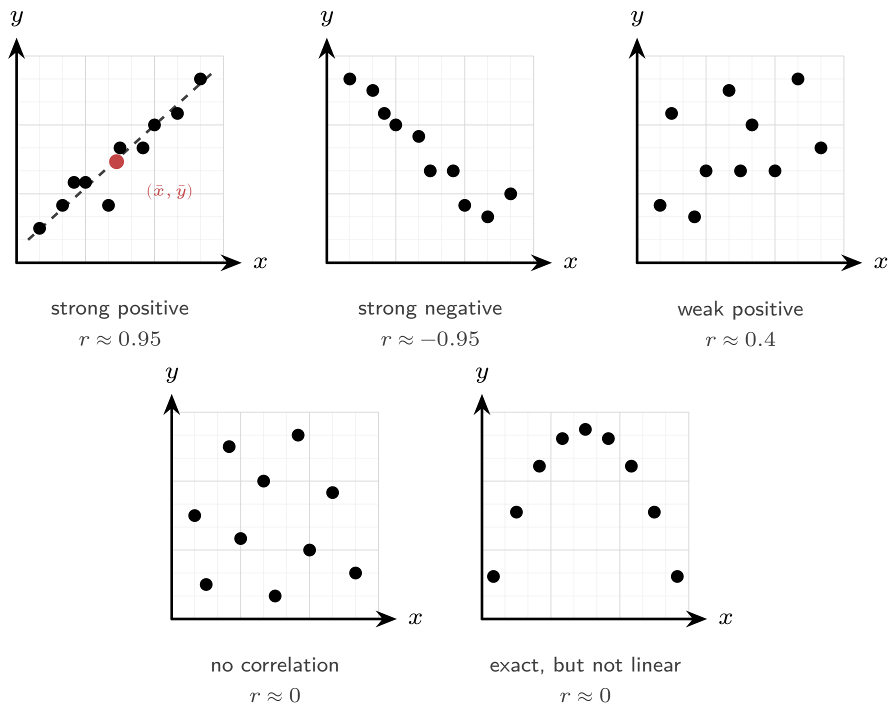


Figure 10.7 Five clouds and the value of r each gives. The sign of r is the direction of the tilt and the size of r is how tightly the points hug a line. The last panel is the warning: the points lie exactly on a curve, so the relationship could not be stronger, yet r is near zero because the curve is not straight.

Read the panels from left to right. The first two tilt clearly, one up and one down, and the points sit close to a line: those are strong correlations, positive and negative. The third tilts upward but the points scatter widely, so the correlation is weak. The fourth has no tilt at all.

The last panel is the one to remember. Its points lie exactly on a curve, so the two variables are perfectly related, yet r is near zero. That is because r only measures how close the points are to a *straight* line.

The first panel also shows a **line of best fit** drawn by eye: a straight line balancing the points above and below it. To position it, first compute the **mean point** $(\bar{x}, \bar{y})$, the mean of the x-values paired with the mean of the y-values, marked in scarlet. Draw your line through it. Every correctly drawn line of best fit passes through the mean point; in §10.6 the calculator makes this exact.

10.5.2 Pearson's correlation coefficient

You can judge correlation well by eye, but not precisely. **Pearson's product-moment correlation coefficient**, written r, turns the strength and direction into a single number between -1 and 1.

> DEF Pearson's correlation coefficient r measures the strength and direction of the **linear** relationship between two variables. It satisfies $-1 \le r \le 1$.

The sign of r gives the direction, and its size gives the strength. The extremes $r = \pm 1$ mean the points lie exactly on a straight line; $r = 0$ means no linear relationship. A rough reading:

$\lvert r \rvert$ **near**	**Interpretation**
1	strong linear correlation
0.5	moderate linear correlation
0	weak or no linear correlation

You compute r with your GDC by entering the two lists. Reading and interpreting r matters far more than the formula behind it.

A linear change of scale leaves r unchanged. Record the heights in metres rather than centimetres, or convert the temperatures from Celsius to Fahrenheit, and every point moves while r stays exactly where it was, because r measures how tightly the cloud hugs a straight line and stretching or shifting an axis does not loosen that grip. Multiplying by a negative number reverses one axis, which flips the sign of r and leaves its size alone.

Those words, strong and weak, are only a rough guide, because how impressive a given r is depends on how much data produced it. With five points a value of $r = 0.7$ is unremarkable, since so few points fall near a line by chance alone. With fifty points the same 0.7 would be hard to explain away. A **critical value** settles the question: it is the smallest $\lvert r \rvert$ that counts as evidence of real correlation for a sample of that size.

KEY Compare $|r|$ with the **critical value** for your sample size n. If $|r|$ is larger, the correlation is taken as evidence of a genuine linear relationship; if it is smaller, the data are consistent with no correlation at all. Critical values shrink as n grows.

EXT Critical values of r belong to *Applications and Interpretation*, not to Analysis and Approaches, so no AA examination will ask for one. They are set out here because they answer the obvious question — how large does r have to be? — and because they make clear that a value of r means nothing until you know how many points produced it.

EXAMPLE 10.13

A study of 10 pairs gives $r = 0.58$. The critical value for $n = 10$ is 0.632. What can be concluded?

Here $|r| = 0.58 < 0.632$, so the correlation does not reach the critical value. There is not enough evidence of a linear relationship in these data.

Notice what this does *not* say. It does not say the two variables are unrelated, only that ten points are too few to demonstrate it. The same $r = 0.58$ from 30 pairs, where the critical value is about 0.361, would support the opposite conclusion.

CAUTION r measures *linear* association only. Data lying perfectly on a parabola can have r near 0, because the relationship, though exact, is not straight. Always look at the scatter diagram before trusting r.

10.5.3 Correlation is not causation

A strong r says two variables move together. It does not say one causes the other. There are at least three quite different reasons a correlation can appear without any causal link, and they must be told apart.

A third variable drives both. Ice-cream sales and drowning deaths rise and fall together across the year. Neither causes the other. Summer heat raises both, because hot weather sends people to the ice-cream shop and to the water. A variable like this, sitting behind both and explaining the link, is called a **confounding variable**.

The arrow points the other way. Hospitals with the most advanced equipment can show the highest death rates. The equipment is not killing patients. The most seriously ill patients are sent to the best-equipped hospitals, so the severity of illness is causing the choice of hospital, not the reverse.

The same reversal turns up in school data. Students who receive extra support often have lower grades than those who do not, and the figures can be read as though the support were doing harm. The direction is the other way round: low grades are why the support was offered in the first place. Judge the support by comparing those students with how they were doing before, not with everyone else.

Coincidence, found by searching. Between 2000 and 2009, cheese consumption per person in the United States tracked the number of people who died by becoming entangled in their bedsheets, with r above 0.94. Nobody thinks these are connected. If you compare thousands of pairs of unrelated variables, some pairs will match closely by chance alone, and a determined search will always find them.

The third case is the one to keep in mind whenever a surprising correlation is reported. A high r is easy to find if you go looking through enough data.

None of this makes correlation useless. The link between smoking and lung cancer was found statistically, in the 1950s, long before anyone could say how a cigarette damages a cell. The correlation was what made the question urgent, and biology then had to supply the mechanism. That is the usual division of labour: mathematics can show that two things move together and how strongly, but the case for cause has to be built outside the mathematics.

CAUTION A high correlation never establishes cause. There may be a hidden third variable, or the link may be coincidence. Causation must be argued from the context, never from r alone.

YOUR TURN 10.5 Correlation

16. Describe the correlation indicated by each value of r.

(a) $r = -0.85$

(b) $r = 0.12$

(c) $r = 0.97$

(d) $r = -0.40$

17. GDC Find Pearson's correlation coefficient for each data set.

(a) $x = 1, 2, 3, 4, 5$ and $y = 2, 5, 4, 7, 9$

(b) $x = 2, 4, 6, 8, 10$ and $y = 5, 9, 10, 14, 17$

18. Interpreting r.

(a) A set of points lies exactly on the parabola $y = x^2$ for $-3 \leq x \leq 3$. Explain why r is close to 0 even though the relationship is exact.

(b) A study finds a strong positive correlation between shoe size and reading ability in children. Explain why shoe size does not cause reading ability.

10.6 Linear Regression

If the scatter diagram shows a clear linear trend, you can do more than measure it. You can draw the line that best fits the data and use it to predict. The question is what "best" should mean.

10.6.1 The least-squares line

For any candidate line, each data point sits some vertical distance above or below it. That distance is the **residual**, the error of the line at that point. The **least-squares regression line** is the line that makes the total of the squared residuals as small as possible. Squaring, again, stops positive and negative errors from cancelling, and it counts large errors much more heavily than small ones.

KEY The **regression line of y on x** has the form

$$y = ax + b,$$

chosen to minimise the sum of squared vertical residuals. It always passes through the mean point $(\bar{x}, \bar{y})$.

You find a and b from your GDC. You must also be able to say what they *mean* in context. The gradient a is the change in y for each one-unit increase in x. The intercept b is the predicted y when $x = 0$. The line always passes through $(\bar{x}, \bar{y})$, which is a useful check and an occasional exam shortcut.

EXAMPLE 10.14

A GDC gives the regression line of exam score y on hours studied x as $y = 6.2x + 41$, valid for students who studied between 2 and 9 hours. Predict the score of a student who studied 5 hours, interpret the values 6.2 and 41 in context, and comment on predicting the score for 20 hours.

For $x = 5$, substitute into the line:

$$y = 6.2(5) + 41 = 31 + 41 = 72.$$

In context, the gradient says each additional hour of study raises the predicted score by 6.2 marks. The intercept 41 is the predicted score of a student who did not study at all. That reading is itself a mild extrapolation, because $x = 0$ lies below the data range.

Predicting at $x = 20$ is unsafe. The data only cover 2 to 9 hours, so 20 lies far outside the known range, and the linear trend may not continue there.

10.6.2 Interpolation and extrapolation

Predicting *inside* the range of the data is **interpolation**, and it is reliable when the correlation is strong. Predicting *outside* the range is **extrapolation**, and it is unreliable. You are assuming the pattern continues where you have no evidence. The further beyond the data you go, the less trustworthy the prediction.

CAUTION A regression line is only trustworthy for prediction within the range of the original data, and only when $|r|$ is reasonably large. Extrapolating far beyond the data, or fitting a line to weakly correlated points, produces predictions worth little.

10.6.3 The two regression lines

There is an asymmetry hidden in "least squares." The line of y on x minimises *vertical* residuals, because it treats y as the quantity to predict from x. If instead you want to predict x from y, you minimise *horizontal* residuals, giving a different line: the **regression line of x on y**.

KEY Use the regression line of y **on** x to predict y from a given x. Use the regression line of x **on** y to predict x from a given y. Both lines pass through $(\bar{x}, \bar{y})$, and they coincide only when $r = \pm 1$.

The rule is simple. Choose the line whose name begins with the quantity you want to predict. Choosing the wrong line is a common and avoidable error.

EXAMPLE 10.15

For a set of bivariate data, $\bar{x} = 10$ and $\bar{y} = 25$. The regression line of y on x is $y = 2x + 5$. Verify that the mean point lies on this line.
Substitute $x = \bar{x} = 10$:

$$y = 2(10) + 5 = 25 = \bar{y}.\checkmark$$

Every least-squares line of y on x passes through $(\bar{x}, \bar{y})$, so this is a quick check on a computed line.

EXAMPLE 10.16

For the same data the regression line of x on y is $x = 0.35y + 1.25$. Estimate x when $y = 30$, and compare with what rearranging the other line would have given.
The quantity wanted is x, so use the line of x on y:

$$x = 0.35(30) + 1.25 = 11.75.$$

Rearranging the y on x line instead would give $x = \dfrac{30-5}{2} = 12.5$, a different answer. Both lines pass through the mean point, but they are not the same line, and only one of them was fitted to predict x.

The two agree only when $r = \pm 1$. Here the product of the gradients is $2 \times 0.35 = 0.7$, and that product is always r^2, so $r = \sqrt{0.7} \approx 0.84$. The further r is from ± 1, the further apart the two lines lie, and the more the wrong choice costs you.

YOUR TURN 10.6 Linear Regression

19. GDC Find the regression line of y on x for each data set.

(a) $x = 1, 2, 3, 4, 5$ and $y = 3, 5, 4, 8, 10$

(b) $x = 2, 4, 6, 8, 10$ and $y = 5, 9, 10, 14, 17$

20. Using a regression line.

(a) A regression line is $y = 2.5x + 4$. Predict y when $x = 6$.

(b) A regression line $y = -1.5x + 30$ was fitted using data with $2 \leq x \leq 15$. Predict y when $x = 8$.

(c) Explain why using that same line to predict y at $x = 40$ is unreliable.

21. Two further results.

(a) The mean point of a data set is $(8, 20)$ and the regression line of y on x has gradient 1.5. Find the equation of the line.

(b) You are given a value of y and want to predict the corresponding x. State which regression line you should use, and why.

Chapter 10 · Statistics

Reflections

TOK Almost everything you are told about the world arrives as a number: your grade, the mass of an atom, how happy a country is. The number is then treated as the thing itself.

Mass really is a quantity, and 12.011 describes carbon fairly. Happiness is not a quantity in that sense. A score of 7.2 is the output of a questionnaire somebody designed, with questions somebody chose, and a grade of 6 compresses two years of work into one digit. So every statistic makes two claims: that the thing measured has a size at all, and that this number captures it. Either can fail while the arithmetic stays perfect.

Which of the numbers used to describe you measure something that was already there, and which create the thing they claim to measure? Variance, meanwhile, can be written in several forms that all return the same value: could mathematics have settled on different formulae, equally true? And if the easiest things to measure are the ones that get counted, does statistics quietly teach us to value what is measurable over what is not?

TOK §10.5 gives you a number for how closely two quantities move together, and §10.6 gives you a line for predicting one from the other. Neither shows that one causes the other.

That limit may be permanent rather than a gap in the method. What we observe is that two things keep company; cause is not something a measurement can display. So can we have knowledge of cause and effect at all, or only of correlation? And when a line is used to describe something real, what decides whether the model is reliable enough to act on: the size of r, the range of the data it was fitted to, or the reasoning that put those two variables together in the first place?

IM The word "statistics" shares a root with "state." It began in the 18th century as the numerical bookkeeping of nations: population, taxes, soldiers, harvests. The mathematics of describing variation grew out of the practical needs of governments counting their citizens, long before it became a tool of science.

RW Galton's "regression toward mediocrity" is now called regression to the mean, and it affects decision-making everywhere. A pilot who flies badly and is scolded usually does better next time. The reason is not the scolding. An extreme performance is, on average, followed by a less extreme one. Mistaking this statistical drift for cause and effect has misled trainers, investors, and doctors for over a century.

The same trap sits in school data. A class that scores unusually low one term tends to score closer to average the next, whatever was changed in between, so any change made after a bad term will appear to have worked. It is also why a drug trial needs a control group: patients enrol when their illness is at its worst, and an illness measured at its worst tends to be measured better next time on its own.

EXT Your calculator returns r without showing its working. The formula behind it is

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \ \sum(y_i - \bar{y})^2}},$$

and it should be read rather than memorised.

Take one term of the numerator, $(x_i - \bar{x})(y_i - \bar{y})$. It asks where a single point sits relative to *both* means. Above average in both, or below average in both, and the product is positive; above in one and below in the other, and it is negative. So each point votes, and the numerator is the total vote.

That total depends on the units, and the denominator divides them out, so r is a pure number. It also forces $-1 \le r \le 1$: the numerator can never exceed the denominator, a fact called the Cauchy–Schwarz inequality, and equality holds exactly when every point lies on one straight line. That is why $r = \pm 1$ means perfect linearity, and nothing else does.

End-of-Chapter Problems — Chapter 10: Statistics

1. The times, in minutes, taken by 40 students to complete a puzzle are summarised below.

Time (min)	$0 \le t < 10$	$10 \le t < 20$	$20 \le t < 30$	$30 \le t < 40$	$40 \le t < 50$
Frequency	4	11	15	8	2

(a) Estimate the mean time. [3]

(b) Write down the modal class. [1]

(c) State the class containing the median. [2]

2. A data set is $12, 15, 18, 19, 21, 22, 24, 25, 27, 28, 30, 46$.

(a) Find the five-number summary. [3]

(b) Determine whether 46 is an outlier, showing your working. [3]

(c) Describe the shape of the distribution. [1]

3. GDC The table shows the number of hours x eight students revised and their test score y.

x	2	3	5	6	8	9	11	12
y	10	14	15	20	24	27	30	34

(a) Find Pearson's correlation coefficient and interpret it. [3]

(b) Find the equation of the regression line of y on x. [2]

(c) Interpret, in context, the gradient and the y-intercept of your line. [2]

(d) Use your line to estimate the score of a student who revised for 7 hours. [2]

(e) Explain whether your line should be used to estimate the score for 25 hours. [2]

4. A set of data has mean 45 and standard deviation 8.

(a) Each value is increased by 12. Write down the new mean and standard deviation. [2]

(b) Instead, each value is multiplied by 1.5. Write down the new mean and standard deviation. [2]

(c) Explain why adding a constant changes the mean but not the standard deviation. [2]

5. A region has populations of 4000, 10 000, and 6000 in three towns A, B, and C. A stratified sample of 100 households is required.

(a) Find the number of households to sample from each town. [3]

(b) State one advantage of stratified sampling over simple random sampling here. [2]

6. The cumulative frequencies of a data set of 60 values are given.

Value $\leq$	10	20	30	40	50
Cumulative frequency	4	16	34	52	60

(a) Draw a cumulative frequency curve. [3]
(b) Use it to estimate the median and the interquartile range. [4]
(c) Estimate the 70th percentile. [2]

7. In a test, 20 students in class A had a mean score of 72, and 30 students in class B had a mean score of 65. Find the mean score of all 50 students combined. [3]

8. A frequency distribution of the values $1, 2, 3, 4$ has frequencies $5, 8, k, 4$. The mean is 2.5. Find the value of k. [4]

9. Two machines fill bottles. A sample from each gives the same mean volume of 500 mL, but machine X has standard deviation 2 mL and machine Y has standard deviation 9 mL.

(a) Explain what this tells you about the two machines. [2]
(b) State which machine is more reliable, with a reason. [2]

10. GDC The table gives the area x (in m^2) and monthly rent y (in \$) of five apartments.

x	10	20	30	40	50
y	14	20	25	33	38

(a) Find the regression line of y on x. [2]
(b) Use it to estimate the rent of a 35 m^2 apartment. [2]
(c) Find the regression line of x on y, and use it to estimate the area of an apartment renting for \$30. [3]

11. GDC A data set is $4, 6, 7, 8, 9, 10, 55$.

(a) Find the mean and the median. [3]
(b) The value 55 is removed. Find the new mean and median. [2]
(c) Comment on which measure of centre is more affected by the outlier, and why. [2]

12. A student's final grade is 40% coursework and 60% final exam. The student scores 70 on coursework and 85 on the exam. Find the final grade. [2]

13. The masses, in grams, of seven eggs are $52, 55, 58, 60, 61, 63, 90$.

(a) Find Q_1, Q_3, and the interquartile range. [3]
(b) Show that 90 is an outlier. [2]
(c) State, with a reason, whether the mean or the median better represents a typical egg. [2]

14. A regression line of y on x passes through the mean point $(6, 19)$ and has gradient -2.

(a) Find the equation of the regression line. [2]
(b) Use it to predict y when $x = 4$. [2]

15. The five-number summaries of test scores in two classes are given below.

	Min	Q_1	Median	Q_3	Max
Class A	32	45	55	62	90
Class B	42	50	58	66	74

(a) Write down the interquartile range for Class A. [1]
(b) Show that the score of 90 in Class A is an outlier. [2]
(c) Compare the two distributions, referring to centre and spread. [2]
(d) State, with a reason, whether the scores in Class B may be normally distributed. [2]

16. GDC The waiting times, in minutes, of a group of customers are summarised below. The estimated mean waiting time is 20 minutes.

Time (min)	$0 \le t < 10$	$10 \le t < 20$	$20 \le t < 30$	$30 \le t < 40$
Frequency	6	k	12	8

(a) Show that $k = 18$. [3]
(b) Write down the modal class. [1]
(c) State the class containing the median. [2]
(d) Use your GDC to estimate the standard deviation. [2]

17. A set of examination marks x has mean 62 and standard deviation 10. The marks are scaled linearly to $y = ax + b$, $a > 0$, so that the scaled marks have mean 70 and standard deviation 8.

(a) Find the values of a and b. [4]
(b) A student's original mark was 75. Find their scaled mark. [1]
(c) The median of the original marks was 61. Write down the median of the scaled marks. [1]

18. GDC The table shows the maximum temperature x (°C) and the number of ice creams sold y on six days.

x (°C)	18	20	23	25	27	30
y	110	125	150	160	175	195

(a) Find Pearson's correlation coefficient r and comment on it. [2]
(b) Find the equation of the regression line of y on x. [2]
(c) Interpret, in context, the gradient of your line. [1]
(d) Estimate the number of ice creams sold on a day with maximum temperature $24\,°$C. [2]

(e) Explain why the line should not be used when the maximum temperature is $40\,^{\circ}$C. [1]

19. For a bivariate data set, the regression line of y on x is $y = 2x + 1$ and the regression line of x on y is $x = 0.4y + 1$.

(a) Find $\bar{x}$ and $\bar{y}$. [4]
(b) Estimate the value of x when $y = 20$. [2]
(c) Explain why the regression line of y on x should not be used in part (b). [1]

20. A supermarket wants to survey its customers.

(a) State the sampling method used when every 20th customer entering the store is selected. [1]
(b) State the sampling method used when customers are grouped by age band and sampled in proportion to the size of each band. [1]
(c) State the sampling method used when the first 30 customers on Monday morning are selected. [1]
(d) Explain why the method in (c) is likely to produce a biased sample. [2]
(e) A school of 320 juniors and 480 seniors is to be surveyed with a stratified sample of 50. Find the number of juniors and seniors in the sample. [2]

21. A group of n students has a mean mark of 68. When 5 students with mean mark 60 join the group, the overall mean falls to 66. Find n. [4]

22.

(a) Match each value of r to one description: $r = -0.92$, $r = -0.35$, $r = 0.08$, $r = 0.85$. Descriptions: strong positive, weak negative, strong negative, no linear correlation. [2]
(b) A study finds a strong positive correlation between the hours children spend watching television and their body mass. Explain why this does not show that television causes weight gain, and suggest a possible third variable that could explain both. [2]
(c) Heights are recorded in centimetres and r is calculated. The heights are converted to metres. State, with a reason, the effect on the value of r. [2]

23. GDC The table gives the monthly mean air temperature in Tokyo, in °C, taken from the Japan Meteorological Agency climatological normals for 1991–2020.

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp (°C)	5.4	6.1	9.4	14.3	18.8	21.9	25.7	26.9	23.3	18.0	12.5	7.7

(a) Find the five-number summary of the twelve values. [4]
(b) Show that the data contain no outliers. [3]
(c) The JMA publishes an annual normal of $15.8\,^{\circ}$C. Calculate the mean of the twelve monthly values and compare it with the published figure. [3]

(d) Describe the shape of the distribution, and explain why this box plot tells you very little about the temperature on any particular day in Tokyo. [3]

24. A data set of n values has mean $\bar{x}$. A new value equal to $\bar{x}$ is added to the set.

(a) Show that the mean is unchanged. [3]

(b) Show that the standard deviation decreases (or stays the same). State the condition for equality. [4]

25. Five positive integers have a mean of 8, a median of 7, and a mode of 4 (occurring exactly twice). Find all possible sets of five integers. [6]

26. The regression line of y on x for a data set is $y = 0.8x + 3$, and the regression line of x on y is $x = 1.05y - 2$.

(a) Both lines pass through $(\bar{x}, \bar{y})$. Find $\bar{x}$ and $\bar{y}$. [4]

(b) It can be shown that the product of the two gradients equals r^2. Find r. [3]

27. A set of n data values has mean $\bar{x}$ and standard deviation σ. Prove algebraically that adding a constant c to every value leaves σ unchanged, starting from the definition $\sigma^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$. [5]

28. A sports club has 900 members, made up of 225 families of exactly four. The membership register lists each family together, always in the same order: father, mother, elder child, younger child. A sample of 45 members is to be taken.

(a) The club secretary takes a systematic sample. Find the sampling interval k, and describe how the 45 members are selected. [2]

(b) State one advantage of systematic sampling over simple random sampling for a register of this size. [2]

(c) Explain why this particular systematic sample is very likely to be badly unrepresentative, and state what the sample would consist of if the random start were 2. [3]

(d) The club has 360 junior members and 540 senior members. Find the number of each in a stratified sample of 45. [2]

(e) All 45 selected members are sent a questionnaire and 18 reply. Explain why the 18 replies may not represent the club, and suggest one thing the secretary could do about it. [3]

29. A city council wants to estimate the mean distance cycled per week by its 12 000 adult residents. Two methods are proposed.

Method A: leave questionnaires at the counter of the city's three bicycle shops and use the forms returned.

Method B: telephone every 200th name on the electoral register of all $12\,000$ residents.

(a) State the sampling method used in Method A, and explain why it will almost certainly overestimate the mean. [3]

(b) State the sampling method used in Method B, and find the size of the sample obtained. [2]

(c) The council instead takes a stratified sample of 300 residents by age band. Using the table below, find the number of residents sampled from each band. [3]

Age band	18–29	30–49	50–69	70+
Residents	3000	4800	2800	1400

(d) Explain why stratifying by age band is a sensible choice for this particular investigation. [2]

(e) The council suggests improving Method A by collecting 1000 returned forms instead of 100. Explain why this would not fix the problem identified in (a). [2]

30. A university has 2400 students, half of them men and half women. A researcher wants to know student opinion on the library's opening hours. She stands at the library entrance and interviews students as they arrive, stopping once she has 40 men and 40 women.

(a) State the sampling method she has used. [1]

(b) Explain why this is not a stratified sample, even though the sample matches the university's split exactly. [2]

(c) Explain why the choice of location makes the sample unsuitable for the question being asked. [2]

(d) Describe a better method for this investigation, and justify your choice. [3]

Challenge Questions

31. **How much can one value move a summary?** Consider the data set

$$3,\ 5,\ 7,\ 9,\ 11.$$

(a) Write down the mean and the median. [2]

(b) The largest value is replaced by k, where $k \geq 9$. Find the mean in terms of k, and write down the median. [4]

(c) Find the value of k for which the mean is 20. [2]

(d) Explain why the median is 7 for *every* $k \geq 9$. [3]

(e) A data set has n values. Explain why changing a single value can move the mean by any amount, however large, but can move the median by at most one place in the ordered list. Hence explain what is meant by calling the median *resistant* and the mean *not resis-*

tant. [5]

(f) State, in terms of n, the smallest number of values that must be changed to make the median as large as you wish. Justify your answer for $n = 5$. [3]

32. GDC **Combining two groups.** Group A has n_1 values with mean $\bar{x}_1$ and variance σ_1^2. Group B has n_2 values with mean $\bar{x}_2$ and variance σ_2^2. The two groups are combined into one set of $n_1 + n_2$ values.

(a) Show that the combined mean is $\bar{x} = \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$. [3]

(b) Group A has 12 values with mean 50, and Group B has 8 values with mean 60. Find the combined mean. [2]

(c) Using $\sum x^2 = n(\sigma^2 + \bar{x}^2)$ for each group, show that the combined variance is

$$\sigma^2 = \frac{n_1(\sigma_1^2 + \bar{x}_1^2) + n_2(\sigma_2^2 + \bar{x}_2^2)}{n_1 + n_2} - \bar{x}^2.$$

[5]

(d) Group A has standard deviation 4 and Group B has standard deviation 3. Find the standard deviation of the combined set. [4]

(e) Show that $\sigma^2 \geq \dfrac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}$, and state the condition for equality. Interpret this result. [4]

33. **Putting data on a common scale.** A data set $x_1, x_2, \dots, x_n$ has mean $\bar{x}$ and standard deviation σ, with $\sigma \neq 0$. Each value is replaced by its *standardised value*

$$z_i = \frac{x_i - \bar{x}}{\sigma}.$$

(a) Show that the mean of the z_i is 0. [4]

(b) Show that the standard deviation of the z_i is 1. [4]

(c) The data set $4, 8, 10, 12, 16$ has mean 10 and standard deviation 4. Write down the five standardised values, and verify your answers to parts (a) and (b). [4]

(d) Student P scores 78 on a test with mean 70 and standard deviation 5. Student Q scores 84 on a different test with mean 75 and standard deviation 8. Determine which student performed better relative to their own group. [4]

(e) Explain why standardised values allow marks from two different tests to be compared, when the raw marks cannot be. [2]

STATISTICS AND PROBABILITY

CHAPTER 11

Probability

11

SYLLABUS 4.5 4.6 4.11 4.13

Probability puts a number on how likely an event is. That single step is what turns risk into something you can measure, compare and act on, so that a decision can rest on more than a feeling about what might happen.

Here is a decision that turns on such a number. A health service screens everyone in an age group for a rare disease. The test is a good one. Of every 100 people who have the disease, it correctly identifies 99. Of every 100 people who do not have it, it correctly clears 95. About one person in a hundred has the disease.

Your result comes back positive. How likely is it that you have the disease?

Most people answer about 99%, and doctors asked the same question often answer the same way. The correct answer is about 17%. The test is not faulty and none of the numbers are unusual. The error is in the reasoning. That 99% says how the test behaves *when the disease is present*, and the quantity you actually want is a different one: how likely the disease is *now that the result is positive*. Those two numbers are not the same, and this chapter builds the rule that connects them.

So probability gives clear answers, and it is easy to apply a clear answer to the wrong question. Two habits guard against that, and both are practised throughout the chapter. The first is to state precisely which event you are asking about. The second is to keep track of where a probability came from: some are calculated from the symmetry of a fair die, others only measured by counting what happened in 200 trials, and the two are not equally trustworthy.

We begin with the simplest experiment there is, a single roll of a die. From there the chapter builds the language of events, the rules for combining them, conditional probability and independence, tree diagrams, and finally Bayes' theorem. When we return to the test in the last section, the answer will follow at once.

11.1 The Language of Probability

Probability measures how likely something is, on a scale from 0 (impossible) to 1 (certain). To compute it we need to be precise about what "something" means.

Five words are used precisely from here on. The die is used to illustrate each one.

- **Experiment**: any process with an uncertain result. Rolling a die is an experiment.
- **Trial**: one repetition of the experiment. One roll is one trial.
- **Outcome**: the result of a single trial, such as a 3.
- **Sample space** U: the set of all possible outcomes. For one die, $U = \{1, 2, 3, 4, 5, 6\}$.
- **Event**: any subset of the sample space, that is, a collection of outcomes you are interested in. "An even number" is the event $\{2, 4, 6\}$.

When every outcome is equally likely, probability is just counting: the fraction of outcomes that belong to the event.

KEY · IB

$$P(A) = \frac{n(A)}{n(U)}$$

Here $n(A)$ is the number of outcomes in event A, and $n(U)$ the number in the whole sample space. Every probability obeys $0 \le P(A) \le 1$.

EXAMPLE 11.1

A fair die is rolled once. Find the probability of (a) an even number, (b) a number greater than 4.

The sample space has $n(U) = 6$ equally likely outcomes.

$$P(\text{even}) = \frac{n(\{2,4,6\})}{6} = \frac{3}{6} = \frac{1}{2}, \qquad P(>4) = \frac{n(\{5,6\})}{6} = \frac{2}{6} = \frac{1}{3}.$$

Counting works only because the die is fair. If the outcomes were not equally likely, this formula would not apply.

11.1.1 Relative frequency

When the outcomes are not equally likely, counting fails, and the probability has to be estimated by experiment. Repeat the trial many times and record how often the event occurs. The fraction of trials in which it happens is the **relative frequency** of the event, and it serves as an experimental estimate of the probability.

KEY After n trials,

$$P(A) \approx \frac{\text{number of trials in which } A \text{ occurred}}{n}$$

The estimate is rough when n is small, and it becomes more stable as n grows. This is the difference between the two kinds of probability you will meet: a *theoretical* probability comes from the structure of the experiment (a fair die, by symmetry), while an *experimental* one comes from data, and is what you fall back on when there is no symmetry to use.

EXAMPLE 11.2

A drawing pin is dropped 200 times and lands point-up 78 times. Estimate the probability that a dropped pin lands point-up.

There is no symmetry argument for a drawing pin, so use the relative frequency:

$$P(\text{point-up}) \approx \frac{78}{200} = 0.39.$$

A further 200 drops would almost certainly not give exactly 78 again, but the estimate becomes more accurate as the number of trials increases.

11.1.2 The complement

Sometimes the event you want is awkward to count, but its opposite is easy. The **complement** of A, written A', is the event that A does *not* happen. Since A either happens or it does not, their probabilities must fill the whole scale.

KEY · IB

$$P(A) + P(A') = 1 \qquad \Longrightarrow \qquad P(A') = 1 - P(A)$$

The complement is the route to take whenever the event itself is hard to count. In particular, whenever a question contains the phrase "at least one," work out the complement first.

EXAMPLE 11.3

A fair coin is tossed three times. Find the probability of getting at least one head.

Counting every way to get one, two, or three heads is possible but slow. The complement of "at least one head" is "no heads at all," a single outcome: TTT.

$$P(\text{at least one head}) = 1 - P(\text{TTT}) = 1 - \frac{1}{8} = \frac{7}{8}.$$

One subtraction replaces a count of seven outcomes. The saving becomes far greater as the number of tosses increases.

11.1.3 Expected number of occurrences

If an event has probability p and you repeat the experiment n times, you expect it to happen about np times.

KEY If an event has probability p and the experiment is repeated n times, the **expected number of occurrences** is

$$\text{expected number} = np.$$

This is not a guarantee. It is a long-run average. If you repeated the whole set of n trials many times and averaged the counts, that average would be close to np.

EXAMPLE 11.4

A fair die is rolled 60 times. How many sixes would you expect?

Each roll gives a six with probability $\frac{1}{6}$, so over 60 rolls:

$$\text{expected sixes} = 60 \times \frac{1}{6} = 10.$$

You will rarely get exactly 10, but the average over many sets of 60 rolls will be close to 10.

YOUR TURN 11.1 The Language of Probability

1.

(a) A fair die is rolled. Find the probability that the number is odd.
(b) A card is drawn from a deck of 52. Find the probability that it is a heart.
(c) A bag holds 4 red, 6 green and 5 blue counters. Find the probability that a counter drawn at random is green.
(d) A letter is chosen at random from the word STATISTICS. Find the probability that it is an S.
(e) A spinner numbered 1 to 8 is spun. Find the probability that it lands on a multiple of 3.

2.

(a) Given $P(A) = 0.35$, find $P(A')$.

(b) A fair die is rolled 90 times. Find the expected number of 5s.
(c) Two fair coins are tossed. Find the probability of at least one tail.

3.

(a) Two fair coins are tossed. List the sample space, and find the probability of exactly one head.
(b) A spinner has five sectors numbered 1 to 5, but it is not fair. In 400 spins it lands on 3 exactly 96 times. Estimate the probability of landing on 3.
(c) Using that estimate, find the expected number of 3s in 250 spins.
(d) Explain why the probability in part (a) can be found without any experiment, while the one in part (b) cannot.

4.

(a) A bag holds 12 counters, 5 of which are red. Find the probability that a counter drawn at random is not red.
(b) The probability that a bus is late on a given day is 0.18. Find the probability that it is not late.
(c) A fair coin is tossed four times. Find the probability of at least one tail.

11.2 Counting the Outcomes HL

The formula $P(A) = \dfrac{n(A)}{n(U)}$ is only as useful as your ability to find the two numbers in it. For a die you list the outcomes. For a committee of four chosen from twelve people you cannot: there are 495 of them. The counting tools of Ch. 2, §2.8 are what make the formula work at that scale, and this section is where they earn their place.

11.2.1 Which tool, and when

Two questions settle it every time. *How many objects are you choosing from, and how many are you taking?* And then the one that decides everything: *does the order matter?*

Order matters when the positions are different from one another — first, second and third place; president, secretary and treasurer; the digits of a PIN. Order does not matter when you end up with a set — a committee, a hand of cards, six lottery numbers. Choosing Ali then Ben gives the same committee as choosing Ben then Ali, and a count that treats those as different has counted the committee twice.

KEY HL Choosing r objects from n distinct objects:

$$\text{order matters:} \quad {}^nP_r = \frac{n!}{(n-r)!}, \qquad \text{order does not:} \quad {}^nC_r = \frac{n!}{r!\,(n-r)!}.$$

Every unordered selection corresponds to $r!$ ordered ones, which is the only difference between the two.

CAUTION Deciding this by instinct is where marks go. Read the question for a word that fixes an order: *arrange*, *rank*, *in a row*, *first and second*. Without one, it is almost always a combination.

11.2.2 Probability by counting

Once both counts use the same tool, the ratio is the probability. Count the sample space with one method, count the event with the same method, and divide.

EXAMPLE 11.5

A committee of 4 is chosen at random from 7 girls and 5 boys. Find the probability that it contains exactly 3 girls.

A committee is a set, so order does not matter and both counts are combinations.

The sample space is every committee of 4 from the 12 students:

$$n(U) = {}^{12}C_4 = 495.$$

For the event, read the requirement as an *and*: 3 of the 7 girls *and* 1 of the 5 boys. And multiplies Ch. 2, §2.8, so the two counts are multiplied:

$$n(A) = {}^7C_3 \times {}^5C_1 = 35 \times 5 = 175.$$

So

$$P(\text{exactly 3 girls}) = \frac{175}{495} = \frac{35}{99} \approx 0.354.$$

Notice the shape of the event count: one combination for each group the question splits the people into, multiplied together. Nearly every counting probability question has that shape.

EXAMPLE 11.6

Five cards are dealt from a standard pack of 52. Find the probability that exactly two of them are kings.

A hand is a set, so combinations again:

$$n(U) = {}^{52}C_5 = 2\,598\,960.$$

Split the pack into the two groups the question cares about: the 4 kings and the 48 non-kings.

Take 2 from the first *and* 3 from the second, so the counts multiply:

$$n(A) = {}^{4}C_2 \times {}^{48}C_3 = 6 \times 17\,296 = 103\,776.$$

$$P(\text{exactly two kings}) = \frac{103\,776}{2\,598\,960} \approx 0.0399.$$

The 48 in the second factor is doing real work. Taking ${}^{4}C_2 \times {}^{50}C_3$ instead would allow a third king into the hand, and "exactly two" would have become "at least two".

TIP Write down $n(U)$ before anything else. It fixes what an outcome *is* for the rest of the question, and most errors in this topic are really two counts made on different definitions of an outcome.

11.2.3 At least one, again

The complement of §11.1 is at its most useful here, because "at least one" spreads over many cases while "none" is a single count.

EXAMPLE 11.7

A box holds 20 components, of which 3 are defective. Four are drawn at random without replacement. Find the probability that at least one is defective.

Counting the cases directly means one defective, or two, or three — three separate counts. The complement is one:

$$P(\text{none defective}) = \frac{{}^{17}C_4}{{}^{20}C_4} = \frac{2380}{4845} = \frac{28}{57}.$$

Therefore

$$P(\text{at least one defective}) = 1 - \frac{28}{57} = \frac{29}{57} \approx 0.509.$$

Three bad components in twenty, and a sample of four is already more likely than not to catch one. That is why sampling works.

11.2.4 When order does matter

EXAMPLE 11.8

Eight athletes run a final, and all orders of finishing are equally likely. Find the probability that a particular three of them take the first three places, in some order.

Here the positions are distinguishable, so the sample space is ordered:

$$n(U) = {}^{8}P_3 = 336$$

ordered ways to fill the three medal positions. The three named athletes can fill them in $3! = 6$

orders, so

$$P = \frac{6}{336} = \frac{1}{56}.$$

The same answer comes from counting unordered: ${}^{8}C_{3} = 56$ possible medal-winning sets, one of which is the named three. Either tool works, provided the top and the bottom use the same one.

The counting principles, factorials, ${}^{n}P_{r}$ and ${}^{n}C_{r}$ are developed in Ch. 2, §2.8. The binomial distribution of Ch. 12 is this section's ${}^{n}C_{r}$ applied to repeated trials.

YOUR TURN 11.2 Counting the Outcomes HL

5. Decide in each case whether order matters, and evaluate the count.

(a) The number of ways to choose 3 books from 9 to take on holiday.
(b) The number of ways to award gold, silver and bronze among 9 finalists.
(c) The number of ways to seat 5 people in a row.
(d) The number of different hands of 5 cards from a pack of 52.

6. A team of 5 is chosen at random from 6 women and 4 men.

(a) Find the number of possible teams.
(b) Find the probability that the team contains exactly 3 women.
(c) Find the probability that the team is all women.
(d) Find the probability that the team contains at least one man.

7. GDC A box contains 15 light bulbs, of which 4 are faulty. Three are chosen at random without replacement.

(a) Find the probability that none is faulty.
(b) Find the probability that at least one is faulty.
(c) Find the probability that exactly two are faulty.

11.3 Combining Events

Real questions rarely involve a single event. You want the probability that a student plays football *or* tennis, or that a card is a king *and* a heart. Two operations cover every case: the **union** $A \cup B$ ("A or B or both") and the **intersection** $A \cap B$ ("A and B together").

The word *or* needs care, because everyday English and mathematics do not use it the same way. "Tea or coffee?" offers you one drink, not both. In mathematics $A \cup B$ always includes the case where both happen. This is called the **non-exclusivity of or**, and it is the reason the addition rule below needs a correction term. When a question really does mean one or the other but not both, it will say so.

A **Venn diagram** makes the structure visible. The rectangle is the sample space, written U and called the **universal set**: everything under discussion lies inside it. Each circle is an event, and every point of the diagram lies in exactly one of four regions. Combining an event with a complement names each region precisely.

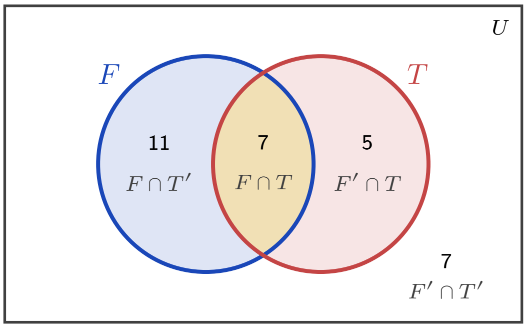


Figure 11.1 The four regions of a two-event Venn diagram, with the class of 30 filled in. Football only is $F \cap T'$, both is $F \cap T$, tennis only is $F' \cap T$, and neither is $F' \cap T'$, the region outside both circles. The four counts add to 30.

Fill a Venn diagram from the middle outwards. Put the overlap in first, then subtract it from each circle total to get the "only" regions, then use the grand total to find the outside. Working the other way round double counts the overlap, which is the same mistake the addition rule is built to correct.

11.3.1 The addition rule

If you add $P(A)$ and $P(B)$ to find $P(A \cup B)$, you count the overlap twice: once in A, once in B. Subtracting it once corrects the double count.

KEY · IB

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

When two events cannot happen together, they are **mutually exclusive**: their intersection contains no outcomes at all, written $A \cap B = \emptyset$, where $\emptyset$ is the **empty set**. Then $P(A \cap B) = 0$ and the rule simplifies.

KEY · IB For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

EXAMPLE 11.9

In a class of 30, 18 students play football, 12 play tennis, and 7 play both. A student is chosen at random. Find the probability that the student plays football or tennis, and the probability that they play neither.

Apply the addition rule directly:

$$P(F \cup T) = \frac{18}{30} + \frac{12}{30} - \frac{7}{30} = \frac{23}{30}.$$

"Neither" is the complement of "football or tennis":

$$P(\text{neither}) = 1 - \frac{23}{30} = \frac{7}{30}.$$

The Venn diagram above shows the same thing: 11 play only football, 7 play both, 5 play only tennis, and 7 play neither, totalling 30.

EXAMPLE 11.10

A single card is drawn from a standard deck of 52. Find the probability that it is a king or a heart.

There are 4 kings and 13 hearts, but the king of hearts is both, so it is counted in each:

$$P(\text{king} \cup \text{heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}.$$

Forgetting to subtract the king of hearts gives $\frac{17}{52}$, which is the most common error in this topic.

Three events need three circles, and the same instruction applies: start in the middle. The centre region belongs to all three events, and every other region is found by subtracting what has already been placed.

EXAMPLE 11.11

In a year group of 40 students, 20 study French, 18 study German and 15 study Spanish. Of these, 8 study French and German, 6 study German and Spanish, 7 study French and Spanish, and 3 study all three. A student is chosen at random. Find the probability that the student studies (a) exactly one of the three languages, (b) none of them.

Start with the 3 in the centre. The 8 who study French and German include those 3, so $8 - 3 =$

5 study those two and no other. In the same way $6-3=3$ study German and Spanish only, and $7-3=4$ study French and Spanish only.
Now the "only" regions. French has 20 altogether, of which $5+3+4 = 12$ are already placed, leaving 8 for French only. German gives $18-(5+3+3)=7$, and Spanish gives $15-(4+3+3)=5$.

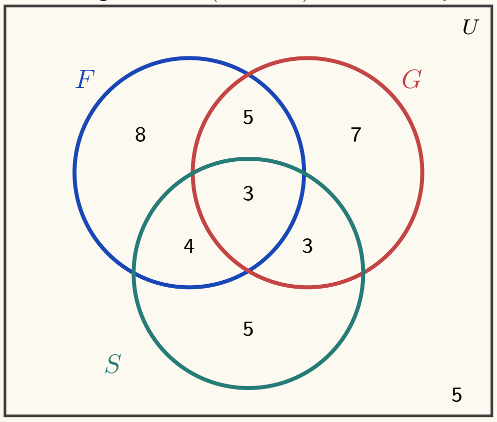


Figure 11.2 The three-circle diagram, filled from the centre outwards. The seven inner counts total 35, so 5 of the 40 students lie outside all three circles.

(a) Exactly one language means the three outer regions: $8+7+5=20$, so

$$P(\text{exactly one}) = \frac{20}{40} = \frac{1}{2}.$$

(b) The seven regions inside the circles total 35, so 5 students study none:

$$P(\text{none}) = \frac{5}{40} = \frac{1}{8}.$$

No formula was used. Once the diagram is filled, every probability is a count divided by 40.

11.3.2 Sample space diagrams and two-way tables

When an experiment has two parts, drawing the whole sample space is often faster than any formula. A **sample space diagram** plots one part on each axis. For two dice, the 36 equally likely outcomes form a grid, and any event is just a set of grid points to count.

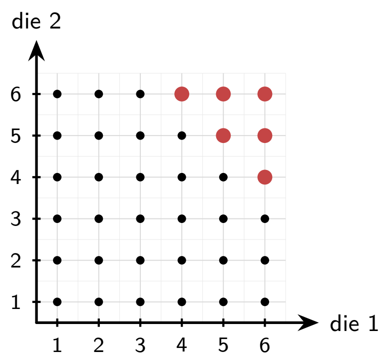


Figure 11.3 The 36 equally likely outcomes of rolling two dice. The six larger scarlet points are the outcomes whose total is at least 10.

The six larger scarlet points are the outcomes with a total of at least 10. There are six of them, so

$$P(\text{total} \geq 10) = \frac{6}{36} = \frac{1}{6}.$$

No rule was needed, only a picture and a count.

A **two-way table** (a table of outcomes) does the same for grouped data: it splits a population by two characteristics at once, and every probability is a cell, row, or column total divided by the grand total.

EXAMPLE 11.12

The table shows the 100 students of a year group by age group and club membership.

	Club	No club	Total
Junior	14	26	40
Senior	21	39	60
Total	35	65	100

A student is chosen at random. Find the probability that the student (a) is a junior in a club, (b) is in a club or is a junior.

(a) Read the single cell where the two conditions meet:

$$P(J \cap C) = \frac{14}{100} = 0.14.$$

(b) Apply the addition rule with the row and column totals:

$$P(J \cup C) = \frac{40}{100} + \frac{35}{100} - \frac{14}{100} = \frac{61}{100} = 0.61.$$

The table already contains every count used in both calculations. We will return to this table in the next section, where it answers conditional questions just as directly.

YOUR TURN 11.3 Combining Events

8.

(a) Given $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, find $P(A \cup B)$.
(b) Events A and B are mutually exclusive, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \cup B)$.
(c) Given $P(A \cup B) = 0.8$, $P(A) = 0.5$ and $P(B) = 0.4$, find $P(A \cap B)$.
(d) Given $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$, determine whether A and B are mutually exclusive.

9.

(a) A card is drawn from a deck of 52. Find the probability that it is a face card or a spade.

(b) In a group of 40 people, 25 like coffee, 20 like tea and 10 like both. Find the probability that a person chosen at random likes coffee or tea.

(c) Two events satisfy $n(A \text{ only}) = 8$, $n(B \text{ only}) = 5$, $n(A \cap B) = 4$ and $n(\text{neither}) = 3$. Find $P(A)$.

(d) Given $n(U) = 50$, $n(A) = 22$, $n(B) = 18$ and $n(A \cap B) = 7$, find $n(A \cap B')$ and $n(A' \cap B')$.

(e) In a group of 30 students, 15 play the piano, 12 the guitar and 9 the drums. Of these, 6 play piano and guitar, 4 play guitar and drums, 5 play piano and drums, and 2 play all three. Find the probability that a student chosen at random plays exactly one instrument, and the probability that they play none.

10. Two fair dice are rolled.

(a) Find the probability that the total is 7.

(b) Find the probability that the two numbers differ by exactly 1.

(c) Find the probability that the two dice show the same number.

11. The table shows 80 people classified by sex and by writing hand.

	Left	**Right**	**Total**
Male	7	33	40
Female	5	35	40
Total	12	68	80

One person is chosen at random.

(a) Find the probability that the person is left-handed.

(b) Find the probability that the person is female and right-handed.

(c) Find the probability that the person is female or left-handed.

11.4 Conditional Probability and Independence

Information changes probability. The chance that a random person has a particular illness is small, but the chance *given* that they tested positive is different. **Conditional probability** is the probability of one event once you know another has happened.

The essential idea is that knowing B has occurred **reduces the sample space** to B alone. Within that smaller set, you ask what fraction also lies in A.

KEY · IB

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) \neq 0$$

Read $P(A \mid B)$ as "the probability of A given B." The denominator $P(B)$ is the new, restricted sample space; the numerator is the part of it where A also holds.

EXAMPLE 11.13

Using the class of 30 from before, a student is chosen at random. Given that they play football, find the probability that they also play tennis.

Restrict to the 18 football players. Of those, 7 also play tennis:

$$P(T \mid F) = \frac{P(T \cap F)}{P(F)} = \frac{7/30}{18/30} = \frac{7}{18}.$$

Note this differs from $P(F \mid T) = \frac{7}{12}$. Conditioning is not symmetric: the order matters.

11.4.1 The multiplication rule

Rearranging the conditional formula gives a rule for the probability that two events both happen.

KEY

$$P(A \cap B) = P(A)\, P(B \mid A)$$

In words: the chance of both is the chance of the first times the chance of the second given the first. This is exactly how you move along the branches of a tree diagram, as the next section shows.

EXAMPLE 11.14

A drawer holds nine pens, four of which are red. Two pens are taken at random, one after the

other, without replacement. Find the probability that both are red.
Let A be "the first pen is red" and B "the second pen is red". Four of the nine pens are red, so $P(A) = \frac{4}{9}$.
For the second factor, assume A has happened. Eight pens remain and three of them are red, so $P(B \mid A) = \frac{3}{8}$. The multiplication rule combines the two:

$$P(A \cap B) = P(A)\,P(B \mid A) = \frac{4}{9} \times \frac{3}{8} = \frac{12}{72} = \frac{1}{6}.$$

The second factor is $\frac{3}{8}$ and not $\frac{4}{9}$ because the first pen is not put back: removing a red pen changes the number of red pens and the total together, so both parts of the fraction drop by one.

11.4.2 Independence

Two events are **independent** when knowing one tells you nothing about the other, so $P(A \mid B) = P(A)$. Substituting this into the multiplication rule gives a cleaner, symmetric test.

KEY · IB A and B are **independent** if and only if

$$P(A \cap B) = P(A)\,P(B).$$

Provided $0 < P(B) < 1$, this is equivalent to $P(A \mid B) = P(A) = P(A \mid B')$, which is not itself printed in the booklet. To test independence, compute both sides of either form and compare. The conditional form states directly what independence means: whether B happened or did not happen, the probability of A is unchanged.

EXAMPLE 11.15

A fair coin and a fair die are tossed together. Let H be "heads" and S be "rolling a six." Are H and S independent?
Here $P(H) = \frac{1}{2}$ and $P(S) = \frac{1}{6}$. The outcome "heads and a six" is one of the 12 equally likely combined outcomes:

$$P(H \cap S) = \frac{1}{12} = \frac{1}{2} \times \frac{1}{6} = P(H)\,P(S).$$

The two sides match, so the events are independent, which fits intuition: a coin cannot influence a die.

The two-way table of §11.3 passes the same test. There, $P(C \mid J) = \frac{14}{40} = 0.35$ and $P(C) = \frac{35}{100} = 0.35$: knowing a student is a junior does not change the chance they are in a club, so age group and club membership are independent.

That table is the exception rather than the rule. Two characteristics recorded on the same group of people are usually linked at least a little, and then the test fails. The calculation is identical when it comes out that way, and so is the shape the answer must take: the two numbers first, then the verdict.

EXAMPLE 11.16

The table classifies the 200 students of a year group by whether they take music and whether they take drama.

	Drama	No drama	Total
Music	30	50	80
No music	20	100	120
Total	50	150	200

A student is chosen at random. Determine whether the events M (takes music) and D (takes drama) are independent.

Read the two single-event probabilities from the totals, and the intersection from the cell where the two conditions meet:

$$P(M) = \frac{80}{200} = 0.4, \qquad P(D) = \frac{50}{200} = 0.25, \qquad P(M \cap D) = \frac{30}{200} = 0.15.$$

Compare the product with the intersection:

$$P(M)\,P(D) = 0.4 \times 0.25 = 0.10 \neq 0.15 = P(M \cap D),$$

so M and D are **not** independent.

The conditional form says the same thing and adds the direction. Restricting to the 80 music students gives $P(D \mid M) = \frac{30}{80} = 0.375$, against $P(D) = 0.25$ for the year group as a whole, so knowing a student takes music raises the chance of drama from a quarter to three eighths.

Independence is an exact statement, so 0.15 and 0.10 being close does not soften the answer. Write both numbers down and then the verdict; a bare "not independent" with only one side computed shows nothing.

CAUTION Mutually exclusive and independent are opposites, not synonyms. If two events with non-zero probability are mutually exclusive, then one happening prevents the other completely, which is the strongest possible dependence. Never assume an "or" situation is also an "independent" one.

YOUR TURN 11.4 Conditional Probability and Independence

12.

(a) Given $P(A \cap B) = 0.2$ and $P(B) = 0.5$, find $P(A \mid B)$.
(b) Given $P(A) = 0.6$ and $P(B \mid A) = 0.3$, find $P(A \cap B)$.
(c) Given $P(A \mid B) = 0.7$ and $P(B) = 0.4$, find $P(A \cap B)$.
(d) Two fair dice are rolled. Given that the first die shows an even number, find the probability that the total is 8.

13.

(a) Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.4$. Find $P(A \cap B)$.
(b) Given $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, determine whether A and B are independent.
(c) Two cards are drawn from a deck of 52 without replacement. Find the probability that both are kings.

14. In a class of 25 students, 15 study Spanish, 10 study art and 6 study both.

(a) Given that a student studies Spanish, find the probability that they also study art.
(b) Given that a student studies art, find the probability that they also study Spanish.
(c) Determine whether studying Spanish and studying art are independent.
(d) A card is drawn from a deck of 52. Given that it is a face card, find the probability that it is a king.

11.5 Tree Diagrams

When events happen in sequence, a **tree diagram** organises every path. Each branch carries the probability of that step, and two rules then do the whole calculation: **multiply along the branches** of a single path (the multiplication rule), and **add across paths** that satisfy the condition (the paths are mutually exclusive).

When each draw is made **with replacement**, the object goes back before the next draw, every stage repeats the same probabilities, and the stages are independent. Care is needed with sequences **without replacement**, where the first outcome changes the probabilities on the second branches.

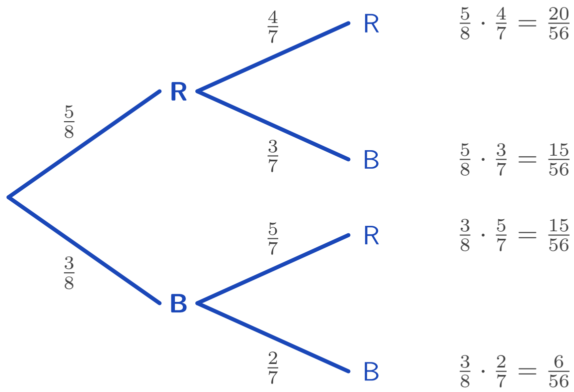


Figure 11.4 A tree diagram for drawing two marbles from 5 red and 3 blue without replacement. The probabilities on the second stage differ from the first, because one marble has already been removed. Multiply along a path; add across paths.

EXAMPLE 11.17

A bag holds 5 red and 3 blue marbles. Two are drawn without replacement. Find the probability that (a) both are red, (b) one of each colour is drawn, (c) at least one is blue.
After drawing one red, only 4 reds remain out of 7 marbles, which is why the second branch changes.
(a) Multiply along the top path:

$$P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

(b) "One of each" covers two paths, red-then-blue and blue-then-red; add them:

$$P(\text{one of each}) = \frac{5}{8} \cdot \frac{3}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{15+15}{56} = \frac{30}{56} = \frac{15}{28}.$$

(c) Use the complement of "at least one blue," which is "both red":

$$P(\text{at least one blue}) = 1 - \frac{5}{14} = \frac{9}{14}.$$

All three rules appear here: multiply along a path, add across alternatives, and take the complement for "at least one."

Putting the marble back changes every second-stage branch. The bag returns to 5 red and 3 blue, so both stages read $\frac{5}{8}$ and $\frac{3}{8}$, and the two draws are independent.

EXAMPLE 11.18

The same bag holds 5 red and 3 blue marbles, but now the first marble is replaced before the second is drawn. Find the probability that both are red, and compare with the answer without replacement.

Every branch keeps its original probability, so

$$P(\text{both red}) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64} \approx 0.391.$$

Without replacement the answer was $\frac{5}{14} \approx 0.357$, which is smaller. Removing a red marble leaves fewer reds for the second draw, so the second red becomes less likely. Replacing it removes that effect.

Replacement also makes the most useful pattern in this chapter available. When the same trial is repeated n times independently, "at least one success" has a complement that is a single product, because "no successes at all" means failure every time.

KEY For n independent trials in which the event has probability p each time,

$$P(\text{at least one}) = 1 - (1-p)^n.$$

EXAMPLE 11.19

A machine part fails on any given day with probability 0.02, independently of other days. Find the probability that it fails at least once during a 30-day month.

Counting the ways to fail once, twice, three times and so on would take all day. The complement is a single product: the part survives all 30 days, each day with probability 0.98.

$$P(\text{at least one failure}) = 1 - (0.98)^{30} = 1 - 0.545 = 0.455.$$

A part that is safe on 98 days out of 100 still has close to an even chance of failing within a month. A small probability repeated often stops being small, which is why maintenance is scheduled by the month and not by the day.

11.5.1 Three stages without replacement

A third draw adds a third stage and nothing else. Each branch still carries the probability of that step given everything before it, and without replacement the denominator falls by one at every draw.

What does change is the counting. An event such as "exactly two of the three" no longer lives on one path or two, but on three, because the odd draw out can come first, second or last. Identify those paths before computing anything.

EXAMPLE 11.20

A bag contains 7 green and 3 yellow counters. Three are drawn at random without replacement.

Find the probability that (a) all three are green, (b) exactly two are green.

(a) One path, with the bag shrinking under it:

$$P(\text{GGG}) = \frac{7}{10} \times \frac{6}{9} \times \frac{5}{8} = \frac{210}{720} = \frac{7}{24}.$$

(b) Exactly two green means one yellow, and the yellow may be drawn first, second or third. The three paths are YGG, GYG and GGY:

$$P(\text{YGG}) = \frac{3}{10} \cdot \frac{7}{9} \cdot \frac{6}{8}, \qquad P(\text{GYG}) = \frac{7}{10} \cdot \frac{3}{9} \cdot \frac{6}{8}, \qquad P(\text{GGY}) = \frac{7}{10} \cdot \frac{6}{9} \cdot \frac{3}{8}.$$

Every one of them is $\frac{126}{720}$, so adding across the three paths gives

$$P(\text{exactly two green}) = 3 \times \frac{126}{720} = \frac{378}{720} = \frac{21}{40} = 0.525.$$

The three products agree because each is built from the same numerators, 7, 6 and 3, over the same denominators $10 \times 9 \times 8$; only the order of the factors differs. So evaluate one path and multiply by the number of orders — and count the orders first, since a missing path is the usual way marks are lost here.

YOUR TURN 11.5 Tree Diagrams

15.

(a) A bag holds 3 white and 2 black balls. Two are drawn with replacement. Find the probability that both are white.

(b) Repeat part (a) for two balls drawn without replacement.

(c) A box holds 6 working and 4 defective bulbs. Two are drawn without replacement. Find the probability that exactly one is defective.

16.

(a) A machine works on any given day with probability 0.9, independently of other days. Find the probability that it works on two consecutive days.

(b) The probability of rain is 0.3. If it rains, the probability of being late is 0.6; if it does not, the probability is 0.1. Find the probability of being late.

(c) A fair die is rolled three times. Find the probability of at least one six.

(d) Using the machine in part (a), find the probability that it fails on at least one of five consecutive days.

17.

(a) A bag holds 4 red and 6 blue counters. Three are drawn without replacement. Find the probability that all three are blue.
(b) For the same three draws, find the probability that exactly two are red.
(c) A component works with probability 0.95, independently of other components. Find the probability that all four components in a circuit work.
(d) Find the probability that at least one of the four components fails.

11.6 Bayes' Theorem HL

We can now return to the screening test from the start of the chapter. About one person in a hundred has the disease. The test correctly identifies 99% of those who have it, and wrongly reports a positive for 5% of those who do not. A positive result has just arrived.

The difficulty is that the test gives you $P(\text{positive} \mid \text{disease})$, but what you actually want is the reverse, $P(\text{disease} \mid \text{positive})$. Conditional probability is not symmetric, so these are different numbers. **Bayes' theorem** is the tool that reverses the condition.

It is built from two facts you already have. The conditional formula gives $P(B \mid A) = \frac{P(A\cap B)}{P(A)}$, and the denominator can be split according to whether B happened or not: $P(A) = P(B)P(A \mid B) + P(B')P(A \mid B')$. Putting these together:

KEY · IB HL

$$P(B \mid A) = \frac{P(B)\,P(A \mid B)}{P(B)\,P(A \mid B) + P(B')\,P(A \mid B')}$$

Read the letters the way the booklet intends them: B is the thing you want to know about, and A is the evidence that has arrived. In the screening test, B is having the disease and A is the positive result.

The numerator is the single path through B; the denominator is every path that leads to A. You are asking what fraction of all the ways to observe A passed through B.

That description is the key to using it. Written out in full the formula is long, but a tree diagram carries the same information in a form that is quicker to set up. Draw the first stage, B and B'. Draw the second stage, A given each of them. Multiply along every path. Then divide the path that passes through B by the total of all the paths that end in A.

TIP Do not memorise the Bayes formula as a string of symbols. Draw the tree, multiply along the paths, and divide the path you want by the total of the paths that match the evidence. It gives the same answer, it is far harder to get wrong under exam pressure, and it earns the same marks.

EXAMPLE 11.21

A disease affects 1% of a population. A test is positive for 99% of those who have it, and (falsely) positive for 5% of those who do not. A person tests positive. Find the probability they actually have the disease.

Let D be "has the disease" and $+$ be "tests positive." We know $P(D) = 0.01$, $P(+ \mid D) = 0.99$, and $P(+ \mid D') = 0.05$. Draw the tree first, then multiply along every path.

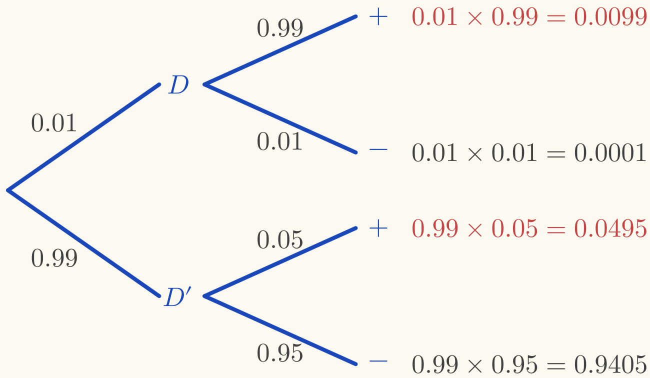


Figure 11.5 The screening test as a tree. Two paths end in a positive result, shown in scarlet. Bayes' theorem divides the scarlet path through D by the total of both scarlet paths.

Only the two scarlet paths match the evidence, so they are the only ones that count. The path through D divided by the total of both gives

$$P(D \mid +) = \frac{(0.01)(0.99)}{(0.01)(0.99) + (0.99)(0.05)} = \frac{0.0099}{0.0099 + 0.0495} = \frac{0.0099}{0.0594} \approx 0.167.$$

That is the formula and the tree saying the same thing: the numerator is one path, the denominator is every path that ends in a positive.

Only about 17%. The reason is the base rate. The disease is so rare that the few true positives are greatly outnumbered by the false positives, which come from the very large healthy majority. The test is good; the disease is simply very rare.

EXAMPLE 11.22

A city runs two taxi companies. 85% of its taxis are Green and 15% are Blue. A taxi is involved in a night-time accident and a witness says it was Blue. Tested under the same conditions, the witness identifies each colour correctly 80% of the time. Find the probability that the taxi really was Blue.

The evidence is the witness's statement, not the colour of the taxi, so put the colour on the first stage and the statement on the second.

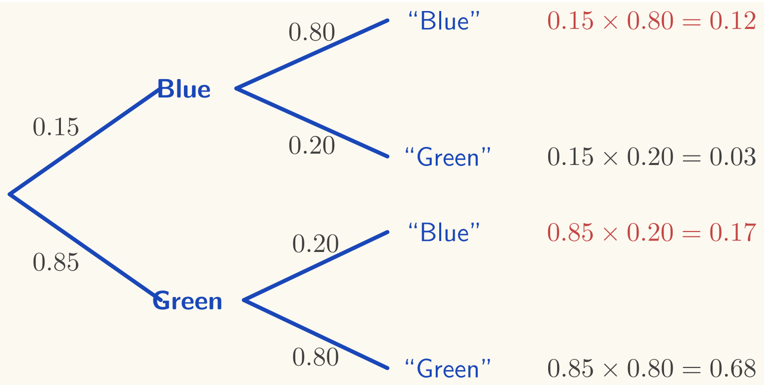


Figure 11.6 The witness problem. The two scarlet paths are the ones on which the witness says "Blue". Note that the larger of the two comes from a Green taxi being misreported.

The witness said "Blue", so only the two scarlet paths remain:

$$P(\text{Blue} \mid \text{“Blue”}) = \frac{0.12}{0.12 + 0.17} = \frac{0.12}{0.29} \approx 0.414.$$

A witness who is right 80% of the time reports "Blue", and the taxi is still more likely to have been Green. There are so many more Green taxis that the 20% of them misreported as Blue (0.17) outnumbers the 80% of Blue taxis reported correctly (0.12).

This is the same base-rate effect as the screening test, in a setting with no medicine in it. Most people asked this question answer 80%, because the witness's reliability is vivid and the proportion of Blue taxis is not. The tree keeps both in view.

11.6.1 Three alternatives

When the first stage has three possibilities instead of two, the denominator simply sums over all three paths. The structure is identical, and the booklet prints this version too.

KEY · IB HL

$$P(B_i \mid A) = \frac{P(B_i)\,P(A \mid B_i)}{P(B_1)\,P(A \mid B_1) + P(B_2)\,P(A \mid B_2) + P(B_3)\,P(A \mid B_3)}$$

Here B_1, B_2 and B_3 are the three possibilities at the first stage, and B_i is whichever one you are asking about. Nothing has changed except the number of terms underneath.

EXAMPLE 11.23

Three machines produce all the parts in a factory: machine A makes 50%, B makes 30%, C makes 20%. Their defect rates are 1%, 2%, and 3% respectively. A part is found to be defective. Find the probability it came from machine C.

Weight each machine's defect rate by its share of production, then take C's share of the total:

$$\begin{aligned} P(C \mid \text{def}) &= \frac{(0.20)(0.03)}{(0.50)(0.01)+(0.30)(0.02)+(0.20)(0.03)} \\ &= \frac{0.006}{0.005+0.006+0.006} = \frac{0.006}{0.017} \approx 0.353. \end{aligned}$$

Machine C makes only a fifth of the parts but, because it is the least reliable, accounts for over a third of the defects.

YOUR TURN 11.6 Bayes' Theorem HL

18.

(a) Using the situation in question 16 part (b), a person is late. Find the probability that it rained.

(b) A disease affects 2% of a population. A test is positive for 95% of those who have it and for 10% of those who do not. Given a positive result, find the probability of having the disease.

(c) Factory A makes 60% of all items, of which 2% are defective; factory B makes the other 40%, of which 5% are defective. Given that an item is defective, find the probability it came from factory B.

19.

(a) In a mailbox, 30% of messages are spam. The word "free" appears in 90% of spam messages and in 5% of the others. A message contains the word "free". Find the probability that it is spam.

(b) A box contains two coins: one fair, and one with heads on both sides. A coin is chosen at random and tossed once, showing heads. Find the probability that it is the two-headed coin.

(c) Of the students in a year group, 70% revise for a test. Of those who revise, 80% pass; of those who do not, 30% pass. A student passes. Find the probability that the student revised.

20. Three suppliers provide the components used by a factory. Supplier A provides 50%, supplier B 30% and supplier C 20%. Of their components, 1%, 3% and 4% respectively are defective.

(a) Find the probability that a component chosen at random is defective.

(b) Given that a component is defective, find the probability that it came from supplier B.

(c) Given that a component is defective, find the probability that it came from supplier A.

(d) Supplier A provides the most components but accounts for the fewest defective ones. Explain why.

Chapter 11 · Probability

Reflections

TOK Bayes' theorem describes how a rational mind should update a belief when new evidence arrives: start from a *prior*, the probability believed before the evidence, weigh the evidence, and arrive at a *posterior*, the probability believed afterwards. Some philosophers argue this is not just a formula but a model of knowledge itself. If that is right, then the disease-test paradox is not a quirk of medicine. It is a warning that evidence means little until you account for how rare the thing you are testing for actually is.

TOK We say a fair die has probability $\frac{1}{6}$ because of its symmetry, and we say a drawing pin lands point-up with probability 0.39 because that is what 200 drops showed. The first number was never measured; the second was never proved.

Are these two the same kind of claim? And when a measured value disagrees with a calculated one, which of the two should be revised?

TOK Probability was built to answer questions about gambling, and casinos use it now to guarantee themselves a profit. The mathematics itself is neutral, but the uses of it are not.

Should a mathematician who works out the odds be held responsible for how those odds are used?

IM Probability began at a gambling table. In 1654 the French writer Antoine Gombaud, the Chevalier de Méré, asked Blaise Pascal why a betting strategy that seemed sound was losing him money. Pascal's correspondence with Pierre de Fermat over that single question laid the foundations of probability theory. A subject now central to physics, finance, and medicine grew out of a dispute about dice.

RW The base-rate paradox has real costs. Courtrooms have convicted people on "one-in-a-million" DNA matches without asking how many people in the population would also match by chance. Mass screening programmes for rare cancers can produce more false alarms than true detections. Understanding $P(D \mid +)$ versus $P(+ \mid D)$ is not only an examination topic. It changes verdicts and medical decisions.

End-of-Chapter Problems — Chapter 11: Probability

1. In a class of 50 students, 28 study Mathematics (M), 20 study Art (A), and 10 study neither.

(a) Find the number of students who study both Mathematics and Art. [3]
(b) Find the probability that a randomly chosen student studies only Art. [2]
(c) Find $P(A \mid M)$. [2]

2. The two-way table shows whether 100 people exercise regularly.

	Exercises	Does not	Total
Male	30	20	50
Female	24	26	50
Total	54	46	100

(a) Find the probability that a person is male and exercises. [2]
(b) Find the probability that a person exercises, given that they are female. [2]
(c) Determine whether the events "male" and "exercises" are independent. [3]

3. A bag contains 7 red and 3 green sweets. Two are drawn without replacement.

(a) Draw a tree diagram for the two draws. [3]
(b) Find the probability that both sweets are red. [2]
(c) Find the probability that both sweets are the same colour. [3]
(d) Find the probability that at least one sweet is green. [2]

4. A condition affects 5% of a population. A screening test is positive for 90% of those who have it, and for 15% of those who do not.

(a) Find the probability that a randomly chosen person tests positive. [3]
(b) Given a positive result, find the probability the person has the condition. [3]
(c) Comment on what your answer says about screening for rare conditions. [2]

5. Events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.3$.

(a) Find $P(A \cup B)$ if A and B are independent. [3]
(b) Find $P(A \cup B)$ if A and B are mutually exclusive. [2]
(c) Explain why A and B cannot be both independent and mutually exclusive here. [2]

6. Two fair dice are rolled and the scores noted.

(a) Find the probability that the sum is 7. [2]
(b) Find the probability that at least one die shows a 6. [2]
(c) Given that at least one die shows a 4, find the probability that the sum is 7. [3]

7. Three suppliers provide components: X supplies 25%, Y supplies 45%, and Z supplies 30%. Their defect rates are 4%, 2%, and 5% respectively.

(a) Find the probability that a randomly chosen component is defective. [3]
(b) Given that a component is defective, find the probability it came from Z. [3]

8. Events A and B are independent. Given $P(A) = 0.4$ and $P(A \cup B) = 0.7$, find $P(B)$. [4]

9. A basketball player makes each free throw independently with probability 0.8. She takes 4 free throws.

(a) Find the probability she makes all four. [2]
(b) Find the probability she misses at least one. [2]

10. At a school, 60% of students study French and 25% study both French and Spanish. Find the probability that a student studies Spanish, given that they study French. [3]

11. In a class of 40 students, the numbers studying two activities A and B are: only A is $2x$, both is x, only B is $x + 4$, and neither is 8.

(a) Find the value of x. [3]
(b) Find $P(A \cap B)$. [1]
(c) Find $P(A \cup B)$. [2]

12. A biased four-sided spinner numbered 1 to 4 was spun 250 times. The results were:

Number	1	2	3	4
Frequency	95	60	55	40

(a) Estimate the probability that the spinner lands on 1. [2]
(b) Estimate the probability that the spinner lands on an odd number. [2]
(c) The spinner is spun a further 60 times. Find the expected number of times it lands on 4. [2]

13. Events A and B satisfy $P(A) = 0.45$, $P(B) = 0.4$, and $P((A \cup B)') = 0.25$.

(a) Find $P(A \cap B)$. [2]
(b) Find the probability that A occurs but B does not. [1]

(c) Find $P(A \mid B)$. [2]

(d) Determine whether A and B are independent. [2]

14. A fair four-sided die (numbered 1–4) and a fair six-sided die (numbered 1–6) are rolled together, and the product of the two scores is recorded.

(a) Represent the outcomes in a sample space diagram. [2]

(b) Find the probability that the product is 12. [2]

(c) Find the probability that the product is odd. [2]

(d) Given that the product is even, find the probability that the four-sided die shows a 4. [3]

15. A football team's key striker is fit for a match with probability 0.85. The team wins with probability 0.7 if she is fit, and with probability 0.4 if she is not.

(a) Draw a tree diagram to represent this information. [2]

(b) Find the probability that the team wins the match. [2]

(c) Given that the team won, find the probability that the striker was fit. [3]

16. A box contains 4 red and 2 green counters. Three counters are drawn at random without replacement.

(a) Find the probability that all three counters are red. [2]

(b) Find the probability that exactly one counter is green. [3]

(c) Find the probability that at least one counter is green. [2]

17. Events A and B satisfy $P(A) = 0.6$, $P(A \mid B) = 0.6$, and $P(A \cup B) = 0.84$.

(a) Explain why A and B are independent. [1]

(b) Find $P(B)$. [3]

(c) Write down $P(B \mid A')$, justifying your answer. [2]

18. The 80 students of a year group are classified by gender and handedness: 10 of the 40 boys and 6 of the 40 girls are left-handed.

(a) Display this information in a two-way table, including totals. [2]

(b) Find the probability that a randomly chosen student is a right-handed girl. [1]

(c) Given that a student is left-handed, find the probability that the student is a boy. [2]

(d) Determine, using conditional probability, whether the events "boy" and "left-handed" are independent. [3]

19. An airline routes 50% of its flights through airport A, 30% through airport B, and 20% through airport C. The probabilities that a flight is delayed are 0.06, 0.03, and 0.10 respectively.

(a) Find the probability that a randomly chosen flight is delayed. [3]

(b) Given that a flight was delayed, find the probability that it was routed through airport A. [3]

20. Events A, B, and C are mutually exclusive and exhaustive, with $P(A) = 2k$, $P(B) = 3k$, and $P(C) = k + 0.1$.

(a) Find the value of k. [2]

(b) Find $P(A \cup B)$. [2]

(c) Find $P(A \mid B')$. [3]

21. GDC An archer hits the target with probability 0.9 on each shot, independently of all other shots.

(a) Find the probability that her first miss is on her third shot. [2]

(b) Find the probability that she misses at least once in five shots. [2]

(c) Find the least number of shots for which the probability of at least one miss exceeds 0.95. [3]

22. A family has two children. Assume each child is equally likely to be a boy or a girl, independently.

(a) List the sample space. [1]

(b) Given that at least one of the children is a boy, find the probability that both are boys. [3]

(c) Given instead that the *elder* child is a boy, find the probability that both are boys. [2]

(d) Explain why the answers to (b) and (c) differ. [1]

23. In a survey of 100 people, 50 like tea, 60 like coffee, and 35 like juice. Also, 30 like tea and coffee, 20 like tea and juice, 25 like coffee and juice, and 10 like all three.

(a) Find the number who like at least one of the three drinks. [3]

(b) Find the number who like none. [1]

(c) Find the number who like exactly one drink. [3]

24. A disease affects 1% of a population. A test is positive for 99% of those with the disease and for 5% of those without. A person tests positive *twice*, the two tests being independent given disease status.

(a) Find the probability the person has the disease after one positive test. [3]

(b) Find the probability the person has the disease after two positive tests. [4]

(c) Comment on the effect of the second test. [1]

25. Events A and B are independent. Prove that A and B' are also independent, that is, show $P(A \cap B') = P(A)\,P(B')$. [5]

26. From a group of 5 men and 4 women, three people are chosen at random. Find the probability that exactly two are women. [5]

27. Of the 200 students at a school, 130 cycle to school (C), 90 travel by train (T), and 40 do both. The remaining students walk.

(a) Draw a Venn diagram showing the number of students in each of the four regions. [3]
(b) Find the probability that a student chosen at random walks to school. [2]
(c) Find $P(C \mid T)$. [3]
(d) Determine whether C and T are independent, justifying your answer. [3]
(e) Two students are chosen at random, without replacement. Find the probability that both of them walk. [3]
(f) State whether your answer to part (e) would be larger or smaller if the two students were chosen with replacement, and explain why. [2]

28. A machine produces components, of which 6% are defective. Each component is inspected. The inspection rejects a defective component with probability 0.95, and wrongly rejects a component that is not defective with probability 0.02.

(a) Draw a tree diagram to represent this information. [3]
(b) Find the probability that a randomly chosen component is rejected. [3]
(c) Find the probability that a rejected component was in fact not defective. [3]
(d) Find the probability that a component which passes inspection is defective. [3]
(e) In a batch of 5000 components, find the expected number that are defective but pass inspection. [2]
(f) Using your answers to parts (d) and (e), comment on how well the inspection protects a customer. [2]

29. A box contains 10 batteries, of which 3 are flat. Batteries are taken from the box one at a time, without replacement, and tested until a working one is found. Let N be the number of batteries tested.

(a) Find $P(N = 1)$. [2]
(b) Find $P(N = 2)$. [3]
(c) Find $P(N \leq 3)$. [4]
(d) Explain why N can never be greater than 4. [2]
(e) Find $P(N = 3)$ and $P(N = 4)$, and verify that the four probabilities sum to 1. [4]

30. A lottery draws 5 numbers at random from 40. A ticket also carries 5 numbers.

(a) Find the number of possible draws. [2]
(b) Find the probability that a ticket matches all 5 numbers. [2]

(c) Find the probability that a ticket matches exactly 3 of the 5. [4]

(d) The organisers propose drawing 5 from 45 instead. Without computing the new probability in full, explain what happens to a ticket's chance of matching all five, and why lotteries are designed this way. [3]

31. A committee of 4 is chosen at random from 5 teachers and 4 students.

(a) Find the number of possible committees. [2]

(b) Find the probability that the committee has exactly 2 teachers and 2 students. [3]

(c) Find the probability that the committee contains at least one teacher and at least one student. [4]

(d) Explain why part (c) is quicker through its complement than by adding the cases directly. [2]

32. EXACT The letters of the word PARALLEL are arranged in a row, all arrangements being equally likely.

(a) Explain why the number of distinct arrangements is $\dfrac{8!}{2!\,3!}$, and evaluate it. [3]

(b) Find the probability that an arrangement begins with the letter P. [4]

(c) Find the probability that the three Ls are together. [4]

Challenge Questions

33. GDC **The birthday problem.** In a group of n people, assume every birthday is equally likely among 365 days and ignore leap years.

(a) For $n = 2$, show that the probability the two birthdays differ is $\dfrac{364}{365}$. [2]

(b) Explain why, for $n = 3$, the probability that all three differ is $\dfrac{364}{365} \times \dfrac{363}{365}$. [3]

(c) Write down an expression for the probability that all n birthdays differ. [3]

(d) Hence find, to three decimal places, the probability that at least two people share a birthday when $n = 10$, $n = 23$ and $n = 50$. [4]

(e) Find the smallest n for which the probability of a shared birthday exceeds $\frac{1}{2}$. [3]

(f) Most people guess that n must be close to 180, because they think of themselves against the other $n - 1$ people. Explain why the right count is the number of *pairs*, ${}^{n}C_2$, and evaluate it for $n = 23$. [3]

(g) Each pair matches with probability $\frac{1}{365}$. Use your answer to part (f) to find the expected number of matching pairs in a group of 23, and comment on how close it is to 1. [3]

(h) A school issues every student a random four-digit code, so there are $10\,000$ codes. Using the same method, find the smallest number of students for which two students sharing a code is more likely than not. Comment on what this means for anyone designing an identifier. [4]

34. Three doors. A prize is behind one of three closed doors. You choose a door. The host knows where the prize is, and opens one of the other two doors, always an empty one. You may keep your door or switch to the last closed door.

(a) Write down the probability that your first choice is correct. [1]

(b) Find the probability of winning if you always switch. Treat the case where your first choice is correct separately from the case where it is not. [5]

(c) State the probability of winning if you always stay, and say which strategy is better. [2]

(d) Now do it with Bayes' theorem. You chose door 1 and the host opened door 3. Write down $P(\text{host opens } 3 \mid \text{prize behind } 1)$, $P(\text{host opens } 3 \mid \text{prize behind } 2)$ and $P(\text{host opens } 3 \mid \text{prize behind } 3)$, and hence find $P(\text{prize behind } 2 \mid \text{host opens } 3)$. [5]

(e) A second host knows nothing, opens one of the two doors you did not choose at random, and happens to reveal an empty one. Repeat the calculation of part (d) for this host, and show that switching now wins with probability $\frac{1}{2}$. [5]

(f) The two hosts performed the same visible action: they opened door 3 and it was empty. Explain what makes the answers differ, and what it is that the first host's action actually tells you. [3]

(g) The game is now played with n doors and one prize, and the knowing host opens every door except your choice and one other. Show that switching wins with probability $\dfrac{n-1}{n}$. [4]

STATISTICS AND PROBABILITY

Chapter 12

Probability Distributions

12

SYLLABUS 4.7 4.8 4.9 4.12 4.14

If you measure the same kind of thing many times, you rarely get the same answer twice. Masses differ from packet to packet, and a repeated experiment gives a different count each time. A single number cannot describe a quantity that behaves this way. What describes it is the list of values the quantity can take, together with how likely each value is. That list is called a **distribution**, and this chapter covers the few that fit a great many situations.

A factory fills bags of rice, each meant to hold 500 grams, and weighs 5000 of them. No two masses are exactly equal. Group the readings, draw them as a graph, and a shape appears: most bags near 500 g, with fewer and fewer on either side, falling away evenly into one smooth hump. The shape is a bell.

The same curve appears where there is no rice at all. Astronomers found it in the *errors* of their telescope readings, and biologists in the heights of people. The reason is always the same: a bag's mass is decided not by one thing but by many, such as the size of the grains, a slight vibration in the machine, the exact moment the valve closes. Each effect is small, and each is as likely to raise the mass as to lower it. When many small effects combine, the result is a bell.

A distribution turns uncertainty into something you can calculate with. From one you can predict an average result, measure how far individual results fall from that average, and put a number on the risk of an extreme value. That is why these curves are used in medicine, manufacturing and finance, and also why care is required: every answer depends on the distribution being the right description, and a wrong choice gives answers that are confident and wrong.

We begin with a single uncertain number taking a few values, move to the distribution used for coin flips and quality control, and end with the bell curve, which appears as their limit.

12.1 Discrete Random Variables

A **random variable** is a number whose value depends on chance. The number of heads in three coin tosses, the score on a die, the number of defective items in a box: each is a rule that assigns a number to a random outcome. We write random variables with capital letters, X, and particular values with lower case, x.

A random variable is **discrete** when it can only take separate, countable values. To describe it fully, you list every value it can take alongside the probability of each. This list is its **probability distribution**.

Here are five discrete random variables, each with the values it can take.

- X = the number of heads in three coin tosses. Values $0, 1, 2, 3$.
- X = the score on one roll of a fair die. Values $1, 2, 3, 4, 5, 6$.
- X = the number of defective items in a box of twenty. Values $0, 1, \ldots, 20$.
- X = the number of students absent from a class of thirty. Values $0, 1, \ldots, 30$.
- X = the number of tosses of a coin until the first head appears. Values $1, 2, 3, \ldots$, with no largest value.

The last one deserves attention. A discrete variable does not have to take a finite number of values. It only has to take separate values that can be listed in order, and that list may continue for ever. What rules a variable out is taking *every* value across a range, as a height or a waiting time does. Those are continuous variables, and §12.4 treats them.

DEF A **discrete random variable** X takes countable values $x_1, x_2, \ldots$ Its **probability distribution** gives $P(X = x_i)$ for each value, subject to

$$P(X = x_i) \geq 0 \qquad \text{and} \qquad \sum_i P(X = x_i) = 1.$$

The second condition is the one you use most often. Since some value must occur, all the probabilities add to exactly 1. This single fact lets you find a missing probability.

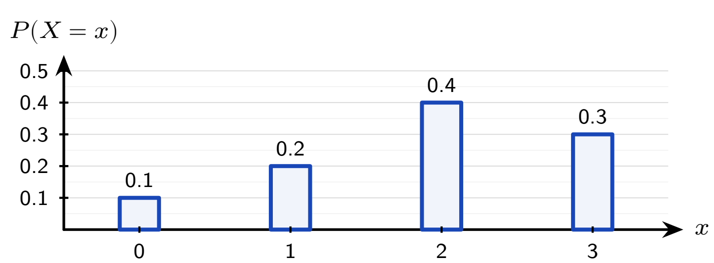


Figure 12.1 The probability distribution of a discrete random variable. Each bar is the probability of one value, and the four bars must total exactly 1.

EXAMPLE 12.1

A discrete random variable X has the distribution below. Find k, then $P(X \geq 3)$.

x	1	2	3	4
$P(X = x)$	0.1	0.3	k	0.2

The probabilities must sum to 1:

$$0.1 + 0.3 + k + 0.2 = 1 \implies k = 0.4.$$

Then $P(X \geq 3) = P(X = 3) + P(X = 4) = 0.4 + 0.2 = 0.6$.

EXAMPLE 12.2

A random variable X takes values $1, 2, 3, 4$ with $P(X = x) = kx$. Find k.
Sum the probabilities and set equal to 1:

$$k(1) + k(2) + k(3) + k(4) = 10k = 1 \implies k = \frac{1}{10}.$$

When the distribution is given by a formula, the total-probability condition still determines the constant.

YOUR TURN 12.1 Discrete Random Variables

1.

(a) For $x = 1, 2, 3$ the probabilities are 0.2, 0.5 and k. Find k.

(b) A variable X has $P(X = x) = 0.15,\ 0.25,\ 0.35,\ k$ for $x = 0, 1, 2, 3$. Find k and $P(X \leq 1)$.

(c) A variable takes values $x = 0, 1, 2, 3$ with $P(X = x) = k(x + 1)$. Find k.

(d) A variable takes values $x = 1, 2, 3, 4$ with $P(X = x) = \dfrac{x^2}{30}$. Find $P(X = 3)$.

(e) For the distribution $P(X = x) = 0.1,\ 0.4,\ 0.3,\ 0.2$ at $x = 0, 1, 2, 3$, find $P(X > 1)$ and $P(1 \leq X \leq 2)$.

2.

(a) Show that $P(X = x) = \frac{1}{18}(4 + x)$ for $x = 1, 2, 3$ defines a probability distribution, and find $P(X \geq 2)$.

(b) A variable takes values $x = 1, 2, 3, 4$ with probabilities 0.1, a, 0.3, b. Given that $P(X \leq 2) = 0.4$, find a and b.

(c) Hence find $P(X > 2)$ for the distribution in part (b).
(d) A variable takes values $x = 1, 2, 3$ with $P(X = x) = k \cdot 2^x$. Find k.

12.2 Expected Value and Variance

If you played a game many times, what would you win on average per play? That long-run average is the **expected value** $E(X)$. It is not the most likely outcome, and it need not be a value that X can take at all. It is the balance point of the distribution, found by weighting each value by its probability.

KEY · IB

$$E(X) = \mu = \sum_i x_i \, P(X = x_i)$$

This is exactly the mean from Ch. 10, with probabilities playing the role of relative frequencies. Over many trials, the relative frequency of each value approaches its probability, so the average approaches $E(X)$.

EXAMPLE 12.3

A game costs \$2 to play. You roll a fair die and win, in dollars, the number shown. Is the game worth playing?
Let X be the amount won. Each face is equally likely:

$$E(X) = \tfrac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \tfrac{21}{6} = 3.5.$$

You expect to win \$3.50 per play but pay \$2, so the expected net gain is \$1.50. Over many games you gain money on average.

That last calculation names an idea the syllabus uses directly. A game is **fair** when the expected net gain is zero, so that neither side profits in the long run. Since the net gain is the winnings minus the stake, a fair stake is exactly the expected winnings.

KEY A game is **fair** when $E(\text{gain}) = 0$, that is, when the stake equals $E(\text{winnings})$.

EXAMPLE 12.4

A fair die is rolled. You win \$10 for a six, \$4 for a four or a five, and nothing otherwise. Find the stake that makes the game fair, and state whether a stake of \$4 favours the player or the organiser.

Let W be the amount won. The three cases have probabilities $\frac{1}{6}$, $\frac{2}{6}$ and $\frac{3}{6}$, so

$$E(W) = 10\left(\tfrac{1}{6}\right) + 4\left(\tfrac{2}{6}\right) + 0\left(\tfrac{3}{6}\right) = \tfrac{10+8}{6} = 3.$$

A stake of \$3 makes the game fair: the expected gain is $3 - 3 = 0$. At a stake of \$4 the expected gain is $3 - 4 = -1$, a loss of one dollar per play on average, so the game favours the organiser. Note that no single play ever produces a loss of exactly \$1; the figure is the average over many plays.

Both calculations so far ran forwards, from a complete distribution to its mean. Questions also run the other way. If a table hides two probabilities and the mean is given, you hold two facts about those unknowns: the probabilities total 1, and the weighted sum of the values is the stated mean. Two facts, two unknowns, so write both as equations and solve them together.

EXAMPLE 12.5

A discrete random variable X has the distribution below, and $E(X) = 1.8$. Find a and b.

x	**0**	**1**	**2**	**3**
$P(X = x)$	0.15	a	b	0.25

The total-probability condition gives one equation:

$$0.15 + a + b + 0.25 = 1 \implies a + b = 0.6.$$

The stated mean gives a second, by weighting each value by its probability:

$$0(0.15) + 1(a) + 2(b) + 3(0.25) = 1.8 \implies a + 2b = 1.05.$$

Subtracting the first equation from the second removes a and leaves $b = 0.45$, so $a = 0.6 - 0.45 = 0.15$.

Check both conditions on the finished table: $0.15+0.15+0.45+0.25 = 1$, and $0(0.15)+1(0.15)+2(0.45) + 3(0.25) = 1.8$, so the pair satisfies the two facts it was built from.

12.2.1 Variance and standard deviation HL

Expected value locates the centre, and **variance** measures the spread, exactly as in descriptive statistics. The variance is the expected squared distance from the mean. A short calculation turns that definition into a form that is easier to use.

KEY · IB HL

$$\text{Var}(X) = E((X-\mu)^2) = E(X^2) - \mu^2, \qquad \text{where } E(X^2) = \sum_i x_i^2\, P(X = x_i)$$

The second form, $E(X^2)-\mu^2$, is usually the quicker one. Find the mean of the squares, then subtract the square of the mean. The standard deviation is $\sigma = \sqrt{\text{Var}(X)}$.

EXAMPLE 12.6

Find the variance and standard deviation of X:

x	**0**	**1**	**2**	**3**
$P(X = x)$	0.1	0.2	0.4	0.3

First the mean:

$$\mu = 0(0.1) + 1(0.2) + 2(0.4) + 3(0.3) = 1.9.$$

Then $E(X^2)$:

$$E(X^2) = 0^2(0.1) + 1^2(0.2) + 2^2(0.4) + 3^2(0.3) = 0.2 + 1.6 + 2.7 = 4.5.$$

So $\text{Var}(X) = 4.5 - 1.9^2 = 4.5 - 3.61 = 0.89$, and $\sigma = \sqrt{0.89} \approx 0.943$.

12.2.2 Linear transformations HL

If a random variable is scaled and shifted, its mean and variance change in a predictable way, exactly as those of a data set do. For constants a and b:

KEY · IB HL

$$E(aX+b) = aE(X) + b, \qquad \text{Var}(aX+b) = a^2\,\text{Var}(X)$$

The shift b moves the mean but leaves the spread unchanged. The scale factor a stretches the mean by a and the variance by a^2, because variance is built from squared distances.

EXAMPLE 12.7

For the variable above, $E(X) = 1.9$ and $\text{Var}(X) = 0.89$. Let $Y = 2X + 3$. Find $E(Y)$ and $\text{Var}(Y)$.

$$E(Y) = 2(1.9) + 3 = 6.8, \qquad \text{Var}(Y) = 2^2(0.89) = 3.56.$$

The $+3$ leaves the variance unchanged. Only the factor of 2 affects it, and it does so squared.

YOUR TURN 12.2 Expected Value and Variance

3.

(a) Find $E(X)$ for $x = 1, 2, 3$ with $P(X = x) = 0.2,\ 0.3,\ 0.5.$

(b) Find $E(X)$ for $x = 0, 1, 2, 3$ with $P(X = x) = 0.1,\ 0.4,\ 0.3,\ 0.2.$

(c) A fair coin pays \$3 for heads and loses \$2 for tails. Find the expected gain per toss.

(d) A game costs \$1. You win \$5 with probability $\frac{1}{8}$, and otherwise lose your stake. Find the expected gain, and state whether the game is fair.

4.

(a) For the distribution in question 3 part (b), find $E(X^2)$ and Var(X).

(b) Given $E(X) = 4$ and Var$(X) = 2$, find $E(3X - 1)$ and Var$(3X - 1)$.

5.

(a) A wheel is spun for a prize. It pays \$8 with probability 0.1, \$2 with probability 0.3, and nothing otherwise. Find the expected amount won, and the stake that would make the game fair.

(b) The stall charges \$2 to spin. Find the expected gain per spin for a player, and the expected loss over 50 spins.

(c) A variable has $E(X) = 3$ and Var$(X) = 5$. Find $E(2X + 1)$, Var$(2X + 1)$ and $E(X^2)$.

12.3 The Binomial Distribution

Many situations consist of the same experiment repeated: ten coin flips, twenty items checked for defects, twelve multiple-choice questions answered by guessing. When each repetition is independent and has only two possible outcomes, the number of "successes" follows one particular distribution, the **binomial**.

Four conditions define it. Check each one before reaching for the distribution.

- **Fixed number of trials.** The value of n is decided in advance, not by the results.
- **Two outcomes only.** Every trial ends as a success or a failure, with nothing in between.

- **Constant probability.** Every trial has the same probability of success p.
- **Independence.** The result of one trial does not change the probabilities for any other.

When all four hold, write $X \sim B(n, p)$.

DEF If X counts the successes in n independent trials each with success probability p, then $X \sim B(n, p)$, and

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}, \qquad r = 0, 1, 2, \ldots, n.$$

Every factor in that formula has a meaning. The term p^r accounts for the r successes, $(1 - p)^{n-r}$ for the remaining trials failing, and ${}^nC_r$ for the number of different orders in which those successes can occur. In practice you evaluate the whole expression on a GDC: use the binomial pdf for $P(X = r)$ and the binomial cdf for $P(X \leq r)$.

EXAMPLE 12.8

A student guesses on a 10-question multiple-choice test, each question having 4 options. Find the probability of (a) exactly 3 correct, (b) at most 2 correct.

Here $X \sim B(10, 0.25)$, since each guess is correct with probability $\frac{1}{4}$.

(a) Using the binomial pdf:

$$P(X = 3) = {}^{10}C_3 (0.25)^3 (0.75)^7 \approx 0.250.$$

(b) Using the binomial cdf:

$$P(X \leq 2) = P(0) + P(1) + P(2) \approx 0.526.$$

Despite 10 questions, scoring 3 or fewer is more likely than not. Guessing rarely produces a good score.

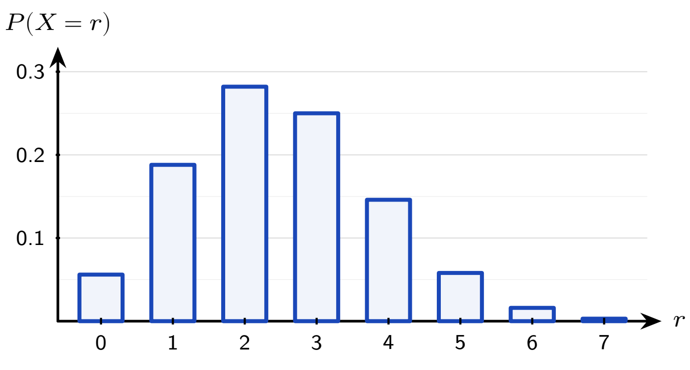


Figure 12.2 The distribution $B(10,\ 0.25)$, the guessing student of the example above. The bars are tallest at $r = 2$ and $r = 3$, and the long tail to the right shows why a high score by guessing is so unlikely.

12.3.1 Mean and variance of the binomial

The mean could be found by summing $r\,P(X = r)$, but a formula gives it directly. For n trials each succeeding with probability p, the results are clean.

KEY · IB For $X \sim B(n, p)$:

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p)$$

The mean is the result you would expect: 10 trials at probability 0.25 give 2.5 successes on average. The variance is largest when $p = 0.5$, which is the value at which a single trial is least predictable.

EXAMPLE 12.9

A factory's items are defective with probability 0.05, independently. A sample of 20 is taken. Find the expected number of defectives and the probability that at least one is defective.

Let $X \sim B(20, 0.05)$. The expected number is

$$E(X) = 20 \times 0.05 = 1.$$

For "at least one," use the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.95)^{20} \approx 0.642.$$

Even with a low defect rate, a sample of 20 is more likely than not to contain a defective item. The probability of "at least one" rises quickly as the sample size increases.

CAUTION Check independence and a fixed n before using the binomial. Drawing items *without replacement* breaks independence, since each draw changes the next probability, so that count is not binomial.

12.3.2 Finding an unknown n or p

Every binomial calculation so far began with n and p and produced a probability. A question can supply the probability instead and ask for one of the parameters. Both reversals start in the same place, at

$$P(X = 0) = {}^{n}C_{0}\, p^{0}(1-p)^{n} = (1-p)^{n},$$

the one term of the distribution that carries no combinatorial factor. It is a single power, so it can be unpicked by logarithms when n is the unknown and by a root when p is.

EXAMPLE 12.10

Each ticket in a charity draw wins a prize with probability 0.1, independently of every other ticket. Find the least number of tickets that must be bought for the probability of winning at least one prize to exceed 0.95.

Let n be the number of tickets, so the number of prizes is $X \sim B(n, 0.1)$ and the complement gives

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.9)^n.$$

The requirement $1 - (0.9)^n > 0.95$ rearranges to $(0.9)^n < 0.05$. Take logarithms of both sides:

$$n \ln 0.9 < \ln 0.05 \implies n > \frac{\ln 0.05}{\ln 0.9} \approx 28.4,$$

where the inequality reverses because $\ln 0.9$ is negative. Tickets are counted in whole numbers, so $n = 29$.

Confirm at both ends: $(0.9)^{29} \approx 0.047$ and $(0.9)^{28} \approx 0.052$, so 29 tickets bring the failure probability below 0.05 and 28 do not.

EXAMPLE 12.11

The number of faulty components in a batch of six is $X \sim B(6, p)$. Given that $P(X = 0) = 0.262$, find p.

With $r = 0$ the binomial formula collapses to a single power:

$$(1 - p)^6 = 0.262.$$

Take the sixth root of both sides:

$$1 - p = 0.262^{1/6} \approx 0.800 \implies p \approx 0.200.$$

The root returns $1 - p$ rather than p, and stopping at 0.800 is the usual slip; substituting back gives $(0.8)^6 \approx 0.262$, which confirms both the value and the final subtraction.

YOUR TURN 12.3 The Binomial Distribution

6. GDC

(a) $X \sim B(8, 0.5)$. Find $P(X = 4)$.
(b) $X \sim B(10, 0.3)$. Find $P(X = 2)$.
(c) $X \sim B(6, 0.4)$. Find $P(X \leq 2)$.
(d) $X \sim B(12, 0.25)$. Find $P(X \geq 1)$.

7.

(a) $X \sim B(20, 0.1)$. Find the mean and the variance.
(b) $X \sim B(15, 0.6)$. Find $E(X)$ and $\text{Var}(X)$.

8. GDC In a factory, 8% of items are defective, independently of one another. A sample of 25 items is taken.

(a) Find the probability that exactly 2 items are defective.
(b) Find the probability that at least 1 item is defective.
(c) Find the expected number of defective items in the sample.
(d) A quiz has 12 questions, each with 5 options, and a student guesses every answer. Find the probability of at least 4 correct answers.
(e) A box holds 10 items, 3 of them defective, and 4 items are drawn without replacement. Explain why the number of defective items drawn is not binomial.

12.4 Continuous Random Variables HL

A discrete variable jumps between separate values. A continuous one, such as a height, a mass or a waiting time, can take any value in a range. You cannot list a probability for each value, because there are infinitely many of them. Something has to replace the list.

The bags of rice show what replaces it. Group the 5000 masses into bars one gram wide. The probability of a bag falling in a given group is then the *area* of that group's bar. Now halve the bar width. There are twice as many bars, each about half as tall, and the total area is unchanged at 1. Halve it again, and again. The tops of the bars settle into a smooth curve, and the area under that curve is still 1.

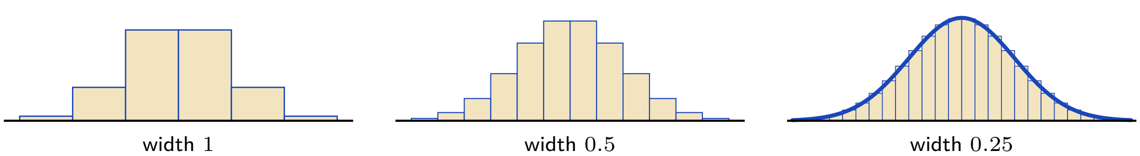


Figure 12.3 The same 5000 masses grouped into bars of decreasing width. Every version encloses total area 1; as the bars narrow, their tops approach a smooth curve. That curve is the density.

So probability for a continuous variable is **area**, and the curve whose area gives it is called a **probability density function**, written $f(x)$. Two conditions make f a valid density: it is never negative, and the total area under it is 1.

DEF HL A **probability density function** $f(x)$ satisfies

$$f(x) \geq 0 \quad \text{for all } x, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1,$$

and probabilities are areas: $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx.$

The symbol $\int$ is new. It is the notation for **the area under a curve**, and $\int_a^b f(x)\,dx$ means the area under $y = f(x)$ between $x = a$ and $x = b$. Ch. 15 defines it properly and shows how to evaluate one; until then, read every integral in this section as an area and nothing more. That reading is enough for everything here, and it is also what the integral really is.

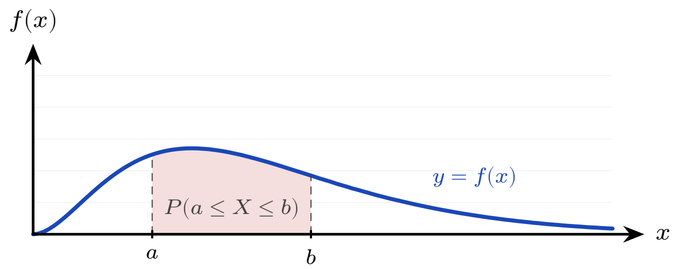


Figure 12.4 For a continuous variable, a probability is an area. The shaded region between a and b is $P(a \leq X \leq b)$, and the whole area under the curve is 1.

One consequence of this looks strange at first.

- A bar of zero width has zero area, so $P(X = c) = 0$ for every single value c.
- This does not mean the value is impossible. It means that no exact value carries a share of the total. A person's height is never exactly 165 cm to unlimited precision, only somewhere in a range near it.
- The endpoints therefore contribute nothing, so $P(a \leq X \leq b)$ and $P(a < X < b)$ are the same number. Here you may be careless about $<$ and $\leq$ in a way you cannot be with a discrete variable.

The expected value and variance carry over from the discrete case, with integrals replacing sums.

KEY · IB HL

$$E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx, \qquad \text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

Compare these with the discrete versions in §12.2. The structure is identical, and only the machinery has changed: a sum over a list of values has become an area over a range of them. As before, the standard deviation is $\sigma = \sqrt{\text{Var}(X)}$, and it is the number to quote when you want a spread in the same units as X.

Three numbers describe where a density sits, and each is read off the curve in its own way.

- The **mode** is the value at which f is highest, that is, the peak of the curve.
- The **median** is the value that cuts the area in half, with 0.5 on each side.
- The **mean** is the balance point, $\int x f(x)\,dx$.

For a symmetric curve all three sit together. For a lopsided one they do not, and the next example finds all three.

EXAMPLE 12.12

A continuous random variable has density $f(x) = kx$ for $0 \le x \le 2$, and $f(x) = 0$ elsewhere. Find k, then $P(X < 1)$, the mean, the variance, and the median.

Total area must be 1:

$$\int_0^2 kx\,dx = k\Big[\tfrac{x^2}{2}\Big]_0^2 = 2k = 1 \implies k = \tfrac{1}{2}.$$

Then

$$P(X<1) = \int_0^1 \tfrac{x}{2}\,dx = \Big[\tfrac{x^2}{4}\Big]_0^1 = \tfrac{1}{4}.$$

The mean:

$$E(X) = \int_0^2 x \cdot \tfrac{x}{2}\,dx = \int_0^2 \tfrac{x^2}{2}\,dx = \Big[\tfrac{x^3}{6}\Big]_0^2 = \tfrac{8}{6} = \tfrac{4}{3}.$$

For the variance, first find $E(X^2)$, then subtract μ^2:

$$E(X^2) = \int_0^2 x^2 \cdot \tfrac{x}{2}\,dx = \int_0^2 \tfrac{x^3}{2}\,dx = \Big[\tfrac{x^4}{8}\Big]_0^2 = 2,$$

$$\mathsf{Var}(X) = 2 - \left(\tfrac{4}{3}\right)^2 = 2 - \tfrac{16}{9} = \tfrac{2}{9}, \qquad \sigma = \sqrt{\tfrac{2}{9}} = \tfrac{\sqrt{2}}{3} \approx 0.471.$$

The **median** m splits the area in half:

$$\int_0^m \tfrac{x}{2}\,dx = \tfrac{1}{2} \implies \tfrac{m^2}{4} = \tfrac{1}{2} \implies m = \sqrt{2}.$$

The **mode** is where f is greatest. Since $f(x) = \frac{x}{2}$ increases across $[0,2]$, the mode is at the right end, $x = 2$.

Collect the three centres: mode 2, median $\sqrt{2} \approx 1.414$, mean $\frac{4}{3} \approx 1.333$. Mean < median < mode: the density rises towards $x = 2$, so its long tail lies to the *left*. That is negative skew, in the language of Ch. 10.

CAUTION The mean, median, and mode of a continuous variable usually differ. The mode maximises $f(x)$; the median halves the area; the mean is the balance point $\int x f(x)\,dx$. Only for a symmetric density do all three coincide.

12.4.1 Piecewise densities

A density need not be a single formula. Any function that is non-negative and encloses total area 1 qualifies, including one built from pieces. The only new requirement is care with the arithmetic: integrals must be split at the joins, and the median may lie in either piece.

EXAMPLE 12.13

A continuous random variable has density

$$f(x) = \begin{cases} \dfrac{2x}{3} & 0 \le x \le 1, \\ \dfrac{3-x}{3} & 1 < x \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

Verify that f is a valid density, find $P(X > 2)$, and find the median.

Both pieces are non-negative, and the total area splits at $x = 1$:

$$\int_0^1 \tfrac{2x}{3}\,dx + \int_1^3 \tfrac{3-x}{3}\,dx = \left[\tfrac{x^2}{3}\right]_0^1 + \tfrac{1}{3}\left[3x - \tfrac{x^2}{2}\right]_1^3 = \tfrac{1}{3} + \tfrac{2}{3} = 1.\checkmark$$

For $P(X > 2)$ only the second piece matters:

$$P(X > 2) = \int_2^3 \tfrac{3-x}{3}\,dx = \tfrac{1}{3}\left[3x - \tfrac{x^2}{2}\right]_2^3 = \tfrac{1}{6}.$$

For the median, first locate it: the area up to $x = 1$ is $\frac{1}{3} < \frac{1}{2}$, so the median lies in the second piece. Collect the remaining area there:

$$\tfrac{1}{3} + \int_1^m \tfrac{3-x}{3}\,dx = \tfrac{1}{2} \implies 3m - \tfrac{m^2}{2} - \tfrac{5}{2} = \tfrac{1}{2} \implies m^2 - 6m + 6 = 0,$$

giving $m = 3 - \sqrt{3} \approx 1.27$ (taking the root inside the interval). The mode is $x = 1$, where the two pieces meet at the peak of the triangle.

12.4.2 Modes by calculus

When a density rises and then falls, its mode lies inside the interval, and you find it by the standard method: differentiate and solve $f'(x) = 0$. A mode can also sit at an endpoint or at a corner, as in the examples above, so always confirm where the maximum actually is before answering.

EXAMPLE 12.14

A continuous random variable has density $f(x) = kx^2(3 - x)$ for $0 \le x \le 3$, and zero elsewhere. Find k, the mode, and $E(X)$.

Total area 1:

$$\int_0^3 k(3x^2 - x^3)\,dx = k\Big[x^3 - \tfrac{x^4}{4}\Big]_0^3 = k\left(27 - \tfrac{81}{4}\right) = \tfrac{27k}{4} = 1 \implies k = \tfrac{4}{27}.$$

For the mode, maximise f:

$$f'(x) = k(6x - 3x^2) = 3kx(2-x) = 0 \implies x = 0 \text{ or } x = 2.$$

Since f rises on $(0,2)$ and falls on $(2,3)$, the mode is $x = 2$. Finally,

$$E(X) = \int_0^3 k(3x^3 - x^4)\,dx = \tfrac{4}{27}\Big[\tfrac{3x^4}{4} - \tfrac{x^5}{5}\Big]_0^3 = \tfrac{4}{27}\cdot\tfrac{243}{20} = \tfrac{9}{5}.$$

Mean 1.8, mode 2: the density is not symmetric, so the balance point sits below the peak.

YOUR TURN 12.4 Continuous Random Variables HL

9.

(a) A density is $f(x) = kx^2$ for $0 \le x \le 3$, and zero elsewhere. Find k.
(b) A density is $f(x) = k$ for $0 \le x \le 4$, and zero elsewhere. Find k and $P(X < 1)$.
(c) A density is $f(x) = \dfrac{x}{8}$ for $0 \le x \le 4$, and zero elsewhere. Find $P(X > 2)$.
(d) For the density in part (c), find $E(X)$.
(e) A density is $f(x) = 3x^2$ for $0 \le x \le 1$, and zero elsewhere. Find the median.

10.

(a) A continuous random variable has density $f(x) = kx(4-x)$ for $0 \le x \le 4$, and zero elsewhere. Find k.
(b) Write down the mode of this density, and justify your answer.
(c) Find $E(X)$.
(d) A second variable has density $f(x) = x$ for $0 \le x \le 1$, $f(x) = \frac{1}{2}$ for $1 < x \le 2$, and zero elsewhere. Verify that the total area is 1, and find $P(X < 1.5)$.
(e) Find the median of the density in part (d).

12.5 The Normal Distribution

If the histogram of bag masses is refined far enough, the bars settle into one particular density. That density is the **normal distribution**, and everything from the last section still applies to it: probability is area, and the total area is 1.

What earns this one curve a section of its own is how often it occurs. Heights, masses, measurement errors and examination marks all follow it closely. The reason was given at the start of the chapter. Whenever a quantity is the sum of many small independent effects, and each effect is as likely to raise the value as to lower it, the normal curve is the result. That situation is so common that the curve appears in almost every area of statistics.

The name is unfortunate, and it misleads students every year. *Normal* here does not mean usual, correct or healthy. It is a nineteenth-century label that survived, and a great many real quantities are not normal at all. Incomes, city populations and waiting times are all strongly skewed. Before using the curve, check that the data actually resemble it.

Every normal curve, whatever its mean and spread, has the same four properties.

- It is symmetric about the mean, so the mean, median and mode are equal and all sit at the peak.
- It has a single maximum, at the centre, and falls away on both sides.
- The tails approach the horizontal axis but never meet it. Every real value therefore has some probability, however small.
- The total area beneath it is 1, as it is for any density.

Two numbers fix a normal distribution completely, and each does a different job.

- The **mean** μ locates the centre. Changing μ slides the whole curve left or right without altering its shape at all.
- The **standard deviation** σ sets the width. Increasing σ stretches the curve sideways, and because the area underneath must stay equal to 1, stretching it sideways also flattens it. A large σ therefore gives a wide, low bell and a small σ a narrow, tall one.

We write $X \sim N(\mu, \sigma^2)$. Note that the second entry is the *variance*, not the standard deviation, so take a square root before using σ. The curve is symmetric about μ, so the mean, median and mode all sit together at the peak.

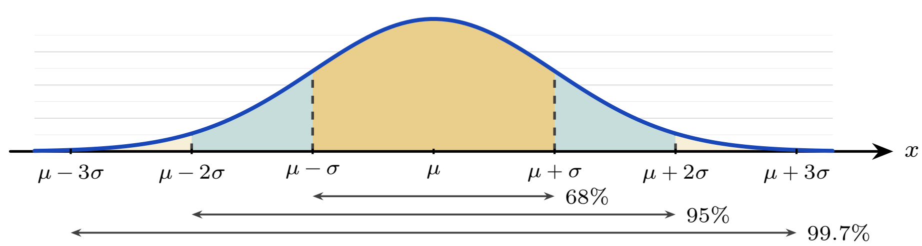


Figure 12.5 The normal curve, symmetric about μ. The three shaded bands reach one, two and three

standard deviations either side of the mean, and hold about 68%, 95% and 99.7% of the values. Almost nothing lies beyond three standard deviations.

This symmetry gives a quick approximation, the **68–95–99.7 rule**. It lets you judge, from μ and σ alone, where most of the data lie, without a calculator.

KEY **The 68–95–99.7 rule.** For $X \sim N(\mu, \sigma^2)$, approximately

$$68\% \text{ of values lie in } \mu \pm \sigma, \qquad 95\% \text{ in } \mu \pm 2\sigma, \qquad 99.7\% \text{ in } \mu \pm 3\sigma.$$

Read in reverse, the same rule measures how surprising a value is. Falling outside two standard deviations happens to about 5% of values, roughly one in twenty. Falling outside three happens to about 0.3%, roughly one in 370. That is why a reading three standard deviations from the mean is treated as a signal that something has changed, rather than as ordinary variation.

The curve has an equation, and although you will not have to use it, seeing it explains where the shape comes from.

KEY **The normal density** (not assessed):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Read the exponent first, because it carries the whole shape. The quantity $(x-\mu)^2$ is a parabola: it is zero at $x = \mu$ and grows as x moves away from the mean, and it depends only on the *distance* from μ, not on the direction. The minus sign turns that growth into decay, since e^{-u} is largest when $u = 0$ and shrinks towards zero as u grows. So the curve is tallest at the mean, falls away at the same rate on both sides, and never quite reaches the axis. The $2\sigma^2$ underneath sets the speed of the fall: a large σ makes the exponent grow slowly, giving a wide, low bell. The factor $\dfrac{1}{\sigma\sqrt{2\pi}}$ in front changes no part of the shape; it is the number that makes the total area exactly 1.

This formula is not in the formula booklet, and no examination question will ask you to use it. Every normal probability in this course is found with a GDC. It is here so that the bell is a consequence of something rather than a shape you are asked to accept.

For exact probabilities you use a GDC, which computes the area under the curve between any two values directly from μ and σ.

EXAMPLE 12.15

The heights of adult women are modelled by $X \sim N(165, 6^2)$ (in cm). Find (a) $P(X < 172)$,

(b) $P(158 < X < 172)$.
Both are areas under the normal curve, read from a GDC with $\mu = 165$, $\sigma = 6$.
(a) $P(X < 172) \approx 0.878$.
(b) $P(158 < X < 172) \approx 0.757$.
The second answer is close to the 68% from the rule of thumb, because 158 and 172 are roughly one standard deviation either side of the mean.

Questions ask for three shapes of area, and a calculator handles all three from the same two entries. An area to the left needs a lower limit far below the mean, an area to the right needs an upper limit far above it, and an area between two values needs both limits as given.

EXAMPLE 12.16

The masses of eggs from a farm are modelled by $X \sim N(58, 4^2)$, in grams. Find (a) $P(X > 65)$, (b) $P(54 < X < 62)$.
Enter $\mu = 58$ and $\sigma = 4$ on a GDC.
(a) A right-hand tail has no natural upper limit, so supply one far above the mean, such as 10^{99}, and take the area from 65 upwards:

$$P(X > 65) \approx 0.0401.$$

(b) Here both limits are given:

$$P(54 < X < 62) \approx 0.683.$$

The limits in (b) sit exactly one standard deviation either side of 58, so the second answer is the 68% of the rule of thumb, computed rather than estimated.

A probability also answers a question about a whole batch. If each of N items follows the same normal distribution independently, the count that falls in some range is binomial with that probability, and its expected value is N times the probability. So the expected number in a batch is a normal calculation followed by one multiplication.

EXAMPLE 12.17

The lifetime of a light bulb is modelled by $X \sim N(1200, 100^2)$, in hours. A hospital installs 400 of these bulbs. Find (a) the probability that a bulb fails before 1000 hours, (b) the expected number of the 400 that fail before 1000 hours.
(a) With $\mu = 1200$ and $\sigma = 100$, a lower limit far below the mean gives

$$P(X < 1000) \approx 0.0228.$$

(b) Multiply the batch size by that probability, carried at more digits than the rounded 0.0228 of part (a):

$$400 \times 0.022750 = 9.100,$$

so about 9 bulbs are expected to fail before 1000 hours.
Round only at the end, and read the answer as a long-run average: a particular batch of 400 may contain any number of early failures.

YOUR TURN 12.5 The Normal Distribution

11. GDC

(a) $X \sim N(50, 10^2)$. Find $P(X < 60)$.
(b) $X \sim N(100, 15^2)$. Find $P(X > 120)$.
(c) $X \sim N(20, 4^2)$. Find $P(16 < X < 24)$.
(d) $Z \sim N(0, 1)$. Find $P(Z < 1.5)$.
(e) $X \sim N(500, 50^2)$. Find $P(X < 400)$.

12. GDC The masses of packets of rice are modelled by $X \sim N(500, 25^2)$, in grams.

(a) Find the probability that a packet weighs more than 540 g.
(b) Find $P(475 < X < 525)$, and compare your answer with the 68% from the rule of thumb.
(c) In a batch of 800 packets, find the expected number weighing more than 540 g.
(d) Write down the range of masses within which about 95% of packets lie.

12.6 Standardization and the Inverse Normal

Every normal distribution is the same bell, just shifted and scaled. **Standardizing** expresses any value as the number of standard deviations it sits from its mean. That number is the **z-value**, and it converts any normal variable to the **standard normal** $Z \sim N(0, 1)$.

KEY · IB

$$z = \frac{x - \mu}{\sigma}$$

Standardizing does two things at once. Subtracting μ moves the centre of the distribution to 0. Dividing by σ then rescales the spread, so that one unit of z means one standard deviation. The units cancel in that division, which is the important part: z is a pure number with no units attached, so a mass in grams and a time in seconds become directly comparable once standardized.

Read a z-value as follows.

- $z > 0$ means the value lies above the mean, and $z < 0$ means it lies below.
- $z = 0$ means the value is exactly the mean.
- The size of z is the distance from the mean, measured in standard deviations. A z-value of 1.5 means "one and a half standard deviations above the mean", whatever the original units were.

The 68–95–99.7 rule of §12.5 now reads very simply. About 68% of values have $|z| < 1$, about 95% have $|z| < 2$, and about 99.7% have $|z| < 3$. Standardizing therefore lets you compare values drawn from different normal distributions, and it is also the key to working backwards from a probability to a value.

EXAMPLE 12.18

Exam scores are $N(60, 12^2)$. A student scores 78. How many standard deviations above the mean is this?

$$z = \frac{78 - 60}{12} = \frac{18}{12} = 1.5.$$

The score sits 1.5 standard deviations above the mean. A student scoring 78 on a different exam with $N(70, 4^2)$ would have $z = 2$, and so be higher *relative to their cohort*, despite the equal raw mark.

12.6.1 The inverse normal

Often the question runs in the other direction. Instead of being given a value and asked for a probability, you are given a probability and asked for the value that produces it. "What score is the 90th percentile?" is a question of this kind. The **inverse normal** on a GDC takes a probability and returns the corresponding value x.

One detail causes most of the errors. A calculator's inverse normal expects the area to the *left* of the unknown value, but examination questions are often phrased in terms of the area to the right. Converting between the two is a single subtraction, and forgetting it produces an answer on the wrong side of the mean.

KEY **Using the inverse normal.**

1. Write down the probability you are given, and decide whether it is the area to the left or to the right of the unknown value.
2. If it is a right-hand area, subtract it from 1. The calculator needs the left-hand area.
3. Enter that left-hand area together with μ and σ.
4. Read off x, then check that it falls on the expected side of the mean.

Step 4 is worth the few seconds it takes. An area greater than 0.5 must return a value above the mean, and an area less than 0.5 must return one below it. This single check catches the subtraction error before it costs marks.

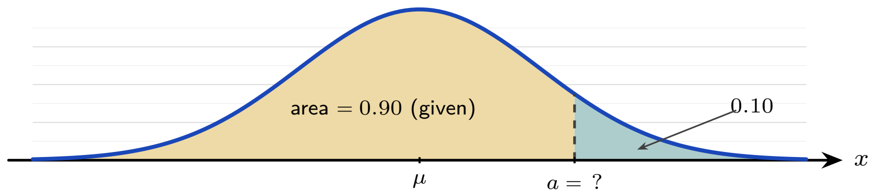


Figure 12.6 The inverse normal reverses the usual question. Instead of being given a and finding the area, you are given the area and must find a. A calculator expects the area to the *left* of a (gold), so a question phrased as a right-hand tail (teal), here 0.10, must be turned into its left-hand partner 0.90 first.

EXAMPLE 12.19

For heights $X \sim N(165, 6^2)$, find the height a below which 90% of women fall.
You need a with $P(X < a) = 0.90$. The inverse normal gives

$$a \approx 172.7 \text{ cm}.$$

Equivalently, the 90th percentile sits at $z \approx 1.282$, so $a = 165 + 1.282(6) \approx 172.7$, matching.

12.6.2 Finding an unknown μ or σ

The inverse normal also lets you recover a missing parameter. Suppose you know one probability and the value attached to it, but either μ or σ is missing. The method has three steps.

1. Use the inverse normal on the *standard* curve, $N(0, 1)$, to turn the probability into a z-value.
2. Substitute that z, the known value x, and whichever parameter you have into $z = \dfrac{x - \mu}{\sigma}$.
3. Solve the resulting equation for the one unknown that remains.

The equation is linear in both μ and σ, so this last step is ordinary algebra.

EXAMPLE 12.20

A machine fills bottles so that volumes are normal with $\sigma = 5$ mL. Given that 80% of bottles contain less than 20 mL, find the mean μ.
Here $P(X < 20) = 0.80$. The inverse normal for the standard curve gives the matching z:

$$z = 0.8416, \qquad \text{so} \qquad 0.8416 = \frac{20 - \mu}{5}.$$

Solving, $20 - \mu = 4.21$, hence $\mu \approx 15.8$ mL.

EXAMPLE 12.21

Scores are normal with mean 50. Given that 10% of scores exceed 60, find σ.

$P(X > 60) = 0.10$ means $P(X < 60) = 0.90$, so $z = 1.282$:

$$1.282 = \frac{60-50}{\sigma} \implies \sigma = \frac{10}{1.282} \approx 7.80.$$

12.6.3 Both parameters unknown

When neither μ nor σ is known, a single probability statement is not enough, because one equation cannot determine two unknowns. Two statements are required.

Rearranging $z = \dfrac{x-\mu}{\sigma}$ into $x = \mu + z\sigma$ shows why this works. Each statement produces one equation of that form, linear in μ and σ, and a pair of linear equations in two unknowns can be solved simultaneously in the usual way.

EXAMPLE 12.22

The masses of a species of fish are normally distributed. It is known that 5% of the fish have mass below 50 g and 10% have mass above 80 g. Find μ and σ.

Translate each statement into a z-value with the inverse normal:

$$P(X < 50) = 0.05 \implies z_1 = -1.645, \qquad P(X < 80) = 0.90 \implies z_2 = 1.282.$$

Each gives an equation $x = \mu + z\sigma$:

$$50 = \mu - 1.645\sigma, \qquad 80 = \mu + 1.282\sigma.$$

Subtracting, $30 = 2.926\sigma$, so $\sigma \approx 10.3$ g. Substituting back, $\mu = 50 + 1.645(10.25) \approx 66.9$ g.

Note the signs carefully: a value *below* the mean has a negative z. One wrong sign here changes both answers, so check each z against the picture before solving.

YOUR TURN 12.6 Standardization and the Inverse Normal

13. GDC

(a) $X \sim N(60, 12^2)$. Find the z-value of $x = 84$.

(b) $X \sim N(70, 5^2)$. Find the value x for which $P(X < x) = 0.95$.

(c) $X \sim N(80, 10^2)$. Find the value x for which $P(X > x) = 0.15$.

14. GDC

(a) $X \sim N(\mu, 8^2)$ with $P(X < 50) = 0.30$. Find μ.

(b) $X \sim N(\mu, 6^2)$ with $P(X > 40) = 0.20$. Find μ.

(c) $X \sim N(100, \sigma^2)$ with $P(X < 90) = 0.25$. Find σ.

15. GDC

(a) Two examinations have marks $A \sim N(65, 10^2)$ and $B \sim N(72, 6^2)$. A student scores 75 in A and 80 in B. Find both z-values, and state which is the better performance relative to the cohort.

(b) $X \sim N(\mu, \sigma^2)$ with $P(X < 20) = 0.10$ and $P(X < 35) = 0.90$. Find μ and σ.

(c) For $X \sim N(50, 8^2)$, find the lower quartile, the upper quartile, and the interquartile range.

Chapter 12 · Probability Distributions

Reflections

TOK In 1846 a Belgian astronomer found this curve in the chest measurements of 5738 soldiers, and drew from it a conclusion the data did not support. He invented *l'homme moyen*, the "average man," and treated this statistical fiction as an ideal that real people deviate from. No actual person has all the average values at once. When we summarise a population by its mean, are we describing a real thing, or creating one? The normal curve is a fact about variation; the "average man" was an interpretation added to it.

TOK A game is called fair when the expected gain is zero. Every casino game fails that test by design, and the house edge is published.

If both sides know the odds, is a game with a built-in edge unfair, or merely unequal? And does the answer change when the player cannot do the arithmetic?

TOK This chapter offers several models for the same kind of situation: a binomial count can be approximated by a normal curve, and a real data set is never exactly either.

What criteria should decide between two models that both fit? Simplicity, accuracy, and agreement with existing theory can point different ways, and the choice is made by people.

RW The bell curve's universality has a name: the central limit theorem. It says that the sum of many independent random effects tends toward a normal distribution, whatever the shape of the individual effects. This is why measurement errors, heights, and the means of large samples are all approximately normal. The pattern was noticed in 1846; the theorem that explains it was not fully proven until the 1930s.

EXT The binomial becoming normal is a theorem, not a loose comparison. Plot $B(n, 0.5)$ for increasing n and the discrete bars take the shape of the continuous bell. Galton built a machine, the quincunx, to show it physically: balls dropped through rows of pegs, each bounce an independent left-or-right trial, pile up at the bottom in a binomial that looks exactly normal. The distributions of this chapter are not separate facts but one story told at increasing scale.

End-of-Chapter Problems — Chapter 12: Probability Distributions

1. A discrete random variable X has the distribution below.

x	1	2	3	4
$P(X = x)$	0.1	k	0.3	0.2

(a) Find k. [2]

(b) Find $E(X)$. [2]

(c) Find Var(X). [3]

2. A charity raffle sells 500 tickets at $2 each. There is one prize of $300 and five prizes of $50.

(a) Find the expected winnings (before cost) for a single ticket. [3]

(b) Hence find the expected gain or loss per ticket, and comment. [2]

3. GDC In a production line, 8% of items are faulty, independently. A random sample of 15 items is taken. Let X be the number of faulty items.

(a) State the distribution of X. [1]

(b) Find $P(X = 0)$. [2]

(c) Find $P(X \geq 2)$. [2]

(d) Find the expected number of faulty items. [1]

4. GDC A fair six-sided die is rolled 30 times. Let X be the number of sixes.

(a) State the distribution of X and find its mean. [2]

(b) Find $P(X = 5)$. [2]

(c) Find $P(X \geq 7)$. [2]

5. A continuous random variable has density $f(x) = kx$ for $0 \leq x \leq 3$, and zero elsewhere.

(a) Find k. [2]

(b) Find $P(X < 2)$. [2]

(c) Find $E(X)$. [3]

(d) Find the median and the mode. [3]

6. GDC The heights of a species of plant are $X \sim N(170, 8^2)$ in cm.

(a) Find $P(X < 160)$. [2]

(b) Find $P(160 < X < 185)$. [2]

(c) Find the proportion of plants taller than 180 cm. [2]

7. GDC Test scores are normally distributed with mean 500 and standard deviation 80.

(a) Find the score at the 90th percentile. [2]

(b) The top 5% of candidates receive a distinction. Find the minimum score for a distinction. [3]

8. GDC A variable X is normally distributed. It is known that $P(X < 40) = 0.10$ and $P(X < 70) = 0.75$.

(a) Write down two equations involving μ and σ using standardized values. [3]

(b) Find μ and σ. [4]

9. The random variable X has $E(X) = 5$ and $\text{Var}(X) = 4$. Let $Y = 2X - 1$.

(a) Find $E(Y)$. [1]

(b) Find $\text{Var}(Y)$ and the standard deviation of Y. [3]

10. GDC Reaction times are $X \sim N(60, 12^2)$ in milliseconds. A reaction is "slow" if it exceeds 75 ms.

(a) Find the probability that a single reaction is slow. [2]

(b) Thirty reactions are recorded independently. Find the expected number of slow reactions. [2]

(c) Find the probability that at least one of the thirty is slow. [3]

11. A continuous random variable has density $f(x) = \dfrac{1}{8}(4 - x)$ for $0 \leq x \leq 4$, and zero elsewhere.

(a) Verify that f is a valid probability density function. [3]

(b) Find $P(X < 2)$. [2]

(c) Find the mode. [1]

12. A discrete random variable X takes the values $1, 2, 3, 4$ with $P(X = 1) = 0.2$, $P(X = 2) = a$, $P(X = 3) = b$, and $P(X = 4) = 0.1$. It is known that $E(X) = 2.4$.

(a) Find the values of a and b. [4]

(b) Find $P(X > 2 \mid X \geq 2)$. [3]

13. A stall game costs $\$c$ to play. A player wins \$10 with probability 0.2, wins \$2 with probability 0.3, and otherwise wins nothing.

(a) Find the expected winnings from a single play. [2]
(b) Find the value of c for which the game is fair. [1]
(c) The stall sets $c = 3$. Find the player's expected loss over 100 plays. [2]

14. GDC A gardener plants 12 seeds. Each seed germinates with probability 0.85, independently of the others. Let X be the number that germinate.

(a) State two conditions that make a binomial model appropriate here. [2]
(b) Find $P(X = 10)$. [2]
(c) Find $P(X \geq 10)$. [2]
(d) Find the mean and standard deviation of X. [2]

15. GDC A darts player hits the bullseye with probability 0.3 on each throw, independently.

(a) In 8 throws, find the probability of exactly 3 bullseyes. [2]
(b) Find the least number of throws needed for the probability of at least one bullseye to exceed 0.98. [3]

16. GDC The random variable $X \sim B(10, p)$ satisfies $P(X = 0) = 0.0282$.

(a) Find the value of p. [3]
(b) Find $P(X = 3)$. [2]
(c) Write down $E(X)$ and Var(X). [2]

17. GDC The lifetime of a battery is $X \sim N(120, 15^2)$, in hours.

(a) Find the probability that a battery lasts more than 140 hours. [2]
(b) Find $P(100 < X < 130)$. [2]
(c) A company ships 10 000 batteries. Find the expected number that last more than 140 hours. [2]

18. GDC The masses of apples are $X \sim N(160, 20^2)$, in grams. The heaviest 8% are sold as premium, and the lightest 5% are rejected.

(a) Find the minimum mass of a premium apple. [2]
(b) Find the maximum mass of a rejected apple. [2]
(c) Write down the proportion of apples that are neither premium nor rejected. [1]

19. GDC Journey times are normally distributed with mean 25 minutes and standard deviation σ. It is known that $P(X < 30) = 0.85$.

(a) Find σ. [3]
(b) Hence write down $P(X < 20)$, justifying your answer. [2]

20. GDC The masses of loaves of bread are normally distributed with unknown mean μ and standard deviation σ. It is known that 2% of loaves have mass below 300 g and 5% have mass above 340 g.

(a) Write down two equations for μ and σ using standardized values. [3]
(b) Find μ and σ. [3]

21. A continuous random variable has density

$$f(x) = \begin{cases} k & 0 \le x \le 1, \\ k(2-x) & 1 < x \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $k = \frac{2}{3}$. [3]
(b) Sketch the graph of f. [2]
(c) Find $P(X > 1.5)$. [2]
(d) Find the median of X. [2]
(e) Find $E(X)$. [3]

22. A continuous random variable has density $f(x) = kx(1-x)$ for $0 \le x \le 1$, and zero elsewhere.

(a) Show that $k = 6$. [2]
(b) Use calculus to find the mode of X. [2]
(c) Find $E(X)$. [2]
(d) Find Var(X). [3]

23. GDC Scores on an examination are $X \sim N(55, 10^2)$. The pass mark is 45.

(a) Find the probability that a randomly chosen candidate passes. [2]
(b) Six candidates are chosen at random. Find the probability that exactly five of them pass. [3]
(c) Find the probability that at least five of them pass. [2]

24. GDC In 1846 a Belgian astronomer published the chest measurements, in inches, of 5738 Scottish militiamen, and used them as evidence that human measurements follow a normal distribution. His table is reproduced below.

Chest (in)	33	34	35	36	37	38	39	40
Frequency	3	18	81	185	420	749	1073	1079
Chest (in)	41	42	43	44	45	46	47	48
Frequency	934	658	370	92	50	21	4	1

(a) Show that the mean chest measurement is 39.8 inches, to 3 significant figures. [3]

(b) The standard deviation of the measurements is 2.05 inches. Assuming $X \sim N(39.8,\, 2.05^2)$, find $P(X < 38.5)$. [2]

(c) Find the proportion of the 5738 soldiers who actually measured 38 inches or less, and compare it with your answer to part (b). [3]

(d) Find the proportion of soldiers whose measurement lies within one standard deviation of the mean, and compare it with the value predicted by the normal model. [3]

(e) Comment on how well the normal distribution describes this real data set. [2]

Challenge Questions

25. **What makes a game fair?** At a fair, a stall offers this game. You pay a stake, roll two fair dice, and are paid \$20 if the total is 12, \$5 if the total is 10 or 11, and nothing otherwise. Let W be the amount paid out.

(a) Write down the probability distribution of W. [3]

(b) Find $E(W)$. [3]

(c) State the stake that would make the game fair, and explain what "fair" means here. [3]

(d) The stall charges \$2. Find the expected profit per game for the stall, and the expected profit over 500 games. [3]

(e) Find Var(W) and the standard deviation of W. [3]

(f) A player who plays ten times often finishes ahead, even though the game favours the stall. Use your answer to part (e) to explain why, and explain why the stall is not worried. [4]

26. GDC **From the binomial to the bell.** The Reflections claim that a binomial distribution comes to look normal as n grows. This question tests that claim, and finds the adjustment that makes it work.

Throughout, let $X \sim B(50, 0.5)$ and let $Y \sim N(\mu, \sigma^2)$ have the same mean and variance as X.

(a) Write down $E(X)$ and Var(X), and hence state μ and σ for Y. [2]

(b) Find $P(20 \le X \le 30)$ using the binomial. [2]

(c) Find $P(20 < Y < 30)$ using the normal. Comment on how well it matches part (b). [3]

(d) X takes whole-number values, while Y takes every value in between. The bar of the histogram for $X = 20$ covers the interval from 19.5 to 20.5. Using this idea, find $P(19.5 < Y < 30.5)$ and compare it with part (b). [4]

(e) Repeat parts (b), (c) and (d) for $X \sim B(10, 0.5)$ and the range $4 \le X \le 6$. [4]

(f) State which of the two normal calculations should be used, and describe how the quality of the approximation changes as n increases. [3]

27. The most likely value. Let $X \sim B(n,p)$ with $0 < p < 1$. This question finds the mode of X, the value it takes most often.

(a) Show that for $r = 1, 2, \ldots, n$,

$$\frac{P(X=r)}{P(X=r-1)} = \frac{(n-r+1)\,p}{r\,(1-p)}.$$

[3]

(b) Hence show that $P(X=r) \geq P(X=r-1)$ if and only if $r \leq (n+1)p$. [3]

(c) Deduce that the mode of X is $\lfloor (n+1)p \rfloor$ when $(n+1)p$ is not an integer, and use this to find the most likely number of successes when $X \sim B(20, 0.3)$. [3]

(d) Explain what happens when $(n+1)p$ is an integer, and illustrate your answer with $X \sim B(9, 0.5)$. [3]

CALCULUS

Chapter 13

Limits & Derivatives

13

SYLLABUS 5.1–5.4 5.6–5.8 5.12 5.13 5.15

A car's speedometer reads 80 km/h. If you ask what that number means, you meet a problem that mathematicians could not solve for two thousand years.

Speed is distance divided by time. But the speedometer shows your speed *right now*, at a single instant. In one instant no time passes, and no distance is covered. Speed at an instant therefore appears to be $0 \div 0$, which has no value. How can the needle point to 80?

The solution is to stop asking about the instant, and to ask instead about short intervals around it.

Measure your average speed over the next hour. Then over the next minute. Then over the next second, and then over the next hundredth of a second. Each interval is shorter than the one before, and each average lies closer to one particular number. That number is the speed at the instant. The averages arrive at it, even though the instant on its own cannot.

This idea is called the **limit**. When a value cannot be reached directly, you find it by asking what nearby values approach. All of calculus is built on this one idea.

The **derivative** follows from the limit. It measures the exact rate at which a quantity changes: how steep a curve is at one point, or how fast an apple is falling at the moment it lands. This chapter builds the derivative from the limit, then develops methods for calculating it quickly.

By the end of this chapter you will evaluate limits, decide when a function is continuous and differentiable, and differentiate from first principles. You will apply the product, quotient and chain rules, differentiate the standard functions of the course, and find the equations of a tangent and a normal. You will also take a second derivative, use its sign to classify a stationary point as a maximum or a minimum, and turn an optimisation question into a stationary-point question. Calculus begins here because a derivative answers two questions at once. Geometrically it is the gradient of a curve at a point. Physically it is a rate of change. Every later application is one of those two readings.

13.1 Limits

A **limit** is the value a function approaches as its input approaches some point. It does not matter what the function does at that point itself. We write

$$\lim_{x \to a} f(x) = L$$

to mean: as x gets **arbitrarily close** (as close as you like) to a, $f(x)$ gets arbitrarily close to L. The words "whatever happens at a" matter. A limit can exist at a point where the function itself is undefined.

For most functions at most points, the limit is found by direct substitution. If f is defined at a and its graph passes through that point unbroken, then the value it approaches is simply the value it takes:

$$\lim_{x \to 2}(x^2 + 3x) = 2^2 + 3(2) = 10.$$

That case is straightforward. The useful cases are the ones where substitution fails.

EXAMPLE 13.1

Find $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

Substituting $x = 3$ gives $\frac{0}{0}$, which is undefined: the function has a hole at $x = 3$. But for every $x \neq 3$ we can factor and cancel:

$$\frac{x^2 - 9}{x - 3} = \frac{(x-3)(x+3)}{x - 3} = x + 3.$$

Away from the hole the function is simply $x + 3$, so as $x \to 3$ it approaches 6:

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = 6.$$

At $x = 3$ the function has no value at all, yet 6 is the value it approaches. The difference between "approaches" and "equals" is what a limit measures.

The form $\frac{0}{0}$ is called **indeterminate**. It does not mean the limit fails to exist. It means more work is needed.

One limit of this kind must be memorised, because the derivatives of the trigonometric functions rest on it:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

When there is no algebraic method, you can *estimate* a limit. Calculate the function at values closer and closer to the target, from both sides. Here is the result for this one, with x in radians:

x	-0.1	-0.01	0.01	0.1
$\dfrac{\sin x}{x}$	0.99833	0.99998	0.99998	0.99833

Both sides approach 1. A table or a zoomed-in graph is a legitimate way to **conjecture** (guess, before proving) the value of a limit. On a calculator paper it is often the fastest way to check an algebraic answer.

13.1.1 Reading a limit off the graph

A graph often shows why a limit takes the value it does, and the two fractions $\dfrac{\sin x}{x}$ and $\dfrac{\cos x}{x}$ make the contrast clear. Both have denominator $x \to 0$, yet they behave completely differently, and the reason is what the *numerator* does at the same moment.

Sketch $y = \sin x$ and $y = x$ on the same axes. Near the origin the two graphs are almost indistinguishable: at $x = 0.1$, $\sin x = 0.0998$. Numerator and denominator shrink to zero together, at the same rate, so their ratio approaches 1.

Now sketch $y = \cos x$ against $y = x$. As $x \to 0$ the numerator approaches 1 while the denominator approaches 0. A fixed quantity divided by something ever smaller grows without bound, so the fraction does not approach any finite value.

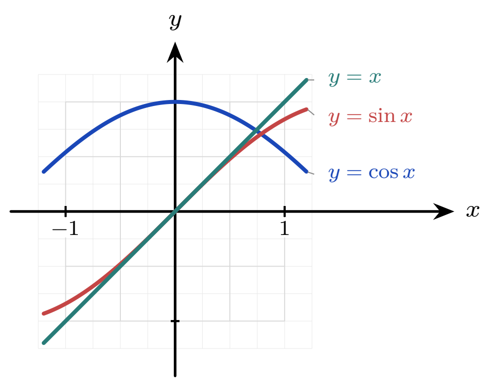


Figure 13.1 Near the origin the graphs of $\sin x$ and x are almost the same, so their ratio tends to 1. But $\cos x$ tends to 1 while x tends to 0, so $\frac{\cos x}{x}$ has no finite limit.

CAUTION Be precise about the second case. As $x \to 0^+$ the fraction $\frac{\cos x}{x}$ increases without bound, but as $x \to 0^-$ it decreases without bound, because x is negative there. The two sides disagree, so $\lim_{x\to 0} \frac{\cos x}{x}$ does not exist. Writing that it "equals infinity" conceals that disagreement, so it is not an acceptable answer.

Not every limit exists. A limit **converges** when the values approach one finite number, and **diverges** when they do not. Two examples show the difference: $\frac{1}{x^2}$ grows without bound as

$x \to 0$, while $\sin x$ oscillates between -1 and 1 as $x \to \infty$ and never settles on any value. You have met this distinction before: the infinite geometric series of Ch. 2 converges exactly when $|r| < 1$, because its partial sums approach a single value, and diverges otherwise.

Deciding which of the two happens as $x \to \infty$ needs a method, because substitution has nothing to substitute. For a ratio of polynomials there is one that always works: divide every term, above and below, by the highest power of x that appears. What is left is a set of constants together with terms like $\frac{1}{x}$ and $\frac{1}{x^2}$, and each of those approaches 0.

EXAMPLE 13.2

Evaluate (a) $\lim_{x\to\infty} \frac{5x-2}{x}$ and (b) $\lim_{x\to\infty} \frac{4x^2-x}{3x^2+5}$.

(a) The highest power is x, so divide both terms of the numerator by it:

$$\lim_{x\to\infty} \frac{5x-2}{x} = \lim_{x\to\infty}\left(5 - \frac{2}{x}\right) = 5 - 0 = 5.$$

(b) The highest power is x^2, so divide all four terms by x^2:

$$\lim_{x\to\infty} \frac{4x^2-x}{3x^2+5} = \lim_{x\to\infty} \frac{4 - \frac{1}{x}}{3 + \frac{5}{x^2}} = \frac{4-0}{3+0} = \frac{4}{3}.$$

Every vanishing term came from a lower power of x than the one you divided by, so what survives is the ratio of the leading coefficients: $\frac{5}{1}$ in (a) and $\frac{4}{3}$ in (b).

CAUTION A limit can fail to exist. If $f(x)$ approaches different values from the left and the right of a, as at a jump, there is no single limit. "Approaches $\frac{0}{0}$" is fixable; "approaches two different things" is not.

YOUR TURN 13.1 Limits

1. Find each limit by direct substitution.

(a) $\lim_{x\to 2}(3x-1)$

(b) $\lim_{x\to 0}(x^2+2x+5)$

(c) $\lim_{x\to 3}(x^2-9)$

(d) $\lim_{x\to 1}\frac{x+1}{x+3}$

2. Each of these gives $\frac{0}{0}$. Factorise, cancel, then take the limit.

(a) $\lim_{x\to 4}\frac{x^2-16}{x-4}$

(b) $\lim_{x\to 1}\frac{x^2-1}{x-1}$

(c) $\lim_{x\to 2}\frac{x^2-x-2}{x-2}$

(d) $\lim_{x\to -3}\frac{x^2-9}{x+3}$

3. GDC Consider $y = \dfrac{\sin x}{x}$.

(a) Write down the value of $\displaystyle\lim_{x\to 0} \frac{\sin x}{x}$.

(b) Graph the curve for $-10 \le x \le 10$ on your GDC, and sketch what you see.

(c) Describe what the graph does at $x = 0$, and what it does as $x \to \pm\infty$.

4. Explain why $\displaystyle\lim_{x\to 0} \frac{|x|}{x}$ does not exist.

5. State whether each of the following converges or diverges as $x \to \infty$, and give the limit where one exists.

(a) $\dfrac{1}{x}$

(b) $\sin x$

(c) x^2

(d) $\dfrac{3x+1}{x}$

6. GDC Estimate $\displaystyle\lim_{x\to 0} \frac{1-\cos x}{x^2}$ by evaluating the expression at $x = 0.1$ and $x = 0.01$ radians. State the exact value you think the limit takes.

13.2 The Derivative

Calculus is the mathematics of change. You already know how to measure change: it is the gradient.

Rise over run gives the change in y for each unit of change in x. If the gradient doubles, the quantity changes twice as fast.

There is one problem. A gradient is defined for a *straight line* only. One number describes the whole line, because a line changes at the same rate everywhere.

Very few real quantities behave like that. A cooling cup, a thrown ball, a growing population: in each case the rate of change is itself changing. So the question "what is the gradient of this curve?" has no single answer. The answer is different at every point.

So ask a smaller question. Do not ask for the gradient of the curve. Ask for the gradient of the curve *at one point*. Once you can answer that for one point, you can answer it for every point, and the rate of change becomes a function of its own.

Here is the method. On the graph of $y = f(x)$, choose your point x and a second point $x + h$ close to it. Join the two points with a straight line, called the **secant**, and find its gradient. You can already calculate that:

$$\frac{f(x+h)-f(x)}{h},$$

the rise over the run. This is the average rate of change over the interval of width h.

The secant gradient is still not the answer. It is an average over an interval of width h, and the question was about a single point.

Now make h smaller and smaller. The second point moves towards the first, and the secant rotates towards a limiting position, meeting the curve at one point only. That line is the **tangent**. Its gradient is the rate of change at a single point, and it is called the **derivative**.

That is the idea. A gradient is defined only for a line, so we find the line that matches the curve at that point, and use its gradient instead.

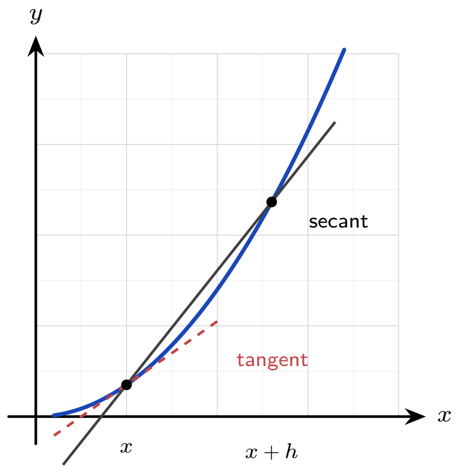


Figure 13.2 The secant joins two points on the curve and measures an average rate of change. As $h \to 0$ the second point moves to the first and the secant approaches the tangent.

DEF · IB HL The **derivative** of f at x, written $f'(x)$, is

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h},$$

when this limit exists. Computing a derivative directly from this definition is called differentiation **from first principles**.

The derivative is itself a function. Give it a value of x and it gives back the gradient of the curve at that value.

13.2.1 Two notations, and where they came from

Calculus was developed independently by two people in the 1670s and 1680s, and each invented his own notation. Both are still in use, and a student of the subject needs to read both.

- **Newton** wrote a dot above the variable: $\dot{x}$ for the rate at which x changes with time, and $\ddot{x}$ for the rate at which *that* changes. He called these quantities *fluxions*, because he pictured a quantity flowing. The dot notation is still standard in mechanics and in physics, where the variable underneath is almost always time.
- **Leibniz** wrote $\dfrac{dy}{dx}$, built from $\Delta y/\Delta x$ for a finite change, with the d marking the limit as the change shrinks to nothing. The related forms $f'(x)$ and $\dfrac{d}{dx}f(x)$ belong to the same tradition, and this course uses all three.

The Leibniz form is the one that carries the most information, because it names both variables. When the quantities are not called x and y, the letters change with them. A volume V changing with radius r has derivative $\dfrac{dV}{dr}$, and a displacement s changing with time t has derivative $\dfrac{ds}{dt}$. Read each as a rate: how fast the top quantity changes per unit of the bottom one. Examination questions use these forms constantly, and choosing the right letters is half of setting the problem up.

One caution about $\dfrac{dy}{dx}$. It is written as a fraction, but it is not one. It is a single symbol standing for the limit of the fraction $\dfrac{\Delta y}{\Delta x}$, and dy and dx have no independent meaning at this level. The notation is nevertheless well designed, because the limit does inherit some of the behaviour of a fraction. That is why the chain rule in §13.6 can be written as $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$, with the du appearing to cancel. The cancelling is a memory aid, not a proof, but it works because the underlying limits behave that way.

Leibniz's notation won out over Newton's for general use, and the reason is practical rather than mathematical: it shows the variable of differentiation, so it extends cleanly to the chain rule, to implicit differentiation, and to integration, where $\int f(x)\,dx$ carries the same dx.

13.2.2 Where derivatives are used

A derivative is a rate of change, and rates of change appear wherever one quantity depends on another.

- **Motion.** Differentiate displacement with respect to time to get velocity, and again to get acceleration. This is how a vehicle's braking distance is calculated, and it is the subject of Ch. 14.
- **Optimisation.** A curve is flat where its derivative is zero, so setting $f'(x) = 0$ locates maxima and minima. Manufacturers use this to find the packaging that uses the least material for a required volume.
- **Medicine.** The rate at which a drug leaves the bloodstream determines the dosing interval, and that rate is a derivative of the concentration with respect to time.
- **Economics.** *Marginal* cost is the derivative of total cost with respect to quantity produced, and it answers what one more unit costs to make.
- **Computer graphics.** A surface is shaded according to the direction it faces, which is found from derivatives of the surface, and this is what makes rendered objects look solid.

Every one of these is the same calculation, applied to a different pair of quantities.

EXAMPLE 13.3

Differentiate $f(x) = x^2 + 4x$ from first principles.

Form the difference quotient and expand:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 + 4(x+h) - x^2 - 4x}{h} = \frac{2xh + h^2 + 4h}{h} = 2x + h + 4.$$

Now let $h \to 0$:

$$f'(x) = \lim_{h \to 0}(2x + h + 4) = 2x + 4.$$

You must cancel the h before taking the limit. While $h \neq 0$ the division is allowed. Only after cancelling is it safe to set h to zero.

EXAMPLE 13.4

Differentiate $f(x) = \dfrac{1}{x}$ from first principles.

$$\frac{f(x+h) - f(x)}{h} = \frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \frac{1}{h} \cdot \frac{x - (x+h)}{x(x+h)} = \frac{-h}{h\,x(x+h)} = \frac{-1}{x(x+h)}.$$

As $h \to 0$:

$$f'(x) = \lim_{h \to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

TIP In examinations, first-principles differentiation is asked only for polynomials. The $\frac{1}{x}$ example above shows how far the definition extends; it is not itself an examination task.

13.2.3 Continuity and differentiability HL

The derivative is a limit, so it can fail to exist. Two ideas are needed to say when.

A function is **continuous** at a point when its limit there exists and equals its value. Informally: the graph passes through the point with no hole and no jump, and you could draw it without lifting the pen.

A function is **differentiable** at a point when the first-principles limit exists. This is stronger. The graph must be unbroken and also *smooth*, with one clear tangent at that point.

Every differentiable function is continuous, but the reverse is not true. The standard example is the absolute value.

EXAMPLE 13.5

Show that $f(x) = |x - 1|$ is continuous at $x = 1$ but not differentiable there.

Continuity is clear: as $x \to 1$ from either side, $|x - 1| \to 0 = f(1)$. The graph is an unbroken V.

For differentiability, form the difference quotient at 1:

$$\frac{f(1+h)-f(1)}{h}=\frac{|h|}{h}=\begin{cases}1 & h>0,\\ -1 & h<0.\end{cases}$$

The quotient approaches 1 from the right and -1 from the left. The two sides disagree, so the limit does not exist and f has no derivative at 1. On the graph, the V has a *corner*, and a corner has no single tangent. A graph can be unbroken and still fail to be smooth.

Note how far this reaches. The function $|x - 1|$ is continuous at every real number, yet that one corner is enough to stop it being differentiable at every real number. A function can be continuous everywhere and still fail to be differentiable everywhere.

TIP Examinations will not ask you to *test* a function for continuity or differentiability. What they expect is the idea itself: you may differentiate only where the curve is smooth.

YOUR TURN 13.2 The Derivative

7. Differentiate each from first principles, showing the limit.

(a) $f(x)=x^2$

(b) $f(x)=3x^2$

(c) $f(x)=x^2+x$

(d) $f(x)=x^2-3x$

8. GDC The definition of the derivative can also be used numerically. For $f(x)=\sin x$, evaluate $\dfrac{f(0+h)-f(0)}{h}$ for $h=0.1$, $h=0.01$ and $h=0.001$, working in radians. State the value the quotient is approaching, and name the standard derivative it confirms.

9. GDC Repeat the same calculation for $f(x)=e^x$ at $x=0$, then at $x=1$. Explain what the two answers suggest about the derivative of e^x.

10. Write down the three notations used in this book for the derivative of $y=f(x)$.

11. Explain why $f(x)=|x|$ has no derivative at $x=0$, even though its graph is unbroken there.

13.3 Differentiating Powers

First principles works, but it is slow. The two worked examples above show a pattern: the x^2 term gave $2x$, and $\frac{1}{x}=x^{-1}$ gave $-x^{-2}$.

In both cases two things happened. The power decreased by one, and the original power became a multiplier in front. This is the **power rule**.

KEY · IB

$$\frac{d}{dx}\,x^n = nx^{n-1}$$

At Standard Level this is used with integer powers. It holds just as well for every rational power, which is the form needed from §13.5 onwards and the reason the next example can differentiate $\sqrt{x}$.

Two more rules let you differentiate any polynomial. A constant multiplier is unchanged, and a sum is differentiated one term at a time:

$$\frac{d}{dx}(c\,f(x)) = c\,f'(x), \qquad \frac{d}{dx}(f(x)+g(x)) = f'(x)+g'(x).$$

A constant on its own has derivative zero, because its graph is a horizontal line with gradient zero.

EXAMPLE 13.6

Differentiate $y = 3x^4 - 2x^2 + 7$.

Differentiate each term with the power rule:

$$\frac{dy}{dx} = 3(4x^3) - 2(2x) + 0 = 12x^3 - 4x.$$

The constant 7 has derivative zero, as every constant does.

EXAMPLE 13.7

Differentiate $y = \sqrt{x} + \dfrac{1}{x^2}$.

Rewrite both terms as powers before differentiating:

$$y = x^{1/2} + x^{-2} \implies \frac{dy}{dx} = \tfrac{1}{2}x^{-1/2} - 2x^{-3} = \frac{1}{2\sqrt{x}} - \frac{2}{x^3}.$$

The power rule requires the function in the form x^n. Convert every root and every reciprocal to a power before differentiating, and the rest of the question becomes routine.

EXAMPLE 13.8

Find the gradient of $y = x^3 - 5x$ at the point where $x = 2$.

Differentiate, then substitute:

$$\frac{dy}{dx} = 3x^2 - 5, \qquad \left.\frac{dy}{dx}\right|_{x=2} = 3(4) - 5 = 7.$$

The derivative is a function. A gradient *at a point* is that function evaluated at the point.

YOUR TURN 13.3 Differentiating Powers

12. Differentiate each of the following.

(a) $y = x^7$
(b) $y = 5x^3 - 2x$
(c) $y = x^4 - 3x^2 + 6$
(d) $y = 4x^5 + x$
(e) $y = 2x^3 - 7x^2 + x - 9$
(f) $y = 6$

13. Rewrite each function as a power of x, then differentiate. In part (f), divide through first.

(a) $y = 2/x$
(b) $y = \sqrt{x}$
(c) $y = x^3 + 1/x$
(d) $y = 1/x^3$
(e) $y = \sqrt[3]{x}$
(f) $y = (x^2 + 1)/x$

14. Find the gradient of each curve at the point given.

(a) $y = x^2 - 5x$ at $x = 3$
(b) $y = x^3 - 4x$ at $x = -1$
(c) $y = \sqrt{x}$ at $x = 4$
(d) $y = 2x^3 + x$ at $x = 1$

15. Find the value or values of x at which each curve has the stated gradient.

(a) $y = x^2 - 6x$, gradient 0
(b) $y = x^3$, gradient 12

16. Explain why $y = \sqrt[4]{x}$ must be rewritten before the power rule can be applied, and state the rewritten form.

13.4 Tangents and Normals

The derivative gives the gradient of the tangent at any point, and a gradient plus a point determines a line. So you can now write the equation of the **tangent** to a curve at a given point. The **normal** is the line perpendicular to the tangent there, with gradient the negative reciprocal.

The tangent and the normal are both straight lines, so both are written with the same equation, the point-gradient form from Ch. 4:

$$y - y_1 = m(x - x_1),$$

where (x_1, y_1) is a known point on the line and m is its gradient. Only two things ever change between the two lines. The point (x_1, y_1) is the same for both, because the tangent and the normal meet the curve at the same place, so all that differs is the gradient m.

KEY At the point (x_1, y_1) on the curve $y = f(x)$, both lines have the form $y - y_1 = m(x - x_1)$, with

$$m_{\text{tangent}} = f'(x_1), \qquad m_{\text{normal}} = -\frac{1}{f'(x_1)}.$$

So every question of this type follows three steps: find y_1 by substituting x_1 into f, find $f'(x_1)$ for the gradient, then put both into the point-gradient form.

On a calculator paper the same job can be done from the screen. A graphic display calculator will evaluate a numerical derivative at a point, and most will draw the tangent there and report its equation. Two cautions come with that convenience. The calculator gives the *tangent* only, so a normal still needs the $-1/f'(x_1)$ step by hand. And a numerical answer earns few marks on its own: show the gradient you used and the point you used it at, then quote the line. Use the screen to check the algebra, not to replace it.

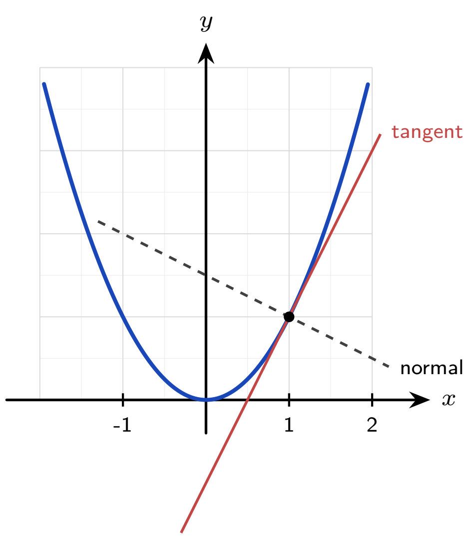


Figure 13.3 At a point on a curve the tangent has gradient $f'(x)$, and the normal is perpendicular to it, with gradient $-1/f'(x)$.

EXAMPLE 13.9

Find the equations of the tangent and normal to $y = x^2 - 3x + 2$ at the point where $x = 2$.
First the point: $y(2) = 4 - 6 + 2 = 0$, so the point is $(2, 0)$.
Then the gradient: $\dfrac{dy}{dx} = 2x - 3$, so at $x = 2$ the tangent gradient is $2(2) - 3 = 1$.
Tangent, through $(2, 0)$ with gradient 1:

$$y - 0 = 1(x - 2) \implies y = x - 2.$$

The normal is perpendicular, so its gradient is -1:

$$y - 0 = -1(x - 2) \implies y = -x + 2.$$

Always find the y-coordinate first. A common error is to write the equation of the line using the x-value alone.

13.4.1 Tangents through a point off the curve

A harder version of the same question names a point the tangent must pass *through*, and that point does not lie on the curve. The three steps above stall immediately, because the point of contact is no longer given, and without it there is no gradient to find.

Turn the missing information into the unknown. Call the point of contact $(a, f(a))$, write the tangent there in terms of a, and then impose the one condition you have: the line passes through the given point. That produces an equation in a alone. For a quadratic curve it is a quadratic equation, so it usually has two roots, and each root is a separate tangent.

EXAMPLE 13.10

Find the equations of both tangents to $y = x^2 - 2x + 3$ that pass through the point $(2, -1)$.
First check where the point sits: at $x = 2$ the curve is at $4 - 4 + 3 = 3$, not -1, so $(2, -1)$ is off the curve and is not the point of contact.
Let the contact point be $(a,\, a^2 - 2a + 3)$. Since $\dfrac{dy}{dx} = 2x - 2$, the tangent there has gradient $2a - 2$, so

$$y - (a^2 - 2a + 3) = (2a - 2)(x - a) \implies y = (2a - 2)x - a^2 + 3.$$

Now force this line through $(2, -1)$:

$$-1 = 2(2a - 2) - a^2 + 3 \implies a^2 - 4a = 0 \implies a(a - 4) = 0,$$

so $a = 0$ or $a = 4$. Substituting each value back into $y = (2a-2)x - a^2 + 3$:

$$a = 0: \quad y = -2x + 3, \qquad\qquad a = 4: \quad y = 6x - 13.$$

Two roots, two tangents, and both survive the check at the external point: $-2(2) + 3 = -1$ and $6(2) - 13 = -1$.

YOUR TURN 13.4 Tangents and Normals

17. Find the equation of the tangent to each curve at the point given.

(a) $y = x^2$ at $(2, 4)$

(b) $y = x^2 - 4x + 3$ at $x = 1$

(c) $y = 1/x$ at $x = 1$

(d) $y = e^x$ at $x = 0$

18. Find the equation of the normal to each curve at the point given.

(a) $y = x^3$ at $(1, 1)$

(b) $y = \sqrt{x}$ at $x = 4$

19. A tangent and a normal are drawn at the same point of a curve. Explain the relationship between their gradients, and state what happens when the tangent is horizontal.

20. The tangent to $y = x^2$ at the point where $x = a$ passes through $(0, -4)$. Find the two possible values of a.

21. Find the coordinates of the point on $y = x^2 - 6x$ at which the tangent is horizontal.

13.5 The Standard Derivatives

The power rule covers algebraic functions. The trigonometric, exponential and logarithmic functions have derivatives of their own. Each can be proved from first principles. You can also find each one by reading gradients from a graph. Do that once before memorising anything.

The method is always the same. Move along the graph of f from left to right. At each point, ask how steep the graph is, and plot that steepness as a height. The curve you plot is f'.

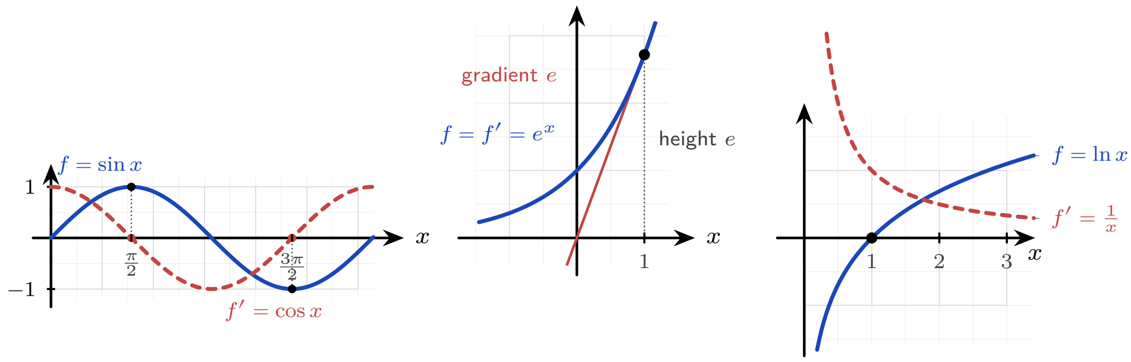


Figure 13.4 Each derivative read from the gradient of its own graph. **Left:** where $\sin x$ is flat, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$, the curve $\cos x$ crosses zero. **Centre:** the tangent to e^x at $x = 1$ has gradient e, which is also the height of the curve there. **Right:** $\ln x$ is steep near 0 and flattens as x grows, and $\frac{1}{x}$ does exactly that.

Read each panel separately.

Sine. At $x = 0$ the sine graph is rising most steeply, so f' starts at its maximum. At the top of the curve, $x = \frac{\pi}{2}$, the graph is flat for an instant, so f' is zero. Past the crest it falls, so f' goes negative. A curve that starts at 1, drops through zero at $\frac{\pi}{2}$ and reaches -1 at π is exactly $\cos x$.

The exponential. Measure the gradient of $y = e^x$ at any point and you get the height of the curve at that point. That is the defining property of e, so here $f' = f$.

The logarithm. Just to the right of 0 the graph of $\ln x$ is almost vertical, so f' is very large. As x increases the curve becomes flatter, so f' decreases towards zero. It stays positive everywhere. That describes $\dfrac{1}{x}$.

Those are the four standard derivatives to learn.

KEY · IB

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}$$

The exponential is the one to look at twice: e^x is its own derivative. More precisely, the functions equal to their own derivative are exactly those of the form $y = Ce^x$, and e^x is the one passing through $(0, 1)$. That property is what makes e important. Note also that the results for $\sin$ and $\cos$ hold only when x is measured in *radians*, and there is a reason for that. The derivative of $\sin x$ is built on the limit $\lim_{x\to 0} \frac{\sin x}{x} = 1$ from §13.1, and that limit equals 1 only in radians. Measured in degrees the same limit is $\frac{\pi}{180}$, so the derivative would come out as $\frac{\pi}{180}\cos x$ instead. Radians are simply the unit that makes the constant equal to 1, which is why calculus uses them throughout.

At Higher Level a wider set of standard derivatives is provided. HL They are listed here for reference and used freely in the rules that follow. The reciprocal and inverse trigonometric functions in the table are defined in Ch. 8; only their derivatives are needed here.

KEY · IB HL

$$\frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\sec x = \sec x \tan x$$

$$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x \qquad \frac{d}{dx}\cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx}a^x = a^x \ln a \qquad \frac{d}{dx}\log_a x = \frac{1}{x \ln a}$$

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

Notice one pair. The derivative of $\arccos$ is the negative of the derivative of $\arcsin$, because the two graphs are reflections of each other.

EXAMPLE 13.11

Differentiate $y = 3\sin x + 2e^x - 5\ln x$.
Apply the standard derivatives term by term:

$$\frac{dy}{dx} = 3\cos x + 2e^x - \frac{5}{x}.$$

Constant multipliers are unchanged, exactly as with powers.

The Higher Level table is used the same way. Read each derivative off it, keep the constant multipliers in place, and differentiate one term at a time.

EXAMPLE 13.12

Differentiate (a) $y = 3\tan x - 2\cos x$, (b) $y = 5^x + \ln x$, (c) $y = \arccos x + 3\arctan x$.
(a)

$$\frac{dy}{dx} = 3\sec^2 x + 2\sin x.$$

The derivative of $\cos x$ is $-\sin x$, so subtracting a cosine adds a sine.
(b)

$$\frac{dy}{dx} = 5^x \ln 5 + \frac{1}{x}.$$

The factor $\ln 5$ is a constant, and it appears because $5^x = e^{x\ln 5}$. Only e^x escapes such a factor, since $\ln e = 1$.
(c)

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}} + \frac{3}{1+x^2}.$$

The leading minus sign is the only thing separating the $\arccos$ row of the table from the $\arcsin$ row. Neither inverse trigonometric derivative contains a trigonometric function; both are algebraic, which is what makes them the results you meet again as integrals in Ch. 15.

YOUR TURN 13.5 The Standard Derivatives

22. Differentiate each of the following.

(a) $y = \sin x + \cos x$
(b) $y = e^x - 3\ln x$
(c) $y = 4\cos x$
(d) $y = \tan x$
(e) $y = \arctan x$
(f) $y = 3^x$

23. Differentiate each of the following. All are in the Higher Level list.

(a) $y = 2\sin x - 5e^x$
(b) $y = \ln x + \tan x$
(c) $y = \sec x$
(d) $y = \arcsin x$

24. Find the gradient of $y = \sin x$ at $x = \dfrac{\pi}{3}$, giving an exact answer.

25. Find the value of x at which the curve $y = e^x$ has gradient 1, and state the significance of that point.

26. The results $\dfrac{d}{dx}\sin x = \cos x$ and $\dfrac{d}{dx}\cos x = -\sin x$ hold only when x is measured in radians. Explain why the choice of unit matters.

13.6 The Rules of Differentiation

Most functions are built from simpler ones, by composing, multiplying or dividing them. Three rules cover these three cases. With the standard derivatives, they let you differentiate almost any function in the course.

13.6.1 The chain rule

A **composite** function is a function inside a function, such as $(2x^2+3)^5$ or $\sin(2x)$. The **chain rule** differentiates it by working from the outside in: differentiate the outer function, then multiply by the derivative of the inner.

KEY · IB If $y = g(u)$ where $u = f(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

The meaning of the rule is that **rates multiply**. Suppose a car's fuel use y depends on its speed u, and its speed depends on time x. If fuel use rises 3 units for every unit of speed, and speed rises 2 units for every unit of time, then fuel use rises $3 \times 2 = 6$ units for every unit of time. That is exactly $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$. A chain of dependencies multiplies its rates, in the same way that gears in a machine multiply their turning ratios.

The chain rule appears in most differentiation questions once the functions stop being simple powers. Two habits make it reliable:

1. **Identify the inner function first**, before differentiating anything. Ask which part is wrapped inside the other.
2. **Multiply by the derivative of the inner function.** Differentiating the outer function alone and stopping there is a common error, so check for it at the end of every chain rule question.

EXAMPLE 13.13

Differentiate (a) $y = (2x^2+3)^5$ and (b) $y = e^{x^2}$.
(a) Outer is u^5, inner is $u = 2x^2+3$:

$$\frac{dy}{dx} = 5(2x^2+3)^4 \cdot 4x = 20x(2x^2+3)^4.$$

(b) Outer is e^u, inner is $u = x^2$:

$$\frac{dy}{dx} = e^{x^2} \cdot 2x = 2x\,e^{x^2}.$$

In each case the second factor, $4x$ and $2x$, is the derivative of the inner function.

The outer function need not be a power or an exponential. Any of the standard derivatives of §13.5 can sit on the outside, and the rule does not change: differentiate the outer function, leave the inner function sitting inside it, then multiply by the inner derivative.

EXAMPLE 13.14

Differentiate (a) $y = \ln(2+\cos x)$ and (b) $y = \sqrt{x^3+2}$.
(a) Outer is $\ln u$, inner is $u = 2+\cos x$. The outer derivative is $\dfrac{1}{u}$, which must be written at $u = 2+\cos x$:

$$\frac{dy}{dx} = \frac{1}{2+\cos x} \cdot (-\sin x) = -\frac{\sin x}{2+\cos x}.$$

(b) Rewrite the root as a power first, $y = (x^3+2)^{1/2}$, so the outer is $u^{1/2}$ and the inner is $u = x^3+2$:

$$\frac{dy}{dx} = \tfrac{1}{2}(x^3+2)^{-1/2} \cdot 3x^2 = \frac{3x^2}{2\sqrt{x^3+2}}.$$

The root itself needs $x^3+2 \geq 0$, and the derivative puts that root in a denominator, so this expression holds for $x^3+2 > 0$, that is $x > -\sqrt[3]{2}$.
Part (a) carries the error worth guarding against: the outer derivative of a logarithm is $\dfrac{1}{\text{inner}}$, so

it is $\dfrac{1}{2+\cos x}$ here, never $\dfrac{1}{x}$.

13.6.2 The product rule

To differentiate a product of two functions, you cannot simply multiply their derivatives. The mistake is a natural one to make, and the reason it fails is instructive.

Take $u = x$ and $v = x$, so that $y = uv = x^2$. The derivative of x^2 is $2x$. But $u' = 1$ and $v' = 1$, and $u'v' = 1$. The two answers do not agree, so multiplying derivatives is simply wrong.

The reason is that a product changes for two separate reasons at the same time.

Suppose both u and v increase. Part of the increase in uv comes from the change in u, multiplied by the whole of v. The rest comes from the change in v, multiplied by the whole of u. The correct rule adds these two parts, which is why each term keeps one factor undifferentiated.

A rectangle makes this concrete. Let u and v be the two side lengths, so that uv is the area. Now stretch both sides slightly. The extra area is a thin strip along the top, of area (change in v) $\times u$, plus a thin strip along the side, of area (change in u) $\times v$, plus a tiny corner square where the two strips meet. That corner is a product of two small quantities, so it vanishes in the limit and contributes nothing. The two strips are the two terms of the product rule.

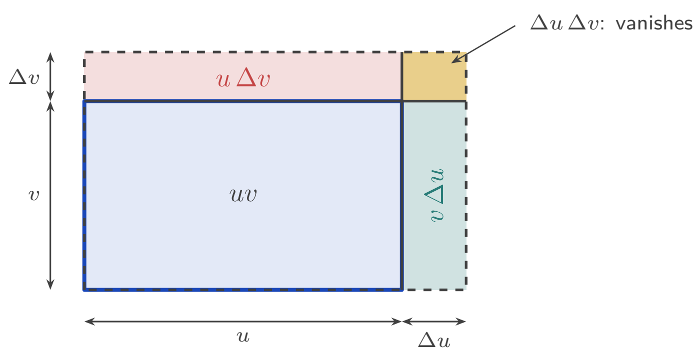


Figure 13.5 The product rule as an area. Growing both sides of a rectangle adds a strip along the top of area $u\,\Delta v$, a strip along the side of area $v\,\Delta u$, and a small corner of area $\Delta u\,\Delta v$. The two strips give the two terms of the rule; the corner is a product of two small quantities and vanishes in the limit.

KEY · IB If $y = uv$, then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

EXAMPLE 13.15

Differentiate $y = (x^2+1)e^x$.
Let $u = x^2+1$ and $v = e^x$, so $u' = 2x$ and $v' = e^x$:

$$\frac{dy}{dx} = (x^2+1)e^x + e^x(2x) = e^x(x^2+2x+1) = (x+1)^2e^x.$$

Factorise the answer where you can. A factorised form makes later work much easier, for example finding where $\frac{dy}{dx} = 0$.

The rules combine, and most examination questions require it. When one factor of a product is itself a composite, the product rule supplies the shape of the answer and the chain rule supplies that factor's derivative. Find each of u' and v' completely before assembling anything.

EXAMPLE 13.16

Differentiate $y = x^3e^{2x}$.
Let $u = x^3$ and $v = e^{2x}$, so $u' = 3x^2$. The second factor is a composite, with outer e^w and inner $w = 2x$, so the chain rule gives

$$v' = e^{2x} \cdot 2 = 2e^{2x}.$$

Now assemble the product rule:

$$\frac{dy}{dx} = x^3\left(2e^{2x}\right) + e^{2x}\left(3x^2\right) = e^{2x}\left(2x^3+3x^2\right) = x^2(2x+3)e^{2x}.$$

The exponential factor is never zero, so the factorised form shows at once that the gradient vanishes only at $x = 0$ and $x = -\frac{3}{2}$.

13.6.3 The quotient rule

For a quotient of two functions, the rule is similar, but the order of the terms now matters because a subtraction is involved.

The rule is not a new idea. A quotient $\dfrac{u}{v}$ is the product $u \times v^{-1}$, so the product and chain rules together produce the quotient rule, and deriving it once shows that nothing new is being assumed.

The minus sign has a plain meaning. Increasing the top of a fraction increases its value, so the u term is positive. Increasing the bottom *decreases* the value, so the v term is negative. The v^2 in the denominator comes from the same source: dividing by v twice, once for the original fraction and once for the rate at which dividing by v itself changes.

KEY · IB If $y = \dfrac{u}{v}$, then

$$\frac{dy}{dx} = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$$

EXAMPLE 13.17

Differentiate $y = \dfrac{x}{x^2+1}$.

Let $u = x$ and $v = x^2+1$, so $u' = 1$ and $v' = 2x$:

$$\frac{dy}{dx} = \frac{(x^2+1)(1) - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}.$$

The order in the numerator is: bottom times derivative of top, minus top times derivative of bottom. Reversing the two terms changes the sign of the answer, and it is a common error.

YOUR TURN 13.6 The Rules of Differentiation

27. Use the chain rule to differentiate each of the following.

(a) $y = (2x+1)^4$ (b) $y = \sin(x^2)$

(c) $y = e^{3x}$ (d) $y = \sqrt{3x+1}$

28. Use the product rule to differentiate each of the following.

(a) $y = x\cos x$ (b) $y = x^2e^x$

(c) $y = x^3\ln x$ (d) $y = x\sin x$

29. Use the quotient rule to differentiate each of the following.

(a) $y = \dfrac{x}{x+1}$ (b) $y = \dfrac{x+1}{x-1}$

(c) $y = \dfrac{\sin x}{x}$ (d) $y = \dfrac{e^x}{x}$

30. Explain, with a counterexample, why the derivative of a product is not the product of the derivatives.

31. Find the gradient of $y = (x^2-1)^3$ at the point where $x = 2$.

32. A student writes $\dfrac{d}{dx}\sin(3x) = \cos(3x)$. Identify the error and give the correct derivative.

13.7 Higher Derivatives and Concavity

The derivative $f'(x)$ is itself a function, so it can be differentiated again. The result is the **second derivative** $f''(x)$, also written $\dfrac{d^2y}{dx^2}$.

If f' measures how fast f changes, then f'' measures how fast that *rate* changes. For motion this is easy to name: f is position, f' is velocity and f'' is acceleration.

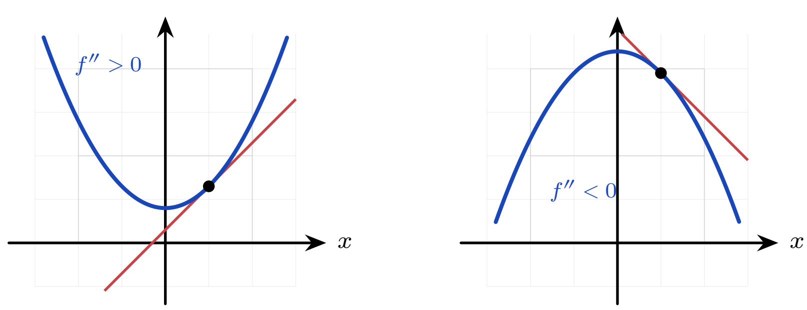
You can continue. Differentiating n times gives the nth derivative, written $\dfrac{d^ny}{dx^n}$ or $f^{(n)}(x)$. You will need that notation for Maclaurin series. See Ch. 16.

Together, the signs of f' and f'' describe the shape of a graph completely. They answer two different questions, so keep them separate.

Sign	f': going up or down?	f'': bending which way?
positive	**increasing**	**concave up** (bends upward)
negative	**decreasing**	**concave down** (bends downward)
zero	**stationary point**: the curve is flat there	**point of inflexion**, *if* the concavity also changes

The two are independent. A curve can be increasing and concave down at the same time: it is rising, but by less and less. It can also be decreasing and concave up: it is falling, but by less and less.

Two pictures make concavity concrete. The first is the tangent line. Where a curve is concave up it lies *above* its own tangents, so a tangent touches the curve and then falls away beneath it. Where the curve is concave down, the curve lies below its tangents. The second is a steering wheel. Drive along the curve from left to right: concave up means you are steering left, concave down means you are steering right, and a point of inflexion is the instant the wheel passes through straight ahead.



concave up: curve *above* its tangent   concave down: curve *below* its tangent

Figure 13.6 The tangent test for concavity. A tangent (scarlet) touches the curve at one point and then leaves it. Where the curve is concave up it stays above every tangent; where it is concave down it stays below.

Notice what f'' is really reporting on. It describes the *slope*, not the height. Concave up means the slope is increasing, and that can happen while the curve is falling, because a slope of -5 increasing to -1 is still an increase. This is what makes concavity useful outside mathematics. If $f(t)$ is the total number of cases in an epidemic, then $f' > 0$ says the outbreak is still growing, while $f'' < 0$ says the growth is slowing. The two together are the difference between bad news and news that is starting to improve. The point of inflexion of f is the moment the daily count peaks, which is why it is the number reported first.

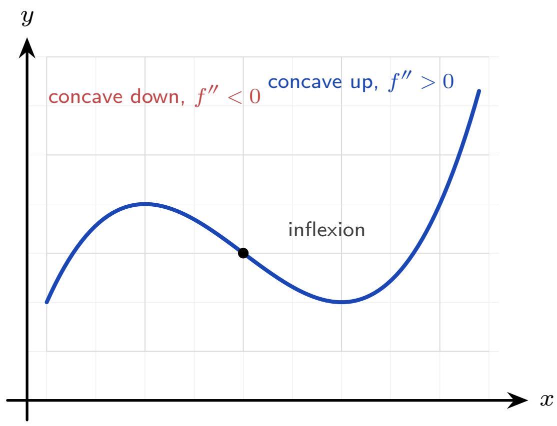


Figure 13.7 The second derivative decides which way the curve bends. Where the bending changes direction the curve has a point of inflexion.

EXAMPLE 13.18

For $f(x) = x^3 - 9x^2$, find $f'(x)$ and $f''(x)$, then state where f is increasing and where it is concave up.

Differentiate twice:

$$f'(x) = 3x^2 - 18x = 3x(x-6), \qquad f''(x) = 6x - 18 = 6(x-3).$$

f is increasing where $f'(x) > 0$, that is where $3x(x-6) > 0$, giving $x < 0$ or $x > 6$.
f is concave up where $f''(x) > 0$, that is $x > 3$.
At $x = 3$ the concavity changes from down to up: this is a **point of inflexion**, which you will study in Ch. 14.

13.7.1 Classifying a stationary point

Solving $f'(a) = 0$ locates the flat points of a curve but says nothing about what kind of point each one is. Flat describes the top of a rise, the bottom of a dip, and the brief pause of $y = x^3$ at the origin equally well.

Concavity settles it, because it reports which way the curve bends there. A curve that is flat and bending upwards is sitting at the bottom of a dip; flat and bending downwards, at the top of a rise. Since the sign of f'' is exactly that bend, one substitution decides the case. This is the **second-derivative test**.

So when $f'(a) = 0$, the sign of $f''(a)$ names the point: positive means concave up and a local minimum, negative means concave down and a local maximum. A zero second derivative leaves the question open, and the sign of f' on each side of a has to settle it instead. The test is developed in full, with the tangent picture behind it and a method for setting the work out, in Ch. 14.

That last case is a genuine gap, not a rare technicality, and it does not mean the point must be an inflexion. Both $y = x^4$ and $y = x^3$ have $f'(0) = 0$ and $f''(0) = 0$, yet the first has a minimum at the origin and the second a horizontal inflexion. Only the sign of f' either side separates them: for x^4 it runs negative then positive, while for x^3 it stays positive throughout.

EXAMPLE 13.19

Find the two stationary points of $f(x) = 2x^3 + 3x^2 - 36x + 4$ and determine the nature of each. Differentiate twice:

$$f'(x) = 6x^2 + 6x - 36 = 6(x+3)(x-2), \qquad f''(x) = 12x + 6.$$

The stationary points are where $f'(x) = 0$, at $x = -3$ and $x = 2$, and their heights are

$$f(-3) = -54 + 27 + 108 + 4 = 85, \qquad f(2) = 16 + 12 - 72 + 4 = -40.$$

Now substitute each into f'':

$$f''(-3) = -36 + 6 = -30 < 0, \qquad f''(2) = 24 + 6 = 30 > 0.$$

So $(-3, 85)$ is a local maximum and $(2, -40)$ is a local minimum.
The word *local* is doing real work: 85 is the largest value of f near $x = -3$, not the largest value of f anywhere, since a cubic with positive leading coefficient climbs without bound to the right.

13.7.2 Optimisation

A great many practical questions ask for a largest or smallest value: the greatest volume from a fixed sheet of card, the least fencing for a required area, the cheapest design. Each becomes a stationary-point question once the quantity is written as a function of one variable.

Two steps are easy to leave out, and both carry marks. State the domain, since the variable is usually a length and cannot take every real value, and use it to discard roots that solve the equation but not the problem. Then justify the nature of the point with f'' rather than asserting it, and finish by answering the question actually asked.

EXAMPLE 13.20

A rectangular poster carries 200 cm^2 of printed text, with a margin of 4 cm at each side and 2 cm at the top and at the bottom. Find the dimensions of the smallest poster that holds it, and state its area.

Let x cm be the width of the printed block. Its area fixes its height at $\dfrac{200}{x}$ cm, so the poster measures $x+8$ by $\dfrac{200}{x}+4$, and

$$A = (x+8)\left(\frac{200}{x}+4\right) = 200 + 4x + \frac{1600}{x} + 32 = 232 + 4x + \frac{1600}{x}.$$

A width is positive, which fixes the domain: $x > 0$.

Differentiate and solve:

$$\frac{dA}{dx} = 4 - \frac{1600}{x^2} = 0 \implies x^2 = 400 \implies x = \pm 20.$$

The root $x = -20$ solves the equation but not the problem, and the domain discards it, leaving $x = 20$. The second derivative confirms the type:

$$\frac{d^2A}{dx^2} = \frac{3200}{x^3}, \qquad \text{so at } x = 20, \quad \frac{d^2A}{dx^2} = 0.4 > 0,$$

so $x = 20$ gives a minimum. The printed block is then 20 cm by $\dfrac{200}{20} = 10$ cm, and the poster is

$$28 \text{ cm} \times 14 \text{ cm}, \qquad A = 392 \text{ cm}^2.$$

Since $x^3 > 0$ across the whole domain, $\dfrac{d^2A}{dx^2}$ is positive everywhere on it, so this local minimum is the smallest poster available anywhere on it.

13.7.3 Reading f, f' and f'' together

A common examination question shows one of these three graphs and asks about another. It helps to see all three at the same time.

Below are $f(x) = x^3 - 3x$ and its two derivatives, drawn one above the other on the same x-axis. The vertical scales are not the same. Only the zeros and the signs matter here.

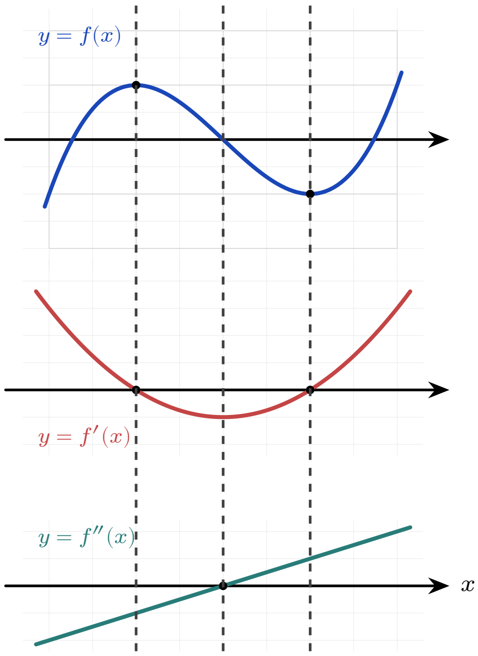


Figure 13.8 f, f' and f'' for $f(x) = x^3 - 3x$, stacked over a shared x-axis. Read down each dashed line: a turning point of f sits above a zero of f', and the inflexion of f sits above a turning point of f' and a zero of f''.

Read down each dashed line. At every dashed line, something happens in all three graphs at once, and the three results below record those links. Examinations test them in both directions: from f to f', and from f' back to f.

- A **turning point of** f sits above a **zero of** f'. At $x = -1$ the curve f has a maximum, and there f' crosses zero from positive to negative. At $x = 1$ the curve has a minimum, and there f' crosses zero from negative to positive.
- An **interval where** f **rises** sits above an **interval where** f' **is positive**, that is, where the graph of f' is above the axis.
- An **inflexion of** f sits above a **turning point of** f', and above a **zero of** f''. In this example all three occur at $x = 0$.

Use the table in either direction. Given any one of the three graphs, you can sketch the other two.

YOUR TURN 13.7 Higher Derivatives and Concavity

33. Find the second derivative of each of the following.

(a) $f(x) = x^3 - 6x^2$

(b) $y = x^4$

(c) $y = \sin x$

(d) $y = e^{2x}$

34. Find the interval or intervals on which each function is increasing.

(a) $f(x) = x^2 - 4x$

(b) $f(x) = x^4 - 2x^2$

(c) $f(x) = \sin x$, for $0 \le x \le 2\pi$

(d) $f(x) = xe^{-x}$

35. Find the interval on which each curve is concave up.

(a) $y = x^3$

(b) $y = x^3 - 3x^2$

(c) $y = \cos x$, for $0 \le x \le 2\pi$

(d) $y = \ln x$, for $x > 0$

36. For $f(x) = x^3 - 3x^2$, find the coordinates of the point of inflexion and confirm that the concavity changes there.

37. The graph of $y = f'(x)$ is a parabola with roots at $x = -2$ and $x = 3$, opening upwards. State the x-coordinates of the stationary points of f, and identify the nature of each.

38. A particle moves so that its displacement after t seconds is $s = 2t^3 - 3t^2$ metres. Find expressions for the velocity and the acceleration, and find the time at which the acceleration is zero.

13.8 L'Hôpital's Rule HL

This chapter began with the indeterminate form $\frac{0}{0}$, and solved it by factorising. Factorising fails as soon as the functions are trigonometric, exponential or logarithmic.

There is a second method for those cases. If a limit gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the derivative gives the answer. The rule is named after the French mathematician Guillaume de l'Hôpital, and the name is said *lo - pee - TAL*: the letter s is silent, and so is the h.

KEY **L'Hôpital's rule.** If $\lim_{x \to a} \dfrac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists.

Differentiate the top and the bottom *separately*. This is not the quotient rule. Then try the limit again, and if it is still indeterminate, apply the rule a second time.

The reason the rule works is short, and it explains why it needs the $\frac{0}{0}$ form.

Suppose $f(a) = 0$ and $g(a) = 0$. Then near $x = a$ each function is close to its own tangent line at a, so

$$f(x) \approx f'(a)(x-a), \qquad g(x) \approx g'(a)(x-a).$$

Divide one by the other. The factor $(x-a)$ appears in both, so it cancels:

$$\frac{f(x)}{g(x)} \approx \frac{f'(a)(x-a)}{g'(a)(x-a)} = \frac{f'(a)}{g'(a)}.$$

So the quotient of the functions approaches the quotient of their gradients. This is why both functions must vanish at a: only then does each one reduce to a multiple of $(x-a)$, and only then does the cancelling work.

The limit $\lim\limits_{x\to 0} \dfrac{\cos x}{x}$ from §13.1 shows what happens when they do not. Here $\cos 0 = 1$, not 0, so the form is $\frac{1}{0}$, which is not indeterminate and not one the rule covers. Differentiating top and bottom anyway would give

$$\lim_{x\to 0} \frac{-\sin x}{1} = 0,$$

and that is simply wrong: the limit does not exist, because the fraction runs to $+\infty$ on one side of 0 and to $-\infty$ on the other. Check the form before you differentiate.

EXAMPLE 13.21

Evaluate (a) $\lim\limits_{x\to 0} \dfrac{\sin x}{x}$ and (b) $\lim\limits_{x\to 0} \dfrac{\ln(1+x)-x}{x^2}$.

(a) Substituting gives $\frac{0}{0}$. Differentiate top and bottom:

$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = \cos 0 = 1.$$

This confirms the limit stated in §13.1.

(b) Substituting gives $\frac{0}{0}$. One application gives $\dfrac{1/(1+x)-1}{2x}$, still $\frac{0}{0}$, so apply the rule again:

$$\lim_{x\to 0} \frac{\ln(1+x)-x}{x^2} = \lim_{x\to 0} \frac{\dfrac{1}{1+x}-1}{2x} = \lim_{x\to 0} \frac{-\dfrac{1}{(1+x)^2}}{2} = -\frac{1}{2}.$$

CAUTION L'Hôpital's rule applies *only* to the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$. Always check that the limit is indeterminate before differentiating. Applying it to a form like $\frac{2}{0}$ or $\frac{3}{5}$ gives a wrong answer.

EXAMPLE 13.22

Evaluate $\lim\limits_{x\to\infty} \dfrac{\ln x}{x}$.

As $x \to \infty$ this is $\frac{\infty}{\infty}$. Differentiate top and bottom:

$$\lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0.$$

The denominator increases without bound while the numerator increases very slowly, so the ratio approaches zero. L'Hôpital's rule makes that comparison exact.

YOUR TURN 13.8 L'Hôpital's Rule HL

39. Evaluate each limit, checking first that it is indeterminate.

(a) $\lim_{x\to 0} \frac{e^x - 1}{x}$

(b) $\lim_{x\to 0} \frac{\sin 2x}{x}$

(c) $\lim_{x\to 0} \frac{\tan x}{x}$

(d) $\lim_{x\to\infty} \frac{\ln x}{x^2}$

40. Each of these needs the rule applied twice. Evaluate them.

(a) $\lim_{x\to 0} \frac{1 - \cos x}{x^2}$

(b) $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$

41. Evaluate $\lim_{x\to\infty} \frac{x^2+1}{2x^2-3}$, first by L'Hôpital's rule and then by dividing every term by x^2. Confirm that the two methods agree.

42. A student evaluates $\lim_{x\to 0} \frac{x+2}{x+1}$ by differentiating top and bottom, obtaining 1. Explain the error and give the correct value.

43. State the two indeterminate forms to which L'Hôpital's rule applies, and explain why the rule must not be used on a form such as $\frac{2}{0}$.

Chapter 13 · Limits & Derivatives

Reflections

TOK The early calculus was built on "infinitesimals," quantities treated as nonzero when dividing and as zero when convenient. The philosopher George Berkeley mocked them in 1734 as "ghosts of departed quantities," and he had a point: the logic was shaky. The method gave correct answers for 150 years before the limit gave it a rigorous foundation. Is a mathematical technique justified by its results, or only by its proof? Calculus worked long before anyone could say exactly why.

TOK A derivative gives a rate of change at a single instant, yet no instant can be measured: every measurement takes time. Is an instantaneous rate something the world contains, or a device invented to describe it? Does it matter, if the device predicts correctly every time?

TOK The expression $\frac{0}{0}$ has no value, and yet a limit of that form often has one exactly. Can a symbol that means nothing still carry information? Where does the meaning sit: in the symbols, or in the process that produced them?

IM Calculus was created twice, independently. Isaac Newton developed his "method of fluxions" in England in the 1660s; Gottfried Leibniz developed his version, with the $\frac{dy}{dx}$ notation we still use, in Germany in the 1670s. A bitter priority dispute split European mathematics for a century. Newton's notation was used in Britain and Leibniz's in the rest of Europe. British mathematics fell behind, partly because it kept the more awkward symbols. Notation is not neutral: the right symbol can speed up a whole field.

RW The derivative is the language of change, so it appears wherever anything varies. Velocity and acceleration in physics, marginal cost in economics, reaction rates in chemistry, growth rates in biology: each is a derivative. When a news report says inflation is "rising but more slowly," it is making a claim about a second derivative, that the rate of change of prices is itself decreasing.

EXT L'Hôpital's rule has a deeper cousin. Near a point, many functions are well approximated by a polynomial, the Maclaurin series: $\sin x \approx x - \frac{x^3}{6}$, $e^x \approx 1 + x + \frac{x^2}{2}$. Substituting these into an indeterminate limit resolves it without differentiating repeatedly. The same problem that L'Hôpital's rule solves leads on to representing functions as infinite polynomials, one of the great ideas of analysis.

End-of-Chapter Problems — Chapter 13: Limits & Derivatives

1. Differentiate $y = e^x \sin x$, and hence find the gradient of the curve at $x = 0$. [4]

2. Find the equation of the tangent to $y = x^2$ that is parallel to the line $y = 6x - 1$. [4]

3. Evaluate $\lim_{x\to 0} \dfrac{e^{2x} - 1}{x}$. [4]

4. Differentiate each function from first principles, showing every step of the limit.

(a) $f(x) = x^3$ [4]

(b) $f(x) = \sqrt{x}$, for $x > 0$. *Multiply the difference quotient by the conjugate.* [4]

(c) $f(x) = e^x$. You may use $\lim_{h\to 0} \dfrac{e^h - 1}{h} = 1$. [3]

(d) $f(x) = \sin x$. You may use the compound-angle formula for $\sin(x + h)$, together with $\lim_{h\to 0} \dfrac{\sin h}{h} = 1$ and $\lim_{h\to 0} \dfrac{\cos h - 1}{h} = 0$. [5]

5. Using the quotient rule on $\tan x = \dfrac{\sin x}{\cos x}$, prove that $\dfrac{d}{dx} \tan x = \sec^2 x$. [4]

6. Consider $f(x) = x^3 - 3x$.

(a) Find $f'(x)$. [1]

(b) Find the coordinates of the points where the tangent is horizontal. [4]

7. GDC The function $f(x) = \dfrac{e^x - 1}{x}$ is not defined at $x = 0$.

(a) Copy and complete the table of values of $f(x)$, giving 5 significant figures: $x = -0.1,\ -0.01,\ 0.01,\ 0.1$. [2]

(b) Suggest the value of $\lim_{x\to 0} f(x)$. [1]

(c) Verify your conjecture using l'Hôpital's rule. [2]

8. Differentiate $f(x) = 3x^2 - 2x$ from first principles. [5]

9. Consider $f(x) = |x - 2|$.

(a) Sketch the graph of f. [1]

(b) Explain briefly why f is continuous at $x = 2$. [2]

(c) Explain why f is not differentiable at $x = 2$. [2]

10. Evaluate $\lim_{x \to 0} \dfrac{1 - \cos 3x}{x^2}$. [5]

11. Find the coordinates of the points on the curve $y = x^3 - 3x$ where the tangent is parallel to the line $y = 9x + 1$. [5]

12. A curve has equation $y = \dfrac{e^{2x}}{1 + x^2}$.

(a) Find $\dfrac{dy}{dx}$. [4]
(b) Find the gradient of the curve at $x = 0$. [1]

13. Evaluate $\lim_{x \to 0} \dfrac{x - \sin x}{x^3}$, justifying each application of L'Hôpital's rule. [5]

14. Consider $f(x) = x^2 - 2x$.

(a) Find $f'(x)$ from first principles. [4]
(b) Hence find the gradient of the curve at $x = 3$. [2]

15. The curve $y = x^2 - 2x$ is considered at the point where $x = 3$.

(a) Find the coordinates of the point. [1]
(b) Find the equation of the tangent at this point. [3]
(c) Find the equation of the normal at this point. [2]

16. Evaluate the following limits.

(a) $\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$ [3]
(b) $\lim_{x \to 0} \dfrac{e^x - 1 - x}{x^2}$ [3]

17. A curve has equation $y = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Find $\dfrac{dy}{dx}$. [3]
(b) Show that the curve has a horizontal tangent at $x = e$. [3]

18. Differentiate each of the following, simplifying your answers.

(a) $y = (2x^3 - 1)^4$ [2] (b) $y = x^2 \ln x$ [2]
(c) $y = \dfrac{\sin x}{x}$ [2] (d) $y = e^{3x} \cos x$ [3]
(e) $y = \sqrt{4x^2 + 1}$ [2] (f) $y = \dfrac{e^x}{x^2 + 1}$ [3]

19. Consider the curve $y = e^x$ at the point where $x = 1$.

(a) Show that the tangent there passes through the origin. [4]

(b) Find the equation of the normal at the same point. [2]

20. Differentiate each of the following.

(a) $y = \arctan(3x)$ [2] (b) $y = \sec(2x)$ [2]

(c) $y = \log_2 x$ [2] (d) $y = \arcsin(x^2)$ [2]

(e) $y = \operatorname{cosec} x$ [2] (f) $y = \cot(3x)$ [2]

21. Consider $f(x) = x^3 - 6x^2 + 9x + 1$.

(a) Find the intervals on which f is increasing. [3]

(b) Find the interval on which f is concave up. [2]

(c) State the x-coordinate of the point of inflexion. [1]

22. Find the equations of *both* tangents to the curve $y = x^2$ that pass through the point $(0, -1)$. [6]

23. Differentiate each of the following, simplifying where possible.

(a) $y = (3x^2 + 1)^5$ [2] (b) $y = x^2 e^x$ [2]

(c) $y = \dfrac{x}{x+1}$ [3] (d) $y = \ln(3x^2 + 2)$ [2]

(e) $y = \dfrac{\cos x}{x^2}$ [3] (f) $y = (x+1)^3(2x-5)^4$ [3]

24. Differentiate each of the following.

(a) $y = \sec x$ [1] (b) $y = \ln(\cos x)$ [3]

(c) $y = x \arctan x$ [3] (d) $y = \arccos(2x)$ [2]

(e) $y = e^{\sin x}$ [2] (f) $y = \dfrac{\arctan x}{x}$ [3]

25. The displacement of a particle is $s(t) = t^3 - 6t^2 + 9t$ metres at time t seconds, for $t \geq 0$.

(a) Find the velocity $v(t) = s'(t)$. [2]

(b) Find the times at which the particle is momentarily at rest. [3]

(c) Find the acceleration at $t = 3$. [2]

26. The derivative of a function f is $f'(x) = (x+2)(x-4)$.

(a) Find the intervals on which f is increasing. [2]

(b) Find the x-coordinates of the turning points of f and determine their nature. [3]

(c) Find the x-coordinate of the point of inflexion of f. [2]

27. Evaluate $\lim_{x\to 0} \frac{\tan x - x}{x^3}$. You may use $\lim_{x\to 0} \frac{\tan x}{x} = 1$ after establishing it. [7]

28. Consider $f(x) = x^3 - 3x^2 - 9x$.

(a) Find $f'(x)$ and the values of x where $f'(x) = 0$. [3]
(b) State the intervals on which f is increasing. [2]
(c) Find $f''(x)$ and the x-coordinate of the point of inflexion. [3]

29. Consider $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find $f'(x)$ and factorise it fully. [2]
(b) Find the coordinates of the two stationary points. [3]
(c) Find $f''(x)$, and use it to determine the nature of each stationary point. [3]
(d) Find the coordinates of the point of inflexion. [2]
(e) State the interval on which f is concave up. [2]

30. Let $g(x) = \frac{2x-1}{x+2}$, $x \neq -2$.

(a) Write down the equations of the vertical and horizontal asymptotes. [2]
(b) Use the quotient rule to show that $g'(x) = \frac{5}{(x+2)^2}$. [3]
(c) Hence explain why g is increasing on every interval of its domain. [2]
(d) Find the equation of the tangent to the curve at the point where $x = 0$. [2]
(e) Find the equation of the normal at the same point. [2]

31. Let $y = x^2 e^{-x}$.

(a) Use the product rule to show that $\frac{dy}{dx} = x(2-x)e^{-x}$. [3]
(b) Find the coordinates of the two stationary points. [3]
(c) Show that $\frac{d^2y}{dx^2} = (x^2 - 4x + 2)e^{-x}$. [3]
(d) Use the second derivative to determine the nature of each stationary point. [2]
(e) Describe the behaviour of y as $x \to \infty$, and state the equation of the asymptote. [2]

32. GDC A closed cylindrical can has volume 500 cm^3. Its radius is r cm and its height is h cm.

(a) Show that $h = \frac{500}{\pi r^2}$. [2]
(b) Hence show that the total surface area is $S = 2\pi r^2 + \frac{1000}{r}$. [3]
(c) Find $\frac{dS}{dr}$. [2]
(d) Find the value of r that minimises S, giving your answer to three significant figures. [3]
(e) Verify, using the second derivative, that this value gives a minimum. [2]
(f) Show that at this radius the height is exactly twice the radius. [2]

Challenge Questions

33. The power rule, proved. Section 13.3 stated the power rule after two examples suggested it. Here you prove it for every positive integer n, using the binomial theorem.

(a) Write down the first three terms of $(x+h)^n$ in ascending powers of h. [3]

(b) Hence show that

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + {}^nC_2 x^{n-2}h + \cdots + h^{n-1},$$

and explain why every term after the first still contains h. [5]

(c) Deduce that $\frac{d}{dx}x^n = nx^{n-1}$ for every positive integer n. [3]

(d) Check your working by expanding in full for $n = 4$. [3]

(e) The rule holds for negative and fractional n as well, but this argument does not reach them. State which step fails when $n = -1$, then confirm that case by differentiating $\frac{1}{x}$ from first principles. [4]

34. GDC **Where the derivative of sine comes from.** The standard derivatives in §13.5 rest on one limit. Here you establish that limit, then use it to prove the first of them.

(a) Evaluate $\dfrac{\sin x}{x}$ at $x = 0.1$, 0.01 and 0.001 radians, and at the matching negative values. Use the results as evidence that $\lim\limits_{x\to 0}\dfrac{\sin x}{x} = 1$. [3]

(b) Show that $\dfrac{\cos x - 1}{x} = \dfrac{-\sin^2 x}{x(\cos x + 1)}$. Hence show that $\lim\limits_{x\to 0}\dfrac{\cos x - 1}{x} = 0$. *(Hint: multiply above and below by $\cos x + 1$)* [5]

(c) The compound-angle identity gives $\sin(x+h) = \sin x\cos h + \cos x \sin h$. Use it to show that

$$\frac{\sin(x+h) - \sin x}{h} = \sin x\left(\frac{\cos h - 1}{h}\right) + \cos x\left(\frac{\sin h}{h}\right).$$

[4]

(d) Hence prove from first principles that $\dfrac{d}{dx}\sin x = \cos x$. [3]

(e) Explain why the proof fails if x is measured in degrees. [2]

35. Two routes to the same limit. An indeterminate limit can be settled by L'Hôpital's rule, or by replacing each function with its Maclaurin series. Here you use both on $\lim\limits_{x\to 0}\dfrac{x - \sin x}{x^3}$ and compare them.

(a) Evaluate the limit by applying L'Hôpital's rule repeatedly, checking at each stage that the form is still indeterminate. [5]

(b) The Maclaurin series for $\sin x$ begins $x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots$. Substitute it into the fraction and simplify, and the limit follows in one step. [4]

(c) Use the series method on $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$, given that $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$. [3]

(d) Give one advantage of each method, and one limit for which the series method would be harder to use. [4]

CALCULUS

Chapter 14

Derivative Applications

14

SYLLABUS 5.8 5.9 5.14

Throw a ball straight up. It rises, it slows, and at the very top it stops for an instant before it falls back. At that instant its velocity is exactly zero. A moment earlier the ball was moving up; a moment later it is moving down. The top of the flight is the one place where it is doing neither.

That instant is the key to a large class of problems. The top of the flight is where the height is greatest, and it is also where the velocity is zero. The same pattern holds everywhere: when a quantity reaches its largest or smallest value, its rate of change is zero at that moment, and the curve flattens.

This is why the derivative is the right tool for finding a best value. The largest volume from a fixed sheet of metal, the least metal for a can, the quickest route, the cheapest design: in each case the answer sits where the derivative is zero. So in this chapter the derivative stops being something you calculate and becomes something you use. You will find the peaks and valleys of a curve and decide which is which, read the shape of a graph from the signs of f' and f'', find a largest or smallest value under a constraint, describe motion in a straight line, and differentiate implicitly when two quantities change together. The course spends this long on it because a derivative is not only a gradient. It is a rate of change, and many real questions ask how fast something is changing, or where a quantity is greatest or least.

The ball at the top of its flight carries the principle: to find an extreme value, find where the rate of change is zero.

14.1 Stationary Points and Their Nature

A **stationary point** is a point where the gradient is zero: the tangent is horizontal. Since the derivative gives the gradient, stationary points are exactly the solutions of $f'(x) = 0$. They are the points where a curve stops rising or stops falling.

For a smooth curve, every local maximum and minimum is a stationary point. The condition $f'(x) = 0$ finds them all. It does not find an extreme value at a corner, where the derivative does not exist: $f(x) = |x|$ has a minimum at $x = 0$ and no stationary point there. See Ch. 13.

DEF A **stationary point** of f occurs where $f'(x) = 0$. At a **local maximum** the curve changes from increasing to decreasing; at a **local minimum**, from decreasing to increasing.

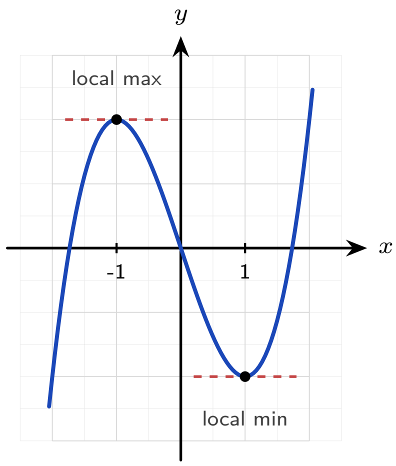


Figure 14.1 At each stationary point the tangent is horizontal. The second derivative decides which is the maximum and which the minimum.

These points deserve a name of their own. A stationary point is a moment of balance. A quantity that has been growing stops growing there, and a quantity that has been falling stops falling. A smooth curve cannot still be rising when it reaches its highest point, so every peak and every valley must be a place where the gradient is zero. That one fact turns a question about *size* into a question about *slope*, and slope is exactly what a derivative measures.

A stationary point is a candidate, however, not an answer. Zero gradient tells you the curve is flat there, and flat can mean a peak, a valley, or only a brief pause on the way past, as happens for $y = x^3$ at the origin. Deciding which is the work of the next two subsections.

The condition then reappears throughout the course and well beyond it. A projectile is at its greatest height when its vertical velocity is zero (§14.4). A manufacturer's profit is largest, and its cost least, where the rate of change with respect to the number of items produced is zero (§14.3). In physics a system rests in equilibrium where its potential energy is stationary. In statistics the mode of a continuous distribution sits where the density function reaches a maximum Ch. 12. When a machine-learning model is trained, the computer is searching for a point where the gradient vanishes. Different subjects, one condition: $f'(x) = 0$.

Finding stationary points is only half the work. The other half is deciding what kind each one is: a maximum, a minimum, or neither. There are two tests.

14.1.1 The first derivative test

The sign of f' tells you whether the curve is rising or falling, so it can be read on either side of the stationary point. Just before a maximum the curve rises, $f' > 0$, and just after it falls, $f' < 0$. A minimum is the reverse. Record the three signs in order and the shape is decided.

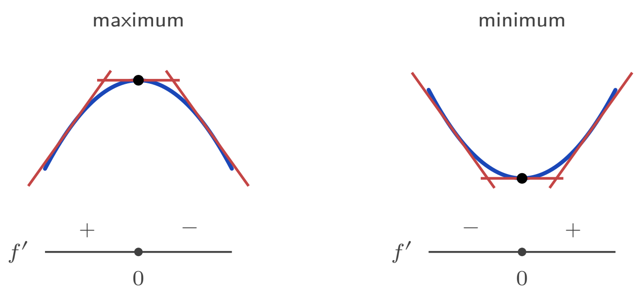


Figure 14.2 The first derivative test. The scarlet strokes are tangents: rising, flat, then falling at a maximum, and the reverse at a minimum. The strip below records the sign of f' on each side.

14.1.2 The second derivative test

This test is usually faster, and it uses the second derivative, which measures **concavity**. Two words name the two cases, and examinations use both. Where $f''(x) > 0$ the curve is **concave up**, bending like a cup that would hold water; where $f''(x) < 0$ it is **concave down**, bending like a cap. The sign of the second derivative is the definition, not a hint. There is a picture behind it. At a minimum the curve bends upward, so it sits *above* its own horizontal tangent, and $f'' > 0$. At a maximum it bends downward, sits *below* the tangent, and $f'' < 0$.

Figure 14.3 The second derivative test. A curve that bends upward lies above its horizontal tangent, and one that bends downward lies below it. That is what the sign of f'' records.

KEY At a stationary point where $f'(x) = 0$:

$$f''(x) > 0 \Rightarrow \text{local minimum}, \qquad f''(x) < 0 \Rightarrow \text{local maximum}.$$

If $f''(x) = 0$ this test gives no answer. Use the sign of f' on either side instead.

EXAMPLE 14.1

Find and classify the stationary points of $f(x) = x^3 + 3x^2 - 9x + 2$.
Differentiate and solve $f'(x) = 0$:

$$f'(x) = 3x^2 + 6x - 9 = 3(x+3)(x-1) = 0 \implies x = -3 \text{ or } x = 1.$$

The second derivative is $f''(x) = 6x + 6$.
At $x = -3$: $f''(-3) = -12 < 0$, a local maximum. $f(-3) = 29$, so $(-3, 29)$.
At $x = 1$: $f''(1) = 12 > 0$, a local minimum. $f(1) = -3$, so $(1, -3)$.
Here the maximum is above the minimum, which is normal. The word "local" means highest or lowest *among nearby points*, not over the whole curve.

The method does not depend on the function being a polynomial. Any function you can differentiate can be searched for stationary points in exactly the same way.

EXAMPLE 14.2

Find and classify the stationary point of $f(x) = x \ln x$, for $x > 0$.
The product rule gives

$$f'(x) = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1.$$

Set $f'(x) = 0$: $\ln x = -1$, so $x = e^{-1} \approx 0.368$.
The second derivative is $f''(x) = \dfrac{1}{x}$, which is positive for every x in the domain. So the point is a local minimum, and

$$f\left(\tfrac{1}{e}\right) = \tfrac{1}{e} \ln \tfrac{1}{e} = -\tfrac{1}{e} \approx -0.368.$$

The minimum is at $\left(\frac{1}{e}, -\frac{1}{e}\right)$. Two details matter here. The domain $x > 0$ comes from the logarithm, not from the question, and it must be stated. And because $f'' > 0$ everywhere, the curve is concave up on its whole domain, so this local minimum is in fact the lowest point of the entire curve.

YOUR TURN 14.1 Stationary Points and Their Nature

1. Find the stationary points of each curve.

(a) $y = x^2 - 6x + 5$

(b) $y = x^3 - 12x$

(c) $y = 2x^3 - 3x^2 - 12x + 1$ (d) $y = x^4 - 2x^2$

2.

(a) Use the second derivative to classify the stationary point of $y = x^2 - 6x + 5$.
(b) Use the second derivative to classify both stationary points of $y = x^3 - 12x$.
(c) Find the stationary point of $y = x + \frac{1}{x}$ for $x > 0$, and state its nature.
(d) A curve has $\frac{dy}{dx} = (x-1)(x-4)$. Write down the x-coordinates of its stationary points, and state the nature of each without differentiating again.

3. The curve $y = x^3$ has a stationary point at the origin.

(a) Show that $x = 0$ is a stationary point.
(b) Explain why the second derivative test cannot classify it.
(c) Use the sign of $\frac{dy}{dx}$ on either side to classify it.

4. Find the stationary point of each curve and determine its nature.

(a) $y = e^x - 3x$
(b) $y = x^2 \ln x$, for $x > 0$
(c) $y = \sin x + \cos x$, for $0 \le x \le 2\pi$ (there are two)
(d) $y = xe^{-2x}$

14.2 Points of Inflexion and Concavity

A stationary point is not the only named feature of a curve. A **point of inflexion** is where the curve changes the *direction of its bend*: from concave up to concave down, or the reverse. It is where f'' changes sign, which requires $f''(x) = 0$.

DEF A **point of inflexion** is a point where the concavity changes sign. For a twice-differentiable function a necessary condition is $f''(x) = 0$, but $f''(x) = 0$ alone is not sufficient: the concavity must actually change.

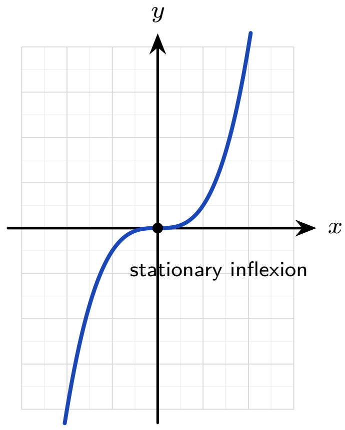


Figure 14.4 At the origin $y = x^3$ has both a horizontal tangent and a change of concavity, so this inflexion is also a stationary point.

An inflexion can happen at any gradient, and the syllabus names both kinds.

In $y = x^3$ above, the inflexion at the origin has **zero gradient**. The tangent is horizontal, so this point is also a stationary point, but it is neither a maximum nor a minimum.

In the next example the inflexion has a **non-zero gradient**. The curve is falling as it passes through.

Only one thing makes a point an inflexion: the concavity changes there. The gradient can take any value.

EXAMPLE 14.3

Find the point of inflexion of $f(x) = x^3 - 3x^2$.
Differentiate twice:

$$f'(x) = 3x^2 - 6x, \qquad f''(x) = 6x - 6.$$

Set $f''(x) = 0$: $6x - 6 = 0 \implies x = 1$. To confirm it is genuine, check that f'' changes sign: for $x < 1$, $f'' < 0$ (concave down); for $x > 1$, $f'' > 0$ (concave up). It changes, so $x = 1$ is a point of inflexion. Since $f(1) = -2$, the point is $(1, -2)$.
Note the gradient there: $f'(1) = 3 - 6 = -3 \neq 0$. This is a *non-stationary* inflexion, the curve changing its bend mid-descent.

14.2.1 Inflexions of other functions

Polynomials are not the only curves that bend both ways. Take $y = \sin x$ and differentiate twice:

$$f'(x) = \cos x, \qquad f''(x) = -\sin x.$$

So $f''(x) = 0$ whenever $\sin x = 0$, that is at $x = 0, \pm\pi, \pm 2\pi, \dots$. At each of these the concavity genuinely changes, because $-\sin x$ changes sign every time the sine crosses the axis. The sine curve therefore has a point of inflexion at *every* multiple of π: infinitely many of them. None is stationary, since $f'(x) = \cos x = \pm 1$ there. Each inflexion is the steepest point on the curve, which is the same picture as before: the curve stops bending one way and starts bending the other exactly where it is climbing or falling fastest.

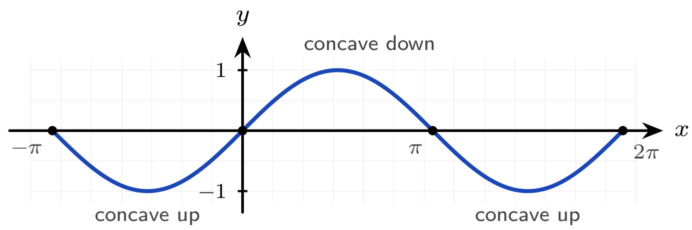


Figure 14.5 Every point where $y = \sin x$ crosses the axis is a point of inflexion: the bend reverses there, and the gradient is ± 1, the steepest the curve ever gets. None of these inflexions is stationary.

Other familiar curves have no point of inflexion at all. For $y = e^x$ every derivative is e^x again, so $f''(x) = e^x > 0$ for every x: the curve is concave up everywhere and its bend never reverses. For $y = \ln x$ we have $f'(x) = \dfrac{1}{x}$ and $f''(x) = -\dfrac{1}{x^2} < 0$ for every $x > 0$, so it is concave down across its whole domain. A parabola is the same story with the sign the other way: for $y = x^2$, $f'' = 2 > 0$ everywhere, so it is concave up everywhere. What all three share is that f'' never changes sign. An inflexion needs the concavity to *change*; if f'' keeps one sign, there is nothing to find.

EXAMPLE 14.4

Find the points of inflexion of $f(x) = e^{-x^2}$.

The chain rule gives $f'(x) = -2xe^{-x^2}$. Differentiating again needs the product rule:

$$f''(x) = -2e^{-x^2} + (-2x)(-2x)e^{-x^2} = \left(4x^2 - 2\right)e^{-x^2}.$$

Since $e^{-x^2} > 0$ for every x, the sign of f'' is decided by $4x^2 - 2$ alone. This is zero when $x^2 = \frac{1}{2}$, so $x = \pm\frac{1}{\sqrt{2}} \approx \pm 0.707$, and it changes sign at each: negative between the two values, positive outside them. Both are therefore genuine points of inflexion, and $f\left(\pm\frac{1}{\sqrt{2}}\right) = e^{-1/2} \approx 0.607$.

So the curve is concave down over the central hump and concave up in each tail. This is the bell shape of Ch. 12, and the same calculation on the standard normal density $e^{-x^2/2}$ puts its inflexions at $x = \pm 1$: exactly one standard deviation from the mean. The points where a bell curve stops curving downward and starts flattening out are not decoration. They mark the standard deviation.

CAUTION $f''(x) = 0$ does not by itself guarantee an inflexion. For $f(x) = x^4$, $f''(0) = 0$, yet the curve is concave up on both sides: $x = 0$ is a minimum, not an inflexion. Always confirm the concavity actually changes.

YOUR TURN 14.2 Points of Inflexion and Concavity

5. Find the point of inflexion of each curve.

(a) $y = x^3 - 6x^2 + 5$
(b) $y = x^3 + 3x^2$
(c) $y = 2x^3 - 9x^2 + 12x$

6.

(a) Explain why $y = x^4$ has no point of inflexion at $x = 0$.
(b) Determine, for $y = x^5$, whether the point where $f'' = 0$ is a genuine inflexion.
(c) State the interval on which $y = x^3 - 6x^2 + 5$ is concave down.

7.

(a) Find the coordinates of the point of inflexion of $y = xe^x$.
(b) Find the x-coordinates of the points of inflexion of $y = 2x + \cos x$ for $0 \le x \le 2\pi$.
(c) Explain why $y = \ln x$ has no point of inflexion.
(d) Show that $y = e^{-x^2/2}$ has points of inflexion at $x = \pm 1$.

14.3 Optimization

Optimisation means finding the largest or smallest value of a real quantity: the greatest volume, the least cost, the shortest time.

Why does the derivative help at all? Without it, the only way to find a best value would be to try inputs one at a time and compare the results. That can suggest an answer but never settles it, because between any two inputs you tried there are infinitely many you did not. The derivative removes the search. A maximum or a minimum leaves a mark that algebra can detect: the curve is flat there, so $f'(x) = 0$. An impossible hunt through infinitely many values becomes one equation to solve, and the handful of solutions are the only candidates to check.

In practice the calculus is the easy part; the work is in setting up the function.

Almost every optimisation question follows the same five steps.

1. **Name the quantity to be optimised** and write an expression for it. At this stage it may contain more than one variable.
2. **Find the constraint**, the second equation linking those variables. It is always in the

question, often as a fixed volume, a fixed length of material, or a fixed perimeter.

3. **Use the constraint to eliminate variables**, until the quantity is a function of one variable only. State the values that variable may take.
4. **Differentiate, set the derivative to zero, and solve.**
5. **Justify the nature of the point**, then answer the question that was actually asked.

Steps 3 and 5 are the ones to watch. Reducing to a single variable is where the algebra goes wrong, and skipping the justification is where marks are dropped at the end.

TIP An optimisation question is not finished when you solve $f'(x) = 0$. Two marks usually depend on what follows: justify the nature of the point, using f'' or the sign of f', and then answer the question that was asked. If the question asks for the maximum volume, give the volume, not the value of x that produces it.

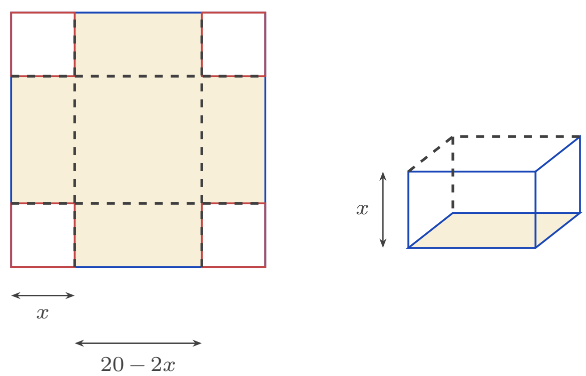


Figure 14.6 Cutting a square of side x from each corner of the card, then folding the flaps up. The base of the box measures $20 - 2x$ each way and its height is x, so $V = x(20 - 2x)^2$.

EXAMPLE 14.5

An open box is made from a 20×20 cm square of card by cutting a square of side x from each corner and folding up the sides. Find the value of x that maximises the volume.

After cutting, the base is $(20 - 2x)$ by $(20 - 2x)$ and the height is x, so

$$V(x) = x(20 - 2x)^2, \qquad 0 < x < 10.$$

Differentiate (expanding the bracket helps):

$$V'(x) = (20 - 2x)^2 + x \cdot 2(20 - 2x)(-2) = (20 - 2x)(20 - 6x).$$

Setting $V'(x) = 0$ gives $x = 10$ (rejected, no box) or $x = \frac{10}{3}$.

Since $V' > 0$ before $\frac{10}{3}$ and $V' < 0$ after, this is a maximum. The maximum volume is

$$V\left(\tfrac{10}{3}\right) = \frac{16000}{27} \approx 593 \text{ cm}^3.$$

The restriction $0 < x < 10$ comes from the geometry. If x is 0 the box has no height; if x is 10

it has no base.

EXAMPLE 14.6

A closed cylindrical can must hold 355 cm^3. Find the radius that minimises the surface area.
Volume fixes the height: $\pi r^2 h = 355$, so $h = \dfrac{355}{\pi r^2}$. The surface area is two circles plus the curved side:

$$S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{710}{r}.$$

Differentiate and set to zero:

$$\frac{dS}{dr} = 4\pi r - \frac{710}{r^2} = 0 \implies r^3 = \frac{710}{4\pi} \implies r \approx 3.84 \text{ cm}.$$

Since $\dfrac{d^2S}{dr^2} = 4\pi + \dfrac{1420}{r^3} > 0$, this is a minimum. Using the constraint to remove h is the step that reduces two variables to one.

CAUTION Setting $f' = 0$ finds only the best values *inside* an interval. If the variable is restricted, for example $0 < r \leq 3$, the best value may sit at an **endpoint** instead, where the derivative need not be zero. HL Always list the candidates: every stationary point in the interval, and both endpoints. Then compare their values and take the largest or the smallest.

EXAMPLE 14.7

For the can above, the manufacturer's machinery cannot roll a radius above 3 cm, so $0 < r \leq 3$. Find the radius that minimises the surface area now.
The unconstrained minimum at $r \approx 3.84$ is no longer allowed. Consider the derivative on $(0, 3]$:

$$\frac{dS}{dr} = 4\pi r - \frac{710}{r^2}.$$

Both terms increase as r increases, so $\dfrac{dS}{dr}$ is increasing on this interval. Testing the largest value is therefore enough: at $r = 3$, $12\pi - \frac{710}{9} \approx -41 < 0$. The derivative is negative at the right-hand end, and smaller still to the left, so it is negative throughout and S is decreasing across the whole interval. The minimum therefore sits at the endpoint $r = 3$, with $S = 18\pi + \frac{710}{3} \approx 293$ cm^2, even though $S'(3) \neq 0$. Whenever the domain is restricted, check its endpoints as well. The condition $f' = 0$ finds only the best values that lie *inside* the interval.

Optimisation is not restricted to areas and volumes either. Any quantity that can be written as a differentiable function of one variable can be maximised or minimised the same way, and in applied work that function is often exponential.

EXAMPLE 14.8

After a dose is given, the concentration of a drug in the blood is modelled by $C(t) = 20t\, e^{-0.5t}$

mg L^{-1}, where $t \geq 0$ is the time in hours. Find when the concentration is greatest, and the maximum concentration.

The product rule gives

$$C'(t) = 20e^{-0.5t} + 20t(-0.5)e^{-0.5t} = 20e^{-0.5t}(1 - 0.5t).$$

Since $e^{-0.5t}$ is never zero, $C'(t) = 0$ requires $1 - 0.5t = 0$, so $t = 2$ hours.

To justify that this is a maximum, look at the sign of C': the factor $e^{-0.5t}$ is always positive, so the sign is that of $1 - 0.5t$, which is positive for $t < 2$ and negative for $t > 2$. The concentration rises, then falls. The maximum is

$$C(2) = 40e^{-1} \approx 14.7 \text{ mg L}^{-1}.$$

The shape can be read directly. The factor t pulls the concentration up as the drug is absorbed, while $e^{-0.5t}$ pulls it down as the body clears it. Early on absorption wins; after two hours clearance does. The maximum is the moment the two rates balance, and that is exactly what $C'(t) = 0$ records.

YOUR TURN 14.3 Optimization

8.

(a) Two numbers have a sum of 20. Find the maximum value of their product, and justify that it is a maximum.

(b) A rectangle has a perimeter of 40 cm. Find the dimensions giving the maximum area.

(c) Find the minimum value of $S = x^2 + \dfrac{16}{x}$ for $x > 0$, justifying that it is a minimum.

9. An open box is made from a 12×12 cm square of card by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$, and state the possible values of x.

(b) Find the value of x that maximises the volume.

(c) Find the maximum volume.

10.

(a) A rectangular field is fenced on three sides, the fourth being a wall, using 100 m of fencing. Find the maximum area.

(b) An open-topped cylindrical tank must hold 1000 cm^3. Show that its surface area is $S = \pi r^2 + \dfrac{2000}{r}$, and find the radius that minimises it.

(c) For the tank in part (b), explain why the answer would change if the manufacturer could not make a radius larger than 5 cm.

11. GDC

(a) The concentration of a drug in the blood is $C(t) = 15t\,e^{-0.3t}$ mg L^{-1}, where $t \geq 0$ is measured in hours. Find the time at which the concentration is greatest, and the maximum concentration.

(b) Find the minimum value of $y = x - \ln x$ for $x > 0$, justifying that it is a minimum.

(c) A rectangle has its base on the x-axis and its two upper vertices on the curve $y = \cos x$, symmetric about the y-axis, with $0 < x < \frac{\pi}{2}$. Show that its area is $A = 2x\cos x$, and find the value of x that maximises A.

14.4 Kinematics

Motion is change of position, so calculus describes it exactly. If a particle's **displacement** from a fixed origin is $s(t)$, then its **velocity** is the rate of change of displacement, $v = \dfrac{ds}{dt}$, and its **acceleration** is the rate of change of velocity.

KEY · IB

$$a = \frac{dv}{dt} = \frac{d^2 s}{dt^2}$$

It helps to read the chain $s \to v \to a$ in ordinary words. Displacement answers *where*. Velocity answers *how fast the where is changing*. Acceleration answers *how fast the how-fast is changing*. In a car, the odometer reads s, the speedometer reads $|v|$, and the push you feel in your seat is a. Each quantity is the derivative of the one before it, which is why acceleration is a rate of a rate, and so a second derivative.

Velocity carries a sign, so it records direction as well as how fast the particle moves. On a straight line there are only two directions, and the sign says which one. **Speed** is the size of the velocity, written $|v|$, and it is never negative. A particle with $v = -8$ is moving in the negative direction at a speed of 8.

The particle is **momentarily at rest** (at rest for an instant) when $v = 0$. This is the stationary-point condition of §14.1 applied to displacement: a turning point on the graph of $s(t)$ is the instant the particle stops and reverses. The ball at the top of its flight is exactly this point.

Speed follows a separate rule, and it is better understood than memorised. Acceleration is a push. If the push acts in the direction the particle is already moving, the particle gains speed; if it acts against the motion, the particle loses speed. In symbols, the particle **speeds up when v and a have the same sign** and slows down when the signs differ. Notice that this does not depend on a being positive: a particle moving in the negative direction with a negative acceleration is speeding up, because the push and the motion agree.

CAUTION $a = 0$ does not mean the particle has stopped. Acceleration is the derivative of velocity, so $a = 0$ marks a stationary point of $v(t)$: the instant when the velocity is greatest or least, not when it is zero. A car cruising at a steady $100\,\mathrm{km\,h^{-1}}$ has zero acceleration and is not stopped at all. The particle is at rest only when $v = 0$.

One quantity needs integration rather than differentiation: the **total distance travelled**, which keeps counting when the particle turns and comes back. Displacement over an interval is $\int v\,dt$ and distance travelled is $\int |v|\,dt$. Both belong to the same syllabus item as the derivatives above, and both are developed in Ch. 15.

EXAMPLE 14.9

A particle moves so that its displacement is $s(t) = t^3 - 12t^2 + 36t$ metres, for $t \geq 0$ seconds. Find when it is at rest, and its acceleration at those times.
Differentiate for velocity:

$$v(t) = 3t^2 - 24t + 36 = 3(t-2)(t-6).$$

The particle is at rest when $v = 0$: at $t = 2$ and $t = 6$ seconds.
Acceleration is $a(t) = 6t - 24$. At $t = 2$, $a = -12\,\mathrm{m\,s^{-2}}$; at $t = 6$, $a = 12\,\mathrm{m\,s^{-2}}$. The sign change of velocity at each rest point means the particle reverses direction there.
Now ask whether the particle is speeding up at $t = 7$. There $v(7) = 3(5)(1) = 15 > 0$ and $a(7) = 6(7) - 24 = 18 > 0$. The two signs are the same, so the particle is speeding up: it moves in the positive direction and the acceleration acts the same way.

Not all motion is polynomial. Anything that repeats, such as a pendulum, a mass on a spring, a wave or a vibrating string, is described by a sine or a cosine.

EXAMPLE 14.10

A particle moves in a straight line with displacement $s = 4\sin 2t$ metres, for $t \geq 0$ seconds. Find its velocity and acceleration, the first time it is at rest, and show that $a = -4s$.
Differentiate twice, using the chain rule each time:

$$v = \frac{ds}{dt} = 8\cos 2t, \qquad a = \frac{dv}{dt} = -16\sin 2t.$$

The particle is at rest when $v = 0$, that is when $\cos 2t = 0$. The first solution is $2t = \frac{\pi}{2}$, so $t = \frac{\pi}{4} \approx 0.785$ s. At that instant $s = 4\sin\frac{\pi}{2} = 4$ m, the greatest displacement it reaches.

Finally, since $s = 4\sin 2t$,

$$-4s = -4(4\sin 2t) = -16\sin 2t = a. \quad \blacksquare$$

Read what this says. The acceleration is proportional to the displacement and points the opposite way: the further the particle is from the origin, the harder it is pulled back. That relation, $a = -k^2 s$, is **simple harmonic motion**, and it describes a pendulum, a mass on a spring and a sound wave alike. The particle is momentarily at rest exactly where the pull on it is strongest, and moving fastest as it passes through the centre, where the pull is zero.

TIP In kinematics questions, state the units with every answer: $\mathrm{m\,s^{-1}}$ for velocity, $\mathrm{m\,s^{-2}}$ for acceleration. A correct number without its unit loses the accuracy mark.

CAUTION Velocity and speed are not the same. A particle with $v = -5$ moves in the negative direction at a speed of 5. Maximum speed can therefore occur where the velocity is most negative, so compare $|v|$, not v.

YOUR TURN 14.4 Kinematics

12. A particle moves in a straight line.

(a) Its displacement is $s = t^2 - 4t + 3$. Find when it is at rest, and its acceleration then.
(b) Its displacement is $s = 2t^3 - 9t^2 + 12t$. Find the times when it is at rest.
(c) Its displacement is $s = t^3 - 3t^2$. Find its velocity and acceleration at $t = 2$.
(d) Its velocity is $v = t^2 - 5t + 6$. Find when it is at rest, and its acceleration at those times.

13. A particle has velocity $v = 3t^2 - 12t + 9$ for $t \geq 0$.

(a) Find the times when it is at rest.
(b) Find its acceleration at those times.
(c) Determine whether the particle is speeding up or slowing down at $t = 1.5$, giving a reason.
(d) A second particle has $v = t - 4$ and $a = 1$ at $t = 2$. State, with a reason, whether it is speeding up or slowing down.

14.

(a) A particle has displacement $s = 3\sin 2t$ metres, $t \geq 0$. Find v and a, and show that $a = -4s$.

(b) For the particle in part (a), find the first time it is at rest and its displacement at that instant.

(c) A particle has velocity $v = 6e^{-0.5t}$ m s^{-1}. Find its acceleration at $t = 0$, and explain why the particle never comes to rest.

(d) A particle has displacement $s = t + 2\cos t$ metres, for $0 \le t \le 2\pi$. Find the times at which it is at rest.

14.5 Implicit Differentiation HL

So far every curve has been given as a function $y = f(x)$, with y written on its own. But an equation such as $x^2 + y^2 = 25$ also defines a curve, without y being isolated.

Implicit differentiation finds the gradient of such a curve directly, without first solving for y.

The method is the chain rule. Treat y as an unknown function of x.

A term in y then differentiates as usual, but it gains an extra factor $\frac{dy}{dx}$, because you are differentiating a function of y with respect to x. For example, $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$.

EXAMPLE 14.11

Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$, and the gradient at $(3, 4)$.
Differentiate both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$ the gradient is $-\frac{3}{4}$. The answer depends on both coordinates, and it must: a circle has a different gradient at every point.

EXAMPLE 14.12

Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$.
The middle term needs the product rule. Differentiating term by term:

$$2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0.$$

Collect the $\dfrac{dy}{dx}$ terms:

$$\frac{dy}{dx}(x+2y) = -(2x+y) \implies \frac{dy}{dx} = -\frac{2x+y}{x+2y}.$$

The procedure is always the same: differentiate, collect the $\frac{dy}{dx}$ terms on one side, factorise, divide.

This means every tangent and normal method from Ch. 13 now works on curves that are not functions.

EXAMPLE 14.13

Find the equation of the tangent to the curve $x^2 + xy + y^2 = 7$ at the point $(1, 2)$.
First confirm the point lies on the curve: $1 + 2 + 4 = 7$. From the previous example,

$$\frac{dy}{dx} = -\frac{2x+y}{x+2y} = -\frac{2+2}{1+4} = -\frac{4}{5} \quad \text{at } (1,2).$$

The tangent is $y - 2 = -\frac{4}{5}(x-1)$, or $4x + 5y = 14$. Implicit differentiation supplies the gradient. The rest is the usual equation of a line through a given point.

The same method handles exponential and trigonometric terms without any change. A term in y still gains a factor $\dfrac{dy}{dx}$, whatever function it sits inside.

EXAMPLE 14.14

The curve C has equation $e^y + xy = 1$. Show that $(0, 0)$ lies on C, and find the gradient there.
At $(0, 0)$: $e^0 + 0 = 1$, so the point lies on C.
Differentiate every term with respect to x. The first needs the chain rule and the second the product rule:

$$e^y \frac{dy}{dx} + \left(y + x\frac{dy}{dx}\right) = 0.$$

Collect the $\dfrac{dy}{dx}$ terms and factorise:

$$\frac{dy}{dx}\,(e^y + x) = -y \implies \frac{dy}{dx} = -\frac{y}{e^y + x}.$$

At $(0, 0)$ this gives $\dfrac{dy}{dx} = \dfrac{0}{1+0} = 0$, so the tangent there is horizontal.
Notice that this curve cannot be rearranged to give y as a formula in x: y appears both inside an exponential and multiplied by x, and no amount of algebra separates them. The curve still has a well-defined gradient at every point, and near $(0, 0)$ it still gives one y for each x — there simply is no expression to write down. Implicit differentiation is not a shortcut here. It is the only method available.

YOUR TURN 14.5 Implicit Differentiation HL

15. Find $\dfrac{dy}{dx}$ for each curve.

(a) $x^2 + y^2 = 16$
(b) $xy = 12$
(c) $x^2 - y^2 = 9$
(d) $x^3 + y^3 = 6$

16. Each of these needs the product rule.

(a) Find $\dfrac{dy}{dx}$ for $x^2 + xy = 5$.
(b) Find $\dfrac{dy}{dx}$ for $x^2y + y^2 = 4$.

17.

(a) Find the gradient of $x^2 + y^2 = 25$ at the point $(4, -3)$.
(b) Find the equation of the tangent to $xy = 4$ at the point $(2, 2)$.
(c) On the circle $x^2 + y^2 = 25$, find the two points at which the tangent is horizontal.
(d) Explain why implicit differentiation gives a gradient in terms of both x and y, while differentiating $y = f(x)$ gives one in terms of x alone.

18. Find $\dfrac{dy}{dx}$ for each curve.

(a) $e^y = x^2 + y$
(b) $\sin y + x^2 = 4$
(c) $x \ln y = 4$
(d) Find the gradient of $e^y + x = 3$ at the point $(2, 0)$.

14.6 Related Rates HL

When two quantities are linked by an equation and both change over time, their rates of change are linked too. **Related rates** problems use the chain rule to connect those rates. You differentiate the equation that relates the quantities with respect to time.

The method has three steps. Write an equation relating the quantities. Differentiate every term with respect to t, so that each variable contributes its own rate. Then substitute the values known at the instant in question.

TIP Substitute the given values only *after* differentiating. Putting $r = 5$ into the volume formula first turns a variable into a constant, and a constant has rate of change zero.

EXAMPLE 14.15

Air is pumped into a spherical balloon at $100\,\text{cm}^3\text{s}^{-1}$. Find the rate at which the radius is increasing when $r = 5$ cm.

Volume and radius are related by $V = \frac{4}{3}\pi r^3$. Differentiate with respect to t:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Substitute $\dfrac{dV}{dt} = 100$ and $r = 5$:

$$100 = 4\pi(25)\frac{dr}{dt} \implies \frac{dr}{dt} = \frac{1}{\pi} \approx 0.318\ \text{cm}\,\text{s}^{-1}.$$

The rate of change of volume is constant, but the radius increases more slowly as the balloon grows, because the same volume is spread over a larger surface.

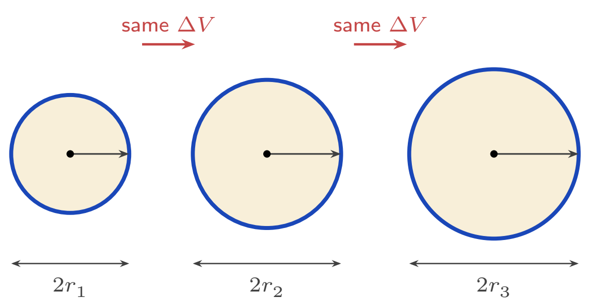


Figure 14.7 Equal volumes of air are added at each step, but the radius grows by less each time. The volume added is spread over the surface, and the surface area $4\pi r^2$ is itself growing, which is why $\frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt}$ falls as r increases.

EXAMPLE 14.16

A 10 m ladder leans against a wall. The foot slides away from the wall at $0.6\,\text{m}\,\text{s}^{-1}$. How fast is the top sliding down when the foot is 6 m from the wall?

Let x be the distance of the foot from the wall and y the height of the top. Then $x^2 + y^2 = 100$. Differentiate with respect to t:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}.$$

When $x = 6$, $y = \sqrt{100 - 36} = 8$. With $\dfrac{dx}{dt} = 0.6$:

$$\frac{dy}{dt} = -\frac{6}{8}(0.6) = -0.45\ \text{m}\,\text{s}^{-1}.$$

The negative sign means the top moves downward, as expected.

EXAMPLE 14.17

A water tank is an inverted cone with its vertex at the bottom, as tall as it is wide: at every level, the water surface has radius $r = \frac{h}{2}$, where h is the water depth. Water flows in at $2\,\mathrm{m}^3\,\mathrm{min}^{-1}$. How fast is the depth rising when $h = 3$ m?

Write the volume in terms of h alone, using the shape constraint $r = \frac{h}{2}$:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}.$$

Differentiate with respect to time:

$$\frac{dV}{dt} = \frac{\pi h^2}{4}\frac{dh}{dt} \implies 2 = \frac{9\pi}{4}\frac{dh}{dt} \implies \frac{dh}{dt} = \frac{8}{9\pi} \approx 0.283\ \mathrm{m\,min}^{-1}.$$

Remove r *before* differentiating. Differentiating an expression that still contains two linked variables is where most errors happen.

The algebra also shows the physics: the same inflow raises the level more slowly as the cone widens.

The quantities linked need not be lengths and volumes. When the link is an angle, the relating equation is trigonometric, and the method is unchanged.

EXAMPLE 14.18

A balloon rises vertically at $3\,\mathrm{m\,s}^{-1}$ from a point 40 m from an observer. Find the rate at which the angle of elevation of the balloon is increasing when the balloon is 40 m high.

Let h be the height of the balloon and θ the angle of elevation. The observer's distance is fixed at 40 m, so

$$\tan\theta = \frac{h}{40}.$$

Differentiate both sides with respect to t. The left side needs the chain rule:

$$\sec^2\theta\,\frac{d\theta}{dt} = \frac{1}{40}\frac{dh}{dt}.$$

When $h = 40$ the triangle is isosceles, so $\theta = \frac{\pi}{4}$ and $\sec^2\theta = 2$. Substituting this and $\dfrac{dh}{dt} = 3$:

$$2\frac{d\theta}{dt} = \frac{3}{40} \implies \frac{d\theta}{dt} = \frac{3}{80} = 0.0375\ \mathrm{rad\,s}^{-1}.$$

The balloon rises at a constant rate, but the angle does not increase at a constant rate. As the balloon climbs, $\sec^2\theta$ grows, so $\dfrac{d\theta}{dt}$ falls: the observer's head turns more and more slowly, even though the balloon never slows down.

YOUR TURN 14.6 Related Rates HL

19.

(a) A square's side grows at $2\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its area when the side is 5 cm.
(b) A circle's radius grows at $3\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its area when $r = 4$ cm.
(c) A cube's side grows at $1\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its volume when the side is 10 cm.
(d) A circle's radius grows at 0.4 cm s^{-1}. Find the rate of increase of its circumference.

20.

(a) A spherical balloon is inflated at $50\,\text{cm}^3\text{s}^{-1}$. Find the rate of change of the radius when $r = 10$ cm.
(b) A ladder 5 m long leans against a vertical wall. Its foot slides away from the wall at $2\,\text{m}\,\text{s}^{-1}$. Find the rate at which the top slides down when the foot is 3 m from the wall.
(c) Water is poured into a cone at $20\,\text{cm}^3\text{s}^{-1}$. The cone's height is always twice its surface radius, so $V = \frac{2}{3}\pi r^3$. Find the rate at which r increases when $r = 5$ cm.

21.

(a) A balloon rises vertically at $2\,\text{m}\,\text{s}^{-1}$ from a point 30 m from an observer. Find the rate of change of the angle of elevation of the balloon when it is 30 m high.
(b) A searchlight rotates at $0.2\,\text{rad}\,\text{s}^{-1}$ and its beam sweeps along a straight wall 50 m away. Find the speed of the spot of light along the wall when the beam makes an angle of $\frac{\pi}{6}$ with the perpendicular to the wall.
(c) A population is modelled by $P = 800e^{0.04t}$, where t is in years. Find the rate of growth when $t = 15$.

Chapter 14 · Derivative Applications

Reflections

TOK The ball is motionless at the top of its flight, but only for an instant, and an instant has no duration. Is being motionless for no time at all a physical state, or an artefact of the mathematics? Does calculus describe motion, or replace it with something we can calculate?

TOK This chapter is full of conditions that are necessary but not sufficient: $f'(x) = 0$ holds at every smooth maximum, and $f''(x) = 0$ at every point of inflexion, yet neither condition guarantees one. What kind of knowledge is a necessary condition? Does knowing where something must be count as knowing where it is?

TOK The can that uses least metal has its height equal to its diameter, but real cans are taller and narrower, because the model ignores printing, shipping and the size of a hand. The mathematics is exact and the conclusion is still wrong. Does mathematically optimal always mean practically best?

IM The idea that nature minimises something appears across physics under many names. In seventeenth-century France, Pierre de Fermat proposed that light travels the path taking the least time, which explains refraction. Euler and Lagrange later generalised this into the calculus of variations and the *principle of least action*, which restates the whole of classical mechanics as an optimisation problem. The belief that the world is in some sense optimal has produced a great deal of physics.

RW The method behind modern machine learning is *gradient descent*, which is this chapter's idea at scale. A model has millions of adjustable numbers, and how wrong its predictions are is a function of those numbers; training means making that function as small as possible. With millions of variables the minimum cannot be found directly, so the computer does what a walker in fog would do: measure the slope where it stands, step down it, and repeat. As the slope flattens the steps shrink and the model settles near a minimum. That is $f' = 0$, applied billions of times.

EXT Implicit differentiation and related rates are the same idea twice. Both are the chain rule applied when the variables depend on something else: on x in the first case, on time in the second.

Carried further it becomes multivariable calculus, where a point has a different rate of change in every direction. Choosing the steepest of them is exactly what gradient descent does.

End-of-Chapter Problems — Chapter 14: Derivative Applications

1. For $f(x) = x^4$, show that $f''(0) = 0$ but that $(0, 0)$ is *not* a point of inflexion. [4]

2. A stone dropped in a pond sends out a circular ripple whose radius grows at $0.5\,\mathrm{m\,s^{-1}}$. Find the rate at which the enclosed area is increasing when the radius is 8 m. [4]

3. A particle's velocity is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]
(b) Find the minimum velocity and the time at which it occurs. [3]

4. A company's profit from selling x units is $P(x) = -2x^2 + 120x - 1000$ dollars. Find the value of x that maximises the profit, justifying that it is a maximum, and state the maximum profit. [5]

5. The curve $\sin y + x^2 = 1$ passes through the point $(1, 0)$. Find the gradient of the curve at that point. [5]

6. A curve has equation $x^2 + xy + y^2 = 3$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y. [3]
(b) Find the equation of the tangent at the point $(1, 1)$. [3]

7. Water drains from an inverted cone whose radius always equals its height. The volume is $V = \frac{1}{3}\pi r^2 h$.

(a) Show that $V = \frac{1}{3}\pi h^3$. [2]
(b) Given the volume decreases at $2\,\mathrm{cm^3 s^{-1}}$, find the rate at which the depth h is falling when $h = 4$ cm. [4]

8. Consider $f(x) = x^3 - 3x$ on the closed interval $0 \leq x \leq 3$.

(a) Find the stationary point(s) in the interval and classify them. [3]
(b) Find the global maximum and minimum of f on the interval. [3]

9. Consider $g(x) = x + \dfrac{4}{x}$ on the interval $1 \leq x \leq 3$.

(a) Find the stationary point of g in the interval and classify it. [4]
(b) Find the maximum value of g on the interval, and state where it occurs. [2]

10. The curve C has equation $x^2 + y^2 - 4x + 2y = 15$.

(a) Show that the point $(0, 3)$ lies on C. [1]

(b) Use implicit differentiation to find $\dfrac{dy}{dx}$. [3]

(c) Find the equation of the tangent to C at $(0, 3)$. [2]

11. A street lamp is 5 m tall. A person of height 1.8 m walks away from it at $1.2\,\mathrm{m\,s^{-1}}$. Find the rate at which the length of the person's shadow is increasing. [6]

12. A water tank is an inverted cone in which the water surface radius is always $\frac{3}{4}$ of the depth h. Water is pumped in at $1.5\,\mathrm{m^3\,min^{-1}}$. Find the rate at which the depth is rising when $h = 2$ m. [6]

13. Find the point on the curve $y = \sqrt{x}$ closest to the point $(4, 0)$, and state the minimum distance. [6]

14. For the curve $x^2 + y^2 = 25$, use implicit differentiation twice to show that $\dfrac{d^2y}{dx^2} = -\dfrac{25}{y^3}$. [6]

15. A closed cylindrical can must hold $500\,\mathrm{cm^3}$.

(a) Show that the surface area is $S = 2\pi r^2 + \dfrac{1000}{r}$. [3]

(b) Find the radius that minimises S, and justify it is a minimum. [4]

16. A rectangular pen of area $200\,\mathrm{m^2}$ is built against a wall, needing fencing on three sides. Let the side perpendicular to the wall be x.

(a) Show that the length of fencing is $L = 2x + \dfrac{200}{x}$. [3]

(b) Find the dimensions that minimise the fencing used. [4]

17. The curve $y = x^3 + ax^2 + bx$ has stationary points at $x = 1$ and $x = -2$.

(a) Find the values of a and b. [4]

(b) Classify each stationary point. [3]

18. An open rectangular tank with a square base of side x must hold $32\,\mathrm{m^3}$.

(a) Show that the surface area (base + four sides) is $A = x^2 + \dfrac{128}{x}$. [3]

(b) Find the value of x that minimises the material used. [4]

19. An open-topped tank with a square base of side x cm is made from exactly $1200\,\mathrm{cm^2}$ of sheet metal.

(a) Show that the volume is $V = \dfrac{1200x - x^3}{4}$. [3]

(b) Find the dimensions that maximise the volume, and the maximum volume. [4]

20. A cone is inscribed in a sphere of radius R. Show that the cone of maximum volume has height $h = \frac{4R}{3}$. [7]

21. A 5 m ladder slides down a wall, its foot moving away at a constant $0.5\,\mathrm{m\,s^{-1}}$.

(a) Find the rate at which the top descends when the foot is 3 m from the wall. [3]

(b) Find the rate at which the area of the triangle formed by the ladder, wall, and ground is changing at that instant. [4]

22. A particle moves along the curve $y = x^2$ so that $\dfrac{dx}{dt} = 2$ at all times. Find the rate at which its distance from the origin is increasing as it passes through the point $(1, 1)$. [7]

23. Consider $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find and classify the stationary points. [4]

(b) Find the point of inflexion. [2]

(c) State the intervals on which f is decreasing. [2]

24. A particle moves with displacement $s(t) = t^3 - 6t^2 + 9t$ metres, $t \geq 0$.

(a) Find the velocity and the times when the particle is at rest. [3]

(b) Find the acceleration when $t = 1$. [2]

(c) Determine whether the particle is speeding up or slowing down at $t = 1.5$. [3]

25. Consider $f(x) = x^3 - 6x^2 + 9x + 2$.

(a) Find and classify the stationary points. [5]

(b) Find the point of inflexion and show that the gradient there is not zero. [3]

26. A particle moves with displacement $s(t) = t^3 - 9t^2 + 24t - 10$ metres, for $0 \leq t \leq 6$ seconds.

(a) Find the times at which the particle is at rest. [3]

(b) Find its acceleration at each of those times. [2]

(c) Find the intervals of time during which the particle's *speed* is increasing. [3]

27. Consider $f(x) = xe^{-x}$.

(a) Find $f'(x)$ and the coordinates of the stationary point. [3]

(b) Determine its nature using the second derivative. [3]

(c) Find the x-coordinate of the point of inflexion. [2]

28. Consider $f(x) = x^4 - 4x^3$.

(a) Find $f'(x)$ and factorise it fully. [2]
(b) Find the coordinates of the stationary points. [3]
(c) Find $f''(x)$, and use it to classify the stationary point at $x = 3$. [3]
(d) Explain why the second derivative test fails at $x = 0$, and use the sign of f' to classify that point instead. [4]
(e) Find the coordinates of any points of inflexion. [3]

29. An open-topped box is made from a 12×12 cm square of card by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$, and state the possible values of x. [3]
(b) Find $\dfrac{dV}{dx}$ and show that it is zero at $x = 2$ and $x = 6$. [3]
(c) Explain why $x = 6$ must be rejected. [2]
(d) Find the maximum volume, justifying that it is a maximum. [4]

30. A particle moves in a straight line with displacement $s = t^3 - 6t^2 + 9t$ metres, for $t \geq 0$ seconds.

(a) Find expressions for the velocity and the acceleration. [3]
(b) Find the times at which the particle is momentarily at rest. [2]
(c) Find the displacement at each of those times. [2]
(d) Find the time at which the acceleration is zero, and state what is happening to the velocity at that instant. [3]
(e) Determine whether the particle is speeding up or slowing down at $t = 4$, giving a reason. [3]

31. The curve C has equation $x^3 + y^3 = 9xy$.

(a) Verify that the point $(2, 4)$ lies on C. [2]
(b) Use implicit differentiation to show that $\dfrac{dy}{dx} = \dfrac{x^2 - 3y}{3x - y^2}$. [4]
(c) Find the gradient of C at $(2, 4)$. [2]
(d) Find the equation of the tangent to C at $(2, 4)$. [3]

32. Consider $f(x) = e^x - 2x$.

(a) Find $f'(x)$ and the coordinates of the stationary point. [3]
(b) Determine the nature of the stationary point. [2]
(c) Explain why f has no point of inflexion. [2]

33. A particle moves in a straight line with displacement $s(t) = 2\sin 3t$ metres, $t \geq 0$ seconds.

(a) Find expressions for the velocity and the acceleration. [2]
(b) Show that $a = -9s$. [2]
(c) Find the first two times at which the particle is at rest. [3]
(d) State the maximum speed of the particle, and the displacement at which it occurs. [3]

34. Consider $f(x) = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. [3]
(b) Find the coordinates of the stationary point and determine its nature. [4]
(c) Show that $f''(x) = \dfrac{2\ln x - 3}{x^3}$, and hence find the x-coordinate of the point of inflexion. [4]

35. A gutter is made by bending a long strip of metal of width 30 cm into three equal sections of 10 cm. The two outer sections are turned up at an angle θ to the horizontal, where $0 < \theta < \frac{\pi}{2}$. The cross-sectional area is $A = 100\sin\theta\,(1 + \cos\theta)$.

(a) Find $\dfrac{dA}{d\theta}$. [3]
(b) Show that A is stationary when $2\cos^2\theta + \cos\theta - 1 = 0$, and find the value of θ that maximises A. [4]
(c) Find the maximum cross-sectional area. [2]

36. The curve C has equation $xe^y + y = 1$.

(a) Verify that the point $(1, 0)$ lies on C. [2]
(b) Show that $\dfrac{dy}{dx} = -\dfrac{e^y}{xe^y + 1}$. [4]
(c) Find the gradient of C at $(1, 0)$. [2]
(d) Find the equation of the tangent to C at $(1, 0)$. [3]

37. A lighthouse stands 2 km from a straight shoreline, and its beam rotates at one revolution per minute. Let θ be the angle between the beam and the perpendicular from the lighthouse to the shore, and let x be the distance from the foot of that perpendicular to the spot of light.

(a) Write x in terms of θ, and state the value of $\dfrac{d\theta}{dt}$ in radians per minute. [2]
(b) Show that $\dfrac{dx}{dt} = 4\pi\sec^2\theta$. [3]
(c) Find the speed of the spot of light along the shore when $\theta = 30^\circ$. [3]
(d) Describe what happens to this speed as $\theta \to 90^\circ$, and explain why. [2]

Challenge Questions

38. The shape of the best can. A closed cylindrical can must hold a fixed volume V. This investigation finds the shape that uses least metal, and shows that the answer does not depend on V.

(a) Show that the surface area is $S = 2\pi r^2 + \dfrac{2V}{r}$. [3]
(b) Find $\dfrac{dS}{dr}$, and show that S is stationary when $r^3 = \dfrac{V}{2\pi}$. [4]
(c) Verify, using the second derivative, that this gives a minimum. [3]
(d) Show that at this radius the height equals the diameter, whatever the value of V. [4]
(e) Real drink cans are taller and narrower than this. Suggest two costs the model leaves out. [2]

39. Closest approach. Finding the nearest point of a curve to a fixed point is an optimisation problem in disguise.

(a) Let $(x, \sqrt{x})$ be a point on $y = \sqrt{x}$ and let D be its distance from $(4, 0)$. Show that $D^2 = x^2 - 7x + 16$. [3]
(b) Explain why minimising D^2 gives the same answer as minimising D, and why using D^2 is easier. [3]
(c) Find the point on the curve closest to $(4, 0)$, and the least distance. [4]
(d) Show that at this closest point the line joining it to $(4, 0)$ is perpendicular to the tangent to the curve. [5]
(e) State the general result that part (d) illustrates. [2]

40. Quantifying curvature. A straight line does not bend; a small circle bends sharply. This investigation builds a number that measures how sharply a curve bends at a point. It is defined by

$$\kappa = \frac{|y''|}{\left(1 + (y')^2\right)^{3/2}},$$

and κ is called the *curvature*.

(a) Show that every straight line has $\kappa = 0$ at every point, and explain why the definition must give this. [3]
(b) For $y = x^2$, find κ as a function of x. Show that the curvature is greatest at the vertex, and find its value there. [4]
(c) Show that $\kappa \to 0$ as $x \to \pm\infty$ for $y = x^2$, and explain what this says about the shape of a parabola far from its vertex. [3]
(d) Use implicit differentiation on $x^2 + y^2 = R^2$ to show that a circle of radius R has $\kappa = \dfrac{1}{R}$ at every point. Explain why a constant curvature is exactly what you should expect. [5]

(e) The *radius of curvature* is defined as $\rho = \dfrac{1}{\kappa}$. Find ρ for $y = x^2$ at the origin, and state the circle that best matches the parabola there. [3]

(f) The second derivative alone will not do as a measure of bending. For $y = x^2$, $y'' = 2$ at every point, yet the curve clearly bends more sharply at the vertex than far out along an arm. State what the denominator $\left(1 + (y')^2\right)^{3/2}$ corrects for, and use your answer to part (b) to support it. [3]

(g) Show that $\kappa = 0$ at every point of inflexion of a twice-differentiable curve. Then explain why a curve with $\kappa = 0$ at one point need not be a straight line, and give an example. [3]

(h) The curve $y = f(x)$ is stretched vertically by a factor $k > 0$, giving $y = k\,f(x)$. At a point where $f'(x) = 0$, show that the curvature of the stretched curve is k times the curvature of the original. Show also that a translation changes κ nowhere. What does this say about judging how sharply a curve bends from a graph drawn with unequal scales on the two axes? [4]

CALCULUS

Chapter 15

Integral Calculus

15

SYLLABUS 5.5 5.9 5.10 5.11 5.15 5.16 5.17

Your car's speedometer tells you how fast you are going at every instant. Suppose you record it for an entire journey. Can you work out how far you travelled, using only the speed?

Over a tiny slice of time, your speed barely changes, so the distance covered in that slice is roughly speed times the slice of time. Add up all the slices and you have the total distance. As the slices shrink toward zero width, the rough sum becomes exact. What you are computing is the area under the speed-time graph.

This is the second great idea of calculus, and it arrives as two ideas rather than one. The first is **reversing a derivative**: given a rate of change, recover the quantity it came from. The second is **accumulating**: adding up a great many thin slices to get a total. The area under a graph is exactly this kind of total.

They have no obvious connection. Reversing a derivative is a problem in algebra. Adding slices is a problem in geometry. Yet the car journey links them, because the distance travelled is both the reverse of the velocity and the area under the velocity graph.

The chapter is built in that order. Section 15.1 does the reversal on its own, with no mention of area. Section 15.2 defines the area on its own, with no mention of derivatives, and then proves that the two are the same calculation. That result is the Fundamental Theorem of Calculus, and once it is in place the accumulated totals, areas between curves, volumes of revolution and distances travelled all become routine.

15.1 Antiderivatives

This chapter reverses differentiation. Given a function f, the task is to find a function whose derivative is f. Such a function is called an **antiderivative** of f.

An antiderivative is never unique. The derivative of a constant is zero, so any constant may be added to an antiderivative without altering its derivative. Antiderivatives therefore occur in families whose members differ by a constant.

DEF F is an **antiderivative** of f when $F'(x) = f(x)$. If F is one antiderivative, then $F(x) + C$ is an antiderivative for every constant C, and these are all of them.

Reversing the power rule means raising the power by one and dividing by the new power.

KEY · IB An antiderivative of x^n is $\dfrac{x^{n+1}}{n+1} + C$, for $n \neq -1$.

The exception $n = -1$ is handled separately in the next section. Constant multiples and sums are reversed term by term, just as they are differentiated term by term.

EXAMPLE 15.1

Find the antiderivatives of $3x^2 - 4x + 1$.

Reverse each term, raising each power by one:

$$F(x) = x^3 - 2x^2 + x + C.$$

Check by differentiating: $\frac{d}{dx}(x^3 - 2x^2 + x + C) = 3x^2 - 4x + 1$. ✓ Differentiating the answer verifies it completely.

Naming the whole family in words each time is cumbersome, so it is given a symbol of its own.

KEY

$$\int f(x)\,dx = F(x) + C \qquad \text{means} \qquad F'(x) = f(x).$$

This is the **indefinite integral** of f, and C is the **constant of integration**. It is easily omitted, and an answer written without it names a single antiderivative rather than the whole family.

15.1.1 Boundary conditions

The $+C$ means that an antiderivative is not determined until further information is supplied. A single condition, such as a point through which the curve passes, fixes C and selects one member of the family.

EXAMPLE 15.2

A curve has gradient $\frac{dy}{dx} = 6x^2 - 2$ and passes through $(1, 4)$. Find its equation.
Antidifferentiate to recover y:

$$y = 2x^3 - 2x + C.$$

Use the point $(1, 4)$: $4 = 2 - 2 + C \implies C = 4$. So $y = 2x^3 - 2x + 4$.

YOUR TURN 15.1 Antiderivatives

1. Find the antiderivatives of each function.

(a) $4x^3 - 6x$

(b) $x^2 + 3x - 5$

(c) $\dfrac{6}{x^3}$

(d) $\sqrt{x}$

(e) $x^2 - \dfrac{1}{x^2}$

(f) $\dfrac{x^3 - 2x}{x}$

(g) $x^{-1/2}$

(h) $(2x)^3$

2.

(a) A curve has $\dfrac{dy}{dx} = 4x - 3$ and passes through $(2, 5)$. Find y.

(b) A curve has $\dfrac{dy}{dx} = 3\sqrt{x}$ and passes through $(4, 20)$. Find y.

(c) A curve has $\dfrac{d^2y}{dx^2} = 6x$, with $\dfrac{dy}{dx} = 1$ and $y = 2$ when $x = 0$. Find y.

(d) Explain why a gradient function alone does not determine a curve, and what extra information does.

(e) Explain why part (d) needed two pieces of information rather than one.

15.2 Definite Integrals and Area

Set antiderivatives aside for the moment. This section takes up a second and apparently unrelated problem; the two are connected only later.

The problem is area. Regions with straight edges have formulas; a region under a curve has none, because its boundary changes direction at every point. The available approach is approximation.

Cut the interval from a to b into n strips, each of width Δx. Replace each strip by a rectangle whose height is the value of the function somewhere in that strip. Every rectangle has area height $\times$ width, so the total is

$$\sum_{i=1}^{n} f(x_i)\,\Delta x.$$

This is not the area. It is a sum of rectangles, and it differs from the area by whatever those rectangles fail to cover or cover in excess. Narrower strips leave less room for that error, and as n increases the sum settles on a single value.

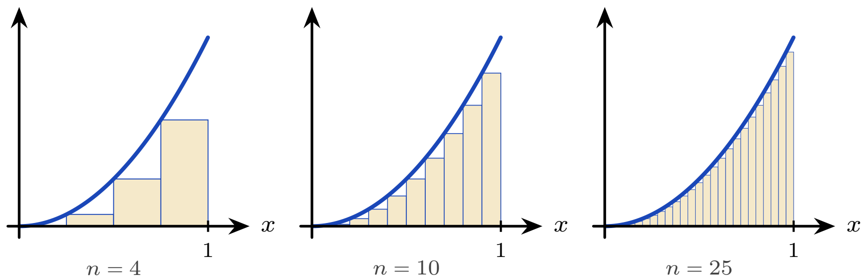


Figure 15.1 Rectangles under $y = x^2$ on $[0, 1]$, taking the height at the left of each strip. Four strips give 0.219, ten give 0.285 and twenty-five give 0.314; the true area is $\frac{1}{3}$. The missing pieces above each rectangle shrink as the strips narrow.

The definition allows the height to be read anywhere in the strip, but two choices are natural: the value at the left end, or the value at the right. Taking both is what makes the argument work, because they err in opposite directions.

For a curve that rises across the interval, as $y = x^2$ does on $[0, 1]$, every left-hand rectangle falls short of the curve and every right-hand one overshoots it. The first sum therefore underestimates the area and the second overestimates it, so the true value is trapped between them. For a curve that falls across the interval the two exchange roles, and the bracketing still holds.

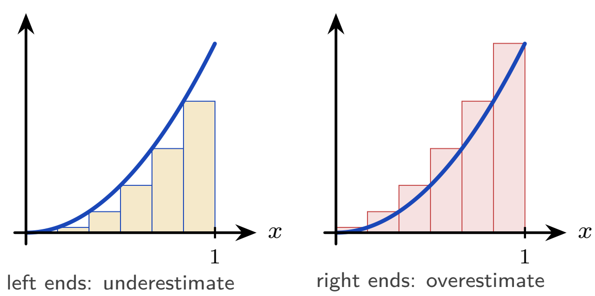


Figure 15.2 The same six strips, with the height of each rectangle read at the left of its strip and then at the right. On a rising curve the left-hand rectangles all sit below it and the right-hand ones all rise above it, so the area lies between the two sums. Narrowing the strips closes the gap from both sides at once.

The numbers behave as the picture predicts.

n	4	10	100	1000
low estimate	0.219	0.285	0.328	0.3328
high estimate	0.469	0.385	0.338	0.3338

The two rows converge, and the value they enclose is $\frac{1}{3}$. That limit is what is meant by the area, and it is called the **definite integral**.

DEF The **definite integral** of f from a to b is the limit of the rectangle sums as the strips become arbitrarily narrow:

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_i f(x_i)\,\Delta x.$$

For $f(x) \geq 0$ it is the area between the curve and the x-axis from a to b.

The notation can now be read directly. The elongated S denotes a sum, $f(x)\,dx$ represents a single rectangle of height $f(x)$ and vanishing width dx, and the integral accumulates them.

As it stands, however, the definition is of no computational use. Nothing in it indicates how such a limit is to be evaluated, and summing a thousand rectangles by hand is not a practical method.

The **Fundamental Theorem of Calculus** supplies the missing method. It states that this limit of sums can be evaluated by antidifferentiation, so the two ideas of the chapter are in fact one calculation.

KEY **The Fundamental Theorem of Calculus.**

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a), \qquad \text{where } F' = f.$$

Reversing a derivative and adding up rectangles give the same answer. The constant C cancels in the subtraction, which is why a definite integral is a plain number while an indefinite one carries a $+C$. Why the theorem is true is shown in §15.2.1; the rest of the chapter needs only the statement.

KEY · IB

$$A = \int_a^b y\,dx \qquad (y \geq 0)$$

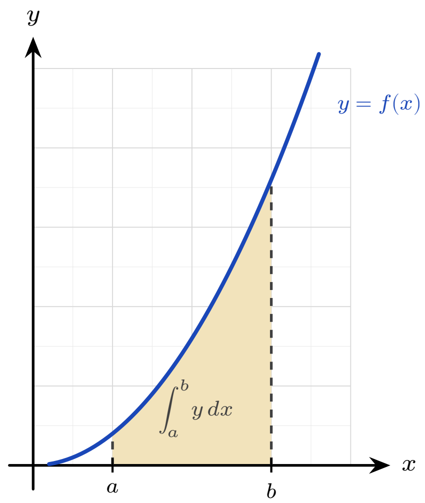


Figure 15.3 For $y \geq 0$ the definite integral measures the shaded area between the curve and the x-axis, from $x = a$ to $x = b$.

EXAMPLE 15.3

Evaluate $\displaystyle\int_0^2 3x^2\,dx$.

$$\int_0^2 3x^2\,dx = \left[x^3\right]_0^2 = 8 - 0 = 8.$$

TIP Your GDC evaluates definite integrals directly, and on a calculator paper that is often all a question requires. Even when analytic working is demanded, the GDC value is a free check of your antiderivative.

15.2.1 The Fundamental Theorem of Calculus (Extension)

This section shows why the Fundamental Theorem is true. It is not examined, and nothing later depends on reading it.

The claim to be justified is that an area, defined as a limit of sums, can be found by antidifferentiation. Throughout, f is continuous with $f(x) \geq 0$.

Suppose the area has a formula, in the way that πr^2 is a formula for the area of a circle. Fix a left end a and let the right end move: write $A(x)$ for the area under the curve from a to x.

If such a function exists, the area of *any* band follows at once by subtraction. The area from

x_1 to x_2 is the area up to x_2 minus the area up to x_1:

$$\text{area from } x_1 \text{ to } x_2 \;=\; A(x_2) - A(x_1).$$

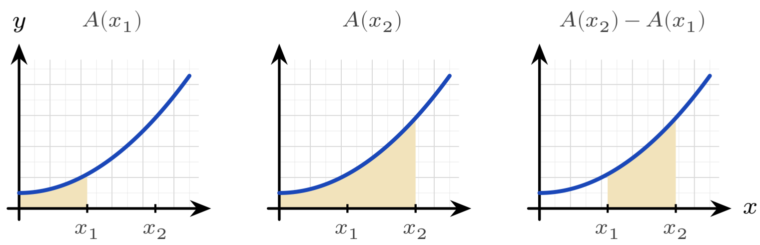


Figure 15.4 If an area function A exists, the area of a band is a difference. Shade up to x_2, remove what was shaded up to x_1, and the band between them is what remains. Finding A would turn every area question into one subtraction.

Nothing is proved yet; this is what would follow *if* A existed. So ask what kind of function A is. When x increases by a small amount h, the extra area $A(x+h) - A(x)$ is one thin strip of width h.

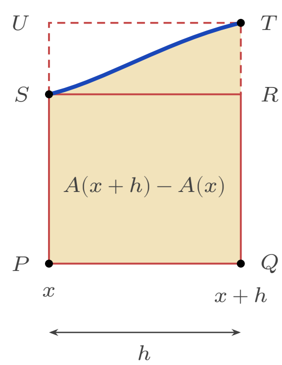


Figure 15.5 One strip, greatly enlarged. Its area is $A(x+h) - A(x)$, and it is trapped: it contains the solid rectangle $PQRS$ and fits inside the dashed rectangle $PQTU$. Both rectangles have width h.

The short rectangle $PQRS$ has the height of the curve at the left of the strip, $f(x)$, and fits inside it. The tall one, $PQTU$, has the height at the right, $f(x+h)$, and contains it. Comparing the three areas,

$$f(x)\,h \;<\; A(x+h) - A(x) \;<\; f(x+h)\,h.$$

Divide through by h:

$$f(x) \;<\; \frac{A(x+h) - A(x)}{h} \;<\; f(x+h).$$

Now let $h \to 0$. The right-hand edge slides back towards the left, so $f(x+h)$ closes on $f(x)$ and the two outer quantities become the same number. The middle is squeezed between them, and it is exactly the definition of $A'(x)$. Therefore

$$A'(x) = f(x).$$

The area function is therefore **an antiderivative of f** — and antiderivatives are something we already know how to find.

The picture is drawn with f rising across the strip. If it falls the two rectangles swap roles, and if it does both, take the smallest and largest values of f on the strip in place of its values at the ends. The squeeze is unchanged.

The computational rule follows at once. If F is any antiderivative of f, then A and F differ by a constant, so $A(x) = F(x) + C$. Setting $x = a$ gives $0 = F(a) + C$, since the area from a to a is nothing, so $C = -F(a)$. Setting $x = b$ then gives

$$\int_a^b f(x)\,dx \;=\; A(b) \;=\; F(b) - F(a),$$

which is the theorem stated in §15.2.

15.2.2 Areas below the axis

Where the curve dips below the x-axis, the integral is negative: it is a signed area. For a true geometric area you integrate the parts separately and take the absolute value of any negative piece.

KEY

$$A = \int_a^b |y|\,dx$$

The booklet writes this second case as $\int y\,dx$ as well, without the modulus. Writing $|y|$ is this book's shorthand for the splitting just described; it is not the printed form, and in an examination you do the splitting yourself.

The distinction is visible as soon as a curve crosses the axis more than once.

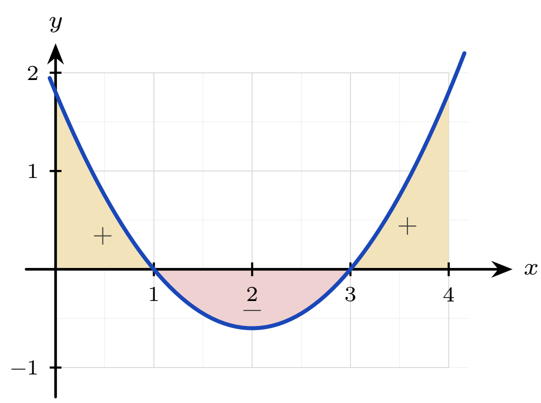


Figure 15.6 A definite integral accumulates *signed* area. The two gold regions lie above the axis and count positively; the scarlet region lies below and counts negatively. A single integral across the whole interval returns the sum of the three signed pieces, which is smaller than the total area enclosed.

EXAMPLE 15.4

Find the area enclosed between $y = x^2 - 4$ and the x-axis from $x = 0$ to $x = 2$.
On $[0, 2]$ the curve lies below the axis, so the integral is negative:

$$\int_0^2 (x^2 - 4)\,dx = \left[\tfrac{x^3}{3} - 4x\right]_0^2 = \tfrac{8}{3} - 8 = -\tfrac{16}{3}.$$

The area is the magnitude, $\dfrac{16}{3}$. An area is never negative, so $-\frac{16}{3}$ answers a different question from the one asked.

CAUTION Before finding an area, check where the curve crosses the x-axis. A curve that goes above and below will have positive and negative parts that partly cancel in a single integral, giving the wrong area. Split at the roots.

YOUR TURN 15.2 Definite Integrals and Area

3. Evaluate each definite integral.

(a) $\displaystyle\int_1^2 4x\,dx$

(b) $\displaystyle\int_0^3 (x^2 + 1)\,dx$

(c) $\displaystyle\int_1^e \frac{1}{x}\,dx$

(d) $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,dx$

(e) $\displaystyle\int_0^{\pi/2} \cos x\,dx$

(f) $\displaystyle\int_{-1}^2 (3x^2 - 2x)\,dx$

4.

(a) Find the area under $y = x^2$ between $x = 1$ and $x = 3$.

(b) Find the area enclosed between $y = x^2 - 1$ and the x-axis.

(c) Find the area under $y = \sin x$ from $x = 0$ to $x = \pi$.

(d) Evaluate $\displaystyle\int_0^{2\pi} \sin x\,dx$, and explain why the answer is not the area between the curve and the axis over that interval.

(e) Find the area between $y = x^3$ and the x-axis from $x = -2$ to $x = 2$.

5. The area under $y = x^2$ from $x = 0$ to $x = 2$ is to be estimated by four rectangles of equal width.

(a) Taking the height of each rectangle at the left of its strip, show that the estimate is 1.75.

(b) Taking the height at the right instead, show that the estimate is 3.75.

(c) Find the exact area, and confirm that it lies between the two estimates.

(d) Repeat part (a) with eight rectangles, and show that the left-hand estimate rises to 2.1875.

(e) State what happens to the two estimates as the number of rectangles increases.

15.3 Standard Integrals

Every derivative established in Ch. 13 can be read in reverse. Doing so converts the table of derivatives into a table of integrals, and turns the question "what is the integral of this function?" into the more tractable "what was differentiated to produce it?"

KEY · IB

$$\int \frac{1}{x}\,dx = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C$$
$$\int \cos x\,dx = \sin x + C, \qquad \int e^x\,dx = e^x + C$$

None of these is a new fact. Each is a derivative already known, read from right to left, and each should be understood rather than memorised.

- $\displaystyle\int \frac{1}{x}\,dx = \ln|x| + C$ fills the one gap the power rule leaves. Raising $n = -1$ by one gives 0, and the rule then asks you to divide by that new power, which is impossible. So the antiderivative of x^{-1} cannot be a power at all. The logarithm supplies it, because $\frac{d}{dx}\ln x = \frac{1}{x}$. The modulus is there because $\frac{1}{x}$ is defined for negative x as well, and on that side $\frac{d}{dx}\ln(-x) = \frac{-1}{-x} = \frac{1}{x}$ by the chain rule: the same derivative on both sides of the origin.
- $\displaystyle\int e^x\,dx = e^x + C$ because e^x is the function equal to its own derivative. A function that is unchanged by differentiating is unchanged by antidifferentiating.
- $\displaystyle\int \cos x\,dx = \sin x + C$ is the direct reversal of $\frac{d}{dx}\sin x = \cos x$, with no sign to watch.
- $\displaystyle\int \sin x\,dx = -\cos x + C$ carries the minus sign that is most often dropped. Differentiating cosine introduces one: $\frac{d}{dx}\cos x = -\sin x$. To finish with $+\sin x$ you must therefore start from $-\cos x$.

The two trigonometric cases are easiest to keep straight as a cycle. Differentiating moves one step clockwise; integrating moves one step back.

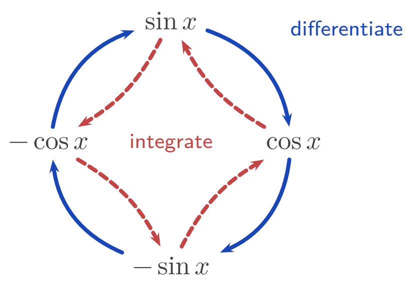


Figure 15.7 The four functions form a closed cycle. Differentiating advances one step clockwise (solid), integrating retreats one step anticlockwise (dashed). Reading the cycle backwards from $\sin x$ lands on $-\cos x$, which is where the minus sign in $\int \sin x\,dx$ comes from.

At HL, three more standard integrals are provided, the reverses of the exponential and inverse-trigonometric derivatives.

KEY · IB HL

$$\int a^x\,dx = \frac{a^x}{\ln a} + C, \qquad \int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\frac{x}{a} + C, \qquad |x| < a$$

These three reward the same reading. Differentiating a^x produces an extra factor: writing $a^x = e^{x\ln a}$ and applying the chain rule gives $\frac{d}{dx}a^x = a^x \ln a$. Dividing by $\ln a$ removes it. When $a = e$ the factor $\ln e$ is 1 and the formula collapses to $\int e^x dx = e^x + C$, as it must.

The two inverse-trigonometric forms come from $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$ and $\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$, with a acting as a scale. Their asymmetry is a common source of error: the arctangent form carries a factor $\frac{1}{a}$ and the arcsine form does not. In each case differentiating x/a contributes a chain-rule factor $\frac{1}{a}$; in the arcsine that factor is cancelled by the a released from the square root, and in the arctangent nothing cancels it.

Reading the rest of the HL derivative table backwards gives four more. They must be learned, because the formula booklet gives only the derivatives:

KEY HL

$$\int \sec^2 x\,dx = \tan x + C, \qquad \int \sec x \tan x\,dx = \sec x + C,$$

$$\int \operatorname{cosec}^2 x\,dx = -\cot x + C, \qquad \int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

15.3.1 Linear composites

Sometimes the function inside is linear, as in e^{2x} or $\cos 3x$. Integrate as usual, then divide by the coefficient of x inside. This reverses the chain rule, whose extra factor was that same coefficient.

KEY

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C, \qquad \text{where } F' = f$$

EXAMPLE 15.5

Find (a) $\int e^{2x}\,dx$, (b) $\int (2x+1)^5\,dx$, (c) $\int \cos 3x\,dx$.

Each has a linear function inside, so integrate and then divide by the coefficient of x:

$$\text{(a) } \tfrac{1}{2}e^{2x} + C, \qquad \text{(b) } \frac{(2x+1)^6}{12} + C, \qquad \text{(c) } \tfrac{1}{3}\sin 3x + C.$$

In (b), reversing the power rule gives $\frac{(2x+1)^6}{6}$, then dividing by the coefficient 2 gives the 12.

EXAMPLE 15.6

Find (a) $\int \frac{1}{x^2+25}\,dx$, (b) $\int \frac{1}{\sqrt{9-x^2}}\,dx$.

Match the standard forms with $a=5$ and $a=3$:

$$\text{(a) } \tfrac{1}{5}\arctan\frac{x}{5} + C, \qquad \text{(b) } \arcsin\frac{x}{3} + C.$$

The only work is recognising a^2 as 25 and as 9. After that the standard form gives the answer.

The linear-composite rule extends to the whole table. For instance, $\int \sec^2(2x+5)\,dx = \frac{1}{2}\tan(2x+5) + C$, the guide's own example: integrate as usual, divide by the inner coefficient.

Definite integrals of these standard forms are evaluated exactly as in §15.2: find the antiderivative, then substitute the limits.

EXAMPLE 15.7

Evaluate (a) $\int_0^{\pi/2} \sin x\,dx$, (b) $\int_1^{e^2} \frac{1}{x}\,dx$.

(a) The antiderivative of $\sin x$ is $-\cos x$, so

$$\int_0^{\pi/2} \sin x\,dx = \Big[-\cos x\Big]_0^{\pi/2} = -\cos\frac{\pi}{2} + \cos 0 = 0 + 1 = 1.$$

The area under one quarter-wave of the sine curve is exactly 1, with no π in the answer.

(b) The antiderivative of $\dfrac{1}{x}$ is $\ln|x|$, and both limits are positive:

$$\int_1^{e^2} \frac{1}{x}\,dx = \Big[\ln x\Big]_1^{e^2} = \ln e^2 - \ln 1 = 2 - 0 = 2.$$

The logarithm hands back the exponent. Stopping at $x = e$ instead would give an area of exactly one, and that is where the number e comes from: it is the upper limit that makes the area under $y = \frac{1}{x}$, starting at $x = 1$, equal to one.

YOUR TURN 15.3 Standard Integrals

6. Find each indefinite integral.

(a) $\displaystyle\int (2\cos x - \sin x)\,dx$ (b) $\displaystyle\int \left(3e^x + \frac{1}{x}\right)dx$

(c) $\displaystyle\int e^{3x}\,dx$ (d) $\displaystyle\int (3x-2)^4\,dx$

(e) $\displaystyle\int \sin 2x\,dx$ (f) $\displaystyle\int \sec^2(4x)\,dx$

(g) $\displaystyle\int \cos(3x+1)\,dx$ (h) $\displaystyle\int \frac{1}{2x+5}\,dx$

7. Find each indefinite integral, matching it to a standard form.

(a) $\displaystyle\int \frac{1}{x^2+9}\,dx$ (b) $\displaystyle\int \frac{1}{\sqrt{16-x^2}}\,dx$

(c) $\displaystyle\int 5^x\,dx$ (d) $\displaystyle\int \frac{1}{x^2+4x+13}\,dx$

(e) $\displaystyle\int \frac{1}{\sqrt{9-4x^2}}\,dx$ (f) $\displaystyle\int \frac{2}{4+x^2}\,dx$

15.4 Rewriting Before Integrating HL

Many integrands do not match a standard form as they stand, but can be rewritten until they do. Four rewritings cover almost every case: divide, complete the square, split into partial fractions, or replace a squared trigonometric function using a double angle. For the first three, deciding between them takes one look at the denominator.

15.4.1 Divide first

When the numerator has degree at least that of the denominator, the fraction is **improper** and no standard form applies. Divide before anything else. For instance $\dfrac{x^2}{x+1} = x - 1 + \dfrac{1}{x+1}$, so the integral is $\frac{x^2}{2} - x + \ln|x+1| + C$. Only when the numerator has the lower degree are the next two methods available.

15.4.2 Completing the square HL

An examination question rarely presents a denominator already in the form $a^2 + x^2$. A quadratic such as $x^2 + 2x + 5$ has that shape, but only after you complete the square.

EXAMPLE 15.8

Find $\displaystyle\int \frac{1}{x^2 + 2x + 5}\,dx$.

Complete the square: $x^2 + 2x + 5 = (x+1)^2 + 4$. This is the arctangent shape with $a = 2$ and the shifted variable $x + 1$:

$$\int \frac{dx}{(x+1)^2 + 4} = \frac{1}{2}\arctan\left(\frac{x+1}{2}\right) + C.$$

Any quadratic with no real roots can be written in the form $a^2 + u^2$ this way. The shift from x to $x + 1$ does not change the integral, because the derivative of $x + 1$ is 1.

15.4.3 Partial fractions HL

When the quadratic *does* factorise, use a different method. Split the fraction into partial fractions, as in Ch. 6. Each piece then has the form $\frac{1}{x+k}$, and integrates to a logarithm.

EXAMPLE 15.9

Find $\displaystyle\int \frac{1}{x^2 + 3x + 2}\,dx$.

Factor and decompose: $\dfrac{1}{(x+1)(x+2)} = \dfrac{1}{x+1} - \dfrac{1}{x+2}$. Then

$$\int \left(\frac{1}{x+1} - \frac{1}{x+2}\right) dx = \ln|x+1| - \ln|x+2| + C = \ln\left|\frac{x+1}{x+2}\right| + C.$$

Factorisable denominator: partial fractions and logarithms. Irreducible denominator: complete the square and arctangent. Choosing between the two is the first decision, and factorising the denominator is what settles it.

15.4.4 Using a double angle

There is no standard integral for $\sin^2 x$ or $\cos^2 x$, and no amount of staring at the table will produce one. Neither is the derivative of anything simple. The way through is to remove the square before integrating, using the double-angle identity from Ch. 8:

$$\cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Rearranged, these give the two rewritings that matter here:

KEY

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}$$

Each right-hand side is a constant plus a linear composite, and both of those integrate immediately.

EXAMPLE 15.10

Find (a) $\int \sin^2 x\,dx$, (b) $\int_0^{\pi} \sin^2 x\,dx$.

(a) Replace the square first:

$$\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx = \frac{1}{2}\int (1-\cos 2x)\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

The $\frac{1}{4}$ comes from the linear composite: integrating $\cos 2x$ gives $\frac{1}{2}\sin 2x$, and the outer $\frac{1}{2}$ halves it again.

(b) Now use the limits:

$$\int_0^{\pi} \sin^2 x\,dx = \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\pi} = \left(\frac{\pi}{2} - 0\right) - (0-0) = \frac{\pi}{2}.$$

The answer comes out cleanly, and there is a reason for that. Over a full arch, $\sin^2 x$ averages exactly $\frac{1}{2}$, so the area is half the width π. The oscillating part, $\sin 2x$, contributes nothing over a whole number of half-periods.

The same rewriting handles $\cos^2 x$, and it is the only way to integrate the square of a sine or cosine at this level. Expect it whenever a volume of revolution involves a trigonometric curve, since the volume formula squares the function.

YOUR TURN 15.4 Rewriting Before Integrating HL

8.

(a) Find $\int \frac{1}{x^2-9}\,dx$ using partial fractions.

(b) Find $\int \frac{4x+5}{(x+1)(x+2)}\,dx$.

(c) Explain how you decide, from the denominator alone, whether to use partial fractions or to complete the square.

9. Rewrite each integrand before integrating.

(a) $\displaystyle\int \frac{x^2+3x}{x}\,dx$

(b) $\displaystyle\int \frac{2x+1}{x}\,dx$

(c) $\displaystyle\int \frac{x^3-1}{x^2}\,dx$

(d) $\displaystyle\int \frac{1}{x^2+6x+13}\,dx$

15.5 Integration by Substitution

Substitution reverses the chain rule. Look for a composite function whose inner function also appears, up to a constant multiple, as a factor outside. Put u equal to that inner function, and a hard integral in x becomes an easy one in u.

The method has four steps. Let u be the inner function. Replace dx using $du = \dfrac{du}{dx}\,dx$. Integrate in u. Then put x back.

EXAMPLE 15.11

Find $\displaystyle\int 2x(x^2+1)^3\,dx$.

Let $u = x^2+1$, so $\dfrac{du}{dx} = 2x$, giving $du = 2x\,dx$. The integral becomes

$$\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+1)^4}{4} + C.$$

The factor $2x$ outside is exactly the $\frac{du}{dx}$ required. Without it the substitution would leave an x behind and fail.

EXAMPLE 15.12

Find $\displaystyle\int \frac{2x}{x^2+1}\,dx$.

With $u = x^2+1$, $du = 2x\,dx$:

$$\int \frac{1}{u}\,du = \ln|u| + C = \ln(x^2+1) + C.$$

An integral of the form $\dfrac{f'(x)}{f(x)}$ always integrates to $\ln|f(x)|$.

CAUTION For a definite integral by substitution, either convert the limits to u-values, or substitute back to x before applying the original limits. Applying the x-limits to a u-expression gives a wrong answer.

EXAMPLE 15.13

Evaluate $\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx$.

Let $u = x^2+1$, so $du = 2x\,dx$. Convert the limits too: $x = 0 \Rightarrow u = 1$ and $x = \sqrt{3} \Rightarrow u = 4$. Then

$$\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx = \frac{1}{2}\int_1^4 \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}\Big[u^{3/2}\Big]_1^4 = \frac{1}{3}(8-1) = \frac{7}{3}.$$

Once the limits are converted, you never return to x at all; the whole calculation finishes in u.

Substitution is at its most useful on exponential and trigonometric integrands, where no rearrangement of the standard table will help.

EXAMPLE 15.14

Find (a) $\int xe^{-x^2}\,dx$, (b) $\int \sin^4 x\cos x\,dx$.

(a) The inner function is $-x^2$, and its derivative $-2x$ appears outside up to the constant -2. Let $u = -x^2$, so $du = -2x\,dx$ and $x\,dx = -\frac{1}{2}\,du$:

$$\int xe^{-x^2}\,dx = -\frac{1}{2}\int e^u\,du = -\frac{1}{2}e^u + C = -\frac{1}{2}e^{-x^2} + C.$$

On its own, e^{-x^2} cannot be integrated in terms of the functions of this course. The factor x is what makes the integral possible, and losing it makes the problem unsolvable.

(b) Here the inner function is $\sin x$, and its derivative $\cos x$ is sitting outside. Let $u = \sin x$, so $du = \cos x\,dx$:

$$\int \sin^4 x\cos x\,dx = \int u^4\,du = \frac{u^5}{5} + C = \frac{\sin^5 x}{5} + C.$$

In both parts the test is the same: is the derivative of the inner function present as a factor? If it is, substitution works. If it is not, look for another method.

YOUR TURN 15.5 Integration by Substitution

10. Find each integral by substitution.

(a) $\int 3x^2(x^3+1)^4\,dx$

(b) $\int xe^{x^2}\,dx$

(c) $\int 2x\sqrt{x^2+1}\,dx$

(d) $\int \frac{x}{(x^2+4)^2}\,dx$

(e) $\displaystyle\int \frac{\ln x}{x}\,dx$

(f) $\displaystyle\int \sin^3 x \cos x\,dx$

11.

(a) Evaluate $\displaystyle\int_0^1 3x^2(x^3+1)^4\,dx$ by converting the limits.

(b) Evaluate the same integral by substituting back to x first, and confirm the answers agree.

(c) Evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x \cos x\,dx$ by converting the limits.

15.6 Integration by Parts HL

Substitution reverses the chain rule; **integration by parts** reverses the product rule. It handles products that substitution cannot, such as xe^x or $\ln x$.

KEY · IB HL

$$\int u\,\frac{dv}{dx}\,dx = uv - \int v\,\frac{du}{dx}\,dx$$

You split the integrand into a part u to differentiate and a part $\dfrac{dv}{dx}$ to integrate. In practice it is easier to work with the same rule written in terms of the two functions you can see in front of you. Writing the integrand as a product uv, and letting $\displaystyle\int v\,dx$ stand for any antiderivative of v:

KEY

$$\int uv\,dx \;=\; u\int v\,dx \;-\; \int\left(\int v\,dx\right)\frac{du}{dx}\,dx$$

Read from left to right this says: integrate the second function, multiply by the first, then subtract the integral of that product with the first function differentiated. Nothing new has happened, but v is never named separately.

Everything now depends on which factor you call u. Choose the one that becomes simpler when differentiated, so that the second integral is easier than the one you started with. The order **ILATE** is a reliable guide: whichever type appears first in this list should be u.

Inverse trigonometric — $\arcsin x$, $\arctan x$

L **Logarithmic** — $\ln x$

A **Algebraic** — $x,\ x^2,\ 3x-1$

T **Trigonometric** — $\sin x,\ \cos x$

E **Exponential** — $e^x,\ 2^x$

In $\int xe^x\,dx$ the factors are algebraic and exponential, so A comes before E and $u = x$. In $\int x\ln x\,dx$ they are logarithmic and algebraic, so L comes before A and $u = \ln x$. The order works because differentiating reduces the first three types to something simpler, while trigonometric and exponential functions repeat themselves however often you differentiate them.

CAUTION ILATE is a guide, not a theorem. It gives the right choice for the integrals you will meet in this course, but the test that always applies is the one behind it: does the second integral end up easier than the first?

EXAMPLE 15.15

Find $\displaystyle\int xe^x\,dx$.

Let $u = x$ (simplifies to 1) and $\dfrac{dv}{dx} = e^x$ (so $v = e^x$):

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Choosing $u = x$ is what makes the new integral, $\int e^x\,dx$, trivial. The other choice would make it worse.

EXAMPLE 15.16

Find (a) $\displaystyle\int (x+1)\sin x\,dx$, (b) $\displaystyle\int_0^{\pi/2} (x+1)\cos x\,dx$.

(a) The factors are algebraic and trigonometric, so ILATE gives $u = x+1$ and $\dfrac{dv}{dx} = \sin x$, hence $v = -\cos x$:

$$\int (x+1)\sin x\,dx = -(x+1)\cos x - \int(-\cos x)\,dx = -(x+1)\cos x + \sin x + C.$$

Watch the two minus signs. Integrating $\sin x$ gives $-\cos x$, and the formula then subtracts the integral of that, which turns the sign back.

(b) Same choice, with $v = \sin x$:

$$\int (x+1)\cos x\,dx = (x+1)\sin x - \int \sin x\,dx = (x+1)\sin x + \cos x + C.$$

Now apply the limits:

$$\int_0^{\pi/2} (x+1)\cos x\,dx = \Big[(x+1)\sin x + \cos x\Big]_0^{\pi/2} = \left(\frac{\pi}{2}+1+0\right) - (0+1) = \frac{\pi}{2}.$$

For a definite integral by parts, find the whole antiderivative first, then substitute the limits once at the end.

EXAMPLE 15.17

Find $\displaystyle\int \ln x\,dx$.

There is only one function, but write $u = \ln x$ and $\dfrac{dv}{dx} = 1$ (so $v = x$):

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C.$$

Taking $\dfrac{dv}{dx} = 1$ is a standard move. It is how both $\ln x$ and $\arctan x$ are integrated.

15.6.1 Repeated integration by parts

If one application of parts leaves an integral that still needs parts, apply it again. Each round of differentiation reduces the power of x by one.

EXAMPLE 15.18

Find $\displaystyle\int x^2e^{-x}\,dx$.

First round, $u = x^2$ and $\dfrac{dv}{dx} = e^{-x}$, so $v = -e^{-x}$:

$$\int x^2e^{-x}\,dx = -x^2e^{-x} + 2\int xe^{-x}\,dx.$$

The remaining integral needs parts a second time, with $u = x$:

$$\int xe^{-x}\,dx = -xe^{-x} + \int e^{-x}\,dx = -e^{-x}(x+1) + C.$$

Substituting that back,

$$\int x^2e^{-x}\,dx = -x^2e^{-x} - 2e^{-x}(x+1) + C = -e^{-x}\left(x^2+2x+2\right) + C.$$

A power x^n takes n rounds of parts to clear, since only the power changes from round to round; the exponential returns unchanged each time.

Something cleverer happens when *neither* factor ever simplifies. For $e^x\sin x$, two rounds of parts bring the original integral back, and you solve for it like an equation.

EXAMPLE 15.19

Find $I = \displaystyle\int e^x \sin x \, dx$.

Parts with $u = \sin x$:

$$I = e^x \sin x - \int e^x \cos x \, dx.$$

Parts again on the new integral, $u = \cos x$:

$$\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I.$$

Substitute back:

$$I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x) \implies I = \tfrac{1}{2} e^x(\sin x - \cos x) + C.$$

The original integral reappeared on the right-hand side, which turned the problem into an equation for I. Watch for this whenever neither factor gets simpler: two rounds of parts bring I back, and algebra finishes the job.

15.6.2 Choosing a method

Five methods are now available, and most marks lost in this topic are lost by starting the wrong one. The choice is settled by looking at the integrand before writing anything.

What you see	Method	Example
A standard form exactly, or with a linear inside	read it off, dividing by the inner coefficient	$\cos 3x$, e^{2x}, $(2x+1)^5$
A function together with a multiple of its own derivative	substitution	$2x(x^2+1)^3$, $\dfrac{2x}{x^2+1}$
A product of two unlike functions, one of which simplifies when differentiated	parts	xe^x, $x \sin x$, $\ln x$
A fraction whose denominator factorises	partial fractions, then logarithms	$\dfrac{1}{x^2+3x+2}$
A fraction whose denominator is a quadratic with no real roots	complete the square, then arctangent	$\dfrac{1}{x^2+2x+5}$

Work down the list in that order, because the earlier tests are quicker to apply and the earlier methods are shorter. Two habits help further. Check first whether the fraction is improper, since dividing may remove the problem entirely. And if a substitution leaves an integral no easier than the one you started with, the substitution was the wrong one; go back rather than pressing on.

YOUR TURN 15.6 Integration by Parts HL

12. Find each integral by parts.

(a) $\displaystyle\int x\cos x\,dx$

(b) $\displaystyle\int xe^{2x}\,dx$

(c) $\displaystyle\int x\ln x\,dx$

(d) $\displaystyle\int \arctan x\,dx$

(e) $\displaystyle\int x^2\ln x\,dx$

(f) $\displaystyle\int x\sec^2 x\,dx$

13.

(a) Evaluate $\displaystyle\int_0^\pi x\sin x\,dx$.

(b) Evaluate $\displaystyle\int_0^1 xe^x\,dx$.

(c) State which factor you chose as u in part (b), and why the other choice would not help.

(d) Use ILATE to say which factor should be u in each of $\displaystyle\int x^2e^x\,dx$, $\displaystyle\int x\arctan x\,dx$ and $\displaystyle\int e^x\sin x\,dx$.

(e) Explain why the third integral in part (d) behaves differently from the first two.

15.7 Areas Between Curves

To find the area between two curves, integrate the difference: the upper curve minus the lower. The signed-area problem disappears, because the difference is positive wherever the first curve is on top, regardless of the axis.

KEY

$$A = \int_a^b \left(y_{\text{top}} - y_{\text{bottom}}\right) dx$$

The limits a and b are usually the x-coordinates where the curves intersect, found by setting them equal.

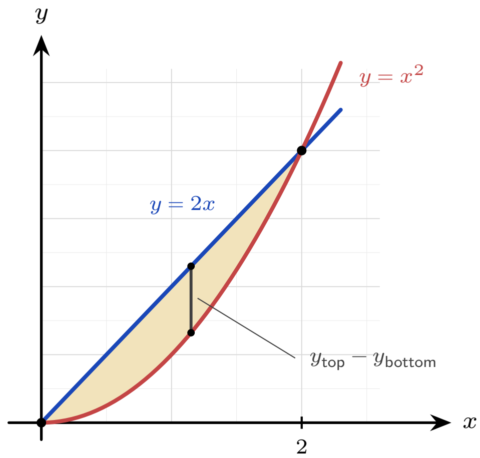


Figure 15.8 The region enclosed between $y = 2x$ and $y = x^2$. The curves meet where $2x = x^2$, at $x = 0$ and $x = 2$, and the line is the upper boundary in between, so the area is $\int_0^2 (2x - x^2)\,dx$.

EXAMPLE 15.20

Find the area enclosed between $y = 2x$ and $y = x^2$.

Intersections: $2x = x^2 \Rightarrow x = 0$ or $x = 2$. On $(0, 2)$ the line $y = 2x$ is above, so

$$A = \int_0^2 (2x - x^2)\,dx = \left[x^2 - \tfrac{x^3}{3}\right]_0^2 = 4 - \tfrac{8}{3} = \tfrac{4}{3}.$$

15.7.1 Area between a curve and the y-axis HL

Sometimes it is easier to integrate with respect to y: express x as a function of y and accumulate horizontal slices. This gives the area between the curve and the y-axis.

KEY · IB HL

$$A = \int_a^b x\,dy$$

EXAMPLE 15.21

Find the area between the curve $x = y^2$ and the y-axis from $y = 0$ to $y = 2$.

$$A = \int_0^2 y^2\,dy = \left[\frac{y^3}{3}\right]_0^2 = \frac{8}{3}.$$

Integrating in y avoids splitting the region, which would be awkward in x.

Nothing in the method requires the curves to be polynomials. The steps are the same for trigonometric and exponential boundaries: find where they cross, decide which is on top, and integrate the difference.

EXAMPLE 15.22

Find the area enclosed between $y = \sin x$ and $y = \cos x$ between their first two intersections for $x \geq 0$.

They meet where $\sin x = \cos x$, that is $\tan x = 1$, so the first two intersections are at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$.

On $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ the sine is above: at $x = \frac{\pi}{2}$, $\sin\frac{\pi}{2} = 1$ while $\cos\frac{\pi}{2} = 0$. So

$$A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx = \Big[-\cos x - \sin x\Big]_{\pi/4}^{5\pi/4}.$$

At the upper limit both $\cos x$ and $\sin x$ equal $-\frac{\sqrt{2}}{2}$, so the bracket is $\sqrt{2}$; at the lower limit both equal $\frac{\sqrt{2}}{2}$ and the bracket is $-\sqrt{2}$:

$$A = \sqrt{2} - \left(-\sqrt{2}\right) = 2\sqrt{2} \approx 2.83.$$

Deciding which curve is on top is the step that carries the sign. Testing one convenient point inside the interval, here $x = \frac{\pi}{2}$, settles it.

YOUR TURN 15.7 Areas Between Curves

14. Find each enclosed area.

(a) Between $y = x$ and $y = x^2$.
(b) Between $y = 4 - x^2$ and the x-axis.
(c) Between $y = 6 - x^2$ and $y = 2$.
(d) Between $x = y^2$ and the y-axis, from $y = 0$ to $y = 3$.
(e) Between $y = x^2$ and $y = 2 - x^2$.

15. Consider the region enclosed by $y = x^2$ and $y = x + 2$.

(a) Find the x-coordinates where the two curves meet.
(b) State which curve is the upper boundary between those values.
(c) Find the area of the enclosed region.

(d) State what answer you would get by integrating $x^2 - (x + 2)$ instead, and explain what has gone wrong.

16. Find each enclosed area.

(a) Under $y = \cos x$, from $x = 0$ to $x = \frac{\pi}{2}$.
(b) Under $y = e^x$, from $x = 0$ to $x = 2$.
(c) Between $y = \sin x$ and the x-axis, from $x = 0$ to $x = \pi$.
(d) Between $y = e^x$ and $y = e^{-x}$, from $x = 0$ to $x = 1$.
(e) Under $y = \dfrac{1}{x}$, from $x = 1$ to $x = 4$.

15.8 Volumes of Revolution HL

Rotate a curve about an axis and it sweeps out a solid. To find the volume, slice the solid into thin disks perpendicular to the axis. Each disk is a cylinder of radius y (the curve's height) and thickness dx, with volume $\pi y^2\,dx$. Summing the disks and letting their thickness shrink turns the sum into an integral.

KEY · IB HL

$$V = \int_a^b \pi y^2\,dx \quad \text{(about the } x\text{-axis)}, \qquad V = \int_a^b \pi x^2\,dy \quad \text{(about the } y\text{-axis)}$$

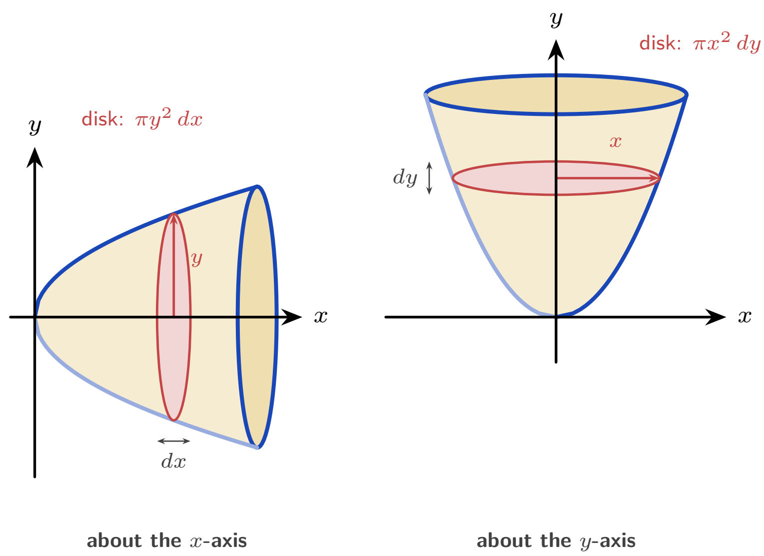


Figure 15.9 The same curve rotated about each axis. Turning it about the x-axis gives disks stacked along x: each has radius y and thickness dx, so its volume is $\pi y^2\,dx$. Turning it about the y-axis gives

disks stacked along y: each has radius x and thickness dy, so its volume is $\pi x^2\,dy$. In both cases the integral adds the disks from one end of the solid to the other.

EXAMPLE 15.23

The curve $y = x^3$ for $0 \le x \le 2$ is rotated about the x-axis. Find the volume.

$$V = \int_0^2 \pi(x^3)^2\,dx = \pi\int_0^2 x^6\,dx = \pi\Big[\tfrac{x^7}{7}\Big]_0^2 = \frac{128\pi}{7}.$$

Square the function *before* integrating; squaring after is wrong.

EXAMPLE 15.24

The region between $y = x^2$ and the y-axis, from $y = 0$ to $y = 1$, is rotated about the y-axis. Find the volume.

About the y-axis, slice into disks of radius x and thickness dy, with $x^2 = y$:

$$V = \int_0^1 \pi x^2\,dy = \pi\int_0^1 y\,dy = \pi\Big[\tfrac{y^2}{2}\Big]_0^1 = \frac{\pi}{2}.$$

The limits are now y-values, and x^2 is rewritten in terms of y.

Read the region carefully. A disk of radius x is swept by the strip running from the y-*axis* out to the curve, so the region rotated is the one between the curve and the y-axis, not the one under the curve. Those are two different regions and in general they give two different volumes.

Because the formula squares the function, a trigonometric curve produces $\sin^2$ or $\cos^2$, and those must be rewritten with a double angle before they can be integrated See §15.4.

EXAMPLE 15.25

The region under $y = \sin 2x$ from $x = 0$ to $x = \frac{\pi}{4}$ is rotated 360° about the x-axis. Find the volume.

Apply the formula, squaring the function first:

$$V = \int_0^{\pi/4} \pi y^2\,dx = \pi\int_0^{\pi/4} \sin^2 2x\,dx.$$

There is no standard integral for $\sin^2 2x$, so replace it using $\sin^2 2x = \dfrac{1-\cos 4x}{2}$:

$$\pi\int_0^{\pi/4} \frac{1-\cos 4x}{2}\,dx = \pi\Big[\frac{x}{2} - \frac{\sin 4x}{8}\Big]_0^{\pi/4} = \pi\left(\frac{\pi}{8} - 0\right) = \frac{\pi^2}{8} \approx 1.23.$$

Two πs appear for different reasons, and they must be kept apart: one comes from the volume formula, the other from the upper limit.

EXAMPLE 15.26

The region under $y = e^x$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

$$V = \pi \int_0^2 (e^x)^2 \, dx = \pi \int_0^2 e^{2x} \, dx = \pi \left[\tfrac{1}{2} e^{2x}\right]_0^2 = \frac{\pi}{2}\left(e^4 - 1\right) \approx 84.2.$$

Squaring first is what turns e^x into e^{2x}, a linear composite. Squaring the integral instead of the function is the usual error, and it gives a different answer entirely.

YOUR TURN 15.8 Volumes of Revolution HL

17. Find each volume of revolution about the x-axis.

(a) $y = x^2$, $0 \le x \le 2$.
(b) $y = \sqrt{x}$, $0 \le x \le 4$.
(c) $y = 2x$, $0 \le x \le 3$.
(d) $y = \dfrac{1}{x}$, $1 \le x \le 3$.
(e) $y = \sin x$, $0 \le x \le \pi$.

18.

(a) The region between $y = x^2$ and the y-axis, from $y = 0$ to $y = 4$, is rotated about the y-axis. Find the volume.
(b) Explain why the answer differs from the earlier volume about the x-axis, even though the same curve is used.
(c) The region between $x = y^2$ and the y-axis, from $y = 0$ to $y = 2$, is rotated about the y-axis. Find the volume.

19. Each of these volumes of revolution about the x-axis needs a rewriting before it can be integrated.

(a) $y = \cos x$, $0 \le x \le \frac{\pi}{2}$.
(b) $y = e^x$, $0 \le x \le 1$.
(c) $y = e^{-x}$, $0 \le x \le \ln 2$.
(d) $y = \sin x$, $0 \le x \le \frac{\pi}{2}$.
(e) Explain why a trigonometric curve always forces the double-angle rewriting of §15.4, while an exponential one does not.

15.9 Kinematics by Integration

In Ch. 14 you differentiated displacement to get velocity, and velocity to get acceleration. Integration reverses those two steps. Integrate acceleration to get velocity, and velocity to get displacement, using an initial condition each time to fix the constant.

KEY · IB

$$\text{displacement} = \int_{t_1}^{t_2} v\,dt, \qquad \text{distance travelled} = \int_{t_1}^{t_2} |v|\,dt$$

Displacement is the signed change in position; distance is the total path length. They differ whenever the particle reverses direction, because backward motion subtracts from displacement but adds to distance. On a velocity–time graph the difference is visible: area below the axis counts as negative for displacement, and as positive for distance.

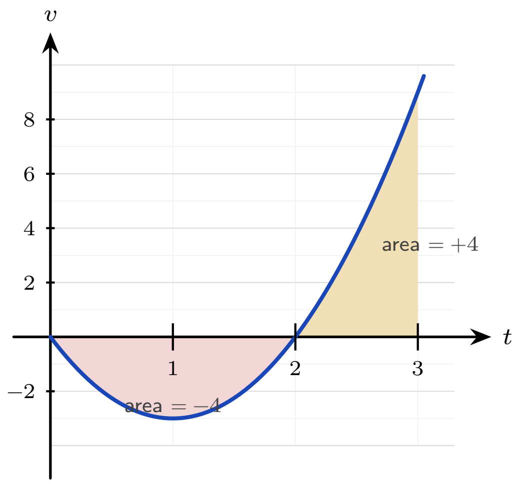


Figure 15.10 The velocity $v = 3t^2 - 6t$ over $0 \le t \le 3$. The particle moves backwards until $t = 2$ and forwards after it. Displacement adds the two signed areas, $-4 + 4 = 0$; distance adds their sizes, $4 + 4 = 8$ m.

EXAMPLE 15.27

A particle has velocity $v(t) = 3t^2 - 6t\,\mathrm{m\,s^{-1}}$. Find its displacement and the distance travelled over $0 \le t \le 3$.

Displacement integrates v directly:

$$\int_0^3 (3t^2 - 6t)\,dt = \Big[t^3 - 3t^2\Big]_0^3 = 27 - 27 = 0.$$

The particle ends where it began. For distance, note $v = 3t(t-2)$ is negative on $(0, 2)$ and positive on $(2,3)$, so split:

$$\left|\int_0^2 v\,dt\right| + \int_2^3 v\,dt = |-4| + 4 = 8 \text{ m}.$$

Zero displacement but 8 m travelled: the particle went out 4 m and came back.

Oscillating motion is handled the same way, and it makes the difference between displacement and distance especially clear.

EXAMPLE 15.28

A particle has velocity $v(t) = 4\cos 2t\,\mathrm{m\,s^{-1}}$. Find its displacement and the distance travelled over $0 \le t \le \frac{\pi}{2}$ seconds.

Displacement integrates v directly. The antiderivative of $4\cos 2t$ is $2\sin 2t$:

$$\int_0^{\pi/2} 4\cos 2t\,dt = \Big[2\sin 2t\Big]_0^{\pi/2} = 2\sin\pi - 2\sin 0 = 0.$$

The particle finishes where it started. For the distance, first find where it turns: $v = 0$ when $\cos 2t = 0$, so $2t = \frac{\pi}{2}$ and $t = \frac{\pi}{4}$. Split the interval there:

$$\int_0^{\pi/4} 4\cos 2t\,dt = \Big[2\sin 2t\Big]_0^{\pi/4} = 2, \qquad \int_{\pi/4}^{\pi/2} 4\cos 2t\,dt = 0 - 2 = -2.$$

Distance adds the sizes: $|2| + |-2| = 4$ m. So the particle travels 2 m out and 2 m back, ending at its starting point.

The turning point is what makes the split necessary, and forgetting it is the standard error: integrating $|v|$ straight through as if it were v would return 0 for the distance as well.

YOUR TURN 15.9 Kinematics by Integration

20. A particle moves in a straight line.

(a) Its velocity is $v = 2t + 1$. Find its displacement over $0 \le t \le 3$.
(b) Its acceleration is $a = 6t$ and $v(0) = 1$. Find $v(t)$.
(c) Its velocity is $v = t^2 - 4$. Find its displacement over $0 \le t \le 3$.
(d) Its velocity is $v = 4 - 2t$. Find the distance travelled over $0 \le t \le 4$.
(e) For the particle in part (d), find the displacement over the same interval, and explain why the two answers differ.
(f) Its velocity is $v = 3t^2 - 12$. Find the time at which it changes direction, and the distance travelled over $0 \le t \le 3$.

21. A particle moves in a straight line.

(a) Its velocity is $v = 3\cos t$ m s^{-1}. Find its displacement over $0 \le t \le \frac{\pi}{2}$.

(b) Its velocity is $v = 6e^{-2t}$ m s^{-1}. Find its displacement over $0 \le t \le 1$.

(c) Its acceleration is $a = \sin t$ m s^{-2} and $v(0) = 0$. Find $v(t)$.

(d) Its velocity is $v = 2\cos t$ m s^{-1}. Find the time in $0 \le t \le \pi$ at which it changes direction, and the distance travelled over that interval.

Chapter 15 · Integral Calculus

Reflections

TOK The proof in §15.2 establishes *that* the area under a curve and the reverse of a derivative are the same calculation. It does not explain *why* the world should be arranged that way. Two ideas invented for different purposes, one geometric and one algebraic, turned out to be one idea.

Does that make the connection a discovery about quantities and their rates, or a consequence of how we chose to define both? Is mathematics invented or uncovered?

IM Long before calculus, Archimedes (3rd century BCE, Syracuse) found the area of a parabolic segment by his "method of exhaustion": filling the region with ever-smaller triangles and summing them. He effectively computed an integral two thousand years before Newton and Leibniz gave the operation a name and a reverse-differentiation shortcut. The idea of accumulation by slicing is older than the algebra we use to carry it out, and his triangles are the ancestors of the rectangles in §15.2.

RW Integration is summation made continuous, so it appears wherever a total is built from a varying rate. The total charge from a varying current, the work done by a changing force, the area of an irregular plot of land, the probability accumulated under a distribution curve (Ch. 12): each is an integral. Whenever a quantity is the running total of a rate, integration computes it.

RW A shape made by spinning a profile about an axis, such as a bottle, a lens or a turbine housing, is fixed completely by that one curve, so its volume is a single integral rather than a measurement. Engineers size fuel tanks and machined parts this way, and the same integral gives the mass once the density is known.

EXT Not every function has an antiderivative in terms of familiar functions: $\int e^{-x^2}\,dx$, central to statistics, cannot be written with powers, exponentials, and trigonometric functions. Yet the definite integral still exists as a perfectly real area, and is computed numerically. The reach of "find the area" is wider than the reach of "reverse the derivative."

End-of-Chapter Problems — Chapter 15: Integral Calculus

1. A curve has $\dfrac{dy}{dx} = 3x^2 - 4x + 1$ and passes through $(1, 2)$.

(a) Find the equation of the curve. [3]

(b) Find the value of y when $x = 2$. [2]

2. Consider $y = x^2 - 4$.

(a) Find the x-intercepts. [2]

(b) Find the area enclosed between the curve and the x-axis. [4]

3. The curves $y = x^2$ and $y = 8 - x^2$ intersect at two points.

(a) Find the coordinates of the intersection points. [3]

(b) Find the area enclosed between the curves. [4]

4.

(a) Using the substitution $u = x^2 + 4$, find $\displaystyle\int \frac{x}{x^2+4}\,dx$. [3]

(b) Hence evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving an exact answer. [3]

5.

(a) Find $\displaystyle\int xe^{2x}\,dx$ using integration by parts. [3]

(b) Evaluate $\displaystyle\int_0^1 xe^{2x}\,dx$, giving an exact answer. [2]

6. The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis.

(a) Write down an integral for the volume. [2]

(b) Find the volume. [2]

7. A particle starts at the origin with velocity $5\,\mathrm{m\,s^{-1}}$ and has acceleration $a = 6t - 4\,\mathrm{m\,s^{-2}}$.

(a) Find the velocity $v(t)$. [2]

(b) Find the displacement $s(t)$. [2]

(c) Find the displacement when $t = 2$. [2]

8. Consider the region enclosed by $y = x^2$, the y-axis, and $y = 4$.

(a) Find the area of the region. [3]

(b) Find the volume when this region is rotated about the y-axis. [3]

9.

(a) Find $\displaystyle\int \frac{1}{x^2+4}\,dx$. [2]

(b) Evaluate $\displaystyle\int_0^2 \frac{1}{x^2+4}\,dx$, giving an exact answer. [3]

10. The velocity of a particle is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]

(b) Find the total distance travelled in the first 3 seconds. [5]

11. A curve has gradient function $\dfrac{dy}{dx} = 4x - 3$ and passes through the point $(2, 5)$.

(a) Find the equation of the curve. [3]

(b) Write down its y-intercept. [1]

12. Find the following integrals.

(a) $\displaystyle\int e^{3x}\,dx$ [2]

(b) $\displaystyle\int (4x-1)^6\,dx$ [2]

(c) $\displaystyle\int \sin 2x\,dx$ [2]

13. Evaluate $\displaystyle\int_1^4 \left(\frac{2}{\sqrt{x}} + 1\right) dx$. [4]

14. Consider the curve $y = x^2 - 1$ on the interval $0 \leq x \leq 2$.

(a) Find where the curve crosses the x-axis in this interval. [1]

(b) Find the total area enclosed between the curve and the x-axis on the interval. [5]

15. Find the area of the region enclosed by the curve $y = 4x - x^2$ and the line $y = x$. [5]

16. Using the substitution $u = x^2 + 1$, evaluate $\displaystyle\int_0^2 x(x^2+1)^3\,dx$. [5]

17. Evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving your answer in the form $a \ln b$. [5]

18. Find $\displaystyle\int \frac{x+7}{(x-1)(x+3)}\,dx$. [6]

19. Find $\displaystyle\int \frac{1}{x^2-4x+13}\,dx$. [5]

20. Find the following integrals.

(a) $\displaystyle\int x\cos x\,dx$ [3]

(b) $\displaystyle\int x^2\ln x\,dx$ [4]

21. The region under $y=\sqrt{x}$ from $x=0$ to $x=4$ is rotated about the x-axis.

(a) Find the volume generated. [3]

(b) The same curve, written $x=y^2$ for $0\le y\le 2$, is rotated about the y-axis. Find the volume generated. [4]

22. A particle moves with velocity $v(t)=t^2-4t+3\,\mathrm{m\,s^{-1}}$ for $0\le t\le 4$.

(a) Find the displacement of the particle over the interval. [3]

(b) Find the total distance travelled. [5]

23. The area under $y=x^2+1$ from $x=0$ to $x=3$ is to be estimated by three rectangles of width 1.

(a) Find the estimate obtained by taking the height of each rectangle at the left of its strip. [3]

(b) Find the estimate obtained by taking the height at the right. [2]

(c) Evaluate $\displaystyle\int_0^3 (x^2+1)\,dx$ and verify that it lies between the two estimates. [3]

(d) State, with a reason, whether the left estimate would increase or decrease if six rectangles were used instead. [2]

24. The curve $y=x^2$ and the line $y=2-x$ enclose a region R.

(a) Find the x-coordinates of the two points of intersection. [3]

(b) Sketch the curve and the line, and shade R. [2]

(c) Show that the area of R is $\dfrac{9}{2}$. [5]

(d) Explain why integrating $y_{\text{curve}}-y_{\text{line}}$ instead would give the wrong sign. [2]

25. This question uses integration by parts throughout.

(a) Find $\displaystyle\int x\ln x\,dx$. [4]

(b) Hence evaluate $\int_1^e x \ln x \, dx$, giving your answer in terms of e. [3]

(c) Evaluate $\int_0^{\pi/2} x \cos x \, dx$. [5]

(d) In part (c), state which factor you took as u, and explain what would go wrong with the other choice. [2]

26. Consider the curve $y = x^3$ for $0 \leq x \leq 1$.

(a) The region between the curve and the x-axis is rotated 360° about the x-axis. Show that the volume is $\frac{\pi}{7}$. [4]

(b) The region between the curve and the y-axis is rotated 360° about the y-axis. Find this volume, giving an exact answer. [5]

(c) State which of the two solids has the greater volume, and explain the answer by describing the two regions. [3]

27. A particle moves in a straight line with acceleration $a = 6 - 2t \, \mathrm{m\,s^{-2}}$, and is at rest at $t = 0$.

(a) Show that $v = 6t - t^2$. [3]

(b) Find the time at which the particle is next at rest. [2]

(c) Find the maximum velocity, and the time at which it occurs. [3]

(d) Find the distance travelled from $t = 0$ until the particle is next at rest. [4]

(e) Explain why, for this motion, the distance travelled equals the displacement. [2]

28. Each of these integrals needs a different method. In each part, name the method before using it.

(a) $\int \frac{x^2 + 1}{x + 1} \, dx$ [5]

(b) $\int_1^2 \frac{2x - 1}{x^2 - x + 1} \, dx$ [4]

(c) $\int_0^{\pi/2} \sin x \cos^2 x \, dx$ [4]

(d) $\int \frac{1}{x^2 - 4x + 13} \, dx$ [4]

29. The region R is enclosed by $y = \sin x$, $y = \cos x$ and the y-axis.

(a) Show that the two curves first meet at $x = \frac{\pi}{4}$. [2]

(b) Find the area of R. [4]

(c) R is rotated 360° about the x-axis. Show that the volume is $\frac{\pi}{2}$. [6]

30. The region under $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$ is rotated 360° about the x-axis.

(a) Write down an integral for the volume. [2]

(b) Use a double angle to show that the volume is $\dfrac{\pi^2}{4}$. [5]

(c) The same region under $y = \sin x$ gives the same volume. Explain why, without integrating. [2]

31. A particle moves in a straight line with velocity $v(t) = 6 \sin 2t\,\mathrm{m\,s^{-1}}$, for $0 \le t \le \pi$ seconds.

(a) Find the times at which the particle is at rest. [3]

(b) Find the displacement over $0 \le t \le \pi$. [3]

(c) Find the total distance travelled over the same interval. [4]

(d) Explain why the two answers differ. [2]

32. A particle starts from rest at the origin with acceleration $a(t) = 4e^{-2t}\,\mathrm{m\,s^{-2}}$.

(a) Show that $v(t) = 2 - 2e^{-2t}$. [3]

(b) Find the displacement over $0 \le t \le 1$. [4]

(c) State the velocity the particle approaches as t grows large, and explain what is happening physically. [2]

33. Consider the region enclosed by $y = e^x$, $y = e^{-x}$ and the line $x = 1$.

(a) Show that the two curves meet at $x = 0$. [1]

(b) Find the area of the region. [4]

(c) The region is rotated 360° about the x-axis. Find the volume, giving your answer in terms of e and π. [5]

34. Each of these integrals needs a different method. In each part, name the method before using it.

(a) $\displaystyle\int xe^{3x}\,dx$ [4]

(b) $\displaystyle\int \cos^2 3x\,dx$ [4]

(c) $\displaystyle\int \frac{\sin x}{\cos x}\,dx$ [3]

(d) $\displaystyle\int_0^{\ln 2} \frac{e^x}{1+e^x}\,dx$ [4]

Challenge Questions

35. Solids you already know. Every volume formula from Ch. 7 can be recovered by rotating a curve. This question derives three of them.

(a) The line $y = r$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\pi r^2 h$, and name the solid. [3]

(b) The line $y = \dfrac{r}{h}x$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\frac{1}{3}\pi r^2 h$. [4]

(c) The semicircle $y = \sqrt{r^2 - x^2}$ for $-r \le x \le r$ is rotated about the x-axis. Show that the volume is $\frac{4}{3}\pi r^3$. [5]

(d) The cone in part (b) has one third of the volume of the cylinder in part (a). Explain where the factor $\frac{1}{3}$ comes from in the integration. [3]

36. When parts goes in a circle. Some integrals return to where they started, and that is what solves them.

(a) Let $I = \displaystyle\int e^x \cos x \, dx$. Apply integration by parts twice, and show that $I = \frac{1}{2}e^x(\sin x + \cos x) + C$. [6]

(b) Explain why choosing $u = e^x$ at the first step and $u = \cos x$ at the second would not have worked. [3]

(c) Find $\displaystyle\int x^2 e^x \, dx$ by applying parts twice, and explain why this one terminates while part (a) does not. [5]

(d) Predict, without calculating, how many rounds of parts $\displaystyle\int x^4 e^x \, dx$ requires. Justify your answer. [2]

37. A reduction formula. Let $I_n = \displaystyle\int_0^{\pi/2} \sin^n x \, dx$ for $n \ge 0$.

(a) Evaluate I_0 and I_1 directly. [3]

(b) Writing $\sin^n x = \sin^{n-1} x \cdot \sin x$ and using integration by parts, show that

$$I_n = \frac{n-1}{n} I_{n-2}, \qquad n \ge 2.$$

[6]

(c) Use the formula to find I_2, I_3, I_4 and I_5 exactly. [4]

(d) Explain why I_n involves π when n is even but not when n is odd. [3]

CHAPTER 16

Differential Equations

16

SYLLABUS 5.18 5.19 HL ONLY

A cup of coffee cools fastest when it is hottest. Pull it from the machine at 80 degrees and it drops quickly at first; near room temperature it barely changes. The *rate* at which it cools depends on how far above the room its temperature currently is.

That sentence is a differential equation. Written in symbols, the rate of cooling $\frac{d\theta}{dt}$ is proportional to the gap $\theta - 20$ between the coffee and the 20-degree room:

$$\frac{d\theta}{dt} = -k(\theta - 20).$$

Notice what kind of equation this is. The unknown is not a number. It is a *function*, $\theta(t)$, and the equation links that function to its own derivative. Most laws of nature have this form, because what we can measure directly is usually a rate: how fast something grows, cools, decays or moves.

A **differential equation** relates a function to its derivatives. To **solve** one is to find the function itself, recovering $\theta(t)$ from a statement about $\frac{d\theta}{dt}$.

That is the point of solving one: it turns a rule about change into a prediction. A law of nature tells you what is happening at this instant; solving its differential equation tells you the whole future.

By the end of this chapter you will be able to solve such an equation exactly by three methods: separating the variables, substituting to make them separate, and using an integrating factor. You will also write a function as an infinite polynomial, called a Maclaurin series. That gives a fourth route to a solution, and a way to approximate a function. When no exact solution exists, you will solve the equation approximately by Euler's method. The coffee equation is solved in the first section.

A calculator is assumed throughout this chapter. The EXACT badge marks the questions whose answer must be left exact.

16.1 Separable Variables

Differential equations appear wherever a law describes a rate of change, which covers most of science. A few examples show the range.

- **Cooling.** $\dfrac{d\theta}{dt} = -k(\theta - \theta_{\text{room}})$ — a hot drink, an engine, or a body found at a crime scene.
- **Radioactive decay.** $\dfrac{dN}{dt} = -\lambda N$ — the basis of carbon dating.
- **Population growth.** $\dfrac{dn}{dt} = kn(M - n)$ — growth that slows as the population fills the space available.
- **Medicine.** $\dfrac{dC}{dt} = -kC$ — a drug leaving the bloodstream, which sets how often a dose must be repeated.
- **Falling through air.** $\dfrac{dv}{dt} = g - kv^2$ — why a skydiver reaches a steady terminal speed.
- **Epidemics.** Three linked differential equations predict how an infection spreads through a population.

Each one says the same thing in different words: the rate is known, and the quantity is wanted.

Two words describe the equations in this chapter, and both are settled by looking at the equation.

- The **order** is the highest derivative that appears. Every equation here contains $\dfrac{dy}{dx}$ and no higher derivative, so all of them are **first order**.
- An equation is **linear** when y and its derivatives appear only to the first power, and are never multiplied by one another. So $\dfrac{dy}{dx} + xy = 1$ is linear, and $\dfrac{dy}{dx} = 2xy^2$ is not.

The three exact methods in this chapter are chosen by which form an equation takes, so these two points are checked first.

16.1.1 Separating the variables

The simplest case can be rearranged so that each variable sits on its own side. A first-order equation is **separable** if it can be written as a function of y times a function of x:

$$\frac{dy}{dx} = g(x)\,h(y).$$

Then you separate and integrate both sides. Treating $\frac{dy}{dx}$ as a ratio is informal but valid here: gather all the y terms with dy and all the x terms with dx.

KEY

$$\frac{dy}{dx} = g(x)\,h(y) \quad\Longrightarrow\quad \int \frac{1}{h(y)}\,dy = \int g(x)\,dx$$

One arbitrary constant appears, giving the **general solution**, a family of curves. An **initial condition** fixes the constant and selects the **particular solution**.

EXAMPLE 16.1

Solve $\frac{dy}{dx} = 2xy^2$ given $y(0) = 1$.

Separate the variables and integrate:

$$\int \frac{1}{y^2}\,dy = \int 2x\,dx \implies -\frac{1}{y} = x^2 + C.$$

Apply $y(0) = 1$: $-1 = 0 + C \implies C = -1$. So $-\frac{1}{y} = x^2 - 1$, giving

$$y = \frac{1}{1 - x^2}.$$

Always apply the initial condition as early as possible; carrying C further than necessary invites errors.

EXAMPLE 16.2

A cup of coffee obeys $\frac{d\theta}{dt} = -k(\theta - 20)$ with $\theta(0) = 80$. Solve for $\theta(t)$.

Separate and integrate:

$$\int \frac{1}{\theta - 20}\,d\theta = \int -k\,dt \implies \ln|\theta - 20| = -kt + C.$$

Exponentiating, $\theta - 20 = Ae^{-kt}$. The condition $\theta(0) = 80$ gives $A = 60$, so

$$\theta(t) = 20 + 60e^{-kt}.$$

The temperature decays exponentially toward the room's 20 degrees, never quite reaching it: exactly the behaviour we observed.

Separation also handles the most famous population model in ecology. In the 1960s, once the cattle disease rinderpest was eliminated around the Serengeti, the wildebeest herd began to grow, and its growth obeyed the **logistic equation**: growth proportional both to the herd itself and to the room left below the plains' carrying capacity.

EXAMPLE 16.3

The Serengeti wildebeest population n (in millions) satisfies $\dfrac{dn}{dt} = kn(1.4-n)$, with $n(0) = 0.25$ in 1963. Solve for $n(t)$.

Separate, and split the left side with the partial fractions of Ch. 6:

$$\int \frac{dn}{n(1.4-n)} = \int k\,dt, \qquad \frac{1}{n(1.4-n)} = \frac{1}{1.4}\left(\frac{1}{n} + \frac{1}{1.4-n}\right).$$

Integrating,

$$\frac{1}{1.4}\ln\left|\frac{n}{1.4-n}\right| = kt + C \implies \frac{n}{1.4-n} = Be^{1.4kt}.$$

The initial condition gives $B = \dfrac{0.25}{1.15} \approx 0.217$, and solving for n:

$$n(t) = \frac{1.4}{1 + 4.6\,e^{-1.4kt}}.$$

This is the logistic S-curve: growth that is nearly exponential at first, then levels off at the carrying capacity of 1.4 million. The bracket $(1.4 - n)$ is what slows it, since it shrinks towards zero as n approaches 1.4. The real herd behaved this way, climbing to about 1.3 million by 1977 and staying near that level since.

YOUR TURN 16.1 Separable Variables

1. Find the general solution of each equation.

(a) $\dfrac{dy}{dx} = \dfrac{x}{y}$

(b) $\dfrac{dy}{dx} = y$

(c) $\dfrac{dy}{dx} = \dfrac{1+y}{1+x}$

(d) $\dfrac{dy}{dx} = e^{x-y}$

2. Find the particular solution of each equation.

(a) $\dfrac{dy}{dx} = xy$ with $y(0) = 2$.

(b) $\dfrac{dy}{dx} = 3x^2y$ with $y(0) = 1$.

(c) $\dfrac{dy}{dx} = y\tan x$ with $y(0) = 4$, for $0 \le x < \frac{\pi}{2}$.

(d) Explain why a general solution contains one arbitrary constant, and what the initial condition does to it.

16.2 Homogeneous Differential Equations

Some equations cannot be separated as they stand, but a substitution makes them separable.

A first-order equation is **homogeneous** if the right-hand side depends only on the ratio $\frac{y}{x}$, and not on x and y separately.

Equations of this kind appear when a problem has no fixed size, only directions and proportions.

- **Pursuit curves.** A ferry always aims at the dock while a current pushes it sideways. At each moment only the *direction* from the ferry to the dock matters, and that direction is set by $\frac{y}{x}$ alone. You will solve the ferry in the challenge problems.
- **Reflector shapes.** The curve of a car headlight or a satellite dish is fixed by requiring every ray from the focus to leave parallel. That condition involves only angles, so the equation depends on the ratio.
- **Scale-free rules.** If doubling both quantities leaves a rule unchanged, the rule can only involve their ratio, and the equation is homogeneous.

The substitution $y = vx$ turns such an equation into a separable one. Since $y = vx$, the product rule gives $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

KEY For $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$, substitute $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

EXAMPLE 16.4

Solve $\frac{dy}{dx} = \frac{x+y}{x}$.

Rewrite the right side in terms of $\frac{y}{x}$: $\frac{x+y}{x} = 1 + \frac{y}{x}$. Substitute $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = 1 + v \implies x\frac{dv}{dx} = 1.$$

This is separable:

$$\int dv = \int \frac{1}{x}\,dx \implies v = \ln|x| + C.$$

Finally replace $v = \frac{y}{x}$:

$$y = x(\ln|x| + C).$$

Here the single v on each side cancelled, but nothing depends on that. Whatever f is, the substitution turns the equation into $x\frac{dv}{dx} = f(v) - v$, with v on one side and x on the other. It is separable however $f(v) - v$ turns out. That is what makes the substitution work.

EXAMPLE 16.5

Solve $\dfrac{dy}{dx} = \dfrac{2x^2 + y^2}{xy}$ for $x > 0$.

Divide the top and bottom by x^2 to show that the right side depends only on the ratio:

$$\frac{2x^2 + y^2}{xy} = \frac{2 + (y/x)^2}{y/x}.$$

Substitute $y = vx$, so $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$:

$$v + x\frac{dv}{dx} = \frac{2 + v^2}{v} \implies x\frac{dv}{dx} = \frac{2 + v^2}{v} - v = \frac{2}{v}.$$

Again the v terms cancel, leaving a separable equation:

$$\int v\,dv = \int \frac{2}{x}\,dx \implies \frac{v^2}{2} = 2\ln x + C.$$

Replace $v = \dfrac{y}{x}$ and clear the fraction:

$$y^2 = 2x^2(2\ln x + C).$$

Checking that the right side really depends only on $\frac{y}{x}$ is the first step every time. Dividing top and bottom by the highest power of x is the quickest way to see it.

YOUR TURN 16.2 Homogeneous Differential Equations

3. Use the substitution $y = vx$ to solve each equation.

(a) $\dfrac{dy}{dx} = \dfrac{2x + y}{x}$

(b) $\dfrac{dy}{dx} = \dfrac{y - x}{x}$

(c) $\dfrac{dy}{dx} = \dfrac{x + 2y}{x}$

(d) $\dfrac{dy}{dx} = \dfrac{y}{x}$ for $x > 0$.

4.

(a) Show that $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{xy}$ is homogeneous, by writing the right-hand side in terms of $\dfrac{y}{x}$.

(b) Hence find its general solution.

16.3 The Integrating Factor

A **linear** first-order differential equation has the form

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

It is usually not separable. The method is to multiply through by a carefully chosen function, the **integrating factor**, which turns the left side into the derivative of a single product.

> KEY · IB For $\dfrac{dy}{dx} + P(x)y = Q(x)$, the integrating factor is
>
> $$I = e^{\int P(x)\,dx}.$$

After multiplying by I, the left side becomes exactly $\dfrac{d}{dx}(Iy)$, by the product rule run backwards. You then integrate both sides directly.

EXAMPLE 16.6

Solve $\dfrac{dy}{dx} + \dfrac{1}{x}y = x^2$ for $x > 0$.

Here $P(x) = \dfrac{1}{x}$, so the integrating factor is

$$I = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x.$$

Multiply through by x:

$$x\frac{dy}{dx} + y = x^3 \implies \frac{d}{dx}(xy) = x^3.$$

The left side has collapsed to a single derivative. Integrate both sides:

$$xy = \frac{x^4}{4} + C \implies y = \frac{x^3}{4} + \frac{C}{x}.$$

The method exists to create that $\dfrac{d}{dx}(xy)$ on the left; everything after is ordinary integration.

EXAMPLE 16.7

Solve $\dfrac{dy}{dx} + y = e^x$.

$P(x) = 1$, so $I = e^{\int 1\,dx} = e^x$. Multiply through:

$$\frac{d}{dx}(e^x y) = e^x \cdot e^x = e^{2x}.$$

Integrate: $e^x y = \frac{1}{2}e^{2x} + C$, hence $y = \frac{1}{2}e^x + Ce^{-x}$.

16.3.1 Where linear equations come from

A linear equation appears whenever a quantity is added and removed at the same time, and either rate may change.

- **Mixing tanks.** Liquid flows into a tank and drains out of it, and the dissolved substance obeys a linear equation. When the volume itself changes, the equation is linear but not separable, so the integrating factor is the method that applies.
- **Drug infusion.** A drug enters the blood at a steady rate through a drip, while the body clears it at a rate proportional to the amount already present.
- **Electrical circuits.** In a circuit with a resistor and a capacitor, the charge satisfies a linear equation driven by the applied voltage.
- **Savings with deposits.** An account earns interest continuously while money is paid in at a steady rate.

The worked example below is a mixing problem of the first kind.

EXAMPLE 16.8

A 100-litre vat in a brewery holds 5 kg of dissolved sugar. Syrup carrying 0.2 kg of sugar per litre flows in at 1 litre per minute, the vat is kept well stirred, and the mixture drains at 2 litres per minute. Find the sugar $S(t)$ in the vat at time t minutes.

The volume shrinks: $V(t) = 100 - t$. Sugar arrives at $0.2 \times 1 = 0.2$ kg min^{-1} and leaves at concentration times outflow, $\dfrac{S}{100-t} \times 2$. So

$$\frac{dS}{dt} = 0.2 - \frac{2S}{100-t} \implies \frac{dS}{dt} + \frac{2}{100-t}S = 0.2.$$

The integrating factor is

$$I = e^{\int \frac{2}{100-t}\,dt} = e^{-2\ln(100-t)} = (100-t)^{-2}.$$

Multiplying through collapses the left side:

$$\frac{d}{dt}\Big(S\,(100-t)^{-2}\Big) = 0.2\,(100-t)^{-2} \implies S\,(100-t)^{-2} = \frac{0.2}{100-t} + C.$$

So $S = 0.2(100-t) + C(100-t)^2$, and $S(0) = 5$ gives $C = -0.0015$:

$$S(t) = 0.2(100-t) - 0.0015(100-t)^2.$$

Sense-check the endpoints: $S(0) = 5$, and $S(100) = 0$, an empty vat holds no sugar. In between, the sugar first *rises*, to about 6.7 kg near $t = 33$, before the draining takes over. That rise is not obvious from the description, and it is the equation rather than intuition that reveals it.

16.3.2 Choosing a method

Three exact methods are now available, and a question rarely says which to use. The form of the equation decides it, and the test is quick.

What the equation looks like	Method	Example
The variables can be split, $\frac{dy}{dx} = g(x)h(y)$	separate and integrate	$\frac{dy}{dx} = 2xy^2$
The right side depends only on $\frac{y}{x}$	substitute $y = vx$	$\frac{dy}{dx} = \frac{x+y}{x}$
Linear in y: $\frac{dy}{dx} + P(x)y = Q(x)$	integrating factor $e^{\int P\,dx}$	$\frac{dy}{dx} + \frac{1}{x}y = x^2$
None of the above	Euler's method, numerically	$\frac{dy}{dx} = x^2 + y^2$

Try them in that order, because each test is quicker than the one after it. Note that the first two overlap: an equation can be both separable and homogeneous, in which case separating directly is the shorter route. A linear equation can overlap too. When Q/P is constant, $\frac{dy}{dx} + P(x)y = Q(x)$ rearranges to $\frac{dy}{dx} = P(x)\left(\frac{Q}{P} - y\right)$ and separates at once, which is how $\frac{dy}{dx} + y = 4$ and the cooling equation of §16.1 are solved. In general, though, Q does not divide P that way, nothing separates, and the integrating factor is what is left.

YOUR TURN 16.3 The Integrating Factor

5. Solve each linear equation.

(a) $\frac{dy}{dx} + y = 4$

(b) $\frac{dy}{dx} + 2y = 6$

(c) $\frac{dy}{dx} + 2y = e^x$

(d) $\frac{dy}{dx} - y = e^{2x}$

6.

(a) Write down the integrating factor for $\dfrac{dy}{dx} + \dfrac{3}{x}y = x^2$, for $x > 0$.

(b) Solve $\dfrac{dy}{dx} + \dfrac{2}{x}y = x$ for $x > 0$.

(c) Solve $\dfrac{dy}{dx} + \dfrac{1}{x}y = x$ with $y(1) = 1$, for $x > 0$.

(d) Explain what the integrating factor is chosen to achieve on the left-hand side.

16.4 Maclaurin Series

Many functions can be written as an infinite polynomial. This idea stands apart from the three methods above, and it has two uses: it approximates a function, and it gives a fourth route to solving a differential equation.

There is a reason a polynomial is wanted at all. A polynomial is built only from adding and multiplying, and those are the two operations a machine can actually perform. Nothing inside a calculator knows what $\sin x$ *is*. When you press the sine key, the calculator evaluates a polynomial instead. Every function outside that small arithmetic world must be turned into one before it can be computed, and a Maclaurin series is how that is done.

The method is to match the function at a single point. Near $x = 0$, we look for a polynomial with the same value as the function, then the same gradient, then the same concavity, then the same higher derivatives. The more derivatives agree, the longer the polynomial stays close to the curve as x moves away from 0. The result is the **Maclaurin series**.

The coefficients are not chosen; they are forced. Suppose the series exists and write it as

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Putting $x = 0$ kills every term but the first, so $a_0 = f(0)$. Differentiating gives $f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots$, and $x = 0$ now leaves $a_1 = f'(0)$. Differentiating again gives $f''(0) = 2a_2$, and once more $f'''(0) = 6a_3 = 3!\,a_3$. Each round of differentiating strips one term and multiplies the next by its own index, so in general $f^{(n)}(0) = n!\,a_n$.

KEY · IB

$$f(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

EXAMPLE 16.9

Find the Maclaurin series for $f(x) = e^x$.

Every derivative of e^x is e^x, and $e^0 = 1$, so $f^{(n)}(0) = 1$ for all n:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This series can be taken as the definition of e^x, and it computes the function to any accuracy you want. It also explains why e^x is its own derivative: differentiate the series term by term and you get the same series back.

The series below are provided in the formula booklet. Each is exact, and truncating it gives a polynomial approximation that is excellent near $x = 0$.

KEY · IB

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

One more expansion is named by the syllabus but printed elsewhere in the booklet, so it is easy to overlook. The binomial family $(1+x)^p$ is a Maclaurin series like any other, and it is the extended binomial theorem of Ch. 2 (booklet 1.10) seen from this side.

KEY For $p \in \mathbb{Q}$ and $|x| < 1$,

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots.$$

Unlike the series above it carries a validity condition, because it does not converge for every x.

EXAMPLE 16.10

Use a Maclaurin series to approximate $\sqrt{1.1}$ to four decimal places.

Write $\sqrt{1.1} = (1+x)^{1/2}$ with $x = 0.1$, which satisfies $|x| < 1$. With $p = \frac{1}{2}$,

$$(1+x)^{1/2} = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \tfrac{1}{16}x^3 - \cdots.$$

Substituting $x = 0.1$ gives

$$1 + 0.05 - 0.00125 + 0.0000625 = 1.0488125.$$

The true value is $\sqrt{1.1} = 1.0488088\ldots$, so four terms agree to four decimal places, 1.0488. Each term is smaller than the last by roughly a factor of x, which is why these series are useful: the smaller x is, the fewer terms you need.

The figure below shows the first three odd-degree truncations of the sine series: the line $y = x$, the cubic, and the quintic. Each new term extends the interval over which the polynomial and the curve agree.

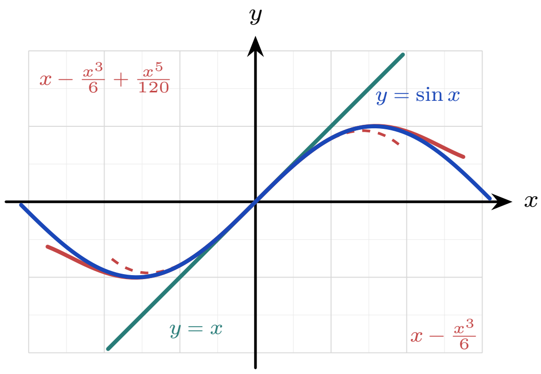


Figure 16.1 The first three odd-degree truncations of the sine series against $y = \sin x$. Each extra term widens the interval on which the polynomial and the curve agree, but every truncation eventually leaves the curve.

The same convergence can be watched numerically. Below are the successive truncations of the sine series at $x = 1$, against the true value $\sin 1 = 0.841471$.

terms up to	x	x^3	x^5	x^7
estimate	1	0.833333	0.841667	0.841468
error	0.159	0.0081	0.0002	0.0000027

Each new term cuts the error by a factor of tens, and that factor grows as more terms are taken: the ratios along this row are about 20, 40 and 70. That is the trade the series offers: a polynomial is simple enough to compute, and taking more terms buys as much accuracy as you are willing to pay for.

16.4.1 Building new series

New series are rarely built from the definition. They are made from the standard five by four operations, all of which the syllabus names.

- **Substitute.** Replacing x with $-x^2$ in the series for e^x gives

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$$

- **Differentiate.** Differentiating the sine series term by term produces exactly the cosine series: the identity $\frac{d}{dx}\sin x = \cos x$ in polynomial form.

- **Integrate.** The geometric series of Ch. 2 gives $\dfrac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots$ for $|x| < 1$, and integrating term by term yields $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$. The booklet's arctangent series is simply the geometric series integrated.
- **Multiply.** For a product, expand each factor and collect powers up to the degree you need.

EXAMPLE 16.11

Find the Maclaurin series of $e^x \sin x$ up to the term in x^3.

Multiply the two standard series, keeping only terms of degree ≤ 3:

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\left(x - \frac{x^3}{6}\right) = x + x^2 + \left(\frac{1}{2} - \frac{1}{6}\right)x^3 + \cdots = x + x^2 + \frac{x^3}{3} + \cdots$$

The work is bookkeeping. Decide the target degree first, and discard anything above it before multiplying rather than after.

16.4.2 Small angles and series from differential equations

Truncating at the first term or two gives the **small-angle approximations** used throughout physics. For θ near 0, in radians,

$$\sin\theta \approx \theta, \qquad \cos\theta \approx 1 - \frac{\theta^2}{2}, \qquad \tan\theta \approx \theta.$$

These are not conveniences invented by physicists; they are the first one or two terms of the series, with the rest discarded. The pendulum formula depends on the first of them. For small swings the restoring force involves $\sin\theta$, and replacing $\sin\theta$ by θ is exactly what turns the equation of motion into simple harmonic motion, which can then be solved.

The series also prices the approximation. The first term discarded is $\frac{\theta^3}{6}$, so that is roughly the error, and the relative error stays under 1% for swings up to about 14°. Beyond that the pendulum's real period drifts away from the textbook formula, by an amount large enough to measure.

A Maclaurin series can also be built directly *from a differential equation*, by differentiating the equation itself to obtain the derivatives at 0. This joins the two halves of the chapter.

EXAMPLE 16.12

A function satisfies $\dfrac{dy}{dx} = 1 + y^2$ with $y(0) = 0$. Find the Maclaurin series of y up to the term in x^3.

Take the derivatives at 0 from the equation itself. First, $y'(0) = 1 + 0 = 1$. Differentiating the equation,

$$y'' = 2y\,y' \implies y''(0) = 0, \qquad y''' = 2(y')^2 + 2y\,y'' \implies y'''(0) = 2.$$

Feed these into the Maclaurin formula:

$$y = 0 + x + \frac{0}{2!}x^2 + \frac{2}{3!}x^3 + \cdots = x + \frac{x^3}{3} + \cdots$$

You may recognise the answer: $y = \tan x$ solves this equation, since $\frac{d}{dx}\tan x = \sec^2 x = 1 + \tan^2 x$, and these are indeed the first terms of its series. The equation was never solved. Each derivative at 0 was read off from the equation itself, one at a time, and that was enough to build the series.

TIP Maclaurin series suggest several good IA topics. Two worth exploring: Madhava's series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$, which comes from setting $x = 1$ in the arctangent series, and why it converges so slowly that a better choice such as $\arctan \frac{1}{\sqrt{3}}$ reaches the same accuracy in a fraction of the terms; or how a calculator really evaluates $\sin x$, by reducing the angle to a small interval first and then using a short truncated series.

EXAMPLE 16.13

Use Maclaurin series to evaluate $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$.

Expand the numerator with the series for e^x:

$$e^x - 1 - x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right) - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

Divide by x^2:

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + \frac{x}{6} + \cdots \xrightarrow{x\to 0} \frac{1}{2}.$$

The series settles an indeterminate limit as neatly as L'Hôpital's rule, and it shows *why* the answer is $\frac{1}{2}$: that number is the leading coefficient.

YOUR TURN 16.4 Maclaurin Series

7. Find the first three nonzero terms of the Maclaurin series for each function.

(a) e^{2x} (b) $\cos x$

(c) $\ln(1 + 2x)$ (d) e^{-x^2}

8. Find the first two nonzero terms of the Maclaurin series for each function.

(a) $\sin 2x$ (b) $\sin(x^2)$

(c) $\arctan 2x$ (d) $\sqrt{1 + x}$

9.

(a) Use the first three terms of the series for e^x to estimate $e^{0.1}$.

(b) Use the first three terms of the cosine series to estimate $\cos 0.2$ to four decimal places.

(c) Compare each estimate with the true value, and state which is the more accurate. Explain why.

(d) Find the first three nonzero terms of the Maclaurin series for $e^x \cos x$.

(e) Differentiate the series $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots$ term by term, and identify the series you obtain.

(f) Differentiate the series for $\ln(1 + x)$ term by term, and verify that the result is the expansion of $\dfrac{1}{1+x}$.

10. A function satisfies $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$.

(a) Write down $y'(0)$.

(b) By differentiating the equation, find $y''(0)$ and $y'''(0)$.

(c) Hence find the Maclaurin series for y up to the term in x^3.

(d) Use the series to estimate $y(0.3)$, and compare it with the Euler estimate 1.362 from the notes.

16.5 Euler's Method

Not every differential equation can be solved exactly, and most of the equations that describe the real world cannot. When no exact solution exists, the solution can still be approximated numerically. **Euler's method** does this by walking along the curve in short straight steps, taking the gradient of each step from the differential equation itself.

The idea is the tangent line of Ch. 13. Near a known point the curve is close to its tangent, so moving a small distance h along the tangent lands close to the curve:

$$y \approx y_0 + \underbrace{f(x_0, y_0)}_{\text{gradient}} \times \underbrace{h}_{\text{step}} .$$

The differential equation supplies the gradient at whatever point you have reached, so the same move can be repeated from the new point, and again from the one after that.

KEY · IB

$$y_{n+1} = y_n + h\, f(x_n, y_n), \qquad x_{n+1} = x_n + h$$

Here h is the fixed **step length**. Each step is exact only at its starting point and drifts from the curve thereafter, so shorter steps stay closer, at the cost of needing more of them.

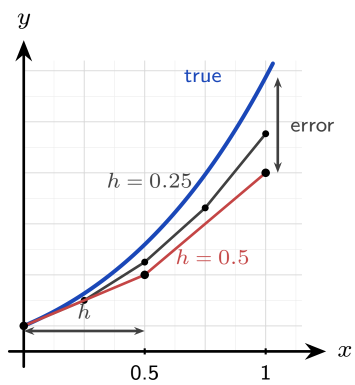


Figure 16.2 Euler's method on $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Each straight step leaves the curve, which bends upward away from it, so the estimate falls short and the gap at $x = 1$ is the error. Halving the step length roughly halves that gap.

The figure shows two things at once. Each step is a straight line, so it leaves a curve that bends away from it, and the gap at the far end is the *error*. The gray path takes steps half as long and stays about twice as close to the curve. Shorter steps give more accuracy, at the cost of more of them.

EXAMPLE 16.14

Given $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with $h = 0.1$ to estimate $y(0.3)$.
Apply $y_{n+1} = y_n + h(x_n + y_n)$ three times. A table is the clearest way to organise the work, and it is what an examiner expects to see.

n	x_n	y_n	$f = x_n + y_n$	$h\,f$
0	0	1	1	0.1
1	0.1	1.1	1.2	0.12
2	0.2	1.22	1.42	0.142
3	0.3	1.362		

Each y_n is the previous one plus the $h\,f$ on the line above. So $y(0.3) \approx 1.362$, against an exact value of about 1.3997.
Euler has undershot, and the reason is visible in the figure: this curve is concave up, so every

tangent lies below it and each step falls short. That is the general rule. A concave-up solution is under-estimated and a concave-down one is over-estimated, so the sign of $\frac{d^2y}{dx^2}$ tells you which way your answer is wrong before you check it.

The method matters most when no exact solution is available. Make the resistance on a falling body depend on the shape of the object and on the air density, which changes with height, and the equation has no solution among the functions of this course, but Euler's method still produces a table of values. This is, in miniature, how weather forecasts are computed: equations too hard to solve exactly, stepped forward numerically, millions of times.

EXAMPLE 16.15

A skydiver falls from rest, and air resistance gives $\dfrac{dv}{dt} = 9.8 - 0.1v^2$ (in m s^{-1} and seconds). Use Euler's method with $h = 0.5$ to estimate the speed after 1.5 seconds, and compare with the terminal velocity.

Apply $v_{n+1} = v_n + 0.5(9.8 - 0.1v_n^2)$ three times from $v_0 = 0$:

$$\begin{aligned} v_1 &= 0 + 0.5(9.8) = 4.9 \\ v_2 &= 4.9 + 0.5(9.8 - 0.1(4.9)^2) = 4.9 + 0.5(7.399) = 8.60 \\ v_3 &= 8.60 + 0.5(9.8 - 0.1(8.60)^2) = 8.60 + 0.5(2.404) = 9.80 \end{aligned}$$

So $v(1.5) \approx 9.80$ m s^{-1}. The terminal velocity is where acceleration dies: $9.8 - 0.1v^2 = 0 \implies v = \sqrt{98} \approx 9.90$ m s^{-1}. After only three steps the estimate is already within 1% of the terminal speed. The physics is visible in the arithmetic: each bracket $9.8 - 0.1v_n^2$ is the net force per unit mass, and it shrinks towards zero as the speed builds.

YOUR TURN 16.5 Euler's Method

11. Use Euler's method for each of the following.

(a) $\dfrac{dy}{dx} = 2x$, $y(0) = 1$; use $h = 0.5$ to estimate $y(1)$.

(b) $\dfrac{dy}{dx} = y$, $y(0) = 1$; use $h = 0.1$ to estimate $y(0.2)$.

(c) $\dfrac{dy}{dx} = x + y$, $y(0) = 1$; use $h = 0.2$ to estimate $y(0.4)$.

(d) $\dfrac{dy}{dx} = xy$, $y(1) = 1$; use $h = 0.1$ to estimate $y(1.2)$.

12.

(a) For $\dfrac{dy}{dx} = xy$ with $y(1) = 2$, take one step with $h = 0.2$ to estimate $y(1.2)$.

(b) For $\dfrac{dy}{dx} = x + y$ with $y(0) = 0$, take two steps with $h = 0.5$ to estimate $y(1)$.

(c) Solve $\frac{dy}{dx} = y$ with $y(0) = 1$ exactly, and compare the exact $y(0.2)$ with your answer to question 11 part (b).

(d) State whether Euler's method over- or under-estimates the solution of part (c), and explain why in terms of the shape of the curve.

Chapter 16 · Differential Equations

Reflections

TOK Euler's method rests on a striking idea: knowing the rule for change at every point, plus a single starting state, determines the entire future of a system. This is mathematical determinism. Yet the same equations can be so sensitive to their starting values that long-term prediction becomes impossible in practice, the phenomenon of chaos. Determined but unpredictable: what does that tell us about the limits of knowledge?

TOK A truncated series is deliberately wrong. It is chosen because it is simple enough to compute, and every extra term bought accuracy at the cost of that simplicity: four terms of the sine series give $\sin 1$ correct to five decimal places, and one term gives it to none.

Is there always a trade-off between accuracy and simplicity? And when a model is knowingly approximate, as every model in this chapter is, what makes it good enough to act on?

IM Both halves of this chapter have a long international lineage.

The series came first from India. Mathematicians of the Kerala school, notably Madhava in the 14th century, found the expansions of $\sin x$, $\cos x$ and $\arctan x$ roughly three centuries before Newton, Leibniz and Maclaurin. The arctangent series, usually named after Gregory and Leibniz, was known in Kerala generations earlier.

Numerical methods have a similar story in the Islamic world. In the 15th century Jamshīd al-Kāshī, working at the observatory in Samarkand, needed $\sin 1^\circ$ to extraordinary precision for his astronomical tables. No formula gave it, so he wrote down an equation the value satisfied and solved it by repeated approximation, each round feeding its answer into the next, reaching sixteen correct decimal places. That is the same bargain Euler's method strikes: when no exact answer is available, iterate towards one and control the error. Two centuries earlier Sharaf al-Dīn al-Ṭūsī had studied when a cubic equation has solutions by examining where its maximum lies, an argument that reads today like the use of a derivative.

Mathematical discovery is rarely the work of one place or one name.

RW Differential equations are the language in which physics is written. Newton's second law $F = ma$ is a differential equation for position; radioactive decay, population growth, electrical circuits, epidemics, and planetary orbits are all governed by them. To predict a system's future is, almost always, to solve a differential equation, exactly when you can and numerically (by methods like Euler's) when you cannot.

EXT The five standard series behave differently at the edges. The series for e^x, $\sin x$ and

$\cos x$ converge for every x, but $\arctan x$ only for $|x| \leq 1$ and $\ln(1+x)$ only for $-1 < x \leq 1$, while the geometric series they are built from needs $|x| < 1$. Each series has a *radius of convergence*, and outside it the polynomial does not merely lose accuracy: it diverges completely. Finding that radius, and explaining why $\ln(1+x)$ has one at all when e^x does not, is the beginning of complex analysis, where the answer turns out to depend on where the function misbehaves in the complex plane. See Ch. 17.

EXT Maclaurin series turn calculus into algebra: integrals with no elementary antiderivative, like $\int e^{-x^2}\,dx$, can be integrated term by term once the function is a series. Series also solve differential equations that no other method in this course can reach, by assuming the answer is a power series and matching coefficients. The infinite polynomial is one of the most far-reaching ideas in all of analysis.

End-of-Chapter Problems

1. A population grows so that $\dfrac{dP}{dt} = 0.2P$, with $P = 500$ when $t = 0$.

(a) Solve the differential equation for $P(t)$. [4]

(b) Find the population when $t = 10$. [2]

2. A body cools according to $\dfrac{d\theta}{dt} = -0.1(\theta - 25)$, with $\theta(0) = 90$ (in °C).

(a) Solve for $\theta(t)$. [4]

(b) Find the temperature after 10 minutes. [2]

(c) State the temperature the body approaches in the long run. [1]

3. Consider $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{xy}$ for $x > 0$.

(a) Use the substitution $y = vx$ to show that $xv\dfrac{dv}{dx} = 1$. [4]

(b) Hence find the general solution. [3]

4. Consider $\dfrac{dy}{dx} + \dfrac{1}{x}y = 3x$ for $x > 0$.

(a) Find the integrating factor. [2]

(b) Solve the equation given $y(1) = 2$. [4]

5. Consider $\dfrac{dy}{dx} = x + y$, with $y(0) = 1$.

(a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$, showing each step. [4]

(b) Solve the equation exactly (integrating factor) and find the true $y(0.5)$. [4]

(c) Comment on the size and direction of the error. [2]

6. Let $f(x) = \cos x$.

(a) Find $f(0), f'(0), f''(0), f'''(0), f^{(4)}(0)$. [3]

(b) Hence write the Maclaurin series of $\cos x$ up to the term in x^4. [2]

(c) Use it to estimate $\cos(0.2)$. [2]

7. EXACT

(a) Write down the Maclaurin series for $\sin x$ up to the term in x^5. [2]

(b) Hence evaluate $\displaystyle\lim_{x\to 0} \frac{x - \sin x}{x^3}$. [3]

8.

(a) Write down the Maclaurin series for e^x up to the term in x^3. [1]

(b) By substituting $-x^2$, find the series for e^{-x^2} up to x^4. [2]

(c) Hence estimate $\int_0^1 e^{-x^2}\,dx$ using terms up to x^4. [4]

9. A bowl of miso soup at 90°C is left in a 25°C room and obeys $\dfrac{d\theta}{dt} = -k(\theta - 25)$. After 10 minutes it has cooled to 65°C.

(a) Solve the differential equation, expressing θ in terms of k and t. [4]

(b) Find k. [2]

(c) Find how long, from the start, the soup takes to reach 40°C. [2]

10. Consider $\dfrac{dy}{dx} = \dfrac{y^2 + xy}{x^2}$ for $x > 0$, with $y(1) = 1$.

(a) Use the substitution $y = vx$ to show that $x\dfrac{dv}{dx} = v^2$. [3]

(b) Hence solve for y in terms of x. [4]

11. Solve each linear equation.

(a) $\dfrac{dy}{dx} - \dfrac{y}{x} = x^2$ for $x > 0$, given $y(1) = 1$. [6]

(b) $\dfrac{dy}{dx} + 2y = e^{-x}$ with $y(0) = 0$. [6]

12. Solve $\dfrac{dy}{dx} + (\tan x)\,y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, given $y(0) = 2$. [7]

13. A fish population n (in thousands) in a lake satisfies the logistic equation $\dfrac{dn}{dt} = 0.4n(2 - n)$, with $n(0) = 0.5$.

(a) Using partial fractions, show that the solution is $n = \dfrac{2}{1 + 3e^{-0.8t}}$. [6]

(b) Find the time at which the population reaches 1500 fish. [3]

14. Consider $\dfrac{dy}{dx} = y - x^2$ with $y(0) = 1$.

(a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$. [5]

(b) The exact solution is concave up on $0 \le x \le 0.5$. State, with a reason, whether your estimate is above or below the true value. [2]

15. A skydiver falls from rest with $\dfrac{dv}{dt} = 9.8 - 0.2v^2$.

(a) Use Euler's method with $h = 0.5$ to estimate $v(1.5)$. [4]

(b) Find the terminal velocity, and comment on the behaviour of the Euler estimates around it. [3]

16. Find the Maclaurin series of $\ln(1+2x)$ up to the term in x^3

(a) by substitution into a standard series; [3]
(b) by computing derivatives from the definition. [3]

17.

(a) Find the Maclaurin series of $e^{-x}\sin x$ up to the term in x^3. [6]
(b) Use your series to estimate $e^{-0.1}\sin 0.1$, and compare it with the value from your calculator. [3]

18. By setting $x = 1$ in the Maclaurin series for $\arctan x$:

(a) write down a series for $\frac{\pi}{4}$; [2]
(b) estimate π using the first four terms; [2]
(c) comment on the rate of convergence. [1]

19. EXACT Use Maclaurin series to evaluate $\lim_{x\to 0} \frac{x - \sin x}{x^3}$. [5]

20. A function satisfies $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$. Find the Maclaurin series of y up to the term in x^3. [7]

21. A drug is infused into the bloodstream at a rate that raises its concentration by r mg L^{-1} per hour, and the body removes it at a rate proportional to the concentration C, so that

$$\frac{dC}{dt} + kC = r,$$

where t is measured in hours and $C(0) = 0$.

(a) Explain why this equation is linear, and write down its integrating factor. [3]
(b) Solve the equation to show that $C = \frac{r}{k}(1 - e^{-kt})$. [4]
(c) A patient is given $r = 6$ and has $k = 0.3$. Find the concentration after 4 hours. [2]
(d) Find the concentration the patient approaches in the long run, and explain what this means for the treatment. [3]
(e) Find how long it takes to reach 90% of that level. [2]
(f) Once that level is reached the infusion is stopped, so that $\frac{dC}{dt} = -kC$. Solve this equation and find how long the concentration then takes to halve. [3]

22. In this question you will build a series for logarithms that converges quickly.

(a) Find the Maclaurin series of $\ln(1+x)$ up to the term in x^4. [4]
(b) Write down the Maclaurin series of $\ln(1-x)$ up to the term in x^4. [2]
(c) Hence show that $\ln\left(\dfrac{1+x}{1-x}\right) = 2\left(x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \cdots\right)$. [3]
(d) By choosing a suitable value of x, use the three terms shown to estimate $\ln 3$. [3]
(e) The series for $\ln(1+x)$ is valid only for $-1 < x \leq 1$. State the value of x that would be needed to reach $\ln 3$ from that series alone, and explain why the series in part (c) is the more useful one here. [2]

23. Water drains from a tank through a hole in its base. The depth h metres at time t minutes satisfies

$$\frac{dh}{dt} = -k\sqrt{h},$$

where k is a positive constant.

(a) Separate the variables to show that $2\sqrt{h} = c - kt$ for some constant c. [3]
(b) Given that the depth is h_0 when $t = 0$, show that $h = \left(\sqrt{h_0} - \frac{1}{2}kt\right)^2$. [2]
(c) A tank starts with $h_0 = 1.44$ and has $k = 0.2$. Find how long it takes to empty. [3]
(d) Find the depth after 6 minutes. [2]
(e) Use Euler's method with steps of 3 minutes to estimate the depth after 6 minutes, and find the error in this estimate. [4]
(f) Describe the shape of the graph of h against t, and use it to explain why the Euler estimate lies below the exact value. [2]

Challenge Questions

24. EXACT **How fast does coffee cool?** The chapter opened with $\dfrac{d\theta}{dt} = -k(\theta - 20)$, where θ is the temperature in degrees and t the time in minutes. A cup is poured at 80°, so $\theta(0) = 80$.

(a) Solve the equation to show that $\theta = 20 + 60e^{-kt}$. [3]
(b) The coffee has cooled to 60° after 10 minutes. Find k exactly, and to three significant figures. [3]
(c) Find the temperature after 20 minutes. [2]
(d) Show that the time taken to reach a temperature θ is $t = \dfrac{1}{k}\ln\left(\dfrac{60}{\theta - 20}\right)$, and explain why the coffee never reaches 20°. [4]
(e) A second cup is poured at 90° in the same room. Show that it takes exactly 10 minutes to cool from 80° to 60°, the same as the first cup. [3]

25. **How wrong is Euler?** Take $\dfrac{dy}{dx} = y$ with $y(0) = 1$, whose exact solution is $y = e^x$, so that $y(1) = e$.

(a) Show that one Euler step of length h gives $y_{n+1} = (1+h)y_n$, and hence that after n steps $y_n = (1+h)^n$. [3]

(b) Taking $h = \dfrac{1}{n}$, deduce that Euler's estimate of $y(1)$ is $\left(1 + \dfrac{1}{n}\right)^n$. [2]

(c) Evaluate this estimate for $n = 2, 4, 8, 16$, and tabulate the error in each case. [4]

(d) Describe what happens to the error each time h is halved, and state what this says about the accuracy of the method. [3]

(e) Write down the limit of $\left(1 + \dfrac{1}{n}\right)^n$ as $n \to \infty$, and explain why Euler's method had to give this answer. [3]

26. The ferry and the current. A ferry sets off from $(1, 0)$ toward a dock at the origin, always aiming straight at the dock, while a river current pushes it in the positive y-direction at a fraction k of the ferry's own speed. Its path satisfies

$$\frac{dy}{dx} = \frac{y - k\sqrt{x^2 + y^2}}{x}.$$

(a) Show the equation is homogeneous, and that $y = vx$ leads to $x\dfrac{dv}{dx} = -k\sqrt{1+v^2}$. [3]

(b) Given that $\displaystyle\int \frac{dv}{\sqrt{1+v^2}} = \ln\left(v + \sqrt{1+v^2}\right) + C$, show that $v + \sqrt{1+v^2} = x^{-k}$. [3]

(c) Hence show that $y = \frac{1}{2}\left(x^{1-k} - x^{1+k}\right)$. [4]

(d) Taking $k = \frac{1}{2}$, find where the ferry lands. [2]

(e) Show that the ferry reaches the dock only when $k < 1$, and describe what happens when $k = 1$ and when $k > 1$. [4]

NUMBER & ALGEBRA

Chapter 17

Complex Numbers

17

SYLLABUS 1.12–1.14 HL ONLY

In the sixteenth century, Italian mathematicians learned to solve cubic equations. Their formula produced something unexpected. It sometimes asked for the square root of a negative number, even when the equation had three perfectly ordinary answers. No number they knew could give a negative square. Yet if they wrote the impossible quantity down and kept calculating, it disappeared before the end, and the answers were correct.

This chapter begins with that impossible quantity. We define a number i with $i^2 = -1$. No real number does this, so i is new. For two centuries mathematicians did not trust it, and the name "imaginary" was chosen as an insult. The name was unfair. Once i is allowed, every polynomial equation has a solution, a result now called the **Fundamental Theorem of Algebra**.

These new numbers are also geometric. Each one is a point in a plane, with the real numbers along one axis. Adding two of them moves a point across the plane. Multiplying two of them rotates and stretches. Multiplying by i rotates by a quarter-turn. In ordinary coordinates a rotation takes several steps; here it is one multiplication. This is why complex numbers are used throughout physics and engineering.

By the end of this chapter you will be able to write a complex number in three forms: $a+bi$, $r\operatorname{cis}\theta$ and $re^{i\theta}$. You will multiply, divide, and find powers and roots in each form. You will also use complex roots to factorise polynomials and to prove trigonometric identities.

Three forms may seem like more than you need. Each one makes a different calculation easy. Adding is simplest in $a+bi$ form. A power that would fill a page of working in Cartesian form is immediate in polar form. Choosing the form before you start is the main skill in this chapter.

We start with that algebra: numbers of the form $a+bi$.

17.1 Cartesian Form

Everything starts from a single definition that no real number satisfies.

> DEF The **imaginary unit** i satisfies $i^2 = -1$. A **complex number** is any $z = a + bi$ with $a, b \in \mathbb{R}$; $a = \mathrm{Re}(z)$ is the **real part** and $b = \mathrm{Im}(z)$ the **imaginary part**. Of these the booklet prints only the form $z = a + bi$.

Complex numbers are added, subtracted and multiplied exactly like binomials in i. There is one extra rule: replace i^2 by -1 wherever it appears.

That one rule is enough to find every power of i, because the powers repeat in a cycle of four:

$$i^1 = i, \qquad i^2 = -1, \qquad i^3 = i^2 \cdot i = -i, \qquad i^4 = (i^2)^2 = 1,$$

and $i^5 = i^4 \cdot i = i$ starts the cycle again. To simplify any power, divide the exponent by 4 and keep only the remainder: $i^{2026} = i^{4(506)+2} = i^2 = -1$.

EXAMPLE 17.1

Given $z = 2 + i$ and $w = 5 - 3i$, find $z + w$ and zw.

Adding is done part by part: $z + w = (2 + 5) + (1 - 3)i = 7 - 2i$. Multiplying, expand and then use $i^2 = -1$:

$$zw = (2 + i)(5 - 3i) = 10 - 6i + 5i - 3i^2 = 10 - i + 3 = 13 - i.$$

The whole of complex arithmetic is ordinary algebra plus the one substitution $i^2 = -1$.

17.1.1 The conjugate and division

Every complex number has a mirror image across the real axis, called its **conjugate**. The conjugate is what makes division possible.

> DEF The **complex conjugate** of $z = a + bi$ is $z^* = a - bi$.

The conjugate matters because a number times its own conjugate is always real: $zz^* = (a + bi)(a - bi) = a^2 + b^2$. This is exactly the tool for division, in the same way that a surd is rationalised. To divide, multiply top and bottom by the conjugate of the denominator, turning the denominator real.

EXAMPLE 17.2

Find $\dfrac{3 + 2i}{1 - i}$.

Multiply numerator and denominator by the conjugate $1+i$:

$$\frac{3+2i}{1-i}\cdot\frac{1+i}{1+i}=\frac{3+3i+2i+2i^2}{1^2+1^2}=\frac{3+5i-2}{2}=\frac{1+5i}{2}=\tfrac{1}{2}+\tfrac{5}{2}i.$$

Multiplying by the conjugate removes i from the denominator every time; the answer is then read off as real and imaginary parts.

One more Cartesian tool: two complex numbers are equal exactly when their real parts agree *and* their imaginary parts agree. A single complex equation therefore carries two real equations at once, and **equating parts** solves for unknowns.

EXAMPLE 17.3

Find the complex number z satisfying $z+2z^*=9+2i$.
Write $z=a+bi$, so $z^*=a-bi$:

$$(a+bi)+2(a-bi)=3a-bi=9+2i.$$

Equate real parts: $3a=9 \implies a=3$. Equate imaginary parts: $-b=2 \implies b=-2$. So $z=3-2i$. One equation was enough for two unknowns, because the real and imaginary parts each give an equation.

17.1.2 The Argand diagram

A complex number carries two real numbers, so it is naturally a *point* in a plane: the horizontal axis for the real part, the vertical for the imaginary. This picture, the **Argand diagram**, is what makes the subject geometric.

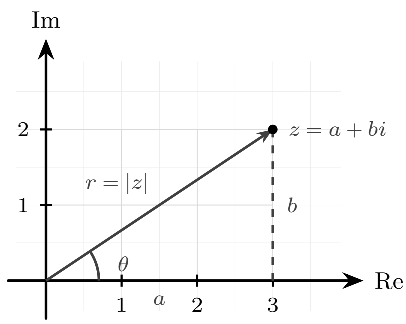


Figure 17.1 A complex number $z=a+bi$ drawn as the point (a,b) of the Argand diagram. Its distance from the origin is the modulus r, and the angle its arrow makes with the positive real axis is the argument θ.

The picture immediately explains the arithmetic. **Addition is vector addition**: to add w to z, place w's arrow tail-to-tip on z's, exactly as in Ch. 9, so $z+w$ is the far corner of the parallelogram the two arrows span. And **conjugation is reflection**: z^* is z flipped across the real axis. This is the picture behind $zz^*=a^2+b^2$ being real, a fact already checked algebraically above.

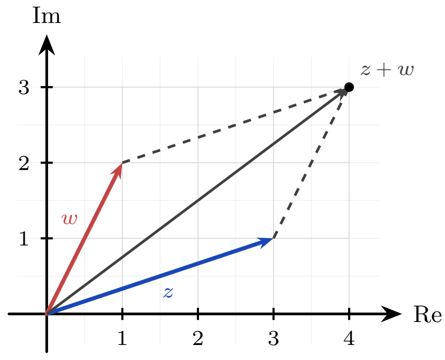


Figure 17.2 Addition is vector addition. Placing w's arrow tail-to-tip on z's puts $z+w$ at the far corner of the parallelogram the two arrows span.

Conjugation is the second picture, and it is the one that explains the symmetry of every polynomial's roots later in the chapter.

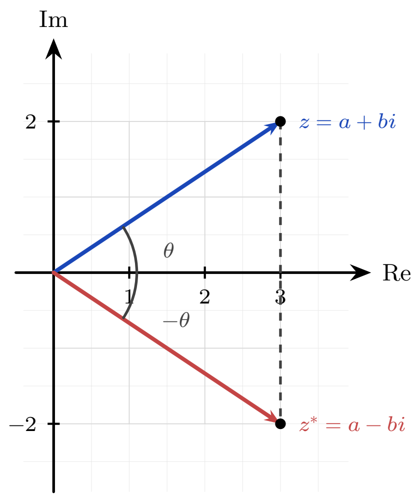


Figure 17.3 The conjugate is a reflection in the real axis. The two numbers share a modulus, and their arguments are equal and opposite. Because conjugating respects addition and multiplication and leaves every real number alone, the roots of a polynomial with real coefficients appear in mirror-image pairs See §17.4.3.

There are two natural quantities to attach to a point in a plane: how far it is from the origin, and in what direction. These are the modulus and the argument, and they are the subject of the next section.

YOUR TURN 17.1 Cartesian Form

1. Evaluate each expression, giving your answer in the form $a + bi$.

(a) $(3+2i)+(1-4i)$ (b) $(5-i)-(2+3i)$
(c) $(2+3i)(1-i)$ (d) $(2+i)(2-i)$

2. Simplify each power of i.

(a) i^3
(b) i^4
(c) i^{15}
(d) i^{100}

3. Let $z = 7 - 4i$.

(a) Write down $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.
(b) Write down z^*.
(c) Evaluate zz^*.
(d) Evaluate $z + z^*$.

4. Write each quotient in the form $a + bi$.

(a) $\dfrac{4+3i}{2-i}$
(b) $\dfrac{1}{1+i}$
(c) $\dfrac{2-3i}{i}$

5. Find the complex number z in each case.

(a) $z + 3z^* = 8 - 4i$
(b) $2z - z^* = 3 + 6i$

17.2 Modulus and Argument

The distance of z from the origin is its **modulus** $|z|$, straight from Pythagoras; the angle its arrow makes with the positive real axis is its **argument** $\arg(z)$, measured anticlockwise.

KEY For $z = a + bi$: $\quad |z| = \sqrt{a^2+b^2}, \qquad \tan(\arg z) = \dfrac{b}{a}.$

CAUTION Finding the argument needs care: $\tan\theta = \frac{b}{a}$ alone cannot tell which quadrant z is in, because tan repeats every π. Always sketch z on the Argand diagram first, then choose the angle in the correct quadrant, conventionally in the range $-\pi < \theta \leq \pi$ (the **principal argument**).

The danger is easiest to see with an example. The numbers $1+i$ and $-1-i$ both give $\frac{b}{a} = 1$, so a calculator returns $\arctan 1 = \frac{\pi}{4}$ for both. Only one of them is correct.

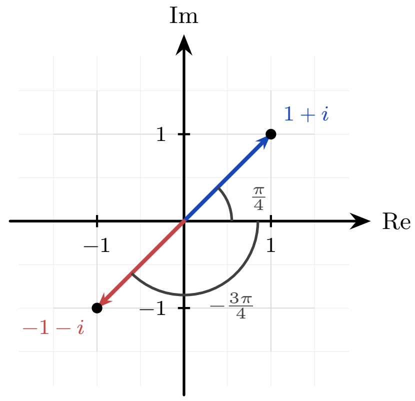


Figure 17.4 Both numbers give $\frac{b}{a}=1$, so both return $\arctan 1 = \frac{\pi}{4}$ from a calculator. The sketch settles it: $1+i$ lies in the first quadrant and has argument $\frac{\pi}{4}$, while $-1-i$ lies in the third and has principal argument $\frac{\pi}{4}-\pi=-\frac{3\pi}{4}$.

The modulus also measures the distance between two points, not only the distance from the origin. Since $z-w$ is the vector running from w to z Ch. 9, its length $|z-w|$ is the distance between the points representing z and w.

Knowing the modulus r and argument θ is enough to rebuild the number, since $a = r\cos\theta$ and $b = r\sin\theta$. This gives the **modulus–argument** (or polar) form, and the equivalent exponential version, **Euler's form**.

KEY · IB **Polar and Euler form:** $z = r(\cos\theta + i\sin\theta) = r\,\mathrm{cis}\,\theta = re^{i\theta}$,
where $r = |z|$ and $\theta = \arg(z)$.

17.2.1 Euler's form

That $e^{i\theta} = \cos\theta + i\sin\theta$ is unexpected at first sight, and you already have the tools to see why it is true. Take the Maclaurin series for e^x from Ch. 16 and substitute $x = i\theta$, using the cycle of powers of i from §17.1:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \cdots\right).$$

The even terms, where i has become ± 1, form the cosine series. The odd terms, which still carry an i, form the sine series. So the exponential function and the trigonometric functions are two views of one function on the complex plane. Setting $\theta = \pi$ gives $e^{i\pi} + 1 = 0$, which links e, i, π, 1 and 0 in a single line. It is a calculation, not a coincidence.

EXAMPLE 17.4

Write $z = -1 - \sqrt{3}\,i$ in polar and Euler form, with $-\pi < \theta \le \pi$.

The modulus is $r = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$. The point $\left(-1, -\sqrt{3}\right)$ lies in the third quadrant, where $\tan\theta = \sqrt{3}$ gives a reference angle of $\frac{\pi}{3}$, so the principal argument is $\theta = \frac{\pi}{3} - \pi =$

$-\frac{2\pi}{3}$. Hence

$$z = 2\left(\cos(-\tfrac{2\pi}{3}) + i\sin(-\tfrac{2\pi}{3})\right) = 2\operatorname{cis}(-\tfrac{2\pi}{3}) = 2e^{-2i\pi/3}.$$

Sketch first, then read off the quadrant. A common error is to take $\arctan$ without checking which quadrant the point is in.

The reverse direction is easier, because no quadrant decision is needed: substitute the angle and evaluate.

EXAMPLE 17.5

Write (a) $4\operatorname{cis}\dfrac{2\pi}{3}$ and (b) $2e^{-i\pi/2}$ in Cartesian form.

(a) Expand the cis and evaluate the two exact values:

$$4\left(\cos\tfrac{2\pi}{3} + i\sin\tfrac{2\pi}{3}\right) = 4\left(-\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i\right) = -2 + 2\sqrt{3}\,i.$$

(b) Euler form is the same thing written differently, so read $r = 2$ and $\theta = -\frac{\pi}{2}$:

$$2\left(\cos(-\tfrac{\pi}{2}) + i\sin(-\tfrac{\pi}{2})\right) = 2(0 - i) = -2i.$$

Going from polar to Cartesian is always this direct. It is the other direction that needs the sketch.

YOUR TURN 17.2 Modulus and Argument

6. Find the modulus of each number.

(a) $3 + 4i$ (b) $5 - 12i$
(c) $1 + i$ (d) $-2i$

7. Find the principal argument of each number, with $-\pi < \theta \le \pi$.

(a) $1 + i$ (b) i
(c) -3 (d) $-1 - i$

8. Write each number in modulus–argument form and in Euler form.

(a) $2i$ (b) -3
(c) $1 - i$ (d) $-1 + \sqrt{3}\,i$

9. Write each number in Cartesian form.

(a) $4\operatorname{cis}\frac{\pi}{3}$ (b) $2e^{-i\pi/4}$
(c) $3\operatorname{cis}\pi$ (d) $e^{i\pi/2}$

10. Let $z = 3 + 4i$ and $w = -1 + 2i$.

(a) Find $|z - w|$, giving an exact answer.
(b) State what $|z - w|$ measures on the Argand diagram.
(c) Verify that $zz^* = |z|^2$.

17.3 Products, Quotients and De Moivre's Theorem

Polar form needs extra notation, and it repays that cost by making multiplication easy. When two complex numbers are multiplied, their moduli multiply and their arguments *add*, which is just the law of exponents applied to $re^{i\theta}$.

KEY $|zw| = |z|\,|w|$ and $\arg(zw) = \arg z + \arg w$; $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ and $\arg\left(\frac{z}{w}\right) = \arg z - \arg w$.

Read geometrically, multiplying by a complex number *rotates* by its argument and *scales* by its modulus. So multiplying by $i = 1 \cdot e^{i\pi/2}$, whose modulus is 1, is a quarter-turn anticlockwise with no stretching at all.

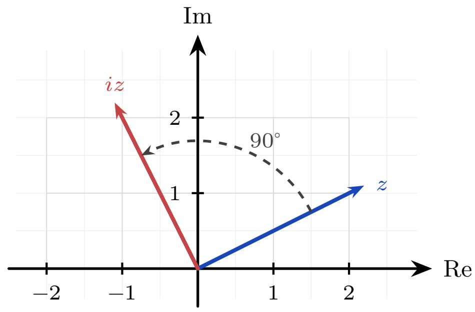


Figure 17.5 Multiplying by i turns a complex number through a quarter-turn anticlockwise about the origin, leaving its modulus unchanged.

EXAMPLE 17.6

Given $z = 6\,\text{cis}\,\frac{2\pi}{3}$ and $w = 3\,\text{cis}\,\frac{\pi}{6}$, find zw and $\frac{z}{w}$ in polar form.
For the product, multiply the moduli and add the arguments:

$$zw = (6)(3)\,\text{cis}\left(\tfrac{2\pi}{3} + \tfrac{\pi}{6}\right) = 18\,\text{cis}\,\tfrac{5\pi}{6}.$$

For the quotient, divide the moduli and subtract the arguments:

$$\frac{z}{w} = \frac{6}{3}\operatorname{cis}\left(\tfrac{2\pi}{3} - \tfrac{\pi}{6}\right) = 2\operatorname{cis}\tfrac{\pi}{2} = 2i.$$

Compare this with doing the same multiplication in Cartesian form, which would need four products and a collection of terms. Once the numbers are in polar form the work is one multiplication and one addition.

If multiplying adds arguments, then raising to a power *multiplies* the argument. This is **De Moivre's theorem**, and everything in the rest of the chapter follows from it.

KEY · IB **De Moivre's theorem:** $\left(r(\cos\theta + i\sin\theta)\right)^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{in\theta} = r^n \operatorname{cis} n\theta$.

The syllabus requires the theorem for $n \in \mathbb{Z}$ and extends it to *rational* exponents. Past the integers the statement has to be read more carefully: a power like $z^{1/n}$ is multi-valued, so $r^n \operatorname{cis} n\theta$ is one of its values rather than the whole answer. That is exactly the story of the nth roots in the next section. For positive integers the theorem is proved by the induction of Ch. 3. The inductive step is one multiplication: assuming $(r\operatorname{cis}\theta)^k = r^k \operatorname{cis} k\theta$, multiplying both sides once more by $r\operatorname{cis}\theta$ scales the modulus to r^{k+1} and, by the compound-angle identities of Ch. 8, adds θ to the argument to give $(k+1)\theta$. Every step is that same identity applied once, and you carry the induction out in full in Problem 22.

EXAMPLE 17.7

The theorem is claimed for every integer exponent. Taking the case $n \in \mathbb{Z}^+$ as proved, show that it also holds for every negative integer n.

Start with the reciprocal of a number of modulus 1. Multiplying adds the arguments, so

$$\operatorname{cis}\phi \cdot \operatorname{cis}(-\phi) = \operatorname{cis}(\phi - \phi) = \operatorname{cis} 0 = 1, \qquad \text{hence} \qquad \frac{1}{\operatorname{cis}\phi} = \operatorname{cis}(-\phi).$$

Dividing by a number of modulus 1 simply reverses its angle.

Now let n be a negative integer, and write $n = -m$ with $m \in \mathbb{Z}^+$. Applying the positive case to m,

$$(r\operatorname{cis}\theta)^n = \frac{1}{(r\operatorname{cis}\theta)^m} = \frac{1}{r^m \operatorname{cis}(m\theta)} = r^{-m}\operatorname{cis}(-m\theta) = r^n \operatorname{cis}(n\theta). \quad \blacksquare$$

The exponent $n = 0$ needs no work either, since $z^0 = 1 = r^0 \operatorname{cis} 0$. So one reciprocal carries the theorem from the positive integers to the whole of $\mathbb{Z}$, which is the range the syllabus asks for.

Raising a complex number to a high power means expanding a bracket again and again in Cartesian form. In polar form there is nothing to expand: raise the modulus to the power, multiply the angle by the power, and the work is finished.

EXAMPLE 17.8

Find $(1+i)^8$.

In polar form $1+i=\sqrt{2}\,\text{cis}\,\frac{\pi}{4}$. By De Moivre,

$$(1+i)^8=(\sqrt{2})^8\,\text{cis}(8\cdot\tfrac{\pi}{4})=16\,\text{cis}(2\pi)=16.$$

Eight multiplications reduce to one power of the modulus and one multiple of the angle.

Draw the powers and you can watch the theorem work. Each multiplication by $1+i$ turns the arrow 45° anticlockwise and stretches it by $\sqrt{2}$, so the powers lie on a spiral that winds outward from the origin:

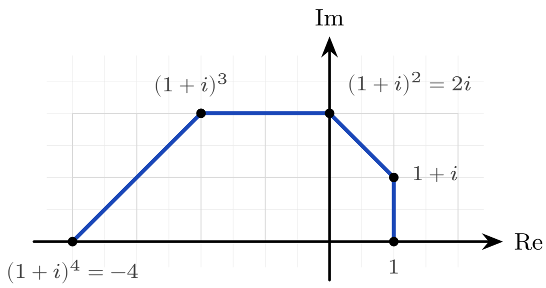


Figure 17.6 The powers $(1+i)^0$ through $(1+i)^4$. Each multiplication turns the arrow through 45° and stretches it by $\sqrt{2}$, so the powers lie on a spiral winding outward from the origin.

Four more steps continue the spiral through the lower half-plane and back to the positive real axis at $(1+i)^8=16$: a full turn of $8\times 45^\circ=360^\circ$, with the modulus grown to $(\sqrt{2})^8=16$. The worked example above is this picture written in algebra.

17.3.1 Numbers on the unit circle

One case of De Moivre is used often enough to state separately. If $|z|=1$ then $z=\text{cis}\,\theta$, and dividing 1 by a number of modulus 1 simply reverses its angle, so $\dfrac{1}{z}=\text{cis}(-\theta)=z^*$. Add the two and the imaginary parts cancel while the real parts double; subtract them and the opposite happens.

KEY For $z=\text{cis}\,\theta$ and any $n\in\mathbb{Z}$:

$$z^n+\frac{1}{z^n}=2\cos n\theta,\qquad z^n-\frac{1}{z^n}=2i\sin n\theta.$$

Read from left to right this converts powers into multiple angles, and that is what makes it useful: an expression like $\cos^6\theta$, awkward to integrate as it stands, becomes a sum of $\cos 2\theta$, $\cos 4\theta$ and $\cos 6\theta$ terms that can be integrated directly. That construction is §17.5.

EXAMPLE 17.9

Let $z = \operatorname{cis}\theta$. Expand $\left(z+\frac{1}{z}\right)\left(z-\frac{1}{z}\right)$ in two ways, and hence express $\sin\theta\cos\theta$ in terms of $\sin 2\theta$.

Taking $n=1$ in the two formulas, $z+\frac{1}{z}=2\cos\theta$ and $z-\frac{1}{z}=2i\sin\theta$, so the product is

$$\left(z+\frac{1}{z}\right)\left(z-\frac{1}{z}\right)=(2\cos\theta)(2i\sin\theta)=4i\sin\theta\cos\theta.$$

Expanding algebraically instead, the two cross terms cancel:

$$\left(z+\frac{1}{z}\right)\left(z-\frac{1}{z}\right)=z^2-1+1-\frac{1}{z^2}=z^2-\frac{1}{z^2}=2i\sin 2\theta,$$

using the second formula again with $n=2$. The two expressions are equal, so

$$4i\sin\theta\cos\theta=2i\sin 2\theta \implies \sin\theta\cos\theta=\frac{\sin 2\theta}{2}.$$

This is the double-angle identity for the sine, from Ch. 8, now obtained without any trigonometric identity. Evaluating one product in two ways is the technique of §17.5.1.

YOUR TURN 17.3 Products, Quotients and De Moivre's Theorem

11. Evaluate each expression, giving your answer in modulus–argument form.

(a) $\left(2\operatorname{cis}\frac{\pi}{6}\right)\left(3\operatorname{cis}\frac{\pi}{3}\right)$

(b) $\dfrac{6\operatorname{cis}\frac{2\pi}{3}}{2\operatorname{cis}\frac{\pi}{3}}$

(c) $\left(\operatorname{cis}\frac{\pi}{5}\right)^5$

(d) $\left(2\operatorname{cis}\frac{\pi}{6}\right)^3$

12. Given $|z|=3$, $|w|=4$, $\arg z=\frac{\pi}{4}$ and $\arg w=\frac{\pi}{6}$, write down

(a) $|zw|$

(b) $\arg(zw)$

(c) $\left|\dfrac{z}{w}\right|$

(d) $\arg\left(\dfrac{z}{w}\right)$

13.

(a) Expand $(\cos\theta+i\sin\theta)^4$ using De Moivre's theorem.

(b) Use De Moivre's theorem to evaluate $(1+i)^6$.

(c) Use De Moivre's theorem to evaluate $\left(1-\sqrt{3}\,i\right)^5$.

14. Let $z=\operatorname{cis}\theta$. Write each expression in terms of a single cosine or sine.

(a) $z+\dfrac{1}{z}$

(b) $z^3+\dfrac{1}{z^3}$

(c) $z^2-\dfrac{1}{z^2}$

17.4 Roots of Complex Numbers

Solving $z^3 = 125$ over the real numbers gives one answer. Over the complex numbers it gives three, and they are not scattered: they sit at equal spacing on a circle. Every root question in this section is answered by that one picture.

The same theorem, used in reverse, finds roots. A nonzero complex number has exactly n distinct nth roots. They all share one modulus, and their arguments are equally spaced. To find them, write the number in polar form, take the nth root of the modulus, divide the argument by n, then step around by $\frac{2\pi}{n}$ until all n are collected.

KEY The nth roots of $z = r\operatorname{cis}\theta$ are $\sqrt[n]{r}\operatorname{cis}\left(\dfrac{\theta + 2\pi k}{n}\right)$, for $k = 0, 1, \ldots, n-1$.

Because the roots share a modulus $\sqrt[n]{r}$ and their arguments differ by equal steps of $\frac{2\pi}{n}$, they sit at the vertices of a **regular n-gon** centred at the origin. One root therefore locates all the others: find any single root, and the rest follow by turning it through $\frac{2\pi}{n}$ at a time.

KEY To solve $z^n = w$:

1. Write w in polar form, $w = r\operatorname{cis}\theta$.
2. The modulus of every root is $\sqrt[n]{r}$.
3. Take $\dfrac{\theta}{n}$ as the first argument, then add $\dfrac{2\pi}{n}$ repeatedly until you have n roots.
4. Convert back to Cartesian form only if the question asks for it.

EXAMPLE 17.10

Find the three cube roots of 125.

Write $125 = 125\operatorname{cis}0$. The modulus of each root is $\sqrt[3]{125} = 5$, and the arguments are $\frac{0+2\pi k}{3}$ for $k = 0, 1, 2$:

$$5\operatorname{cis}0 = 5, \qquad 5\operatorname{cis}\tfrac{2\pi}{3} = -\frac{5}{2} + \frac{5\sqrt{3}}{2}i, \qquad 5\operatorname{cis}\tfrac{4\pi}{3} = -\frac{5}{2} - \frac{5\sqrt{3}}{2}i.$$

The three roots form an equilateral triangle on the circle of radius 5, one real and a conjugate pair. Only the first argument needed any thought; the other two came from adding $\frac{2\pi}{3}$.

That example hid one step, because 125 has argument 0 and dividing 0 by 3 changes nothing. When the number has a non-zero argument, the division does real work.

EXAMPLE 17.11

Find the three cube roots of $-27i$, giving each in polar form and in Cartesian form.

First write $-27i$ in polar form. It lies on the negative imaginary axis, so its modulus is 27 and its argument is $-\frac{\pi}{2}$:

$$-27i = 27\operatorname{cis}\left(-\tfrac{\pi}{2}\right).$$

Each root has modulus $\sqrt[3]{27} = 3$. The first argument is the given argument divided by 3:

$$\frac{-\pi/2}{3} = -\frac{\pi}{6}.$$

The others follow by adding $\frac{2\pi}{3}$ each time:

$$-\frac{\pi}{6}, \qquad -\frac{\pi}{6} + \frac{2\pi}{3} = \frac{\pi}{2}, \qquad \frac{\pi}{2} + \frac{2\pi}{3} = \frac{7\pi}{6} \equiv -\frac{5\pi}{6}.$$

The third has been reduced by 2π to bring it back into $-\pi < \theta \leq \pi$. In polar form the roots are $3\operatorname{cis}\left(-\frac{\pi}{6}\right)$, $3\operatorname{cis}\frac{\pi}{2}$ and $3\operatorname{cis}\left(-\frac{5\pi}{6}\right)$, and converting each,

$$\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \qquad 3i, \qquad -\frac{3\sqrt{3}}{2} - \frac{3}{2}i.$$

Check the easiest one: $(3i)^3 = 27i^3 = -27i$. ✓ Note that adding $\frac{2\pi}{3}$ repeatedly can carry an argument past π; subtract 2π to return it to the principal range if the question asks for principal arguments.

17.4.1 Roots of Unity

One case matters more than the rest, and examinations use it most often: $w = 1$. Solving $z^n = 1$ gives the **roots of unity**. These are n points on the *unit* circle. One of them is 1, and together they sit at the vertices of a regular n-gon.

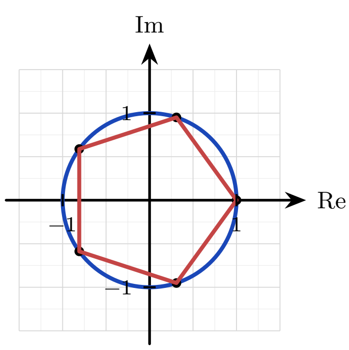


Figure 17.7 The five fifth roots of unity, at the vertices of a regular pentagon inscribed in the unit circle with one vertex at 1.

One root generates all the others. Write $\omega = \operatorname{cis}\dfrac{2\pi}{n}$ for the first root anticlockwise from 1. Then $\omega^k = \operatorname{cis}\dfrac{2\pi k}{n}$, which is the kth root round the circle. Every power of ω is again a root, because $\left(\omega^k\right)^n = \left(\omega^n\right)^k = 1^k = 1$. The n roots are therefore

$$1,\ \omega,\ \omega^2,\ \ldots,\ \omega^{n-1},$$

one number and its powers, stepping round the polygon and returning to the start.

The case $n = 3$ is worth carrying in memory. Converting each root to Cartesian form, the cube roots of unity are 1, $\omega = \operatorname{cis}\frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\omega^2 = \operatorname{cis}\frac{4\pi}{3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$: a real root and a conjugate pair on the unit circle.

They also **sum to zero**. There are two ways to see this. Geometrically, the n arrows point symmetrically in every direction, so they cancel. Algebraically, the geometric series of Ch. 2 gives the result directly, since $\omega \neq 1$:

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega - 1} = 0.$$

The same conclusion follows from Ch. 6. The polynomial $z^n - 1$ has no z^{n-1} term, so the sum of its roots is zero. For $n = 3$ this gives the identity $1 + \omega + \omega^2 = 0$, which appears whenever cube roots of unity are used.

EXAMPLE 17.12

Let $\omega = \operatorname{cis}\dfrac{2\pi}{3}$, so that $\omega^3 = 1$ and $1+\omega+\omega^2 = 0$. Simplify (a) $\omega^{10} + \omega^{20}$, and (b) $(1+\omega)(1+\omega^2)$.

(a) A power of ω is reduced by dividing the exponent by 3 and keeping the remainder, since $\omega^3 = 1$. Here $10 = 3(3) + 1$ and $20 = 3(6) + 2$, so

$$\omega^{10} + \omega^{20} = \omega + \omega^2 = -1,$$

the last step because $1 + \omega + \omega^2 = 0$.

(b) Expand, and use $\omega^3 = 1$ again:

$$(1+\omega)(1+\omega^2) = 1 + \omega^2 + \omega + \omega^3 = (1 + \omega + \omega^2) + 1 = 0 + 1 = 1.$$

Questions of this type turn on two facts: $\omega^3 = 1$ reduces any power of ω, and $1 + \omega + \omega^2 = 0$ removes what is left. Apply them in that order.

These evenly spaced points have a large practical use. The *discrete Fourier transform* separates a signal, such as a sound, into the pure frequencies it is made of. It does this by sampling the signal against exactly these n points on the circle. A fast method of computing it exploits the symmetry of the polygon. That method is why a phone can compress music and process sound as it arrives.

17.4.2 Square Roots Without Polar Form

Polar form finds roots of every order, but it has one drawback. It needs the argument, and unless the angle is one of the standard ones, $\arctan$ returns a decimal. The answer is then no longer exact. For *square* roots there is a second route that stays in Cartesian form and keeps the surds intact. Assume the root is $a + bi$, square it, and equate real and imaginary parts as in §17.1.

EXAMPLE 17.13

Find the square roots of $7 + 24i$ in the form $a + bi$.
Let $(a + bi)^2 = 7 + 24i$. Expanding the left side gives $a^2 - b^2 + 2abi$, so equating parts,

$$a^2 - b^2 = 7 \qquad \text{and} \qquad 2ab = 24.$$

The second gives $b = \dfrac{12}{a}$. Substituting into the first,

$$a^2 - \frac{144}{a^2} = 7 \implies a^4 - 7a^2 - 144 = 0 \implies (a^2 - 16)(a^2 + 9) = 0.$$

Since a is real, $a^2 = 16$, so $a = \pm 4$ and correspondingly $b = \pm 3$. The square roots are $4+3i$ and $-4 - 3i$, that is $\pm(4 + 3i)$.

Two features of that calculation repeat every time. The substitution always produces a **quadratic in a^2**, and the negative solution for a^2 is always rejected, because a is real. The two answers always differ by a sign, since square roots lie at opposite ends of a diameter, half a turn apart.

CAUTION Use Cartesian equating for square roots when the answer should be exact. For cube roots and higher, go back to polar form: equating parts would give a cubic or an equation of even higher degree, while De Moivre handles roots of any order directly.

17.4.3 Complex Roots of Polynomials

Every nth root above was a solution of $z^n = w$, which is a polynomial equation of degree n with exactly n solutions. That is not a coincidence of this one family. This is the **Fundamental Theorem of Algebra**: every polynomial of degree $n \geq 1$ has exactly n roots in the complex numbers, counted with multiplicity. The roots of a polynomial need not lie on the real line, but none of them can lie outside the complex plane.

The simplest case has already appeared, in a different form. In Ch. 4, a quadratic with negative discriminant had "no solutions." It always had two roots; they were simply not on the real line.

EXAMPLE 17.14

Solve $z^2 - 4z + 13 = 0$.
The quadratic formula works unchanged:

$$z = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i.$$

In Ch. 4 a negative discriminant ended the question. Here it simply produces an i: the two roots are $2 + 3i$ and $2 - 3i$, a conjugate pair placed symmetrically above and below the real axis.

That symmetry is no accident. When a polynomial has *real* coefficients, its non-real roots always come in conjugate pairs.

KEY **Conjugate root theorem.** If a polynomial has real coefficients and $z = a + bi$ is a root, then $z^* = a - bi$ is also a root.

This is why a cubic with real coefficients always has at least one real root: its three roots are either all real, or one real root plus a conjugate pair. A non-real root never appears alone in a polynomial with real coefficients.

EXAMPLE 17.15

Given that $2+i$ is a root of a polynomial with real coefficients, state another root and find a real quadratic factor.

By the conjugate root theorem, $2 - i$ is also a root. The two together give a quadratic factor with real coefficients:

$$(z - (2 + i))(z - (2 - i)) = (z - 2)^2 - i^2 = z^2 - 4z + 5.$$

A conjugate pair of roots always multiplies to a real quadratic, here $z^2 - 4z + 5$, which can then be divided out to find the remaining real root.

YOUR TURN 17.4 Roots of Complex Numbers

15.

(a) Write down a square root of -9.
(b) State the modulus of each cube root of 27.
(c) State how many distinct fifth roots a nonzero complex number has.
(d) State the shape formed by the nth roots of a complex number on the Argand diagram.

16. Roots of unity.

(a) State the three cube roots of unity, in Cartesian form.
(b) State the four fourth roots of unity.
(c) State the six sixth roots of unity, in modulus–argument form.
(d) Let ω be a cube root of unity with $\omega \neq 1$. Show that $1 + \omega + \omega^2 = 0$.

17. Find, in modulus–argument form,

(a) the three cube roots of 8;
(b) the four fourth roots of 16;

(c) the three cube roots of i.

18. Find the square roots of each number in the form $a+bi$ by equating real and imaginary parts.

(a) $5+12i$
(b) $-8+6i$
(c) $-5+12i$

19. Solve each equation.

(a) $z^2+9=0$
(b) $z^2+4=0$
(c) $z^2-2z+5=0$
(d) $z^2+2z+2=0$

20.

(a) Given that $3+i$ is a root of a polynomial with real coefficients, state another root.
(b) Write down a quadratic with real coefficients whose roots are $2i$ and $-2i$.
(c) Write down a quadratic with real coefficients whose roots are $1+i$ and $1-i$.

21.

(a) The equation $z^2+bz+c=0$ has real coefficients and $1+2i$ as a root. Find b and c.
(b) Explain why a cubic with real coefficients must have at least one real root.

17.5 Trigonometric Identities from De Moivre

The compound-angle rules of Ch. 8 give $\cos 2\theta$ and $\sin 2\theta$ without much work. For $\cos 3\theta$ they take considerably more. Each multiple angle has to be built from the one before it, so the work grows with every step. One expansion with complex numbers gives any multiple angle at once, and it gives the sine identity and the cosine identity together.

The method comes from one observation. De Moivre's theorem says that $(\cos\theta+i\sin\theta)^n$ and $\cos n\theta+i\sin n\theta$ are the same number. So one number has been written in two ways. Expand the first way with the binomial theorem of Ch. 2. The two ways must then match in both parts: the real part of one equals the real part of the other, and the imaginary parts match in the same way.

KEY To find an identity for $\cos n\theta$ or $\sin n\theta$:

1. Expand $(\cos\theta + i\sin\theta)^n$ by the binomial theorem.
2. Collect the terms into a real part and an imaginary part, using $i^2 = -1$.
3. Equate the real part to $\cos n\theta$ and the imaginary part to $\sin n\theta$.
4. Replace $\sin^2\theta$ by $1 - \cos^2\theta$, or the reverse, to leave one function only.

Each expansion is compared in two parts, so it always produces two identities at once. Divide one identity by the other and you have a third: the imaginary part over the real part is $\tan n\theta$, and the real part over the imaginary part is $\cot n\theta$.

EXAMPLE 17.16

Use De Moivre's theorem to show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, and find a matching identity for $\sin 3\theta$.

Write $c = \cos\theta$, $s = \sin\theta$, and expand by the binomial theorem:

$$(c + is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = (c^3 - 3cs^2) + i(3c^2 s - s^3).$$

By De Moivre this equals $\cos 3\theta + i\sin 3\theta$, so the parts must match separately. Real parts, using $s^2 = 1 - c^2$:

$$\cos 3\theta = c^3 - 3c(1 - c^2) = 4\cos^3\theta - 3\cos\theta.$$

Imaginary parts, using $c^2 = 1 - s^2$:

$$\sin 3\theta = 3(1 - s^2)s - s^3 = 3\sin\theta - 4\sin^3\theta.$$

One expansion gives two identities, and the same method works for any n. Results that took several steps with compound angles in Ch. 8 now follow from a single binomial expansion.

A third identity comes from the same expansion and needs no new work: divide one part by the other.

EXAMPLE 17.17

Use the expansion above to show that $\cot 3\theta = \dfrac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}$.

Take the two parts of the expansion *before* either was rewritten in a single function:

$$\cos 3\theta = c^3 - 3cs^2, \qquad \sin 3\theta = 3c^2 s - s^3.$$

Since $\cot 3\theta$ is the first divided by the second,

$$\cot 3\theta = \frac{c^3 - 3cs^2}{3c^2 s - s^3}.$$

Now divide every term, top and bottom, by s^3. Each division turns a c over an s into a $\cot\theta$:

$$\cot 3\theta = \frac{\dfrac{c^3}{s^3} - \dfrac{3cs^2}{s^3}}{\dfrac{3c^2s}{s^3} - \dfrac{s^3}{s^3}} = \frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}. \qquad \blacksquare$$

Two points make this reliable. Use the parts in their original mixed form, since replacing s^2 by $1 - c^2$ first would leave a mixture that no division tidies. And divide by the highest power present, here s^3, so that every term becomes a power of a single ratio.

17.5.1 Powers of Cosine and Sine

The method above turns a multiple angle into powers of $\cos\theta$. The same idea works in the other direction. A power such as $\cos^6\theta$ becomes a sum of multiple angles, and that is what makes it possible to integrate in Ch. 15.

The tool is the pair of results from §17.3. For $z = \operatorname{cis}\theta$,

$$z + \frac{1}{z} = 2\cos\theta, \qquad z^n + \frac{1}{z^n} = 2\cos n\theta,$$

and the same expressions with a minus sign give $2i\sin\theta$ and $2i\sin n\theta$. So raise $z + \frac{1}{z}$ to a power and expand it. Each pair of terms then combines into a cosine of a multiple angle.

EXAMPLE 17.18

Express $\cos^6\theta$ in terms of $\cos 2\theta$, $\cos 4\theta$ and $\cos 6\theta$.

Start from $2\cos\theta = z + \dfrac{1}{z}$ and raise both sides to the sixth power, with binomial coefficients $1, 6, 15, 20, 15, 6, 1$:

$$64\cos^6\theta = \left(z + \frac{1}{z}\right)^6 = z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}.$$

Now pair the terms that are reciprocals of each other:

$$64\cos^6\theta = \left(z^6 + \frac{1}{z^6}\right) + 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) + 20 = 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20.$$

Divide by 64:

$$\cos^6\theta = \tfrac{1}{32}\cos 6\theta + \tfrac{3}{16}\cos 4\theta + \tfrac{15}{32}\cos 2\theta + \tfrac{5}{16}.$$

The pairing is what makes the method work. In the expansion, the power of z in the first term is the reciprocal of the power in the last term, and the same holds for the second term and the second from last. Every pair therefore combines into a cosine.

TIP A power of $\sin\theta$ works the same way, starting from $2i\sin\theta = z - \frac{1}{z}$. Take one extra care. Raising i to the power leaves a factor that must be divided out at the end, and an odd power gives sines rather than cosines. Check your answer by putting $\theta = 0$.

TIP Complex numbers offer several good IA topics. Three directions to explore: iterate $z \mapsto z^2 + c$ for different c and investigate which starting points escape to infinity (this is how the Mandelbrot set is defined); investigate which regular polygons can be constructed with ruler and compass, a question Gauss answered at nineteen using exactly the roots of unity above; or measure phase shift in a simple AC circuit and model it with rotating complex numbers.

YOUR TURN 17.5 Trigonometric Identities from De Moivre

22. Use De Moivre's theorem and the binomial theorem for each part.

(a) Show that $\cos 2\theta = 2\cos^2\theta - 1$.
(b) Find a matching identity for $\sin 2\theta$.
(c) Show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$.
(d) Hence, or otherwise, express $\tan 2\theta$ in terms of $\tan\theta$.

23. Let $z = \operatorname{cis}\theta$. Use $z + \dfrac{1}{z} = 2\cos\theta$ for each part.

(a) Express $\cos^2\theta$ in terms of $\cos 2\theta$.
(b) Express $\sin^2\theta$ in terms of $\cos 2\theta$.
(c) Express $\cos^3\theta$ in terms of $\cos 3\theta$ and $\cos\theta$.

24.

(a) Show that $\sin^3\theta = \frac{1}{4}(3\sin\theta - \sin 3\theta)$.
(b) Express $\cos^5\theta$ in terms of cosines of multiple angles.

Chapter 17 · Complex Numbers

Reflections

TOK Does the naming shape the difficulty? "Imaginary" was coined as a dismissal and "real" as its opposite, and both names outlived the objection by centuries. Would these numbers have been accepted sooner if they had been called something neutral, such as "lateral"? If a change of vocabulary can change how readily an idea is believed, how much of what we call mathematical knowledge is carried by its language rather than its content?

TOK Euler's identity $e^{i\pi} + 1 = 0$ is routinely called the most beautiful equation in mathematics. What is being judged when a mathematician calls a result beautiful, and is that judgement evidence of anything? Beauty in mathematics is often described as economy, surprise, or unexpected connection, all of which this identity has.

IM The name "imaginary" was an insult. Descartes coined it in the seventeenth century to dismiss these numbers as fictions, and Leibniz called them "an amphibian between being and non-being." Their rehabilitation was slow and international. The Norwegian surveyor Caspar Wessel and the Swiss bookkeeper Jean-Robert Argand independently drew the plane picture that made them concrete. It was the German mathematician Carl Friedrich Gauss who finally made them respectable: he gave the first accepted proof of the Fundamental Theorem of Algebra, and he popularised the term "complex" itself. A number system that began in suspicion became, within two centuries, essential.

RW Complex numbers are the natural language of anything that oscillates. An alternating current, a radio wave or a vibration is a rotating complex number, and its real part is the signal you measure. Engineers add and scale these "phasors" directly, in place of pages of trigonometry. Quantum mechanics goes further: a particle's state is a complex-valued wave, and i sits at the centre of the equation that moves it. Phone signal processing, power-grid impedance and quantum computing all run on the numbers of this chapter.

EXT De Moivre's theorem hints at a whole hidden unity. The identity $e^{i\theta} = \cos\theta + i\sin\theta$ says that the exponential function and the trigonometric functions, which look nothing alike on the real line, are the same function seen along different directions in the complex plane. Growth and oscillation, e^x and $\sin x$, are one thing viewed two ways. Push this further and you reach complex analysis, the study of functions of a complex variable. Such functions turn out to be far more tightly constrained than real ones, and problems about ordinary integers, such as how the prime numbers are distributed, are answered by studying functions over the complex plane.

End-of-Chapter Problems

1. Given $z = 3 + 4i$ and $w = 1 - 2i$:

(a) find $z + w$; [1]
(b) find zw; [2]
(c) find $\frac{z}{w}$; [2]
(d) find $|z|$. [1]

2. Express $\dfrac{2+i}{3-i}$ in the form $a + bi$. [4]

3. Find the modulus and the principal argument of

(a) $z = -1 + i$; [3]
(b) $z = \sqrt{3} - i$. [3]

4. Write $z = 1 + \sqrt{3}\,i$ in modulus–argument form and in Euler form. [4]

5. Use De Moivre's theorem to evaluate

(a) $(1 + \sqrt{3}\,i)^6$; [3]
(b) $(1 + i)^{10}$; [3]
(c) $\left(\dfrac{1+i}{\sqrt{2}}\right)^{12}$. [3]

6. Find the square roots of each number in the form $a + bi$.

(a) $3 + 4i$ [4]
(b) $-5 + 12i$ [4]

7. Given $z = 2\,\mathrm{cis}\,\frac{\pi}{3}$:

(a) find z^2; [2]
(b) find z^3. [2]

8. The equation $z^2 + bz + c = 0$ has real coefficients and $1 - 2i$ as a root. Find b and c. [4]

9. Find the three cube roots of $8i$, giving each

(a) in modulus–argument form; [3]
(b) in exact Cartesian form. [3]

10. Simplify

(a) i^{2026}; [2]
(b) i^{-7}. [2]

11. Find all solutions of $z^3 = 27$, giving each in modulus–argument form. [4]

12. Find real numbers x and y such that $(x + yi)(2 - i) = 3 + i$. [4]

13.

(a) Express -8 in the form $re^{i\theta}$, with $-\pi < \theta \leq \pi$. [3]
(b) The complex number z has $|z| = 2$ and $\arg z = \frac{\pi}{6}$. Write z in Cartesian form. [3]

14.

(a) Find the four fourth roots of -16, giving each in the form $a + bi$. [5]
(b) Find all solutions of $z^4 = 1$, and show that they sum to zero. [4]

15. Solve $2z^2 + 2z + 5 = 0$. [4]

16. Given $z = \cos\theta + i\sin\theta$, show that $z + \dfrac{1}{z} = 2\cos\theta$. [4]

17. The points representing 1, i, -1 and $-i$ are plotted on an Argand diagram. State the shape they form and find its area. [3]

18. Given that $2 + 3i$ is a root of a polynomial with real coefficients, find a real quadratic factor of that polynomial. [4]

19. Solve $z^2 = 2i$, giving the roots in the form $a + bi$. [4]

20. Given $z = 2 + 2i$, find $|z|$ and $\arg z$, and hence write z in Euler form. [4]

21. Find the complex number z satisfying $2z - z^* = 3 + 12i$. [4]

22. Prove De Moivre's theorem, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, for all $n \in \mathbb{Z}^+$ by mathematical induction. [7]

23. Use De Moivre's theorem to show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$. [6]

24. Use De Moivre's theorem to show that $\tan 3\theta = \dfrac{3t - t^3}{1 - 3t^2}$, where $t = \tan\theta$. [7]

25. One cube root of a complex number w is $1+\sqrt{3}\,i$. Find w and the other two cube roots. [6]

26. GDC In this question you will investigate the fifth roots of unity, the solutions of $z^5 = 1$.

(a) Write down the five roots in the form $\operatorname{cis}\theta$. [3]
(b) Show that the five roots sum to zero, using the sum of a geometric series. [3]
(c) Let $\omega = \operatorname{cis}\frac{2\pi}{5}$. Show that every fifth root of unity is a power of ω. [2]
(d) Deduce that $\omega+\omega^2+\omega^3+\omega^4=-1$. [2]
(e) The five roots are plotted on an Argand diagram and joined to form a regular pentagon. Show that its area is $\frac{5}{2}\sin\frac{2\pi}{5}$, and evaluate this to three significant figures. [5]

27. Let $z = \operatorname{cis}\theta$.

(a) Show that $z^n + \dfrac{1}{z^n} = 2\cos n\theta$ for any $n \in \mathbb{Z}^+$. [3]
(b) Expand $\left(z+\dfrac{1}{z}\right)^4$ using the binomial theorem, and hence show that $16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$. [5]
(c) Deduce that $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$. [2]
(d) Hence evaluate $\displaystyle\int_0^{\pi/2} \cos^4\theta\,d\theta$ exactly. [4]
(e) Explain what this method achieved that integrating $\cos^4\theta$ directly would not. [2]

28. The polynomial $P(z) = z^4 - 2z^3 + 14z^2 - 18z + 45$ has real coefficients, and $1+2i$ is a root.

(a) Write down another root of $P(z)$, and give a reason. [2]
(b) Find a real quadratic factor of $P(z)$. [3]
(c) Find the other quadratic factor. [3]
(d) Hence write down all four roots of $P(z)$. [3]
(e) The four roots are plotted on an Argand diagram. Describe the symmetry of the picture, and state the feature of $P(z)$ that guarantees it. [3]

Challenge Questions

29. GDC **Why engineers use complex numbers: AC circuits.** In a direct-current circuit, voltage and current are related by $V = IR$. In an alternating-current circuit both oscillate, and a component can also shift the current's *timing* relative to the voltage. Engineers handle size and timing together by replacing resistance with a complex **impedance** Z, so that

$V = IZ$ still holds. For a resistor, an inductor and a capacitor in series,

$$Z = R + i\left(\omega L - \frac{1}{\omega C}\right),$$

where ω is the angular frequency of the supply. The real part is resistance; the imaginary part is called the *reactance*.

(a) A circuit has $R = 30\,\Omega$ and reactance $40\,\Omega$, so $Z = 30 + 40i$. Find $|Z|$ and $\arg Z$. [3]
(b) The current is $I = 4\operatorname{cis} 0$ amps. Find $V = IZ$ in polar form. [2]
(c) $|Z|$ and $\arg Z$ each mean something physical. Using (a) and (b), say what each one tells you about the voltage. [3]
(d) Show that Z is real exactly when $\omega = \dfrac{1}{\sqrt{LC}}$. This frequency is called **resonance**. [3]
(e) A circuit has $L = 0.05$ H and $C = 20\,\mu$F. Find its resonant frequency ω, and verify that the two reactances cancel there. [3]
(f) Explain why $|Z|$ is smallest at resonance, whatever R is, and hence why turning the dial on a radio selects one station. [3]

30. Adding two waves. Two loudspeakers play the same pure note. At a listening point the two arrivals are $A\cos\omega t$ and $B\cos(\omega t + \phi)$, where ϕ is the phase difference caused by the extra distance one sound has travelled. Adding them with trigonometry is unpleasant; complex numbers make it addition of arrows.

(a) Show that $A\cos\omega t + B\cos(\omega t+\phi) = \operatorname{Re}\Big[(A + B\operatorname{cis}\phi)\operatorname{cis}\omega t\Big]$. [3]
(b) Deduce that the combined sound is a wave of the same frequency, with amplitude $|A + B\operatorname{cis}\phi|$ and phase $\arg(A + B\operatorname{cis}\phi)$. [2]
(c) Two speakers each have amplitude 3 and the phase difference is $\frac{\pi}{3}$. Find the amplitude and phase of the combined sound, in exact form. [4]
(d) Now let both amplitudes equal A. Show that the combined amplitude is $2A\cos\dfrac{\phi}{2}$ and the phase is $\dfrac{\phi}{2}$, for $-\pi < \phi < \pi$. [4]
(e) Use (d) to explain what a listener hears when $\phi = 0$ and when $\phi = \pi$, and why noise-cancelling headphones work by generating a wave with $\phi = \pi$. [3]

31. From a circle to an aeroplane wing. The map $w = z + \dfrac{1}{z}$ is called the **Joukowski transformation**. Early aerodynamicists used it because it turns circles, whose airflow is easy to calculate, into wing-shaped curves, whose airflow is not.

(a) Let $z = \operatorname{cis}\theta$, so $|z| = 1$. Show that $w = 2\cos\theta$, and hence that the unit circle maps onto the line segment from -2 to 2. [4]
(b) Now let $z = r\operatorname{cis}\theta$ with $r > 1$. Writing $w = x + iy$, show that

$$x = \left(r + \tfrac{1}{r}\right)\cos\theta, \qquad y = \left(r - \tfrac{1}{r}\right)\sin\theta,$$

and deduce that $\dfrac{x^2}{\left(r+\frac{1}{r}\right)^2} + \dfrac{y^2}{\left(r-\frac{1}{r}\right)^2} = 1.$ [5]

(c) That equation describes an ellipse with semi-axes $r + \frac{1}{r}$ and $r - \frac{1}{r}$. Find both for $r = 2$. [2]

(d) Describe what happens to the ellipse as $r \to 1^+$, and reconcile your answer with part (a). [3]

(e) A wing needs a blunt front and a sharp trailing edge. Explain why a circle passing *through* $z = 1$ rather than around it produces a sharp point, using the fact that $\dfrac{dw}{dz} = 1 - \dfrac{1}{z^2}$ vanishes there, and that where it vanishes the map *doubles* angles instead of preserving them. [3]

32. How a phone hears a chord. A microphone records a sound as a list of numbers, one per sampling instant. To find which frequencies are present, the recording is compared against the roots of unity. For four samples x_0, x_1, x_2, x_3, the **discrete Fourier transform** is

$$X_k = \sum_{n=0}^{3} x_n\, i^{-nk}, \qquad k = 0, 1, 2, 3,$$

where i is the primitive fourth root of unity $\operatorname{cis}\frac{2\pi}{4}$. Each X_k measures how strongly the frequency k is present.

(a) Write down the four fourth roots of unity and verify that they sum to zero. [2]

(b) A steady tone with no variation gives samples $(1, 1, 1, 1)$. Compute X_0, X_1, X_2, X_3. [4]

(c) Explain, using part (a), why every X_k with $k \neq 0$ had to vanish in (b), and say what X_0 measures. [3]

(d) A signal that rises and falls once across the four samples gives $(1, 0, -1, 0)$. Compute X_0, X_1, X_2, X_3. [4]

(e) Interpret your answer to (d): which frequency is present, and what does $X_0 = 0$ tell you about the signal? [3]

33. The edge of the Mandelbrot set. Fix a complex number c, start at $z_0 = 0$, and iterate

$$z_{n+1} = z_n^2 + c.$$

For some c the sequence stays bounded forever; for others it runs away to infinity. The set of c for which it stays bounded is the **Mandelbrot set**, and a computer draws it by testing one c per pixel.

(a) Take $c = i$. Compute z_1, z_2, z_3 and z_4, and show that the sequence then repeats with period 2. Is i in the set? [4]

(b) Take $c = 1$. Compute z_1, z_2, z_3, z_4 and say what is happening. [2]

(c) Show that if $|z| > 2$ and $|z| \geq |c|$ then $\left|z^2 + c\right| > |z|$. *(Hint: $|z^2 + c| \geq |z|^2 - |c|$)* [4]

(d) Deduce that once the sequence reaches modulus greater than 2 it never returns, so it escapes. Explain why a computer can therefore stop testing a pixel as soon as $|z_n| > 2$.

[4]

(e) Show that every c in the Mandelbrot set satisfies $|c| \leq 2$, so the whole picture fits inside a circle of radius 2. [2]

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Second edition, September 2026